

PART X - MIGRATION, DELIVERY, AND FINAL ARCHITECTURE

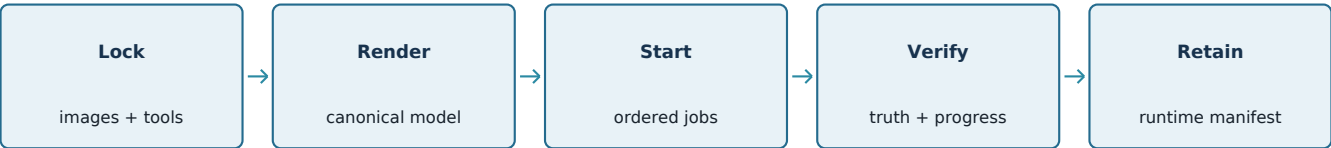
55. The Reproducible Docker Compose Runtime

One command is reproducible only when it creates the declared topology, pinned software, deterministic data, readiness order and evidence interfaces - and retains enough identity to prove what actually ran.

CHAPTER DECISION

Package the NEXUS-1 distributed slice as a fail-closed Docker Compose runtime. Core services remain profile-free; observability, evidence and tools are explicit profiles. Database migration and seed jobs complete before applications start, while a second readiness oracle verifies durable progress beyond container health.

PROOF PATH 55-A - FROM SOURCE TREE TO ADMITTED RUNTIME



GUARANTEE LEDGER 55-A

Evidence	Allowed claim	Required proof	Cannot claim
image lock	exact OCI content requested	digest references	host is identical
Compose model	declared topology rendered	canonical config digest	all services became ready
seed receipt	named corpus is present	seed ID + canonical digest	production-like data
runtime manifest	named local run admitted	fresh probes + identities	production readiness

EVIDENCE VERDICT

SPECIFIED | RUNTIME PENDING. This chapter defines the executable assets and gates; no repository build or retained Compose run is fabricated.

ORDINARY DEVELOPER PROBLEM

It worked yesterday because yesterday was a different runtime

A developer runs Compose and sees green containers. Another developer runs the same file and receives different image tags, a database with old volumes, a new random seed and a worker that started before migration. Both commands succeeded, yet they did not create the same experiment.

FAILURE LEDGER 55-1

Shortcut	Hidden variation	Correction
image: latest	publisher moved the tag	reviewed digest lock
docker compose up	unknown profiles and stale volumes	typed command + manifest
sleep 30	machine-dependent timing	health + stable readiness
random demo data	different identities/order	canonical seed corpus
all ports published	host collisions and bypass paths	minimal ingress inventory
green health	backlog or migration may lag	durable experiment-ready probe

PLAIN-LANGUAGE PRINCIPLE

Reproducibility is a controlled equivalence claim. It does not mean identical clock timings; it means the differences that matter are declared, bounded and retained.

CHAPTER BOUNDARY

Local runtime here; extraction strategy next

This chapter owns local composition, image and configuration identity, data bootstrap, startup order, evidence interfaces and lifecycle commands. Chapter 56 will use that runtime to implement a reversible strangler seam and coexistence with the modular path.

DECISION RECORD

ADR-055 - Compose is a reproducibility envelope, not an operations platform

NEXUS-1 uses Docker Compose to make the educational distributed slice runnable with one admitted command. The source model, digest lock, migrations, deterministic seed, readiness oracle and evidence manifest are versioned together. Compose mechanisms do not replace domain invariants, production orchestration or retained runtime proof.

DECISION CONSEQUENCES 55-B

Choice	Benefit	Cost / boundary
one typed launcher	stable user contract	launcher must be tested
digest-pinned images	reviewable content identity	repinning is deliberate maintenance
service-owned databases	ownership visible in topology	more containers and migrations
one-shot bootstrap jobs	ordered schema/data admission	restart semantics need care
profile overlays	fault/tooling authority isolated	rendered combinations need tests
manifest after verification	claim is inspectable	missing evidence fails closed

NON-GOAL

The chapter does not present Compose as production scheduling, secret distribution, high availability, backup, multi-host networking or capacity evidence.

REPRODUCIBILITY CONTRACT

Five equivalences define the claim

EQUIVALENCE MODEL 55-C

Dimension	Must remain equivalent	Allowed difference
topology	logical services, networks, volumes, ownership	container IDs and IPs
software	image/config/migration/seed digests	pull/build duration
data	canonical synthetic facts and order	physical page layout
readiness	dependency order and admitted predicates	time inside budgets
evidence	probe endpoints, labels and target map	trace/span IDs

EXECUTABLE ASSET 55-A - EQUIVALENCE IDENTITY

```
public sealed record RuntimeIdentity(
    string SourceRevision,
    string ComposeModelSha256,
    string ImageLockSha256,
    string ConfigurationSha256,
    string MigrationSetSha256,
    string SeedManifestSha256,
    string TopologyContractVersion);

public static bool IsEquivalent(RuntimeIdentity a, RuntimeIdentity b) =>
    a.ComposeModelSha256 == b.ComposeModelSha256 &&
    a.ImageLockSha256 == b.ImageLockSha256 &&
    a.ConfigurationSha256 == b.ConfigurationSha256 &&
    a.MigrationSetSha256 == b.MigrationSetSha256 &&
    a.SeedManifestSha256 == b.SeedManifestSha256 &&
    a.TopologyContractVersion == b.TopologyContractVersion;
```

DELIBERATE EXCLUSION

SourceRevision is retained for lineage but is not alone the runtime equivalence predicate: documentation-only commits may differ while all executable identities remain equal.

ONE-COMMAND CONTRACT

The launcher owns a stable verb vocabulary

EXECUTABLE ASSET 55-B - PUBLIC COMMAND SURFACE

```
# admitted happy path
dotnet run --project eng/Nexus.Runtime -- \
  up --profile observability --seed phase0-v1

# inspect without mutation
dotnet run --project eng/Nexus.Runtime -- status
dotnet run --project eng/Nexus.Runtime -- doctor

# retain named volumes / explicitly destroy synthetic volumes
dotnet run --project eng/Nexus.Runtime -- down
dotnet run --project eng/Nexus.Runtime -- reset --confirm-synthetic-data

# Chapter 54-compatible runtime
dotnet run --project eng/Nexus.Runtime -- \
  up --profile observability --profile evidence --seed phase0-v1
```

COMMAND GUARANTEES 55-D

Verb	Mutation	Postcondition
doctor	none	host prerequisites admitted or diagnostics returned
up	create/start	experiment-ready manifest retained
status	none	actual vs expected inventory reported
down	stop/remove runtime	named data volumes retained
reset	remove runtime + volumes	only guarded synthetic project destroyed

EXIT-CODE RULE

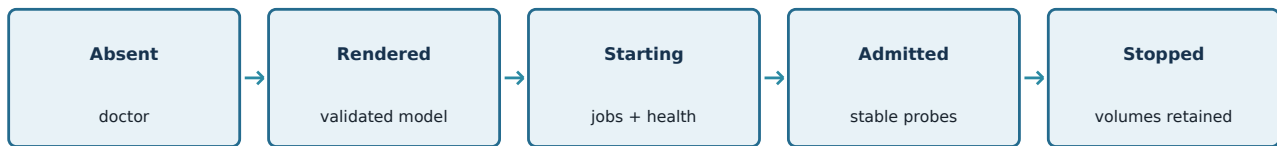
Zero means the verb's declared postcondition is true. A merely successful child process cannot turn a failed readiness or evidence check into exit code zero.

FROM SERVICES TO RUNTIME

LAUNCHER STATE MACHINE

Every transition is explicit and recoverable

RUNTIME LIFECYCLE 55-B



TRANSITION FAILURES 55-E

Transition	Failure result	Required artifact
Absent -> Rendered	INVALID_CONFIGURATION	diagnostic + source paths
Rendered -> Starting	START_FAILED	Compose events/log tail
Starting -> Admitted	READINESS_TIMEOUT	last probe snapshot
Admitted -> Stopped	STOP_INCOMPLETE	actual container inventory
Stopped -> Absent	RESET_REFUSED/FAILED	guard decision + inventory

EXECUTABLE ASSET 55-C - TYPED OUTCOMES

```

public enum RuntimeExit
{
    Admitted = 0, InvalidArguments = 2, PrerequisiteMissing = 3,
    InvalidConfiguration = 4, StartFailed = 5,
    ReadinessTimeout = 6, EvidenceInvalid = 7,
    ResetRefused = 8, CleanupIncomplete = 9
}

public sealed record RuntimeOutcome(
    RuntimeExit Exit, string Code, string Message,
    string? ManifestPath, IReadOnlyList<Diagnostic> Diagnostics);
  
```

TOPOLOGY CONTRACT

The runtime inventory is reviewed before YAML

CORE TOPOLOGY 55-F

Logical service	Role	Owned dependency
alarm-api	commands and alarm truth	alarm-db
alarm-worker	Inbox/outbox and broker work	alarm-db + RabbitMQ
rootcause-api	case commands and truth	rootcause-db
rootcause-worker	facts and policy execution	rootcause-db + RabbitMQ
projection-worker	reporting projection	projection-db + RabbitMQ
projection-api	published reporting reads	projection-db (read only)
query-bff	external query composition	internal APIs + projection-api

PLATFORM AND JOBS 55-G

Component	Purpose	Lifecycle
three database containers	service-owned persistence	long-running
rabbitmq	public fact transport	long-running
otel-collector	OTLP ingress and routing	observability profile
prometheus / tempo / grafana	metrics, traces, inspection	observability profile
three migration jobs	schema authority	complete successfully
seed-bootstrap	canonical synthetic corpus	complete successfully

TOPOLOGY RULE

A schema is not shared merely because containers share a host or network. Each service principal reaches only its owned database; migration principals are separate and short-lived.

OWNERSHIP MADE VISIBLE

Every durable store has one runtime writer boundary

DATABASE MAP 55-H

Database service	Runtime writer	Migration job	Forbidden access
alarm-db	alarm-api/worker	alarm-migrate	rootcause/projection/BFF SQL
rootcause-db	rootcause-api/worker	rootcause-migrate	alarm/projection/BFF SQL
projection-db	projection-worker writes; projection-api reads	projection-migrate	command-service access

EXECUTABLE ASSET 55-D - CONNECTION BOUNDARY

```
services:
  alarm-api:
    environment:
      ConnectionStrings_AlarmDbFile: /run/secrets/alarm_runtime_dsn
    secrets: [alarm_runtime_dsn]
    networks: [alarm-data, service-mesh]

  rootcause-api:
    environment:
      ConnectionStrings_RootCauseDbFile: /run/secrets/rootcause_runtime_dsn
    secrets: [rootcause_runtime_dsn]
    networks: [rootcause-data, service-mesh]
```

TESTABLE PROHIBITION

Rendered configuration tests reject any application attached to another service's data network or granted another database secret. Naming alone is not evidence of isolation.

NETWORKS AND INGRESS

Internal reachability is not host publication

NETWORK INVENTORY 55-I

Network	Members	Host exposure
edge	query-bff	BFF HTTP only
service-mesh	BFF, APIs, workers, broker, collector	none
alarm-data	alarm API/worker/migrate + alarm DB	none
rootcause-data	rootcause API/worker/migrate + DB	none
projection-data	projection API/worker/migrate + DB	none
observability	collector, Prometheus, Tempo, Grafana	Grafana only

EXECUTABLE ASSET 55-E - EXPLICIT LOCAL PORTS

```
services:
  query-bff:
    ports: ["127.0.0.1:${NEXUS_BFF_PORT:-8080}:8080"]
  grafana:
    profiles: [observability]
    ports: ["127.0.0.1:${NEXUS_GRAFANA_PORT:-3000}:3000"]
  rabbitmq-tools:
    profiles: [tools]
    ports: ["127.0.0.1:${NEXUS_RABBIT_UI_PORT:-15672}:15672"]
```

BYPASS RULE

Application APIs, databases, OTLP receivers and the broker protocol are not published to the host by default. Tools that need them join the appropriate internal network.

VOLUMES AND PERSISTENCE

Named volumes make retention a command decision

VOLUME CATALOGUE 55-J

Volume	Owner	Retained by down?	Removed by reset?
alarm-data	alarm-db	yes	yes
rootcause-data	rootcause-db	yes	yes
projection-data	projection-db	yes	yes
rabbit-data	rabbitmq	yes	yes
prometheus-data	Prometheus	yes	yes
grafana-data	Grafana	yes	yes

EXECUTABLE ASSET 55-F - STABLE VOLUME DECLARATIONS

```
volumes:
  alarm-data: { labels: { nexus1.data: synthetic } }
  rootcause-data: { labels: { nexus1.data: synthetic } }
  projection-data: { labels: { nexus1.data: synthetic } }
  rabbit-data: { labels: { nexus1.data: synthetic } }
  prometheus-data: { labels: { nexus1.data: synthetic } }
  grafana-data: { labels: { nexus1.data: synthetic } }
```

NO ANONYMOUS TRUTH

Durable components use named volumes. Anonymous volumes are rejected because a later up cannot reliably identify and remount them.

IMAGE IDENTITY

Tags explain; digests select

A human-readable tag may remain beside a digest, but every executed image reference resolves to immutable OCI content. The repository stores a reviewed lock, and the launcher rejects missing, duplicate or tag-only entries.

EXECUTABLE ASSET 55-G - IMAGE LOCK SHAPE

```
{
  "schema": "nexus1.images/v1",
  "images": {
    "alarm-api": "registry.example/nexus/alarm-api:0.1.0@sha256:<64-hex>",
    "query-bff": "registry.example/nexus/query-bff:0.1.0@sha256:<64-hex>",
    "rabbitmq": "rabbitmq:4-management@sha256:<64-hex>",
    "otel-collector": "otel/opentelemetry-collector-contrib:<tag>@sha256:<64-hex>",
    "prometheus": "prom/prometheus:<tag>@sha256:<64-hex>",
    "tempo": "grafana/tempo:<tag>@sha256:<64-hex>",
    "grafana": "grafana/grafana:<tag>@sha256:<64-hex>"
  }
}
```

PLACEHOLDER BOUNDARY

The references above specify the lock format, not real digests. Publication replaces every <64-hex> token with a registry-resolved, reviewed digest; the verifier refuses placeholders.

IMAGE LOCK VALIDATION

The launcher rejects mutable or incomplete software identity

EXECUTABLE ASSET 55-H - STRICT REFERENCE VERIFIER

```
private static readonly Regex Pinned = new(
    @"^[a-z0-9./:_-]+@sha256:[a-f0-9]{64}$",
    RegexOptions.CultureInvariant | RegexOptions.Compiled);

public static void Validate(ImageLock value, ISet<string> required)
{
    var keys = value.Images.Keys.ToHashSet(StringComparer.Ordinal);
    Require(keys.SetEquals(required), "image lock inventory mismatch");
    foreach (var (logical, reference) in value.Images)
    {
        Require(Pinned.IsMatch(reference), $"{logical}: image is not pinned");
        Require(!reference.Contains("<", StringComparison.Ordinal),
            $"{logical}: placeholder survived");
    }
}
```

LOCK GATES 55-K

Gate	Rejects
exact logical key set	undeclared or missing images
sha256 syntax	tag-only and malformed references
canonical JSON digest	format/order ambiguity
registry allowlist	unexpected source registries
platform inventory	unsupported host architecture

UPDATE RULE

Repinning is reviewed as a software change. The old and new digest, upstream release reason, vulnerability impact and clean-room result belong in the change evidence.

APPLICATION IMAGE IDENTITY

Local source is built once, then addressed immutably

EXECUTABLE ASSET 55-I - DETERMINISTIC BUILD LABELS

```
ARG SOURCE_REVISION
ARG BUILD_VERSION
LABEL org.opencontainers.image.revision=$SOURCE_REVISION \
    org.opencontainers.image.version=$BUILD_VERSION \
    org.opencontainers.image.source="https://github.com/gregory82gr/Nexus-1-phase-0"

FROM mcr.microsoft.com/dotnet/aspnet:<tag>@sha256:<runtime-digest>
WORKDIR /app
COPY --from=publish /out/ ./
USER 1654
ENTRYPOINT ["dotnet", "Nexus.Alarm.Api.dll"]
```

BUILD IDENTITY 55-L

Value	Source	Runtime check
source revision	version control	OCI label + manifest
package lock	locked restore	digest retained
base image	Dockerfile digest	resolved image inventory
application image	post-build RepoDigest	Compose rendered model
configuration	canonical safe projection	startup identity endpoint

BUILD BOUNDARY

Chapter 55 specifies identity and consumption. Chapter 57 will own the CI gates that produce, scan, sign where applicable and publish these images.

BASE COMPOSE MODEL

A project name and required variables remove ambient defaults

EXECUTABLE ASSET 55-J - COMPOSE.YAML ROOT

```
name: ${NEXUS_PROJECT:?project name required}

x-runtime: &runtime
  init: true
  read_only: true
  security_opt: ["no-new-privileges:true"]
  tmpfs: ["/tmp"]
  restart: unless-stopped
  labels:
    nexus1.runtime: "phase0"
    nexus1.topology-contract: "v1"

services: {}
networks:
  edge: {}
  service-mesh: { internal: true }
  alarm-data: { internal: true }
  rootcause-data: { internal: true }
  projection-data: { internal: true }
  observability: { internal: true }
```

EXTENSION-FIELD RULE

Anchors reduce repetition but do not hide authority. Rendered-model tests inspect the expanded result, because reviews of a YAML anchor alone cannot prove every service inherited the intended settings.

APPLICATION SERVICE PATTERN

Health, identity and least authority travel together

EXECUTABLE ASSET 55-K - ALARM API SERVICE

```
services:
  alarm-api:
    <<: *runtime
    image: ${ALARM_API_IMAGE:?image lock required}
    environment:
      ASPNETCORE_URLS: http://+:8080
      OTEL_EXPORTER_OTLP_ENDPOINT: http://otel-collector:4317
      ConnectionStrings__AlarmDbFile: /run/secrets/alarm_runtime_dsn
    secrets: [alarm_runtime_dsn]
    networks: [service-mesh, alarm-data]
    expose: ["8080"]
    healthcheck:
      test: ["CMD", "/app/healthcheck", "http://localhost:8080/health/ready"]
      interval: 5s
      timeout: 2s
      retries: 12
      start_period: 15s
    depends_on:
      alarm-migrate: { condition: service_completed_successfully }
```

PROBE PACKAGING

The image includes a small, versioned healthcheck executable. The runtime does not assume curl, wget or a shell exists in the application image; telemetry availability is verified without making an optional profile a core startup dependency.

DATABASE SERVICE PATTERN

Health proves connectivity, not schema admission

EXECUTABLE ASSET 55-L - OWNED SQL SERVER CONTAINER

```
services:
  alarm-db:
    image: ${NEXUS_SQL_IMAGE:?image lock required}
    entrypoint: ["/nexus/sql-entrypoint"]
    environment:
      ACCEPT_EULA: "Y"
      MSSQL_PID: Developer
      NEXUS_SA_PASSWORD_FILE: /run/secrets/alarm_sa_password
    secrets: [alarm_sa_password]
    networks: [alarm-data]
    volumes: ["alarm-data:/var/opt/mssql"]
    healthcheck:
      test:
        - CMD-SHELL
        - >>
          SQLCMDPASSWORD=$(cat /run/secrets/alarm_sa_password)
          /opt/mssql-tools18/bin/sqlcmd -C -S localhost -U sa
          -Q "SELECT 1" -b
      interval: 5s
      timeout: 3s
      retries: 24
      start_period: 20s
    labels:
      nexus1.owner: alarm-management
      nexus1.data: synthetic
```

TWO DIFFERENT CLAIMS

The pinned local SQL image adds a reviewed entrypoint that reads NEXUS_SA_PASSWORD_FILE before execing SQL Server; it is a local secret adapter, not an invented vendor variable. Database health admits the engine. The migration receipt separately admits the expected schema.

BROKER SERVICE PATTERN

RabbitMQ health and topology are verified separately

EXECUTABLE ASSET 55-M - BROKER

```
services:
  rabbitmq:
    image: ${RABBITMQ_IMAGE:?image lock required}
    hostname: rabbitmq
    environment:
      RABBITMQ_DEFAULT_USER_FILE: /run/secrets/rabbit_user
      RABBITMQ_DEFAULT_PASS_FILE: /run/secrets/rabbit_password
    secrets: [rabbit_user, rabbit_password]
    networks: [service-mesh]
    volumes: ["rabbit-data:/var/lib/rabbitmq"]
    healthcheck:
      test: ["CMD", "rabbitmq-diagnostics", "-q", "ping"]
      interval: 5s
      timeout: 5s
      retries: 18
      start_period: 15s
    labels:
      nexus1.target: broker
      nexus1.fault-injection: denied
```

BROKER ADMISSION 55-M

Probe	Question
container health	is the node responsive?
declaration receipt	do expected exchanges/queues/bindings exist?
publisher control	can a confirm be received?
consumer control	can one synthetic message be acknowledged?

TELEMETRY PROFILE

The Collector is the only application telemetry ingress

EXECUTABLE ASSET 55-N - OBSERVABLE PIPELINE

```
services:
  otel-collector:
    image: ${OTEL_COLLECTOR_IMAGE:?image lock required}
    profiles: [observability]
    command: ["--config=/etc/otelcol/config.yaml"]
    configs: [otel_config]
    networks: [service-mesh, observability]
    expose: ["4317", "4318", "13133"]
    healthcheck:
      test: ["CMD", "/otelcol-contrib", "validate", "--config=/etc/otelcol/config.yaml"]
      interval: 10s
      timeout: 5s
      retries: 6

configs:
  otel_config: { file: ./deploy/otel/collector.yaml }
```

PIPELINE CONTRACT 55-N

Signal	Receiver	Processor	Destination
traces	OTLP	memory limiter + batch	Tempo
metrics	OTLP	memory limiter + batch	Prometheus exporter
logs	OTLP	redaction + batch	debug/file evidence sink

VALIDATION NUANCE

Configuration validation proves the Collector can parse the model. A separate freshness probe must prove that a control signal traversed the active pipeline.

BFF INGRESS

The browser sees one application boundary

EXECUTABLE ASSET 55-O - QUERY BFF

```
services:
  query-bff:
    <<: *runtime
    image: ${QUERY_BFF_IMAGE:?image lock required}
    environment:
      Services__Alarm: http://alarm-api:8080
      Services__RootCause: http://rootcause-api:8080
      Services__Projection: http://projection-api:8080
    networks: [edge, service-mesh]
    ports: ["127.0.0.1:${NEXUS_BFF_PORT:-8080}:8080"]
    depends_on:
      alarm-api: { condition: service_healthy }
      rootcause-api: { condition: service_healthy }
      projection-api: { condition: service_healthy }
    healthcheck:
      test: ["CMD", "/app/healthcheck", "http://localhost:8080/health/ready"]
```

NO DATABASE SHORTCUT

The BFF composes APIs and published read contracts. It never receives database credentials, joins data networks or queries command-service tables.

PROFILES

Core is unconditional; optional authority is named

PROFILE CATALOGUE 55-O

Profile	Adds	Default?	Authority
core	profile-free services	yes	ordinary runtime
observability	collector/backends/Grafana	explicit	read telemetry
tools	management UIs and inspectors	explicit	manual inspection
evidence	runner, proxy and fault labels	explicit	controlled mutation

EXECUTABLE ASSET 55-P - EXPLICIT PROFILE SELECTION

```

var admitted = new HashSet<string>(StringComparer.Ordinal)
    { "observability", "tools", "evidence" };

foreach (var profile in options.Profiles)
    Require(admitted.Contains(profile), $"unknown profile: {profile}");

// Core services carry no profiles property and therefore always start.
var args = options.Profiles
    .Order(StringComparer.Ordinal)
    .SelectMany(p => new[] { "--profile", p })
    .ToArray();

```

PROFILE DEPENDENCY RULE

An optional service and every optional dependency it needs are assigned compatible profiles; rendered profile combinations are validated, not inferred from YAML indentation.

EVIDENCE OVERLAY

Automatic restart is disabled when restart itself is the experiment

EXECUTABLE ASSET 55-Q - COMPOSE.EVIDENCE.YAML

```

services:
  rabbitmq:
    restart: "no"
    labels:
      nexus1.target: broker
      nexus1.fault-injection: allowed
  alarm-worker:
    restart: "no"
    labels:
      nexus1.target: alarm-consumer
      nexus1.fault-injection: allowed
  rootcause-worker:
    restart: "no"
    labels:
      nexus1.target: rootcause-consumer
      nexus1.fault-injection: allowed
  projection-worker:
    restart: "no"
    labels:
      nexus1.target: projection-consumer
      nexus1.fault-injection: allowed
  query-bff:
    restart: "no"
    labels:
      nexus1.target: query-service
      nexus1.fault-injection: allowed
  toxiproxy:
    image: ${TOXIPROXY_IMAGE:?image lock required}
    profiles: [evidence]
    networks: [service-mesh, alarm-data, rootcause-data, projection-data]

```

EXPERIMENT TRUTH

If restart: unless-stopped remains active, a kill-consumer campaign may measure the restart policy rather than backlog behavior. The rendered evidence model must prove restart is disabled for fault targets.

CANONICAL ACTUATOR TARGETS

Chapter 54 names resolve through labels, never guessed containers

TARGET MAP 55-P

Experiment target	Compose service	Required label	Independent probe
alarm-consumer	alarm-worker	fault-injection=allowed	container not running
broker	rabbitmq	fault-injection=allowed	publisher path fails
alarm-database	toxiproxy route	fault-injection=allowed	toxic active
projection-delay	projection barrier	evidence=true	checkpoint stops
query-service	query-bff	fault-injection=allowed	epoch changes

EXECUTABLE ASSET 55-R - UNIQUE TARGET RESOLUTION

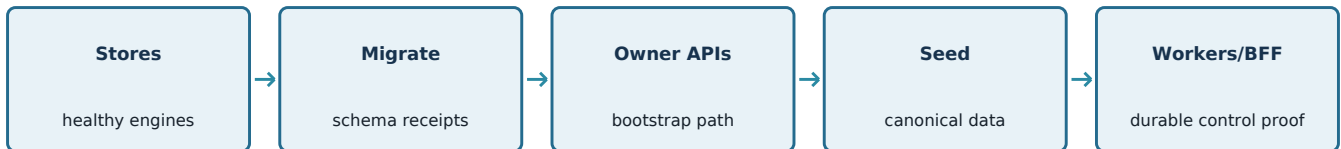
```
var matches = inventory.Containers
    .Where(c => c.Project == project)
    .Where(c => c.Labels.GetValueOrDefault("nexus1.target") == logicalTarget)
    .Where(c => c.Labels.GetValueOrDefault("nexus1.fault-injection") == "allowed")
    .ToArray();

Require(matches.Length == 1,
    $"target '{logicalTarget}' resolved to {matches.Length} containers");
return matches[0];
```

READINESS DEPENDENCY GRAPH

Start order follows authority and completed work

DEPENDENCY PATH 55-C



DEPENDENCY EDGES 55-Q

Dependent	Dependency	Condition
migration jobs	owned database	service_healthy
seed-bootstrap	migrations, broker declaration + owner APIs	completed + service_healthy
owner APIs	owned migration	service_completed_successfully
workers/BFF	seed + required APIs	completed + service_healthy
runtime verifier	BFF + workers + telemetry	service_healthy, then semantic probes

NO DEPENDENCY CYCLE

Telemetry export must not become a hard prerequisite for database migration. The observability profile is verified, but core business startup remains diagnosable if a backend is unavailable.

COMPOSE DEPENDENCY CONDITIONS

Healthy services and successful jobs are different edges

EXECUTABLE ASSET 55-S - ORDERED STARTUP

```

services:
  alarm-migrate:
    image: ${ALARM_MIGRATOR_IMAGE:?image lock required}
    restart: "no"
    depends_on:
      alarm-db: { condition: service_healthy }

  seed-bootstrap:
    image: ${SEED_IMAGE:?image lock required}
    restart: "no"
    depends_on:
      alarm-migrate: { condition: service_completed_successfully }
      rootcause-migrate: { condition: service_completed_successfully }
      projection-migrate: { condition: service_completed_successfully }
      topology-declare: { condition: service_completed_successfully }
      alarm-api: { condition: service_healthy }
      rootcause-api: { condition: service_healthy }

  alarm-worker:
    depends_on:
      seed-bootstrap: { condition: service_completed_successfully }
      rabbitmq: { condition: service_healthy }

```

COMPLETED MEANS ZERO

A migration or seed container exiting is not success. Only exit code zero satisfies `service_completed_successfully`; a non-zero job blocks dependents and remains visible in diagnostics.

MIGRATION AUTHORITY

Applications never migrate on startup

MIGRATION CONTRACT 55-R

Principal	May	May not
migration	DDL + migration history	serve requests or consume facts
runtime	DML within owned schema	DDL or migration history writes
inspection	bounded read-only diagnostics	write or own schema

EXECUTABLE ASSET 55-T - MIGRATION JOB ARGUMENTS

```

services:
  alarm-migrate:
    image: ${ALARM_MIGRATOR_IMAGE:?image lock required}
    command:
      - migrate
      - --connection-file=/run/secrets/alarm_migration_dsn
      - --expected-set=${ALARM_MIGRATION_SET_SHA256:?required}
      - --receipt=evidence/alarm-migration.json
    secrets: [alarm_migration_dsn]
    volumes: ["runtime-evidence:/evidence"]
    networks: [alarm-data]
    read_only: true
    tmpfs: ["/tmp"]

```

ONE WRITER

Only one alarm-migrate instance runs under the launcher lease. Provider-level migration locking remains defense in depth, not a substitute for orchestration ownership.

MIGRATION RECEIPT

Schema admission is machine-readable and fail-closed

EXECUTABLE ASSET 55-U - MIGRATION EVIDENCE

```
public sealed record MigrationReceipt(
    string Service,
    string DatabaseIdentity,
    string ExpectedSetSha256,
    string[] AppliedBefore,
    string[] AppliedNow,
    string[] PendingAfter,
    DateTimeOffset CompletedUtc,
    int ExitCode);

public static void Admit(MigrationReceipt r, string expected)
{
    Require(r.ExitCode == 0, $"{r.Service}: migration failed");
    Require(r.ExpectedSetSha256 == expected, $"{r.Service}: set drift");
    Require(r.PendingAfter.Length == 0, $"{r.Service}: pending migrations");
}
```

MIGRATION RERUN 55-S

Existing state	Action	Outcome
empty database	apply complete ordered set	receipt lists AppliedNow
same complete set	validate only	zero-change success
unknown migration	stop	SCHEMA_DIVERGED
expected migration missing	stop	MIGRATION_SET_MISMATCH

DETERMINISTIC SEED MANIFEST

Synthetic data is a versioned input, not startup improvisation

EXECUTABLE ASSET 55-V - SEED MANIFEST

```
{
  "schema": "nexus1.seed/v1",
  "seedId": "phase0-v1",
  "clockUtc": "2026-01-15T09:00:00Z",
  "tenantId": "00000000-0000-0000-0000-000000000101",
  "alarms": [
    {
      "id": "00000000-0000-0000-0000-000000000101",
      "occurredUtc": "2026-01-15T08:55:00Z",
      "code": "P-101-HI"
    }
  ],
  "rootCauseCases": [
    {
      "id": "00000000-0000-0000-0000-0000000002001",
      "alarmId": "00000000-0000-0000-0000-000000000101"
    }
  ]
}
```

DETERMINISM RULES 55-T

Field	Rule
identities	fixed UUIDs within a named corpus
time	manifest clock, never host Now
ordering	canonical key/order before hashing and issue
messages	derived stable IDs and causation
digest	RFC-style canonical JSON projection + SHA-256

DETERMINISTIC DATA CONSTRUCTION

Randomness is injected only when the scenario names it

EXECUTABLE ASSET 55-W - SEED CLOCK AND IDENTITY

```
public sealed record SeedContext(
    string SeedId, DateTimeOffset ClockUtc, byte[] CanonicalBytes,
    string Sha256);

public static Alarm CreateAlarm(SeedAlarm row, SeedContext context)
{
    var occurred = DateTimeOffset.ParseExact(
        row.OccurredUtc, "0", CultureInfo.InvariantCulture,
        DateTimeStyles.RoundtripKind);
    Require(occurred <= context.ClockUtc, "future seed alarm");
    return Alarm.RestoreForSeed(
        AlarmId.From(Guid.Parse(row.Id)), row.Code, occurred,
        importedAt: context.ClockUtc);
}

// Forbidden in seed code: Guid.NewGuid(), Random.Shared, UtcNow.
```

WORKLOAD VARIATION

A fault campaign may use a declared integer seed to generate repeatable workload variation. The base corpus remains canonical; generated requests retain the workload algorithm version and seed.

SEED ADMISSION STATE MACHINE

Same identity is a no-op; same name with different bytes is a conflict

SEED STATES 55-U

Database receipt	Requested manifest	Decision
absent	phase0-v1 / digest A	apply in owner transactions
phase0-v1 / digest A	phase0-v1 / digest A	validate and no-op
phase0-v1 / digest A	phase0-v1 / digest B	SEED_ID_CONFLICT
phase0-v2 / digest B	phase0-v1 / digest A	SEED_DOWNGRADE_REFUSED
partial receipt	any	SEED_INCOMPLETE; reset or repair explicitly

EXECUTABLE ASSET 55-X - IDEMPOTENT DECISION

```
return existing switch
{
    null => SeedDecision.Apply,
    { Completed: true } r when r.SeedId == requested.SeedId &&
        r.ManifestSha256 == requested.Sha256 => SeedDecision.ValidateNoOp,
    { SeedId: var id } when id == requested.SeedId =>
        throw new SeedConflictException(id),
    { Completed: false } => throw new IncompleteSeedException(),
    _ => throw new SeedTransitionRefusedException()
};
```

SEED BOOTSTRAP JOB

Bootstrap uses public ownership paths, not cross-database SQL

EXECUTABLE ASSET 55-Y - SEED JOB

```

services:
  seed-bootstrap:
    image: ${SEED_IMAGE:?image lock required}
    command:
      - seed
      - --manifest=/seed/${NEXUS_SEED_ID:?seed required}.json
      - --alarm=http://alarm-api:8080/internal/bootstrap
      - --rootcause=http://rootcause-api:8080/internal/bootstrap
      - --receipt=/evidence/seed.json
    configs: [seed_phase0_v1]
    secrets: [bootstrap_token]
    volumes: ["runtime-evidence:/evidence"]
    networks: [service-mesh]
    restart: "no"

configs:
  seed_phase0_v1: { file: ./deploy/seed/phase0-v1.json }

```

BOOTSTRAP ENDPOINT BOUNDARY

The internal endpoints exist only in Local/Evidence composition, require a short-lived bootstrap credential and enforce seed state. They are absent from production endpoint inventory.

READINESS VOCABULARY

Five layers prevent a green container from overclaiming

READINESS LADDER 55-V

Layer	Question	Evidence
running	did the process start?	container state
healthy	does the local health contract pass?	healthcheck result
application-ready	can the service accept work?	/health/ready
durable-ready	did control work reach owned truth?	owner snapshots + receipts
experiment-ready	are labels/probes/telemetry/fault paths admitted?	profile-specific oracle

LIVENESS DISCIPLINE

Liveness remains process-local. A broker or database outage can make readiness fail without making liveness fail and causing a restart storm.

EXECUTABLE ASSET 55-Z - ENDPOINT MAPPING

```

app.MapHealthChecks("/health/live", new() {
  Predicate = check => check.Tags.Contains("live") });
app.MapHealthChecks("/health/ready", new() {
  Predicate = check => check.Tags.Contains("ready") });

```


STABLE READINESS ORACLE

Admission requires consecutive fresh observations

EXECUTABLE ASSET 55-AA - BOUNDED STABLE POLL

```
public async Task<ReadinessSnapshot> AwaitStableAsync(
    RuntimeContract expected, TimeSpan budget, CancellationToken token)
{
    var deadline = clock.GetTimestamp() + clock.ToTicks(budget);
    ReadinessSnapshot? last = null;
    var stable = 0;
    while (clock.GetTimestamp() < deadline)
    {
        token.ThrowIfCancellationRequested();
        last = await probes.CaptureAsync(expected, token);
        stable = last.AllRequiredFreshAndPassing ? stable + 1 : 0;
        if (stable >= 3) return last;
        await clock.Delay(TimeSpan.FromSeconds(2), token);
    }
    throw new RuntimeReadinessException(last);
}
```

REQUIRED SEMANTIC PROBES 55-W

Probe	Pass condition
migration	three expected-set receipts, no pending migration
seed	requested ID/digest admitted by all owners
broker	topology digest + publisher/consumer control
projection	control event checkpoint reaches high-water
telemetry	control trace visible and pipeline fresh when profile enabled

COMPOSE WAIT

--wait is necessary, then the NEXUS oracle finishes the claim

EXECUTABLE ASSET 55-AB - UNDERLYING START COMMAND

```
docker compose \
  --project-name nexus1-local \
  --file deploy/compose.yaml \
  --file deploy/compose.local.yaml \
  --profile observability \
  up --detach --wait --wait-timeout 180 --remove-orphans
```

WAIT BOUNDARY 55-X

Mechanism	What it proves	Next check
depends_on	creation order / declared conditions	job receipts
healthcheck	container-local predicate	fresh app identity
up --wait	project reached running/healthy	durable oracle
runtime verifier	topology/data/progress/evidence stable	manifest admission

NO ARBITRARY SLEEP

A timeout is a maximum budget, never a substitute for a predicate. Failure retains the last actual snapshot, failing component, logs and Compose state.

TYPED LAUNCHER OPTIONS

Ambient environment becomes explicit input

EXECUTABLE ASSET 55-AC - COMMAND OPTIONS

```
public sealed record UpOptions(
    string Project,
    string SeedId,
    IReadOnlySet<string> Profiles,
    TimeSpan StartBudget,
    bool PullMissing,
    string OutputDirectory);

public static UpOptions Defaults() => new(
    Project: "nexus1-local",
    SeedId: "phase0-v1",
    Profiles: new HashSet<string> { "observability" },
    StartBudget: TimeSpan.FromMinutes(3),
    PullMissing: false,
    OutputDirectory: "artifacts/runtime");
```

INPUT POLICY 55-Y

Input	Rule
project	lowercase safe pattern; explicit evidence prefix when needed
seed	must exist in seed catalogue
profiles	allowlisted and sorted
ports	bounded local values; preflight ownership check
output	inside repository artifact root; no traversal

ORCHESTRATOR PIPELINE

One command is a sequence of admitted postconditions

EXECUTABLE ASSET 55-AD - UP ORCHESTRATION

```
public async Task<RuntimeOutcome> UpAsync(
    UpOptions options, CancellationToken token)
{
    await using var lease = await leases.AcquireAsync(options.Project, token);
    var inputs = await preflight.AdmitAsync(options, token);
    var rendered = await compose.RenderAndValidateAsync(inputs, token);
    await images.AdmitAsync(rendered.Images, inputs.ImageLock, token);

    var start = await compose.UpWaitAsync(rendered, options.StartBudget, token);
    if (!start.Succeeded)
        return await evidence.StartFailureAsync(inputs, rendered, start, token);

    var ready = await readiness.AwaitStableAsync(
        inputs.Contract, options.StartBudget, token);
    return await evidence.AdmitRuntimeAsync(inputs, rendered, ready, token);
}
```

CANCELLATION RULE

User cancellation stops waiting and retains diagnostics; it does not implicitly delete volumes. A separately bounded cleanup path may stop newly created containers when the command owns that transition.

SAFE PROCESS EXECUTION

Arguments are tokens; output is bounded and redacted

EXECUTABLE ASSET 55-AE - CHILD PROCESS RUNNER

```

var start = new ProcessStartInfo("docker")
{
    RedirectStandardOutput = true,
    RedirectStandardError = true,
    UseShellExecute = false,
    CreateNoWindow = true
};
foreach (var argument in arguments)
    start.ArgumentList.Add(argument);

using var process = Process.Start(start)
    ?? throw new InvalidOperationException("docker did not start");
var stdout = ReadBoundedAsync(process.StandardOutput, 1_000_000, token);
var stderr = ReadBoundedAsync(process.StandardError, 1_000_000, token);
await process.WaitForExitAsync(token);
return new ProcessReceipt(process.ExitCode,
    redactor.Apply(await stdout), redactor.Apply(await stderr));

```

NO SHELL INTERPOLATION

The launcher never concatenates a user-supplied project, path, profile or target into a shell command. `ProcessStartInfo.ArgumentList` preserves token boundaries.

PREFLIGHT DOCTOR

Failures are found before containers mutate host state

DOCTOR CHECKS 55-Z

Check	Evidence	Failure code
Docker Engine reachable	version/info receipt	ENGINE_UNAVAILABLE
Compose feature floor	parsed semantic version	COMPOSE_TOO_OLD
host architecture	platform identity	PLATFORM_UNSUPPORTED
required source/config files	path + digest inventory	INPUT_MISSING
secrets readable and private	metadata only	SECRET_INVALID
published ports free/owned	listener inventory	PORT_CONFLICT
disk/memory floor	host capacity snapshot	CAPACITY_INSUFFICIENT

EXECUTABLE ASSET 55-AF - FEATURE ADMISSION

```

var version = await docker.ComposeVersionAsync(token);
Require(version >= new SemanticVersion(2, 20, 0),
    $"Docker Compose {version} lacks admitted wait/condition behavior");
Require(await docker.EngineRespondsAsync(token), "Docker Engine unavailable");
Require(paths.All(File.Exists), "runtime input missing");

```

VERSION POLICY

The actual minimum is stored with the runtime contract and raised only with an explicit feature reason and clean-room evidence.

CANONICAL COMPOSE RENDERING

Validate the merged model that Docker will apply

EXECUTABLE ASSET 55-AG - RENDER BEFORE UP

```
docker compose \  
  --project-name nexus1-local \  
  --file deploy/compose.yaml \  
  --file deploy/compose.local.yaml \  
  --profile observability \  
  config --format json > artifacts/runtime/compose.rendered.json  
  
docker compose \  
  --project-name nexus1-local \  
  --file deploy/compose.yaml \  
  --file deploy/compose.local.yaml \  
  --profile observability \  
  config --quiet
```

RENDERED-MODEL GATES 55-AA

Inventory	Gate
services	exact core + selected profile set
images	all match admitted lock references
ports	only BFF/Grafana/tooling allowlist
networks	no cross-owner data attachment
volumes	all durable mounts named and labelled synthetic
secrets	per-service least-access inventory

LOCAL SECRETS

Compose grants files per service; evidence retains no values

EXECUTABLE ASSET 55-AH - SECRET DECLARATIONS

```
secrets:  
  alarm_runtime_dsn:  
    file: ${NEXUS_SECRET_DIR:?secret directory required}/alarm-runtime-dsn  
  alarm_migration_dsn:  
    file: ${NEXUS_SECRET_DIR:?secret directory required}/alarm-migration-dsn  
  rabbit_password:  
    file: ${NEXUS_SECRET_DIR:?secret directory required}/rabbit-password  
  
services:  
  alarm-api:  
    secrets: [alarm_runtime_dsn]  
  alarm-migrate:  
    secrets: [alarm_migration_dsn]
```

SECRET EVIDENCE 55-AB

Retain	Never retain
logical name and consuming service	secret bytes
source type and file metadata policy	absolute developer home path
presence/permission validation result	connection string or password
optional one-way version identifier	hash of low-entropy password

LOCAL-ONLY QUALIFICATION

Compose secret files improve least access and prevent ordinary environment dumps. They are not a production secret manager or automatic rotation mechanism.

CONFIGURATION IDENTITY

Hash a canonical safe projection, not secret values

EXECUTABLE ASSET 55-AI - IDENTITY PROJECTION

```
public sealed record RuntimeConfigurationIdentity(  
    string Environment,  
    string[] EnabledProfiles,  
    IReadOnlyDictionary<string, string> PolicyDigests,  
    IReadOnlyDictionary<string, string> EndpointNames,  
    IReadOnlyDictionary<string, bool> SecretPresence,  
    string CanonicalSha256);  
  
var identity = new RuntimeConfigurationIdentity(  
    Environment: "LocalEvidence",  
    EnabledProfiles: profiles.Order(StringComparer.Ordinal).ToArray(),  
    PolicyDigests: policyDigests,  
    EndpointNames: endpointNames,  
    SecretPresence: requiredSecrets.ToDictionary(x => x, File.Exists),  
    CanonicalSha256: "computed-after-canonical-serialization");
```

VALUE-FREE EVIDENCE

Endpoint names identify logical dependencies, not embedded credentials. Secret evidence records presence and access policy; it does not hash guessable secret material.

RUNTIME MANIFEST

Admission is retained as a signed-shaped, digest-verifiable record

EXECUTABLE ASSET 55-AJ - MANIFEST CONTRACT

```
public sealed record RuntimeManifest(  
    string Schema,  
    string RunId,  
    string Project,  
    RuntimeIdentity Identity,  
    string[] Profiles,  
    ServiceEvidence[] Services,  
    MigrationReceipt[] Migrations,  
    SeedReceipt Seed,  
    ReadinessSnapshot Readiness,  
    EvidenceInterface[] Interfaces,  
    DateTimeOffset AdmittedUtc,  
    string Verdict);  
  
public sealed record ServiceEvidence(  
    string LogicalName, string ContainerId, string ImageDigest,  
    string Health, string ConfigurationSha256, string BuildRevision);
```

MANIFEST RULES 55-AC

Rule	Reason
canonical JSON + SHA-256 sidecar	tamper-evident handoff
atomic write then rename	no partial admitted manifest
one run directory	logs, model, receipts and snapshots join
no secret values	evidence is safe to retain
verdict last	admission follows all mandatory sources

STATUS AND DRIFT INSPECTION

Expected and actual topology are compared by logical identity

EXECUTABLE ASSET 55-AK - STATUS RESULT

```
public sealed record RuntimeStatus(
    string Project,
    RuntimeState State,
    RuntimeIdentity? Expected,
    RuntimeIdentity? Actual,
    ServiceStatus[] Services,
    DriftFinding[] Findings);

public enum DriftKind
{
    MissingService, UnexpectedService, ImageMismatch,
    Unhealthy, NetworkMismatch, VolumeMismatch,
    ConfigurationMismatch, SeedMismatch, Unknown
}
```

STATUS EXAMPLES 55-AD

Actual observation	Result
all required services healthy; identities equal	ADMITTED
container exists with old image digest	DRIFTED / ImageMismatch
project absent	ABSENT, not failure
unknown orphan service	DRIFTED / UnexpectedService
probe unavailable	UNKNOWN, never inferred healthy

DOWN AND RESET

Stopping is ordinary; destroying data is guarded

EXECUTABLE ASSET 55-AL - LIFECYCLE MAPPING

```
// down: named volumes remain
docker compose --project-name nexus1-local \
  --file deploy/compose.yaml down --remove-orphans

// reset: only after explicit synthetic-data guard
docker compose --project-name nexus1-local \
  --file deploy/compose.yaml down --volumes --remove-orphans
```

EXECUTABLE ASSET 55-AM - RESET GUARD

```
public static void AuthorizeReset(ResetOptions o, ProjectInventory actual)
{
    Require(o.ConfirmSyntheticData, "explicit confirmation missing");
    Require(Regex.IsMatch(o.Project, @"^nexus1-(local|evidence-[a-z0-9-]+)$"),
        "unsafe project name");
    Require(actual.Volumes.Count > 0, "no resolved target volumes");
    Require(actual.Volumes.All(v => v.Labels["nexus1.data"] == "synthetic"),
        "non-synthetic or unlabelled volume");
}
```

NO BROAD DELETION

The launcher resolves exact project resources first. It never deletes by an empty variable, wildcard, home directory, workspace root or unverified label query.

SECOND-UP IDEMPOTENCY

Rerunning the command converges or reports drift

SECOND-UP MATRIX 55-AE

Existing runtime	Requested runtime	Decision
same identities, admitted	same	verify; no destructive reset
same project, image change	reviewed new lock	recreate affected containers; retain data
same project, seed conflict	different corpus bytes	refuse and explain reset/migration path
unknown containers	same topology	refuse or explicit remove-orphans policy
incomplete bootstrap	same	diagnose; never silently mark admitted

EXECUTABLE ASSET 55-AN - CONVERGENCE GATE

```
var before = await status.CaptureAsync(options.Project, token);
if (before.State == RuntimeState.Admitted &&
    before.Actual is not null && IsEquivalent(before.Actual, requested))
    return await readiness.ReverifyAndRetainAsync(before, token);

Require(!before.Findings.Any(f => f.Kind == DriftKind.Unknown),
    "runtime contains unknown drift");
return await StartOrRecreateKnownResourcesAsync(requested, before, token);
```

CLEAN-ROOM REPRODUCTION CAMPAIGN

Two independent resets must produce equivalent admitted identities

CAMPAIGN 55-AF

Step	Action	Retained result
1	doctor on clean supported host	host/tool identity
2	up observability + phase0-v1	manifest A
3	execute control workflow	owner/trace/projection snapshot A
4	guarded reset	resource absence receipt
5	repeat up with same inputs	manifest B
6	repeat control workflow	snapshot B
7	compare admitted equivalence	verdict + qualified timing differences

EXECUTABLE ASSET 55-AO - EQUIVALENCE ASSERTION

```
Assert.True(IsEquivalent(runA.Identity, runB.Identity));
Assert.Equal(runA.Seed.ManifestSha256, runB.Seed.ManifestSha256);
Assert.Equal(runA.Readiness.TopologyDigest, runB.Readiness.TopologyDigest);
Assert.Equal(runA.Control.OwnerStateDigest, runB.Control.OwnerStateDigest);
Assert.Equal(runA.Control.ProjectionDigest, runB.Control.ProjectionDigest);
// Container IDs, IPs, trace IDs and elapsed milliseconds are intentionally unequal.
```

CHAPTER 54 CAMPAIGN PATH

The evidence profile makes failure experiments executable without changing names

EXECUTABLE ASSET 55-AP - CAMPAIGN ENTRY

```
dotnet run --project eng/Nexus.Runtime -- \
  up --project nexus1-evidence-run54 \
  --profile observability --profile evidence \
  --seed phase0-v1

dotnet test tests/Nexus.FailureEvidence.Tests \
  --filter "Category=FailureEvidence" \
  --settings deploy/evidence/ch54.runsettings
```

HANDOFF CHECKS 55-AG

Chapter 54 requirement	Chapter 55 runtime asset
typed experiment catalogue	evidence runner + scenario filter
canonical target names	unique Compose labels
state probes	internal networks + read-only evidence credentials
telemetry probes	Collector/backends + freshness control
synthetic workload	phase0-v1 manifest + algorithm seed
safe cleanup	project prefix, lease, reset guard

EVIDENCE-PROFILE SAFETY

Fault authority exists only inside the isolated project

EVIDENCE GATES 55-AH

Gate	Required value
project	nexus1-evidence-<run>
data	seed receipt classification synthetic
profiles	observability + evidence
target labels	exact allowlist and unique resolution
restart policy	no for kill/restart targets
control socket	runner-side narrow adapter only
runtime manifest	fresh ADMITTED before campaign

SOCKET BOUNDARY

Application services never mount the Docker socket. If a runner adapter requires Engine control, it is external to the application trust boundary, allowlisted, time-bounded and absent from core/production composition.

EXECUTABLE ASSET 55-AQ - PRE-CAMPAIGN ADMISSION

```
Require(manifest.Project.StartsWith("nexus1-evidence-", StringComparison.Ordinal));
Require(manifest.Verdict == "ADMITTED");
Require(manifest.Seed.Classification == "synthetic");
Require(manifest.Profiles.Order().SequenceEqual(new[] { "evidence", "observability" }));
Require(targets.SetEquals(requiredExperimentTargets));
```


STATIC TOPOLOGY TESTS

The rendered model is an architecture surface

EXECUTABLE ASSET 55-AR - OWNERSHIP GATE

```
[Fact]
public async Task Applications_do_not_cross_data_networks()
{
    var model = await fixture.RenderAsync("observability");
    foreach (var service in model.ApplicationServices)
    {
        var allowed = contract.AllowedNetworksFor(service.Name);
        service.Networks.Should().BeSubsetOf(allowed);
        service.Secrets.Should().BeSubsetOf(
            contract.AllowedSecretsFor(service.Name));
    }
}

[Fact]
public async Task Only_allowlisted_ports_are_published() =>
    (await fixture.RenderAsync("observability"))
        .PublishedPorts.Should().BeEquivalentTo(contract.LocalPublishedPorts);
```

STATIC SUITE 55-AI

Test	Rejects
digest-only images	mutable tag or placeholder
exact service set	missing/undeclared component
ownership networks	cross-service data access
secret consumers	excess credential authority
profile matrix	invalid optional dependency
evidence overlay	restart or label drift

RUNTIME SMOKE TESTS

A green project must move one control fact end to end

EXECUTABLE ASSET 55-AS - ADMITTED RUNTIME FIXTURE

```
[Collection("NEXUS-1 Compose runtime")]
public sealed class RuntimeSmokeTests(ComposeRuntime runtime)
{
    [Fact]
    public async Task Control_alarm_reaches_owner_broker_and_projection()
    {
        var receipt = await runtime.Bff.CreateControlAlarmAsync(ControlAlarm.Fixed);
        var state = await runtime.Probes.AwaitStableAsync(
            s => s.AlarmCommitted(receipt.AlarmId) &&
                s.OutboxConfirmed(receipt.FactId) &&
                s.ProjectionContains(receipt.AlarmId),
            TimeSpan.FromSeconds(45));
        state.ConservationViolations.Should().BeEmpty();
    }
}
```

SMOKE SUITE 55-AJ

Test	Claim
second up	idempotent verification/convergence
control workflow	durable cross-boundary progress
telemetry control	active profile exports a fresh trace
down/up	named volumes retain expected seed/state
reset/up	clean bootstrap reproduces equivalence

DELIBERATELY BROKEN RUNTIME CAMPAIGN

A useful verifier must reject plausible green Compose projects

BROKEN VARIANTS 55-AK

Mutation	Compose appearance	Required rejection
replace digest with tag	containers healthy	IMAGE_NOT_PINNED
remove projection migration edge	worker may start	DEPENDENCY_CONTRACT_DRIFT
change seed bytes, keep seed ID	job exits non-zero	SEED_ID_CONFLICT
publish alarm database port	database healthy	PORT_SURFACE_DRIFT
enable worker restart in evidence	campaign may look recovered	EVIDENCE_RESTART_DRIFT
drop telemetry control trace	apps healthy	TELEMETRY_NOT_FRESH

EXECUTABLE ASSET 55-AT - MUTATION ASSERTION

```
[Theory]
[MemberData(nameof(BrokenRuntimeCases.All))]
public async Task Verifier_rejects_broken_runtime(BrokenRuntimeCase broken)
{
    var model = await fixture.RenderMutatedAsync(broken.Mutation);
    var result = await verifier.VerifyAsync(model, broken.Observations);
    result.Verdict.Should().Be(RuntimeVerdict.Invalid);
    result.Findings.Should().Contain(f => f.Code == broken.ExpectedCode);
}
```

FAIL-CLOSED EVIDENCE VERIFIER

Unknown is not equivalent to passed

EXECUTABLE ASSET 55-AU - ADMISSION VERDICT

```
public RuntimeVerdict Verify(RuntimeManifest m, RuntimeContract c)
{
    var findings = new List<Finding>();
    RequireExact(m.Services.Select(x => x.LogicalName), c.RequiredServices, findings);
    RequirePinnedImages(m.Services, c.ImageLock, findings);
    RequireMigrationReceipts(m.Migrations, c.MigrationSets, findings);
    RequireSeed(m.Seed, c.Seed, findings);
    RequireFreshStableReadiness(m.Readiness, c.Readiness, findings);
    RequireInterfaces(m.Interfaces, c.EvidenceInterfaces, findings);

    if (findings.Any(f => f.Severity is Severity.Error or Severity.Unknown))
        return RuntimeVerdict.Invalid;
    return RuntimeVerdict.Admitted;
}
```

EVIDENCE SOURCE RULES 55-AL

Source	Freshness / integrity
rendered model	generated for this run and digest-retained
Engine inventory	project-filtered after up --wait
migration/seed receipts	atomic files, expected run inputs
owner probes	captured inside stable window
telemetry query	control identity + pipeline freshness

REPOSITORY LAYOUT

The one-command runtime is a versioned product surface

EXECUTABLE ASSET 55-AV - EXPECTED PATHS

```
deploy/
  compose.yaml
  compose.local.yaml
  compose.evidence.yaml
  images.lock.json
  topology.contract.json
  otel/collector.yaml
  prometheus/prometheus.yaml
  grafana/provisioning/...
  seed/phase0-v1.json
  evidence/ch54.runsettings
eng/Nexus.Runtime/
tests/Nexus.Runtime.TopologyTests/
tests/Nexus.Runtime.SmokeTests/
artifacts/runtime/<run-id>/
```

OWNERSHIP 55-AM

Asset	Change owner / review
Compose/topology	runtime architecture + affected service owners
image lock	delivery/security review
migrations	single database owner
seed corpus	domain owners + test/evidence review
telemetry config	observability owner + service owners
launcher/verifier	runtime engineering

STANDARDS AND SOURCE LEDGER

Compose supplies mechanisms; NEXUS-1 defines admission semantics

SOURCE LEDGER 55-AN

Official source	Used for	Local qualification
Docker Compose up	--wait and timeout	semantic oracle follows
Compose startup order	healthy/completed conditions	receipt verification follows
Compose profiles	optional services	core remains profile-free
Compose config	canonical rendered model	local contract tests inspect it
Compose secrets	per-service file grants	local-only, value-free evidence
Docker image digests	immutable content reference	reviewed lock and repin process
OpenTelemetry Collector	receiver/processor/exporter pipelines	fresh control signal required
ASP.NET Core health checks	liveness/readiness separation	durable readiness remains external

VERSION RETENTION

The publication checkpoint retains CLI/Engine versions, resolved image digests, source revision and input hashes. Documentation describes capability; only the retained run can prove execution.

ANTI-PATTERN REVIEW

Ten convenient local-runtime claims remain invalid

REJECTED CLAIMS 55-AO

Claim	Why invalid
same YAML means same runtime	variables, overlays, images and volumes vary
version tag is pinned	tags can move
container running means ready	dependency or startup may be incomplete
--wait proves the workflow	it observes running/health contracts
migration on app startup is simpler	authority and races are hidden
random seed is realistic	experiments become incomparable
all ports help debugging	bypass and collision surface expands
environment variables are harmless evidence	they may contain secrets
down gives a clean database	named volumes remain by default
local success proves production	topology and operations differ

REVIEW QUESTION

Can another supported host create an equivalent admitted runtime from the same revision and inputs, then rerun a named failure campaign without manual repair? If not, reproducibility remains unproved.

COMPLETION LEDGER

What Chapter 55 now specifies

COMPLETION LEDGER 55-AP

Obligation	Asset	Status
one-command lifecycle	typed launcher and exit vocabulary	SPECIFIED
topology/ownership	service/network/secret/volume contracts	SPECIFIED
software identity	digest lock and verifier	SPECIFIED
ordered startup	health/completed dependency DAG	SPECIFIED
schema/data	migration and deterministic seed receipts	SPECIFIED
readiness	stable durable and experiment oracle	SPECIFIED
Ch54 execution path	evidence profile and target map	SPECIFIED
runtime proof	fresh clean-room artifacts	PENDING

OPEN RESIDUALS 55-AQ

Residual	Consequence / owner
no executable repository checkpoint	listings remain compile-oriented
placeholder image digests	must be resolved before execution
no retained clean-room runs	reproducibility claim remains pending
host resource floors educational	measure on supported hosts
no CI-built images	Chapter 57 owns production of artifacts

EVIDENCE VERDICT

SPECIFIED | RUNTIME PENDING. The topology, image/config identity, migrations, deterministic seed, readiness path, lifecycle safety and Ch54-compatible evidence profile are defined; no successful Compose run is invented.

HANDOFF TO CHAPTER 56

Extracting Without a Big-Bang Rewrite

Chapter 56 will place the reproducible distributed slice beside the existing modular path. It will implement the first strangler seam, explicit traffic routing, coexistence rules, shadow/compare evidence, reversible cutover and a removal decision that preserves the old path until evidence justifies retirement.

BOUNDARY CARRIED FORWARD 55-AR

Chapter 55 leaves	Chapter 56 must add
admitted Compose runtime	coexisting modular and distributed paths
stable BFF ingress	explicit route/feature control
deterministic control corpus	shadow and reconciliation comparisons
service-owned databases	one-writer migration states
runtime/evidence manifests	cutover and reversal receipts
down/reset lifecycle	retirement only after retained evidence

NEXT INVARIANT

Extraction is safe only when every request and durable write has one declared authority, the old and new paths can coexist without dual truth, and reversal remains possible until evidence admits retirement.

FINAL CHAPTER RULE

Reproducibility is an admitted equivalence, not a successful command

NEXUS-1 calls the local runtime reproducible only when the same pinned inputs render the declared topology, complete owned migrations, admit the canonical seed, reach stable durable readiness, expose the reviewed evidence interfaces and retain a manifest that another clean host can compare.

PART X OPENING

The runtime is now specified as a repeatable migration laboratory. The remaining chapters must use it to extract reversibly, build and gate images, roll out compatible change, translate only proven needs to Kubernetes and assemble the final demonstrator.

PART X - MIGRATION, DELIVERY, AND FINAL ARCHITECTURE

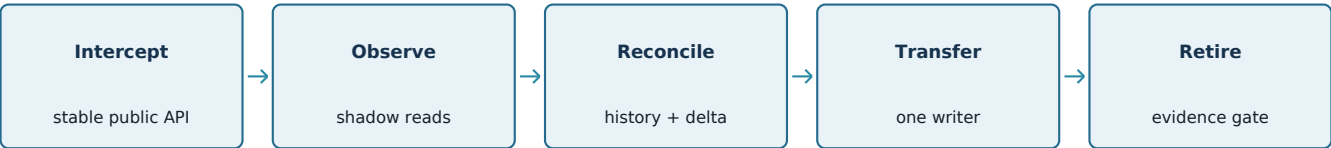
56. Extracting Without a Big-Bang Rewrite

The first service boundary is extracted only when traffic, write authority, historical state, public facts, comparison evidence and reversal are controlled as one migration system.

CHAPTER DECISION

Introduce one strangler seam at the stable edge for the Alarm capability. Preserve the modular implementation as the initial authority, shadow only side-effect-free reads, move deterministic plant-unit cohorts through explicit stages, and transfer each aggregate to exactly one writer through an epoch-based authority ledger.

PROOF PATH 56-A - FROM COEXISTENCE TO ADMITTED EXTRACTION



GUARANTEE LEDGER 56-A

Evidence	Allowed claim	Required proof	Cannot claim
route receipt	request used named path	plan digest + decision	target committed
shadow comparison	two reads were compared	normalized pair + freshness	write parity
authority ledger	one writer was selected	aggregate + epoch + state	rollback is safe
cutover manifest	declared gates passed	reconcile + drain + probes	legacy is removable

EVIDENCE VERDICT

SPECIFIED | RUNTIME PENDING. The seam, stages, authority model, comparisons and reversal gates are defined; no live migration or retained cutover is claimed.

ORDINARY DEVELOPER PROBLEM

The endpoint moved, but ownership did not

A team adds a proxy and sends ten percent of requests to a new Alarm service. Reads appear correct, so commands are split by the same percentage. One retry reaches the old path after the new path timed out. Both stores accept a write, both publish a fact, and the migration now has two truths.

FAILURE LEDGER 56-1

Shortcut	Failure	Correction
percentage-route commands	same aggregate changes writer	stable authority partition + ledger
fallback after timeout	first target may have committed	retry same authority with idempotency
dual-write from edge	partial success creates divergence	one writer + backfill/delta
compare raw JSON	ordering/time noise	versioned semantic normalizer
green new service	history or outbox may lag	reconciliation + high-water proof
delete old code after pilot	reversal disappears	retirement evidence window

PLAIN-LANGUAGE PRINCIPLE

A strangler is not a traffic trick. It is a temporary architecture that makes coexistence safer than replacement while evidence accumulates.

CHAPTER BOUNDARY

Extraction semantics here; delivery gates next

This chapter owns the first seam, routing stages, semantic translation, state synchronization, write-authority transfer, reversal and retirement decision. Chapter 57 will build the CI gates and container images that make these assets repeatable delivery outputs.

DECISION RECORD

ADR-056 - One public seam, one writer, reversible stages

NEXUS-1 places a temporary migration router behind the stable external API. Queries may be shadowed because the shadow credential is read-only and its response is discarded. Commands are never duplicated or randomly re-routed: a deterministic partition selects the initial authority and an aggregate ledger makes that authority sticky across retries and later operations.

DECISION CONSEQUENCES 56-B

Choice	Benefit	Cost / boundary
stable edge contract	clients do not migrate twice	router becomes critical
plant-unit cohorts	bounded deterministic pilot	uneven cohort size
aggregate authority ledger	one writer across retries	temporary control data
shadow reads only	comparison without effects	cannot prove commands
fenced writer transfer	no dual truth	brief bounded write pause
retained legacy path	reversal remains possible	duplicate operational surface

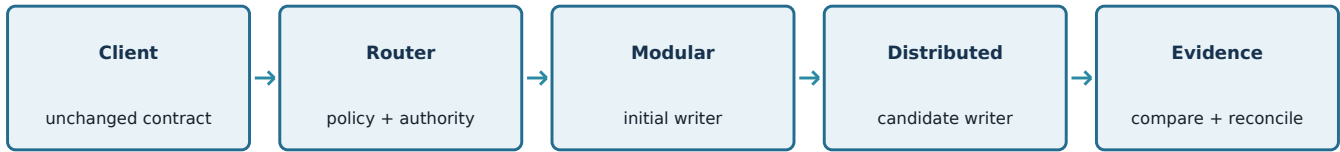
NON-GOAL

This chapter does not migrate every bounded context, guarantee zero downtime, prove production scale or authorize immediate rollback after unreplicated distributed writes.

STRANGLER MODEL

Replacement grows behind a facade while the old path stays useful

MIGRATION SHAPE 56-B



TEMPORARY RESPONSIBILITIES 56-C

Component	Owns	Must not own
migration router	route decision and receipt	Alarm business rules
anti-corruption layer	contract translation	durable truth
authority ledger	writer/epoch per aggregate	domain state
backfill/delta jobs	historical transfer	live command acceptance
evidence verifier	stage verdict	route mutation

TEMPORARY DOES NOT MEAN INFORMAL

Every migration component has an owner, contract, removal condition and evidence interface. Otherwise temporary code becomes the next permanent ambiguity.

FIRST EXTRACTION SLICE

AlarmManagement is narrow enough to route and rich enough to prove

SLICE CONTRACT 56-D

Surface	Included	Excluded for first seam
commands	create and acknowledge alarm	bulk import and administration
queries	get alarm and unit alarm list	cross-context reports
facts	AlarmRaisedV1, AlarmAcknowledgedV1	private domain events
data	Alarm aggregate + Inbox/Outbox	RootCause and projection truth
cohort	synthetic PlantUnit U-01	other units

EXECUTABLE ASSET 56-A - STABLE PUBLIC ROUTES

```

POST /api/v1/units/{unitId}/alarms
GET  /api/v1/units/{unitId}/alarms
GET  /api/v1/alarms/{alarmId}
POST /api/v1/alarms/{alarmId}/acknowledgements
  
```

Required command headers:

Idempotency-Key: UUID

If-Match: aggregate version for state-changing follow-up commands

Responses retain the same V1 public contract regardless of authority.

ROUTE-KEY DESIGN

Creation exposes unitId in the route, so routing never buffers or parses a command body. Follow-up commands resolve alarmId through the authority ledger.

SEAM INVENTORY

Each interaction class has a different migration technique

SEAM MATRIX 56-E

Interaction	Technique	Safety rule
HTTP query	primary + optional shadow	shadow result never served
HTTP command	single selected authority	no fallback after dispatch
legacy domain event	outbox interception/translation	public contract only
historical rows	snapshot + delta backfill	no cross-store live write
operator route change	versioned stage transition	preconditions + receipt

AUTHORITY QUESTIONS 56-F

Question	Mandatory answer
Who accepts this command?	one target named before forwarding
Who owns the aggregate now?	ledger target and epoch
What if response is lost?	retry same target with same idempotency key
Can the old path still write it?	no after authority transfer
Can route config change mid-request?	no; immutable plan snapshot

STABLE PUBLIC CONTRACT

Clients address the capability, not its current implementation

EXECUTABLE ASSET 56-B - V1 CREATE CONTRACT

```

public sealed record CreateAlarmV1(
    Guid AlarmId,
    string Code,
    DateTimeOffset OccurredUtc,
    string Severity,
    string SourceSignal);

public sealed record AlarmAcceptedV1(
    Guid AlarmId,
    long Version,
    string Status,
    DateTimeOffset AcceptedUtc);

public sealed record PublicErrorV1(
    string Code, string Message, string CorrelationId,
    IReadOnlyDictionary<string, string[]>? Validation);

```

CONTRACT INVARIANTS 56-G

Invariant	Both paths
AlarmId supplied by caller/edge	same identity before dispatch
UTC and enum validation	same rejection vocabulary
idempotency key required	same replay response
ETag/version semantics	same concurrency conflict
error envelope	no implementation leakage

EDGE COMPOSITION

YARP forwards; NEXUS-1 decides

EXECUTABLE ASSET 56-C - EDGE REGISTRATION

```
builder.Services.AddHttpForwarder();
builder.Services.AddSingleton<RoutePlanStore>();
builder.Services.AddScoped<MigrationRouteDecider>();
builder.Services.AddScoped<IWriteAuthorityLedger, SqlWriteAuthorityLedger>();
builder.Services.AddSingleton<MigrationTransformer>();

var app = builder.Build();
app.UseAuthentication();
app.UseAuthorization();
app.MapMethods("/api/v1/{**path}",
    new[] { "GET", "HEAD", "POST", "PUT", "PATCH", "DELETE" },
    MigrationForwardEndpoint.HandleAsync)
    .RequireAuthorization("NexusOperator");
```

RESPONSIBILITY BOUNDARY 56-H

YARP supplies	Migration layer supplies
HTTP forwarding mechanics	route and authority decision
request/response streaming	operation classification
transforms and error feature	internal migration headers
connection reuse	stage and destination policy

DEFENSE IN DEPTH

The edge authenticates and authorizes the public request. Each destination also authenticates the forwarded service identity and revalidates authorization-sensitive command data.

TYPED ROUTE MODEL

A route decision is data, not a branch hidden in middleware

EXECUTABLE ASSET 56-D - ROUTE VOCABULARY

```
public enum MigrationStage
{
    LegacyOnly, ShadowRead, DistributedReadPilot,
    DistributedReadFull, WritePilot, DistributedWriter,
    ReversalPending, LegacyRetired
}

public enum RouteTarget { LegacyModule, DistributedAlarm }
public enum OperationClass { Query, CreateCommand, AggregateCommand }

public sealed record RouteDecision(
    Guid RequestId,
    OperationClass Operation,
    RouteTarget Primary,
    RouteTarget? Shadow,
    string Reason,
    long PlanVersion,
    string PlanSha256,
    long? AuthorityEpoch);
```

NO IMPLICIT DEFAULT

Unknown operation, missing route key, unavailable plan or unresolved authority returns a migration error. It never silently chooses whichever target is healthy.

VERSIONED ROUTE PLAN

The full plan changes atomically

EXECUTABLE ASSET 56-E - CANONICAL PLAN

```
public sealed record AlarmMigrationPlan(  
    long Version,  
    MigrationStage Stage,  
    ImmutableHashSet<UnitId> PilotUnits,  
    int ShadowSamplePerMille,  
    bool CommandsEnabled,  
    DateTimeOffset EffectiveUtc,  
    string PreviousPlanSha256,  
    string Sha256);  
  
public sealed record RoutePlanSnapshot(  
    AlarmMigrationPlan Plan,  
    long LoadedGeneration,  
    DateTimeOffset LoadedUtc);
```

PLAN GATES 56-I

Field	Rule
version	strictly monotonic
stage	legal transition only
pilot units	known synthetic cohort; no duplicates
shadow sample	0..1000; queries only
commands enabled	true only after write gates
digest chain	previous plan identity must match

PLAN VALIDATION

Invalid or stale routing policy fails closed

EXECUTABLE ASSET 56-F - TRANSITION VALIDATOR

```
public static void ValidateTransition(  
    AlarmMigrationPlan before, AlarmMigrationPlan after)  
{  
    Require(after.Version == before.Version + 1, "route version gap");  
    Require(after.PreviousPlanSha256 == before.Sha256, "digest chain break");  
    Require(Allowed[before.Stage].Contains(after.Stage), "illegal stage transition");  
    Require(after.ShadowSamplePerMille is >= 0 and <= 1000,  
        "invalid shadow sample");  
    Require(!after.CommandsEnabled || after.Stage is  
        MigrationStage.WritePilot or MigrationStage.DistributedWriter,  
        "commands enabled before writer transfer");  
}
```

REJECTED TRANSITIONS 56-J

From	To	Reason
LegacyOnly	WritePilot	no shadow/read/reconcile evidence
ShadowRead	LegacyRetired	no writer transfer
WritePilot	LegacyOnly	distributed writes may exist
any	same version, changed bytes	policy identity conflict

PER-REQUEST PLAN SNAPSHOT

One request cannot change stage halfway through

EXECUTABLE ASSET 56-G - IMMUTABLE DECISION INPUT

```
public async ValueTask<RouteDecision> DecideAsync(
    RouteContext request, CancellationToken token)
{
    RoutePlanSnapshot snapshot = plans.Current;
    ValidateRequestAgainst(snapshot.Plan, request);

    return request.Operation switch
    {
        OperationClass.Query => DecideQuery(snapshot.Plan, request),
        OperationClass.CreateCommand =>
            await DecideCreateAsync(snapshot.Plan, request, token),
        OperationClass.AggregateCommand =>
            await DecideExistingAsync(snapshot.Plan, request, token),
        _ => throw new UnroutableRequestException()
    };
}
```

RELOAD RULE

A new plan is parsed, validated and published atomically. In-flight requests keep their captured version; new requests use the next snapshot and retain its digest in route receipts.

STABLE PILOT PARTITION

Plant-unit cohorts are deterministic; commands are never percentage-random

EXECUTABLE ASSET 56-H - COHORT DECISION

```
private static RouteTarget TargetForCreate(
    AlarmMigrationPlan plan, UnitId unit)
{
    if (!plan.CommandsEnabled)
        return RouteTarget.LegacyModule;

    return plan.Stage switch
    {
        MigrationStage.WritePilot when plan.PilotUnits.Contains(unit) =>
            RouteTarget.DistributedAlarm,
        MigrationStage.DistributedWriter => RouteTarget.DistributedAlarm,
        _ => RouteTarget.LegacyModule
    };
}
```

WHY NOT PERCENTAGE COMMANDS? 56-K

Percentage behavior	Risk
request identity changes	retry may select another writer
user identity changes	same aggregate may cross authority
percentage configuration changes	existing aggregate moves silently
hash implementation changes	cohort membership drifts

FEATURE-MANAGEMENT BOUNDARY

Percentage or targeting filters may enable UI/read experiments. Write authority remains domain-keyed and ledger-backed.

WRITE-AUTHORITY LEDGER

Every aggregate has one sticky writer and one monotonic epoch

EXECUTABLE ASSET 56-I - AUTHORITY RECORD

```
public sealed record WriteAuthority(
    AlarmId AlarmId,
    UnitId UnitId,
    RouteTarget Writer,
    long Epoch,
    AuthorityState State,
    Guid ReservationId,
    Guid FirstRequestId,
    DateTimeOffset ReservedUtc,
    DateTimeOffset? ConfirmedUtc,
    string PlanSha256,
    byte[] RowVersion);

public enum AuthorityState
{
    Reserved, Confirmed, TransferFenced, ReversalPending, Retired
}
```

LEDGER INVARIANTS 56-L

Invariant	Enforcement
one row per AlarmId	unique key
writer changes only under fence	state + epoch transition
epoch strictly increases	conditional update
request retries reuse reservation	unique idempotency key
no delete during migration	append audit + terminal state

RESERVATION BEFORE FORWARDING

A lost response cannot cause target fallback

EXECUTABLE ASSET 56-J - CREATE AUTHORITY RESERVATION

```
public async Task<WriteAuthority> ReserveCreateAsync(
    AlarmId alarmId, UnitId unitId, Guid requestId,
    RouteTarget proposed, AlarmMigrationPlan plan,
    CancellationToken token)
{
    var existing = await store.FindAsync(alarmId, token);
    if (existing is not null)
    {
        Require(existing.UnitId == unitId, "alarm/unit conflict");
        Require(existing.FirstRequestId == requestId, "alarm identity reused");
        return existing;
    }

    return await store.InsertReservedAsync(new WriteAuthority(
        alarmId, unitId, proposed, 1, AuthorityState.Reserved,
        Guid.NewGuid(), requestId, clock.UtcNow, null,
        plan.Sha256, Array.Empty<byte>()), token);
}
```

RESERVED IS STICKY

If forwarding times out, the reservation remains bound to the same writer. A retry sends the same AlarmId and Idempotency-Key to that target; the edge never tries the other path as a fallback.

CONFIRMING AUTHORITY

Target acceptance completes the reservation, idempotently

EXECUTABLE ASSET 56-K - CONFIRMATION

```
public async Task ConfirmAsync(
    WriteAuthority reserved, TargetReceipt receipt,
    CancellationToken token)
{
    Require(receipt.AlarmId == reserved.AlarmId, "target identity mismatch");
    Require(receipt.IdempotencyKey == reserved.FirstRequestId,
        "target request mismatch");
    Require(receipt.Writer == reserved.Writer, "target authority mismatch");

    await store.CompareExchangeAsync(
        reserved,
        reserved with {
            State = AuthorityState.Confirmed,
            ConfirmedUtc = receipt.CommittedUtc
        }, token);
}
```

AMBIGUOUS OUTCOMES 56-M

Observation	Action
2xx + matching receipt	confirm authority
known validation rejection	retain reservation as rejected audit; no other writer
timeout / connection loss	query target idempotency receipt, then retry same target
identity mismatch	quarantine request and migration stage

COMMAND ROUTE ALGORITHM

Creation chooses once; later commands look up authority

EXECUTABLE ASSET 56-L - COMMAND DECISION

```
private async Task<RouteDecision> DecideCommandAsync(
    AlarmMigrationPlan plan, RouteContext request,
    CancellationToken token)
{
    WriteAuthority authority = request.Operation switch
    {
        OperationClass.CreateCommand => await ledger.ReserveCreateAsync(
            request.AlarmId, request.UnitId, request.RequestId,
            TargetForCreate(plan, request.UnitId), plan, token),
        OperationClass.AggregateCommand => await ledger.RequireAsync(
            request.AlarmId, token),
        _ => throw new UnroutableRequestException()
    };

    Require(authority.State is AuthorityState.Reserved
        or AuthorityState.Confirmed, "aggregate is write-fenced");
    return RouteDecision.ForAuthority(request, plan, authority);
}
```

ROUTE TRUTH

The current plan selects only new aggregate placement. It cannot move an existing Alarm implicitly; the ledger is authoritative for every follow-up command.

DIRECT FORWARDING

The selected destination is explicit and the forwarder result is inspected

EXECUTABLE ASSET 56-M - YARP FORWARDING

```
public static async Task HandleAsync(
    HttpContext context,
    MigrationRouteDecider routes,
    IHttpForwarder forwarder,
    MigrationProxyClient client,
    MigrationTransformer transformer)
{
    RouteDecision decision = await routes.DecideAsync(
        RouteContext.From(context), context.RequestAborted);
    context.Items[RouteDecision.ItemKey] = decision;

    string destination = client.DestinationFor(decision.Primary);
    ForwarderError error = await forwarder.SendAsync(
        context, destination, client.Invoker,
        client.RequestConfig, transformer);

    if (error != ForwarderError.None)
        RouteFailureRecorder.Record(context, decision, error);
}
```

MECHANISM BOUNDARY

IHttpForwarder performs forwarding and returns a typed error. NEXUS-1 remains responsible for route classification, destination selection, authority, receipts and retry semantics.

MIGRATION REQUEST TRANSFORM

Client-supplied migration headers are removed and replaced

EXECUTABLE ASSET 56-N - TRUSTED INTERNAL HEADERS

```
public override async ValueTask TransformRequestAsync(
    HttpContext context, HttpRequestMessage proxyRequest,
    string destinationPrefix, CancellationToken token)
{
    await base.TransformRequestAsync(
        context, proxyRequest, destinationPrefix, token);

    var decision = (RouteDecision)context.Items[RouteDecision.ItemKey]!;
    proxyRequest.Headers.Remove("X-Nexus-Route-Plan");
    proxyRequest.Headers.Remove("X-Nexus-Authority-Epoch");
    proxyRequest.Headers.TryAddWithoutValidation(
        "X-Nexus-Route-Plan", decision.PlanSha256);
    if (decision.AuthorityEpoch is long epoch)
        proxyRequest.Headers.TryAddWithoutValidation(
            "X-Nexus-Authority-Epoch", epoch.ToString(CultureInfo.InvariantCulture));
    proxyRequest.Headers.Host = null;
}
```

TRUST BOUNDARY

Destinations accept these headers only from the authenticated edge identity on the internal network. The headers carry evidence identity, not authorization by themselves.

FORWARDING FAILURE SEMANTICS

A failed command is not replayed against another implementation

FAILURE POLICY 56-N

Failure	Query	Command
target unavailable before send	fail selected path; optional safe primary policy	fail; retain authority
timeout after send	report unavailable	ambiguous; query receipt/retry same target
5xx response	return qualified error	return error; no other writer
transform/config error	fail closed	fail closed and quarantine
client cancellation	cancel shadow and primary	retain reservation/receipt state

EXECUTABLE ASSET 56-O - NO CROSS-TARGET FALLBACK

```
if (decision.Operation is not OperationClass.Query &&
    error is not ForwarderError.None)
{
    await ambiguousCommands.RecordAsync(
        decision.RequestId, decision.Primary,
        decision.AuthorityEpoch, error, cancellationToken);
    // Never invoke Alternate(decision.Primary).
}
```

RETRY OWNERSHIP

Client or edge retry uses the same request identity and resolved authority. Transport retry policy cannot choose a different implementation.

FEATURE FLAGS

Flags gate stages; they do not replace the authority ledger

FLAG USE 56-O

Flag	Allowed use	Forbidden use
MigrationRouterEnabled	enable stable facade	bypass ledger for commands
ShadowReadsEnabled	allow bounded query shadow	shadow commands
DistributedReadPilot	enable named read cohort	random writer selection
EmergencyWriteFence	stop new writes	move authority automatically

EXECUTABLE ASSET 56-P - CONTEXTUAL READ GATE

```
public sealed record MigrationContext(UnitId UnitId, string Operation);

bool shadowEnabled = await featureManager.IsEnabledAsync(
    "ShadowReadsEnabled", new MigrationContext(unitId, "query"));

// The result may allow a shadow read. It never determines command writer.
RouteTarget writer = await authorityLedger.RequireWriterAsync(alarmId, token);
```

SNAPSHOT RULE

Feature and route-plan results are captured once per request. Evidence retains the evaluated variant/flag identity without treating a mutable percentage as durable ownership.

QUERY ROUTING

The served path is explicit even when a shadow also runs

EXECUTABLE ASSET 56-Q - QUERY DECISION

```
private RouteDecision DecideQuery(
    AlarmMigrationPlan plan, RouteContext request)
{
    RouteTarget primary = plan.Stage switch
    {
        MigrationStage.DistributedReadPilot
            when plan.PilotUnits.Contains(request.UnitId) =>
                RouteTarget.DistributedAlarm,
        MigrationStage.DistributedReadFull or
        MigrationStage.WritePilot or
        MigrationStage.DistributedWriter => RouteTarget.DistributedAlarm,
        _ => RouteTarget.LegacyModule
    };

    RouteTarget? shadow = ShouldShadow(plan, request)
        ? Other(primary) : null;
    return RouteDecision.ForQuery(request, plan, primary, shadow);
}
```

FRESHNESS QUALIFICATION

A distributed read may be correct but behind. The response includes its projection checkpoint, and the comparison distinguishes acceptable declared lag from semantic mismatch.

SHADOW-READ SAFETY

Only side-effect-free operations may enter the shadow lane

SHADOW ADMISSION 56-P

Gate	Required value
HTTP method	GET or HEAD
destination credential	read-only and separate
endpoint inventory	declared side-effect-free
request body	none
timeout	shorter than primary budget
capacity	bounded queue and concurrency
response	never returned to client

EXECUTABLE ASSET 56-R - ADMISSION GUARD

```
public static void AuthorizeShadow(RouteContext request, ShadowEndpoint endpoint)
{
    Require(request.Method is "GET" or "HEAD", "shadow method is unsafe");
    Require(endpoint.SideEffectFree, "endpoint not admitted for shadowing");
    Require(endpoint.CredentialClass == "read-only", "shadow credential can write");
    Require(request.ContentLength is null or 0, "shadow body not allowed");
}
```

BOUNDED SHADOW DISPATCHER

Comparison load cannot threaten the served path

EXECUTABLE ASSET 56-S - ISOLATED QUEUE

```
var queue = Channel.CreateBounded<ShadowWork>(new BoundedChannelOptions(256)
{
    FullMode = BoundedChannelFullMode.DropWrite,
    SingleReader = false,
    SingleWriter = false
});

public bool TrySchedule(ShadowWork work)
{
    bool accepted = queue.Writer.TryWrite(work);
    metrics.ShadowScheduled.Add(accepted ? 1 : 0,
        new("outcome", accepted ? "accepted" : "dropped"));
    return accepted;
}

// Workers use a dedicated HttpMessageInvoker, read-only identity,
// concurrency semaphore and timeout shorter than the primary request.
```

ISOLATION DIMENSIONS 56-Q

Dimension	Primary	Shadow
client response	authoritative	discarded
connection pool	serving pool	dedicated pool
timeout	route budget	strict sub-budget
failure	visible to client	evidence only
load shedding	normal policy	drop first

DETERMINISTIC SHADOW SAMPLING

The same request identity makes the same sampling decision

EXECUTABLE ASSET 56-T - STABLE SAMPLE

```
public static bool IsSelected(Guid requestId, int permille)
{
    Require(permille is >= 0 and <= 1000, "invalid permille");
    Span<byte> bytes = stackalloc byte[16];
    requestId.TryWriteBytes(bytes, bigEndian: true, out _);
    Span<byte> digest = stackalloc byte[32];
    SHA256.HashData(bytes, digest);
    uint bucket = BinaryPrimitives.ReadUInt32BigEndian(digest) % 1000;
    return bucket < permille;
}
```

SAMPLING EVIDENCE 56-R

Retain	Reason
request identity	reproduce decision
algorithm version	detect implementation drift
permille and plan digest	reproduce policy
selected / not selected	denominator integrity
dropped by capacity	avoid survivorship bias

RESPONSE CANONICALIZATION

Compare meaning, not serialization accidents

EXECUTABLE ASSET 56-U - NORMALIZED ALARM

```
public sealed record CanonicalAlarm(  
    Guid AlarmId,  
    string UnitId,  
    string Code,  
    string Severity,  
    string Status,  
    DateTimeOffset OccurredUtc,  
    string[] EvidenceIds,  
    long DomainVersion,  
    long ProjectionCheckpoint);  
  
public interface IResponseNormalizer  
{  
    string Version { get; }  
    CanonicalAlarm Normalize(ReadOnlySpan<byte> json);  
}
```

NORMALIZATION RULES 56-S

Field	Rule
timestamps	UTC instant equality; format ignored
sets	sort by stable identity
enums	map legacy aliases to public vocabulary
links	exclude host-specific base URI
checkpoint	compare as freshness, not business value
unknown field	retain finding; do not silently discard

SEMANTIC COMPARISON

Equivalent, lagging, mismatched and incomparable remain distinct

EXECUTABLE ASSET 56-V - COMPARISON VERDICT

```
public enum ComparisonVerdict  
{  
    Equivalent, AcceptableLag, SemanticMismatch,  
    PrimaryUnavailable, ShadowUnavailable, Incomparable  
}  
  
public sealed record ComparisonResult(  
    Guid RequestId,  
    ComparisonVerdict Verdict,  
    string NormalizerVersion,  
    string PrimaryDigest,  
    string? ShadowDigest,  
    FieldDifference[] Differences,  
    TimeSpan? ProjectionLag,  
    string PlanSha256);
```

EXECUTABLE ASSET 56-W - CORE COMPARATOR

```
if (primary.BusinessFieldsEqual(shadow))  
    return primary.ProjectionCheckpoint == shadow.ProjectionCheckpoint  
        ? ComparisonVerdict.Equivalent  
        : WithinLagBudget(primary, shadow)  
            ? ComparisonVerdict.AcceptableLag  
            : ComparisonVerdict.SemanticMismatch;  
return ComparisonVerdict.SemanticMismatch;
```

NO GREEN AVERAGING

A high overall match rate cannot hide invariant-level mismatches. Identity, status, version and evidence-reference differences are individually blocking.

COMPARISON PRIVACY AND RETENTION

Evidence keeps semantics without retaining sensitive payloads

RETENTION CONTRACT 56-T

Artifact	Retained form
normalized values	field-level digest or synthetic value
differences	field name, class and bounded diagnostic
raw payload	off by default; encrypted short retention if explicitly admitted
identity	opaque request/alarm IDs in evidence store
authorization data	policy/result identity, never bearer token

EXECUTABLE ASSET 56-X - SAFE DIFFERENCE

```
public sealed record FieldDifference(  
    string Field,  
    DifferenceClass Class,  
    string PrimarySha256,  
    string ShadowSha256,  
    string? SafeDiagnostic);  
  
public enum DifferenceClass  
{  
    Identity, DomainValue, Status, Version,  
    EvidenceReference, Freshness, Representation  
}
```

SYNTHETIC RUNTIME

The Chapter 55 local runtime uses deterministic synthetic data, so full normalized pairs may be retained for its evidence campaigns. Production policy would be stricter and is not inferred here.

ANTI-CORRUPTION LAYER

Legacy semantics stop at the adapter

TRANSLATION OWNERSHIP 56-U

Direction	Input	Output
public -> legacy	CreateAlarmV1	legacy command and codes
legacy -> public	legacy DTO/error	AlarmAcceptedV1/PublicErrorV1
legacy fact -> public	module notification	AlarmRaisedV1
public -> distributed	V1 contract	application command
distributed -> public	application outcome	V1 response/error

BUSINESS-RULE BOUNDARY

The adapter translates names, shapes and error vocabularies. It does not decide severity, closure, acknowledgement or evidence invariants; those remain inside the authoritative implementation.

CONTRACT SYMMETRY

Both adapters are tested against one public corpus

A versioned request/response/error corpus is executed against the modular adapter and the distributed API. The corpus asserts public behavior; it does not require either internal model to look the same.

LEGACY ADAPTER

The old module remains behind a narrow interface

EXECUTABLE ASSET 56-Y - LEGACY FACADE

```
public sealed class LegacyAlarmFacade(
    LegacyAlarmModule module,
    LegacyAlarmTranslator translator) : IAlarmPath
{
    public async Task<AlarmAcceptedV1> CreateAsync(
        UnitId unitId, CreateAlarmV1 request,
        IdempotencyKey key, CancellationToken token)
    {
        LegacyCreateAlarm command = translator.ToLegacy(unitId, request, key);
        LegacyAlarmResult result = await module.ExecuteAsync(command, token);
        return translator.ToPublic(result);
    }
}
```

LEGACY CONTAINMENT 56-V

Allowed	Forbidden
call module application facade	new service references legacy entities
translate legacy errors	public contract exposes legacy codes
read legacy owner state for backfill	distributed runtime writes legacy tables
publish through legacy outbox	direct broker publish inside transaction

DISTRIBUTED ADAPTER

The new service speaks its own application language

EXECUTABLE ASSET 56-Z - DISTRIBUTED ENDPOINT MAPPING

```
app.MapPost("/internal/v1/units/{unitId}/alarms", async (
    string unitId,
    CreateAlarmV1 body,
    HttpRequest request,
    ISender sender,
    CancellationToken token) =>
{
    var key = IdempotencyKey.Require(request.Headers["Idempotency-Key"]);
    var command = new CreateAlarm(
        UnitId.From(unitId), AlarmId.From(body.AlarmId),
        AlarmCode.From(body.Code), body.OccurredUtc,
        Severity.Parse(body.Severity), SignalId.From(body.SourceSignal), key);
    CreateAlarmOutcome outcome = await sender.Send(command, token);
    return PublicAlarmMapper.ToHttp(outcome);
}).RequireAuthorization("MigrationEdge");
```

NO SHARED MODEL

The edge shares the public V1 contract, not the distributed domain types. Mapping occurs at the endpoint/application boundary.

DATA COEXISTENCE

Stores coexist; tables are never shared

DATA RULES 56-W

State	Legacy store	AlarmManagementDb
before backfill	writer + query source	empty/candidate schema
shadow read	writer	read-only imported copy + deltas
read pilot	writer	query source for pilot
write pilot	writer for legacy units	writer for transferred units
distributed full	read-only reversal window	writer

FORBIDDEN COEXISTENCE 56-X

Technique	Why rejected
both paths update same table	ownership is not extracted
edge transaction spans both stores	no atomic local boundary
database trigger copies live writes	hidden coupling and failure
new service reads legacy schema directly	legacy model leaks inward
dual-write and compare later	partial success already broke truth

SNAPSHOT PLUS DELTA

Historical copy and ongoing change are separated

BACKFILL PATH 56-C



HIGH-WATER CONTRACT 56-Y

Marker	Meaning
H0	legacy commit sequence captured before snapshot scan
snapshot receipt	range, row count and canonical digest
delta checkpoint	last legacy public fact applied
H1	legacy commit sequence at comparison start
caught-up	distributed checkpoint >= H1 for stable window

NO TIMESTAMP CURSOR

Backfill uses a monotonic owner-controlled commit sequence, not wall-clock timestamps that may collide, drift or reorder.

BACKFILL READER

Rows are paged by stable key under a declared snapshot boundary

EXECUTABLE ASSET 56-AA - KEYSSET PAGE

```
public async IEnumerable<LegacyAlarmRow> ReadSnapshotAsync(
    long highWater, int pageSize,
    [EnumeratorCancellation] CancellationToken token)
{
    long afterSequence = 0;
    while (true)
    {
        LegacyAlarmRow[] page = await db.Alarms.AsNoTracking()
            .Where(x => x.CommitSequence > afterSequence
                && x.CommitSequence <= highWater)
            .OrderBy(x => x.CommitSequence)
            .Take(pageSize)
            .ToArrayAsync(token);
        if (page.Length == 0) yield break;
        foreach (var row in page) yield return row;
        afterSequence = page[^1].CommitSequence;
    }
}
```

READER GATES 56-Z

Gate	Reason
AsNoTracking	bounded source memory
keyset paging	no offset drift
commit sequence <= H0	stable snapshot boundary
canonical order	repeatable digest
cancellation + bounded batch	operational control

IDEMPOTENT IMPORTER

Same source version is a no-op; conflicting history stops the run

EXECUTABLE ASSET 56-AB - IMPORT DECISION

```
public async Task ImportAsync(
    ImportedAlarm candidate, CancellationToken token)
{
    ImportedAlarm? existing = await store.FindAsync(candidate.AlarmId, token);
    if (existing is null)
    {
        await store.InsertAsync(candidate, token);
        return;
    }

    if (existing.SourceVersion == candidate.SourceVersion &&
        existing.CanonicalSha256 == candidate.CanonicalSha256)
        return;

    throw new HistoricalConflictException(
        candidate.AlarmId, existing.SourceVersion, candidate.SourceVersion);
}
```

IMPORT AUTHORITY

Imported rows are marked legacy-origin and remain non-writable while the legacy path owns the aggregate. Transfer changes authority only after reconciliation and a write fence.

DELTA EVENT INTERCEPTION

Legacy commits publish public facts through its existing outbox

EXECUTABLE ASSET 56-AC - LEGACY FACT TRANSLATION

```
public AlarmRaisedV1 Translate(LegacyAlarmRaised source) => new(  
    MessageId: PublicMessageId.FromLegacy(source.OutboxId),  
    AlarmId: source.AlarmId,  
    UnitId: source.UnitCode,  
    Code: legacyCodes.ToPublic(source.AlarmCode),  
    Severity: legacySeverity.ToPublic(source.Priority),  
    OccurredUtc: source.OccurredUtc,  
    AggregateVersion: source.AlarmVersion,  
    SourceCommitSequence: source.CommitSequence,  
    AuthorityEpoch: 0,  
    ContractVersion: 1);
```

INTERCEPTION RULES 56-AA

Rule	Purpose
translate after legacy commit	no uncommitted public fact
stable MessageId from outbox	idempotent redelivery
source version/sequence	order and gap detection
public contract only	no legacy entity leakage
Inbox at importer	one applied delta

BACKFILL RECEIPT

Every copied range and delta checkpoint is retained

EXECUTABLE ASSET 56-AD - TRANSFER EVIDENCE

```
public sealed record BackfillReceipt(  
    Guid RunId,  
    UnitId UnitId,  
    long SnapshotHighWater,  
    Guid? LastAlarmId,  
    long RowsRead,  
    long RowsInserted,  
    long RowsNoOp,  
    string CanonicalSetSha256,  
    long DeltaCheckpoint,  
    DateTimeOffset CompletedUtc,  
    string Verdict);
```

RECEIPT VERDICT 56-AB

Verdict	Meaning
COMPLETE	snapshot exhausted and delta caught up
PARTIAL	safe checkpoint retained; no cutover
CONFLICT	same identity differs; quarantine
GAP	missing source sequence; rebuild evidence
INVALID	input identity or digest unknown

RECONCILIATION ORACLE

Counts are useful; per-aggregate semantics decide

EXECUTABLE ASSET 56-AE - RECONCILIATION RESULT

```
public sealed record ReconciliationResult(
    Guid RunId,
    UnitId UnitId,
    long LegacyHighWater,
    long DistributedCheckpoint,
    long Compared,
    long Equivalent,
    long Mismatched,
    long MissingLegacy,
    long MissingDistributed,
    InvariantFinding[] Findings,
    string LegacySetSha256,
    string DistributedSetSha256,
    ReconciliationVerdict Verdict);
```

BLOCKING INVARIANTS 56-AC

Invariant	Required equality
identity conservation	same AlarmIds for admitted range
status/version	same current domain state
evidence references	same canonical evidence IDs
public fact conservation	one expected fact per transition
authority	legacy before fence; distributed after transfer

STABLE CONVERGENCE

One matching scan is not enough for cutover

EXECUTABLE ASSET 56-AF - AWAIT RECONCILIATION STABILITY

```
public async Task<ReconciliationResult> AwaitStableAsync(
    UnitId unitId, TimeSpan budget, CancellationToken token)
{
    var deadline = clock.GetTimestamp() + clock.ToTicks(budget);
    ReconciliationResult? last = null;
    var consecutive = 0;
    while (clock.GetTimestamp() < deadline)
    {
        last = await reconciler.CompareAsync(unitId, token);
        consecutive = last.Verdict == ReconciliationVerdict.Equivalent
            ? consecutive + 1 : 0;
        if (consecutive >= 3) return last;
        await clock.Delay(TimeSpan.FromSeconds(5), token);
    }
    throw new ReconciliationTimeoutException(last);
}
```

FRESHNESS FIRST

The oracle first proves distributed checkpoint $\geq$ captured legacy high-water. It never labels two equally stale snapshots equivalent.

MIGRATION STAGE MACHINE

Each stage removes one uncertainty before adding authority

STAGES 56-AD

Stage	Served reads	Writer	Evidence gate
LegacyOnly	legacy	legacy	baseline
ShadowRead	legacy	legacy	semantic comparison
DistributedReadPilot	pilot distributed	legacy	read SLO + parity
DistributedReadFull	distributed	legacy	backfill + delta stable
WritePilot	distributed	split by unit/ledger	fence + command campaign
DistributedWriter	distributed	distributed	stable recovery window
LegacyRetired	distributed	distributed	dependency absence proof

NO STAGE SKIPPING

A later-stage configuration without the earlier evidence manifests is invalid even if the services are healthy.

STAGE ADMISSION CATALOGUE

Transitions consume named, fresh evidence

TRANSITION GATES 56-AE

Transition	Mandatory proof
LegacyOnly -> ShadowRead	read-only shadow authority + capacity isolation
ShadowRead -> ReadPilot	comparison coverage and zero blocking mismatch
ReadPilot -> ReadFull	pilot SLO/error/freshness within budgets
ReadFull -> WritePilot	stable reconciliation + drained legacy outbox + fence plan
WritePilot -> DistributedWriter	command/idempotency/failure campaigns
DistributedWriter -> LegacyRetired	reversal window complete + no dependencies

EXECUTABLE ASSET 56-AG - EVIDENCE GATE

```
public static void Admit(StageTransition request, EvidenceSet evidence)
{
    Require(evidence.PlanSha256 == request.Before.Sha256, "plan drift");
    Require(evidence.RuntimeVerdict == "ADMITTED", "runtime not admitted");
    Require(evidence.Required.All(x => x.Fresh && x.Verified),
        "missing or stale evidence");
    Require(!evidence.Findings.Any(x => x.Blocking), "blocking finding");
    ValidateTransition(request.Before, request.After);
}
```

WRITER CUTOVER PLAN

A short fence makes the authority transfer observable

CUTOVER STEPS 56-AF

Step	Action	Postcondition
1. arm	validate runtime, plan, cohort and lease	exclusive transition
2. fence	reject new cohort commands	no new legacy writes
3. drain	legacy requests/outbox to zero	no pending old effects
4. catch up	apply deltas and reconcile Hcut	stores equivalent
5. transfer	increment authority epoch for cohort	distributed writer
6. probe	one idempotent control command	new path commits/publishes
7. open	remove distributed write fence	cohort admitted

BOUNDED INTERRUPTION

The write fence is visible and time-bounded. Reads continue from the reconciled distributed path; commands receive a retryable migration response rather than entering either writer ambiguously.

COHORT AUTHORITY TRANSFER

Epoch changes under one lease and conditional update

EXECUTABLE ASSET 56-AH - TRANSFER COMMAND

```
public async Task<AuthorityTransferReceipt> TransferUnitAsync(
    UnitId unitId, long expectedEpoch,
    string reconciliationSha256, CancellationToken token)
{
    await using var lease = await leases.AcquireAsync(unitId, token);
    Require(await fences.IsClosedAsync(unitId, token), "writes not fenced");
    Require(await outbox.IsDrainedAsync(unitId, token), "legacy outbox not drained");
    Require(await evidence.IsAdmittedAsync(reconciliationSha256, token),
        "reconciliation not admitted");

    long changed = await ledger.TransferConfirmedAsync(
        unitId, RouteTarget.LegacyModule,
        RouteTarget.DistributedAlarm, expectedEpoch, expectedEpoch + 1, token);
    Require(changed == await ledger.CountUnitAsync(unitId, token),
        "partial authority transfer");
    return await receipts.RetainAsync(unitId, expectedEpoch + 1, token);
}
```

ATOMICITY SCOPE

The control database changes cohort authority transactionally. Domain stores are already reconciled under the fence; no distributed transaction is attempted across them.

WRITE-PILOT EXECUTION

New and existing aggregates follow different decisions

PILOT ROUTING 56-AG

Request	U-01 pilot	Other units
create new Alarm	distributed; ledger epoch 1	legacy; ledger epoch 1
command existing Alarm	authority ledger	authority ledger
get by AlarmId	ledger target or distributed projection	ledger/legacy
list by UnitId	distributed read	legacy read

EXECUTABLE ASSET 56-AI - DESTINATION INVARIANT

```
[Fact]
public async Task Existing_alarm_never_moves_when_plan_percentage_changes()
{
    WriteAuthority before = await fixture.CreateUnderLegacyAsync(UnitId.U01);
    await fixture.EnableDistributedWriterForAsync(UnitId.U01);
    RouteDecision decision = await fixture.DecideAcknowledgeAsync(before.AlarmId);
    decision.Primary.Should().Be(before.Writer);
    decision.AuthorityEpoch.Should().Be(before.Epoch);
}
```

HISTORICAL MIGRATION

Existing alarms move only through the explicit fenced transfer operation. A stage or feature change alone never rewrites their authority.

ROLLBACK SAFETY

Reversal depends on what the new writer has already accepted

REVERSAL MATRIX 56-AH

Current stage	Immediate route reversal?	Required action
ShadowRead	yes	disable shadow
DistributedReadPilot	yes	serve legacy; retain findings
DistributedReadFull	yes if legacy reads remain current	verify legacy freshness
WritePilot before first new write	yes under fence	restore epoch/route receipt
WritePilot after new writes	no	fence, reverse-sync, reconcile, transfer
LegacyRetired	no ordinary reversal	new migration/restore procedure

HONEST ROLLBACK

A route flip is not rollback when the old path lacks new durable state. After distributed writes, reversal is a second controlled migration.

REVERSAL ORCHESTRATION

The same fence-transfer protocol runs in the opposite direction

EXECUTABLE ASSET 56-AJ - REVERSAL OUTLINE

```
public async Task ReverseAsync(UnitId unitId, CancellationToken token)
{
    await fences.CloseAsync(unitId, "reversal", token);
    await distributedOutbox.AwaitDrainedAsync(unitId, token);
    ReverseBackfillReceipt copy = await reverseBackfill.CopyAsync(unitId, token);
    ReconciliationResult equal = await reverseReconcile.AwaitStableAsync(
        unitId, budgets.Reversal, token);
    Require(copy.Complete && equal.Verdict == ReconciliationVerdict.Equivalent,
        "reversal state not equivalent");
    await authorities.TransferAsync(
        unitId, RouteTarget.DistributedAlarm, RouteTarget.LegacyModule, token);
    await legacyControlProbe.ExecuteAsync(unitId, token);
    await fences.OpenAsync(unitId, RouteTarget.LegacyModule, token);
}
```

CAPABILITY CONSTRAINT

Reversal is available only while the legacy model can represent every distributed state and contract version. An incompatible change must block retirement or require an explicit downgrade mapping.

EMERGENCY WRITE FENCE

Stop new ambiguity before attempting recovery

EXECUTABLE ASSET 56-AK - FENCED RESPONSE

```
if (await fences.IsClosedAsync(route.UnitId, context.RequestAborted))
{
    context.Response.StatusCode = StatusCodes.Status503ServiceUnavailable;
    context.Response.Headers.RetryAfter = "5";
    await context.Response.WriteAsJsonAsync(new PublicErrorV1(
        "MIGRATION_WRITE_FENCED",
        "Writes are temporarily paused for a controlled migration transition.",
        Activity.Current?.TraceId.ToString() ?? context.TraceIdentifier,
        null), context.RequestAborted);
    return;
}
```

FENCE RULES 56-AI

Rule	Reason
unit-scoped	other cohorts continue
commands only	reads remain available
expiry + owner	no forgotten indefinite fence
manual open requires receipt	recovery is audited
visible metric/log	operators see migration state

IDEMPOTENCY ACROSS PATHS

The public key and AlarmId survive every retry and transfer

EXECUTABLE ASSET 56-AL - TARGET IDEMPOTENCY RECORD

```
public sealed record CommandDeduplication(  
    IdempotencyKey Key,  
    AlarmId AlarmId,  
    string RequestSha256,  
    long AuthorityEpoch,  
    int HttpStatus,  
    string ResponseJson,  
    DateTimeOffset CompletedUtc);  
  
public static void ValidateReplay(CommandDeduplication existing,  
    AlarmId alarmId, string requestSha256, long epoch)  
{  
    Require(existing.AlarmId == alarmId, "idempotency identity conflict");  
    Require(existing.RequestSha256 == requestSha256, "idempotency body conflict");  
    Require(existing.AuthorityEpoch == epoch, "authority epoch conflict");  
}
```

EPOCH IN DEDUPLICATION

A command replay from an obsolete authority epoch is rejected and investigated. It cannot silently mutate the new writer after transfer.

PUBLIC FACTS DURING COEXISTENCE

Both possible writers emit one contract, but only for owned aggregates

PUBLISHER RULES 56-AJ

Rule	Proof
writer matches authority epoch	ledger/command receipt
same V1 public schema	contract corpus
globally unique MessageId	origin + outbox identity
aggregate version monotonic	consumer gap/order checks
origin retained as metadata	migration diagnostics
consumer idempotent	Inbox unique MessageId

EXECUTABLE ASSET 56-AM - AUTHORITY-AWARE ENVELOPE

```
public sealed record PublicFactEnvelope<T>(  
    Guid MessageId,  
    string Contract,  
    int ContractVersion,  
    Guid AggregateId,  
    long AggregateVersion,  
    long AuthorityEpoch,  
    string Writer,  
    DateTimeOffset OccurredUtc,  
    T Payload);
```

CONSUMER COEXISTENCE

RootCause and projections do not care which valid writer emitted the fact

EXECUTABLE ASSET 56-AN - CONSUMER AUTHORITY GUARD

```
public async Task HandleAsync(
    PublicFactEnvelope<AlarmRaisedV1> message,
    CancellationToken token)
{
    if (await inbox.ContainsAsync(message.MessageId, token)) return;

    AuthorityView authority = await authorityCache.RequireAsync(
        AlarmId.From(message.AggregateId), token);
    Require(message.AuthorityEpoch == authority.Epoch, "stale writer fact");
    Require(message.Writer == authority.WriterName, "wrong writer fact");

    await unitOfWork.ExecuteAsync(async () =>
    {
        await projections.ApplyAsync(message.Payload, token);
        await inbox.CompleteAsync(message.MessageId, token);
    }, token);
}
```

AUTHORITY CACHE QUALIFICATION

The consumer uses a versioned replicated authority view. Missing authority is UNKNOWN and parks the message; it is not interpreted as legacy or distributed by default.

DUPLICATE-PUBLICATION DETECTOR

One logical transition must not appear from both writers

EXECUTABLE ASSET 56-AO - CONSERVATION KEY

```
public readonly record struct TransitionKey(
    Guid AggregateId, long AggregateVersion, string Contract);

public async Task ObserveAsync(PublicFactHeader fact, CancellationToken token)
{
    var key = new TransitionKey(
        fact.AggregateId, fact.AggregateVersion, fact.Contract);
    ObservedTransition? prior = await store.FindAsync(key, token);
    if (prior is not null && prior.MessageId != fact.MessageId)
        throw new DuplicateTransitionException(
            key, prior.Writer, fact.Writer, prior.MessageId, fact.MessageId);
    await store.InsertIfAbsentAsync(key, fact, token);
}
```

CONSERVATION FINDINGS 56-AK

Finding	Meaning
same MessageId redelivered	expected transport duplicate
different MessageId, same transition	possible dual publication
version gap	missing fact or out-of-order arrival
obsolete authority epoch	stale writer escaped fence

MIGRATION OBSERVABILITY

Every request exposes route, stage, authority and comparison identity

TELEMETRY 56-AL

Signal	Required attributes / dimensions
route span	stage, plan version, primary, operation
authority event	aggregate, writer, epoch, reservation state
shadow metric	selected, scheduled, dropped, verdict
reconciliation metric	cohort, mismatch class, checkpoint lag
cutover log	step, receipt digest, operator, outcome

EXECUTABLE ASSET 56-AP - BOUNDED ROUTE METRICS

```
routeCounter.Add(1,
    new("migration.stage", decision.StageName),
    new("migration.primary", decision.Primary.ToString()),
    new("migration.operation", decision.Operation.ToString()),
    new("migration.outcome", outcome));

Activity.Current?.SetTag("migration.plan.version", decision.PlanVersion);
Activity.Current?.SetTag("migration.authority.epoch", decision.AuthorityEpoch);
// AlarmId and RequestId remain trace/log fields, never metric labels.
```

ROUTE AND CUTOVER MANIFEST

The migration claim is machine-readable

EXECUTABLE ASSET 56-AQ - EXTRACTION EVIDENCE

```
public sealed record ExtractionManifest(
    string Schema,
    Guid TransitionId,
    string RuntimeManifestSha256,
    string BeforePlanSha256,
    string AfterPlanSha256,
    MigrationStage From,
    MigrationStage To,
    UnitId[] Cohort,
    ComparisonSummary Comparison,
    BackfillReceipt Backfill,
    ReconciliationResult Reconciliation,
    DrainReceipt Drain,
    AuthorityTransferReceipt? Transfer,
    ControlProbeReceipt ControlProbe,
    DateTimeOffset CompletedUtc,
    string Verdict);
```

MANIFEST VERDICT 56-AM

Verdict	Meaning
ADMITTED	all required fresh evidence passed
REJECTED	valid evidence contradicts transition
INVALID	required evidence missing/stale/tampered
REVERSED	controlled reversal completed
PENDING	design or incomplete runtime execution

FROM SERVICES TO RUNTIME

UNIT AND POLICY TESTS

Routing rules are executable architecture

EXECUTABLE ASSET 56-AR - DECISION MATRIX TEST

```
[Theory]
[InlineData(MigrationStage.LegacyOnly, false, "LegacyModule")]
[InlineData(MigrationStage.ShadowRead, false, "LegacyModule")]
[InlineData(MigrationStage.WritePilot, true, "DistributedAlarm")]
public async Task Create_route_is_deterministic(
    MigrationStage stage, bool commands, string expected)
{
    var plan = Plans.For(stage, commands, pilotUnits: [UnitId.U01]);
    RouteDecision first = await fixture.DecideCreateAsync(plan, UnitId.U01, Fixed.Request);
    RouteDecision retry = await fixture.DecideCreateAsync(plan, UnitId.U01, Fixed.Request);
    first.Primary.ToString().Should().Be(expected);
    retry.Primary.Should().Be(first.Primary);
    retry.AuthorityEpoch.Should().Be(first.AuthorityEpoch);
}
```

UNIT SUITE 56-AN

Test	Rejects
legal stage graph	stage skipping
plan digest chain	stale/edited policy
sticky authority	writer drift across retries
shadow admission	unsafe method/credential
normalizer corpus	representation false positives
rollback matrix	unsafe immediate reversal

CONTRACT AND COMPONENT TESTS

Both paths must satisfy the same public behavior

EXECUTABLE ASSET 56-AS - SHARED CONTRACT CORPUS

```
public abstract class AlarmPathContractTests
{
    protected abstract Task<IAAlarmPath> CreatePathAsync();

    [Fact]
    public async Task Duplicate_create_returns_same_public_receipt()
    {
        IAAlarmPath path = await CreatePathAsync();
        AlarmAcceptedV1 first = await path.CreateAsync(
            Fixed.Unit, Fixed.Create, Fixed.Idempotency, default);
        AlarmAcceptedV1 second = await path.CreateAsync(
            Fixed.Unit, Fixed.Create, Fixed.Idempotency, default);
        second.Should().Be(first);
    }
}
```

PROVIDER-BACKED SUITE 56-AO

Surface	Proof
authority store	unique reservation and CAS epoch
legacy outbox	fact only after commit
distributed Inbox/Outbox	idempotent fact/effect
backfill	resume/no-op/conflict/gap
reconciliation	missing/mismatch/stale detection

PRODUCTION-PATH COEXISTENCE TESTS

The real edge, both paths and broker run together

EXECUTABLE ASSET 56-AT - ROUTED COMMAND TEST

```
[Collection("NEXUS-1 extraction runtime")]
public sealed class CoexistenceTests(ExtractionRuntime runtime)
{
    [Fact]
    public async Task Pilot_command_has_one_writer_and_one_public_fact()
    {
        await runtime.TransitionToWritePilotAsync(UnitId.U01);
        AlarmAcceptedV1 accepted = await runtime.Edge.CreateAsync(Fixed.Create);
        DistributedSnapshot distributed = await runtime.Distributed.GetAsync(accepted.AlarmId);
        LegacySnapshot legacy = await runtime.Legacy.TryGetAsync(accepted.AlarmId);
        distributed.Exists.Should().BeTrue();
        legacy.Exists.Should().BeFalse();
        (await runtime.Broker.FactsForAsync(accepted.AlarmId)).Should().ContainSingle();
    }
}
```

RUNTIME SCENARIOS 56-AP

Scenario	Required result
legacy baseline	all traffic old path
shadow mismatch	client served; transition blocked
write timeout	retry same new writer; one fact
fenced transfer	no command accepted during epoch change
safe read reversal	route restored with retained finding

DELIBERATELY BROKEN EXTRACTION CAMPAIGNS

The verifier must detect a plausible but unsafe migration

BROKEN CAMPAIGNS 56-AQ

Mutation	Expected rejection
percentage-route commands	UNSTABLE_WRITE_PARTITION
fallback to legacy after distributed timeout	CROSS_AUTHORITY_RETRY
shadow credential can POST	SHADOW_CAN_MUTATE
skip delta high-water	RECONCILIATION_NOT_FRESH
transfer epoch with non-empty legacy outbox	LEGACY_NOT_DRAINED
two facts for same aggregate version	DUAL_PUBLICATION
retire with active legacy caller	RETIREMENT_DEPENDENCY_PRESENT

EXECUTABLE ASSET 56-AU - MUTATION ASSERTION

```
[Theory]
[MemberData(nameof(BrokenExtractionCases.All))]
public async Task Evidence_gate_rejects_unsafe_case(BrokenExtractionCase broken)
{
    ExtractionEvidence evidence = await fixture.ExecuteAsync(broken);
    StageGateResult result = verifier.Verify(broken.Transition, evidence);
    result.Verdict.Should().NotBe(StageVerdict.Admitted);
    result.Findings.Should().Contain(x => x.Code == broken.ExpectedCode);
}
```

RETIREMENT DEPENDENCY INVENTORY

Zero routed traffic is necessary but not sufficient

LEGACY DEPENDENCY SCAN 56-AR

Dependency	Retirement proof
HTTP routes	no route targets legacy for capability
module callers	static architecture inventory empty
database reads/writes	query/permission audit empty
legacy outbox	drained and producer disabled
scheduled jobs	removed or redirected
operational dashboards/alerts	new ownership installed
reversal window	expired by admitted decision

EXECUTABLE ASSET 56-AV - RETIREMENT GATE

```
public static void AdmitRetirement(RetirementEvidence e)
{
    Require(e.RouteCountToLegacy == 0, "legacy route remains");
    Require(e.StaticCallers.Length == 0, "legacy caller remains");
    Require(e.DatabaseAccesses.Length == 0, "legacy data access remains");
    Require(e.OutboxPending == 0, "legacy outbox not empty");
    Require(e.ObservationWindow >= e.RequiredWindow, "window too short");
    Require(e.ReversalDecision == "EXPLICITLY_CLOSED", "reversal still required");
}
```

STANDARDS AND SOURCE LEDGER

Patterns and tools supply mechanisms; the migration owns proof

SOURCE LEDGER 56-AS

Official source	Used for	Local qualification
Strangler Fig pattern	facade and incremental replacement	one-writer stages added
Anti-Corruption Layer	semantic translation boundary	no business rules in adapter
YARP direct forwarding	dynamic destination and typed errors	NEXUS route/authority owns policy
YARP routes/config	stable route/cluster concepts	canonical plan digest retained
.NET feature management	contextual read/stage gates	never durable command authority
integration-event guidance	public facts between systems	Inbox/Outbox + epoch evidence

EVIDENCE BOUNDARY

Pattern documentation explains why incremental replacement is useful. Only the retained NEXUS-1 comparison, reconciliation, authority and cutover manifests can prove this extraction.

ANTI-PATTERN REVIEW

Twelve migration shortcuts remain unsafe

REJECTED CLAIMS 56-AT

Claim	Why invalid
ten percent of commands is a pilot	writer is not sticky
fallback improves reliability	first writer may have committed
dual-write keeps stores synchronized	partial success creates two truths
shared tables simplify coexistence	ownership is not extracted
raw JSON diff proves parity	representation noise or hidden semantics
shadow success proves commands	writes and invariants were not exercised
row counts match	per-aggregate values may differ
latest timestamp is a cursor	ties and clock drift
feature flag owns routing	mutable flag lacks aggregate authority
green health admits cutover	history/outbox may lag
route flip is always rollback	legacy may lack new writes
zero traffic means remove	jobs/data/callers may remain

REVIEW QUESTION

Can the team name the writer, route plan, authority epoch, state checkpoint and safe reversal action for any Alarm at any migration stage? If not, coexistence remains ambiguous.

COMPLETION LEDGER

What Chapter 56 now specifies

COMPLETION LEDGER 56-AU

Obligation	Asset	Status
first strangler seam	stable Alarm V1 facade	SPECIFIED
traffic routing	typed plan and YARP forwarding	SPECIFIED
one writer	aggregate authority ledger + epoch	SPECIFIED
read comparison	bounded shadow + semantic diff	SPECIFIED
data coexistence	snapshot/delta/reconciliation	SPECIFIED
cutover/reversal	fence, drain, transfer, reverse	SPECIFIED
retirement	dependency and evidence gate	SPECIFIED
executed migration	retained runtime manifests	PENDING

OPEN RESIDUALS 56-AV

Residual	Consequence / owner
no executable repository checkpoint	listings remain compile-oriented
no retained shadow corpus	semantic parity unproved
no measured fence duration	availability budget educational
no executed reverse backfill	post-write reversal unproved
images and packages not CI-produced	Chapter 57 owns gates

EVIDENCE VERDICT

SPECIFIED | RUNTIME PENDING. The facade, route stages, authority ledger, anti-corruption adapters, shadow comparison, backfill/reconciliation, cutover, reversal and retirement gates are complete as design; no successful migration is invented.

HANDOFF TO CHAPTER 57

CI Quality Gates and Container Images

Chapter 57 will turn the reproducible runtime and reversible extraction assets into a delivery pipeline. It must restore and compile deterministically, run unit, architecture, component, contract, security, image and end-to-end gates, and produce containers with explicit source revision, version, dependency and digest identity.

BOUNDARY CARRIED FORWARD 56-AW

Chapter 56 leaves	Chapter 57 must add
public contract corpus	contract gate and compatibility report
route/authority tests	unit and architecture CI gates
provider-backed coexistence tests	ephemeral component stage
Compose extraction campaigns	end-to-end pipeline stage
runtime/extraction manifests	artifact retention and provenance
compile-oriented image identities	built, scanned and digest-pinned images

NEXT INVARIANT

A migration artifact is deliverable only when the same source and dependency identities pass every required gate and produce an immutable image whose provenance joins the runtime and extraction evidence.

FINAL CHAPTER RULE

Extraction ends with proven ownership, not redirected traffic

NEXUS-1 calls the Alarm boundary extracted only when the public seam is stable, every command has one writer, history and deltas reconcile, facts are conserved, stage transitions and reversal are retained, and the old path has no remaining dependency after an explicit evidence window.

PART X PROGRESS

Chapter 55 made the laboratory reproducible; Chapter 56 made change incremental and reversible. The next chapters must make delivery repeatable, rollout compatible, orchestration proportional and the final demonstrator honest.

PART X - MIGRATION, DELIVERY, AND FINAL ARCHITECTURE

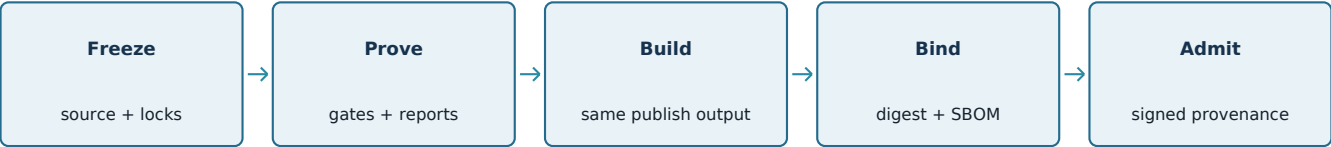
57. CI Quality Gates and Container Images

A candidate becomes deliverable only when one immutable source identity passes every required gate and produces digest-addressed images, SBOM, provenance and retained evidence from the same build graph.

CHAPTER DECISION

Use one fail-closed delivery graph. Pull requests may restore, compile, test and build untrusted local images without secrets. Only a trusted commit that repeats the required gates may push an image, and promotion always refers to its registry digest rather than rebuilding or trusting a mutable tag.

PROOF PATH 57-A - FROM SOURCE TO ADMITTED IMAGE



GUARANTEE LEDGER 57-A

Evidence	Allowed claim	Required binding	Cannot claim
gate receipt	named check passed	commit + input digest	image passed
image digest	registry bytes identified	manifest sha256	runtime is healthy
SBOM	components were enumerated	subject digest	no vulnerability exists
provenance	builder and inputs recorded	subject + workflow	production approved

EVIDENCE VERDICT

SPECIFIED | RUNTIME PENDING. The gates, identities, container policy and attestation joins are defined; no repository run, registry push, scan result or signed attestation is invented.

ORDINARY DEVELOPER PROBLEM

Everything was green, but nothing identifies what shipped

A pull request restores floating packages, builds again inside Docker, runs a subset of tests and pushes latest. The tag later points elsewhere. A scan report names a tag rather than a digest, the compose test used a local image, and no retained record proves that the tested binaries are the bytes in the registry.

FAILURE LEDGER 57-B

Shortcut	Failure	Correction
implicit restore in every step	dependency graph may change	one locked restore
test then rebuild in Docker	different binary lineage	publish once; COPY output
success with zero tests	false green	expected inventory gate
scan image:latest	subject is mutable	scan registry digest
push from pull request	untrusted code reaches credentials	separate trust zones
rebuild per environment	promotion changes artifact	promote one digest

PLAIN-LANGUAGE PRINCIPLE

CI is a proof assembly line. A green check has meaning only when its inputs, subject, verdict and retained output can be joined to the exact artifact being considered.

CHAPTER BOUNDARY

Build and admission here; rollout next

This chapter owns source/toolchain identity, dependency locks, gate ordering, test discovery, image construction, SBOM, vulnerability policy, provenance and digest promotion. Chapter 58 owns compatibility windows, rollout order, database expansion and rollback safety.

DECISION RECORD

ADR-057 - Build once, prove once, promote by digest

DECISION CONSEQUENCES 57-C

Choice	Benefit	Cost / boundary
locked source and dependencies	repeatable graph	intentional update workflow
one publish output	tests and image share binaries	larger retained artifact
fail-closed gate catalogue	missing evidence cannot pass	more pipeline plumbing
untrusted/trusted split	secrets avoid pull requests	trusted commit repeats gates
digest-only promotion	immutable subject	tags become discovery aliases
SBOM + provenance	supply-chain traceability	verification policy required

NON-GOAL

The chapter does not claim bit-for-bit reproducibility across arbitrary builders, a vulnerability-free image, SLSA level attainment, production authorization or successful deployment. Those require retained executions and an explicit trust policy.

CLAIM LADDER

Compile, test, image and release are different claims

CLAIM LADDER 57-D

Level	Subject	Minimum evidence	Does not prove
restored	dependency graph	locks accepted	source compiles
compiled	source revision	warning-clean build	tests discovered
tested	published binaries	typed test receipts	image policy
imaged	OCI manifest digest	config/layers/labels	runtime scenario
attested	image digest	SBOM + provenance	deployment safety
candidate	evidence bundle	all required joins admitted	rollout is reversible

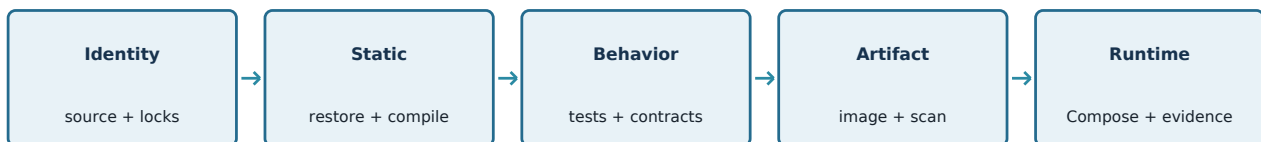
NO TRANSITIVE GREEN

A later green gate does not repair an earlier missing gate. An image scan cannot prove unit tests; an end-to-end scenario cannot prove dependency locks; provenance cannot make an unsafe workflow safe.

DELIVERY GRAPH

The graph is ordered by evidence dependency

GATE GRAPH 57-E - FAIL CLOSED



REQUIRED ORDER 57-F

Gate	Consumes	Emits
identity	checkout	BuildIdentity
restore	identity + lock files	RestoreReceipt
compile	restored graph	PublishManifest
test families	same binaries + fixtures	TestReceipts
image	publish manifest + pinned bases	ImageReceipt
security	image digest + SBOM	SecurityVerdict
runtime	digest-pinned Compose	RuntimeManifest
candidate	all receipts	CandidateManifest

PIPELINE TRUST ZONES

Untrusted code may prove; only trusted code may publish

TRUST-ZONE MATRIX 57-G

Event	Secrets	Allowed outputs	Forbidden
pull_request	none	reports + local OCI archive	push, sign, deploy
merge to protected main	scoped token	digest push + attestations	mutable deploy
version tag	scoped token	candidate alias for existing digest	rebuild
manual promotion	environment approval	digest reference only	tag-only subject

THREAT BOUNDARY

A workflow triggered by untrusted pull-request code must not receive registry credentials, signing authority or an environment token. Re-running the trusted commit is duplication of proof, not permission to reuse untrusted artifacts blindly.

BUILD IDENTITY

Every receipt carries one canonical identity

EXECUTABLE ASSET 57-A - IDENTITY RECORD

```
public sealed record BuildIdentity(  
    string Repository,  
    string CommitSha,  
    string SourceTreeSha256,  
    string ToolchainLockSha256,  
    string PackageLocksSha256,  
    string ContractCorpusSha256,  
    string DockerInputsSha256,  
    string WorkflowSha256,  
    string BuilderImageIdentity,  
    string Configuration,  
    string TargetFramework,  
    string RuntimeIdentifier)  
{  
    public string CanonicalDigest() =>  
        CanonicalJson.Sha256(this);  
}
```

IDENTITY EXCLUSIONS 57-H

Not identity	Reason
branch name	moves to another commit
latest tag	registry alias is mutable
workflow run number	identifies execution, not inputs
cache key alone	cache is an accelerator, not authority
semantic version alone	may be assigned to different bytes

IDENTITY ADMISSION

A malformed or dirty identity stops before restore

EXECUTABLE ASSET 57-B - IDENTITY VALIDATOR

```
public static BuildIdentity Admit(BuildIdentity x)  
{  
    Require(Regex.IsMatch(x.CommitSha, "[0-9a-f]{40}$"),  
        "SOURCE_REVISION_INVALID");  
    Require(AllSha256(x.SourceTreeSha256, x.ToolchainLockSha256,  
        x.PackageLocksSha256, x.ContractCorpusSha256,  
        x.DockerInputsSha256, x.WorkflowSha256),  
        "INPUT_DIGEST_INVALID");  
    Require(x.Configuration == "Release", "CONFIGURATION_NOT_RELEASE");  
    Require(!string.IsNullOrEmpty(x.BuilderImageIdentity),  
        "BUILDER_IMAGE_UNIDENTIFIED");  
    Require(x.TargetFramework == "net8.0", "TFM_UNEXPECTED");  
    Require(x.RuntimeIdentifier == "linux-x64", "RID_UNEXPECTED");  
    return x;  
}
```

DIRTY TREE RULE

Local reproduction may run from a dirty tree, but its identity is explicitly NON-CANDIDATE. Candidate evidence requires an exact checked-out commit and a source-tree digest derived after checkout.

REPOSITORY CONTRACT

Pipeline assets are versioned beside the code

EXECUTABLE ASSET 57-C - DELIVERY TREE

```
/.github/workflows/ci-pr.yml
/.github/workflows/ci-main.yml
/global.json
/Directory.Build.props
/Directory.Packages.props
/packages.lock.json
/eng/ci/gates.json
/eng/ci/run-gate.sh
/eng/ci/verify-actions.py
/eng/ci/verify-evidence.py
/eng/container/base-images.lock.json
/eng/container/verify-image.py
/eng/security/vulnerability-policy.json
/contracts/v1/corpus/
/deploy/compose/compose.evidence.yaml
/Dockerfile
/.dockerignore
```

EXACT PATH RULE

A chapter listing is a proposed asset until the repository contains the same path and retained execution. The final edition must replace this PENDING boundary with repository and run identities.

TOOLCHAIN LOCK

SDK, workload and base images are inputs, not ambience

TOOLCHAIN LOCK 57-I

Input	Required identity	Gate
.NET SDK	global.json exact feature-band version	dotnet --info match
NuGet sources	nuget.config digest + allowlist	unexpected feed rejected
BuildKit frontend	Dockerfile syntax digest	frontend material retained
SDK base image	repository@sha256	tag-only FROM rejected
ASP.NET base image	repository@sha256	tag-only FROM rejected
scanner	tool/image digest + policy version	unidentified scan rejected

SERVICING POLICY

A lock is not permission to remain stale. Toolchain and base-image updates are reviewed changes that regenerate evidence; CI never updates them implicitly while judging an unrelated commit.

SDK BOOTSTRAP

The runner installs the exact SDK through a hashed bootstrap

EXECUTABLE ASSET 57-D0 - BOOTSTRAP-DOTNET.SH

```
set -euo pipefail
SDK_VERSION="$(python3 -c \
'import json; print(json.load(open("global.json"))["sdk"]["version"]'))"
INSTALLER_SHA256="$(python3 -c \
'import json; print(json.load(open("eng/ci/toolchain.lock.json"))["dotnetInstallerSha256"]'))"
INSTALLER="artifacts/tooling/dotnet-install.sh"

curl --proto 'https' --tlsv1.2 --fail --location \
https://dot.net/v1/dotnet-install.sh --output "$INSTALLER"
printf '%s %s\n' "$INSTALLER_SHA256" "$INSTALLER" | sha256sum --check
bash "$INSTALLER" --version "$SDK_VERSION" \
--install-dir "$RUNNER_TEMP/dotnet" --no-path
"$RUNNER_TEMP/dotnet/dotnet" --info > artifacts/toolchain/dotnet-info.txt
```

BOOTSTRAP TRUST

global.json selects the exact SDK, while toolchain.lock.json identifies the installer bytes. Updating either is a reviewed dependency change. The runner's preinstalled SDK is never accepted merely because it is convenient.

LOCKED RESTORE

One restore establishes the dependency graph

EXECUTABLE ASSET 57-D - RESTORE GATE

```
set -euo pipefail
test -z "$(git status --porcelain)" || exit 41
dotnet --info > artifacts/toolchain/dotnet-info.txt

dotnet restore NEXUS1.sln \
--locked-mode \
--configfile nuget.config \
-p:RestoreIgnoreFailedSources=false \
-p:RestoreNoCache=true

git diff --exit-code -- \
packages.lock.json '**/packages.lock.json'
```

RESTORE OUTCOMES 57-J

Outcome	Verdict
lock exactly satisfies project	RESTORE_ADMITTED
lock would change	PACKAGE_LOCK_DRIFT
feed unavailable	RESTORE_SOURCE_UNAVAILABLE
unexpected package source	PACKAGE_SOURCE_UNTRUSTED
implicit later restore	PIPELINE_GRAPH_AMBIGUOUS

PACKAGE GRAPH EVIDENCE

A lock file is validated as a set, not merely present

EXECUTABLE ASSET 57-E - DEPENDENCY RECEIPT

```
public sealed record PackageGraphReceipt(  
    string NuGetConfigSha256,  
    string CentralPackagesSha256,  
    string CombinedLocksSha256,  
    int DirectCount,  
    int TransitiveCount,  
    string[] Sources,  
    string TargetFramework,  
    string RuntimeIdentifier);  
  
Require(receipt.Sources.SequenceEqual(AllowedSources),  
    "PACKAGE_SOURCE_SET_CHANGED");  
Require(receipt.DirectCount > 0, "PACKAGE_GRAPH_EMPTY");  
Require(receipt.CombinedLocksSha256 == identity.PackageLocksSha256,  
    "PACKAGE_LOCK_IDENTITY_MISMATCH");
```

VULNERABILITY DISTINCTION

Locked restore proves which versions were selected. A separate security gate evaluates known advisories for that exact graph. Neither claim substitutes for the other.

DETERMINISTIC COMPILE PROFILE

The build embeds source identity and normalizes CI paths

EXECUTABLE ASSET 57-F - DIRECTORY.BUILD.PROPS

```
<Project>  
  <PropertyGroup>  
    <TargetFramework>net8.0</TargetFramework>  
    <Nullable>enable</Nullable>  
    <ImplicitUsings>enable</ImplicitUsings>  
    <Deterministic>true</Deterministic>  
    <TreatWarningsAsErrors>true</TreatWarningsAsErrors>  
    <ContinuousIntegrationBuild  
      Condition="'$(GITHUB_ACTIONS)' == 'true'">true</ContinuousIntegrationBuild>  
    <SourceRevisionId Condition="'$(SourceRevisionId)' == ''>  
      unknown-local  
    </SourceRevisionId>  
    <IncludeSourceRevisionInInformationalVersion>true</IncludeSourceRevisionInInformationalVersion>  
  </PropertyGroup>  
</Project>
```

CANDIDATE GUARD

unknown-local is acceptable for developer builds but rejected by the candidate validator. CI passes the admitted 40-hex revision and verifies the emitted informational version.

COMPILE GATE

Restore does not happen again behind the build

EXECUTABLE ASSET 57-G - COMPILE AND PUBLISH ONCE

```
dotnet build NEXUS1.sln \
  --configuration Release \
  --no-restore \
  -p:SourceRevisionId="$GITHUB_SHA" \
  -p:ContinuousIntegrationBuild=true \
  -bl:artifacts/build/nexus1.binlog

dotnet publish src/AlarmManagement/AlarmManagement.csproj \
  --configuration Release \
  --no-restore --no-build \
  -p:SourceRevisionId="$GITHUB_SHA" \
  --output artifacts/publish/alarm-management
```

COMPILE RECEIPT 57-K

Field	Meaning
sourceRevisionId	embedded full commit
publishTreeSha256	canonical file-tree digest
binlogSha256	diagnostic build record
warningCount	must be zero under policy
assemblyInventory	expected outputs present

WARNING POLICY

Warnings are either fixed or owned explicitly

WARNING POLICY 57-L

Category	Default	Exception evidence
compiler/analyzer	error	code + owner + expiry
nullable	error	narrow suppression with justification
package advisory	security gate	risk acceptance + expiry
generated source	scoped rule	generator identity
obsolete API	error	migration issue + deadline

EXECUTABLE ASSET 57-H - EXCEPTION SCHEMA

```
public sealed record GateException(
    string Code,
    string Scope,
    string Owner,
    string Reason,
    DateOnly ExpiresOn,
    string ApprovalCommit);

Require(x.ExpiresOn >= clock.Today, "EXCEPTION_EXPIRED");
Require(FullSha(x.ApprovalCommit), "EXCEPTION_APPROVAL_INVALID");
Require(AllowedCodes.Contains(x.Code), "EXCEPTION_CODE_FORBIDDEN");
```

TEST CATALOGUE

Every claimed test family has an expected inventory

REQUIRED TEST FAMILIES 57-M

Family	Subject	Required proof
Domain	aggregates and policies	deterministic outcomes
Application	ports and orchestration	calls/effects/order
Architecture	layer and ownership rules	forbidden dependency mutations
Component	real provider adapters	transaction/concurrency
Contract	HTTP/event corpora	golden + rejected bytes
Security	authorization and data exposure	deny paths + redaction
Runtime	digest-pinned Compose	scenario + recovery evidence

INVENTORY RULE

A test command returning zero is a pipeline failure. The catalogue records project path, expected framework, minimum discovered count, shard, evidence schema and owner.

ZERO-DISCOVERY GATE

Exit code zero is not enough

EXECUTABLE ASSET 57-I - TEST INVENTORY VERIFIER

```
public static void Admit(TestRun run, TestExpectation expected)
{
    Require(run.ExitCode == 0, "TEST_PROCESS_FAILED");
    Require(run.Discovered >= expected.MinimumDiscovered,
        "TEST_DISCOVERY_BELOW_MINIMUM");
    Require(run.Executed == run.Discovered - run.ExplicitlySkipped,
        "TEST_EXECUTION_INCOMPLETE");
    Require(run.Failed == 0, "TEST_FAILURE");
    Require(run.ResultFiles.Length > 0, "TEST_EVIDENCE_MISSING");
    Require(run.IdentityDigest == expected.IdentityDigest,
        "TEST_IDENTITY_MISMATCH");
}
```

FALSE-GREEN CASES 57-N

Case	Stable failure
wrong project glob	TEST_PROJECT_SET_CHANGED
adapter absent	TEST_DISCOVERY_BELOW_MINIMUM
all tests filtered	TEST_FILTER_EMPTY
stale TRX reused	TEST_IDENTITY_MISMATCH
shard missing	TEST_SHARD_INCOMPLETE

FAST GATES

Unit and architecture tests fail before infrastructure cost

EXECUTABLE ASSET 57-J - FAST TEST STAGE

```
dotnet test tests/NEXUS1.Domain.Tests/NEXUS1.Domain.Tests.csproj \
  -c Release --no-restore --no-build \
  --logger "trx;LogFileName=domain.trx" \
  --results-directory artifacts/tests/domain

dotnet test tests/NEXUS1.Architecture.Tests/NEXUS1.Architecture.Tests.csproj \
  -c Release --no-restore --no-build \
  --logger "trx;LogFileName=architecture.trx" \
  --results-directory artifacts/tests/architecture

dotnet run --project eng/ci/TestReceipt --no-build -- \
  verify eng/ci/gates.json artifacts/tests
```

COST ORDER

Cheap deterministic gates run first, but no gate is silently omitted because a later one is expensive. Requiredness belongs to the catalogue, not to a developer's intuition.

PROVIDER-BACKED COMPONENT GATE

Real persistence semantics run in an isolated lease

COMPONENT LEASE 57-O

Property	Requirement
database	declared provider image digest
isolation	unique database/schema per run
migration	same artifact planned for runtime
seed	deterministic fixture digest
cleanup	bounded and receipt-producing
evidence	provider/version/connection class, never secret

EXECUTABLE ASSET 57-K - COMPONENT-STAGE ADMISSION

```
await using ProviderLease lease = await factory.CreateAsync(identity, token);
await lease.ApplyMigrationsAsync(token);
ComponentRun run = await runner.ExecuteAsync(lease, token);

Require(run.ProviderDigest == lease.ProviderDigest,
  "PROVIDER_IDENTITY_MISMATCH");
Require(run.TestReceipt.Failed == 0, "COMPONENT_TEST_FAILED");
Require(run.RollbackProbe.Passed, "TRANSACTION_ROLLBACK_UNPROVED");
Require(run.ConcurrencyProbe.Passed, "CONCURRENCY_UNPROVED");
```

CONTRACT GATE

Compatibility is evaluated against retained public bytes

CONTRACT SURFACES 57-P

Surface	Positive corpus	Negative corpus
Alarm HTTP V1	request/response golden JSON	unknown member/status drift
AlarmFloodDetected.v1	canonical envelope bytes	fingerprint/type/schema mutation
RootCauseCaseOpened.v1	producer and causality bytes	lineage/order mutation
BFF projection	stable client view	private field exposure
health	safe status schema	exception/secret leakage

EXECUTABLE ASSET 57-L - CORPUS RECEIPT

```
public sealed record ContractCorpusReceipt(
    string CorpusSha256,
    string ConsumerBaselineSha256,
    int GoldenCases,
    int RejectedCases,
    string[] SchemaIds,
    TestReceipt Producer,
    TestReceipt Consumer);

Require(x.CorporSha256 == identity.ContractCorpusSha256,
    "CONTRACT_CORPUS_IDENTITY_MISMATCH");
Require(x.GoldenCases > 0 && x.RejectedCases > 0,
    "CONTRACT_CORPUS_INCOMPLETE");
```

COMPATIBILITY REPORT

A schema diff is classified by actual consumers

COMPATIBILITY CLASSES 57-Q

Change	Default verdict	Required next action
add optional response field	review	old/new consumer corpus
remove or rename field	reject	new version or compatibility window
widen enum without tolerant reader	reject	consumer upgrade first
change event meaning	reject	new event contract
internal DTO change	not public	architecture boundary proves isolation

CHAPTER 58 BRIDGE

This gate detects compatibility obligations. Chapter 58 will decide rollout order and the duration of overlap; CI must not infer that a syntactically additive change is operationally safe.

ROUTE AND AUTHORITY REGRESSION

Extraction safety becomes a mandatory gate

EXECUTABLE ASSET 57-M - CHAPTER 56 GATE

```
[Theory]
[MemberData(nameof(MigrationStages.All))]
public async Task Every_command_has_one_sticky_writer(MigrationStage stage)
{
    var first = await fixture.RouteCreateAsync(stage, AlarmId.A17);
    var retry = await fixture.RouteRetryAsync(stage, AlarmId.A17,
        first.IdempotencyKey);

    retry.Writer.Should().Be(first.Writer);
    retry.AuthorityEpoch.Should().Be(first.AuthorityEpoch);
    fixture.CommandInvocations.For(AlarmId.A17).Writers
        .Should().ContainSingle();
}
```

MUTATION OBLIGATIONS 57-R

Mutation	Expected rejection
percentage route a command	UNSTABLE_WRITE_PARTITION
fallback after timeout	CROSS_AUTHORITY_RETRY
transfer before outbox drain	LEGACY_NOT_DRAINED
retire with caller	RETIREMENT_DEPENDENCY_PRESENT

SECURITY GATE FAMILIES

Security is several subjects, not one scanner

SECURITY MATRIX 57-S

Gate	Subject	Fail-closed evidence
dependency	locked NuGet graph	advisories + policy
workflow	YAML and actions	permissions + immutable pins
source	code/config	static findings + secret patterns
container	digest + filesystem	packages/CVEs/config
contract	HTTP/events/logs	authorization + disclosure tests
provenance	attestation statement	builder/source/subject policy

NO UNIVERSAL GREEN

A clean source scan says nothing about packages in the runtime base image. A clean image scan says nothing about workflow token permissions. Each gate has a separate subject and receipt.

WORKFLOW HARDENING

Actions, permissions and triggers are code dependencies

EXECUTABLE ASSET 57-N - FAIL-CLOSED WORKFLOW POLICY

```
public static void Verify(WorkflowModel w)
{
    Require(w.TopLevelPermissions == PermissionSet.ReadOnly,
        "WORKFLOW_DEFAULT_PERMISSION_EXCESSIVE");
    Require(!w.UsesPullRequestTargetWithCheckout,
        "UNTRUSTED_PRIVILEGED_CHECKOUT");
    foreach (ActionUse use in w.Actions)
        Require(Regex.IsMatch(use.Revision, "[0-9a-f]{40}$"),
            "ACTION_NOT_PINNED_TO_FULL_SHA");
    foreach (Job job in w.Jobs)
        Require(job.Permissions.IsSubsetOf(job.RequiredPermissions),
            "JOB_PERMISSION_EXCESSIVE");
}
```

WORKFLOW CHECKS 57-T

Check	Reason
full action commit	tag can move
read-only default token	least privilege
no PR secrets	untrusted source boundary
environment approval	promotion authority
CODEOWNERS for workflows	review delivery changes

PULL-REQUEST WORKFLOW

The untrusted path has no publishing authority

EXECUTABLE ASSET 57-O - CI-PR.YML CORE

```
name: ci-pr
on:
  pull_request:
permissions:
  contents: read
concurrency:
  group: ci-pr-${{ github.event.pull_request.number }}
  cancel-in-progress: true
jobs:
  prove:
    runs-on: ubuntu-24.04
    timeout-minutes: 45
    steps:
      - uses: actions/checkout@11bd71901bbe5b1630ceea73d27597364c9af683
        with:
          persist-credentials: false
      - name: Verify and run local gates
        env:
          CANDIDATE_MODE: "false"
        run: ./eng/ci/run-gate.sh pr
```

PINNED EXAMPLE

The checkout revision is an immutable, reviewed dependency. The pin ledger records its upstream tag and source for maintenance; CI evaluates the full SHA, never the comment or tag.

TRUSTED WORKFLOW

Publishing authority is narrow and job-local

EXECUTABLE ASSET 57-P - TRUSTED JOB PERMISSIONS

```
name: ci-main
on:
  push:
    branches: [ main ]
permissions:
  contents: read
jobs:
  candidate:
    runs-on: ubuntu-24.04
    timeout-minutes: 60
    permissions:
      contents: read
      packages: write
      id-token: write
      attestations: write
    steps:
      - uses: actions/checkout@11bd71901bbe5b1630ceea73d27597364c9af683
        with:
          persist-credentials: false
      - run: ./eng/ci/run-gate.sh candidate
```

CONDITIONAL PLATFORM FEATURE

Attestation publication is enabled only when the repository plan and visibility support it. If unavailable, the candidate remains PENDING_ATTESTATION rather than silently weakening the policy.

SECRET BOUNDARY

Build arguments and logs are not secret stores

SECRET RULES 57-U

Surface	Allowed	Forbidden
pull-request job	no secrets	registry/signing/cloud token
Docker build	BuildKit secret mount	ARG/ENV credential
logs	digest and credential class	token/connection value
test fixtures	ephemeral synthetic values	production copy
attestation	secret identity only when safe	secret value

EXECUTABLE ASSET 57-Q - BUILDKIT SECRET USE

```
# syntax=docker/dockerfile:1
RUN --mount=type=secret,id=nuget_config,target=/run/secrets/nuget.config \
    dotnet restore NEXUS1.sln \
        --configfile /run/secrets/nuget.config \
        --locked-mode
```

PREFERRED NEXUS PATH

The candidate image normally copies an already admitted publish tree and performs no private-feed restore inside Docker. The secret-mount pattern is retained for cases where a controlled builder must restore.

CACHE BOUNDARY

A cache may accelerate proof but never define it

CACHE RULES 57-V

Cache	Key material	Post-restore validation
NuGet packages	OS + SDK + lock digest	locked restore still runs
BuildKit layers	Docker/context/base inputs	image digest is recomputed
test fixtures	fixture + provider digest	seed receipt compared
reports	never restored as current evidence	new run identity required

POISONING DEFENSE

Untrusted runs do not write caches later consumed by privileged jobs unless the cache namespace and content validation make the trust transition explicit.

PUBLISH-TREE MANIFEST

The exact files tested become the image input

EXECUTABLE ASSET 57-R - CANONICAL PUBLISH TREE

```
public static PublishManifest Create(string root, BuildIdentity identity)
{
    FileReceipt[] files = Directory.EnumerateFiles(root, "*",
        SearchOption.AllDirectories)
        .OrderBy(path => Path.GetRelativePath(root, path),
            StringComparer.Ordinal)
        .Select(path => FileReceipt.From(root, path))
        .ToArray();
    Require(files.Length > 0, "PUBLISH_TREE_EMPTY");
    return new PublishManifest(identity.CanonicalDigest(), files,
        CanonicalJson.Sha256(files));
}
```

MANIFEST EXCLUSIONS 57-W

Excluded	Reason
test result files	not runtime payload
source and .git	not required in runtime
developer secrets	must never enter context
temporary/cache files	unstable and unnecessary
runtime evidence	created after execution

CONTAINER BUILD CONTRACT

Docker consumes admitted output and pinned bases

EXECUTABLE ASSET 57-S - RUNTIME-ONLY DOCKERFILE

```
# syntax=docker/dockerfile:1
ARG RUNTIME_IMAGE
FROM ${RUNTIME_IMAGE} AS runtime
ARG SOURCE_REVISION
ARG BUILD_VERSION
WORKDIR /app
COPY artifacts/publish/alarm-management/ ./
LABEL org.opencontainers.image.revision=$SOURCE_REVISION \
    org.opencontainers.image.version=$BUILD_VERSION \
    org.opencontainers.image.source="https://github.com/gregory82gr/Nexus-1-phase-0"
USER $APP_UID
EXPOSE 8080
ENTRYPOINT ["dotnet", "Nexus1.AlarmManagement.Edge.dll"]
```

BASE-IMAGE RULE

CI supplies RUNTIME_IMAGE as repository@sha256 from base-images.lock.json. A tag-only FROM, an unrecorded digest or a build stage that recompiles the application is rejected.

CONTAINER RUNTIME POLICY

Small is useful; explicit is mandatory

IMAGE CONFIGURATION 57-X

Property	Policy
user	numeric non-root APP_UID
entrypoint	exec-form dotnet process
filesystem	no source, package cache or secret
ports	documented 8080 only
labels	source, revision, version, created, licenses
health	orchestrator invokes endpoint; image need not contain curl
platform	linux/amd64 until another platform is proven

NO COSMETIC HARDENING

A non-root label is not proof that the configured process runs non-root. CI inspects the image config and executes a container probe that reports the effective UID.

BUILD CONTEXT

The context digest makes hidden inputs visible

EXECUTABLE ASSET 57-T - .DOCKERIGNORE

```
.git
.github
**/bin
**/obj
**/TestResults
artifacts/*
!artifacts/publish
artifacts/publish/*
!artifacts/publish/alarm-management
!artifacts/publish/alarm-management/**
!artifacts/publish/alarm-management.manifest.json
secrets
*.user
*.pfx
*.key
```

EXECUTABLE ASSET 57-U - CONTEXT ADMISSION

```
ContextReceipt context = ContextHasher.Create(".", ".dockerignore");
Require(context.Files.All(AllowedContextFile),
    "DOCKER_CONTEXT_UNEXPECTED_FILE");
Require(context.SecretPatternMatches == 0,
    "DOCKER_CONTEXT_SECRET_CANDIDATE");
Require(context.PublishTreeSha256 == publish.TreeSha256,
    "DOCKER_CONTEXT_PUBLISH_MISMATCH");
```

REPRODUCIBLE TIMESTAMPS

The commit timestamp is an input, not wall-clock noise

EXECUTABLE ASSET 57-V - SOURCE EPOCH

```
SOURCE_DATE_EPOCH=$(git show -s --format=%ct "$GITHUB_SHA")
export SOURCE_DATE_EPOCH
test "$SOURCE_DATE_EPOCH" -gt 0

docker buildx build \
  --platform linux/amd64 \
  --build-arg SOURCE_REVISION="$GITHUB_SHA" \
  --build-arg BUILD_VERSION="$BUILD_VERSION" \
  --build-arg RUNTIME_IMAGE="$RUNTIME_IMAGE" \
  --output type=oci,dest=artifacts/images/alarm-management.oci.tar,rewrite-timestamp=true \
  .
```

HONEST REPRODUCIBILITY CLAIM

SOURCE_DATE_EPOCH removes common timestamp noise. NEXUS-1 still reports a rebuild comparison separately; it does not declare bit-for-bit reproducibility merely because this variable was set.

BUILD AND PUSH

Attestations and image share the registry digest

EXECUTABLE ASSET 57-W - CANDIDATE BUILD

```
IMAGE="ghcr.io/gregory82gr/nexus1-alarm-management"
docker buildx build \
  --platform linux/amd64 \
  --tag "$IMAGE:sha-$GITHUB_SHA" \
  --build-arg RUNTIME_IMAGE="$RUNTIME_IMAGE" \
  --build-arg SOURCE_REVISION="$GITHUB_SHA" \
  --build-arg BUILD_VERSION="$BUILD_VERSION" \
  --attest type=sbom \
  --attest type=provenance,mode=min,version=v1 \
  --metadata-file artifacts/images/build-metadata.json \
  --push .
```

PROVENANCE MODE

mode=min is the safe default for a public educational repository because it retains source revision, materials and builder without exposing build-argument values. Secrets never travel through ARG in either mode.

DIGEST EXTRACTION

The registry digest replaces the tag as authority

EXECUTABLE ASSET 57-X - IMAGE RECEIPT

```
public static ImageReceipt Admit(BuildMetadata m, BuildIdentity identity)
{
    string digest = m.ContainerImageDigest;
    Require(Regex.IsMatch(digest, "^sha256:[0-9a-f]{64}$"),
        "IMAGE_DIGEST_INVALID");
    Require(m.SourceRevision == identity.CommitSha,
        "IMAGE_REVISION_MISMATCH");
    Require(m.Platforms.SequenceEqual(["linux/amd64"]),
        "IMAGE_PLATFORM_SET_CHANGED");
    return new ImageReceipt(m.ImageName, digest, m.ConfigDigest,
        m.ManifestMediaType, identity.CanonicalDigest());
}
```

TAG SEMANTICS 57-Y

Reference	Use
image@sha256:...	promotion, scan, Compose and attestation subject
sha-<commit>	human discovery alias
version tag	release discovery alias after admission
latest	not used by evidence or deployment

SBOM EVIDENCE

The bill of materials is inspected, not merely generated

EXECUTABLE ASSET 57-Y - SBOM ADMISSION

```
public static void Admit(SbomDocument sbom, ImageReceipt image)
{
    Require(sbom.SubjectDigest == image.ManifestDigest,
        "SBOM_SUBJECT_MISMATCH");
    Require(sbom.Format == "SPDX-2.3", "SBOM_FORMAT_UNSUPPORTED");
    Require(sbom.Packages.Count > 0, "SBOM_EMPTY");
    Require(sbom.Packages.All(x => x.HasIdentity),
        "SBOM_COMPONENT_IDENTITY_MISSING");
    Require(sbom.Relationships.Count > 0, "SBOM_RELATIONSHIPS_EMPTY");
}
```

SCANNER SCOPE

The final-stage SBOM describes runtime contents. Build-time dependencies need their own builder/material evidence or explicit BuildKit stage scanning; they are not assumed to be present because the final image is small.

VULNERABILITY POLICY

Findings are evaluated against a versioned, expiring policy

VULNERABILITY VERDICT 57-Z

Finding	Default	Exception
critical, fix available	reject	none for candidate
high, fix available	reject	time-bounded approval
no fix available	risk review	owner + mitigation + expiry
unknown severity	review	cannot count as clean
suppressed CVE	verify match	exact package/range/reason
scanner error	reject	not zero findings

EXECUTABLE ASSET 57-Z - POLICY DECISION

```
SecurityVerdict Decide(ScanReceipt scan, VulnerabilityPolicy policy)
{
    Require(scan.SubjectDigest == image.ManifestDigest,
        "SCAN_SUBJECT_MISMATCH");
    Require(scan.Completed, "SCAN_INCOMPLETE");
    Require(scan.DatabaseIdentity is not null,
        "ADVISORY_DATABASE_UNIDENTIFIED");
    policy.RejectExpiredExceptions(clock.Today);
    return policy.Evaluate(scan.Findings);
}
```

IMAGE CONFIGURATION GATE

OCI metadata and effective runtime are both checked

EXECUTABLE ASSET 57-AA - IMAGE POLICY

```
ImageInspection x = inspector.Read(image.ManifestDigest);
Require(x.User.IsNumeric && x.User.Value != 0, "IMAGE_RUNS_AS_ROOT");
Require(x.Entrypoint.SequenceEqual(
    ["dotnet", "Nexus1.AlarmManagement.Edge.dll"]),
    "IMAGE_ENTRYPOINT_CHANGED");
Require(x.Labels["org.opencontainers.image.revision"] == identity.CommitSha,
    "IMAGE_LABEL_REVISION_MISMATCH");
Require(x.ExposedPorts.SequenceEqual(["8080/tcp"]),
    "IMAGE_PORT_SET_CHANGED");
Require(!x.Paths.Any(ForbiddenRuntimePath),
    "IMAGE_FORBIDDEN_CONTENT");
```

FORBIDDEN RUNTIME PATHS 57-AA

Pattern	Reason
/.git, /src	source leakage
/root/.nuget	package/cache leakage
*.pfx, *.key	credential material
appsettings.Production.json	environment value baked in

CONTAINER CONTRACT TEST

The image starts as a non-root process and exposes the expected seam

EXECUTABLE ASSET 57-AB - BOUNDED SMOKE PROBE

```
await using ContainerLease lease = await factory.StartAsync(
    image.ManifestDigest,
    new ContainerLimits(MemoryMb: 256, Cpu: 0.5), token);

Require(await lease.ReadUidAsync(token) != 0, "CONTAINER_ROOT_EFFECTIVE");
await probes.AwaitStartupAsync(lease, TimeSpan.FromSeconds(30), token);
HttpResponseMessage live = await lease.GetAsync("/health/live", token);
Require(live.StatusCode == HttpStatusCode.OK, "LIVENESS_FAILED");
SafeHealthDocument body = await live.ReadSafeHealthAsync(token);
Require(!body.ContainsDependencyDetails, "HEALTH_DISCLOSURE");
```

PROBE BOUNDARY

A single container smoke test proves entrypt, process identity and basic endpoint behavior. It does not prove database, broker or distributed scenarios; those belong to the digest-pinned Compose gate.

COMPOSE END-TO-END GATE

The runtime uses the registry digests just admitted

EXECUTABLE ASSET 57-AC - DIGEST LOCK

```
services:
  alarm-management:
    image: ${ALARM_IMAGE_NAME}@${ALARM_IMAGE_DIGEST}
  root-cause:
    image: ${ROOTCAUSE_IMAGE_NAME}@${ROOTCAUSE_IMAGE_DIGEST}
  query-bff:
    image: ${BFF_IMAGE_NAME}@${BFF_IMAGE_DIGEST}
```

EXECUTABLE ASSET 57-AD - RUNTIME GATE

```
./eng/runtime/up.sh --identity artifacts/candidate/images.lock.json
./eng/runtime/await-stable.sh --timeout 120
./eng/scenarios/run.sh alarm-to-root-cause-v1
./eng/scenarios/run.sh extraction-coexistence-v1
./eng/runtime/capture-evidence.sh artifacts/runtime
./eng/runtime/down.sh --bounded
./eng/ci/verify-evidence.py artifacts/runtime
```

NO LOCAL SUBSTITUTION

Compose refuses an image without a sha256 digest and records every resolved manifest. A developer-built tag may be used in NON-CANDIDATE mode only and can never satisfy this gate.

END-TO-END SCENARIO

The scenario asserts public meaning, not service uptime

SCENARIO RECEIPT 57-AB

Step	Required observation
operator command	stable BFF request accepted once
Alarm commit	owner state + pending public fact
broker transfer	fact delivered with identity
RootCause commit	Inbox + case + outbox atomic
query projection	delayed truth reaches expected version
trace	correlation and causation path complete
cleanup	runtime exits with bounded receipt

DELAYED CONSISTENCY

The gate waits on an explicit convergence condition and records elapsed time. It never inserts a fixed sleep and never reads another service's private database to declare success.

EXTRACTION CAMPAIGN GATE

Chapter 56 coexistence becomes executable evidence

CAMPAIGNS 57-AC

Campaign	Required proof
legacy authority	all commands stay legacy; shadow is read-only
read pilot	semantic comparison within freshness budget
write pilot	transferred aggregate has one distributed writer
timeout retry	same idempotency key and authority epoch
fenced transfer	drain, reconcile, transfer, control probe
pre-write reversal	route restores without state copy
post-write reversal	reverse backfill and reconcile required

EXECUTABLE ASSET 57-AE - CAMPAIGN ADMISSION

```
Require(run.ImageDigests.SequenceEqual(candidate.ImageDigests),
  "RUNTIME_IMAGE_SET_MISMATCH");
Require(run.AuthorityViolations == 0, "MULTIPLE_WRITERS_OBSERVED");
Require(run.DuplicatePublicFacts == 0, "DUAL_PUBLICATION");
Require(run.Reconciliation.Mismatches == 0,
  "RECONCILIATION_MISMATCH");
Require(run.Shutdown.Bounded, "RUNTIME_SHUTDOWN_UNBOUNDED");
```

RUNTIME EVIDENCE IDENTITY

Every observation joins back to the candidate

EXECUTABLE ASSET 57-AF - RUNTIME EVIDENCE ENVELOPE

```
public sealed record RuntimeGateReceipt(
  string CandidateDigest,
  string ComposeDigest,
  IReadOnlyDictionary<string, string> ImageDigests,
  string SeedDigest,
  string ScenarioCorpusDigest,
  string RuntimeManifestSha256,
  string TraceBundleSha256,
  string MetricsSnapshotSha256,
  string LogBundleSha256,
  DateTimeOffset StartedUtc,
  DateTimeOffset CompletedUtc,
  RuntimeGateVerdict Verdict);
```

VALUE-FREE IDENTITY

The receipt names evidence bundles by digest. Sensitive runtime values stay inside access-controlled artifacts and are never copied into image labels, metric dimensions or ordinary candidate metadata.

PROVENANCE POLICY

An attestation is parsed against expectations

EXECUTABLE ASSET 57-AG - PROVENANCE VERIFIER

```
public static void Verify(ProvenanceV1 p, CandidatePolicy expected)
{
    Require(p.PredicateType == "https://slsa.dev/provenance/v1",
        "PROVENANCE_SCHEMA_UNEXPECTED");
    Require(p.Subjects.Single().Digest == expected.ImageDigest,
        "PROVENANCE_SUBJECT_MISMATCH");
    Require(p.BuildDefinition.SourceRevision == expected.CommitSha,
        "PROVENANCE_SOURCE_MISMATCH");
    Require(p.RunDetails.BuilderId == expected.BuilderId,
        "PROVENANCE_BUILDER_UNTRUSTED");
    Require(p.Materials.ContainsAll(expected.MaterialDigests),
        "PROVENANCE_MATERIALS_INCOMPLETE");
}
```

LEVEL DISCIPLINE

NEXUS-1 records which provenance properties were observed. It does not claim a SLSA level unless every requirement for that level, including builder isolation and authenticity, is independently evaluated.

ARTIFACT ATTESTATION

The registry subject is name plus digest, never tag

ATTESTATION JOINS 57-AD

Statement	Subject	Required join
build provenance	image manifest digest	commit + workflow + materials
SBOM	same image digest	component inventory
security verdict	same image + SBOM	scanner database + policy
runtime receipt	candidate digest set	Compose + scenario

PLATFORM AVAILABILITY

GitHub artifact attestations can bind a container image name and digest when the repository and plan support them. The local CandidateManifest remains mandatory and reports publication state explicitly.

CANDIDATE MANIFEST

One document joins every gate without copying its payload

EXECUTABLE ASSET 57-AH - CANDIDATE RECORD

```
public sealed record CandidateManifest(
    BuildIdentity Identity,
    string RestoreReceiptSha256,
    string PublishManifestSha256,
    IReadOnlyList<TestEvidenceRef> Tests,
    IReadOnlyList<ImageReceipt> Images,
    string SbomAttestationDigest,
    string ProvenanceAttestationDigest,
    string SecurityVerdictSha256,
    string RuntimeGateReceiptSha256,
    string GateCatalogueSha256,
    CandidateVerdict Verdict);
```

CANDIDATE VERDICTS 57-AE

Verdict	Meaning
REJECTED	one required gate failed
INCOMPLETE	required evidence missing or stale
PENDING_ATTESTATION	build passed; publication unsupported/unavailable
ADMITTED	all declared candidate gates joined

CANDIDATE ADMISSION

Every reference is resolved and rehashed

EXECUTABLE ASSET 57-AI - FAIL-CLOSED ADMISSION

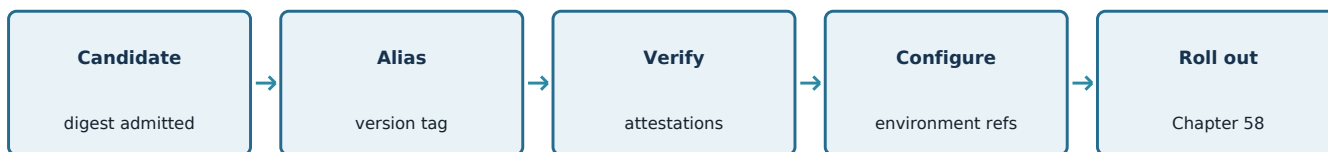
```
CandidateVerdict Admit(CandidateManifest c, CandidatePolicy p)
{
    Require(c.Identity.CanonicalDigest() == p.IdentityDigest,
        "CANDIDATE_IDENTITY_MISMATCH");
    EvidenceSet e = resolver.ResolveAndHash(c.AllReferences());
    Require(e.RequiredKinds.SetEquals(p.RequiredEvidence),
        "CANDIDATE_EVIDENCE_SET_INCOMPLETE");
    Require(e.All(x => x.IdentityDigest == p.IdentityDigest),
        "CROSS_BUILD_EVIDENCE");
    Require(e.All(x => x.Verdict == "ADMITTED"),
        "GATE_NOT_ADMITTED");
    Require(c.Images.All(DigestAddressed), "MUTABLE_IMAGE_REFERENCE");
    return CandidateVerdict.Admitted;
}
```

NO STALE EVIDENCE

A report copied from an earlier run fails because its BuildIdentity digest differs. Filename, branch and semantic version never authorize reuse.

PROMOTION MODEL

Environments receive a digest, not another build

PROMOTION 57-AF - NO REBUILD**PROMOTION PROHIBITIONS 57-AG**

Prohibited	Why
docker build in deployment job	artifact changes after proof
deploy image:latest	subject can move
copy image without digest verification	registry transfer unproved
patch config into image	environment creates new artifact
sign a tag	signature subject ambiguous

REBUILD COMPARISON

Repeatability is measured on a clean second builder

REBUILD COMPARISON 57-AH

Comparison	Interpretation
publish tree equal	.NET output reproducible for tested inputs
image manifest equal	OCI output bit-for-bit equal
only provenance differs	expected execution metadata separation
layer differs	inspect timestamp/order/base/cache inputs
package graph differs	lock or feed integrity failure

EXECUTABLE ASSET 57-AJ - COMPARISON RESULT

```
public sealed record RebuildComparison(  
    string IdentityDigest,  
    string FirstPublishTree,  
    string SecondPublishTree,  
    string FirstImageDigest,  
    string SecondImageDigest,  
    string[] DifferingPaths,  
    bool PublishEqual,  
    bool ImageEqual);
```

CLAIM BOUNDARY

A single equal pair is retained evidence for that input and builder policy. It is not a timeless guarantee that every future rebuild will be identical.

GATE ORCHESTRATION

Timeouts, cancellation and cleanup are first-class

EXECUTABLE ASSET 57-AK - BOUNDED GATE RUNNER

```
public async Task<GateReceipt> RunAsync(GateDefinition gate,  
    BuildIdentity identity, CancellationToken pipelineToken)  
{  
    using var timeout = new CancellationTokenSource(gate.Timeout);  
    using var linked = CancellationTokenSource.CreateLinkedTokenSource(  
        pipelineToken, timeout.Token);  
    DateTimeOffset started = clock.UtcNow;  
    try  
    {  
        GateOutput output = await gate.ExecuteAsync(linked.Token);  
        return verifier.Admit(gate, identity, started, output);  
    }  
    finally { await gate.CleanupAsync(CancellationToken.None); }  
}
```

CLEANUP LIMIT

Cleanup has its own bounded implementation and emits a receipt. Cancellation never converts an unknown database, broker or registry outcome into success.

PARALLELISM AND SHARDS

Parallel work rejoins under one identity

PARALLEL GATE RULES 57-AI

Rule	Enforcement
independent families may run concurrently	DAG dependency set
each shard has expected inventory	zero/missing shard rejected
shared provider state forbidden	unique lease per shard
coverage merged by module	same binaries and source digest
one failed shard fails family	no average verdict

EXECUTABLE ASSET 57-AL - SHARD JOIN

```
Require(receipts.Select(x => x.ShardId).ToHashSet()
    .SetEquals(expectation.RequiredShards), "TEST_SHARD_INCOMPLETE");
Require(receipts.All(x => x.IdentityDigest == identity.CanonicalDigest()),
    "TEST_SHARD_IDENTITY_MISMATCH");
Require(receipts.All(x => x.Verdict == GateVerdict.Admitted),
    "TEST_SHARD_FAILED");
```

COVERAGE MEANING

Coverage is a diagnostic, not the behavioral verdict

COVERAGE POLICY 57-AJ

Measure	Use	Cannot replace
line/branch by module	detect unexercised code	invariant assertions
changed-lines coverage	review focused risk	full regression
mutation score	test sensitivity	provider/runtime evidence
trend	detect sudden loss	absolute quality claim

NEXUS-1 RULE

A threshold failure blocks the configured gate. Passing the threshold never upgrades a test family beyond the behavior its assertions actually observe.

EVIDENCE RETENTION

Small manifests stay longer than heavy diagnostics

RETENTION CLASSES 57-AK

Class	Examples	Policy
candidate	manifest, image digests, attestations	release lifetime
gate	TRX, contract/security reports	review + audit window
runtime	logs, traces, metrics, DB receipts	bounded evidence window
diagnostic	binlog, raw scanner output	shorter failure analysis
sensitive	restricted evidence payload	access + deletion policy

MANIFEST DURABILITY

Deleting a heavy bundle does not permit its claim to become unverifiable. The retained manifest preserves subject digest, evidence digest, verdict, schema and retention decision.

LOCAL REPRODUCTION

The developer runs the same gate program without candidate authority

EXECUTABLE ASSET 57-AM - LOCAL ENTRY POINTS

```
./eng/ci/run-gate.sh identity
./eng/ci/run-gate.sh fast
./eng/ci/run-gate.sh component
./eng/ci/run-gate.sh contract
./eng/ci/run-gate.sh image-local
./eng/ci/run-gate.sh runtime-local
./eng/ci/verify-evidence.py artifacts
```

LOCAL VERSUS CANDIDATE 57-AL

Property	Local	Candidate
dirty tree	allowed, labeled	rejected
registry push	forbidden	trusted job only
attestation	local draft	platform-published
evidence schema	same	same
gate logic	same versioned program	same

FAILURE CODE CATALOGUE

Pipeline failures are stable, searchable evidence

FAILURE CATALOGUE 57-AM

Code	Meaning
PACKAGE_LOCK_DRIFT	restore would change selected graph
TEST_DISCOVERY_BELOW_MINIMUM	expected tests absent
CONTRACT_CORPUS_IDENTITY_MISMATCH	wrong public corpus
ACTION_NOT_PINNED_TO_FULL_SHA	mutable workflow dependency
CROSS_BUILD_EVIDENCE	receipt belongs to another input
IMAGE_LABEL_REVISION_MISMATCH	image/source join broken
SBOM_SUBJECT_MISMATCH	inventory describes another image
PROVENANCE_BUILDER_UNTRUSTED	builder outside policy
RUNTIME_IMAGE_SET_MISMATCH	Compose tested other bytes

DELIBERATELY BROKEN CI CAMPAIGNS

The verifier must reject plausible false greens

BROKEN CAMPAIGNS 57-AN

Mutation	Expected rejection
delete a package lock entry	PACKAGE_LOCK_DRIFT
filter all architecture tests	TEST_FILTER_EMPTY
reuse prior contract report	CROSS_BUILD_EVIDENCE
change action SHA to tag	ACTION_NOT_PINNED_TO_FULL_SHA
build again inside Docker	PUBLISH_TREE_IDENTITY_MISMATCH
scan sha tag, not digest	SCAN_SUBJECT_MISMATCH
SBOM from another image	SBOM_SUBJECT_MISMATCH
run Compose with local tag	RUNTIME_IMAGE_SET_MISMATCH
remove one shard receipt	TEST_SHARD_INCOMPLETE
expire CVE exception	EXCEPTION_EXPIRED

EXECUTABLE ASSET 57-AN - MUTATION ASSERTION

```
[Theory]
[MemberData(nameof(BrokenPipelineCases.All))]
public async Task Candidate_gate_rejects_false_green(BrokenPipelineCase c)
{
    CandidateManifest manifest = await fixture.RunAsync(c);
    CandidateDecision result = verifier.Verify(manifest);
    result.Verdict.Should().NotBe(CandidateVerdict.Admitted);
    result.Findings.Should().Contain(x => x.Code == c.ExpectedCode);
}
```

STANDARDS AND SOURCE LEDGER

Official mechanisms are qualified by NEXUS-1 evidence rules

SOURCE LEDGER 57-AO

Primary source	Mechanism used	Local qualification
.NET SDK/MSBuild reference	CI path normalization and source revision	candidate identity verifier
NuGet locked restore	reject dependency drift	source allowlist + receipt
GitHub secure use	least privilege and full-SHA actions	untrusted/trusted split
Docker Build attestations	SBOM and SLSA provenance	subject and policy verification
Docker reproducible builds	SOURCE_DATE_EPOCH	clean rebuild comparison
SLSA provenance v1	builder/material/subject model	no level claim without audit

EVIDENCE BOUNDARY

Documentation defines mechanisms and formats. Only a retained NEXUS-1 run can prove that the repository used those mechanisms, that every gate executed and that the admitted registry digest matches the runtime evidence.

ANTI-PATTERN REVIEW

Twelve delivery shortcuts remain unsafe

REJECTED CLAIMS 57-AP

Claim	Why invalid
restore succeeded, so dependencies are fixed	lock/source may drift
build succeeded, so tests passed	different subject
test command was green	zero discovery possible
coverage is high, so behavior is proven	assertion quality unknown
image tag identifies release	tag can move
Docker rebuilt the same source	binary lineage changed
SBOM means secure	inventory is not a verdict
zero CVEs means scan succeeded	scanner may have failed
provenance means trusted	builder policy unverified
signed means deployable	signature says no runtime safety
Compose was green	could use other images
promote by rebuild	artifact changes after proof

REVIEW QUESTION

Can a reviewer start from one image digest and resolve its commit, dependency locks, publish tree, test receipts, SBOM, security verdict, provenance and Compose evidence without trusting a tag or filename? If not, the candidate is not joined.

COMPLETION LEDGER

What Chapter 57 now specifies

COMPLETION LEDGER 57-AQ

Obligation	Asset	Status
identity	BuildIdentity + input digests	SPECIFIED
restore/compile	locked graph + one publish tree	SPECIFIED
test gates	catalogue + zero-discovery verifier	SPECIFIED
security	workflow/dependency/image policies	SPECIFIED
container	pinned base + non-root + OCI identity	SPECIFIED
SBOM/provenance	digest-bound attestations	SPECIFIED
runtime	digest-pinned Compose campaigns	SPECIFIED
executed CI	retained external run	PENDING

OPEN RESIDUALS 57-AR

Residual	Consequence / owner
no repository checkpoint	listings remain compile-oriented
no registry subject	image and attestation claims pending
no advisory database receipt	security verdict pending
no clean rebuild pair	bit equality unproved
no protected rollout	Chapter 58 owns deployment safety

EVIDENCE VERDICT

SPECIFIED | RUNTIME PENDING. The delivery graph, identity joins, gate catalogue, container policy, SBOM, provenance, mutation campaigns and candidate admission are complete as design; no green CI run or published image is fabricated.

HANDOFF TO CHAPTER 58

Rollout, Database Change, and Rollback

Chapter 58 will consume an admitted digest without rebuilding it. It must demonstrate backward-compatible HTTP/event/database change, consumer-before-producer ordering, expand-and-contract migration, feature control, staged exposure, rollback boundaries and the cases where new durable writes make rollback unsafe.

BOUNDARY CARRIED FORWARD 57-AS

Chapter 57 leaves	Chapter 58 must add
immutable image digest	environment-specific rollout reference
contract compatibility report	ordered producer/consumer window
migration artifacts in publish graph	expand/backfill/contract execution
candidate and runtime receipts	stage observations and rollback decision
feature controls as versioned inputs	safe activation sequence
known image/provenance identity	no-rebuild promotion and reversal

NEXT INVARIANT

A rollout is reversible only while every active binary, public contract and durable data shape remains compatible with the previous writer and reader; after incompatible truth is committed, recovery requires a forward or compensating migration, not a tag flip.

FINAL CHAPTER RULE

A green pipeline is a joined proof, not a row of check marks

NEXUS-1 calls an image a candidate only when the exact source and locked inputs passed every required gate, the tested publish tree became the digest-addressed image, SBOM and provenance name that digest, and the same digest set produced retained runtime evidence.

PART X PROGRESS

Chapter 55 made the laboratory reproducible, Chapter 56 made extraction reversible, and Chapter 57 made delivery identity explicit. Chapter 58 must preserve those identities while code, contracts and data coexist during rollout.

PART X - MIGRATION, DELIVERY, AND FINAL ARCHITECTURE

58. Rollout, Database Change, and Rollback

A release is safe only when old and new binaries, contracts and durable data can coexist in an explicitly ordered compatibility window whose evidence determines whether reversal is still possible.

CHAPTER DECISION

Promote the admitted Chapter 57 image digest without rebuilding it. Expand durable and public contracts before depending on them, make consumers tolerant before producers emit a new version, expose the compatible binary in bounded stages, and close rollback only through an explicit evidence-backed decision.

PROOF PATH 58-A - ONE CANDIDATE THROUGH A COMPATIBILITY WINDOW**GUARANTEE LEDGER 58-A**

Evidence	Allowed claim	Cannot claim
compatibility matrix	named combinations were judged/tested	all clients upgraded
migration receipt	artifact changed the expected database	backfill is complete
stage receipt	one bounded cohort passed its gates	whole environment is healthy
rollback decision	named prior state is reachable or unsafe	data can be uncommitted

EVIDENCE VERDICT

SPECIFIED | RUNTIME PENDING. The rollout controller, compatibility rules, migrations, tests and receipts are compile-oriented reference assets. No environment rollout, database mutation or production recovery is claimed.

ORDINARY DEVELOPER PROBLEM

A healthy new pod can still make rollback impossible

The new image starts, readiness turns green and a percentage flag enables a changed writer. Minutes later an old consumer rejects the new event, the database contains rows the old binary cannot interpret, and an operator changes the image tag back. Compute was reversed; committed truth was not.

FAILURE LEDGER 58-B

Shortcut	Hidden failure	Correction
producer first	old consumer rejects V2	consumer-first capability receipt
ALTER and switch together	old binary loses its shape	expand / coexist / contract
percentage flag for writes	one aggregate gets mixed semantics	stable keyed cohort + epoch
app migrates on startup	replicas race and startup couples to DDL	one privileged migrator
rollback means old image	new durable truth remains	classify reversibility first

PLAIN-LANGUAGE PRINCIPLE

Rollout is a controlled period of coexistence. If coexistence is not designed and tested, rollback is merely hope with an older tag.

CHAPTER BOUNDARY

Deployment safety here; Kubernetes translation next

This chapter owns compatibility, rollout order, database expansion, backfill, feature activation, observation and reversal. Chapter 59 will translate the already-proved runtime and these controls into minimal educational Kubernetes resources; it may not invent new runtime behavior.

DECISION RECORD

ADR-058 - Expand, coexist, verify, then contract

DECISION CONSEQUENCES 58-C

Choice	Benefit	Cost / boundary
one admitted digest	artifact identity survives environments	environment references are separate evidence
versioned public contracts	strict V1 consumers stay valid	overlap and retirement inventory
additive database first	old binaries remain operable	temporary duplicate representation
consumer-before-producer	new facts are understood before emitted	capability receipts required
staged, stable cohorts	bounded blast radius	longer observation time
explicit rollback closure	operators know the last safe point	contract delayed to later release

NON-GOAL

The chapter does not promise zero downtime for every SQL operation, automatic safety from a feature-flag product, reversible arbitrary data loss, or successful production rollout. The provider, workload and retained execution determine those claims.

CLAIM LADDER

Compatible, deployed, exposed and reversible are separate claims

CLAIM LADDER 58-D

Level	Subject	Minimum evidence	Does not prove
planned	candidate + target	signed RolloutPlan	schema is ready
expanded	database/public readers	migration + capability receipts	new writer is safe
coexisting	old/new binaries	mixed-version tests	full exposure
observed	one stage	fresh health + business gates	next stage will pass
reversible	named prior state	rollback classifier + rehearsal	contract is removable
contracted	retired shape	absence proof + migration	old binary can return

NO TRANSITIVE SAFETY

A green readiness probe cannot prove event compatibility. A successful additive migration cannot prove old jobs stopped. A completed backfill cannot prove the old representation may be dropped.

CONCRETE CHANGE SLICE

One AlarmManagement change crosses every seam

The worked slice introduces NormalizedSourceAlarmId for vendor alarm identities. The server derives it with policy alarm-id/2; V2 HTTP responses and AlarmRaisedV2 expose it, while V1 contracts remain byte-shape stable. AlarmManagementDb gains a nullable normalized_source_alarm_id, the compatible binary reads it with a deterministic fallback and writes both representations in one local transaction.

CHANGE-SET INVENTORY 58-E

Surface	Old	Compatibility state	Later contract
HTTP	/api/v1/alarms	V1 and V2 side by side	V1 removed after caller zero
event	AlarmRaisedV1	consumers read V1+V2; producer emits one	V1 emission retired
database	source_alarm_id	nullable normalized column + fallback	old column remains unless semantics permit
behavior	alarm-id/1	stable cohort chooses policy	alarm-id/2 becomes baseline

OWNERSHIP RULE

Both columns are private representations inside one AlarmManagementDb transaction. They are not two independently authoritative stores. The aggregate and one owner still decide truth once.

FROM SERVICES TO RUNTIME

COMPATIBILITY MATRIX

The matrix names what may coexist

BINARY / SCHEMA MATRIX 58-F

Binary	S0 old schema	S1 expanded	S2 backfilled	S3 contracted
B0 old	supported	supported	supported	FORBIDDEN
B1 compatible	FORBIDDEN	supported	supported	supported if old read retained
B2 new-only	FORBIDDEN	FORBIDDEN	supported	supported

PUBLIC-CONTRACT MATRIX 58-G

Participant	Reads V1	Reads V2	Emits V1	Emits V2
C0 old consumer	yes	no	n/a	n/a
C1 compatible consumer	yes	yes	n/a	n/a
P0 old producer	n/a	n/a	yes	no
P1 compatible producer	n/a	n/a	one selected version	one selected version

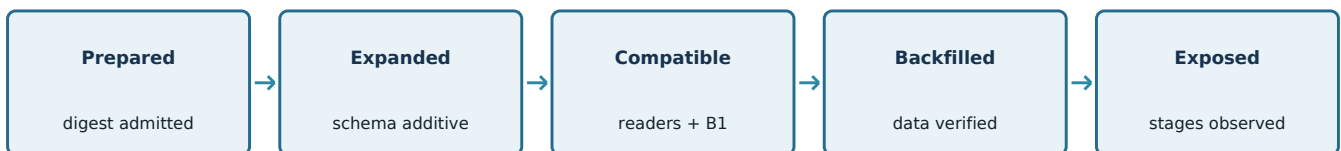
ADMISSION RULE

A stage is illegal when any routed request, message or row can reach a participant whose matrix cell is FORBIDDEN. Replicas existing but receiving no work do not satisfy capability; routing and worker subscriptions are part of the proof.

ROLLOUT STATE MACHINE

Progress is monotonic; recovery is an explicit transition

STATE MACHINE 58-H



CONTROLLED STATES 58-I

State	Entry gate	Permitted recovery
Prepared	plan and target bind	cancel plan
Expanded	migration receipt	keep additive schema; return B0
ConsumerReady	all subscriber capabilities	return old producer
CompatibleBinary	B1 mixed-version tests	return B0
Backfilled	checkpoint + reconciliation	B0 or B1 old-read
WritePilot	stable cohort + dual representation	disable pilot; B0 remains readable
RollbackClosed	new-only truth decision	forward/compensating only
Contracted	absence proof + contract migration	no old binary

TYPED ROLLOUT PLAN

The controller consumes identities, not mutable names

EXECUTABLE ASSET 58-A - ROLLOUT PLAN

```
public sealed record RolloutPlan(  
    Guid PlanId,  
    long Epoch,  
    string Environment,  
    string CandidateManifestSha256,  
    string AlarmImageDigest,  
    string MigrationArtifactSha256,  
    string ContractCorpusSha256,  
    string CompatibilityMatrixSha256,  
    string FeaturePlanSha256,  
    RolloutStage RequestedStage,  
    DateTimeOffset NotBeforeUtc,  
    TimeSpan ObservationWindow,  
    string ApprovedBy,  
    string Signature);  
  
public enum RolloutStage  
{  
    Prepared = 10, Expanded = 20, ConsumerReady = 30,  
    CompatibleBinary = 40, Backfilled = 50, ReadPilot = 60,  
    WritePilot = 70, FullExposure = 80,  
    RollbackClosed = 90, Contracted = 100  
}
```

DIGEST RULE

AlarmImageDigest is the registry manifest digest admitted in Chapter 57. The environment may add a deployment reference, replica count and target configuration, but it must not rebuild or silently substitute the image.

PLAN ADMISSION

A stale, incomplete or regressive plan fails closed

EXECUTABLE ASSET 58-B - PLAN VALIDATOR

```
public static void Admit(  
    RolloutPlan plan,  
    EnvironmentState current,  
    CandidateManifest candidate,  
    TimeProvider clock)  
{  
    Require(plan.Epoch == current.Epoch + 1, "ROLLOUT_EPOCH_STALE");  
    Require(plan.RequestedStage >= current.Stage, "STAGE_REGRESSION");  
    Require(plan.NotBeforeUtc <= clock.GetUtcNow(), "PLAN_NOT_ACTIVE");  
    Require(plan.ObservationWindow >= TimeSpan.FromMinutes(10),  
        "OBSERVATION_WINDOW_TOO_SHORT");  
    Require(plan.CandidateManifestSha256 == candidate.Digest,  
        "CANDIDATE_BINDING_MISMATCH");  
    Require(plan.AlarmImageDigest == candidate.Images["alarm-management"],  
        "IMAGE_DIGEST_SUBSTITUTED");  
    Require(plan.MigrationArtifactSha256 == candidate.Migrations["alarm"],  
        "MIGRATION_ARTIFACT_SUBSTITUTED");  
    Require(VerifySignature(plan), "ROLLOUT_PLAN_SIGNATURE_INVALID");  
}
```

CONCURRENCY RULE

The environment state advances with compare-and-swap on Epoch. Two controllers may read, but only one can commit the next stage. A retry reloads the state and re-evaluates every gate.

STAGE RECEIPT

Every transition records subject, observations and decision

EXECUTABLE ASSET 58-C - STAGE RECEIPT

```
public sealed record StageReceipt(
    Guid PlanId,
    long Epoch,
    RolloutStage From,
    RolloutStage To,
    string CandidateManifestSha256,
    string EnvironmentStateBeforeSha256,
    string EnvironmentStateAfterSha256,
    IReadOnlyList<string> GateReceiptSha256s,
    DateTimeOffset ObservationStartedUtc,
    DateTimeOffset ObservationEndedUtc,
    StageVerdict Verdict,
    string? FailureCode,
    string ControllerIdentity,
    string ReceiptSha256);

public enum StageVerdict { Admitted, Rejected, Aborted }
```

RECEIPT JOIN 58-J

Field	Question answered
candidate digest	which bytes were exposed?
before/after state	what changed?
gate digests	which evidence authorized it?
observation interval	when were signals judged?
controller identity	which authority decided?

NO-REBUILD PROMOTION

Environment references preserve the admitted digest

EXECUTABLE ASSET 58-D - PROMOTION ADMISSION

```
public sealed record EnvironmentImageReference(
    string Environment,
    string Service,
    string RegistryDigest,
    string CandidateManifestSha256,
    string DeploymentReferenceSha256);

public static EnvironmentImageReference Promote(
    CandidateManifest candidate, string environment,
    SignedDeploymentReference reference)
{
    string digest = candidate.Images["alarm-management"];
    Require(reference.ImageDigest == digest, "NO_REBUILD_VIOLATION");
    Require(reference.CandidateManifestSha256 == candidate.Digest,
        "CANDIDATE_REFERENCE_MISMATCH");
    Require(reference.Environment == environment, "TARGET_MISMATCH");
    return new(environment, "alarm-management", digest,
        candidate.Digest, reference.Digest);
}
```

TAG BOUNDARY

A tag may help a human discover a candidate, but neither a rollout plan nor a runtime receipt uses it as the subject. Reversal likewise selects a previously admitted digest; it does not rebuild an old commit.

PREFLIGHT

Inspect before mutating

EXECUTABLE ASSET 58-E - ENVIRONMENT PREFLIGHT

```
public sealed record RolloutPreflight(
    string CurrentImageDigest,
    string DatabaseIdentity,
    string CurrentMigration,
    string SchemaFingerprint,
    IReadOnlySet<string> ActiveBinaryCapabilities,
    IReadOnlySet<string> ActiveConsumerCapabilities,
    long PendingFacts,
    long DispatchLag,
    DateTimeOffset ObservedAtUtc);

static void RequireExpandedPreconditions(
    RolloutPreflight x, RolloutPlan plan, TimeProvider clock)
{
    Require(x.DatabaseIdentity == "AlarmManagementDb", "DATABASE_TARGET_WRONG");
    Require(x.CurrentMigration == "202608010900_Baseline", "MIGRATION_BASE_WRONG");
    Require(x.PendingFacts == 0 || x.DispatchLag < 1000, "OUTBOX_LAG_HIGH");
    Require(clock.GetUtcNow() - x.ObservedAtUtc < TimeSpan.FromMinutes(2),
        "PREFLIGHT_STALE");
}
```

READ-ONLY FIRST

Preflight queries the target, current migration history, active capability inventory and pending-fact lag. A mismatch stops before DDL, traffic or flags change.

HTTP COMPATIBILITY

V1 remains stable while V2 becomes independently testable

EXECUTABLE ASSET 58-F - SIDE-BY-SIDE ENDPOINTS

```
RouteGroupBuilder v1 = app.MapGroup("/api/v1/alarms");
v1.MapGet("/{id:guid}", async (Guid id, IAlarmQueries q, CancellationToken ct) =>
    (await q.Find(id, ct)) is { } x
    ? Results.Ok(AlarmHttpV1.From(x))
    : Results.NotFound());

RouteGroupBuilder v2 = app.MapGroup("/api/v2/alarms");
v2.MapGet("/{id:guid}", async (Guid id, IAlarmQueries q, CancellationToken ct) =>
    (await q.Find(id, ct)) is { } x
    ? Results.Ok(AlarmHttpV2.From(x))
    : Results.NotFound());

public sealed record AlarmV1(Guid Id, string SourceAlarmId, string State);
public sealed record AlarmV2(Guid Id, string SourceAlarmId,
    string NormalizedSourceAlarmId, string State, string NormalizationPolicy);
```

STRICT-READER RULE

V1 is not changed by adding a member, because Chapter 33's strict corpus treats unknown members as a contract decision. V2 has its own route, DTO, schema and corpus.

CALLER RETIREMENT

A route is removed only after an evidence window reaches zero

EXECUTABLE ASSET 58-G - CALLER CAPABILITY RECEIPT

```
public sealed record HttpCallerReceipt(
    string CallerIdentity,
    string RouteVersion,
    string DeploymentDigest,
    long SuccessfulCalls,
    long RejectedCalls,
    DateTimeOffset WindowStartUtc,
    DateTimeOffset WindowEndUtc,
    string EvidenceSha256);

static void AdmitV1Retirement(IEnumerable<HttpCallerReceipt> receipts,
    DateTimeOffset now)
{
    var current = receipts.Where(x => now - x.WindowEndUtc < TimeSpan.FromDays(1));
    Require(current.Any(), "CALLER_INVENTORY_EMPTY");
    Require(current.All(x => x.RouteVersion != "v1" || x.SuccessfulCalls == 0),
        "V1_CALLERS_REMAIN");
    Require(current.All(x => x.RejectedCalls == 0), "CALLER_ERRORS_PRESENT");
}
```

ABSENCE PROOF

No V1 calls in one short sample is not retirement evidence. The required window covers the slowest scheduled caller, retry horizon and supportable offline client declared by the ownership catalogue.

EVENT COMPATIBILITY

A new event version is a new contract, not an edited V1

EVENT EVOLUTION 58-K

Change	Same V1?	Required action
clarify documentation only	yes	corpus text update if bytes unchanged
add normalized member	no for strict readers	publish AlarmRaisedV2
rename or remove member	no	new version + translation
change ordering/identity semantics	no	new event type and policy identity

EXECUTABLE ASSET 58-H - VERSIONED FACTS

```
public sealed record AlarmRaisedV1(
    Guid AlarmId, string SourceSystem, string SourceAlarmId,
    DateTimeOffset OccurredAtUtc);

public sealed record AlarmRaisedV2(
    Guid AlarmId, string SourceSystem, string SourceAlarmId,
    string NormalizedSourceAlarmId, string NormalizationPolicy,
    DateTimeOffset OccurredAtUtc);
```

SCHEMA-REGISTRY BOUNDARY

Compatibility modes from Avro-oriented registries are useful vocabulary, not proof for NEXUS-1 JSON. The versioned JSON corpus and actual consumer implementations decide compatibility here.

CONSUMER-FIRST DEPLOYMENT

Compatible consumers translate both versions into one command

EXECUTABLE ASSET 58-I - DUAL READER

```
public sealed class AlarmRaisedConsumer(IAAlarmIntake intake)
{
    public Task Consume(PublicFactEnvelope envelope, CancellationToken ct) =>
        envelope.EventType switch
        {
            "nexus1.alarm-management.alarm-raised.v1" =>
                intake.Accept(Translate(envelope.Read<AlarmRaisedV1>()), ct),
            "nexus1.alarm-management.alarm-raised.v2" =>
                intake.Accept(Translate(envelope.Read<AlarmRaisedV2>()), ct),
            _ => throw new UnsupportedPublicFactException(envelope.EventType)
        };

    static AcceptAlarm Translate(AlarmRaisedV1 x) => new(
        x.AlarmId, x.SourceSystem, x.SourceAlarmId,
        AlarmIdNormalizer.V1(x.SourceAlarmId), x.OccurredAtUtc);

    static AcceptAlarm Translate(AlarmRaisedV2 x) => new(
        x.AlarmId, x.SourceSystem, x.SourceAlarmId,
        RequirePolicy(x.NormalizedSourceAlarmId, x.NormalizationPolicy),
        x.OccurredAtUtc);
}
```

CAPABILITY RECEIPT

The producer remains on V1 until every routed subscriber identity reports the V1+V2 corpus digest and has passed replay, redelivery and poison-message tests with its deployed digest.

PRODUCER ACTIVATION

One semantic transition produces exactly one public fact

EXECUTABLE ASSET 58-J - EXCLUSIVE EVENT SELECTION

```
public PendingPublicFactRow CreateRaised(
    Alarm alarm, EventEmissionPlan plan)
{
    EventVersion version = plan.ForPartition(alarm.ScopeKey);
    return version switch
    {
        EventVersion.V1 => PendingFacts.Create(
            "nexus1.alarm-management.alarm-raised.v1",
            AlarmRaisedV1.From(alarm)),
        EventVersion.V2 => PendingFacts.Create(
            "nexus1.alarm-management.alarm-raised.v2",
            AlarmRaisedV2.From(alarm)),
        _ => throw new UnreachableException()
    };
}

public enum EventVersion { V1, V2 }
```

CONSERVATION RULE

Do not publish V1 and V2 for the same transition. Dual publication creates two facts, two message identities and ambiguous deduplication. The producer chooses exactly one version after consumer readiness is proved.

CONTRACT CORPUS

Compatibility is executable in both directions that matter

EXECUTABLE ASSET 58-K - CORPUS MATRIX TEST

```
[Theory]
[MemberData(nameof(CompatibilityCases))]
public void Corpus_is_admitted(ContractCase c)
{
    byte[] body = File.ReadAllBytes(c.Path);
    ContractVerdict verdict = c.Reader.Read(body);
    Assert.Equal(c.Expected, verdict);
}

public static TheoryData<ContractCase> CompatibilityCases => new()
{
    Case("v1/valid.json", Readers.V1, Accepted),
    Case("v1/valid.json", Readers.Compatible, Accepted),
    Case("v2/valid.json", Readers.V1, UnsupportedVersion),
    Case("v2/valid.json", Readers.Compatible, Accepted),
    Case("v2/unknown-policy.json", Readers.Compatible, Rejected),
    Case("v2/unknown-member.json", Readers.Compatible, Rejected)
};
```

RETAINED OUTPUTS

The report records reader digest, corpus digest, case count and every verdict. Zero discovery, omitted negative cases or a report bound to another candidate fail admission.

DATABASE EXPANSION

The first migration is additive and old-binary compatible

EXECUTABLE ASSET 58-L - EF CORE EXPAND MIGRATION

```
public partial class AddNormalizedSourceAlarmId : Migration
{
    protected override void Up(MigrationBuilder m)
    {
        m.AddColumn<string>(
            name: "normalized_source_alarm_id",
            table: "alarm_observations",
            type: "nvarchar(160)",
            maxLength: 160,
            nullable: true);

        m.CreateIndex(
            name: "ix_alarm_observations_normalized_source_alarm_id",
            table: "alarm_observations",
            column: "normalized_source_alarm_id",
            filter: "[normalized_source_alarm_id] IS NOT NULL");
    }

    protected override void Down(MigrationBuilder m)
    {
        m.DropIndex("ix_alarm_observations_normalized_source_alarm_id",
            "alarm_observations");
        m.DropColumn("normalized_source_alarm_id", "alarm_observations");
    }
}
```

ONLINE-CHANGE BOUNDARY

Nullable and additive is a compatibility property, not a universal no-lock guarantee. Provider metadata locks, index build cost, table size and workload are measured in a representative provider-backed rehearsal before the change is admitted.

MIGRATION ARTIFACT

Migrations are built with the candidate and applied separately

EXECUTABLE ASSET 58-M - RETAINED MIGRATION OUTPUTS

```
dotnet ef migrations bundle \
--project src/AlarmManagement/Infrastructure \
--startup-project src/AlarmManagement/Migrator \
--configuration Release \
--self-contained \
--target-runtime linux-x64 \
--output artifacts/migrations/alarm-management

dotnet ef migrations script 0 \
--project src/AlarmManagement/Infrastructure \
--startup-project src/AlarmManagement/Migrator \
--configuration Release \
--idempotent \
--output artifacts/migrations/alarm-management-idempotent.sql

sha256sum artifacts/migrations/alarm-management* \
> artifacts/migrations/alarm-management.sha256
```

APPLICATION BOUNDARY

The HTTP process never calls Database.Migrate at startup. A dedicated migration identity, narrow database principal and serialized execution apply the admitted artifact before compatible replicas receive traffic.

SERIALIZED MIGRATION AUTHORITY

One session owns the provider lock and the migration

EXECUTABLE ASSET 58-N - SQL SERVER APPLICATION LOCK

```
DECLARE @result int;
EXEC @result = sys.sp_getapplock
    @Resource = N'nexus1:AlarmManagementDb:migration',
    @LockMode = 'Exclusive',
    @LockOwner = 'Session',
    @LockTimeout = 30000;
IF @result < 0
    THROW 51058, 'MIGRATION_LEASE_UNAVAILABLE', 1;

-- The migrator executes on this same open connection/session.
-- In finally, on the same session:
EXEC sys.sp_releaseapplock
    @Resource = N'nexus1:AlarmManagementDb:migration',
    @LockOwner = 'Session';
```

AUTHORITY SPLIT 58-L

Identity	Allowed	Forbidden
runtime principal	read/write owned application tables	DDL, migration history changes
migration principal	declared DDL + history during window	serve requests, publish facts
backfill principal	bounded update + checkpoint	schema contract, cross-database access

MIGRATION RUNNER

Lock lifetime and connection lifetime are the same

EXECUTABLE ASSET 58-O - DEDICATED MIGRATOR

```
public sealed class AlarmMigrationRunner(
    AlarmManagementDbContext db,
    ISqlApplicationLock migrationLock,
    IMigrationEvidenceSink evidence)
{
    public async Task<MigrationReceipt> Run(CancellationTokens ct)
    {
        DbConnection connection = db.Database.GetDbConnection();
        await connection.OpenAsync(ct);
        await using var lease = await migrationLock.AcquireOn(
            connection, "nexus1:AlarmManagementDb:migration",
            TimeSpan.FromSeconds(30), ct);

        string before = await History.Head(db, ct);
        await db.Database.MigrateAsync(ct); // same DbContext and connection
        string after = await History.Head(db, ct);
        MigrationReceipt receipt = await Evidence.Create(before, after, db, ct);
        await evidence.Append(receipt, ct);
        return receipt;
    }
}
```

TRANSACTION QUALIFICATION

Migration transactions are provider- and operation-dependent. The runner records the generated SQL, provider version, transactional exceptions and observed history; it does not describe every multi-statement migration as atomically reversible.

MIGRATION RECEIPT

DDL success is joined to target, artifact and observed schema

EXECUTABLE ASSET 58-P - MIGRATION EVIDENCE

```
public sealed record MigrationReceipt(
    Guid PlanId,
    string DatabaseIdentity,
    string ProviderIdentity,
    string MigrationArtifactSha256,
    string FromMigration,
    string ToMigration,
    string SchemaFingerprintBefore,
    string SchemaFingerprintAfter,
    DateTimeOffset StartedUtc,
    DateTimeOffset CompletedUtc,
    MigrationVerdict Verdict,
    string? FailureCode,
    string ReceiptSha256);

static void Admit(MigrationReceipt x, RolloutPlan plan)
{
    Require(x.PlanId == plan.PlanId, "MIGRATION_PLAN_MISMATCH");
    Require(x.MigrationArtifactSha256 == plan.MigrationArtifactSha256,
        "MIGRATION_BYTES_MISMATCH");
    Require(x.Verdict == MigrationVerdict.Applied, "MIGRATION_NOT_APPLIED");
    Require(x.ToMigration == "202608100900_AddNormalizedSourceAlarmId",
        "MIGRATION_HEAD_WRONG");
}
```

COMPATIBLE READER

New code tolerates pre-backfill nulls deterministically

EXECUTABLE ASSET 58-Q - READ FALLBACK

```
public static AlarmObservation ToDomain(AlarmObservationRow row)
{
    string normalized = row.NormalizedSourceAlarmId
        ?? AlarmIdNormalizer.V1(row.SourceAlarmId);
    string policy = row.NormalizedSourceAlarmId is null
        ? "alarm-id/1-fallback"
        : row.NormalizationPolicy ?? "alarm-id/2";

    return new AlarmObservation(
        new ObservationId(row.ObservationId),
        row.SourceSystem,
        row.SourceAlarmId,
        normalized,
        policy,
        row.OccurredAtUtc);
}
```

FALLBACK BOUNDARY

Fallback is versioned domain behavior, not an incidental string expression scattered through queries. Its tests fix casing, Unicode, whitespace and invalid-input outcomes so backfill and online writes cannot diverge.

COMPATIBLE WRITER

One local commit maintains old and new representations

EXECUTABLE ASSET 58-R - DUAL REPRESENTATION IN ONE OWNER

```
internal static AlarmObservationRow From(
    AcceptedAlarmObservation accepted,
    NormalizationPlan plan)
{
    string normalized = AlarmIdNormalizer.Apply(
        accepted.SourceAlarmId, plan.PolicyIdentity);
    return new AlarmObservationRow
    {
        ObservationId = accepted.Id.Value,
        SourceSystem = accepted.SourceSystem,
        SourceEventId = accepted.SourceEventId,
        SourceAlarmId = accepted.SourceAlarmId,           // B0-readable
        NormalizedSourceAlarmId = normalized,             // B1/B2-readable
        NormalizationPolicy = plan.PolicyIdentity,
        OccurredAtUtc = accepted.OccurredAtUtc,
        RecordedAtUtc = accepted.RecordedAtUtc
    };
}

// Owned truth and pending public fact still enter one SaveChangesAsync.
```

NOT DUAL TRUTH

The values are committed by one AlarmManagementDb unit of work and one owner. If an update cannot preserve both representations atomically, B0 rollback is not safe and the write stage must not open.

BACKFILL MODEL

The backfill is resumable, keyset-paged and idempotent

EXECUTABLE ASSET 58-S - CHECKPOINT AND BATCH

```
internal sealed class BackfillCheckpointRow
{
    public required string Operation { get; init; }
    public long LastObservationSequence { get; set; }
    public long RowsExamined { get; set; }
    public long RowsChanged { get; set; }
    public required string PolicyIdentity { get; init; }
    public DateTimeOffset UpdatedAtUtc { get; set; }
    public Guid ConcurrencyVersion { get; set; }
}

public sealed record BackfillBatch(
    long AfterSequence,
    long ThroughSequence,
    int Examined,
    int Changed,
    TimeSpan Duration,
    string BatchDigest);
```

CURSOR RULE

The cursor is an immutable owner-local ObservationSequence, not OFFSET and not a timestamp. New rows are already dual-written; the backfill only repairs rows below the captured high-water mark.

BACKFILL EXECUTION

Row changes and checkpoint commit together

EXECUTABLE ASSET 58-T - BOUNDED BATCH

```
public async Task<BackfillBatch?> RunBatch(int size, CancellationToken ct)
{
    BackfillCheckpointRow cp = await checkpoints.Lock("alarm-id/2", ct);
    long highWater = await checkpoints.HighWater("alarm_observations", ct);
    AlarmObservationRow[] rows = await db.AlarmObservations
        .Where(x => x.ObservationSequence > cp.LastObservationSequence
            && x.ObservationSequence <= highWater)
        .OrderBy(x => x.ObservationSequence)
        .Take(size)
        .ToArrayAsync(ct);
    if (rows.Length == 0) return null;

    foreach (var row in rows.Where(x => x.NormalizedSourceAlarmId is null))
    {
        row.NormalizedSourceAlarmId = AlarmIdNormalizer.V2(row.SourceAlarmId);
        row.NormalizationPolicy = "alarm-id/2";
        cp.RowsChanged++;
    }
    cp.RowsExamined += rows.Length;
    cp.LastObservationSequence = rows[^1].ObservationSequence;
    cp.UpdatedAtUtc = clock.GetUtcNow();
    cp.ConcurrencyVersion = Guid.NewGuid();
    await db.SaveChangesAsync(ct);
    return BackfillBatchReceipt.From(rows, cp);
}
```

CRASH BEHAVIOR

A crash before SaveChanges changes neither rows nor cursor; a retry repeats the batch. A crash after commit resumes after the committed key. Reapplying normalization to an already-populated row is forbidden unless its policy identity matches.

BACKFILL CONTROL

Throughput is subordinate to foreground health

EXECUTABLE ASSET 58-U - THROTTLED LOOP

```
public async Task Run(BackfillLimits limits, CancellationToken ct)
{
    using var deadline = CancellationTokenSource.CreateLinkedTokenSource(ct);
    deadline.CancelAfter(limits.MaximumDuration);

    while (!deadline.IsCancellationRequested)
    {
        ForegroundHealth health = await healthProbe.Observe(deadline.Token);
        if (!health.Within(limits.MaxP95Latency, limits.MaxLockWait))
        {
            await delay.Delay(limits.Cooldown, deadline.Token);
            continue;
        }

        BackfillBatch? batch = await worker.RunBatch(limits.BatchSize,
            deadline.Token);
        if (batch is null) break;
        await receipts.Append(batch, deadline.Token);
        await delay.Delay(limits.InterBatchDelay, deadline.Token);
    }
}
```

REQUIRED LIMITS 58-M

Limit	Failure code
maximum duration	BACKFILL_WINDOW_EXPIRED
batch size	BACKFILL_BATCH_UNBOUNDED
lock-wait ceiling	FOREGROUND_LOCK_PRESSURE
foreground latency ceiling	FOREGROUND_LATENCY_HIGH

BACKFILL RECONCILIATION

Completion means content agrees, not merely cursor at end

EXECUTABLE ASSET 58-V - RECONCILIATION RECEIPT

```
public sealed record BackfillReconciliation(
    long HighWaterSequence,
    long EligibleRows,
    long PopulatedRows,
    long NullRows,
    long PolicyMismatchRows,
    string OrderedProjectionSha256,
    DateTimeOffset ObservedAtUtc,
    string ReceiptSha256);

static void Admit(BackfillReconciliation x, TimeProvider clock)
{
    Require(x.EligibleRows == x.PopulatedRows, "BACKFILL_COUNT_MISMATCH");
    Require(x.NullRows == 0, "BACKFILL_NULLS_REMAIN");
    Require(x.PolicyMismatchRows == 0, "BACKFILL_POLICY_MISMATCH");
    Require(IsSha256(x.OrderedProjectionSha256), "BACKFILL_DIGEST_INVALID");
    Require(clock.GetUtcNow() - x.ObservedAtUtc < TimeSpan.FromMinutes(5),
        "BACKFILL_RECONCILIATION_STALE");
}
```

RACE CLOSURE

The high-water cohort is reconciled, while all later rows are required to come from the compatible dual writer. A separate query proves no post-high-water nulls before ReadPilot opens.

FEATURE PLAN

Feature control is a versioned rollout input

EXECUTABLE ASSET 58-W - IMMUTABLE FEATURE PLAN

```

public sealed record FeaturePlan(
    long Epoch,
    string PolicyIdentity,
    int ReadBasisPoints,
    int WriteBasisPoints,
    string CohortKey,
    DateTimeOffset EffectiveUtc,
    DateTimeOffset ExpiresUtc,
    string RolloutPlanSha256,
    string Signature)
{
    public string Digest => CanonicalJson.Sha256(this);
}

static void Admit(FeaturePlan x, RolloutPlan rollout, TimeProvider clock)
{
    Require(x.RolloutPlanSha256 == CanonicalJson.Sha256(rollout),
        "FEATURE_PLAN_ROLLOUT_MISMATCH");
    Require(x.CohortKey == "plant-unit", "UNSTABLE_COHORT_KEY");
    Require(x.WriteBasisPoints <= x.ReadBasisPoints, "WRITE_EXCEEDS_READ");
    Require(clock.GetUtcNow() < x.ExpiresUtc, "FEATURE_PLAN_EXPIRED");
}

```

LIBRARY QUALIFICATION

A feature-management library can evaluate filters and variants, but NEXUS-1 still binds policy, cohort key, expiry and rollout epoch. A mutable remote percentage is not authority to change durable semantics.

STABLE COHORT

The same plant unit receives the same decision for the epoch

EXECUTABLE ASSET 58-X - DETERMINISTIC ALLOCATION

```

public static bool Included(
    string plantUnitId, FeaturePlan plan, int basisPoints)
{
    Require(basisPoints is >= 0 and <= 10_000, "BASIS_POINTS_INVALID");
    byte[] input = Encoding.UTF8.GetBytes(
        $"{plan.EPOCH}|{plan.PolicyIdentity}|{plantUnitId}");
    byte[] hash = SHA256.HashData(input);
    uint bucket = BinaryPrimitives.ReadUInt32BigEndian(hash.AsSpan(0, 4))
        % 10_000;
    return bucket < basisPoints;
}

public sealed record RequestFeatureSnapshot(
    long Epoch, string PlanSha256, string PlantUnitId,
    bool ReadV2, bool WriteV2, DateTimeOffset CapturedUtc);

```

SNAPSHOT RULE

One request, command or message captures one feature snapshot. It is not re-evaluated halfway through a transaction. Long-lived aggregate workflows persist the chosen policy identity with their durable state.

FEATURE SAFETY CLASSES

Flags control exposure; they do not redefine ownership

FEATURE-CONTROL MATRIX 58-N

Control	Safe use	Forbidden use
read path	select V2 projection with fallback	hide data required for correctness
write path	stable cohort while old form also maintained	random per-request durable format
event version	partition-stable one-version selection	emit both versions for one fact
writer authority	none	choose between two writers
database contract	none	drop column because flag is 100%

CHAPTER 56 CONTINUITY

Writer authority still comes from the route/authority ledger established during extraction. Feature state may change behavior inside the admitted writer; it never creates a second authority.

ORDERED ROLLOUT

Capabilities precede dependencies

EXECUTION ORDER 58-O

Order	Action	Required gate
1	admit candidate and target plan	digests and signatures join
2	apply additive database migration	schema fingerprint S1
3	deploy V1+V2 consumers	subscriber capability complete
4	deploy compatible B1 binary	mixed B0/B1 runtime test
5	backfill and reconcile	content/count/freshness pass
6	read pilot then write pilot	stage observations pass
7	full exposure	bounded windows all pass
8	declare rollback closed if needed	unsafe-truth decision recorded
9	contract in later release	absence proofs complete

STAGED EXPOSURE

Each stage limits blast radius and has a hold period

EXPOSURE STAGES 58-P

Stage	Target	Minimum hold	Advance gate
shadow	B1 receives mirrored reads, no effects	30 min	projection equality
canary	one internal plant unit	30 min	technical + business SLO
pilot	1% stable units	2 h	no compatibility errors
broad	10% then 50%	one peak period each	fresh trend within budget
full	100%	declared observation window	retained final receipt

SAFE-DEPLOYMENT PRINCIPLE

Small increments, explicit health models and hold periods reduce blast radius. Stage size and duration are workload inputs, not universal constants; the table is the chapter's concrete AlarmManagement plan.

RUNTIME ADMISSION

Readiness opens only for a compatible environment

EXECUTABLE ASSET 58-Y - ROLLOUT-AWARE READINESS

```
public sealed class RolloutCompatibilityHealthCheck(
    IEnvironmentState state,
    IDatabaseShapeReader database,
    IFeaturePlanStore features) : IHealthCheck
{
    public async Task<HealthCheckResult> CheckHealthAsync(
        HealthCheckContext context, CancellationToken ct)
    {
        EnvironmentState expected = await state.Read(ct);
        DatabaseShape actual = await database.Read(ct);
        FeaturePlan plan = await features.Read(ct);

        if (!actual.Supports(expected.BinaryCapabilities))
            return HealthCheckResult.Unhealthy("SCHEMA_INCOMPATIBLE");
        if (plan.Epoch != expected.Epoch)
            return HealthCheckResult.Unhealthy("FEATURE_EPOCH_MISMATCH");
        return HealthCheckResult.Healthy();
    }
}
```

PROBE BOUNDARY

This readiness check is scoped, bounded and value-safe. Liveness remains process-local. Neither check applies migrations, performs backfills or declares a rollout stage successful.

OBSERVATION GATE

Technical and business signals share one time window

EXECUTABLE ASSET 58-Z - STAGE GATE

```
public sealed record StageObservation(
    TimeRange Window,
    long Requests,
    long CompatibilityRejects,
    long DuplicatePublicFacts,
    long MissingPublicFacts,
    double ErrorRate,
    TimeSpan P95Latency,
    long BackfillNullRows,
    long OutboxLag,
    DateTimeOffset CollectedUtc);

static void Admit(StageObservation x, StageBudget b, TimeProvider clock)
{
    Require(x.Window.Duration >= b.MinimumWindow, "WINDOW_TOO_SHORT");
    Require(x.Requests >= b.MinimumRequests, "SAMPLE_TOO_SMALL");
    Require(x.CompatibilityRejects == 0, "CONTRACT_REJECTION");
    Require(x.DuplicatePublicFacts == 0, "DUAL_PUBLICATION");
    Require(x.MissingPublicFacts == 0, "PUBLIC_FACT_LOSS");
    Require(x.ErrorRate <= b.MaximumErrorRate, "ERROR_BUDGET_EXCEEDED");
    Require(x.P95Latency <= b.MaximumP95, "LATENCY_BUDGET_EXCEEDED");
    Require(clock.GetUtcNow() - x.CollectedUtc < b.Freshness, "SIGNALS_STALE");
}
```

STAGE CONTROLLER

Advance, hold and abort are deterministic outcomes

EXECUTABLE ASSET 58-AA - CONTROLLER DECISION

```

public async Task<StageReceipt> Evaluate(
    RolloutPlan plan, RolloutStage target, CancellationToken ct)
{
    EnvironmentState before = await state.Read(ct);
    Require(target == before.Stage.Next(), "STAGE_SKIP_FORBIDDEN");

    GateReceipt[] gates = await catalogue.RequiredFor(target)
        .SelectAwait(x => runner.Run(x, plan, ct)).ToArrayAsync(ct);
    StageVerdict verdict = gates.All(x => x.Verdict == GateVerdict.Pass)
        ? StageVerdict.Admitted
        : StageVerdict.Rejected;

    EnvironmentState after = verdict == StageVerdict.Admitted
        ? await state.CompareExchange(before, before.Advance(target), ct)
        : before;
    return await receipts.Create(plan, before, after, gates, verdict, ct);
}

```

NO SKIP

An operator may abort or hold, but cannot jump from Expanded to FullExposure. Emergency actions have their own signed receipt and still run the rollback classifier.

ROLLBACK CLASSIFICATION

Reversibility is computed against a named target

EXECUTABLE ASSET 58-AB - ROLLBACK DECISION

```

public sealed record RollbackDecision(
    Guid PlanId,
    RolloutStage CurrentStage,
    string CurrentImageDigest,
    string TargetImageDigest,
    bool BinaryCanReadCurrentSchema,
    bool BinaryCanReadCurrentRows,
    bool ConsumersCanReadCurrentFacts,
    bool OldRepresentationMaintained,
    bool IncompatibleTruthCommitted,
    RollbackAction Action,
    IReadOnlyList<string> RequiredOperations,
    string EvidenceSha256);

public enum RollbackAction
{
    ReplaceBinaryOnly,
    DisableFeatureThenReplace,
    ForwardRepair,
    CompensatingMigration,
    RestoreAndReconcile,
    Forbidden
}

```

ROLLBACK MATRIX**The safe action changes as durable truth changes****ROLLBACK MATRIX 58-Q**

Current state	Old image?	Schema action	Required data action
Prepared	safe	none	none
Expanded	safe	leave additive S1	none
CompatibleBinary	safe	leave S1	none
Backfilled / ReadPilot	safe	leave S1	retain backfilled values
WritePilot, dual form	safe after flag off	leave S1	verify old form maintained
new-only facts or rows	UNSAFE	do not tag-flip	reverse project or compensate
Contracted S3	FORBIDDEN	forward migration	restore compatibility first

ADDITIVE ROLLBACK

Dropping the new nullable column during an incident is usually unnecessary risk. Returning B0 while leaving S1 in place preserves optional data and avoids another DDL operation under pressure.

ROLLBACK EXECUTION**Feature, traffic and binary change in dependency order****EXECUTABLE ASSET 58-AC - SAFE BINARY REVERSAL**

```
public async Task<RollbackReceipt> ExecuteSafeBinaryRollback(
    RollbackDecision decision, CancellationToken ct)
{
    Require(decision.Action is RollbackAction.ReplaceBinaryOnly
        or RollbackAction.DisableFeatureThenReplace, "ROLLBACK_UNSAFE");
    Require(decision.BinaryCanReadCurrentSchema, "OLD_BINARY_SCHEMA_INCOMPATIBLE");
    Require(decision.BinaryCanReadCurrentRows, "OLD_BINARY_DATA_INCOMPATIBLE");
    Require(decision.ConsumersCanReadCurrentFacts, "FACTS_ALREADY_INCOMPATIBLE");

    if (decision.Action == RollbackAction.DisableFeatureThenReplace)
        await features.ApplySignedKillPlan(decision.PlanId, ct);
    await router.StopNewWriteCohorts(ct);
    await dispatcher.DrainBounded(ct);
    await deployment.SelectDigest(decision.TargetImageDigest, ct);
    await runtime.VerifyMixedState(ct);
    return await evidence.Record(decision, ct);
}
```

DRAIN BOUNDARY

Drain is bounded and does not discard accepted work. If the pending-fact backlog cannot reach its limit, rollback holds or follows the explicitly rehearsed recovery path; it does not delete outbox rows.

UNSAFE ROLLBACK**A tag flip cannot erase incompatible durable truth****IRREVERSIBILITY TRIGGERS 58-R**

Trigger	Why old state cannot resume
V2-only event already emitted	old subscriber may reject or misinterpret retained delivery
new-only row representation	B0 cannot reconstruct required old field
destructive merge/split	information needed for reverse mapping was discarded
old column/index dropped	old SQL/model is structurally invalid
external effect under V2 semantics	side effect cannot be uncalled

REQUIRED RESPONSE

Freeze exposure, preserve evidence, classify affected identities, and execute a forward repair, reverse projection, compensating migration or restore-and-reconcile plan. The incident receipt must say why binary rollback alone is forbidden.

FORWARD AND COMPENSATING REPAIR

Recovery restores compatibility without pretending history vanished

EXECUTABLE ASSET 58-AD - REPAIR PLAN

```
public sealed record CompatibilityRepairPlan(
    Guid IncidentId,
    string CurrentSchemaFingerprint,
    string TargetReadableShape,
    string AffectedIdentityQuerySha256,
    string ReverseProjectionPolicy,
    string RepairArtifactSha256,
    bool StopsNewWrites,
    bool PreservesOriginalRows,
    ReconciliationPolicy Reconciliation,
    string ApprovedBy,
    string Signature);

static void Admit(CompatibilityRepairPlan x)
{
    Require(x.StopsNewWrites, "REPAIR_WITH_CONCURRENT_WRITES");
    Require(x.PreservesOriginalRows, "REPAIR_DESTROYS_EVIDENCE");
    Require(IsDeterministic(x.ReverseProjectionPolicy), "REVERSE_POLICY_AMBIGUOUS");
    Require(x.Reconciliation.RequiresIdentityAndCount,
        "REPAIR_RECONCILIATION_WEAK");
}
```

COMPENSATION BOUNDARY

A compensating public fact records a new domain event; it is not message deletion. A database repair appends audit evidence and reconciles identities before traffic reopens.

ROLLBACK CLOSURE

The last safe point is a recorded decision, not folklore

EXECUTABLE ASSET 58-AE - CLOSURE RECEIPT

```
public sealed record RollbackClosureReceipt(
    Guid PlanId,
    RolloutStage Stage,
    string LastBinaryRollbackTargetDigest,
    string CompatibilityMatrixSha256,
    string BackfillReconciliationSha256,
    string ConsumerCapabilityInventorySha256,
    string Reason,
    DateTimeOffset EffectiveUtc,
    string ApprovedBy,
    string ReceiptSha256);

static void CloseRollback(RollbackClosureReceipt x, EnvironmentState state)
{
    Require(state.Stage == RolloutStage.FullExposure, "EXPOSURE_NOT_COMPLETE");
    Require(x.Reason is "NEW_ONLY_WRITE" or "V2_ONLY_FACT" or "CONTRACT_DDL",
        "ROLLBACK_CLOSURE_REASON_UNKNOWN");
    Require(AllEvidencePresent(x), "ROLLBACK_CLOSURE_EVIDENCE_MISSING");
}
```

HUMAN CONSEQUENCE

After EffectiveUtc the runbook no longer presents old-image selection as recovery. It links directly to the admitted forward/compensating repair path.

CONTRACT MIGRATION

Destructive cleanup belongs to a later candidate

EXECUTABLE ASSET 58-AF - CONTRACT PRECONDITIONS

```
public static void AdmitContractMigration(ContractEvidence x)
{
    Require(x.ActiveB0Instances == 0, "OLD_BINARY_ACTIVE");
    Require(x.ActiveOldWorkers == 0, "OLD_WORKER_ACTIVE");
    Require(x.V1CallsAcrossRetirementWindow == 0, "V1_CALLER_REMAINS");
    Require(x.V1FactsAcrossRetentionWindow == 0, "V1_FACT_REDELIVERY_REMAINS");
    Require(x.OldColumnReads == 0, "OLD_COLUMN_READ_REMAINS");
    Require(x.OldColumnWrites == 0, "OLD_COLUMN_WRITE_REMAINS");
    Require(x.BackupRestoreRehearsalPassed, "RESTORE_REHEARSAL_MISSING");
    Require(x.ContractArtifactIsLaterCandidate, "CONTRACT_SAME_RELEASE");
}
```

DELETE LESS

If `source_alarm_id` remains semantically useful audit evidence, it is retained even after V2 becomes baseline. Contract is not a ritual of dropping every old column; it removes only proven-dead representation whose loss is acceptable.

DATABASE RECOVERY BOUNDARY

Backup restore is not a standalone rollback

RESTORE DECISION 58-S

Question	Required answer
recovery point	which committed writes will be absent?
broker position	which facts will redeliver or disappear?
external effects	which cannot be restored with the database?
writer freeze	how are concurrent writes stopped and fenced?
reconciliation	how are rows, facts and effects joined after restore?
identity	which backup, key, database and artifact are used?

RESTORE RULE

A point-in-time restore may recover database bytes while making broker and external systems inconsistent. It is admitted only as one step of a rehearsed restore-and-reconcile plan with explicit accepted data loss.

PUBLIC-FACT CONSERVATION

Rollout must preserve one accepted transition to one intent

EXECUTABLE ASSET 58-AG - CONSERVATION GATE

```
public sealed record FactConservationObservation(  
    long AcceptedDomainTransitions,  
    long PendingFactRowsCreated,  
    long DistinctSemanticFacts,  
    long PublishedMessages,  
    long ConsumerApplications,  
    long DuplicateSemanticFacts,  
    long MissingSemanticFacts);  
  
static void Admit(FactConservationObservation x)  
{  
    Require(x.AcceptedDomainTransitions == x.PendingFactRowsCreated,  
        "DOMAIN_TO_INTENT_LOSS");  
    Require(x.PendingFactRowsCreated == x.DistinctSemanticFacts,  
        "DUAL_EVENT_VERSION_PUBLICATION");  
    Require(x.DuplicateSemanticFacts == 0, "SEMANTIC_DUPLICATE");  
    Require(x.MissingSemanticFacts == 0, "SEMANTIC_FACT_MISSING");  
    // PublishedMessages may exceed facts because transport is at-least-once.  
}
```

TRANSPORT QUALIFICATION

ConsumerApplications count first successful application per semantic identity. Redelivery may increase PublishedMessages but must not create another domain transition.

FAILURE CAMPAIGNS

The laboratory attacks rollout assumptions deliberately

CAMPAIGN CATALOGUE 58-T

Fault	Expected refusal / recovery
V2 producer before one consumer	CONSUMER_CAPABILITY_INCOMPLETE
two migrators	MIGRATION_LEASE_UNAVAILABLE
B0 routed after contract	OLD_BINARY_SCHEMA_INCOMPATIBLE
stale feature epoch	FEATURE_EPOCH_MISMATCH
non-sticky write percentage	UNSTABLE_COHORT_KEY
crash before checkpoint commit	same batch safely repeats
crash after checkpoint commit	resume after committed key
dual V1+V2 publication	DUAL_EVENT_VERSION_PUBLICATION
rollback after new-only truth	ROLLBACK_UNSAFE
contract with dormant scheduled job	OLD_WORKER_ACTIVE

PROVIDER-BACKED COMPONENT TESTS

The real provider decides DDL, locks and transaction behavior

EXECUTABLE ASSET 58-AH - MIGRATION/BACKFILL SUITE

```
[Collection("sqlserver-rollout")]
public sealed class AlarmRolloutDatabaseTests(SqlServerLease lease)
{
    [Fact] public Task B0_reads_rows_after_expand() =>
        Scenario.Start(lease).ApplyBaseline().SeedB0Rows()
            .ApplyExpand().RunB0ReadAssertions();

    [Fact] public Task Backfill_resumes_after_commit_boundary() =>
        Scenario.Start(lease).ApplyExpand().SeedMixedRows()
            .CrashBackfillAfterCommittedBatch().Resume()
            .AssertNoNulls().AssertProjectionDigest();

    [Fact] public Task Second_migrator_is_refused() =>
        Scenario.Start(lease).HoldMigrationSessionLock()
            .RunSecondMigrator().AssertFailure("MIGRATION_LEASE_UNAVAILABLE");

    [Fact] public Task Contract_refuses_old_binary() =>
        Scenario.Start(lease).ApplyContract()
            .RunB0ReadAssertionsExpectingStructuralFailure();
}
```

PROVIDER IDENTITY

The receipt records SQL Server image digest, compatibility level, collation, EF Core version, generated SQL digest and database seed digest. An in-memory provider cannot replace these tests.

MIXED-BINARY RUNTIME TESTS

The production composition path proves coexistence

EXECUTABLE ASSET 58-AI - DEPLOYMENT SCENARIO

```
[Fact]
public async Task B0_and_B1_coexist_without_fact_loss()
{
    await using RolloutLab lab = await RolloutLab.Start(
        schema: SchemaStage.Expanded,
        alarmImages: [Images.AlarmB0, Images.AlarmB1],
        consumers: ConsumerCapability.V1AndV2);

    await lab.RouteStableCohort(readV2BasisPoints: 1_000,
        writeV2BasisPoints: 100);
    await lab.Submit(WorkLoad.MixedAlarms(count: 2_000));
    await lab.WaitForQuiescence(Bounds.OutboxDrain);

    lab.AssertNoCompatibilityRejects();
    lab.AssertExactlyOneSemanticFactPerAcceptedTransition();
    lab.AssertB0CanReadEveryCommittedRow();
    lab.AssertStableCohortAssignments();
}
```

PATH IDENTITY

The lab uses the Chapter 57 image digests and the same Program.cs composition root. Test-only endpoints, alternate serializers and substitute databases are forbidden.

ROLLBACK RUNTIME TESTS

Safe and unsafe reversals are both executable assertions

EXECUTABLE ASSET 58-AJ - ROLLBACK CAMPAIGNS

```
[Theory]
[InlineData(RolloutStage.Expanded, RollbackAction.ReplaceBinaryOnly)]
[InlineData(RolloutStage.Backfilled, RollbackAction.ReplaceBinaryOnly)]
[InlineData(RolloutStage.WritePilot,
    RollbackAction.DisableFeatureThenReplace)]
public Task Compatible_stages_rehearse_expected_rollback(
    RolloutStage stage, RollbackAction expected) =>
    RollbackScenario.At(stage).WithDualRepresentation()
        .Execute().AssertAction(expected).AssertB0Readable();

[Fact]
public Task V2_only_truth_refuses_binary_rollback() =>
    RollbackScenario.At(RolloutStage.RollbackClosed)
        .WithV2OnlyFactAlreadyPublished()
        .Classify().AssertAction(RollbackAction.ForwardRepair)
        .AssertFailureCode("ROLLBACK_UNSAFE");
```

NEGATIVE PROOF

A test that expects refusal is as important as a successful rollback. It prevents a runbook or controller from advertising a destructive tag flip after the compatibility window has closed.

EVIDENCE MANIFEST

One manifest joins plan, stages, data and reversal boundary

EXECUTABLE ASSET 58-AK - ROLLOUT EVIDENCE BUNDLE

```
public sealed record RolloutEvidenceManifest(
    Guid PlanId,
    string CandidateManifestSha256,
    string EnvironmentReferenceSha256,
    string CompatibilityMatrixSha256,
    string MigrationReceiptSha256,
    string ConsumerCapabilityInventorySha256,
    string BackfillReconciliationSha256,
    string FeaturePlanSha256,
    IReadOnlyList<string> StageReceiptSha256s,
    string FactConservationReceiptSha256,
    string RollbackDecisionSha256,
    string? RollbackClosureReceiptSha256,
    EvidenceVerdict Verdict,
    string ManifestSha256);

public enum EvidenceVerdict { SpecifiedRuntimePending, Admitted, Rejected }
```

FAIL-CLOSED BUNDLE

Missing, stale or cross-plan evidence rejects the bundle. A complete SPECIFIED manifest remains RUNTIME PENDING until retained executions populate every required receipt.

EVIDENCE VALIDATOR

A receipt from another plan cannot fill this plan's gap

EXECUTABLE ASSET 58-AL - JOIN VALIDATOR

```
public static void Admit(RolloutEvidenceManifest m, RolloutPlan plan,
    IEnumerable<StageReceipt> stages)
{
    Require(m.PlanId == plan.PlanId, "EVIDENCE_PLAN_MISMATCH");
    Require(m.CandidateManifestSha256 == plan.CandidateManifestSha256,
        "EVIDENCE_CANDIDATE_MISMATCH");
    Require(m.CompatibilityMatrixSha256 == plan.CompatibilityMatrixSha256,
        "EVIDENCE_MATRIX_MISMATCH");
    Require(m.FeaturePlanSha256 == plan.FeaturePlanSha256,
        "EVIDENCE_FEATURE_PLAN_MISMATCH");

    StageReceipt[] ordered = stages.OrderBy(x => x.To).ToArray();
    Require(ordered.Select(x => x.To).SequenceEqual(
        RequiredStages.Through(plan.RequestedStage)), "STAGE_RECEIPT_GAP");
    Require(ordered.All(x => x.PlanId == plan.PlanId), "CROSS_PLAN_EVIDENCE");
    Require(ordered.All(x => x.Verdict == StageVerdict.Admitted),
        "STAGE_NOT_ADMITTED");
}
```

OPERATIONAL TELEMETRY

Signals carry rollout dimensions without leaking values

TELEMETRY CONTRACT 58-U

Signal	Required dimensions	Forbidden dimension
request	plan epoch, binary digest prefix, route version, cohort	raw alarm id
consumer	event version, consumer digest, verdict	payload
database	schema stage, migration id, lock wait bucket	connection secret
backfill	operation, batch, cursor bucket, counts	row values
rollback	decision, current/target digest, reason code	free-form sensitive data

CARDINALITY RULE

PlantUnitId is used for stable allocation but is not emitted as an unbounded metric label. The evidence store may retain a protected cohort membership digest for audit.

SOURCE QUALIFICATION LEDGER

External guidance is adopted with explicit local boundaries

PRIMARY-SOURCE MAPPING 58-V

Guidance	Applied here	Local qualification
Azure safe deployment practices	stages, health model, hold and rollback plan	stage budgets remain NEXUS workload inputs
EF Core applying migrations	bundle/idempotent script as reviewed artifacts	target state and provider rehearsal still required
SQL Server sp_getapplock	serialized migrator session	same database/session/owner must be retained
.NET feature management	filters/variants vocabulary	signed epoch and stable cohort added locally
schema compatibility concepts	reader/writer direction	JSON corpus, not Avro claims, is authoritative

FRESHNESS

These references inform the chapter's design as of 10 August 2026. The final implementation pins actual package/provider versions and retains generated SQL and executed reports.

ANTI-PATTERN CATALOGUE

Convenient shortcuts that close the safety window

DO-NOT-DO LIST 58-W

Anti-pattern	Why rejected
edit V1 JSON in place	strict readers and corpus no longer identify one contract
dual-publish V1 and V2	duplicates one semantic transition
migrate from every replica	authority and DDL serialization become ambiguous
backfill with OFFSET	concurrent writes shift pages and repeat/skip rows
flip writes randomly per request	one aggregate receives mixed durable semantics
drop additive schema during incident	adds DDL risk without restoring truth
rollback tag after new-only writes	compute reverses but data does not
contract in same release	no retained coexistence window
declare zero callers from missing telemetry	absence of evidence becomes false evidence

COMPLETION MATRIX

What Chapter 58 now makes reviewable

COMPLETION MATRIX 58-X

Concern	Executable asset	Status
no-rebuild promotion	RolloutPlan + EnvironmentImageReference	SPECIFIED
HTTP/event coexistence	versioned endpoints, dual reader, exclusive producer	SPECIFIED
database expansion	EF migration + serialized migrator	SPECIFIED
backfill	keyset worker + checkpoint + reconciliation	SPECIFIED
feature control	signed epoch + stable cohort + request snapshot	SPECIFIED
staged rollout	state machine + observation gate + receipts	SPECIFIED
rollback	classifier + safe executor + unsafe repair path	SPECIFIED
runtime truth	provider/mixed-binary/failure campaign reports	PENDING

RESIDUAL OBLIGATIONS

Real repository paths, package versions, SQL Server image digest, generated SQL, traffic budgets, cohort policy, retention windows, backups and rehearsal receipts remain environment-owned evidence. The manuscript does not promote them from PENDING by assertion.

HANDOFF TO CHAPTER 59

Translate proved runtime boundaries; do not let YAML invent them

BOUNDARY CARRIED FORWARD 58-Y

Chapter 58 leaves	Chapter 59 must map
admitted service image digest	container image by digest
stable edge and worker responsibilities	minimal Deployments and Services
validated configuration and secret references	ConfigMap/Secret references without values
startup/liveness/readiness semantics	three distinct probes
dedicated migration authority	one controlled migration execution
resource and shutdown observations	requests, limits and termination budget
staged rollout receipts	one controlled educational rollout

NEXT INVARIANT

A Kubernetes manifest is justified only when every workload, Service, configuration reference, probe, resource boundary and rollout control traces to evidence already produced by the runtime and rollout laboratory; YAML cannot create proof the runtime never earned.

FINAL CHAPTER RULE

Rollback is a property of coexistence, not of image history

NEXUS-1 calls a rollout reversible only while the selected prior binary can read the current schema and rows, every active consumer can interpret facts already emitted, old representation remains valid, and no irreversible external effect has crossed the boundary.

PART X PROGRESS

Chapter 57 bound delivery to immutable evidence. Chapter 58 preserves that identity while binaries, contracts and data coexist. Chapter 59 must express only those already-proved runtime needs as minimal, explicitly educational Kubernetes resources.

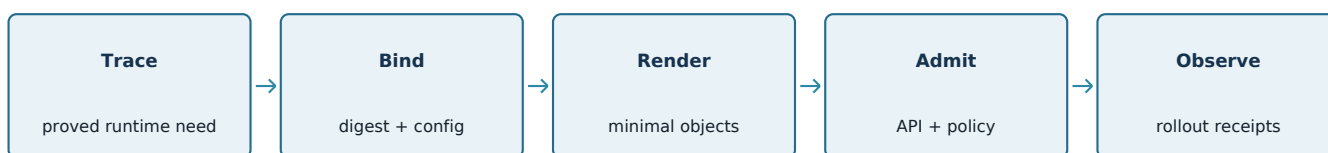
PART X - MIGRATION, DELIVERY, AND FINAL ARCHITECTURE

59. Translating Proven Needs to Kubernetes

Kubernetes may schedule, discover, probe and replace the already-proved NEXUS-1 workloads; it may not create service ownership, migration safety, readiness semantics or rollout evidence that the runtime never earned.

CHAPTER DECISION

Render a minimal, value-free Kubernetes package from the admitted candidate, rollout plan, resource budget and cluster capability profile. Use Deployments for the seven stateless application roles, ClusterIP Services only for callable HTTP roles, one separately authorized migration Job, explicit configuration and secret references, three distinct probes, bounded shutdown and one evidence-controlled rollout.

PROOF PATH 59-A - EVIDENCE BEFORE YAML**GUARANTEE LEDGER 59-A**

Kubernetes evidence	Allowed claim	Cannot claim
rendered object digest	exact desired objects identified	cluster accepted them
server dry-run receipt	API/admission accepted shape	Pods became ready
Deployment receipt	named replicas reached declared state	workflow is correct
Job receipt + migration receipt	one admitted migration completed	rollback is safe

EVIDENCE VERDICT

SPECIFIED | RUNTIME PENDING. The manifests, renderer, validators, rollout driver and failure campaigns are compile-oriented assets. No cluster, image pull, migration Job, Pod readiness or controlled rollout is invented.

ORDINARY DEVELOPER PROBLEM

The YAML is valid, but the architecture disappeared

A team converts every Compose service into a Deployment, copies connection strings into ConfigMaps, exposes every API, adds one generic /health probe and lets each replica migrate on startup. Kubernetes accepts the documents. The platform now automates the exact ambiguity the book spent fifty-eight chapters removing.

FAILURE LEDGER 59-B

Shortcut	Hidden failure	Correction
one Deployment per container name	roles and evidence were not traced	runtime-to-object ledger
tagged image	candidate identity can move	digest-only renderer
ConfigMap for credentials	confidential values become configuration	Secret reference only
one /health probe	restart and traffic semantics collapse	startup/live/ready split
migration initContainer	every replica becomes schema authority	one controlled Job
kubectl rollout undo	data and contracts may be irreversible	Chapter 58 classifier first

PLAIN-LANGUAGE PRINCIPLE

Kubernetes is a mechanism host. The architecture remains in the ownership, contracts, data, health and evidence rules that constrain how those mechanisms may be used.

CHAPTER BOUNDARY

Translation here; final distributed proof next

This chapter owns the minimal Kubernetes mapping, render/admission workflow and one controlled educational rollout. Chapter 60 will execute and judge the complete operator-to-delayed-reporting scenario across the final service, contract, data, security, observability, failure and deployment catalogues.

DECISION RECORD

ADR-059 - Kubernetes is a deployment projection

DECISION CONSEQUENCES 59-C

Choice	Benefit	Cost / boundary
generated from admitted locks	no tag or config substitution	renderer becomes governed code
raw minimal API objects	mechanisms remain visible	some repetition is intentional
separate API and worker workloads	lifecycle/scaling remain independent	more Deployments
ClusterIP HTTP Services only	no accidental public bypass	external ingress remains out of scope
one migration Job	schema authority stays explicit	migrator OCI packaging required
application evidence after rollout	green Pods cannot become proof	controller must join two planes

NON-GOAL

The chapter does not deliver a production operations platform, managed SQL or RabbitMQ, ingress/TLS, autoscaling, high availability, backup, service mesh, policy engine, GitOps controller, multi-cluster operation or secret-management system.

CLAIM LADDER

Parsed, admitted, scheduled and proved are different claims

CLAIM LADDER 59-D

Level	Subject	Minimum evidence	Does not prove
rendered	object documents	no tokens + canonical digest	API validity
server-admitted	target cluster API	server dry-run + admission receipt	resources exist
applied	object UIDs/generations	apply inventory	workloads ready
available	Deployment/Job conditions	bounded watch + image IDs	domain flow
runtime-admitted	application evidence	readiness + conservation + traces	future rollout safe
reversible	named prior digest	Chapter 58 RollbackDecision	kubectl history is enough

NO TRANSITIVE GREEN

AvailableReplicas cannot prove the database principal owns only one database. A completed Job cannot prove the expected schema fingerprint. A successful Service lookup cannot prove authorization.

INHERITED RUNTIME TOPOLOGY

Seven application roles, not an arbitrary service explosion

RUNTIME-TO-KUBERNETES MAP 59-E

Logical role	Workload	Service?	Owned dependency
alarm-api	Deployment	ClusterIP	AlarmManagementDb
alarm-worker	Deployment	none	AlarmManagementDb + broker
rootcause-api	Deployment	ClusterIP	RootCauseDb
rootcause-worker	Deployment	none	RootCauseDb + broker
projection-api	Deployment	ClusterIP	ReportingDb read
projection-worker	Deployment	none	ReportingDb write + broker
query-bff	Deployment	ClusterIP	three internal APIs

PRESERVED BOUNDARY

Audit and Compliance remain public-fact subscribers in the broader architecture, but the Chapter 55 admitted core topology has seven roles. Chapter 59 does not create extra Deployments merely because candidates exist in the catalogue.

WHAT DOES NOT ENTER THE PACKAGE

Stateful platforms stay explicit dependencies

EXCLUSION LEDGER 59-F

Component	Kubernetes representation	Reason
SQL Server databases	endpoint + per-owner Secret reference	no storage/backup/HA evidence
RabbitMQ	endpoint + scoped credential reference	no broker operations design
telemetry backend	OTLP endpoint configuration	collector/backend operation is environment-owned
public ingress	none; BFF remains ClusterIP	TLS/DNS/edge ownership not proved
HPA	none	no admitted scaling metric or bounds
PDB	none	replica/failure-domain availability not proved
service mesh	none	not required by runtime evidence

EDUCATIONAL ACCESS

A supported lab may use `kubectl port-forward` to the BFF. That convenience is recorded in the run receipt and is not described as production ingress.

REPOSITORY CONTRACT

Templates, locks, generated objects and evidence are separate

EXECUTABLE ASSET 59-A - KUBERNETES TREE

```
deploy/kubernetes/  
  templates/  
    namespace.yaml  
    service-accounts.yaml  
    configmap.yaml  
  deployments/  
    alarm-api.yaml  
    alarm-worker.yaml  
    rootcause-api.yaml  
    rootcause-worker.yaml  
    projection-api.yaml  
    projection-worker.yaml  
    query-bff.yaml  
  services/internal-http.yaml  
  jobs/alarm-migrate.yaml  
  policies/internal-ingress.yaml  
  profile/kubernetes-capabilities.json  
  policy/manifest-policy.json  
eng/Nexus.Kubernetes/  
  tests/Nexus.Kubernetes.ManifestTests/  
  tests/Nexus.Kubernetes.RuntimeTests/  
docs/runbooks/ch59-controlled-rollout.md  
artifacts/kubernetes/<plan-id>/
```

GENERATED-ONLY EXECUTION

Files under templates contain typed tokens and are not applicable. Only the renderer's immutable artifacts/kubernetes/<plan-id> output may reach validation or a cluster.

CLUSTER CAPABILITY PROFILE

The target is identified before objects are rendered

EXECUTABLE ASSET 59-B - CLUSTER PROFILE

```
public sealed record KubernetesCapabilityProfile(
    string ClusterIdentity,
    string ServerVersion,
    string ApiResourceInventorySha256,
    string AdmissionPolicySha256,
    bool ServerSideDryRun,
    bool AppsV1Deployment,
    bool BatchV1Job,
    bool NetworkingV1NetworkPolicy,
    bool NetworkPolicyEnforced,
    string Namespace,
    DateTimeOffset ObservedAtUtc,
    string ReceiptSha256);

static void Admit(KubernetesCapabilityProfile x, TimeProvider clock)
{
    Require(x.ServerSideDryRun, "SERVER_DRY_RUN_UNAVAILABLE");
    Require(x.AppsV1Deployment && x.BatchV1Job, "REQUIRED_API_MISSING");
    Require(x.Namespace == "nexus1-phase0", "NAMESPACE_TARGET_MISMATCH");
    Require(clock.GetUtcNow() - x.ObservedAtUtc < TimeSpan.FromHours(1),
        "CLUSTER_PROFILE_STALE");
}
```

VERSION BOUNDARY

The manuscript uses stable core APIs, but it does not claim compatibility with every Kubernetes distribution or version. The repository pins kubectl and records the target server/API inventory for each run.

ARTIFACT PRECONDITIONS

Every executable role needs an admitted OCI subject

EXECUTABLE ASSET 59-C - ROLE IMAGE LOCK

```
public sealed record KubernetesImageLock(
    string CandidateManifestSha256,
    IReadOnlyDictionary<string, string> RoleImages,
    string LockSha256);

static readonly string[] RequiredRoles =
[
    "alarm-api", "alarm-worker", "alarm-migrator",
    "rootcause-api", "rootcause-worker",
    "projection-api", "projection-worker", "query-bff"
];

static void Admit(KubernetesImageLock x, CandidateManifest candidate)
{
    Require(x.CandidateManifestSha256 == candidate.Digest,
        "CANDIDATE_IMAGE_LOCK_MISMATCH");
    Require(x.RoleImages.Keys.ToHashSet().SetEquals(RequiredRoles),
        "ROLE_IMAGE_INVENTORY_MISMATCH");
    Require(x.RoleImages.Values.All(IsPinnedDigest), "ROLE_IMAGE_NOT_PINNED");
    Require(x.RoleImages.All(pair => candidate.Contains(pair.Value)),
        "ROLE_IMAGE_NOT_IN_CANDIDATE");
}
```

HONEST PACKAGING GAP

Chapter 57 admitted image receipts and Chapter 58 admitted a migration bundle artifact. A Kubernetes Job additionally needs a runnable migrator OCI digest. Until the delivery graph emits and admits that role image, rendering stops with `ROLE_IMAGE_INVENTORY_MISMATCH`.

RENDER INPUT

One document binds candidate, rollout, resources and cluster

EXECUTABLE ASSET 59-D - DEPLOYMENT PROJECTION INPUT

```
public sealed record KubernetesRenderInput(
    Guid PlanId,
    long RolloutEpoch,
    string CandidateManifestSha256,
    string RolloutPlanSha256,
    string FeaturePlanSha256,
    string ConfigurationIdentity,
    string ResourceBudgetSha256,
    string KubernetesImageLockSha256,
    string ClusterCapabilityProfileSha256,
    string Namespace,
    string Environment,
    string Signature);

static void Admit(KubernetesRenderInput x, RolloutPlan rollout)
{
    Require(x.PlanId == rollout.PlanId, "KUBE_PLAN_MISMATCH");
    Require(x.RolloutEpoch == rollout.Epoch, "KUBE_EPOCH_MISMATCH");
    Require(x.CandidateManifestSha256 == rollout.CandidateManifestSha256,
        "KUBE_CANDIDATE_MISMATCH");
    Require(x.Namespace == "nexus1-phase0", "KUBE_NAMESPACE_UNEXPECTED");
    Require(VerifySignature(x), "KUBE_RENDER_INPUT_SIGNATURE_INVALID");
}
```

FAIL-CLOSED RENDERER

Tokens are replaced from typed locks, never ambient shell state

EXECUTABLE ASSET 59-E - RENDER COMMAND

```
dotnet run --project eng/Nexus.Kubernetes -- render \
  --candidate artifacts/candidate/manifest.json \
  --rollout artifacts/rollout/plan.json \
  --images deploy/kubernetes/images.lock.json \
  --resources artifacts/capacity/resource-budget.json \
  --cluster deploy/kubernetes/profile/kubernetes-capabilities.json \
  --output artifacts/kubernetes/<plan-id>

dotnet run --project eng/Nexus.Kubernetes -- verify \
  --input artifacts/kubernetes/<plan-id> \
  --policy deploy/kubernetes/policy/manifest-policy.json
```

EXECUTABLE ASSET 59-F - UNRESOLVED-TOKEN GATE

```
static void VerifyRendered(string text, ManifestPolicy policy)
{
    Require(!Regex.IsMatch(text, "@[A-Z0-9_]+@"),
        "MANIFEST_TOKEN_SURVIVED");
    Require(!text.Contains("image: latest", OrdinalIgnoreCase),
        "MUTABLE_IMAGE_REFERENCE");
    Require(!text.Contains("stringData:", Ordinal),
        "SECRET_VALUE_IN_MANIFEST");
    Require(CanonicalYaml.Sha256(text) == policy.ExpectedManifestSha256,
        "RENDERED_MANIFEST_DIGEST_MISMATCH");
}
```

METADATA IDENTITY

Selectors stay stable; evidence identities stay annotations

METADATA CONTRACT 59-G

Location	Field	Purpose
stable label	app.kubernetes.io/name	logical product
stable label	nexus1.io/workload	selector and inventory
stable label	nexus1.io/role	api, worker, bff, migrator
annotation	nexus1.io/image-digest	full OCI subject
annotation	nexus1.io/candidate-sha256	Chapter 57 join
annotation	nexus1.io/rollout-plan-sha256	Chapter 58 join
annotation	nexus1.io/configuration-sha256	Chapter 10 join

CARDINALITY RULE

Full digests are annotations and evidence fields, not metric labels. Pod-template annotations intentionally change when the admitted image, configuration or rollout plan changes, causing a new ReplicaSet.

NAMESPACE

A namespace is an inventory boundary, not data ownership

EXECUTABLE ASSET 59-G - NAMESPACE

```
apiVersion: v1
kind: Namespace
metadata:
  name: nexus1-phase0
  labels:
    app.kubernetes.io/part-of: nexus1
    nexus1.io/environment-class: educational-phase0
  annotations:
    nexus1.io/boundary: >-
      Educational demonstrator; not operational or safety-class software.
```

NAMESPACE CAN/CANNOT 59-H

Can support	Cannot prove
object naming, quota/policy attachment, inventory	service owns its database
RBAC and Secret scoping	network isolation by itself
cleanup and evidence queries	tenant or safety boundary

SERVICE IDENTITIES

Pods receive no Kubernetes API token by default

EXECUTABLE ASSET 59-H - SERVICE ACCOUNTS

```
apiVersion: v1
kind: ServiceAccount
metadata:
  name: alarm-api
  namespace: nexus1-phase0
automountServiceAccountToken: false
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: alarm-worker
  namespace: nexus1-phase0
automountServiceAccountToken: false
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: alarm-migrator
  namespace: nexus1-phase0
automountServiceAccountToken: false
```

IDENTITY BOUNDARY

These ServiceAccounts identify Pods to Kubernetes. Database and broker authorization still use separately scoped runtime or migration credentials. No RoleBinding is created because the application does not call the Kubernetes API.

NON-CONFIDENTIAL CONFIGURATION

An immutable ConfigMap carries values safe to inspect

EXECUTABLE ASSET 59-I - GENERATED CONFIGMAP

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: nexus1-runtime-@@CONFIG_SHA12@@
  namespace: nexus1-phase0
  annotations:
    nexus1.io/configuration-sha256: "@@CONFIG_SHA256@@"
immutable: true
data:
  Environment__Identity: "phase0-kubernetes-lab"
  Broker__Host: "rabbitmq.dependencies.svc.cluster.local"
  Broker__Port: "5672"
  Telemetry__OtlpEndpoint: "http://otel.dependencies.svc.cluster.local:4317"
  Rollout__PlanSha256: "@@ROLLOUT_PLAN_SHA256@@"
  Rollout__Epoch: "@@ROLLOUT_EPOCH@@"
```

VALUE-FREE EVIDENCE

The evidence projection may retain keys, object name and digest, but not Secret values. Changing an immutable ConfigMap creates a new digest-suffixed object and an explicit Pod-template change.

SECRET REFERENCES

The repository names requirements, never credentials

SECRET-REFERENCE CATALOGUE 59-I

Workload	Secret name	Required keys
alarm-api	alarm-api-runtime	alarm-db-dsn
alarm-worker	alarm-worker-runtime	alarm-db-dsn, broker-credential
alarm-migrator	alarm-migrator-runtime	alarm-db-migration-dsn
rootcause-api	rootcause-api-runtime	rootcause-db-dsn
rootcause-worker	rootcause-worker-runtime	rootcause-db-dsn, broker-credential
projection-api	projection-api-runtime	projection-db-read-dsn
projection-worker	projection-worker-runtime	projection-db-write-dsn, broker-credential

EXECUTABLE ASSET 59-J - REFERENCE, NOT SECRET VALUE

```
volumes:
- name: runtime-secrets
  secret:
    secretName: alarm-api-runtime
    optional: false
containers:
- name: alarm-api
  env:
    - name: ConnectionStrings_AlarmDbFile
      value: /var/run/secrets/nexus1/alarm-db-dsn
  volumeMounts:
    - name: runtime-secrets
      mountPath: /var/run/secrets/nexus1
      readOnly: true
```

SECRET BOUNDARY

Kubernetes Secrets are not a complete secret-management platform and may be stored unencrypted unless the cluster is configured otherwise. The chapter requires at-rest protection, least privilege and value-free evidence, but does not claim to configure them.

ALARM API DEPLOYMENT

One workload maps one already-proved role

EXECUTABLE ASSET 59-K - ALARM-API METADATA AND STRATEGY

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: alarm-api
  namespace: nexus1-phase0
  labels:
    app.kubernetes.io/part-of: nexus1
    nexus1.io/workload: alarm-api
    nexus1.io/role: api
spec:
  replicas: 2
  revisionHistoryLimit: 3
  minReadySeconds: 30
  progressDeadlineSeconds: 600
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 0
      maxSurge: 1
  selector:
    matchLabels:
      nexus1.io/workload: alarm-api
```

REPLICA QUALIFICATION

Two replicas and these rollout values are the educational Alarm API profile, not production capacity evidence. The ResourceBudget and Chapter 53 load receipt must replace them before a stronger claim is made.

ALARM API POD TEMPLATE

Digest, config and rollout identities enter the template

EXECUTABLE ASSET 59-L - POD IDENTITY AND CONTAINER

```

template:
  metadata:
    labels:
      app.kubernetes.io/part-of: nexus1
      nexus1.io/workload: alarm-api
      nexus1.io/role: api
    annotations:
      nexus1.io/candidate-sha256: "@@CANDIDATE_SHA256@@"
      nexus1.io/rollout-plan-sha256: "@@ROLLOUT_PLAN_SHA256@@"
      nexus1.io/configuration-sha256: "@@CONFIG_SHA256@@"
      nexus1.io/image-digest: "@@ALARM_API_DIGEST@@"
  spec:
    serviceAccountName: alarm-api
    automountServiceAccountToken: false
    terminationGracePeriodSeconds: 45
    containers:
      - name: alarm-api
        image: "@@ALARM_API_IMAGE@@"
        imagePullPolicy: IfNotPresent
        ports:
          - name: http
            containerPort: 8080
            protocol: TCP

```

PULL POLICY BOUNDARY

The digest selects immutable registry content; IfNotPresent only controls pulling behavior. Runtime evidence still records the container imageID observed by the node and compares it with the admitted digest.

ALARM API CONFIGURATION AND SECURITY

Explicit keys and a restricted process surface

EXECUTABLE ASSET 59-M - ENV AND SECURITYCONTEXT

```

env:
  - name: ASPNETCORE_URLS
    value: http://+:8080
  - name: Rollout_Epoch
    valueFrom:
      configMapKeyRef:
        name: nexus1-runtime-@@CONFIG_SHA12@@
        key: Rollout_Epoch
  - name: ConnectionStrings_AlarmDbFile
    value: /var/run/secrets/nexus1/alarm-db-dsn
securityContext:
  allowPrivilegeEscalation: false
  runAsNonRoot: true
  capabilities:
    drop: ["ALL"]
  seccompProfile:
    type: RuntimeDefault
volumeMounts:
  - name: runtime-secrets
    mountPath: /var/run/secrets/nexus1
    readOnly: true

```

FILESYSTEM QUALIFICATION

readOnlyRootFilesystem is not asserted until the image is tested with all .NET temporary, diagnostics and certificate paths. A hardening field is not added merely because it looks desirable.

THREE PROBE SEMANTICS

Startup, liveness and readiness keep their Chapter 11 meanings

EXECUTABLE ASSET 59-N - PROBE MAPPING

```

startupProbe:
  httpGet: { path: /health/startup, port: http }
  periodSeconds: 2
  timeoutSeconds: 1
  failureThreshold: 30
livenessProbe:
  httpGet: { path: /health/live, port: http }
  periodSeconds: 10
  timeoutSeconds: 1
  failureThreshold: 3
readinessProbe:
  httpGet: { path: /health/ready, port: http }
  periodSeconds: 5
  timeoutSeconds: 2
  failureThreshold: 2
  successThreshold: 1

```

PROBE EFFECTS 59-J

Probe	Question	Kubernetes effect
startup	has bounded initialization completed?	holds liveness/readiness until success
liveness	is this process locally irrecoverable?	restart container
readiness	may this instance receive work now?	remove/add endpoint

PROBE BUDGETS

Numbers come from observed runtime bounds

EXECUTABLE ASSET 59-O - BUDGET VALIDATOR

```

public static void Admit(ProbePlan p, RuntimeBounds b)
{
    TimeSpan startupBudget = p.Startup.Period * p.Startup.FailureThreshold;
    Require(startupBudget >= b.P99Startup, "STARTUP_PROBE_BUDGET_TOO_SMALL");
    Require(p.Live.Path == "/health/live", "LIVENESS_PATH_WRONG");
    Require(p.Ready.Path == "/health/ready", "READINESS_PATH_WRONG");
    Require(!p.Live.Dependencies.Any(), "LIVENESS_HAS_REMOTE_DEPENDENCY");
    Require(p.Ready.Timeout >= b.P99ReadyCheck, "READINESS_TIMEOUT_TOO_SMALL");
    Require(p.Ready.FailureThreshold * p.Ready.Period >= b.TransientFaultWindow,
        "READINESS_EVICTS_TRANSIENT_FAULT_TOO_FAST");
}

```

MECHANISM QUALIFICATION

Kubernetes defines what probe outcomes do. NEXUS-1 defines what each endpoint means, which dependencies are allowed and which measured bounds make the timings credible.

RESOURCE BOUNDARIES

Requests and limits are deployment inputs, not decorative defaults

EXECUTABLE ASSET 59-P - RESOURCE BUDGET

```
public sealed record WorkloadResourceBudget(
    string Workload,
    string CpuRequest,
    string CpuLimit,
    string MemoryRequest,
    string MemoryLimit,
    string LoadReceiptSha256,
    string RuntimeProfileSha256,
    DateTimeOffset MeasuredUtc);

static void Admit(WorkloadResourceBudget x, TimeProvider clock)
{
    Require(ParseCpu(x.CpuRequest) > 0, "CPU_REQUEST_MISSING");
    Require(ParseCpu(x.CpuLimit) >= ParseCpu(x.CpuRequest), "CPU_LIMIT_INVALID");
    Require(ParseBytes(x.MemoryLimit) >= ParseBytes(x.MemoryRequest),
        "MEMORY_LIMIT_INVALID");
    Require(IsSha256(x.LoadReceiptSha256), "RESOURCE_LOAD_RECEIPT_MISSING");
    Require(clock.GetUtcNow() - x.MeasuredUtc < TimeSpan.FromDays(30),
        "RESOURCE_BUDGET_STALE");
}
```

SCHEDULING BOUNDARY

A request informs scheduling and a limit constrains consumption. Neither value proves latency, availability or cost. The controller joins them to the exact workload/image/load profile that produced the measurement.

RESOURCE MANIFEST

The renderer inserts admitted quantities

EXECUTABLE ASSET 59-Q - RESOURCES AND SECRET VOLUME

```
resources:
  requests:
    cpu: "@@ALARM_API_CPU_REQUEST@"
    memory: "@@ALARM_API_MEMORY_REQUEST@"
  limits:
    cpu: "@@ALARM_API_CPU_LIMIT@"
    memory: "@@ALARM_API_MEMORY_LIMIT@"
volumes:
- name: runtime-secrets
  secret:
    secretName: alarm-api-runtime
    optional: false
  items:
  - key: alarm-db-dsn
    path: alarm-db-dsn
```

RESOURCE FAILURE VISIBILITY 59-K

Condition	Required evidence
Pending / unschedulable	scheduler reason + requested capacity
CPU throttling	container CPU throttled time + latency
OOMKilled	termination reason + working set + limit
eviction	node pressure + Pod QoS + accepted work reconciliation

INTERNAL SERVICE

A stable name selects only ready API Pods

EXECUTABLE ASSET 59-R - ALARM-API SERVICE

```

apiVersion: v1
kind: Service
metadata:
  name: alarm-api
  namespace: nexus1-phase0
  labels:
    app.kubernetes.io/part-of: nexus1
    nexus1.io/workload: alarm-api
spec:
  type: ClusterIP
  selector:
    nexus1.io/workload: alarm-api
  ports:
    - name: http
      protocol: TCP
      port: 8080
      targetPort: http

```

SERVICE BOUNDARY

A Service provides discovery and routing to selected endpoints. It does not authorize the caller, encrypt traffic, expose the API publicly or prove every selected Pod runs the admitted image. Those are separate controls and receipts.

WORKER DEPLOYMENT

A broker worker is scheduled but never given an HTTP Service

EXECUTABLE ASSET 59-S - ALARM-WORKER DIFFERENCES

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: alarm-worker
  namespace: nexus1-phase0
spec:
  replicas: 1
  strategy:
    type: RollingUpdate
    rollingUpdate: { maxUnavailable: 0, maxSurge: 1 }
  selector:
    matchLabels: { nexus1.io/workload: alarm-worker }
  template:
    metadata:
      labels:
        nexus1.io/workload: alarm-worker
        nexus1.io/role: worker
    spec:
      serviceAccountName: alarm-worker
      automountServiceAccountToken: false
      terminationGracePeriodSeconds: 60
      containers:
        - name: alarm-worker
          image: "@@ALARM_WORKER_IMAGE@@"
          # same three health endpoints; kubelet probes Pod IP directly
          # database and broker credentials are mounted from alarm-worker-runtime

```

SCALING BOUNDARY

Worker replicas remain one until partitioning, consumer concurrency, ordering and duplicate-application evidence admit another count. Kubernetes can create replicas; it cannot make the message semantics safe.

FROM SERVICES TO RUNTIME

BFF AND CALLABLE API INVENTORY

Only HTTP request roles receive Services

SERVICE INVENTORY 59-L

Service	Selected Deployment	Callers	External?
alarm-api	alarm-api	query-bff	no
rootcause-api	rootcause-api	query-bff	no
projection-api	projection-api	query-bff	no
query-bff	query-bff	lab port-forward	no Ingress

EXECUTABLE ASSET 59-T - BFF DOWNSTREAM IDENTITIES

```
env:
- name: Downstream_AlarmBaseUri
  value: http://alarm-api.nexus1-phase0.svc.cluster.local:8080
- name: Downstream_RootCauseBaseUri
  value: http://rootcause-api.nexus1-phase0.svc.cluster.local:8080
- name: Downstream_ProjectionBaseUri
  value: http://projection-api.nexus1-phase0.svc.cluster.local:8080
```

NO BROWSER ORCHESTRATION

The browser reaches the BFF only. Kubernetes DNS changes discovery mechanics; it does not move workflow orchestration into the UI or permit database access from the BFF.

INVENTORY VALIDATOR

The generated set must equal the proved topology

EXECUTABLE ASSET 59-U - EXACT OBJECT INVENTORY

```
static void Admit(KubernetesObjectSet set, TopologyContract topology)
{
    RequireExact(set.Deployments,
    [
        "alarm-api", "alarm-worker",
        "rootcause-api", "rootcause-worker",
        "projection-api", "projection-worker", "query-bff"
    ], "DEPLOYMENT_INVENTORY_DRIFT");

    RequireExact(set.Services,
    ["alarm-api", "rootcause-api", "projection-api", "query-bff"],
    "SERVICE_INVENTORY_DRIFT");
    Require(set.Services.All(x => x.Type == "ClusterIP"),
    "EXTERNAL_SERVICE_FORBIDDEN");
    Require(!set.Services.Any(x => x.SelectsRole("worker")),
    "WORKER_SERVICE_FORBIDDEN");
    Require(set.Jobs.Count(x => x.Role == "migrator") == 1,
    "MIGRATION_JOB_CARDINALITY");
}
```

NO BROAD GENERATOR

The renderer cannot turn any project folder into a Deployment. The reviewed topology contract is the source of the allowed object inventory.

BOUNDED SHUTDOWN

SIGTERM consumes an observed drain budget

EXECUTABLE ASSET 59-V - TERMINATION BUDGET JOIN

```
public sealed record TerminationPlan(
    string Workload,
    TimeSpan StopAcceptingBudget,
    TimeSpan InFlightDrainBudget,
    TimeSpan EvidenceFlushBudget,
    TimeSpan SafetyMargin,
    int TerminationGracePeriodSeconds,
    string ShutdownProbeReceiptSha256);

static void Admit(TerminationPlan x)
{
    TimeSpan required = x.StopAcceptingBudget + x.InFlightDrainBudget
        + x.EvidenceFlushBudget + x.SafetyMargin;
    Require(TimeSpan.FromSeconds(x.TerminationGracePeriodSeconds) >= required,
        "TERMINATION_GRACE_TOO_SMALL");
    Require(IsSha256(x.ShutdownProbeReceiptSha256),
        "SHUTDOWN_EVIDENCE_MISSING");
}
```

NO SLEEP HOOK

The manifest does not add a preStop sleep. ASP.NET Core already receives termination and executes its bounded StopAsync path. A preStop action is added only for a proved coordination need, because it consumes the same grace period.

MIGRATION PACKAGING GATE

A Job cannot execute a loose CI artifact by implication

MIGRATION ARTIFACT BRIDGE 59-M

Required subject	Chapter source	Admission
migration code + model snapshot	Chapter 16/58	artifact digest retained
generated SQL/bundle	Chapter 58	migration artifact SHA-256
runnable migrator OCI	delivery graph extension	image receipt + SBOM + provenance
migration database principal	Chapter 14/58	Secret key reference exists
serialized provider lock	Chapter 58	component test receipt

FAIL-CLOSED DECISION

If alarm-migrator has no admitted OCI digest, Chapter 59 emits KUBE_MIGRATOR_IMAGE_MISSING and no Job. Copying a bundle at apply time or downloading it from an unbound URL would break the Chapter 57 subject chain.

CONTROLLED MIGRATION JOB

One bounded execution holds schema authority

EXECUTABLE ASSET 59-W - ALARM MIGRATION JOB

```
apiVersion: batch/v1
kind: Job
metadata:
  name: alarm-migrate-@@ROLLOUT_EPOCH@@
  namespace: nexus1-phase0
  annotations:
    nexus1.io/plan-id: "@@PLAN_ID@"
    nexus1.io/migration-artifact-sha256: "@@MIGRATION_SHA256@"
spec:
  completions: 1
  parallelism: 1
  backoffLimit: 0
  activeDeadlineSeconds: 300
  ttlSecondsAfterFinished: 86400
  template:
    metadata:
      labels: { nexus1.io/role: migrator, nexus1.io/owner: alarm-management }
    spec:
      serviceAccountName: alarm-migrator
      automountServiceAccountToken: false
      restartPolicy: Never
      containers:
        - name: migrator
          image: "@@ALARM_MIGRATOR_IMAGE@"
          args: ["--plan", "@@PLAN_ID@", "--expected-head", "@@MIGRATION_HEAD@"]
```

RETRY BOUNDARY

backoffLimit: 0 makes orchestration failure visible, but the provider lock and idempotent migration semantics remain mandatory. A Job controller is not exactly-once execution.

MIGRATION SEQUENCE

Job completion is necessary but not sufficient

EXECUTABLE ASSET 59-X - MIGRATION ADMISSION

```
public async Task<MigrationStageReceipt> Expand(Cancellation token ct)
{
    RenderedJob job = renderer.RequireMigratorImage();
    ServerDryRunReceipt dryRun = await cluster.DryRun(job, ct);
    Require(dryRun.Admitted, "MIGRATION_JOB_NOT_ADMITTED");

    ObjectIdentity applied = await cluster.Apply(job, ct);
    JobCondition condition = await cluster.WaitForJob(
        applied.Uid, TimeSpan.FromMinutes(6), ct);
    Require(condition == JobCondition.Complete, "MIGRATION_JOB_FAILED");

    MigrationReceipt receipt = await evidence.ReadMigrationReceipt(
        job.PlanId, ct);
    Chapter58MigrationAdmission.Admit(receipt, rollout);
    return MigrationStageReceipt.Join(applied, condition, receipt);
}
```

NO INITCONTAINER

The API and worker Pod templates contain no migration initContainer. Replica count, rescheduling and rollout must never multiply schema authority.

NETWORK REACHABILITY

ClusterIP is discovery; policy is a separate capability

EXECUTABLE ASSET 59-Y - BFF TO ALARM API INGRESS

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: alarm-api-ingress
  namespace: nexus1-phase0
spec:
  podSelector:
    matchLabels: { nexus1.io/workload: alarm-api }
  policyTypes: [Ingress]
  ingress:
    - from:
        - podSelector:
            matchLabels: { nexus1.io/workload: query-bff }
      ports:
        - { protocol: TCP, port: 8080 }
```

ENFORCEMENT QUALIFICATION

A NetworkPolicy object has effect only when the cluster networking implementation enforces it. The capability profile records a deliberate deny/allow probe; otherwise the policy remains declarative intent and the rollout is rejected for profiles that require it.

DATA OWNERSHIP REMAINS CREDENTIAL-BASED

Network policy narrows reachability but does not create ownership

DATABASE BOUNDARY 59-N

Workload	Mounted key	Database permission
alarm-api/worker	alarm-db-dsn	AlarmManagement runtime read/write
alarm-migrator	alarm-db-migration-dsn	AlarmManagement DDL/history
rootcause-api/worker	rootcause-db-dsn	RootCause runtime read/write
projection-worker	projection-db-write-dsn	Projection write
projection-api	projection-db-read-dsn	Projection read only
query-bff	none	no database login

THREE CONTROLS

Object inventory, network reachability and database grants are checked independently. Passing any one cannot fill missing evidence in another.

ROLLINGUPDATE MAPPING

Deployment mechanics implement only the replica replacement step

ROLLOUT RESPONSIBILITY SPLIT 59-O

Concern	Kubernetes	NEXUS-1 controller
replace Pods	Deployment/ReplicaSet	selects admitted digest
keep ready capacity	maxUnavailable/maxSurge	chooses budget from evidence
wait for availability	conditions/rollout status	pins generation and deadline
consumer before producer	not cross-Deployment semantics	orders capability stages
database expansion	Job lifecycle	admits migration receipt
business safety	not known	application observation gates
rollback safety	revision history only	Chapter 58 RollbackDecision

IMPORTANT DISTINCTION

A successful RollingUpdate says the Deployment controller reached its desired state. It does not say V2 consumers were deployed first, a backfill reconciled, or old binaries can read current rows.

SERVER-SIDE PREFLIGHT

Admission is tested without persisting objects

EXECUTABLE ASSET 59-Z - PREFLIGHT COMMANDS

```
kubectl version --output=json \
> artifacts/kubernetes/<plan-id>/cluster-version.json

kubectl apply --server-side \
--dry-run=server \
--field-manager=nexus1-rollout \
--namespace=nexus1-phase0 \
--filename=artifacts/kubernetes/<plan-id>/objects.yaml \
--output=yaml \
> artifacts/kubernetes/<plan-id>/server-admitted.yaml

kubectl diff --server-side \
--field-manager=nexus1-rollout \
--filename=artifacts/kubernetes/<plan-id>/objects.yaml \
> artifacts/kubernetes/<plan-id>/server-diff.txt
```

DRY-RUN BOUNDARY

Server dry-run exercises target API defaulting and admission but does not schedule, pull images or run probes. kubectl diff is retained as a reviewed plan; its non-zero 'differences found' outcome is normalized separately from execution failure.

CONTROLLED ROLLOUT DRIVER

Object generation and application stage are pinned

EXECUTABLE ASSET 59-AA - ONE STAGE

```
public async Task<KubernetesStageReceipt> ApplyStage(
    KubernetesStage stage, RolloutPlan rollout, CancellationToken ct)
{
    RenderedObjectSet objects = manifests.ForStage(stage);
    ServerDryRunReceipt admission = await cluster.DryRun(objects, ct);
    Require(admission.ObjectSetSha256 == objects.Digest,
        "SERVER_DRY_RUN_SUBJECT_MISMATCH");

    AppliedObject[] applied = await cluster.ApplyServerSide(
        objects, fieldManager: "nexus1-rollout", ct);
    Require(applied.All(x => x.GenerationObserved), "GENERATION_NOT_OBSERVED");

    WorkloadReceipt[] workloads = await cluster.WaitForExactWorkloads(
        objects.Workloads, rollout.ObservationWindow, ct);
    Require(workloads.All(x => x.ImageIdMatchesLock), "RUNTIME_IMAGE_MISMATCH");
    return KubernetesStageReceipt.Join(stage, admission, applied, workloads);
}
```

NO LATEST-REVISION RACE

The driver watches each applied object UID and generation. A second writer that advances a Deployment makes the receipt CONCURRENT_DEPLOYMENT_CHANGE instead of silently watching a different latest revision.

CROSS-WORKLOAD ORDER

The Chapter 58 stages remain the authority

KUBERNETES EXECUTION ORDER 59-P

Stage	Objects	Required external receipt
Foundation	Namespace, accounts, config refs, policies	cluster profile admitted
Expanded	alarm migration Job	MigrationReceipt S1
ConsumerReady	rootcause/projection workers	V1+V2 capability inventory
CompatibleBinary	alarm worker/API	mixed B0/B1 evidence
ReadPilot	BFF/API template with read cohort	projection equality window
WritePilot	feature plan reference change	dual-representation conservation
FullExposure	same digest, 100% signed plan	Chapter 58 observation gates

NO DEPLOYMENT DEPENDENCY GRAPH

Kubernetes does not supply semantic startup ordering among these Deployments. The rollout driver waits for the application evidence required by one stage before rendering/applying the next.

APPLICATION EVIDENCE AFTER AVAILABILITY

Kubernetes and runtime receipts are joined, not conflated

EXECUTABLE ASSET 59-AB - STAGE EVIDENCE

```

public sealed record KubernetesStageReceipt(
    Guid PlanId,
    long Epoch,
    string ObjectSetSha256,
    string ClusterIdentity,
    IReadOnlyList<ObjectReceipt> Objects,
    IReadOnlyList<WorkloadReceipt> Workloads,
    string ServerDryRunSha256,
    string RuntimeObservationSha256,
    string Chapter58StageReceiptSha256,
    StageVerdict Verdict,
    string ReceiptSha256);

static void Admit(KubernetesStageReceipt x, RolloutPlan plan)
{
    Require(x.PlanId == plan.PlanId && x.Epoch == plan.Epoch,
        "KUBE_STAGE_PLAN_MISMATCH");
    Require(x.Workloads.All(w => w.Ready && w.ImageIdMatchesLock),
        "KUBE_WORKLOAD_NOT_ADMITTED");
    Require(IsSha256(x.RuntimeObservationSha256),
        "APPLICATION_OBSERVATION_MISSING");
    Require(x.Verdict == StageVerdict.Admitted, "KUBE_STAGE_REJECTED");
}

```

ROLLBACK MAPPING

Kubernetes history is not a rollback decision

EXECUTABLE ASSET 59-AC - ROLLBACK ADMISSION

```

public async Task<KubernetesRollbackReceipt> Rollback(
    RollbackDecision decision, CancellationToken ct)
{
    Require(decision.Action is RollbackAction.ReplaceBinaryOnly
        or RollbackAction.DisableFeatureThenReplace, "KUBE_ROLLBACK_UNSAFE");
    Require(imageLock.Contains(decision.TargetImageDigest),
        "ROLLBACK_DIGEST_NOT_ADMITTED");

    if (decision.Action == RollbackAction.DisableFeatureThenReplace)
        await featurePlans.ApplySignedKillPlan(decision.PlanId, ct);

    RenderedObjectSet target = renderer.ForRollbackDecision(decision);
    KubernetesStageReceipt replaced = await ApplyStage(
        KubernetesStage.Rollback, rollout, ct);
    RuntimeObservation runtime = await observations.VerifyRollback(decision, ct);
    return KubernetesRollbackReceipt.Join(decision, replaced, runtime);
}

```

FORBIDDEN CONVENIENCE

The runbook does not use kubectl rollout undo as independent authority. It may select a historical Pod template, but it cannot decide whether current rows, events and effects remain compatible with that binary.

CONFIGURATION ROLLOUT

A new identity causes a deliberate Pod-template revision

EXECUTABLE ASSET 59-AD - IMMUTABLE CONFIG TRANSITION

```
public RenderedObjectSet ChangeConfiguration(
    ConfigurationIdentity oldIdentity,
    ConfigurationIdentity next,
    SignedConfigurationPlan plan)
{
    Require(next.Digest == plan.ConfigurationSha256,
        "CONFIGURATION_PLAN_MISMATCH");
    Require(next.ProviderManifestSha256 == plan.ProviderManifestSha256,
        "CONFIGURATION_PROVIDER_DRIFT");

    string name = $"nexus1-runtime-{next.Digest[..12]}";
    ConfigMap map = templates.ConfigMap.WithName(name).With(next.SafeValues);
    Deployment[] workloads = templates.Workloads
        .Select(x => x.ReferenceConfig(name, next.Digest)).ToArray();
    return RenderedObjectSet.Create(map, workloads, plan);
}
```

RELOAD BOUNDARY

The chapter chooses revision-by-identity for this package. Mounted ConfigMap update timing is not used as an atomic rollout mechanism, and a process never sees half an admitted configuration plan.

SECRET ROTATION BOUNDARY

Secret change and application reload are separate evidence

ROTATION SEQUENCE 59-Q

Step	Required proof
create new value outside repository	secret authority receipt; no value in evidence
create/version Secret object	name, key set, metadata digest only
preflight new credential	own database/broker only
render new reference	Pod template digest changes
roll compatible Pods	readiness and authentication pass
revoke old credential	no old Pod/reference remains

NO HIDDEN HOT RELOAD

A mounted Secret may update, but the application and provider determine when it is observed. NEXUS-1 uses an explicit versioned reference and rollout unless a tested atomic reload path exists.

SCALING BOUNDARY

Replica changes require message and data evidence

WHY HPA REMAINS ABSENT 59-R

Workload	Missing proof before automation
HTTP APIs	metric, target, latency/error coupling, database capacity
workers	partition/order semantics, prefetch, lease and duplicate bounds
BFF	downstream concurrency and timeout budget
projection worker	single-writer/order constraint and rebuild behavior

EXECUTABLE ASSET 59-AE - REPLICA CHANGE RECEIPT

```
public sealed record ReplicaPlan(  
    string Workload,  
    int From,  
    int To,  
    string CapacityModelSha256,  
    string ConcurrencySafetyReceiptSha256,  
    string DatabaseBudgetSha256,  
    string ApprovedBy,  
    string Signature);
```

MANUAL IS NOT UNGOVERNED

The educational profile may set fixed replica counts. Changing them still creates a signed plan and reruns topology, capacity and conservation gates.

STATIC MANIFEST TESTS

Policy violations fail before cluster contact

EXECUTABLE ASSET 59-AF - OBJECT POLICY TESTS

```
[Fact]  
public void Generated_package_preserves_runtime_contract()  
{  
    KubernetesObjectSet set = fixture.Render(Fixed.AdmittedInput);  
    set.AssertExactDeploymentInventory(Topology.SevenRoles);  
    set.AssertExactServices(Topology.FourHttpRoles);  
    set.AssertAllImagesPinnedByDigest();  
    set.AssertNoSecretValues();  
    set.AssertNoExternalService();  
    set.AssertNoMigrationInitContainer();  
    set.AssertOneMigrationJob();  
    set.AssertDistinctStartupLiveReadyProbes();  
    set.AssertResourceBudgetReferences();  
    set.AssertTerminationBudgets();  
}
```

PARSER RULE

Tests parse Kubernetes objects and inspect typed fields. They do not grep indentation or assume document order. The canonical object-set digest is independent of YAML key order.

SERVER ADMISSION TESTS

The target API and admission chain judge the rendered package

EXECUTABLE ASSET 59-AG - ADMISSION FIXTURE

```
[Collection("kubernetes-admission")]
public sealed class KubernetesAdmissionTests(ClusterLease cluster)
{
    [Fact]
    public async Task Package_is_admitted_without_persistence()
    {
        RenderedObjectSet set = fixture.Render(Fixed.AdmittedInput);
        ServerDryRunReceipt receipt = await cluster.ServerSideDryRun(set);
        receipt.RejectedObjects.Should().BeEmpty();
        receipt.AdmittedObjectCount.Should().Be(set.Count);
        receipt.ObjectSetSha256.Should().Be(set.Digest);
    }

    [Fact]
    public async Task Missing_required_Secret_is_visible_before_rollout()
    {
        PreflightReceipt receipt = await cluster.PreflightSecretReferences(
            fixture.Render(Fixed.MissingAlarmSecret));
        receipt.Findings.Should().ContainCode("SECRET_REFERENCE_UNRESOLVED");
    }
}
```

SERVER DRY-RUN BOUNDARY

Admission can validate policy and schema without persistence, but Secret existence and authorization may require separate read-only preflight. Neither test pulls an image.

RUNTIME KUBERNETES TESTS

A disposable cluster repeats the proved control flow

EXECUTABLE ASSET 59-AH - RUNTIME SCENARIO

```
[Fact]
public async Task Digest_pinned_slice_reaches_delayed_projection()
{
    await using KubernetesLab lab = await KubernetesLab.Start(
        profile: Profiles.EducationalPhase0,
        package: Fixed.RenderedPackage);

    await lab.ApplyFoundation();
    await lab.RunMigrationsAndAdmitReceipts();
    await lab.DeployConsumersBeforeProducers();
    await lab.DeployApisAndBff();
    await lab.WaitForApplicationReadiness(Bounds.Ready);

    ControlReceipt control = await lab.Bff.Submit(ControlAlarm.Fixed);
    RuntimeObservation result = await lab.AwaitProjection(
        control.AlarmId, Bounds.EndToEnd);
    result.AssertOneSemanticFactPerTransition();
    result.AssertTraceConnected();
    result.AssertEveryPodImageMatchesLock();
}
```

NOT PRODUCTION PROOF

A disposable-cluster result proves this package under the recorded cluster/profile/load. It does not prove managed-cluster security, failure domains, capacity, backup or operations.

DELIBERATELY BROKEN PACKAGE

The verifier rejects plausible Kubernetes green

FAILURE CAMPAIGNS 59-S

Mutation	Required rejection
replace digest with tag	MUTABLE_IMAGE_REFERENCE
add LoadBalancer to alarm-api	EXTERNAL_SERVICE_FORBIDDEN
put DSN in ConfigMap	CONFIDENTIAL_KEY_IN_CONFIGMAP
remove startupProbe	PROBE_CONTRACT_INCOMPLETE
readiness points to /health/live	READINESS_PATH_WRONG
add migration initContainer	MIGRATION_AUTHORITY_MULTIPLIED
worker receives Service	WORKER_SERVICE_FORBIDDEN
resource token unresolved	MANIFEST_TOKEN_SURVIVED
second controller changes generation	CONCURRENT_DEPLOYMENT_CHANGE
undo after V2-only fact	KUBE_ROLLBACK_UNSAFE

KUBERNETES EVIDENCE MANIFEST

One bundle joins desired objects, observed cluster and runtime

EXECUTABLE ASSET 59-AI - DEPLOYMENT EVIDENCE

```
public sealed record KubernetesEvidenceManifest(
    Guid PlanId,
    string CandidateManifestSha256,
    string RolloutPlanSha256,
    string ClusterCapabilityProfileSha256,
    string KubernetesImageLockSha256,
    string ResourceBudgetSha256,
    string RenderedObjectSetSha256,
    string ServerDryRunReceiptSha256,
    string ServerDiffSha256,
    string MigrationStageReceiptSha256,
    IReadOnlyList<string> KubernetesStageReceiptSha256s,
    string RuntimeObservationSha256,
    string FailureCampaignSha256,
    EvidenceVerdict Verdict,
    string ManifestSha256);
```

EVIDENCE LOCATION

Kubernetes object status is transient operational state, not the evidence archive. Receipts retain object UUIDs/generations, conditions, imageIDs and value-free diagnostics outside the cluster with the plan identity.

EVIDENCE ADMISSION

Cross-plan or stale receipts cannot complete the bundle

EXECUTABLE ASSET 59-AJ - FAIL-CLOSED VALIDATOR

```
public static void Admit(KubernetesEvidenceManifest m,
    KubernetesRenderInput input, IEnumerable<KubernetesStageReceipt> stages)
{
    Require(m.PlanId == input.PlanId, "KUBE_EVIDENCE_PLAN_MISMATCH");
    Require(m.CandidateManifestSha256 == input.CandidateManifestSha256,
        "KUBE_EVIDENCE_CANDIDATE_MISMATCH");
    Require(m.ResourceBudgetSha256 == input.ResourceBudgetSha256,
        "KUBE_EVIDENCE_RESOURCE_MISMATCH");
    Require(stages.All(x => x.PlanId == input.PlanId), "CROSS_PLAN_STAGE_RECEIPT");
    Require(stages.Select(x => x.Epoch).Distinct().Single() == input.RolloutEpoch,
        "CROSS_EPOCH_STAGE_RECEIPT");
    Require(stages.All(x => x.Verdict == StageVerdict.Admitted),
        "KUBE_STAGE_NOT_ADMITTED");
    Require(IsSha256(m.RuntimeObservationSha256), "KUBE_RUNTIME_EVIDENCE_MISSING");
    Require(IsSha256(m.FailureCampaignSha256), "KUBE_FAILURE_CAMPAIGN_MISSING");
}
```

SOURCE QUALIFICATION LEDGER

Kubernetes mechanisms receive local engineering constraints

PRIMARY-SOURCE MAPPING 59-T

Official mechanism	Used for	NEXUS-1 qualification
Deployment	declarative ReplicaSet replacement	cross-workload order remains external
Service	stable HTTP discovery	not authorization or public ingress
startup/live/ready probes	restart and endpoint effects	endpoint meanings come from Ch11
requests/limits	scheduling and enforcement	values require Ch53 receipts
Job	bounded one-shot workload	provider lock/idempotency remain mandatory
ConfigMap/Secret	nonsecret config / confidential reference	no secret platform claim
NetworkPolicy	L3/L4 flow intent	CNI enforcement must be proved

FRESHNESS

The mechanism review reflects official Kubernetes documentation checked on 10 August 2026. The repository must pin the actual client/server profile and retain generated API/defaulting output.

ANTI-PATTERN CATALOGUE

YAML shortcuts that falsify the architecture

REJECTED PATTERNS 59-U

Anti-pattern	Why rejected
one Pod with API, worker and migrator	lifecycle and schema authority collapse
one shared Secret for all services	database/broker ownership cannot be enforced
LoadBalancer on every API	BFF bypass and new public surface
liveness checks SQL or broker	dependency outage becomes restart storm
startup sleeps	machine timing replaces readiness evidence
ConfigMap values edited in place	configuration identity and atomic rollout disappear
HPA before concurrency proof	replication may break ordering and capacity
kubectl status as domain proof	platform health replaces conservation evidence
rollout undo as data rollback	durable truth is not reverted
Helm/mesh/operator by default	tool catalogue replaces minimal proved needs

COMPLETION MATRIX

What Chapter 59 now makes reviewable

COMPLETION MATRIX 59-V

Concern	Asset	Status
topology translation	seven Deployments + four Services	SPECIFIED
identity	candidate/role-image/render locks	SPECIFIED
configuration/secrets	immutable safe config + references	SPECIFIED
health/shutdown	three probes + termination budget	SPECIFIED
resources	receipt-bound request/limit tokens	SPECIFIED
migration	packaging gate + serialized Job	SPECIFIED; IMAGE PENDING
rollout/rollback	stage driver + Ch58 decision join	SPECIFIED
runtime Kubernetes proof	cluster/admission/runtime receipts	PENDING

RESIDUAL OBLIGATIONS

The repository still needs admitted per-role OCI digests including the migrator, a pinned cluster profile, actual Secret provisioning, measured resource/probe budgets, CNI enforcement proof, server-admission output and retained runtime/failure receipts.

HANDOFF TO CHAPTER 60

The final demonstrator must join every evidence plane

BOUNDARY CARRIED FORWARD 59-W

Chapter 59 leaves	Chapter 60 must prove
digest-bound Kubernetes object set	actual admitted workload inventory
migration and rollout stage receipts	complete operator-to-report execution
stable BFF and service discovery	HTTP/event path and authorization
database/broker Secret boundaries	data ownership and public-fact conservation
three probes and resource budgets	fresh health/capacity observations
cluster and application evidence joins	final catalogues and verdict
explicit Kubernetes non-goals	which candidates still remain modules

NEXT INVARIANT

The final demonstrator is credible only when one admitted operator command can be traced through ownership, contracts, messages, persistence, telemetry, failure and recovery to delayed reporting truth, while every missing, stale or cross-run evidence item fails the final verdict closed.

FINAL CHAPTER RULE

YAML may carry proof identities; it cannot manufacture proof

NEXUS-1 admits a Kubernetes package only when every object traces to a proved runtime responsibility, every image and configuration is immutable and joined to one plan, platform conditions are combined with application evidence, and rollback remains governed by current durable truth.

PART X PROGRESS

Chapter 57 made delivery identity explicit, Chapter 58 made coexistence and rollback reviewable, and Chapter 59 projected those needs into minimal Kubernetes mechanisms. Chapter 60 must now judge the complete distributed demonstrator as one evidence chain.

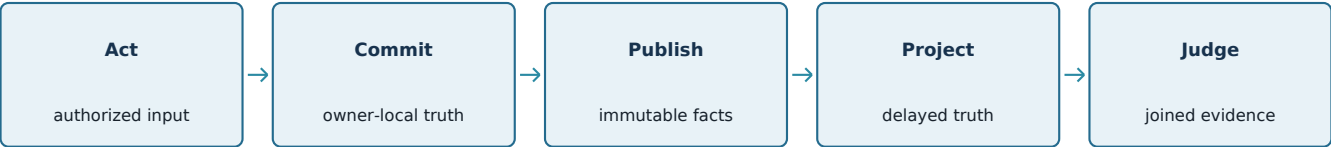
PART X - MIGRATION, DELIVERY, AND FINAL ARCHITECTURE

60. The Final NEXUS-1 Distributed Demonstrator

The final chapter does not celebrate a diagram. It asks whether one admitted operator action can be followed, without invented certainty, through authorization, owner-local commits, immutable public facts, delayed projection, telemetry, failure and recovery to one fresh reporting observation.

CHAPTER DECISION
Define one reproducible final scenario and one fail-closed verdict. Join the exact service, HTTP, event, data-ownership, security, observability, failure, deployment and evidence catalogues to the same RunId, build identity and golden coordinates. Report SPECIFIED | RUNTIME PENDING until real repository, provider, broker, trace, deployment and recovery artifacts exist.

PROOF PATH 60-A - ONE SCENARIO, MANY EVIDENCE PLANES



FINAL CLAIM LADDER 60-A

Claim	Minimum proof	Current manuscript status
architecture is coherent	catalogues and invariants agree	SPECIFIED
implementation exists	clean build + discovered tests + artifact identities	PENDING
scenario converges	provider/broker/HTTP/trace receipts	PENDING
failure recovery holds	named campaigns and post-recovery conservation	PENDING
production ready	operations, safety, capacity, HA, backup and governance	NOT CLAIMED

EVIDENCE VERDICT
SPECIFIED | RUNTIME PENDING. This chapter completes the engineering contract and the honest decision surface. It does not fabricate a successful run.

ORDINARY DEVELOPER PROBLEM

Every component is green, but nobody can prove the story

A dashboard shows healthy processes, a queue depth of zero and one reporting row. None of those facts alone establishes that the row came from the authorized operator action, that duplicates caused no second effect, that every database owner stayed inside its boundary, or that the observation is fresh. The final demonstrator therefore judges coordinates and conserved effects, not screenshots.

PLAUSIBLE BUT INSUFFICIENT EVIDENCE 60-B

Observation	What it proves	What remains unknown
HTTP 202	Alarm accepted a local outcome	broker or downstream progress
queue empty	no ready messages at one instant	loss, acknowledgement or dead-letter state
Pods ready	declared readiness currently passed	domain flow and ownership
reporting row exists	some projection state exists	lineage, contiguity and freshness
trace is visible	one sampled execution was exported	database conservation and unsampled paths

PLAIN-LANGUAGE PRINCIPLE

The final result is not one green light. It is a vector of independently grounded observations that agree on one run and one causal story.

DECISION RECORD

ADR-060 - final acceptance is an evidence join

DECISION CONSEQUENCES 60-C

Choice	Benefit	Cost / boundary
one immutable RunId	cross-plane evidence cannot be mixed silently	every producer must carry it
fixed golden coordinates	failures are diagnosable and repeatable	not a production traffic model
owner-local checkpoints	partial progress remains truthful	no global completion transaction
exact public bytes	contract drift becomes observable	canonical corpus is governed
freshness-bearing final read	delayed truth is not mislabeled	BFF must expose uncertainty
fail-closed verdict	missing evidence cannot become Passed	incomplete runs remain visible

NON-GOAL

This is a Phase-0 educational demonstrator. It is not a certified industrial control, safety, compliance, security or production operations platform.

SCOPE CONTRACT

One action begins at the Alarm edge and ends at Reporting

SCENARIO BOUNDARY 60-D

Inside the final proof	Outside the final proof
five authorized alarm observations	physical detector correctness
Alarm flood decision and outbox	real plant actuation
RabbitMQ delivery and redelivery	broker HA administration
RootCause case and verdict	scientific truth of diagnosis
Reporting stream positions 1 and 2	synchronous global truth
BFF current-as-observed response	production UI usability certification
provider/broker/crash fault campaigns	all cloud or regional failures

SCENARIO WORDING

Operator action means an authenticated call through the published Alarm HTTP contract. Final observation means a freshness-bearing Reporting projection read through the BFF. Neither phrase implies direct database access.

GOLDEN RUN IDENTITY

Every artifact belongs to one immutable execution

EXECUTABLE ASSET 60-A - FINAL RUN IDENTITY

```
public sealed record FinalRunIdentity(  
    Guid RunId,  
    string BuildIdentity,  
    string CandidateManifestSha256,  
    string DeploymentManifestSha256,  
    string ConfigurationIdentity,  
    string GoldenCorpusSha256,  
    string EnvironmentIdentity,  
    DateTimeOffset StartedAtUtc,  
    string Signature);  
  
static void Admit(FinalRunIdentity x)  
{  
    Require(x.RunId != Guid.Empty, "FINAL_RUN_ID_MISSING");  
    Require(IsSha256(x.BuildIdentity), "FINAL_BUILD_ID_INVALID");  
    Require(IsSha256(x.CandidateManifestSha256), "FINAL_CANDIDATE_INVALID");  
    Require(IsSha256(x.GoldenCorpusSha256), "FINAL_CORPUS_INVALID");  
    Require(VerifySignature(x), "FINAL_RUN_SIGNATURE_INVALID");  
}
```

IDENTITY BOUNDARY

RunId joins evidence; CorrelationId joins business work; trace_id joins one observed execution. They are related coordinates, not interchangeable identifiers.

IMMUTABLE RUN PLAN

The scenario is planned before the first request

EXECUTABLE ASSET 60-B - PLAN SHAPE

```
public sealed record FinalScenarioPlan(
    FinalRunIdentity Identity,
    string Profile,
    string EntryContract,
    string FinalReadContract,
    IReadOnlyList<ExpectedCheckpoint> Checkpoints,
    IReadOnlyList<RequiredCampaign> FailureCampaigns,
    TimeSpan EndToEndBound,
    TimeSpan ProjectionFreshnessBound,
    string PlanSha256);

static void Admit(FinalScenarioPlan p)
{
    Require(p.Profile == "Phase0Core", "FINAL_PROFILE_UNEXPECTED");
    Require(p.Checkpoints.Select(x => x.Id).Distinct().Count()
        == p.Checkpoints.Count, "FINAL_CHECKPOINT_DUPLICATE");
    Require(p.FailureCampaigns.Count > 0, "FINAL_FAILURE_SET_EMPTY");
    Require(p.PlanSha256 == Sha256.Canonical(p with { PlanSha256 = "" }),
        "FINAL_PLAN_DIGEST_MISMATCH");
}
```

NO MID-RUN EDITING

Changing a timeout, expected digest, enabled subscriber or failure campaign creates a new plan and RunId. A failing run is never repaired by mutating its acceptance criteria.

PREFLIGHT

Known clean state is proved, not assumed

EXECUTABLE ASSET 60-C - PREFLIGHT GATE

```
await preflight.RequireAsync(
    run.Identity,
    BuildState.ExactCandidate(),
    DeploymentState.ExactAdmittedInventory(),
    MigrationState.ExactSetAndFingerprint(),
    DatabaseState.GoldenCoordinatesAbsent(),
    BrokerState.TopologyExactAndGoldenQueuesEmpty(),
    ConfigurationState.ExactIdentity(),
    SecurityState.PrincipalsAndAclInventoryFresh(),
    TelemetryState.ControlsSignalsFresh(),
    EvidenceState.EmptyWritableAndRunScoped(),
    ct);
```

PREFLIGHT FAILURES 60-E

Finding	Required result
golden business row already present	reject; do not delete silently
ready or unacked golden message	reject and retain broker snapshot
candidate/deployment digest mismatch	reject as cross-build state
migration fingerprint drift	reject before traffic
telemetry control stale	reject observability claim

FROM SERVICES TO RUNTIME

AUTHENTICATED PRODUCTION ENTRY

Five observations enter the real Alarm HTTP contract

EXECUTABLE ASSET 60-D - OPERATOR ACTION

```
using HttpClient client = alarmHost.CreateAuthorizedClient(
    capability: "alarm.observation.submit",
    detectorScope: "site-ath-17/line-02",
    runId: plan.Identity.RunId);

foreach (AlarmObservationV1 request in Golden.AlarmObservations)
{
    using var message = HttpContracts.PostJson(
        "/api/v1/alarm-observations", request);
    message.Headers.Add("Idempotency-Key", request.IdempotencyKey);
    message.Headers.Add("traceparent", request.TraceParent);
    HttpResponseMessage response = await client.SendAsync(message, ct);
    response.StatusCode.Should().Be(HttpStatusCode.Accepted);
}
```

GOLDEN INPUT COORDINATES 60-F

Coordinate	Value
CorrelationId	request-am-20482
scope	site-ath-17 / line-02
fifth request idempotency key	idem-am-7f9c2a
accepted observation	alarm-am-9005
threshold / flood decisions	decision-am-9184 / flood-am-6031

AUTHORIZATION AND REPLAY

Identity, scope and bytes are checked before business work

HTTP ADMISSION MATRIX 60-G

Input	Result	Durable effect
valid capability + scope + new key	202 Accepted	one accepted outcome
same key + same canonical request	stored 202 outcome	no second decision
same key + changed body	409 identity conflict	none
capability absent	403 problem response	none
scope mismatch	403 problem response	none
malformed contract	400 problem response	none

EXECUTABLE ASSET 60-E - EDGE ORACLE

```
await alarmDb.AssertIdempotencyAsync("idem-am-7f9c2a", x => x
    .HasCanonicalRequestSha256(Golden.FifthRequestSha256)
    .HasStoredStatus(202)
    .HasOneBusinessDecision("decision-am-9184"));

await auditTrail.AssertRejectedAsync(Golden.UnauthorizedAttempt,
    code: "CAPABILITY_DENIED", secretValuesPresent: false);
```

TX-A - ALARM OWNERSHIP

The local commit justifies the HTTP response

EXECUTABLE ASSET 60-F - ALARM TRANSACTION ORACLE

```
await alarmDb.AssertTransactionAsync("TX-A", tx => tx
    .HasAcceptedObservation("alarm-am-9005")
    .HasFloodDecision("flood-am-6031", "decision-am-9184")
    .HasPendingOutbox(
        "019bc4d4-5dc1-7b8d-9d48-542d25315356")
    .HasCanonicalBodySha256(
        "7cf98319c20a5915d5710bbcb5878fe51cdae89a5b68081c135fd0e962c276a")
    .HasNoPublishedTimestamp());

await foreignStores.AssertGoldenRowsAbsentAsync(
    "RootCauseDb", "ReportingDb", "AuditDb", "ComplianceDb");
```

ALLOWED CLAIM

AlarmManagement has durably accepted the observation, made one deterministic flood decision and stored one exact publication intent. Broker acceptance and downstream effects remain unproved at this checkpoint.

BROKER OUTAGE AT THE SEAM

A 202 response survives loss of transport availability

EXECUTABLE ASSET 60-G - OUTBOX OUTAGE CAMPAIGN

```
await broker.StopAsync(ct);
HttpResponseMessage accepted = await operatorClient.SubmitAsync(
    Golden.FifthObservation, ct);
accepted.StatusCode.Should().Be(HttpStatusCode.Accepted);

await alarmDb.AssertOutboxAsync(Golden.FloodMessageId,
    state: "Pending", dispatchCount: 0);
await rootCauseDb.AssertCaseCountAsync(Golden.CaseId, 0);

await broker.StartAsync(ct);
await alarmWorker.WaitForDispatchAsync(Golden.FloodMessageId, ct);
await brokerCapture.AssertOneCanonicalBodyAsync(
    Golden.FloodMessageId, Golden.FloodBodySha256, ct);
```

RECOVERY RULE

Recovery publishes the stored bytes. It does not rebuild the event from current rows or ask the operator to resubmit the command.

FIRST PUBLIC FACT

AlarmFloodDetected.v1 is admitted as captured bytes

PUBLIC FACT IDENTITY 60-H

Field	Golden value
MessageId	019bc4d4-5dc1-7b8d-9d48-542d25315356
event type	nexus1.alarm-management.alarm-flood-detected.v1
routing key	alarm-management.alarm-flood-detected.v1
canonical length	1003 UTF-8 bytes
body SHA-256	7cf98319c20a5915d5710bbcb5878fe51cdae89a5b68081c135fd0e962c276a
correlation / causation	request-am-20482 / decision-am-9184

EXECUTABLE ASSET 60-H - TRANSPORT CAPTURE

```
CapturedDelivery flood = await brokerCapture.RequiredAsync(
    Golden.FloodMessageId, ct);
flood.Body.Length.Should().Be(1003);
Sha256.Hex(flood.Body).Should().Be(Golden.FloodBodySha256);
flood.Properties.MessageId.Should().Be(Golden.FloodMessageId);
flood.Redelivered.Should().BeFalse();
```

TX-RC-O - ROOTCAUSE OPENS

The consumer owns its inbox, case and next intent

EXECUTABLE ASSET 60-I - ROOTCAUSE OPEN ORACLE

```
await rootCauseDb.AssertTransactionAsync("TX-RC-O", tx => tx
    .HasInboxReceipt(Golden.FloodMessageId)
    .HasCase(Golden.CaseId, sourceDecisionId: "flood-am-6031")
    .HasPendingOutbox(Golden.CaseOpenedMessageId)
    .HasCaseStreamPosition(1));

await alarmDb.AssertUnchangedSinceAsync("TX-A");
await reportingDb.AssertPositionAsync(Golden.CaseStreamId, atMost: 1);
```

ROOTCAUSE OPEN COORDINATES 60-I

Coordinate	Value
Caseld	019bc4d4-6dfe-7bcb-9de8-3af71979285a
CaseOpened MessageId	019bc4d4-6ea0-7d14-a57a-3e8643c55f13
stream / position	root-cause/case/<Caseld> / 1
correlation / causation	request-am-20482 / Flood MessageId

EVIDENCE-BOUND VERDICT

Diagnosis is deterministic over frozen evidence

VERDICT BASIS 60-J

Input	Golden value	Authority
ordered evidence	five admitted evidence keys	RootCause
policy identity	root-cause/verdict/3	configuration policy
verdict code	COMMON_POWER_SUPPLY_INSTABILITY	RootCause
confidence	0.94	bounded decision semantics
VerdictIdentity	sha256:a2451a...fb1734c17	canonical decision basis

EXECUTABLE ASSET 60-J - DETERMINISTIC REPLAY

```
VerdictIdentity first = VerdictPolicy.Identify(Golden.VerdictBasis);
VerdictIdentity replay = VerdictPolicy.Identify(
    Golden.VerdictBasis.ReorderedIntoCanonicalSequence());
first.Should().Be(replay);

Golden.VerdictBasis.WithEvidenceRemoved(1)
    .Invoking(VerdictPolicy.Identify)
    .Should().Throw<EvidenceMembershipRejected>();
```

TX-RC-V - VERDICT OWNERSHIP

Verdict and publication intent commit together

EXECUTABLE ASSET 60-K - VERDICT TRANSACTION ORACLE

```
await rootCauseDb.AssertTransactionAsync("TX-RC-V", tx => tx
    .HasVerdict(Golden.VerdictId, "COMMON_POWER_SUPPLY_INSTABILITY")
    .HasOrderedEvidenceCount(5)
    .HasClosedCase(Golden.CaseId)
    .HasPendingOutbox(Golden.VerdictMessageId)
    .HasCaseStreamPosition(2));

await foreignStores.AssertNoWritesByAsync(
    principal: "rootcause-runtime",
    stores: ["AlarmDb", "ReportingDb", "AuditDb", "ComplianceDb"]);
```

VERDICT COORDINATES 60-K

Coordinate	Value
VerdictId	019bc4d4-81c0-7ea1-82d5-4f9b8719d0a1
Verdict MessageId	019bc4d4-82a4-77f2-ae06-09b172aa59db
causation	decision-rc-7184
stream position	2

SECOND PUBLIC FACT

RootCauseVerdictIssued.v1 preserves direct causation

EXECUTABLE ASSET 60-L - VERDICT BODY ADMISSION

```
CapturedDelivery verdict = await brokerCapture.RequiredAsync(
    "019bc4d4-82a4-77f2-ae06-09b172aa59db", ct);
verdict.Route.Should().Be(
    "root-cause.root-cause-verdict-issued.v1");
verdict.Body.Length.Should().Be(1157);
Sha256.Hex(verdict.Body).Should().Be(
    "192e5e73c5a35d438e2ee9edb0a9e7044293c29bf321ba1abe34f191d8b44c00");
Envelope.Read(verdict.Body).CorrelationId
    .Should().Be("request-am-20482");
Envelope.Read(verdict.Body).CausationId
    .Should().Be("decision-rc-7184");
```

LINEAGE RULE

Correlation remains constant for the business journey. Causation names the direct decision or public fact that caused the next boundary action; it is not flattened to the original HTTP request.

INDEPENDENT FAN-OUT

Subscribers progress without global completion

VERDICT SUBSCRIBER VECTOR 60-L

Subscriber	Owner-local outcome	Blocks final delayed read?
Reporting	inbox + ledger + summary + checkpoint	yes for core scenario
Audit	inbox + immutable evidence append	no in admitted seven-role runtime
Compliance	inbox + Pending review	no in admitted seven-role runtime
RL Advisory	optional private advisory outcome	never

RECONCILIATION

Chapters 34-36 specify AuditDb and ComplianceDb transactions, but Chapters 55 and 59 admit only seven application roles and three databases. Therefore Audit and Compliance are semantically complete subscriber candidates, not members of the final deployable core until their images, migrations, principals, topology and fault receipts are added.

TX-REP - DELAYED TRUTH

Reporting advances only through contiguous stream positions

EXECUTABLE ASSET 60-M - PROJECTION CONVERGENCE

```
await reportingDb.AssertProjectionAsync(
    projection: "root-cause-case-summary",
    generation: activeGeneration,
    streamId: Golden.CaseStreamId,
    expectedPosition: 2,
    expectedMessageId: Golden.VerdictMessageId,
    assertRow: row => row
        .HasStatus("VerdictIssued")
        .HasVerdict(Golden.VerdictId)
        .HasCode("COMMON_POWER_SUPPLY_INSTABILITY")
        .HasConfidence(0.94m));

await reportingDb.AssertGapCountAsync(Golden.CaseStreamId, 0);
```

ORDERING BOUNDARY

The broker may deliver position 2 before position 1. Reporting may buffer the gap, but it cannot publish a contiguous watermark of 2 until both positions are durably admitted in order.

FINAL OPERATOR READ

The BFF returns an observation with freshness evidence

EXECUTABLE ASSET 60-N - FINAL HTTP ORACLE

```
Observed<RootCauseCaseView> result =
    await bff.GetRootCauseCaseAsync(Golden.CaseId, ct);

result.Value.Status.Should().Be("VerdictIssued");
result.Value.VerdictId.Should().Be(Golden.VerdictId);
result.Evidence.LastContiguousPosition.Should().Be(2);
result.Evidence.GapCount.Should().Be(0);
result.Evidence.UpdateStatus.Should().Be("current-as-observed");
result.Evidence.ProjectionAge.Should()
    .BeLessThanOrEqualTo(plan.ProjectionFreshnessBound);
result.Evidence.ObservedAtUtc.Should().NotBe(default);
```

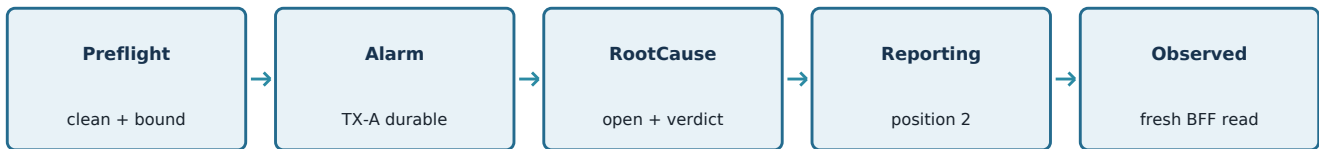
TRUTH WORDING

The source-authoritative facts live in AlarmManagementDb and RootCauseDb. The BFF returns Reporting's delayed observation with its age and watermark; it never claims a distributed snapshot or universally current truth.

SCENARIO STATE MACHINE

Partial progress is a valid, explainable state

STATE MODEL 60-B



TERMINAL AND NONTERMINAL STATES 60-M

State	Meaning	Permitted next action
Pending	valid checkpoint not yet observed	bounded wait / inspect
Failed	explicit invariant or identity violation	stop and retain evidence
Unknown	observation path could not decide	stop; never assume success
Proved	required receipt exists and coordinates agree	advance
Passed	every mandatory checkpoint is Proved	seal final manifest

CONVERGENCE VECTOR

Completion remains decomposable

EXECUTABLE ASSET 60-O - FINAL CHECKPOINTS

```
public sealed record FinalConvergence(
    Checkpoint AlarmCommit,
    Checkpoint FloodPublication,
    Checkpoint RootCauseOpenCommit,
    Checkpoint CaseOpenedPublication,
    Checkpoint VerdictCommit,
    Checkpoint VerdictPublication,
    Checkpoint ReportingPositionTwo,
    Checkpoint BffObservation,
    Checkpoint AuthorizationCampaign,
    Checkpoint RecoveryCampaigns,
    Checkpoint TraceContinuity,
    Checkpoint OwnershipConservation);

public bool IsComplete => Required.All(x => x.State == Proved);
```

NO GLOBAL CALLBACK

Reporting, Audit, Compliance or the BFF never call RootCause to declare the distributed flow complete. The evidence harness observes independent owner-local states.

CAUSALITY CATALOGUE

Business lineage and trace lineage agree without collapsing

GOLDEN CAUSAL CHAIN 60-N

Node	Identity	Direct cause
operator request	request-am-20482	authenticated actor intent
Alarm decision	decision-am-9184	fifth admitted observation
Flood fact	019bc4d4-5dc1-...	decision-am-9184
RootCause open	CaseId / stream position 1	Flood MessageId
Verdict decision	decision-rc-7184	frozen case evidence
Verdict fact	019bc4d4-82a4-...	decision-rc-7184
Reporting position 2	case stream / 2	Verdict MessageId

TRACE CONTEXT

W3C traceparent is propagated at HTTP and messaging boundaries when valid. CorrelationId and CausationId remain explicit domain coordinates even when sampling drops spans or a replay creates a new processing trace.

TRACE CONTINUITY

One exported trace is checked as a graph

EXECUTABLE ASSET 60-P - TRACE GRAPH ORACLE

```
TraceGraph trace = await traceStore.RequireAsync(run.RunId, ct);
trace.RequireRoot("POST /api/v1/alarm-observations");
trace.RequireEdge("alarm.commit", "outbox.dispatch");
trace.RequireLinkedConsumer(Golden.FloodMessageId, "rootcause.consume");
trace.RequireEdge("rootcause.verdict.commit", "verdict.dispatch");
trace.RequireLinkedConsumer(Golden.VerdictMessageId, "reporting.consume");
trace.RequireEdge("GET /api/v1/root-cause-cases/{id}", "projection.read");
trace.RequireAttributes(
    "nexus.run_id", "nexus.correlation_id", "messaging.message.id");
trace.RejectSecretOrPayloadAttributes();
```

SAMPLING BOUNDARY

A complete golden trace proves propagation and instrumentation for this admitted run. It does not prove that every production request is sampled or that traces are a durable business ledger.

CONSERVATION MODEL

Every accepted fact has bounded durable consequences

CONSERVATION EQUATIONS 60-O

Coordinate	Expected final count	Arbiter
accepted alarm-am-9005	1	AlarmManagementDb unique constraint
flood-am-6031	1	Alarm decision identity
Flood outbox body	1 semantic body; dispatch attempts >= 1	AlarmManagementDb
RootCause case	1	source decision unique constraint
Verdict	1	case/policy/evidence identity
Reporting ledger positions	2 contiguous	ReportingDb checkpoint
BFF business writes	0	database privilege inventory

EXECUTABLE ASSET 60-Q - CONSERVATION VERDICT

```
ConservationResult result = await conservation.MeasureAsync(run.RunId, ct);
Require(result.AlarmDecisions == 1, "ALARM_EFFECT_COUNT_DRIFT");
Require(result.RootCauseCases == 1, "CASE_EFFECT_COUNT_DRIFT");
Require(result.Verdicts == 1, "VERDICT_EFFECT_COUNT_DRIFT");
Require(result.ReportingPositions.SequenceEqual([1L, 2L]),
    "REPORTING_CONTIGUITY_DRIFT");
```

FINAL SERVICE CATALOGUE

Runtime roles are grouped by ownership, not container count

ADMITTED CORE SERVICE CATALOGUE 60-P

Boundary	Runtime roles	Authority	Status
AlarmManagement	alarm-api + alarm-worker	alarm truth, decisions, outbox	CORE; runtime pending
RootCause	rootcause-api + rootcause-worker	cases, evidence, verdicts, outbox	CORE; runtime pending
Reporting	projection-worker + projection-api	derived ledger, checkpoint, read model	CORE; runtime pending
Query edge	query-bff	authorized composition; no domain truth	EDGE; runtime pending

COUNTING RULE

Seven Deployments represent three service boundaries plus one edge role. API and worker separation is a lifecycle decision, not evidence of seven microservices.

CANDIDATE SERVICE CATALOGUE

Semantically strong candidates do not enter the runtime by prose

CANDIDATE CATALOGUE 60-Q

Candidate	Evidence already specified	Missing admission evidence	Decision
Audit	immutable append, inbox, AuditDb constraints	image, migration, principal, topology, runtime campaigns	REMAIN MODULE/CANDIDATE
Compliance	review aggregate, inbox, ComplianceDb constraints	image, migration, principal, topology, human workflow ops	REMAIN MODULE/CANDIDATE
RL Advisory	deterministic optional adapter and abstention	public contract, owner, SLO, store, economics	REMAIN OPTIONAL MODULE

NO RETROACTIVE DEMOTION

Audit and Compliance remain valid architectural candidates and can run as modules/subscribers in a broader host. The decision says only that the final seven-role deployable core cannot claim them as independently deployed services without the omitted operational artifacts.

PROTECTED CORE CATALOGUE

Some boundaries should deliberately remain modules

MODULE DECISION 60-R

Capability	Placement	Reason
Instrumentation adapters	AlarmManagement module	close to acquisition semantics; no independent owner
ReactorFleet context	protected core/domain module	no independent contract or store proof
DigitalTwin model	protected analytical module	model versioning does not imply a service
simulation policy	RootCause/analysis module	deterministic policy input, not network authority
workflow runner	test/evidence harness	Chapter 37 process manager owns production coordination
UI composition	Query edge	no domain writes or truth ownership

EXTRACTION TEST

A module earns independent distribution only after contract, operations, deployment, data, security and economic evidence agree. Process isolation alone is not a service boundary.

FINAL HTTP CATALOGUE

Public and internal surfaces are explicit

HTTP CONTRACT CATALOGUE 60-S

Surface	Owner	Caller	Semantics
POST /api/v1/alarm-observations	AlarmManagement	authorized operator/adapter	idempotent acceptance; 202 local durability
RootCause command APIs	RootCause	authorized workflow/operator	owner-local case/evidence/verdict changes
GET projection case view	Reporting	query-bff	delayed observation + watermark/freshness
GET BFF root-cause case	Query edge	operator UI/client	authorized composition; partial status visible
/health/startup live ready	each host	platform	distinct lifecycle questions

CONTRACT BOUNDARY

Internal does not mean unauthenticated. The catalogue records route, version, principal, capability, scope, timeout, error shape and owning team; database schemas are never HTTP contracts.

FINAL EVENT CATALOGUE

Activated facts are separated from candidates

PUBLIC FACT CATALOGUE 60-T

Fact	Producer	Consumers in core	Status
AlarmFloodDetected.v1	AlarmManagement	RootCause	ACTIVATED SPECIFICATION
RootCauseCaseOpened.v1	RootCause	Reporting	ACTIVATED SPECIFICATION
RootCauseVerdictIssued.v1	RootCause	Reporting; candidate Audit/Compliance	ACTIVATED SPECIFICATION
RecommendationGenerated.v1	RL Advisory	none	NOT ACTIVATED

EVENT GOVERNANCE 60-U

Required field	Rule
MessageId	producer-owned immutable identity
EventType + schemaVersion	allowlisted; consumer rejects unknown breaking form
CorrelationId / CausationId	business journey / direct cause
ContentFingerprint + body digest	canonical meaning / exact transported bytes
producer stream position	owner order; consumer advances contiguously

FINAL DATA-OWNERSHIP CATALOGUE

Each writer has one database authority boundary

DATA OWNERSHIP 60-V

Store	Runtime writer	Reads	Forbidden
AlarmManagementDb	alarm-api/worker	Alarm owner APIs	RootCause/Reporting/BFF SQL
RootCauseDb	rootcause-api/worker	RootCause owner APIs	Alarm/Reporting/BFF SQL
ReportingDb	projection-worker	projection-api read-only	command-service writes
AuditDb candidate	Audit module/consumer	authorized audit query	RootCause/Compliance writes
ComplianceDb candidate	Compliance module/consumer	authorized workflow query	RootCause/Audit writes

MIGRATION AUTHORITY

Runtime principals cannot apply schema changes. Each admitted database has a separate, short-lived migration principal and a digest-bound migration receipt before traffic.

DATABASE ACCESS PROOF

Ownership is tested by both success and denial

EXECUTABLE ASSET 60-R - PRINCIPAL MATRIX PROBE

```
foreach (DatabasePrincipal principal in catalogue.RuntimePrincipals)
{
    await provider.AssertCanExecuteOwnedOperationsAsync(principal, ct);
    foreach (DatabaseTarget foreign in catalogue.ForeignTargets(principal))
        await provider.AssertPermissionDeniedAsync(principal, foreign, ct);
}

await provider.AssertRuntimeDdlDeniedAsync("alarm-runtime", ct);
await provider.AssertRuntimeDdlDeniedAsync("rootcause-runtime", ct);
await provider.AssertRuntimeDdlDeniedAsync("reporting-runtime", ct);
```

EVIDENCE RULE

A configuration review proves declared intent. Provider-backed denied operations prove that the target environment enforced the boundary for the tested principals.

FINAL SECURITY CATALOGUE

Trust is explicit at every entry and hop

SECURITY CATALOGUE 60-W

Boundary	Credential / identity	Authorization	Evidence
operator -> Alarm	authenticated principal	capability + detector scope	allow/deny HTTP corpus
BFF -> internal APIs	service identity	audience + query capability	token/audience tests
worker -> broker	scoped broker principal	exchange/queue ACL	permission probes
host -> database	owner runtime principal	least DML; no foreign DB/DDL	provider denial matrix
migrator -> database	short-lived migration principal	owned schema DDL only	migration receipt
evidence runner	separate test authority	allowlisted targets and time bound	signed campaign plan

SECRETS AND EVIDENCE

Proof artefacts remain value-free

SENSITIVE-DATA RULES 60-X

Surface	Allowed	Forbidden
configuration manifest	key names, provider order, identity digest	secret value or DSN
logs	bounded IDs, codes, run/correlation	tokens, passwords, event payloads
traces	route templates, span links, IDs	raw bodies or credentials
metrics	bounded service/operation/result labels	Messageld, Caseld, tenant or payload
failure bundle	redacted diagnostics and digests	environment dump

EXECUTABLE ASSET 60-S - EVIDENCE REDACTION GATE

```
EvidenceScanResult scan = EvidenceScanner.Scan(bundle);
Require(scan.SecretFindings.Count == 0, "FINAL_EVIDENCE_SECRET_FOUND");
Require(scan.UnboundedMetricLabels.Count == 0,
"FINAL_METRIC_CARDINALITY_UNBOUNDED");
Require(scan.RawPayloadFindings.Count == 0,
"FINAL_RAW_PAYLOAD_EXPOSED");
```

FINAL OBSERVABILITY CATALOGUE

Signals answer distinct operational questions

OBSERVABILITY CATALOGUE 60-Y

Signal	Required observation	Not a substitute for
trace	one connected golden graph with links	durable business ledger
outbox metrics	pending age, lease conflicts, dispatch result	row-level audit
consumer metrics	delivery, retry, reject, quarantine	semantic effect count
projection metrics	watermark, gap count, source/projection age	source truth
HTTP metrics	rate, latency, result code, auth rejection	payload inspection
health	startup/live/ready state and age	end-to-end acceptance
structured logs	bounded diagnostic events with RunId	metrics or traces

FRESHNESS AND SERVICE LEVELS

Delayed truth needs age, backlog and gap semantics

FINAL RUNTIME INDICATORS 60-Z

Indicator	Golden acceptance	Failure code
end-to-end duration	<= plan.EndToEndBound	FINAL_FLOW_TIMEOUT
projection age	<= plan.ProjectionFreshnessBound	FINAL_PROJECTION_STALE
case stream gap count	0	FINAL_PROJECTION_GAP
outbox oldest pending age	below admitted bound after recovery	FINAL_OUTBOX_STALLED
quarantined golden messages	0	FINAL_GOLDEN_QUARANTINED
trace control freshness	inside telemetry bound	FINAL_TRACE_PIPELINE_STALE

NO INVENTED SLO

The chapter defines the measurement and comparison mechanism. Numeric production objectives remain environment- and workload-specific until capacity and operating evidence approve them.

FINAL FAILURE CATALOGUE

Each campaign states the conserved truth and recovery oracle

MANDATORY CAMPAIGNS 60-AA

Campaign	Injection	Must remain true	Recovery proof
F-OUTBOX	broker unavailable after TX-A	202 + pending exact body	one dispatch; no new decision
F-DUP	duplicate/redelivered facts	one semantic effect	receipts may increase
F-POISON	invalid body/contract	no business write	visible quarantine + safe replay
F-GAP	deliver position 2 before 1	watermark does not skip	contiguous position 2
F-CRASH	kill consumer before/after commit	transaction arbitration	bounded convergence
F-DB	owner DB transient failure	foreign owners unchanged	retry with new unit of work
F-AUTH	missing capability/scope	zero durable effect	bounded rejection evidence
F-TELEM	collector unavailable	business path governed separately	signal loss visible

DUPLICATE AND CRASH CAMPAIGN

Transport attempts may multiply; business effects may not

EXECUTABLE ASSET 60-T - REDELIVERY ORACLE

```
await broker.DeliverAsync(Golden.FloodBody, times: 3, ct);
await fault.KillAsync("rootcause-worker", CrashPoint.AfterCommitBeforeAck, ct);
await runtime.RestartAsync("rootcause-worker", ct);

await rootCauseDb.AssertInboxReceiptCountAsync(Golden.FloodMessageId, 1);
await rootCauseDb.AssertCaseCountAsync(Golden.CaseId, 1);
await rootCauseDb.AssertOutboxBodyCountAsync(Golden.CaseOpenedMessageId, 1);
await broker.AssertDeliveryAttemptsAtLeastAsync(Golden.FloodMessageId, 2);
```

TWO-KEY ARBITRATION

Inbox MessageId prevents transport replay. A semantic unique key prevents a new MessageId carrying the same business identity from creating a second effect. Both constraints are required.

POISON AND REPLAY CAMPAIGN

A bad message becomes visible evidence, not an endless loop

EXECUTABLE ASSET 60-U - QUARANTINE PROOF

```
PoisonDelivery poison = Golden.VerdictBody.WithInvalidEvidenceDigest();
await broker.PublishRawAsync(poison, ct);

await quarantine.AssertAsync(poison.MessageId, x => x
    .HasReason("verdict-evidence-summary-invalid")
    .HasOriginalBodySha256(Sha256.Hex(poison.Body))
    .HasNoSecretValues());
await reportingDb.AssertNoBusinessEffectAsync(poison.MessageId);

ReplayReceipt replay = await replay.RequestAsync(
    poison.MessageId, approvedCorrectedBody: Golden.VerdictBody, ct);
replay.RequireNewDeliveryAttemptSameBusinessIdentity();
await reportingDb.AssertOneSemanticEffectAsync(Golden.VerdictId);
```

PROJECTION GAP CAMPAIGN

Out-of-order delivery cannot manufacture current truth

EXECUTABLE ASSET 60-V - GAP AND REPAIR

```
await broker.DeliverAsync(Golden.VerdictBody, ct); // position 2 first
await reportingDb.AssertCheckpointAsync(Golden.CaseStreamId, 0);
await reportingDb.AssertGapAsync(Golden.CaseStreamId, missing: 1);

Observed<RootCauseCaseView> delayed = await bff.GetRootCauseCaseAsync(
    Golden.CaseId, ct);
delayed.Evidence.UpdateStatus.Should().Be("gap-detected");

await broker.DeliverAsync(Golden.CaseOpenedBody, ct); // position 1
await reportingDb.WaitForContiguousPositionAsync(
    Golden.CaseStreamId, 2, ct);
await bff.RequireCurrentAsObservedAsync(Golden.CaseId, ct);
```

USER-VISIBLE DEGRADATION

The BFF returns partial/stale/unknown evidence explicitly when allowed by contract. It does not hide a detected gap behind the last good row.

TELEMETRY OUTAGE CAMPAIGN

Observability failure is visible without becoming business authority

TELEMETRY SEPARATION 60-AB

Condition	Business path	Final evidence verdict
collector unavailable	may continue if policy permits	PENDING/FAILED observability checkpoint
trace missing	owner commits remain independently measurable	cannot claim trace continuity
metrics stale	health/readiness follow explicit policy	cannot claim capacity/freshness
logs unavailable	no compensating business rollback	diagnostic completeness fails

NO TRANSITIVE FAILURE

A telemetry outage does not erase committed domain truth. Conversely, committed truth does not make the observability checkpoint pass.

FINAL DEPLOYMENT CATALOGUE

Compose and Kubernetes project the same core responsibilities

DEPLOYMENT EQUIVALENCE 60-AC

Role	Compose	Kubernetes	Invariant
alarm-api/worker	2 services	2 Deployments + 1 Service	one Alarm DB owner
rootcause-api/worker	2 services	2 Deployments + 1 Service	one RootCause DB owner
projection-worker/api	2 services	2 Deployments + 1 Service	one writer; read-only API
query-bff	1 service	1 Deployment + 1 Service	no domain DB
migrations	3 completion jobs	controlled digest-bound jobs	runtime cannot migrate
stateful dependencies	local containers	environment-owned endpoints	same ownership/contract

EQUIVALENCE BOUNDARY

Equivalent means the same application responsibilities, identities and evidence obligations. It does not mean Compose and Kubernetes have identical networking, scheduling, failure domains or operations.

DEPLOYMENT ADMISSION

Platform readiness and application proof are joined

EXECUTABLE ASSET 60-W - DEPLOYMENT RECEIPT JOIN

```
DeploymentObservation deployment = await platform.ObserveAsync(plan, ct);
deployment.RequireExactRoles(Topology.SevenRoles);
deployment.RequireImages(candidate.RoleImageDigests);
deployment.RequireConfiguration(run.ConfigurationIdentity);
deployment.RequireMigrations(migrationSet.Digests);
deployment.RequireDistinctStartupLiveReadyProbes();

FinalConvergence convergence = await scenario.ExecuteAsync(plan, ct);
Require(deployment.RunId == convergence.RunId,
    "FINAL_DEPLOYMENT_SCENARIO_RUN_MISMATCH");
Require(deployment.IsAdmitted && convergence.IsComplete,
    "FINAL_PLATFORM_OR_APPLICATION_INCOMPLETE");
```

KUBERNETES BOUNDARY

Available replicas, Service discovery and completed Jobs are required platform observations. The scenario still needs authorization, contract, persistence, trace, failure and delayed-truth evidence.

FINAL EVIDENCE CATALOGUE

Every claim names an artifact, digest and freshness rule

EVIDENCE CATALOGUE 60-AD

Claim family	Required artifact	Freshness / identity
build/tests	binlog + TRX/JUnit + discovery manifest	exact BuildIdentity; nonzero discovery
deployment	Compose/Kubernetes inventory and receipts	exact candidate/config/run
database	migration + row/conservation snapshots	provider identity + RunId
broker	topology + captured bodies + queue states	broker identity + MessageIds
HTTP/security	request/response corpus + auth matrix	contract/policy identity
observability	trace export + metric/log snapshots	inside admitted time bounds
failure/recovery	campaign plan, injection and oracle receipts	same run/build/environment

EVIDENCE MANIFEST

The final JSON cannot say Passed by omission

EXECUTABLE ASSET 60-X - SEALED FINAL MANIFEST

```
public sealed record FinalDemonstratorManifest(
    Guid RunId,
    string BuildIdentity,
    string CandidateManifestSha256,
    string DeploymentReceiptSha256,
    string ServiceCatalogueSha256,
    string HttpCatalogueSha256,
    string EventCatalogueSha256,
    string DataOwnershipCatalogueSha256,
    string SecurityCatalogueSha256,
    string ObservabilityCatalogueSha256,
    string FailureCatalogueSha256,
    string ConvergenceVectorSha256,
    IReadOnlyList<EvidenceItem> Evidence,
    FinalVerdict Verdict,
    IReadOnlyList<string> Missing,
    string ManifestSha256);
```

CURRENT HONEST RECORD

Verdict = PENDING; Missing includes repository implementation, clean build output, provider-backed database receipts, broker capture, deployment receipt, trace export and executed recovery campaigns.

FAIL-CLOSED FINAL VALIDATOR

Missing, stale or cross-run evidence is a failure

EXECUTABLE ASSET 60-Y - FINAL ADMISSION

```
public static void Admit(FinalDemonstratorManifest m, FinalScenarioPlan p)
{
    Require(m.RunId == p.Identity.RunId, "FINAL_RUN_MISMATCH");
    Require(m.BuildIdentity == p.Identity.BuildIdentity,
        "FINAL_BUILD_MISMATCH");
    Require(m.Evidence.Count > 0, "FINAL_EVIDENCE_EMPTY");
    Require(m.Evidence.All(x => x.RunId == m.RunId),
        "FINAL_CROSS_RUN_EVIDENCE");
    Require(m.Evidence.All(x => x.BuildIdentity == m.BuildIdentity),
        "FINAL_CROSS_BUILD_EVIDENCE");
    Require(m.Evidence.All(x => !x.IsStaleAt(p.Identity.StartedAtUtc)),
        "FINAL_STALE_EVIDENCE");
    Require(p.Checkpoints.All(c => m.Evidence.Proves(c)),
        "FINAL_CHECKPOINT_UNPROVED");
    Require(m.Missing.Count == 0, "FINAL_MISSING_EVIDENCE");
    Require(m.Verdict == FinalVerdict.Passed, "FINAL_VERDICT_NOT_PASSED");
}
```

ZERO-DISCOVERY AND STALE-GREEN GATES

A test runner can succeed while proving nothing

EXECUTABLE ASSET 60-Z - EVIDENCE QUALITY GATE

```
foreach (TestFamily family in RequiredFamilies.All)
{
    TestReceipt receipt = manifest.RequireTestFamily(family);
    Require(receipt.Discovered > 0, $"ZERO_DISCOVERY:{family}");
    Require(receipt.Executed == receipt.Discovered,
        $"PARTIAL_EXECUTION:{family}");
    Require(receipt.Failed == 0 && receipt.Skipped == 0,
        $"NONPASSING_TESTS:{family}");
    Require(receipt.BuildIdentity == run.BuildIdentity,
        $"STALE_TEST_BUILD:{family}");
}
```

REQUIRED TEST FAMILIES 60-AE

Family	Primary claim
Domain/Application	deterministic policy and effect order
Architecture	dependency and ownership boundaries
Provider component	constraints, transactions, permissions
Contract	HTTP/event compatibility and exact corpus
Production-path host	real composition and middleware
Broker/distributed	delivery, duplicate, poison, gap
Deployment/failure	runtime topology, recovery and evidence

FINAL ACCEPTANCE CHECKLIST

The manuscript separates specified work from executed proof

ACCEPTANCE MATRIX 60-AF

Requirement	Specified asset	Runtime status
clean checkout builds	build graph + identity	PENDING
all test families pass	family catalogue + zero-discovery gate	PENDING
Compose starts full core	Chapter 55 runtime contract	PENDING
operator-to-report scenario repeats	FinalScenarioPlan + convergence	PENDING
one writer per database	principal matrix + denial probes	PENDING
broker outage preserves commit	F-OUTBOX	PENDING
duplicates produce one effect	F-DUP + conservation	PENDING
poison visible/replay safe	F-POISON	PENDING
breaking contract rejected	contract corpus tests	PENDING
projection delay visible	gap/freshness contract	PENDING

ACCEPTANCE CHECKLIST CONTINUED

Security, trace, failure and documentation also need receipts

ACCEPTANCE MATRIX 60-AG

Requirement	Specified asset	Runtime status
unauthorized access rejected	F-AUTH + auth corpus	PENDING
correlation/causation cross boundaries	causal catalogue	PENDING
one trace is complete	trace graph oracle	PENDING
failure campaigns match expected	failure catalogue/oracles	PENDING
diagrams match executable topology	catalogue digest comparison	PENDING
technical guarantees name proof	evidence catalogue	SPECIFIED
Phase-0 boundary is explicit	scope/non-goals	SPECIFIED
conditional candidates stay conditional	candidate decision record	SPECIFIED
demonstrated/inferred/unproved split	claim maturity ledger	SPECIFIED

RELEASE RULE

No row changes from PENDING to PASSED because the manuscript is detailed. Only attached, digest-bound execution evidence may change a runtime status.

CLAIM MATURITY LEDGER

Predicted, specified, demonstrated and falsified are distinct

MATURITY VOCABULARY 60-AH

State	Meaning	Allowed wording
PREDICTED	architecture expects a property	should / is intended to
SPECIFIED	executable contract and oracle are defined	the design requires
DEMONSTRATED	attached run proves the property in one profile	the recorded run showed
FALSIFIED	experiment contradicted the claim	the claim failed under...
REFINED	boundary or claim changed after evidence	the model now limits...
UNPROVED	required evidence is absent or stale	pending / not claimed

CURRENT CHAPTER MATURITY

The architecture and proof harness are SPECIFIED. The repository implementation, full deployment, final run and failure campaigns remain UNPROVED/PENDING in this manuscript artifact.

ARCHITECTURE PREDICTIONS RETAINED

The final model preserves the book's strongest deductions

PREDICTION LEDGER 60-AI

Prediction	Final architectural form	Proof still required
ownership before distribution	three core data owners + edge	provider permission receipts
facts cross boundaries	three activated V1 public contracts	broker capture and contract runs
local commit before publication	outbox at Alarm and RootCause	outage/crash evidence
reads become delayed observations	Reporting ledger + BFF freshness	gap/freshness run
failures reveal boundaries	campaign-specific conservation oracles	executed fault laboratory
deployment carries identities	candidate/config/migration/run digests	Compose/Kubernetes receipts

REFINEMENTS MADE EXPLICIT

The book changed its mind where evidence boundaries demanded it

REFINEMENT LEDGER 60-AJ

Earlier temptation	Final refinement
schema count equals service count	schemas are inventory; ownership evidence defines boundaries
one process per service	API/worker roles share a service when authority and data owner agree
HTTP success means distributed success	202 proves only the owner-local Alarm outcome
queue empty means complete	owner-local rows, captured bytes, checkpoints and freshness are joined
Kubernetes green means runtime proved	platform and application evidence remain separate planes
every candidate deserves extraction	Audit, Compliance and RL remain conditional without operations evidence
latest data means current truth	BFF says current-as-observed with age and watermark

ANTI-PATTERN CATALOGUE

Final-demo shortcuts that manufacture certainty

REJECTED PATTERNS 60-AK

Anti-pattern	Why rejected
manual clicking without plan	input, identity and timing cannot be reproduced
query all databases for the oracle	evidence harness becomes a production coupling blueprint
sleep then assert final row	timing replaces bounded state observation
reserialize captured events	exact contract bytes are no longer tested
delete stale rows automatically	preflight hides contamination
one global completed flag	partial progress and owner authority disappear
reuse yesterday's green TRX	build/run identity is stale
trace IDs as metric labels	unbounded cardinality and leakage
claim Audit/Compliance deployed	catalogue contradicts seven-role topology
call Phase-0 production ready	operations and safety claims exceed evidence

REPOSITORY EXECUTION CONTRACT

One command produces one immutable evidence directory

EXECUTABLE ASSET 60-AA - FINAL RUNNER

```
dotnet run --project eng/Nexus.FinalDemonstrator -- run \  
  --plan deploy/scenarios/final-phase0.json \  
  --candidate artifacts/candidate/manifest.json \  
  --deployment artifacts/deployment/admitted.json \  
  --golden tests/Nexus.GoldenCorpus/final-v1.json \  
  --output artifacts/final/<run-id>  
  
dotnet run --project eng/Nexus.FinalDemonstrator -- verify \  
  --manifest artifacts/final/<run-id>/manifest.json \  
  --policy eng/evidence/final-policy.json
```

EXECUTABLE ASSET 60-AB - INTENDED EVIDENCE TREE

```
artifacts/final/<run-id>/  
  identity.json          preflight.json  
  http/                  broker/  
  databases/             telemetry/  
  failures/              deployment/  
  tests/                 catalogues/  
  convergence.json       manifest.json
```

PROPOSED PATH

These are the required repository contracts, not a claim that the paths already exist. Their status remains PENDING until committed and executed under the recorded BuildIdentity.

OFFICIAL MECHANISM ANCHORS

Standards explain mechanisms; local evidence carries guarantees

PRIMARY-SOURCE MAPPING 60-AL

Official source	Mechanism used	NEXUS-1 qualification
W3C Trace Context	traceparent/tracestate propagation	domain correlation and causation remain explicit
OpenTelemetry context propagation	span context across HTTP/messaging	sampling cannot become business evidence
RabbitMQ reliability guide	publisher confirms and consumer acknowledgements	outbox/inbox and semantic constraints remain mandatory
Kubernetes probe documentation	startup/liveness/readiness effects	endpoint meanings come from Chapter 11

FRESHNESS

The mechanism review was checked against official sources on 10 August 2026. A future build must pin the actual framework, broker, database and cluster versions used by its evidence.

COMPLETION MATRIX

What Chapter 60 completes and what it deliberately leaves open

COMPLETION MATRIX 60-AM

Concern	Chapter result	Status
final scenario	operator action through delayed BFF truth	SPECIFIED
service/API/event/data catalogues	exact core and candidate boundaries	SPECIFIED
security/observability/failure catalogues	entry, signal and recovery oracles	SPECIFIED
deployment/evidence catalogues	Compose/Kubernetes and fail-closed joins	SPECIFIED
candidate decisions	core, edge, conditional services and modules	SPECIFIED
repository implementation	clean build and discovered tests	PENDING
executed final scenario	provider/broker/HTTP/trace/deployment bundle	PENDING
production readiness	operations, safety, HA, backup, governance	NOT CLAIMED

RESIDUAL OBLIGATIONS

Implement the named assets, pin the environment, execute every required test and fault campaign, retain value-free receipts, and let the validator decide. A reader should be able to reproduce a failure as precisely as a success.

FINAL DECISION ON SERVICE EXTRACTION

Distribute only the boundaries that earned it

ADR-060 FINAL BOUNDARY DECISION 60-AN

Capability	Final decision	Reason
AlarmManagement	ADMIT core service boundary	independent commands, truth, store, failures and scale
RootCause	ADMIT core service boundary	independent evidence/verdict authority and public facts
Reporting	ADMIT derived service boundary	independent projection lifecycle, store and freshness contract
Query BFF	ADMIT edge deployment; not domain service	public composition/security boundary; owns no truth
Audit	KEEP module/candidate for current runtime	semantic design exists; deployment/data operations evidence absent
Compliance	KEEP module/candidate for current runtime	workflow design exists; independent operations evidence absent
RL Advisory	KEEP optional module	no activated public contract or extraction economics
protected analytical/acquisition capabilities	KEEP modules	no independently earned ownership boundary

DECISION MATURITY

These are architectural placement decisions. Runtime admission for even the core remains PENDING until execution evidence exists.

PART X AND MANUSCRIPT HANDOFF

The architecture ends where evidence begins

The book began with a modular monolith, refused to equate schemas with services, extracted ownership before transport, made public facts explicit, accepted delayed truth, treated failures as experiments, and bound delivery to immutable identities. The final demonstrator now expresses that journey as one reviewable scenario whose uncertainty cannot be edited away.

PART X CLOSURE 60-AO

Chapter	Irreversible contribution
57	reproducible build, test and artifact identity
58	coexistence, cutover and evidence-aware rollback
59	minimal Kubernetes projection of proved runtime needs
60	one final evidence join and honest boundary verdict

BACK-MATTER HANDOFF

The Epilogue will distinguish what the architecture predicted, what implementation evidence demonstrated, which failures refined the model, and what remains educational or unproved. Appendices will consolidate the solution map, catalogues, migration matrix, failure matrix, trust boundaries and reproduction commands.

FINAL CHAPTER RULE

One engineering model survived the transition

NEXUS-1 is not complete because every candidate became a microservice or because every deployment object turned green. It is complete as an engineering model when ownership, contracts, time, failure and proof remain coherent from the first operator request to the final delayed observation—and when the answer is allowed to remain Pending whenever evidence is absent.

FINAL INVARIANT

The final demonstrator is credible only when one admitted operator command can be traced through ownership, contracts, messages, persistence, telemetry, failure and recovery to delayed reporting truth, while every missing, stale or cross-run evidence item fails the final verdict closed.

CURRENT VERDICT

MANUSCRIPT CHAPTER COMPLETE | ARCHITECTURE SPECIFIED | RUNTIME EVIDENCE PENDING | PRODUCTION READINESS NOT CLAIMED

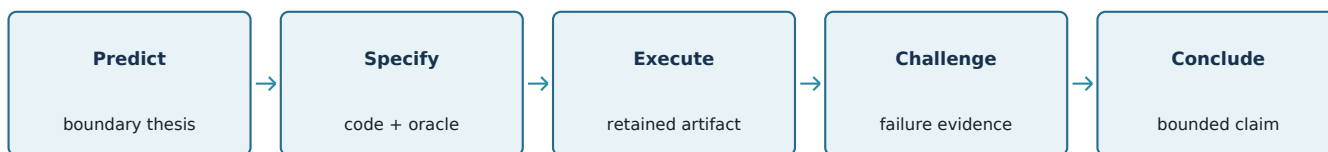
EPILOGUE

What the Runtime Has Actually Proved

The honest conclusion is not that NEXUS-1 became a production microservice platform. It is that one coherent engineering model can cross from modular boundaries into a fully specified distributed runtime without surrendering ownership, language, time, failure or the right to say: evidence is still missing.

EPILOGUE DECISION

Separate architectural prediction, executable specification, attached demonstration, falsification/refinement and remaining uncertainty. Preserve the Chapter 60 verdict: the manuscript is complete through the final demonstrator, the architecture is specified, runtime evidence is pending and production readiness is not claimed.

THE FINAL REASONING PATH**CLOSING CLAIM LADDER E-1**

Layer	Question	Current answer
architecture	is the distributed model coherent?	YES - SPECIFIED
implementation	does the named repository build and run?	PENDING
failure	did the executed campaigns conserve truth?	PENDING
production	is the platform operationally ready?	NOT CLAIMED

STANDING RULE

Nothing claims to exist that does not. The conclusion becomes stronger, not weaker, when it refuses to turn a detailed plan into an executed fact.

THE QUESTION AT THE HARDER SCALE

Can the system remain understandable when certainty is no longer local?

The earlier NEXUS-1 volumes established physical meaning, domain language, persistence, contracts and modular execution. Distribution changes the question. A local transaction can no longer settle the whole story; a remote answer may be late; transport may repeat; a dependency may disappear; a projection may be useful and still not be current. Understanding therefore has to survive across owners, clocks and failures.

WHAT DISTRIBUTION CHANGES E-2

Before distribution	After distribution	Required discipline
one commit can settle a use case	several owner-local commits advance independently	convergence vector
a reference can be a foreign key	remote identity has no local existence guarantee	typed ID + local observation
a read can follow a write directly	projection may lag or contain a gap	freshness and watermark
an exception can unwind the call	facts may already be durable elsewhere	retry/repair/compensation
logs describe one process	causality crosses HTTP, broker and workers	correlation + causation + trace

ANSWER IN ONE SENTENCE

The system remains understandable only when every boundary says what it owns, what it observed, what it cannot know yet and which evidence would change that answer.

CLAIM VOCABULARY

Prediction, specification and demonstration are not synonyms

EVIDENCE MATURITY E-3

State	Meaning	Permitted conclusion
PREDICTED	the architecture expects a property	a hypothesis worth testing
SPECIFIED	mechanism, boundary and oracle are explicit	the design is reviewable
DEMONSTRATED	retained execution evidence supports one profile	the recorded run showed
FALSIFIED	an experiment contradicted the claim	the claim does not survive that condition
REFINED	the claim or boundary was narrowed	the updated model is more precise
UNPROVED	required evidence is absent, stale or unrelated	remain PENDING
NOT CLAIMED	outside educational scope	no inference permitted

BOOK-WIDE CORRECTION

Compile-oriented listings, exact commands and rigorous test shapes are specifications until the named build and execution receipts are attached. Detail is not a substitute for execution.

EVIDENCE DOMAINS

No single green artifact can carry the final conclusion

EVIDENCE PLANES E-4

Plane	Can establish	Cannot establish alone
source/build	what code and packages were selected	runtime behaviour
tests	named claims under fixtures	deployment equivalence
database	owner-local durable effects	broker delivery
broker	routing, body and acknowledgement evidence	semantic business uniqueness
HTTP/security	contract and authorization outcomes	downstream convergence
telemetry	one observed execution path	durable truth
deployment	admitted workloads and configuration	domain correctness
failure campaign	behaviour under one controlled disturbance	all operational risk

FINAL JOIN

A Passed verdict requires these planes to agree on the same RunId, BuildIdentity, configuration, golden coordinates and freshness window. Cross-run evidence fails closed.

WHAT THE ARCHITECTURE PREDICTED - OWNERSHIP

Distribution should follow authority, not schema count

The seventeen schemas inherited from the earlier data volumes were an inventory of concerns, not a ready-made service map. The book predicted that only boundaries with stable language, write authority, contracts, failure behaviour and operational reason would survive extraction. That prediction shaped three core owners and one edge role in the final runtime topology.

OWNERSHIP PREDICTION E-5

Boundary	Independent authority	Final placement
AlarmManagement	alarm observations, flood decisions, publication intent	core service boundary
RootCause	cases, evidence membership, verdicts, public facts	core service boundary
Reporting	derived ledger, checkpoint, freshness-bearing read model	derived service boundary
Query BFF	authorized composition only	edge deployment; owns no truth
Audit / Compliance	valid candidate semantics	modules/candidates in current runtime

WHAT THIS PROVES

The manuscript demonstrates a coherent ownership decision. Provider-enforced permissions and independent runtime operation remain PENDING until executed evidence exists.

WHAT THE ARCHITECTURE PREDICTED - PUBLIC LANGUAGE

Private domain events could not simply cross the network

PUBLIC-LANGUAGE OUTCOME E-6

Prediction	Final specified form	Evidence state
producer owns public meaning	explicit V1 HTTP/event contracts	SPECIFIED
identity survives retries	MessageId and idempotency keys	SPECIFIED; runtime pending
journey and direct cause differ	CorrelationId + CausationId	SPECIFIED; trace pending
bytes matter	canonical fingerprint + transported-body digest	SPECIFIED; capture pending
evolution is governed	compatibility corpus and breaking mutation	SPECIFIED; run pending

ACTIVATED SPECIFICATION

AlarmFloodDetected.v1, RootCauseCaseOpened.v1 and RootCauseVerdictIssued.v1 are the activated public contracts of the architecture. RecommendationGenerated.v1 remains a design candidate, not a present fact.

WHAT THE ARCHITECTURE PREDICTED - LOCAL CERTAINTY

A local commit must survive transport failure

The architecture rejected a distributed transaction and instead paired domain state with an outbox row inside the owning database. Consumers pair an inbox receipt with their owner-local effect. This predicts at-least-once transport with effectively-once business consequences where the semantic keys are correctly chosen.

LOCAL-CERTAINTY LEDGER E-7

Boundary	Atomic local work	Independent uncertainty
Alarm TX-A	observation + flood decision + Pending fact	broker/downstream progress
RootCause TX-RC-O	inbox + case + CaseOpened intent	Reporting progress
RootCause TX-RC-V	verdict + evidence + VerdictIssued intent	subscriber completion
Reporting TX-REP	inbox + ledger + summary + checkpoint	source-authoritative now

CURRENT EVIDENCE BOUNDARY

The transaction shapes and oracles are specified. Broker-outage, crash-after-commit and provider-backed conservation receipts are not attached, so survival remains a runtime hypothesis.

WHAT THE ARCHITECTURE PREDICTED - DELAYED TRUTH

A useful view can be correct and still be late

Reporting owns an observation of upstream facts, not a distributed snapshot. Its watermark, gap count, source age and projection age are part of the contract because the absence of those coordinates would let an old row impersonate current truth. The BFF therefore says current-as-observed rather than globally current.

TRUTH VOCABULARY E-8

Term	Owner	Honest statement
alarm truth	AlarmManagement	what the alarm owner durably accepted
verdict truth	RootCause	what the evidence-bound policy issued
reporting observation	Reporting	what contiguous source positions currently support
operator view	Query BFF	authorized composition with explicit freshness

PREDICTION RETAINED

The network did not make truth disappear. It made the location, age and authority of each statement part of its meaning.

WHAT THE ARCHITECTURE PREDICTED - PARTIAL FAILURE

Incomplete progress is normal distributed state

FAILURE-AWARE PREDICTION E-9

Condition	Valid intermediate state	Required visibility
broker unavailable	owner commit + Pending outbox	pending age and recovery
consumer offline	durable queue backlog; no subscriber row	queue and owner-local state
duplicate delivery	extra attempt; one business effect	transport vs semantic counters
poison message	quarantine; zero business write	reason, digest and replay authority
position 2 before 1	gap buffered; watermark does not skip	missing position and age
telemetry unavailable	truth may commit; observation incomplete	separate evidence failure

NO GLOBAL COMPLETION

No service calls the producer to declare the distributed journey complete. The evidence harness observes a convergence vector of independent, durable coordinates.

WHAT THE ARCHITECTURE PREDICTED - OPERATIONS

A service boundary includes the cost of explaining it

The candidate map predicted that deployment independence without health, telemetry, security, migration, failure and rollback ownership would be a hollow extraction. The later Parts made those obligations visible. API/worker separation, distinct probes, service-owned migrations, bounded resilience and immutable delivery identities are consequences of that prediction.

OPERATIONAL CONSEQUENCE E-10

Concern	Specified mechanism	Current claim
lifecycle	startup/live/ready + bounded shutdown	SPECIFIED
identity	candidate/config/image/migration/run digests	SPECIFIED
capacity	measured budgets and resource receipts	PENDING
resilience	timeout/retry/circuit/bulkhead boundaries	SPECIFIED; run pending
rollout	compatibility stages and rollback classifier	SPECIFIED; deployment pending
observability	logs, metrics, traces and evidence joins	SPECIFIED; exports pending

WHAT THE IMPLEMENTATION MATERIAL ESTABLISHES

The manuscript moved the architecture beyond diagrams

MATERIAL CONTRIBUTION E-11

Material	What is now reviewable
compile-oriented C#	domain/application/adaptor responsibilities and failure outcomes
EF Core and SQL shapes	transactions, constraints, migrations and permissions
HTTP/event corpora	versioned public language, bytes and compatibility
workers and broker topology	dispatch, acknowledgement, retry, poison and ordering
Compose/Kubernetes projections	exact roles, dependencies, probes, secrets and rollout
test and fault oracles	what evidence would accept or reject every critical claim
final catalogues	one consistent map of ownership, contracts, trust and evidence

PRECISE CONCLUSION

The book has demonstrated that its architecture can be expressed as a complete, falsifiable implementation contract. It has not yet demonstrated that the named repository realizes that contract.

WHAT HAS NOT BEEN DEMONSTRATED**The missing artifacts are concrete, not philosophical****RUNTIME EVIDENCE STILL REQUIRED E-12**

Claim	Missing retained artifact
clean implementation exists	source revision + restore/build binlog
test families execute	nonzero discovery + passing TRX/JUnit
databases enforce ownership	provider permission and transaction receipts
broker semantics hold	topology, confirm, acknowledgement and captured-body evidence
final scenario converges	one RunId-bound HTTP/database/broker observation bundle
trace is continuous	exported span graph for the same run
failure recovery works	injection and post-recovery conservation receipts
delivery is reproducible	image/config/migration/deployment identities

FAIL-CLOSED CONSEQUENCE

Until these artifacts are attached, the final manifest's Missing collection is nonempty and its verdict cannot be Passed.

THE FAILURE EXPERIMENTS**The manuscript challenged shortcuts even before runtime execution**

The failure chapters are valuable because they change the questions a design must answer. A broker outage asks whether local truth depends on transport. Redelivery asks whether identity is semantic or merely physical. Projection delay asks whether the user can see uncertainty. A crash between commit and acknowledgement asks which durable store arbitrates the outcome.

SPECIFIED CAMPAIGN SET E-13

Campaign	Architectural question	Execution status
broker outage	does the outbox preserve committed intent?	PENDING
duplicate/redelivery	does one fact create one business effect?	PENDING
poison + replay	is invalid input visible and recoverable safely?	PENDING
projection gap	can delayed truth avoid skipping order?	PENDING
consumer crash	does the transaction remain final arbiter?	PENDING
database transient	does retry create a fresh unit of work?	PENDING
authorization denial	is zero effect enforced at the owner?	PENDING
telemetry outage	does evidence fail separately from business truth?	PENDING

WHAT THE FAILURE ANALYSIS FALSIFIED

Several comfortable assumptions could not survive review

FALSIFICATION LEDGER E-14

Rejected assumption	Why it fails	Refined rule
HTTP success means workflow success	only one owner committed	return local outcome; observe convergence
queue empty means complete	loss, ack and durable effects are unknown	join broker and store evidence
retry makes remote calls reliable	amplification and duplicate effects	bounded retry + idempotency + budget
one health endpoint is enough	restart and traffic questions differ	startup/live/ready separation
latest row is current truth	projection may be stale or gapped	age + watermark + status
Kubernetes green proves architecture	platform status lacks domain evidence	join platform and application planes
rollback always restores safety	durable data/contracts may be irreversible	classify before rollback

MATURITY NOTE

These assumptions are architecturally falsified by counterexample and invariant analysis. The environment-specific recovery claims still require executed campaigns.

WHAT THE MODEL REFINED

The final architecture is smaller and more precise than the candidate list

BOUNDARY REFINEMENT E-15

Initial possibility	Final decision	Reason
seventeen schemas -> seventeen services	rejected	schema organization is not independent authority
five strong candidates all deployed now	narrowed	final admitted topology contains three owners + edge
Audit as independent service	module/candidate	semantic design exists; deployment evidence absent
Compliance as independent service	module/candidate	workflow semantics exist; operations evidence absent
RL Advisory as service	optional module	no activated public output or extraction economics
BFF as domain service	edge only	composition/security boundary owns no truth

A SMALLER ARCHITECTURE IS NOT A RETREAT

Keeping a capability inside a module is a successful decision when independent distribution would add failure, latency and operational cost without independent authority.

THE ADMITTED CORE

Three owners and one edge carry the final demonstrator

FINAL PLACEMENT E-16

Boundary	Architectural status	Runtime status
AlarmManagement	ADMITTED CORE OWNER	PENDING
RootCause	ADMITTED CORE OWNER	PENDING
Reporting	ADMITTED DERIVED OWNER	PENDING
Query BFF	ADMITTED EDGE; NO DOMAIN TRUTH	PENDING
Audit	MODULE / SERVICE CANDIDATE	NOT IN CORE TOPOLOGY
Compliance	MODULE / SERVICE CANDIDATE	NOT IN CORE TOPOLOGY
RL Advisory	OPTIONAL MODULE	PUBLIC OUTPUT NOT ACTIVATED

NO STATUS INHERITANCE

Architectural admission does not imply runtime admission. A candidate's detailed chapter does not silently place it in the deployment inventory.

SECURITY AND TRUST

The architecture defined trust boundaries without claiming security maturity

SECURITY CONCLUSION E-17

Specified	Still unproved / out of scope
operator and service identities	production identity lifecycle
capability, scope and owner-local enforcement	penetration hardening and zero-trust maturity
database and broker least-access principals	environment-specific ACL enforcement
secret references and value-free evidence	complete secret-management platform
negative authorization and configuration tests	certified security assurance
bounded audit/security events	regulatory compliance certification

HONEST BOUNDARY

The Phase-0 model is security-aware, not security-certified. Its value lies in making trust decisions explicit and testable at the owning boundary.

OBSERVABILITY AND EVIDENCE

Telemetry explains an execution; it does not become the execution

SIGNAL CONCLUSION E-18

Signal	Architectural use	Limit
structured logs	bounded diagnostic events with domain meaning	not a durable ledger
metrics	rates, age, gaps, retries and capacity	no high-cardinality business IDs
traces	one causal execution graph	sampling may omit paths
health	lifecycle and traffic eligibility	not end-to-end correctness
evidence manifests	digest-bound claim admission	only as strong as attached artifacts

IMPORTANT DISTINCTION

Observable does not mean proved. A signal becomes evidence only when its identity, freshness, scope and relationship to the claim are validated.

DELIVERY AND MIGRATION

Deployment mechanisms did not become the architecture

Compose makes the educational runtime reproducible; Kubernetes projects the already-earned responsibilities into a scheduler. The strangler seam, migration authority, compatibility windows, evidence-aware rollout and rollback classifier preserve the architecture while its deployment form changes.

DELIVERY CONCLUSION E-19

Mechanism	Allowed claim	Disallowed shortcut
Docker Compose	declared local topology and startup contract	production equivalence
container image	immutable packaged role identity	correct runtime behaviour
migration job	one explicit schema authority action	automatic reversibility
Kubernetes Deployment	admitted desired workload and rollout state	domain-flow correctness
rollback plan	approved action for a classified state	universal undo

THE COST LEDGER

Every distributed benefit acquired an operating cost

BENEFIT AND COST E-20

Benefit	Cost introduced	Required counter-discipline
independent change	version and rollout coordination	compatibility and evidence identities
owner-local persistence	eventual cross-owner convergence	outbox/inbox and repair
failure isolation	more partial states	state vectors and observability
independent scaling	resource and ordering risk	measured budgets and concurrency proof
public contracts	evolution responsibility	producer ownership and contract tests
deployment independence	more security and operational surfaces	principals, runbooks and gates

ENGINEERING JUDGEMENT

The book does not argue that microservices are better. It shows the conditions under which selected boundaries may justify their additional cost.

WHAT THE READER MAY TRUST

The strongest deliverable is the chain of reviewable obligations

READER TRUST BOUNDARY E-21

The reader may trust	The reader must not infer
the ownership and contract decisions are explicit	the databases were provisioned and permission-tested
critical code and configuration shapes are concrete	the listings were compiled in a named repository build
failure oracles state conserved and forbidden effects	the failures were injected successfully
the final scenario has exact coordinates and acceptance rules	the scenario passed end to end
conditional candidates remain identified as conditional	future extraction is approved
the Phase-0 boundary is repeated consistently	the system is production or safety ready

WHY THIS MATTERS

A falsifiable architecture is useful before execution because it tells implementers exactly what must be built and tells reviewers exactly what evidence would reject it.

WHAT REMAINS TO BE DONE

The final manuscript and the final runtime are different projects

REMAINING WORK E-22

Track	Next obligation	Completion signal
book	Epilogue, Appendices A-H, glossary, notation, sources, index	complete back matter
editorial	consistency audit and page-aware final layout	no drift or unresolved references
repository	implement named solution and pin package versions	clean checkout build
runtime	Compose/Kubernetes execution and final scenario	RunId-bound evidence bundle
failure	execute all mandatory campaigns	conservation and recovery receipts
publication	exact artifact paths, build identity and reproduction commands	final checkpoint

TWO HONEST ENDPOINTS

The book may become a complete reference manuscript while runtime evidence remains Pending. It may claim demonstrated runtime behaviour only after the repository track produces retained artifacts.

THE PHASE-0 BOUNDARY

The demonstrator teaches decisions; it does not operate a plant

NEXUS-1 uses a nuclear-plant-inspired domain because it forces the architecture to respect causality, evidence, time, responsibility and the cost of ambiguity. That seriousness must never be confused with operational authority. The software remains disconnected from a real facility and is not safety-class, certified or suitable for control.

EXPLICIT NON-CLAIMS E-23

Not claimed	Why it stays outside the conclusion
safety-class behaviour	no safety lifecycle, certification or operational integration
production platform readiness	no retained operations, capacity, HA, backup or support evidence
scientific correctness of diagnosis	the demonstrator proves software semantics, not physical truth
regulatory compliance	workflow examples are educational
AI decision authority	deterministic NEXUS-1 policy remains authoritative; AI may advise/explain

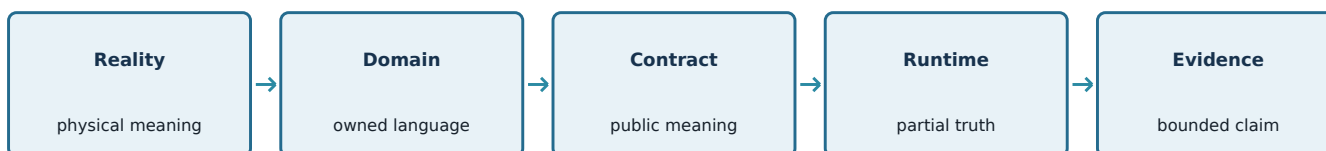
STANDING BOUNDARY

The domain raises the standard of thought. It does not expand the software's authority.

POSITION IN THE NEXUS-1 SERIES

The subject did not change when the system became distributed

From Grid to Core began with physical meaning. The domain, schema, entity, context, contract and container volumes repeatedly asked who may state a fact and how that fact can be challenged. From Flow to Services asked which boundaries deserved distribution. This volume asked whether those boundaries could preserve their meaning when truth became delayed and failure became partial.

SERIES REASONING CONTINUITY**SERIES THESIS**

The series is not a catalogue of technologies. It is one sustained engineering inquiry into whether a system can remain understandable as its representations, responsibilities and uncertainties multiply.

THE ANSWER

Yes - but only if uncertainty becomes part of the design

NEXUS-1 remains understandable in the specified distributed model because each owner keeps its own truth, public facts carry explicit identity and causation, projections expose delay, failures preserve intermediate state, security is rechecked at the owner, and evidence is not allowed to borrow certainty from another run. Remove those disciplines and the same network becomes an architecture of plausible stories.

CONDITIONS OF UNDERSTANDABILITY E-24

Condition	Without it
explicit ownership	two services may claim or mutate the same truth
versioned public language	private implementation shapes leak across teams
owner-local atomicity	committed truth and publication intent diverge
idempotent consumption	transport repetition becomes business repetition
freshness-bearing observations	old state impersonates current truth
failure evidence	recovery remains a hopeful narrative
bounded claims	architecture presentation outruns implementation

FINAL JUDGEMENT

One coherent engineering model survived a controlled transition

The architecture did not survive because every original candidate became a service. It survived because the design was willing to keep Audit, Compliance, RL Advisory and protected analytical capabilities inside modules when independent operational evidence was absent. It did not survive because every runtime test passed. It survived because the proof contract makes an unsupported pass impossible.

FINAL JUDGEMENT LEDGER E-25

Question	Conclusion
Did domain language survive distribution?	YES - in the specified ownership and contract model
Did local certainty remain bounded?	YES - transactions and publication intent are owner-local
Did delayed truth become explicit?	YES - watermark, gaps and age are part of the read contract
Did failure become reviewable?	YES - campaigns and conservation oracles are specified
Did the reference runtime prove these claims?	NOT YET - retained execution evidence is absent
Is production readiness established?	NO - explicitly outside the present evidence

FINAL VERDICT

MANUSCRIPT COMPLETE THROUGH EPILOGUE | ARCHITECTURE SPECIFIED | RUNTIME EVIDENCE PENDING |
PRODUCTION READINESS NOT CLAIMED

HANDOFF TO THE BACK MATTER

The argument closes; the reference apparatus now begins

The remaining pages will not introduce new architectural claims. They will make the existing ones easier to navigate, reproduce and challenge. The appendices consolidate the solution map, service boundaries, public contracts, database ownership, failures, trust boundaries, commands and technical-material traceability. The glossary, notation, sources and index then make the volume usable as a reference.

NEXT SEQUENCE E-26

Next item	Purpose
Appendix A - Solution and Project Map	one navigable repository and dependency topology
Appendix B - Service Catalogue	one final boundary and responsibility inventory
Appendix C - API and Event Contract Catalogue	one versioned public-language reference
Appendices D-H	data, failures, security, reproduction and traceability
reference apparatus	glossary, notation, sources and working index
final audits	consistency and reproduction checkpoints

CLOSING RULE

A distributed system becomes understandable not when uncertainty disappears, but when ownership, time, failure and evidence prevent uncertainty from pretending to be certainty.

THE BOOK'S LAST WORD

Evidence is what keeps engineering honest

This volume ends without a fabricated green badge. That is not an unfinished conclusion. It is the practical expression of its thesis: engineering remains credible only when the strength of the wording never exceeds the strength of the evidence. The architecture now offers a complete path from service candidate to runtime proof; implementation must still walk that path.

LAST WORD

NEXUS-1 has not eliminated uncertainty. It has made uncertainty owned, timed, observable, challengeable and incapable of silently becoming truth.

APPENDIX A

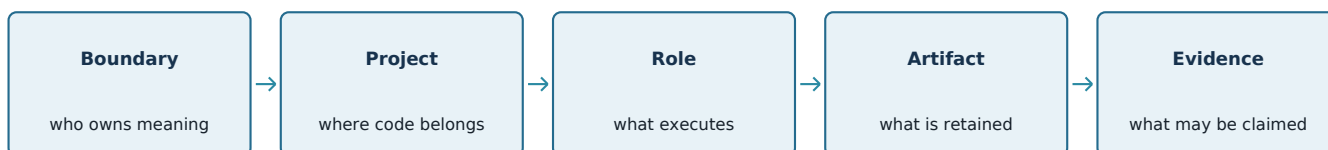
Solution and Project Map

A normative, navigable map from the NEXUS-1 architectural decisions to solution projects, runtime roles, owned stores, tests, deployment material and retained evidence - without pretending that a proposed path already exists in the repository.

APPENDIX DECISION

Use one map with two independent coordinates: architectural placement and repository presence.

AlarmManagement, RootCause and Reporting are admitted owners; Query BFF is edge only; Audit, Compliance and RL Advisory remain modules or candidates. Every physical path remains PENDING until a retained inventory proves it.

HOW TO READ THE MAP**APPENDIX A CLAIM BOUNDARY**

Question	Answer
Is this the final target topology?	YES - NORMATIVE FOR THE COMPANION IMPLEMENTATION
Does every named path exist now?	PENDING - REPOSITORY INVENTORY NOT ATTACHED
Does the target repository build and run?	PENDING - BUILD/RUNTIME RECEIPTS NOT ATTACHED
Does this appendix activate new services?	NO - IT CONSOLIDATES FINAL BOOK DECISIONS

STANDING RULE

Nothing claims to exist that does not. A precise target map is a specification; only the retained repository inventory can promote a path to PRESENT.

PURPOSE AND SCOPE

The map is an executable boundary contract

A repository tree is useful only when it agrees with authority, runtime isolation and evidence. This appendix therefore maps each capability through six questions: who owns the language, which assembly contains the rule, which executable role hosts it, which database it may write, which tests challenge it and which artifact proves the claim.

MAP DIMENSIONS A-1

Dimension	Map answers	Failure prevented
semantic	which boundary owns a decision or fact	shared domain truth
source	which project may contain the code	layer leakage
runtime	which process executes the project	hidden deployment coupling
data	which principal and migrator touch a store	cross-owner writes
verification	which test family challenges the claim	green-but-irrelevant suites
evidence	which immutable output carries the result	unsupported completion claims

PHASE-0 BOUNDARY

The companion is an educational demonstrator, not safety-class or operational software. The map makes the implementation falsifiable; it does not confer production readiness.

STATUS GRAMMAR

Placement and presence never collapse into one status

TWO-AXIS LEDGER A-2

Axis	Allowed values	Meaning
architectural placement	CORE OWNER / DERIVED OWNER / EDGE / MODULE / CANDIDATE / EXCLUDED	where the capability belongs
repository presence	PRESENT / PARTIAL / PENDING / ABSENT-BY-DESIGN	what the retained tree proves
execution evidence	PASSED / FAILED / PENDING / NOT CLAIMED	what a named run established

PROMOTION RULES A-3

From	To	Required evidence
PENDING path	PRESENT	inventory entry + exact relative path + content digest
PARTIAL project	PRESENT	declared outputs, references and required assets all found
architecture SPECIFIED	runtime PASSED	same-build restore/build/test/deploy/run evidence
candidate	admitted runtime role	authority, economics, image, principal, migration, probes and failure proof

FAIL-CLOSED DEFAULT

Unknown, missing, stale or cross-build evidence maps to PENDING. It never maps to Passed by omission.

RECONCILIATION WITH THE PROJECT DESCRIPTION

The original candidate tree is refined by the completed argument

The project description deliberately began with five strong service candidates. Chapters 31-60 later distinguished semantic completeness from deployment admission. Appendix A preserves that history while making the final topology unambiguous.

CANDIDATE-TO-FINAL RECONCILIATION A-4

Original location	Final target placement	Reason
Services/AlarmManagement	Services/AlarmManagement	admitted owner with API, worker and database
Services/RootCause	Services/RootCause	admitted owner with API, worker and database
Services/Reporting	Services/Reporting	admitted derived owner with worker, read API and database
Services/Audit	Modules/Audit	semantics specified; independent runtime evidence absent
Services/Compliance	Modules/Compliance	workflow specified; independent runtime evidence absent
optional RL service	Modules/RIAdvisory	public output not activated
OperatorConsole.Bff	Edge/Query.Bff	authorized composition; no domain truth

NO SILENT DEMOTION

Audit and Compliance remain first-class architectural material and service candidates. Moving them under Modules records their current operational status; it does not erase their language, policies or tests.

REPOSITORY ROOT

One publication coordinate, one companion root

NORMATIVE REPOSITORY COORDINATE A-1

upstream: <https://github.com/gregory82gr/Nexus-1-phase-0>

```
artifacts/  
  from-services-to-runtime/  
    Nexus1.Runtime.sln  
    global.json  
    Directory.Build.props  
    Directory.Packages.props  
  src/  
  tests/  
  deploy/  
  docs/  
  eng/  
  artifacts/
```

ROOT CONTRACT A-5

Item	Responsibility	Claim state
Nexus1.Runtime.sln	complete build/test project inventory	PENDING
global.json	selected SDK feature band	PENDING
Directory.Build.props	shared compiler/analyzer policy	PENDING
Directory.Packages.props	central package version ownership	PENDING
eng/	deterministic orchestration and validation	PENDING

PATH RULE

All later paths are relative to artifacts/from-services-to-runtime/. The spelling is normative; platform-specific absolute paths are evidence metadata, not source identity.

FROM SERVICES TO RUNTIME

TOP-LEVEL SOLUTION MAP

Source, proof, deployment and evidence have separate homes

NORMATIVE TARGET TREE A-2

```
src/                # product code and technical building blocks
  BuildingBlocks/
  Contracts/
  Services/
  Edge/
  Modules/
  Hosts/
tests/              # claim-focused verification projects
  Unit/ Architecture/ Component/ Contract/
  Integration/ EndToEnd/ FailureScenarios/ Deployment/
deploy/              # Compose and Kubernetes projections
docs/                # ADRs, catalogues, diagrams, runbooks, evidence specs
eng/                 # restore/build/test/package/deploy/evidence entry points
artifacts/           # generated, immutable run outputs; never hand-authored truth
```

TOP-LEVEL OWNERSHIP A-6

Folder	May contain	Must not contain
src	buildable product projects	run receipts or secrets
tests	oracles, fixtures and harnesses	production business implementations
deploy	environment projections and policies	application source
docs	decisions and human navigation	fabricated execution results
eng	repeatable commands and validators	hidden one-off state
artifacts	generated evidence bundles	source-of-truth configuration

SERVICE PROJECT TEMPLATE

Every admitted owner exposes the same dependency shape

OWNER PROJECT TEMPLATE A-3

```
src/Services/<Owner>/
  Nexus1.<Owner>.Domain/
  Nexus1.<Owner>.Application/
  Nexus1.<Owner>.Infrastructure/
  Nexus1.<Owner>.Api/
  Nexus1.<Owner>.Worker/
  Nexus1.<Owner>.Migrator/
```

PROJECT RESPONSIBILITIES A-7

Project	Owns	May depend on
Domain	aggregates, values, policies, private events	small abstractions only
Application	use cases, ports, outcomes, transaction intent	Domain + producer contracts where needed
Infrastructure	EF, broker, HTTP, telemetry adapters	Application + Domain + contracts
Api	HTTP edge and composition root	Application + Infrastructure
Worker	message/scheduled execution composition	Application + Infrastructure
Migrator	schema authority executable	Infrastructure migration assembly only

DEPLOYMENT RULE

Api, Worker and Migrator are executable roles. Domain, Application and Infrastructure are assemblies, never independently deployed services.

ALARMMANAGEMENT MAP

The alarm owner begins the distributed slice

ALARMMANAGEMENT PROJECTS A-8

Project	Primary material	Runtime consumer
AlarmManagement.Domain	observations, flood policy, decision invariants	API and worker through Application
AlarmManagement.Application	raise/observe commands, outcomes, ports	alarm-api / alarm-worker
AlarmManagement.Infrastructure	AlarmManagementDb, outbox, broker adapters	alarm-api / alarm-worker / migrator
AlarmManagement.Contracts	AlarmFloodDetected.v1 and owned HTTP DTOs	external consumers
AlarmManagement.Api	write/read HTTP surface	alarm-api
AlarmManagement.Worker	outbox dispatch and owner background work	alarm-worker
AlarmManagement.Migrator	database migrations only	alarm-migrator job

AUTHORITY BOUNDARY

AlarmManagement may publish the accepted meaning of an alarm flood. It cannot create RootCause cases, Reporting projections or candidate-module effects inside its transaction.

ROOTCAUSE MAP

The verdict owner consumes facts without borrowing the alarm database

ROOTCAUSE PROJECTS A-9

Project	Primary material	Runtime consumer
RootCause.Domain	case, evidence membership, verdict policy	API and worker through Application
RootCause.Application	open/attach/issue use cases and ports	rootcause-api / rootcause-worker
RootCause.Infrastructure	RootCauseDb, inbox/outbox, broker adapters	API / worker / migrator
RootCause.Contracts	CaseOpened.v1, VerdictIssued.v1, owned HTTP DTOs	Reporting and candidate modules
RootCause.Api	owner HTTP surface	rootcause-api
RootCause.Worker	AlarmFlood consumer + outbox dispatch	rootcause-worker
RootCause.Migrator	database migrations only	rootcause-migrator job

NO SHARED TRUTH

RootCause consumes sufficient public facts and typed remote identities. A project reference, DbContext or SQL connection to AlarmManagementDb is forbidden.

REPORTING MAP

The derived owner stores observations and exposes their age

REPORTING PROJECTS A-10

Project	Primary material	Runtime consumer
Reporting.Domain	ledger/checkpoint/gap/freshness invariants	worker and API through Application
Reporting.Application	projection handlers and freshness queries	projection-worker / projection-api
Reporting.Infrastructure	ReportingDb, inbox, replay and query adapters	worker / API / migrator
Reporting.Contracts	freshness-bearing read DTOs only	Query BFF
Reporting.Worker	ordered fact consumption and repair	projection-worker
Reporting.Api	read-only projection HTTP surface	projection-api
Reporting.Migrator	database migrations only	reporting-migrator job

READ-ONLY API

Reporting.Api may read ReportingDb through its application ports. It has no command that manufactures upstream alarm or verdict truth and no endpoint that mutates source-authoritative state.

EDGE MAP

Query BFF composes authorized views and owns no truth

EDGE PROJECT SHAPE A-4

```
src/Edge/Query.Bff/  
  Nexus1.QueryBff.Application/  
  Nexus1.QueryBff.Infrastructure/  
  Nexus1.QueryBff.Api/  
  
# deliberately absent  
Nexus1.QueryBff.Domain/  
Nexus1.QueryBff.Persistence/  
Nexus1.QueryBff.Migrator/
```

BFF CONSTRAINTS A-11

May	Must not
authenticate operator and enforce composition authorization	own an aggregate or domain database
call owner APIs with bounded budgets	read owner databases directly
return partial/unknown/freshness coordinates	rename stale projection data as current truth
emit edge telemetry and security events	publish producer-owned business facts

PLACEMENT

Query BFF is one executable role: query-bff. Its independent deployment is justified by client composition and trust concerns, not by domain authority.

PUBLIC CONTRACT PROJECTS

Contracts are producer-owned language, not a shared domain model

ACTIVATED CONTRACT TREE A-5

```
src/Contracts/
  Nexus1.Messaging.Contracts/      # envelope primitives only
  AlarmManagement.Contracts/      # producer: AlarmManagement
  RootCause.Contracts/            # producer: RootCause
  Reporting.Contracts/            # producer: Reporting read API

docs/catalogs/candidates/
  audit-contract-candidate.yaml
  compliance-contract-candidate.yaml
  recommendation-generated-candidate.yaml
```

CONTRACT CONTENT RULE A-12

Allowed	Forbidden
versioned DTOs, event records, enum wire values	aggregates, repositories or DbContexts
schema identity and compatibility metadata	consumer-specific persistence rows
envelope primitives and canonical serialization profile	service-internal domain events
owned HTTP request/response grammar	shared business service interfaces

CANDIDATE RULE

Audit, Compliance and RecommendationGenerated contract sketches remain documentation candidates until a producer is admitted. A candidate schema is not a compiled public promise.

BUILDING BLOCKS

Shared code stays smaller than shared understanding

PERMITTED BUILDING BLOCKS A-13

Project	Permitted material	Admission test
Nexus1.Abstractions	Clock, ID generation, Result primitives	technical and policy-free
Nexus1.Messaging	envelope, canonical bytes, dispatch abstractions	no producer payload meaning
Nexus1.Observability	trace/log/metric helpers	no business decision
Nexus1.Hosting	health/lifecycle/configuration plumbing	no service-specific option
Nexus1.Testing	test-only fixtures and evidence helpers	never referenced by product projects

FORBIDDEN SHARED MATERIAL A-14

Material	Correct owner
Alarm, RootCauseCase, Verdict, ComplianceReview	their semantic module/service
DbContext or repository base with business assumptions	owner Infrastructure
one shared integration event union	producer-owned contract project
global retry/resilience policy	caller and operation boundary
shared database transaction coordinator	excluded from architecture

EXTRACTION SMELL

If changing one business rule requires a BuildingBlocks release, the code is not a building block; it is displaced domain ownership.

FROM SERVICES TO RUNTIME

MODULES AND CANDIDATES

Semantically complete does not mean independently deployed

CANDIDATE MODULE TREE A-6

```
src/Modules/
  Audit/
    Nexus1.Audit.Domain/
    Nexus1.Audit.Application/
    Nexus1.Audit.Infrastructure/
  Compliance/
    Nexus1.Compliance.Domain/
    Nexus1.Compliance.Application/
    Nexus1.Compliance.Infrastructure/
  RLAdvisory/
    Nexus1.RLAdvisory.Application/
    Nexus1.RLAdvisory.Infrastructure/
```

CANDIDATE PLACEMENT A-15

Capability	Current placement	Missing for independent role
Audit	MODULE / SERVICE CANDIDATE	image, principal, store/migration decision, probes, fault receipts
Compliance	MODULE / SERVICE CANDIDATE	same operational ownership proof
RL Advisory	OPTIONAL MODULE	activated public output, data authority and extraction economics
DigitalTwin / Instrumentation / ReactorFleet	PROTECTED MODULES	outside Phase-0 distributed slice

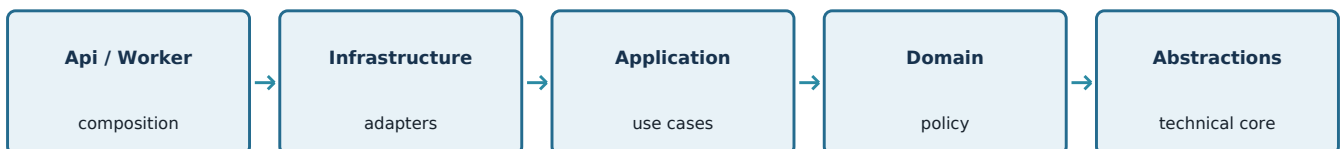
NO ACCIDENTAL HOST

Candidate modules have no Api, Worker or Migrator project in the admitted solution map. Adding one is an architectural change that requires a decision record and admission evidence.

PROJECT DEPENDENCY DIRECTION

References point inward; public contracts cross owners

ALLOWED DEPENDENCY DIRECTION



REFERENCE RULES A-16

From	May reference	Must not reference
Domain	technical abstractions	Application, Infrastructure, contracts, hosts
Application	own Domain; required producer contracts	Infrastructure or executable hosts
Infrastructure	own Application/Domain; external contracts	another owner's Infrastructure
Api / Worker	own Application + Infrastructure	another owner's Domain/DbContext
contract project	envelope primitives	any Domain/Application/Infrastructure
test project	explicit system under test	production reverse reference

CROSS-OWNER RULE

One owner learns another owner's public language through its producer-owned contract assembly or captured contract corpus - never through a reference to private code.

FORBIDDEN PROJECT EDGES

The solution graph must reject convenient shortcuts

FORBIDDEN-EDGE CATALOGUE A-17

Forbidden edge	Why	Gate
RootCause.* -> AlarmManagement.Infrastructure	shared database/persistence authority	architecture test
Reporting.* -> RootCause.Domain	private semantic coupling	architecture test
QueryBff.* -> any owner Infrastructure	database bypass	reference + namespace test
Contracts -> Services or Modules	public grammar depends on implementation	project graph test
Domain -> EF/RabbitMQ/ASP.NET assemblies	policy depends on adapter technology	package/namespace test
product -> Nexus1.Testing	test code leaks into runtime	reference test
core hosts -> candidate module hosts	silent topology expansion	solution/runtime inventory join

REVIEW POLICY

A forbidden edge fails the build even when all behaviour tests pass. It is an architectural defect, not a style warning.

NAMESPACES AND ASSEMBLY IDENTITIES

Names expose owner, layer and public status

NAMING GRAMMAR A-18

Kind	Pattern	Example
private assembly	Nexus1.<Owner>.<Layer>	Nexus1.RootCause.Application
public contract	<Owner>.Contracts	AlarmManagement.Contracts
runtime role	lowercase-kebab	rootcause-worker
database	<Owner>Db	RootCauseDb
migration job	<owner>-migrator	reporting-migrator
container image	nexus1/<role>:<digest-bound-tag>	nexus1/alarm-api:<tag>
test project	Nexus1.<Surface>.Tests	Nexus1.Contracts.Tests

NAMESPACE RULES A-19

Rule	Rationale
folder and root namespace agree	inventory and assembly scanning stay deterministic
no .Shared business namespace	ownership remains searchable
V1 appears at public grammar boundary	wire compatibility is explicit
Internal marks non-public adapters/helpers	surface does not grow accidentally
Generated code has a dedicated namespace	source provenance is visible

RUNTIME-ROLE MAP

Seven roles execute; three jobs migrate

ADMITTED RUNTIME ROLES A-20

Role	Entry project	Writes	Public surface
alarm-api	AlarmManagement.Api	AlarmManagementDb	HTTP
alarm-worker	AlarmManagement.Worker	AlarmManagementDb	broker/health
rootcause-api	RootCause.Api	RootCauseDb	HTTP
rootcause-worker	RootCause.Worker	RootCauseDb	broker/health
projection-worker	Reporting.Worker	ReportingDb	broker/health
projection-api	Reporting.Api	none through read principal	HTTP
query-bff	QueryBff.Api	none	HTTP

MIGRATION JOBS A-21

Job	Migration assembly	Target
alarm-migrator	AlarmManagement.Infrastructure	AlarmManagementDb
rootcause-migrator	RootCause.Infrastructure	RootCauseDb
reporting-migrator	Reporting.Infrastructure	ReportingDb

INVENTORY ASSERTION

The final core topology contains exactly these seven long-running roles and three bounded migration jobs. Audit, Compliance and RL Advisory are not silently counted.

DATABASE AND PRINCIPAL MAP

Each store has one writer family and separate migration authority

STORE OWNERSHIP A-22

Store	Runtime writer	Reader	Migration authority
AlarmManagementDb	alarm-api / alarm-worker	owner-local read paths	alarm-migrator principal
RootCauseDb	rootcause-api / rootcause-worker	owner-local read paths	rootcause-migrator principal
ReportingDb	projection-worker	projection-api read principal	reporting-migrator principal

DATABASE PROHIBITIONS A-23

Prohibition	Verification
no cross-database query, view, synonym or foreign key	catalog/dependency inventory
no runtime DDL grant	provider principal test
no migrator account in API/worker secrets	deployment/config inventory
no Reporting write from its API	permission + endpoint tests
no shared application login across owners	provider principal inventory

EVIDENCE BOUNDARY

The project map specifies credentials and authority. Provider-enforced grants remain PENDING until the exact environment receipts are retained.

TEST SOLUTION MAP

Test folders name proof surfaces, not confidence levels

NORMATIVE TEST TREE A-7

```
tests/
  Unit/           # deterministic Domain/Application claims
  Architecture/   # project, package, namespace and topology rules
  Component/      # owner + real provider/broker adapter boundary
  Contract/       # HTTP/event corpora and compatibility mutations
  Integration/    # several real adapters in one controlled profile
  EndToEnd/       # admitted seven-role slice
  FailureScenarios/ # injection, conservation and recovery
  Deployment/     # Compose/Kubernetes manifest equivalence
```

TEST PLACEMENT RULE A-24

Family	May fake	Must be real
Unit	all ports and time	Domain/Application policy
Architecture	runtime execution	compiled graph/inventory
Component	unrelated remote systems	named provider/adapter
Contract	producer runtime	canonical corpus and reader/writer
EndToEnd	external identity issuer when profile permits	production composition paths
FailureScenarios	nothing named by the disturbance	fault, effect and recovery oracle

PROOF-SURFACE OWNERSHIP

Every critical claim has one primary home and one cross-check

CLAIM-TO-PROJECT MAP A-25

Claim	Primary test project	Cross-check
flood decision is deterministic	AlarmManagement.Unit.Tests	component persistence
duplicate flood opens one case	RootCause.Component.Tests	end-to-end broker count
public bytes remain compatible	Contracts.Tests	captured-body fingerprint
projection watermark never skips a gap	Reporting.Unit/Component.Tests	failure scenario
BFF exposes stale/partial truth	QueryBff.Contract.Tests	end-to-end delayed projection
only owners write their databases	Security/Component.Tests	provider grant inventory
topologies are equivalent	Deployment.Tests	runtime inventory manifest

ZERO-DISCOVERY GATE

A test project with zero discovered tests fails the evidence pipeline. An empty green suite cannot prove an architectural claim.

DEPLOYMENT PROJECT MAP

Compose and Kubernetes are two projections of one role inventory

DEPLOYMENT TREE A-8

```

deploy/
  topology/runtime-inventory.yaml
  compose/
    compose.yaml
    compose.override.development.yaml
  kubernetes/
    base/
      overlays/phase0/
    policies/
      network/
      resources/
      disruption/
    schemas/
      runtime-inventory.schema.json

```

PROJECTION EQUIVALENCE A-26

Coordinate	Compose	Kubernetes	Must agree
role	service	Deployment/Job	logical role id
image	image digest	image digest	immutable digest
config	env/secret refs	ConfigMap/Secret refs	configuration identity
probe	healthcheck	startup/live/ready	meaning and bounds
dependency	condition/order	policy/init readiness	admission semantics
resource	limits profile	requests/limits	declared budget

SINGLE SOURCE

runtime-inventory.yaml is the logical topology source. Compose and Kubernetes are validated projections; neither becomes an undocumented second architecture.

HOST PROJECTS

Operational executables are explicit and narrowly authorized

HOSTS TREE A-9

```

src/Hosts/
  Nexus1.RuntimeVerifier/
  Nexus1.FinalDemonstrator/
  Nexus1.EvidenceCollector/

# service-owned migrators remain under each service;
# there is no universal DbContext-owning migration host.

```

HOST RESPONSIBILITIES A-27

Host	May do	Must not do
RuntimeVerifier	read health/topology/config identities	manufacture business success
FinalDemonstrator	issue one scenario and observe convergence	write databases directly
EvidenceCollector	copy/redact/hash named artifacts	infer missing evidence
service migrator	apply one owner's migrations	start long-running business work

MIGRATIONRUNNER REFINEMENT

The project description proposed one MigrationRunner host. The final map uses three service-owned migrator entry projects so schema authority cannot collapse into a universal privileged process.

DOCUMENTATION AND EVIDENCE MAP

Human navigation and machine evidence are related but not interchangeable

DOCUMENTATION TREE A-10

```
docs/
  adr/                # accepted/rejected architectural decisions
  catalogs/           # owners, roles, contracts, stores, failures
  diagrams/           # source + rendered forms
  runbooks/           # operator actions and reversal conditions
  evidence/           # schemas, claim catalogue, redaction rules
  traceability/       # chapter -> asset -> test -> artifact links

artifacts/evidence/<RunId>/
  manifest.json build/ tests/ runtime/ database/
  broker/ http/ telemetry/ failures/ security/ summary/
```

DOCUMENTATION TRUTH RULES A-28

Artifact	Authority
ADR	why a decision was made; never proof that code follows it
catalogue	declared inventory; must be compared with discovered inventory
diagram	view over a machine-checkable map; drift is a failure
runbook	bounded procedure; execution receipt proves its use
evidence manifest	join of immutable run artifacts; no hand-edited Passed state

ENGINEERING ENTRY POINTS

One orchestration layer records every consequential command

ENGINEERING TREE A-11

```
eng/
  Nexus.Build/        # restore, build, test, pack
  Nexus.Inventory/    # discover solution/projects/roles/stores
  Nexus.Contracts/    # corpora, fingerprints, compatibility
  Nexus.Deployment/   # render and compare projections
  Nexus.Evidence/     # manifest, joins, freshness, redaction
  scripts/
    nexus.ps1
    nexus.sh
```

REQUIRED COMMAND IDENTITIES A-29

Entry	Output identity
restore	SDK + lock files + package graph
build	revision + binlog + assembly digests
test	nonzero discovery + result/log/coverage artifacts
inventory	projects + references + roles + principals + paths
deploy	rendered manifests + image/config identities
demonstrate	RunId-bound scenario observations
verify-evidence	claim verdicts and Missing collection

REPRODUCTION BOUNDARY

Appendix G will fix exact command syntax. Appendix A fixes where command implementations and outputs belong.

ARTIFACT OUTPUT MAP

Generated output is partitioned by run and evidence plane

RUN ARTIFACT TREE A-12

```
artifacts/evidence/<RunId>/
  manifest.json
  identity/{source,build,packages,images,config,migrations}.json
  build/{restore.log,build.binlog,inventory.json}
  tests/{unit,architecture,component,contract,e2e,failure,deployment}/
  runtime/{topology,health,lifecycle}/
  database/{principals,transactions,convergence}/
  broker/{topology,deliveries,settlements,quarantine}/
  http/{requests,responses,authorization}/
  telemetry/{logs,metrics,traces}/
  summary/{claims.json,missing.json,verdict.json}
```

OUTPUT INVARIANTS A-30

Invariant	Failure result
one RunId and BuildIdentity across joined evidence	cross-run join rejected
raw evidence immutable after capture	digest mismatch -> FAILED
derived summaries name their inputs	untraceable summary -> PENDING
secret values absent; references retained	leak -> security failure
missing plane listed explicitly	final verdict cannot be Passed

VERSION AND PACKAGE OWNERSHIP

Central versions do not create central business ownership

VERSION LOCATIONS A-31

Version	Owner file	Consumer
SDK	global.json	all .NET projects
NuGet packages	Directory.Packages.props	projects through PackageReference
contract schema	producer contract metadata/catalogue	producer + consumers
database migration	service migration assembly/history	one owner store
container image	build manifest/registry digest	deployment projections
configuration schema	role-owned options + catalogue	role startup validator
book evidence schema	docs/evidence/schemas	collector/verifier

NO FLOATING PROOF

Evidence records resolved SDK, package, image, configuration and migration identities. A branch name, latest tag or unpinned range is not a reproducible build coordinate.

BUILD GRAPH

Compilation order follows dependency order, not deployment order

DETERMINISTIC BUILD GRAPH



BUILD-STAGE GATES A-32

Stage	Required gate
restore	locked graph, approved sources, no unowned version override
compile	warnings policy, analyzers, deterministic output
architecture	project/package/namespace/runtime inventory agreement
unit/contract	nonzero discovery and canonical corpora
component	named provider profile and isolated resources
package/image	digest and provenance manifest
deployment	logical inventory equals rendered projections

PARALLELISM RULE

Independent owner projects may build and test in parallel after shared technical and contract inputs are fixed. Runtime startup order does not justify a source dependency.

ARCHITECTURE TESTS

The map becomes a build-breaking specification

EXECUTABLE ASSET A-1 - GRAPH POLICY SKETCH

```
[Fact]
public void Product_graph_obeyes_owner_and_layer_rules()
{
    var graph = SolutionGraph.Discover(RepositoryRoot.Find());

    graph.RequireNoCycle();
    graph.Forbid("*.Domain", "/*.Infrastructure", "/*.Api", "/*.Worker");
    graph.Forbid("*.Contracts", "Nexus1.*.Domain", "Nexus1.*.Application");
    graph.ForbidCrossOwnerPrivateReferences();
    graph.RequireExecutables(RuntimeInventory.ExpectedEntryProjects);
    graph.RequireNoUnexpectedExecutables();
}
```

ARCHITECTURE SUITE A-33

Test	Rejects
ProjectReferencePolicy	forbidden layer/owner edges
PackagePolicy	technology package in Domain or unowned override
NamespacePolicy	Shared business namespaces and public leakage
RuntimeInventoryPolicy	missing or extra executable role
DatabaseOwnershipPolicy	cross-owner connection/migration registration
ContractOwnershipPolicy	consumer-defined copy masquerading as producer grammar

SOLUTION INVENTORY VALIDATOR

Declared and discovered trees must agree exactly

EXECUTABLE ASSET A-2 - FAIL-CLOSED INVENTORY JOIN

```
public static InventoryVerdict Compare(
    TargetMap target,
    DiscoveredSolution actual)
{
    var missing = target.RequiredProjects.Except(actual.Projects).Order().ToArray();
    var extra = actual.ProductProjects.Except(target.AllowedProjects).Order().ToArray();
    var wrongEdges = actual.References.Where(target.IsForbidden).Order().ToArray();
    var wrongRoles = target.RuntimeRoles.SymmetricExcept(actual.RuntimeRoles);

    return missing.Length == 0 && extra.Length == 0 &&
        wrongEdges.Length == 0 && wrongRoles.Count == 0
        ? InventoryVerdict.Passed(actual.Identity)
        : InventoryVerdict.Failed(missing, extra, wrongEdges, wrongRoles);
}
```

DISCOVERY SOURCES A-34

Coordinate	Discovered from
projects and references	solution + evaluated MSBuild graph
packages	assets/lock files after restore
executables	OutputType + composition entry points
runtime roles	runtime-inventory.yaml
stores/migrations	registrations + migration assemblies + deploy secrets
tests	test SDK discovery output

NO DIRECTORY-ONLY PROOF

A folder name is insufficient. The validator evaluates the build graph and joins it to runtime and database inventories.

PRESENCE LEDGER

The target is complete; physical presence remains unproved

CURRENT PATH STATUS A-35

Target group	Architectural state	Repository state
solution/build roots	REQUIRED	PENDING
AlarmManagement projects	ADMITTED CORE OWNER	PENDING
RootCause projects	ADMITTED CORE OWNER	PENDING
Reporting projects	ADMITTED DERIVED OWNER	PENDING
Query BFF projects	ADMITTED EDGE	PENDING
Audit / Compliance modules	MODULE / CANDIDATE	PENDING
RL Advisory module	OPTIONAL; OUTPUT NOT ACTIVATED	PENDING
tests/deploy/docs/eng	REQUIRED PROOF APPARATUS	PENDING

WHY EVERY PATH IS PENDING

No retained repository tree, evaluated project graph or build identity is attached to this volume. The appendix refuses to infer physical presence from compile-oriented listings in the manuscript.

MAP-DRIFT CONTROL

A changed tree is an architectural event when authority moves

CHANGE CLASSIFICATION A-36

Change	Required action
rename without semantic change	update target map, links, inventory snapshot and traceability
new project inside existing layer	prove dependency and packaging need
new executable role	ADR + topology/economics/security/operations admission
new database or writer	ownership/migration/principal/failure review
new public contract	producer ownership + version + corpus + compatibility gate
module becomes service	full candidate-to-runtime admission evidence
diagram changes only	reject until machine map changes first

DRIFT VERDICTS A-37

Observed state	Verdict
map and inventory identical	CONSISTENT
declared project missing	FAILED
unexpected product executable	FAILED
candidate executable appears in deploy only	FAILED - SILENT ADMISSION
evidence too old for current build	PENDING

CHAPTER-TO-MAP TRACEABILITY

The solution tree is the index of the book's engineering claims

PRIMARY TRACE MAP A-38

Book material	Primary target area
Chapters 1-12 - boundary/layers/config/testing	src structure + Architecture/Unit tests
Chapters 13-18 - database extraction	service Infrastructure/Migrator + Component tests
Chapters 19-30 - communication	Contracts + Messaging + Contract/Failure tests
Chapters 31-42 - end-to-end slice	Services/Edge/Modules + EndToEnd
Chapters 43-48 - security	identity/config/policies + Security evidence
Chapters 49-54 - observability/resilience	Observability + runtime/failure artifacts
Chapters 55-60 - delivery/demonstrator	deploy + eng + Hosts + evidence summary
Epilogue	claim catalogue and final Missing collection

APPENDIX H BOUNDARY

Appendix A provides the structural join. Appendix H will enumerate each technical asset, its claim, implementation path, test and evidence artifact.

ACCEPTANCE CHECKLIST

A repository satisfies Appendix A only when the map is discovered, not assumed

ACCEPTANCE GATES A-39

Gate	Required result
root	all normative roots and identity files present
graph	no cycles, forbidden references or unowned package overrides
owners	three owner families and one edge match the final decisions
roles	exactly seven long-running roles and three migrator jobs
contracts	producer-owned projects/corpora; no candidate activation
stores	three owners, separated principals and migration authority
tests	all required families present with nonzero discovery
deployment	Compose/Kubernetes projections equal logical inventory
evidence	immutable RunId/BuildIdentity bundle with explicit Missing
navigation	catalogues, diagrams and trace links resolve

CURRENT VERDICT

TARGET MAP COMPLETE | REPOSITORY PRESENCE PENDING | BUILD EVIDENCE PENDING | RUNTIME EVIDENCE PENDING | PRODUCTION READINESS NOT CLAIMED

ANTI-PATTERNS

Trees can conceal coupling as easily as they reveal it

REJECTED STRUCTURES A-40

Anti-pattern	Why rejected	Correction
one project per folder label	names without dependency enforcement	evaluated graph + tests
Common.Domain	shared business language loses owner	owner-local Domain projects
all DTOs in Shared.Contracts	producer and evolution authority disappear	producer-owned contracts
one universal Worker	failure/lifecycle ownership collapses	owner-specific workers
one privileged MigrationRunner	cross-store schema authority	service-owned migrators
BFF persistence cache as truth	edge becomes hidden database owner	freshness-bearing owner reads
candidate folders under Services	topology appears admitted	Modules + explicit status
screenshots as inventory	stale and not machine-checkable	manifest + digest + render

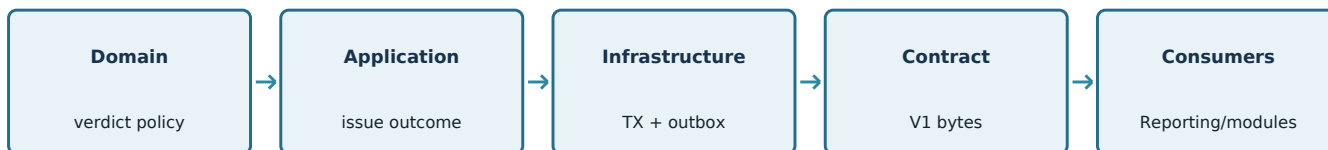
REVIEW QUESTION

If a path disappeared, which authority, runtime role, proof surface or evidence join would become ambiguous? If the answer is none, the path is likely accidental complexity.

WORKED NAVIGATION EXAMPLE

Trace one verdict fact through the complete map

ROOTCAUSEVERDICTISSUED.V1 NAVIGATION



NAVIGATION COORDINATES A-41

Question	Map destination
who may issue the verdict?	Nexus1.RootCause.Domain
who commits it and its public intent?	RootCause.Infrastructure -> RootCauseDb
who owns the public grammar?	RootCause.Contracts
which role publishes it?	rootcause-worker
who derives operator-visible state?	projection-worker -> ReportingDb
who may compose the response?	query-bff through Reporting.Api
where is idempotency challenged?	Component/Contract/EndToEnd/FailureScenarios
what proves the journey?	one RunId evidence bundle; currently PENDING

MAP VALUE

The reader can navigate from business meaning to source, process, store, consumer, test and evidence without granting any layer more authority than it owns.

APPENDIX A CHECKPOINT

The final solution map is coherent, bounded and falsifiable

WHAT APPENDIX A FIXES A-42

Coordinate	Final decision
core owners	AlarmManagement, RootCause, Reporting
edge	Query BFF; no domain truth or database
candidates/modules	Audit, Compliance, RL Advisory and protected capabilities
runtime	seven roles + three service-owned migration jobs
data	three owner databases with separated principals
public language	producer-owned activated contracts; candidates remain documentation
verification	eight named test families + zero-discovery gate
delivery	one logical inventory projected to Compose and Kubernetes
evidence	RunId/BuildIdentity bundle; physical implementation still PENDING

APPENDIX A VERDICT

SOLUTION MAP SPECIFIED | PROJECT GRAPH SPECIFIED | PHYSICAL REPOSITORY PRESENCE PENDING | EXECUTION EVIDENCE PENDING | PRODUCTION READINESS NOT CLAIMED

HANDOFF TO APPENDIX B

The tree is fixed; the next appendix inventories the services

Appendix A established where each kind of material belongs and which executables may exist. Appendix B will now catalogue each admitted owner, derived owner, edge and candidate module by purpose, language, commands, public contracts, dependencies, stores, failure modes, security obligations, telemetry and admission status.

NEXT SEQUENCE A-43

Next item	Purpose
Appendix B - Service Catalogue	one final responsibility and runtime-status ledger
Appendix C - API and Event Contract Catalogue	versioned public-language reference
Appendix D - Database Ownership and Migration Catalogue	writer, reader, principal and migration authority
Appendices E-H	failures, security, reproduction and traceability

CARRY-FORWARD RULE

A service entry in Appendix B must point back to this map. If it cannot name a valid project family, runtime placement and evidence status, it is not an admitted service.

APPENDIX A LAST WORD

A solution map is credible only when reality can disagree with it

The target is now precise enough to be discovered, compared and rejected. That is its purpose. The next implementation checkpoint must replace every PENDING presence entry with an observed path or an explicit absence - never with optimism.

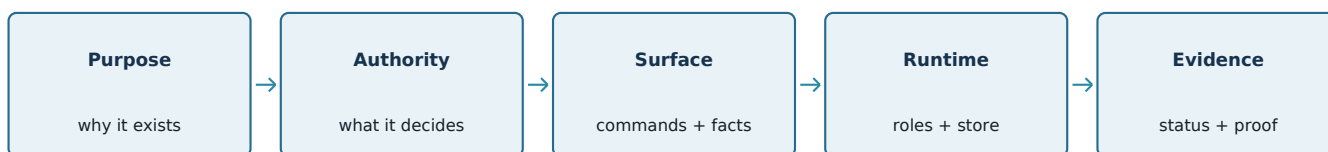
APPENDIX B

Service Catalogue

The final responsibility and runtime-status ledger for every admitted owner, derived owner, edge deployment, candidate service and protected module in the NEXUS-1 Phase-0 distributed slice.

APPENDIX DECISION

Catalogue service boundaries by authority, not by process count. AlarmManagement, RootCause and Reporting are admitted boundaries; Query BFF is an edge deployment; Audit, Compliance, RL Advisory and the durable Process Manager remain modules or candidates. Architectural admission is SPECIFIED. Repository presence and runtime proof remain PENDING.

CATALOGUE READING PATH**CATALOGUE BOUNDARY B-1**

Question	Answer
How many service boundaries are admitted?	3 - AlarmManagement, RootCause, Reporting
How many edge boundaries are admitted?	1 - Query BFF; owns no domain truth
How many long-running runtime roles are admitted?	7 - API/worker separation does not create more services
Are candidate modules deployed services?	NO - independent runtime admission remains absent

STANDING RULE

A detailed service card is an architectural specification. It becomes an observed runtime catalogue only when the same BuildIdentity and RunId bind inventory, deployment and execution receipts.

CATALOGUE PURPOSE

A service card is a compact ownership proof

The catalogue does more than name processes. Each entry must explain why independent distribution is justified, what language the boundary owns, which local transaction settles its decisions, what it publishes or consumes, how it fails, how it is secured and which evidence could falsify its status.

REQUIRED CARD DIMENSIONS B-2

Dimension	Required answer	Prevents
purpose	one bounded business or derived responsibility	generic service labels
authority	commands, invariants and explicit exclusions	shared decisions
public surface	HTTP and producer-owned facts	private-model leakage
runtime	roles, lifecycle and composition roots	container-count confusion
data	store, writer and migration principal	cross-owner persistence
operations	health, telemetry, failures and recovery	hollow extraction
evidence	tests, artifacts and current status	prose-only admission

PHASE-0 BOUNDARY

The catalogue is for an educational demonstrator. It does not claim safety-class operation, licensing suitability, real-plant connectivity, high availability or production readiness.

COUNTING GRAMMAR

Service, role, project and module are different units

COUNTING RULES B-3

Term	Definition	NEXUS-1 example
service boundary	independent authority and owned durable truth	RootCause
derived service	independent lifecycle over copied observations	Reporting
edge deployment	client-facing security/composition; no domain truth	Query BFF
runtime role	one executable lifecycle within a boundary	rootcause-worker
migration job	short-lived schema authority	reporting-migrator
module	cohesive capability inside a host/solution boundary	Audit in current core
candidate service	semantically plausible; admission incomplete	Compliance

COUNTING RESULT

Seven Deployments represent three service boundaries plus one edge. Three migration jobs are operational authorities, not additional services.

CATALOGUE STATUS MODEL

Placement, presence and proof remain independent

THREE STATUS AXES B-4

Axis	Values used here	Current result
placement	CORE OWNER / DERIVED OWNER / EDGE / MODULE / CANDIDATE	SPECIFIED
repository presence	PRESENT / PARTIAL / PENDING / ABSENT-BY-DESIGN	PENDING
runtime evidence	DEMONSTRATED / FALSIFIED / PENDING / NOT CLAIMED	PENDING

MATURITY VOCABULARY B-5

State	Allowed wording
PREDICTED	the architecture expects...
SPECIFIED	the design requires...
DEMONSTRATED	the attached run showed...
FALSIFIED	the claim failed under...
REFINED	the model now limits...
UNPROVED	pending / not claimed

NO INHERITANCE

A service can be architecturally admitted and still have PENDING repository and runtime status. A module can be semantically complete and still remain outside the deployable core.

FINAL CATALOGUE AT A GLANCE

Three owners, one edge and four conditional placements

BOUNDARY CATALOGUE B-6

Boundary	Placement	Owned truth	Runtime
AlarmManagement	ADMITTED CORE OWNER	alarms, flood decisions, publication intent	2 roles; PENDING
RootCause	ADMITTED CORE OWNER	cases, evidence, verdicts, publication intent	2 roles; PENDING
Reporting	ADMITTED DERIVED OWNER	ledger, checkpoints, freshness-bearing view	2 roles; PENDING
Query BFF	ADMITTED EDGE	none	1 role; PENDING
Audit	MODULE / SERVICE CANDIDATE	candidate immutable evidence	not in core
Compliance	MODULE / SERVICE CANDIDATE	candidate review workflow	not in core
RL Advisory	OPTIONAL MODULE	advisory result only	output not activated
Process Manager	EXTENSION CANDIDATE	workflow progress only	not in core

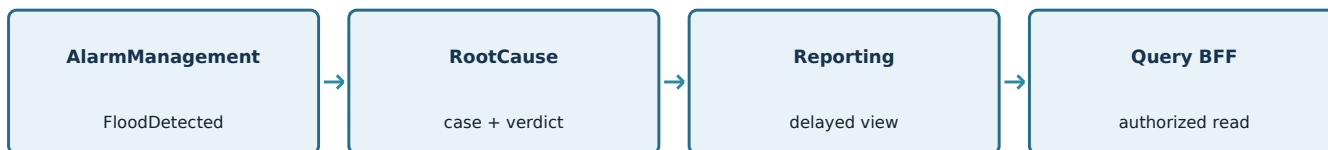
FINAL TOPOLOGY AUTHORITY

This table follows the Chapter 60 boundary decision and Appendix A. Earlier chapters may specify richer candidates; they do not silently expand the final seven-role inventory.

CORE INTERACTION MAP

Facts cross owners; commands stop at the owner

ADMITTED PHASE-0 PATH



INTERACTION CATALOGUE B-7

From	To	Mechanism	Meaning
operator/adaptor	AlarmManagement	authorized HTTP command	accept one alarm observation
AlarmManagement	RootCause	AlarmFloodDetected.v1	producer-confirmed flood fact
RootCause	Reporting	CaseOpened.v1 + VerdictIssued.v1	owner case stream
Query BFF	Reporting API	authorized bounded HTTP query	current-as-observed view
platform	every role	startup/live/ready probes	lifecycle control only

NO GLOBAL COMPLETION CALLBACK

Reporting and the BFF never call RootCause to declare the journey complete. The evidence harness observes independent durable coordinates.

ALARMMANAGEMENT - PROFILE

The alarm owner accepts observations and decides flood meaning

SERVICE CARD B-8

Coordinate	AlarmManagement
purpose	own accepted alarm observations and deterministic flood decisions
ubiquitous language	Observation, Alarm, FloodDecision, PolicyIdentity, PendingPublicFact
authority	accept/reject observation; determine whether one flood fact exists
explicit exclusions	no RootCause case, verdict, Reporting projection or compliance decision
local durability	one AlarmManagementDb transaction
architectural placement	ADMITTED CORE OWNER
repository/runtime status	PENDING / PENDING

OWNER PROMISE

A successful command proves only that AlarmManagement durably accepted its own outcome. It does not prove broker publication or downstream convergence.

FROM SERVICES TO RUNTIME

ALARMMANAGEMENT - COMMANDS AND INVARIANTS

One input becomes one replayable owner decision

COMMAND CATALOGUE B-9

Command/surface	Required semantics	Local outcome
POST /api/v1/alarm-observations	authorized, bounded, idempotent acceptance	202 after local durability
replay flood policy	same canonical inputs + policy identity	same DecisionIdentity
query accepted result	owner-local observation/decision status	no downstream inference

INVARIANT CATALOGUE B-10

Invariant	Enforcement
one idempotency key resolves to one accepted request meaning	unique semantic key + digest conflict
decision uses canonical ordered evidence/observations	Domain policy
decision and Pending fact agree	single database transaction
environment, retry count and broker state do not change meaning	decision basis excludes adapter noise
published bytes equal staged bytes	outbox body digest

ALARMMANAGEMENT - RUNTIME AND DATA

Two roles share one authority boundary

RUNTIME ROLES B-11

Role	Responsibility	Durable access
alarm-api	HTTP admission, authorization, command handling, owner reads	AlarmManagementDb DML
alarm-worker	outbox lease, exact publish, recovery/background work	AlarmManagementDb DML
alarm-migrator	bounded schema migration job	owned schema DDL

DATA OWNERSHIP B-12

Data family	Authority
observations and alarm state	AlarmManagement only
flood decision and decision identity	AlarmManagement only
outbox state and exact public body	AlarmManagement only
inbound HTTP idempotency receipt	AlarmManagement only
RootCause/Reporting identifiers	references after publication; never foreign rows

PRINCIPAL BOUNDARY

Runtime roles receive least DML for AlarmManagementDb and no DDL or foreign-store access. The migrator principal is short-lived and separate.

ALARMMANAGEMENT - PUBLIC LANGUAGE

AlarmFloodDetected.v1 is its one activated public fact

PRODUCED FACT B-13

Coordinate	Decision
event	AlarmFloodDetected.v1
producer	AlarmManagement
core consumer	RootCause
meaning	one durable flood decision occurred under a named policy
identity	producer-owned MessageId + semantic fact key
order	AlarmManagement producer stream coordinate where declared
status	ACTIVATED SPECIFICATION; runtime delivery PENDING

PUBLIC CONTRACT EXCLUSIONS B-14

Carries	Does not carry
sufficient flood context, decision identity, source coordinates	Alarm aggregate or EF row
correlation, causation, occurrence and fingerprint	broker retry/delivery state as business meaning
versioned stable public values	RootCause case or verdict assumptions

EVOLUTION OWNER

AlarmManagement owns the writer and compatibility decision. RootCause owns only its supported-reader set and refusal behaviour.

ALARMMANAGEMENT - DEPENDENCIES AND OPERATIONS

Local truth survives unavailable transport

DEPENDENCY CATALOGUE B-15

Dependency	Used by	Failure meaning
AlarmManagementDb	API + worker	owner commands/dispatch cannot progress
RabbitMQ	worker	committed Pending facts accumulate
identity provider	API	unauthenticated commands rejected
telemetry pipeline	both roles	observation degrades; truth remains separate

OPERATIONAL SIGNALS B-16

Surface	Required observation
health	startup configuration/migration identity; process liveness; role readiness
metrics	accept result, decision result, outbox pending age, lease/publish result
logs	bounded decision, commit, lease and dispatch events
traces	HTTP -> commit and commit -> dispatch causal link

NO TRANSITIVE READINESS

Alarm API readiness does not require RootCause, Reporting or candidate modules. Downstream availability is not a precondition for owner-local truth.

ALARMMANAGEMENT - FAILURE AND EVIDENCE

Its conserved quantity is the accepted decision

FAILURE CATALOGUE B-17

Condition	Must remain true	Recovery proof
broker unavailable after commit	decision + exact Pending body durable	one eventual dispatch; no new decision
duplicate HTTP request	one semantic acceptance/effect	matching receipt or conflict
DB transient	no partial local state	fresh unit of work retry
lost publisher confirm	outbox not falsely Published	safe replay with same body
worker shutdown	lease/settlement bounded	remaining item visible

ADMISSION EVIDENCE B-18

Claim	Required artifact
owner graph exists	project/inventory + clean build
transaction is atomic	provider-backed rollback receipt
outbox survives outage	F-OUTBOX campaign
public bytes are stable	contract corpus + broker capture
principal is bounded	allowed and denied provider operations

CURRENT STATUS

ALARM BOUNDARY SPECIFIED | REPOSITORY PENDING | RUNTIME PENDING

ROOTCAUSE - PROFILE

The evidence owner opens cases and issues verdicts

SERVICE CARD B-19

Coordinate	RootCause
purpose	own cases, admitted evidence membership and evidence-bound verdicts
ubiquitous language	Case, EvidenceKey, Verdict, PolicyIdentity, VerdictIdentity
authority	open case, attach evidence, issue or refuse verdict
explicit exclusions	no alarm decision, compliance approval, projection truth or audit history
local durability	RootCauseDb transactions with inbox/outbox
architectural placement	ADMITTED CORE OWNER
repository/runtime status	PENDING / PENDING

CONSUMER BOUNDARY

RootCause receives sufficient public Alarm meaning. It never reads AlarmManagementDb or treats a foreign private type as evidence.

ROOTCAUSE - COMMANDS AND INVARIANTS

Evidence membership is frozen with the verdict

COMMAND CATALOGUE B-20

Command	Authority	Outcome family
OpenRootCauseCase	RootCause policy after admitted flood	Opened / Existing / Rejected
AttachEvidence	authorized case mutation	Attached / Duplicate / Closed
IssueRootCauseVerdict	approved policy + actor capability	Issued / Existing / Abstained / Rejected
owner queries	RootCause truth only	case/evidence/verdict observation

INVARIANT CATALOGUE B-21

Invariant	Enforcement
one flood business identity opens one case	semantic unique key
closed case cannot change evidence membership	aggregate state transition
verdict requires nonempty ordered evidence	Domain policy
policy and evidence revision are frozen	VerdictBasis + digest
case/verdict and public intent commit together	RootCauseDb transaction

ROOTCAUSE - RUNTIME AND DATA

The worker consumes Alarm facts and publishes RootCause facts

RUNTIME ROLES B-22

Role	Responsibility	Durable access
rootcause-api	authorized case/evidence/verdict commands and reads	RootCauseDb DML
rootcause-worker	Alarm consumer, inbox, outbox relay, recovery	RootCauseDb DML
rootcause-migrator	bounded schema migration job	owned schema DDL

DATA OWNERSHIP B-23

Data family	Authority
RootCauseCase and evidence membership	RootCause only
verdict, decision basis and identity	RootCause only
inbox receipts and inbound lineage	RootCause processing evidence
outbox bodies and producer stream positions	RootCause public intent
Alarm identifiers	remote coordinates; no foreign existence guarantee

TRANSACTION BOUNDARY

RootCause case/verdict work, inbox effect and staged public fact can share RootCauseDb. Broker acknowledgement and subscriber effects remain outside.

ROOTCAUSE - PUBLIC LANGUAGE

One producer stream carries case opening and verdict issuance

PRODUCED FACTS B-24

Fact	Meaning	Core consumer	Status
RootCauseCaseOpened.v1	case durably opened from one flood	Reporting	ACTIVATED SPECIFICATION
RootCauseVerdictIssued.v1	evidence-bound verdict durably issued	Reporting	ACTIVATED SPECIFICATION

CONSUMED FACT B-25

Fact	Producer	Local reaction	Forbidden inference
AlarmFloodDetected.v1	AlarmManagement	open/resolve one case effect	Alarm private rows or current state

CANDIDATE SUBSCRIBERS

Audit and Compliance are legitimate candidate consumers of VerdictIssued. Their queues and local semantics do not place them in the admitted seven-role core.

ROOTCAUSE - DEPENDENCIES AND OPERATIONS

Subscriber outages cannot veto verdict truth

DEPENDENCY CATALOGUE B-26

Dependency	Used by	Failure meaning
RootCauseDb	API + worker	owner commands/consumption stop or retry
RabbitMQ	worker	input delivery or output dispatch delays
identity provider	API	unauthorized command rejected
telemetry pipeline	both roles	evidence plane may fail independently

OPERATIONAL SIGNALS B-27

Surface	Required observation
health	configuration/migration compatibility, process liveness, role readiness
metrics	consumer result, duplicate/conflict, case/verdict result, outbox age
logs	admission reason, decision identity, commit and settlement
traces	linked Flood consume -> commit -> RootCause dispatch

NO SUBSCRIBER DEPENDENCY

Audit, Compliance, Reporting and Query BFF are not RootCause startup or verdict-commit dependencies.

ROOTCAUSE - FAILURE AND EVIDENCE

Transport attempts may multiply; case and verdict effects may not

FAILURE CATALOGUE B-28

Condition	Must remain true	Recovery proof
duplicate/redelivered flood	one inbox effect and one case	attempt count may increase
crash after commit before ack	transaction remains arbiter	redelivery resolves receipt
poison public body	zero business mutation	visible quarantine + authorized replay
broker unavailable after verdict	verdict + Pending exact body	eventual one semantic publication
DB transient	foreign owners unchanged	fresh unit of work retry

ADMISSION EVIDENCE B-29

Claim	Required artifact
idempotent consumer	provider inbox/semantic-key receipts
verdict determinism	Domain replay and mutation tests
two public contracts	canonical corpus + compatibility tests
crash safety	F-DUP/F-CRASH campaigns
owner isolation	foreign permission-denial probes

CURRENT STATUS

ROOTCAUSE BOUNDARY SPECIFIED | REPOSITORY PENDING | RUNTIME PENDING

REPORTING - PROFILE

The derived owner makes delayed truth explicit

SERVICE CARD B-30

Coordinate	Reporting
purpose	maintain an independently operated observation of RootCause public facts
ubiquitous language	LedgerEntry, ProjectionGeneration, Watermark, Gap, Freshness
authority	apply supported facts contiguously and describe observation status
explicit exclusions	no source verdict, alarm decision, compliance decision or command-side existence
local durability	ReportingDb projection transactions
architectural placement	ADMITTED DERIVED OWNER
repository/runtime status	PENDING / PENDING

TRUTH VOCABULARY

Reporting owns what it has observed, the order it has applied and the age/gaps of that observation. It does not own source-authoritative now.

REPORTING - PROJECTION CONTRACT

A broad binding feeds a narrow reducer allowlist

CONSUMED FACTS B-31

Fact	Producer	Projection action
RootCauseCaseOpened.v1	RootCause	create/open summary at stream position 1
RootCauseVerdictIssued.v1	RootCause	apply verdict fields at stream position 2

PROJECTION ADMISSION B-32

Input state	Action
known type/schema + valid bytes	translate to immutable ProjectionEvent
valid duplicate	resolve stored receipt/ledger
same MessageId, changed bytes	identity conflict; never overwrite
unknown/breaking contract	terminal evidence or quarantine; no reducer
position arrives ahead	buffer gap; watermark does not skip

CONTRACT-SET IDENTITY

Adding a public fact or mapper changes the projection contract-set digest and requires an explicit rebuild/migration decision.

REPORTING - RUNTIME, DATA AND API

One writer role and one read-only API share a derived boundary

RUNTIME ROLES B-33

Role	Responsibility	Durable access
projection-worker	admit facts, inbox, ledger, reduce, gap/repair	ReportingDb writer
projection-api	freshness-bearing read contract	ReportingDb read-only
reporting-migrator	bounded schema migration job	owned schema DDL

OWNED DATA B-34

Data family	Meaning
immutable event ledger + inbox	what exact public bytes were accepted
case summary	current projection generation's observation
stream checkpoint + gap buffer	contiguous progress and missing positions
projection status samples	heartbeat/backlog observations at a named time
rebuild generation	which reducer/contract set produced the view

READ-ONLY SURFACE

Projection API cannot issue upstream commands or write derived business state. Its database principal is read-only.

REPORTING - FRESHNESS CONTRACT

A value travels with evidence about its currency

READ RESPONSE EVIDENCE B-35

Coordinate	Question answered
source occurred time	when the producer says the fact occurred
last applied time	when Reporting committed its observation
last contiguous position	how far this stream is complete
gap start/count	which earlier fact is missing
heartbeat/backlog sample age	whether processing appears operational
projection generation	which build/reducer created the row
update status	current-as-observed / delayed / gap / unknown / failed

NO INVENTED SLO

The catalogue fixes measurements and semantics. Numeric production objectives remain environment- and workload-specific until capacity and operating evidence approves them.

REPORTING - DEPENDENCIES AND OPERATIONS

Source availability and projection currency are separate

DEPENDENCY CATALOGUE B-36

Dependency	Used by	Failure meaning
ReportingDb	worker + API	projection or reads unavailable
RabbitMQ	worker	new observations delayed; prior view remains qualified
RootCause public contracts	worker	unsupported change is rejected, not guessed
telemetry pipeline	both roles	status evidence may become incomplete

OPERATIONAL SIGNALS B-37

Surface	Required observation
health	migration/config, process, broker/DB readiness by role
metrics	watermark, gap count, source age, projection age, backlog
logs	admission, gap, rebuild, repair and query status
traces	Verdict consume -> projection commit -> BFF read

QUIET STREAM RULE

No recent event does not by itself mean failure. Heartbeat, backlog, gap and source position must be interpreted separately.

REPORTING - FAILURE AND EVIDENCE

A gap must become visible before it becomes current

FAILURE CATALOGUE B-38

Condition	Must remain true	Recovery proof
position 2 before 1	watermark stays before gap	apply 1 then advance contiguously
duplicate fact	one ledger/effect	receipt resolves attempt
poison fact	zero reducer mutation	quarantine + safe replay
worker outage	prior view remains labelled by age/status	bounded catch-up
rebuild	generation never mixes rows	atomic generation switch

ADMISSION EVIDENCE B-39

Claim	Required artifact
ordered progress	provider gap/checkpoint component tests
staleness visible	contract corpus + delayed scenario
read-only API	permission and endpoint denial tests
repair safe	rebuild/replay receipts
projection converges	RunId-bound final scenario

CURRENT STATUS

REPORTING BOUNDARY SPECIFIED | REPOSITORY PENDING | RUNTIME PENDING

QUERY BFF - PROFILE

The edge composes authorized observations and owns no truth

EDGE CARD B-40

Coordinate	Query BFF
purpose	present client-specific authorized views with partial/freshness semantics
language	Observation, UpdateStatus, Partial, Unknown, AuthorizationContext
authority	authenticate, authorize, budget calls and shape the response
explicit exclusions	no aggregate, domain verdict, event production or database
runtime role	query-bff
architectural placement	ADMITTED EDGE; NOT A DOMAIN SERVICE
repository/runtime status	PENDING / PENDING

EDGE PROMISE

A BFF 200 means an authorized response was composed under its contract. It does not certify source freshness or distributed completion unless the returned evidence says so.

QUERY BFF - COMPOSITION CONTRACT

Current-as-observed is a response state, not a slogan

COMPOSITION CATALOGUE B-41

Input	BFF action	Returned meaning
fresh contiguous Reporting view	map allowed fields	current-as-observed
old view without known gap	include age/status	delayed
known projection gap	preserve last safe view if contract allows	gap-detected / partial
not yet observed	distinguish from source absence	unknown / not-observed-yet
dependency timeout	apply bounded partial policy	partial/unavailable; never fabricated

BFF PROHIBITIONS B-42

Forbidden	Reason
direct owner database query	bypasses contracts and principals
cross-service transaction	no common commit exists
cache without source/freshness coordinates	stale data impersonates truth
business fact publication	edge owns no producer meaning
verdict/compliance decision	authority belongs elsewhere

QUERY BFF - SECURITY, FAILURE AND EVIDENCE

Authorization failure must produce zero domain effect

OPERATIONAL CATALOGUE B-43

Concern	Required behaviour
identity	validate operator token and downstream service identity
authorization	capability, audience and permitted query scope
timeout	bounded per dependency and overall request budget
partial failure	explicit status; no false global snapshot
telemetry	route-template latency/result/auth; no payload or high-cardinality IDs
health	edge lifecycle and required dependency policy; not domain correctness

ADMISSION EVIDENCE B-44

Claim	Required artifact
no database/business writes	inventory + privilege + effect probes
authorization enforced	allow/deny HTTP corpus
stale/partial honest	contract + delayed/gap scenarios
budgets bounded	fault/timeout receipts

CURRENT STATUS

QUERY EDGE SPECIFIED | REPOSITORY PENDING | RUNTIME PENDING

AUDIT - CANDIDATE PROFILE

Immutable evidence semantics are strong; service admission is incomplete

CANDIDATE CARD B-45

Coordinate	Audit
purpose	append an immutable local observation of accepted verdict facts
candidate language	AuditEntry, SourceFact, BodyDigest, AppendIdentity
candidate input	RootCauseVerdictIssued.v1
candidate local effect	inbox receipt + immutable append in AuditDb candidate
explicit exclusions	cannot approve, correct or replace the RootCause verdict
public output in final core	none
placement	MODULE / SERVICE CANDIDATE; NOT IN CORE TOPOLOGY

INDEPENDENT PROGRESS

Audit may be offline without rolling back RootCause. That semantic isolation is specified; an independent Audit deployment is not demonstrated.

AUDIT - ADMISSION GAP

A candidate database and queue do not yet make a service

SPECIFIED CANDIDATE ASSETS B-46

Asset	State
strict VerdictIssued admission	SPECIFIED
inbox and semantic append identity	SPECIFIED
immutable AuditDb constraints	SPECIFIED AS CANDIDATE
subscriber-offline independence tests	SPECIFIED
query/reporting semantics	NOT ACTIVATED FOR CORE

MISSING INDEPENDENT-SERVICE EVIDENCE B-47

Obligation	Current state
host image and exact runtime role	PENDING
migration job and principal	PENDING
topology/deployment projection	PENDING
startup/live/ready semantics	PENDING
capacity, fault and recovery campaigns	PENDING
security and operational owner	PENDING

DECISION

KEEP MODULE / SERVICE CANDIDATE. Do not add audit-api, audit-worker or audit-migrator to the core inventory.

COMPLIANCE - CANDIDATE PROFILE

Review workflow authority is valid but not admitted as a core service

CANDIDATE CARD B-48

Coordinate	Compliance
purpose	open and govern a review from an accepted RootCause verdict
candidate language	ComplianceReview, ReviewState, ReviewPolicy, HumanDecision
candidate input	RootCauseVerdictIssued.v1
candidate local effect	inbox receipt + one Pending review in ComplianceDb candidate
explicit exclusions	cannot recalculate RootCause meaning or make Alarm decisions
public output in final core	none activated
placement	MODULE / SERVICE CANDIDATE; NOT IN CORE TOPOLOGY

HUMAN BOUNDARY

Opening a review is not approval, closure or regulatory certification. Human workflow operations remain a separate obligation.

COMPLIANCE - CONTRACT RECONCILIATION

ComplianceReviewOpened.v1 remains a designed extension, not a core fact

Chapter 37 specified a minimal ComplianceReviewOpened.v1 so a durable Process Manager could observe review creation. The final Chapter 60 public-fact catalogue and Appendix A admit only three core facts and no Compliance runtime role. The catalogue therefore preserves the contract design without activating it.

RECONCILIATION B-49

Coordinate	Final catalogue state
payload and producer semantics	DESIGN-SPECIFIED IN PART VII
producer project/runtime role	NOT ADMITTED IN FINAL CORE
core consumer	NONE
broker binding	NOT PART OF FINAL SEVEN-ROLE INVENTORY
compatibility promise	NOT ACTIVATED
future admission	requires Compliance + Process Manager decisions together

NO ERASURE

The design remains useful technical material. The status correction prevents an earlier extension scenario from masquerading as a final deployed contract.

COMPLIANCE - ADMISSION GAP

Workflow semantics need operational ownership

SPECIFIED CANDIDATE ASSETS B-50

Asset	State
strict VerdictIssued admission	SPECIFIED
one review per source verdict	SPECIFIED
review aggregate and candidate constraints	SPECIFIED AS CANDIDATE
subscriber-offline independence	SPECIFIED
review-opened public fact	DESIGNED; NOT ACTIVATED

MISSING INDEPENDENT-SERVICE EVIDENCE B-51

Obligation	Current state
host image, roles, migration and principals	PENDING
human assignment/escalation operations	PENDING
authorization and sensitive-data policy	PENDING
topology, probes and SLO evidence	PENDING
failure/recovery and retention campaigns	PENDING

DECISION

KEEP MODULE / SERVICE CANDIDATE. Do not add compliance-api, compliance-worker or compliance-migrator to the core inventory.

RL ADVISORY - OPTIONAL MODULE

Advice remains conditional, explainable and non-authoritative

OPTIONAL CARD B-52

Coordinate	RL Advisory
purpose	produce a bounded recommendation or abstain
authority	advisory only; no alarm, verdict, compliance or control decision
inputs	approved snapshots/features under explicit versioning
output	recommendation + policy/model identity + explanation/abstention
public fact	RecommendationGenerated.v1 - NOT ACTIVATED
store/role	none admitted
placement	OPTIONAL MODULE

ADMISSION OBLIGATIONS B-53

Required before extraction	Why
independent data authority	avoid a model becoming a hidden copy of owner truth
activated consumer need and contract	no public fact without a purpose
abstention and deterministic fallback	advice cannot become mandatory authority
model/policy lifecycle and evidence	reproducibility and explanation
latency/cost/failure economics	network isolation must earn its cost

DURABLE PROCESS MANAGER - EXTENSION CANDIDATE

Workflow progress is a valid authority, but no final runtime role was admitted

EXTENSION CARD B-54

Coordinate	Alarm-to-Compliance Process Manager
purpose	record expected facts, deadlines, lineage and completion progress
authority	workflow phase and overdue state only
explicit exclusions	no alarm, verdict, review or audit decision
designed store	WorkflowDb
designed inputs	FloodDetected, CaseOpened, VerdictIssued, ReviewOpened
final core role/store	NONE
placement	EXTENSION CANDIDATE; NOT IN SEVEN-ROLE CORE

FINAL-CORE DISTINCTION

The FinalDemonstrator observes a convergence vector for evidence. It is not a production coordinator and does not create WorkflowDb or issue owner commands.

PROCESS MANAGER RECONCILIATION

Part VII is retained as a falsifiable extension design

DESIGNED STRENGTHS B-55

Strength	Value
durable phase and exact lineage	partial progress becomes explainable
public-fact-only observation	no foreign database polling
duplicate/early-delivery handling	restart and ordering resilience
deadline ownership	lateness does not become business failure
repair from retained facts	workflow state can be reconstructed

MISSING ADMISSION EVIDENCE B-56

Obligation	State
final authority/economic decision	PENDING
runtime role, image and principal	PENDING
WorkflowDb migration and operations	PENDING
activated Compliance producer/contract	PENDING
failure, timeout, repair and capacity runs	PENDING

DECISION

PRESERVE THE DESIGN | DO NOT COUNT IT AS PRESENT | REQUIRE A NEW ADMISSION ADR BEFORE DEPLOYMENT

PROTECTED MODULES

Some capabilities deliberately remain close to their authority

PROTECTED CATALOGUE B-57

Capability	Placement	Reason
Instrumentation adapters	AlarmManagement/protected acquisition module	close to acquisition semantics; no independent owner
ReactorFleet	protected core/domain module	no independent contract/store proof
DigitalTwin model	protected analytical module	model versioning is not service authority
simulation policy	RootCause/analysis module	deterministic policy input, not network authority
UI composition	Query edge	no domain writes or truth ownership
final scenario runner	test/evidence harness	observes proof; not production coordination

EXTRACTION TEST

A capability earns independent distribution only after contract, data, operations, deployment, security, failure and economic evidence agree. Process isolation alone is insufficient.

SYNCHRONOUS DEPENDENCY MATRIX

Core commands do not form a distributed call chain

ALLOWED SYNCHRONOUS CALLS B-58

Caller	Target	Purpose	Not allowed
operator/adaptor	alarm-api	submit observation	infer downstream completion
operator/workflow	rootcause-api	authorized owner command/query	write RootCauseDb indirectly without policy
query-bff	projection-api	freshness-bearing read	global snapshot claim
platform	all roles	health/lifecycle	business correctness inference
migrator	owned database	schema change	foreign schema or runtime work

FORBIDDEN SYNCHRONOUS COUPLING B-59

Shape	Reason
Alarm commit waits for RootCause	downstream availability vetoes owner truth
RootCause verdict waits for subscribers	fan-out becomes distributed transaction
Reporting calls source to validate each row	derived owner loses independence
BFF queries databases	contract and least-privilege bypass

FROM SERVICES TO RUNTIME

PUBLIC-LANGUAGE MATRIX

Three facts are activated; all other public language is conditional

EVENT CATALOGUE B-60

Fact	Producer	Core consumer	Final status
AlarmFloodDetected.v1	AlarmManagement	RootCause	ACTIVATED SPECIFICATION
RootCauseCaseOpened.v1	RootCause	Reporting	ACTIVATED SPECIFICATION
RootCauseVerdictIssued.v1	RootCause	Reporting	ACTIVATED SPECIFICATION
ComplianceReviewOpened.v1	Compliance candidate	Process Manager candidate	DESIGNED; NOT ACTIVATED
RecommendationGenerated.v1	RL Advisory	none	NOT ACTIVATED

HTTP OWNERSHIP B-61

Surface	Owner	Semantics
alarm observation command	AlarmManagement	idempotent owner-local acceptance
case/evidence/verdict commands	RootCause	owner-local mutation under policy
projection query	Reporting	delayed observation + freshness
operator query	Query BFF	authorized client composition

APPENDIX C BOUNDARY

Appendix B fixes producer, consumer and status. Appendix C will provide the exact API/event contract catalogue and version rules.

STORE AND WRITER MATRIX

Only three database owners enter the final core

DATA CATALOGUE B-62

Store	Writer	Reader	Status
AlarmManagementDb	alarm-api / alarm-worker	owner-local	CORE; runtime pending
RootCauseDb	rootcause-api / rootcause-worker	owner-local	CORE; runtime pending
ReportingDb	projection-worker	projection-api read-only	CORE; runtime pending
AuditDb	Audit candidate	candidate query	CANDIDATE ONLY
ComplianceDb	Compliance candidate	candidate workflow	CANDIDATE ONLY
WorkflowDb	Process Manager candidate	candidate operations	EXTENSION ONLY

MIGRATION RULE

The final runtime admits exactly three service-owned migration jobs. Candidate stores have no admitted migrator, principal or deployment receipt.

SECURITY AND TRUST MATRIX

Every entry and hop has an identity and a denial case

TRUST CATALOGUE B-63

Boundary	Identity	Authorization	Negative proof
operator -> Alarm	operator principal	capability + detector scope	deny wrong/missing scope
operator -> RootCause	operator/workflow principal	case/evidence/verdict capability	zero effect on denial
BFF -> Reporting	service identity	audience + query capability	token/audience corpus
worker -> broker	role principal	exchange/queue ACL	foreign route denial
role -> database	owner runtime principal	least DML; no DDL/foreign store	provider denial matrix
migrator -> database	short-lived migration principal	owned schema DDL	foreign/runtime denial

CANDIDATE BOUNDARY

Candidate modules do not receive production credentials merely because their semantic models exist.

OBSERVABILITY MATRIX

Signals describe distinct parts of service behaviour

SIGNAL OWNERSHIP B-64

Boundary	Primary bounded signals	Must not become
AlarmManagement	accept/decision/commit/outbox age/dispatch	downstream-completion claim
RootCause	admission/case/verdict/inbox/outbox	Alarm or subscriber truth
Reporting	watermark/gap/source age/projection age/backlog	source-authoritative now
Query BFF	auth/result/latency/partial status	business ledger
candidate modules	module-local diagnostics where hosted	evidence of independent deployment

CARDINALITY AND SECRECY B-65

Signal	Bounded rule
metrics	service/role/operation/result only; no MessageId/Caseld
logs	bounded IDs/codes; no tokens, secrets or raw bodies
traces	route templates and span links; no payloads
evidence	digests and redacted references; no environment dump

FAILURE-ISOLATION MATRIX

A service boundary is visible when another boundary fails

ISOLATION CATALOGUE B-66

Failure	Unaffected authority	Expected degraded state
RabbitMQ unavailable	committed Alarm/RootCause truth	Pending outbox; delayed consumers
RootCause offline	Alarm commits/publication intent	queue backlog; no case yet
Reporting offline	Alarm and RootCause truth	stale/unknown BFF observation
Query BFF offline	all owner stores/workers	client query unavailable only
Audit/Compliance absent	final core convergence	candidate side effects unavailable
telemetry unavailable	policy-governed business path	evidence checkpoint incomplete

CONSERVATION RULE

A failure campaign passes only when it proves both the expected degraded state and the conserved owner-local truth under the same RunId.

LIFECYCLE MATRIX

Startup, liveness and readiness answer different questions

ROLE LIFECYCLE B-67

Role family	Startup	Liveness	Readiness
API	valid config + compatible owned schema	process can make progress	can safely admit its traffic
worker	valid config + contract/topology identity	pump/control loop alive	can safely consume/dispatch
projection API	config + read schema	process-local	read contract can be served
Query BFF	auth/config contract	process-local	edge policy permits traffic
migrator	identity + target admission	bounded job progress	not applicable as traffic endpoint

DEPENDENCY RULE

Readiness may include a role's necessary immediate dependency under explicit policy. It must not recursively require every downstream service.

MACHINE-READABLE CATALOGUE

Human service cards project from one validated manifest

NORMATIVE CATALOGUE SHAPE B-1

```
catalogueVersion: 1
runtimeInventory: deploy/topology/runtime-inventory.yaml
boundaries:
- id: svc.alarm-management
  placement: admitted-core-owner
  roles: [alarm-api, alarm-worker]
  migrationJobs: [alarm-migrator]
  ownedStores: [AlarmManagementDb]
  produces: [AlarmFloodDetected.v1]
  consumes: []
  repositoryPresence: pending
  runtimeEvidence: pending
- id: edge.query-bff
  placement: admitted-edge
  roles: [query-bff]
  ownedStores: []
  produces: []
  runtimeEvidence: pending
```

MANIFEST RULES B-68

Rule	Reason
stable boundary/role IDs	joins survive display-name changes
sorted canonical serialization	digest is reproducible
status fields explicit	absence cannot mean Passed
candidate entries included	conditional scope remains visible
generated human tables	book/runtime catalogue drift can be detected

CATALOGUE VALIDATOR

The final boundary decision becomes a fail-closed gate

EXECUTABLE ASSET B-1 - CATALOGUE ADMISSION

```
public static CatalogueVerdict Validate(ServiceCatalogue c)
{
    RequireUnique(c.Boundaries.Select(x => x.Id));
    RequireExact(c.AdmittedRoles, Topology.SevenRoles);
    RequireExact(c.AdmittedMigrationJobs, Topology.ThreeMigrators);
    RequireSingleWriter(c.OwnedStores);

    foreach (var fact in c.ActivatedFacts)
        RequireSingleAdmittedProducer(fact, c);

    foreach (var candidate in c.Candidates)
        Require(candidate.AdmittedRoles.Count == 0,
            $"CANDIDATE_ROLE_ACTIVATED:{candidate.Id}");

    Require(c.Edge("edge.query-bff").OwnedStores.Count == 0,
        "BFF_STORE_FORBIDDEN");
    return CatalogueVerdict.Passed(c.CanonicalSha256());
}
```

CROSS-CATALOGUE JOINS B-69

Join	Failure
service roles <-> runtime inventory	missing/extra executable
owned stores <-> principal/migration catalogue	writer authority drift
facts <-> contract catalogue	unowned or inactive public grammar
dependencies <-> deployment manifests	undeclared runtime edge
claims <-> evidence manifest	unproved service status

ADMISSION GATES

No candidate becomes a service by adding a container

CANDIDATE-TO-SERVICE GATES B-70

Gate	Required evidence
semantic	stable language, commands, invariants and explicit exclusions
contract	real consumers, producer ownership and compatibility corpus
data	owned store or explicit stateless authority; migration/retention decision
runtime	roles, images, composition, lifecycle and bounded shutdown
security	identities, authorization, secrets and denial probes
operations	telemetry, SLO mechanism, capacity and on-call ownership
failure	outage, duplicate, poison, crash and recovery evidence
economics	independent scale/change/failure benefit exceeds distribution cost

DECISION EVIDENCE

Passing these gates changes the catalogue, Appendix A topology, Appendix C contracts, Appendix D data ownership and deployment inventory in one reviewed change.

DRIFT AND CHANGE CONTROL

Service catalogue changes are architecture changes

CHANGE CLASSIFICATION B-71

Change	Required review
new runtime role inside admitted owner	lifecycle, capacity, image and topology
new command or fact	authority, contract, security and compatibility
new store/writer	ownership, migration, principal and recovery
new synchronous dependency	availability, timeout, retry and partial-result policy
module -> service	all eight admission gates
service -> module	contract/data retirement and migration plan
status -> demonstrated	fresh RunId/BuildIdentity evidence

AUTOMATIC DRIFT FAILURES B-72

Observed mismatch	Verdict
unexpected executable role	FAILED
candidate role in deployment	FAILED - SILENT ADMISSION
activated fact without admitted producer	FAILED
store with two boundary writers	FAILED
catalogue digest differs from final manifest	PENDING/FAILED

CATALOGUE ACCEPTANCE CHECKLIST

The service map is complete only when every status is explainable

ACCEPTANCE GATES B-73

Gate	Required result
boundary count	three owners + one edge; candidates listed separately
role count	exactly seven long-running roles
migration count	exactly three service-owned jobs
store count	three core owner databases; candidate stores conditional
public facts	three activated specifications; candidates explicitly inactive
authority	every command/fact/store has one owner and exclusions
dependencies	no hidden cross-database or global transaction edge
operations	health, signals, failure and recovery obligations named
evidence	current repository/runtime statuses remain PENDING
coherence	Process Manager and ComplianceReviewOpened reconciled

APPENDIX B VERDICT

SERVICE RESPONSIBILITIES SPECIFIED | CORE/CANDIDATE BOUNDARY RECONCILED | REPOSITORY PRESENCE
PENDING | RUNTIME EVIDENCE PENDING | PRODUCTION READINESS NOT CLAIMED

ANTI-PATTERNS

A catalogue fails when names hide authority

REJECTED CATALOGUE PATTERNS B-74

Anti-pattern	Why rejected	Correction
count containers as services	API/worker lifecycle becomes fake authority	count owner boundaries
list every module as candidate service	distribution becomes default	require admission gates
omit explicit exclusions	overlap remains invisible	state must-not-own fields
mark designed contract activated	consumers rely on nonexistent promise	separate design/activation
use database name as service reason	schema inventory is not language authority	prove commands/invariants
global dependency health	one outage drains the whole topology	role-local readiness policy
hand-edit Passed status	evidence provenance is lost	validator-derived verdict

REVIEW QUESTION

If this boundary disappeared, which decision, durable truth, contract evolution, failure recovery or operational obligation would lose a single owner? If none, it may not be a service.

APPENDIX B CHECKPOINT

The final core is intentionally smaller than the design space

FINAL RESPONSIBILITY LEDGER B-75

Placement	Entries	Final meaning
admitted core owners	AlarmManagement, RootCause	source-authoritative command services
admitted derived owner	Reporting	freshness-bearing copied observation
admitted edge	Query BFF	authorized composition; no truth/store
module/service candidates	Audit, Compliance	semantics retained; runtime not admitted
optional module	RL Advisory	advice/abstention; output inactive
extension candidate	Process Manager	workflow progress design; no final role/store
protected modules	acquisition/analytical/core capabilities	deliberately not distributed

CURRENT HONEST RECORD

The service catalogue is complete as an architectural specification. No service card may claim an observed build, deployment or recovery result until the corresponding evidence bundle is attached.

HANDOFF TO APPENDIX C

The owners are fixed; the next appendix fixes their public language

Appendix B established who may decide, write, publish, consume and compose. Appendix C will catalogue the exact HTTP and event contracts: routes, methods, DTOs, envelope fields, versions, compatibility rules, producer ownership, consumer allowlists, canonical corpora, fingerprints and activation status.

NEXT SEQUENCE B-76

Next item	Purpose
Appendix C - API and Event Contract Catalogue	one exact versioned public-language reference
Appendix D - Database Ownership and Migration Catalogue	writers, readers, principals and schema authority
Appendix E - Failure Scenario and Recovery Catalogue	disturbance, conservation and recovery oracles
Appendices F-H	security, reproduction and traceability

APPENDIX B CONTROL TOTALS B-77

Coordinate	Final count / state
admitted service boundaries	3
admitted edge deployments	1
long-running runtime roles	7
service-owned migration jobs	3
core owner databases	3
activated public facts	3 specifications; execution PENDING
conditional placements	Audit, Compliance, RL Advisory, Process Manager

CARRY-FORWARD RULE

Every contract in Appendix C must point to exactly one producer or HTTP owner in this catalogue.
Designed-but-inactive language remains visibly inactive.

APPENDIX B LAST WORD

A boundary earns distribution by owning consequences

The catalogue does not reward a capability for having a folder, database sketch or container. It records the harder test: one owner for meaning, one bounded runtime obligation and evidence strong enough to let reality disagree.

FINAL STATUS

CATALOGUE COMPLETE AS SPECIFICATION | CORE COUNT FIXED | CONDITIONAL CAPABILITIES PRESERVED |
REPOSITORY AND RUNTIME EVIDENCE PENDING

APPENDIX C

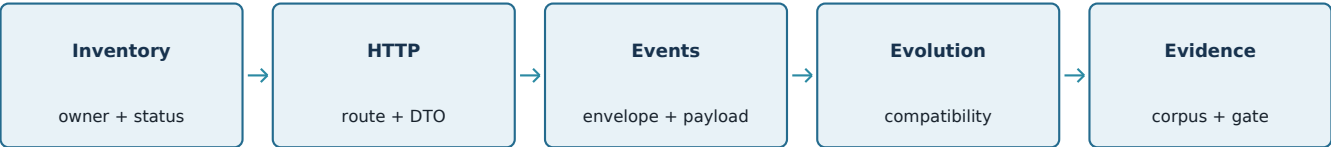
API and Event Contract Catalogue

The exact public language of the final NEXUS-1 core: HTTP routes, request and response DTOs, event envelopes, producer ownership, consumer allowlists, compatibility rules, canonical byte corpora and evidence status.

APPENDIX DECISION

A contract is admitted only when one owner, one versioned coordinate, one strict reader profile and one falsifiable corpus agree. This appendix freezes the canonical target contracts. Repository presence, cross-language conformance and deployed execution remain PENDING unless an attached receipt proves otherwise.

CATALOGUE READING PATH



CONTROL TOTALS C-1

Coordinate	Final value
HTTP owners represented	AlarmManagement, RootCause, Reporting, Query BFF and host lifecycle surfaces
activated public facts	3 - AlarmFloodDetected.v1, RootCauseCaseOpened.v1, RootCauseVerdictIssued.v1
inactive candidates	RecommendationGenerated.v1 and later module facts remain outside the final core
evidence posture	SPECIFIED; repository and runtime conformance PENDING

PURPOSE AND SCOPE

The catalogue freezes promises, not implementations

A public contract is the smallest stable promise that another independently changing component may rely on. It is not a controller signature, database row, generated client class, broker delivery attempt or current aggregate snapshot. The catalogue therefore binds meaning, coordinates, bytes and refusal behaviour while leaving private implementation free to evolve.

CONTRACT CATEGORIES C-2

Category	Authority	What is frozen
owner HTTP command	domain owner	method, route, V1 DTO, idempotency and outcomes
owner HTTP query	domain owner	authorized observation of owner-local truth
derived HTTP query	Reporting / BFF	observation plus watermark and freshness evidence
public event	producer	event type, payload, envelope, route and exact corpus
lifecycle endpoint	host	safe startup/live/ready vocabulary

EXCLUDED

Private Domain events, ORM entities, SQL schemas, queue delivery tags, trace spans, retry counters and deployment objects are not public business contracts.

CLAIM AND ACTIVATION STATUS

Specified, activated and demonstrated are different claims

STATUS GRAMMAR C-3

State	Meaning	Allowed wording
DESIGNED	shape exists but is not admitted to final traffic	candidate / extension
ACTIVATED SPECIFICATION	contract belongs to the final architectural path	must be implemented and tested
PRESENT	versioned repository artifact exists	file/package/route inventory proves presence
DEMONSTRATED	same identity executed against retained corpus	receipt names build, run and corpus
FALSIFIED	observed result contradicts a claimed contract	gate fails with retained evidence

CURRENT BOUNDARY

The three public facts and the HTTP targets below are authoritative specifications. Their physical repository inventory and runtime execution remain PENDING in this manuscript environment.

FINAL CONTRACT INVENTORY

Every contract points to exactly one owner

CANONICAL INVENTORY C-4

Contract	Owner	Consumer / caller	Status
POST /api/v1/alarm-observations	AlarmManagement	authorized adapter/operator	ACTIVATED SPECIFICATION
RootCause owner API V1	RootCause	authorized workflow/operator	CANONICAL TARGET
Reporting projection query V1	Reporting	Query BFF	CANONICAL TARGET
GET /api/v1/root-cause-cases/{caseld}	Query BFF	authorized UI/client	ACTIVATED TARGET
AlarmFloodDetected.v1	AlarmManagement	RootCause	ACTIVATED SPECIFICATION
RootCauseCaseOpened.v1	RootCause	Reporting	ACTIVATED SPECIFICATION
RootCauseVerdictIssued.v1	RootCause	Reporting	ACTIVATED SPECIFICATION
startup/live/ready	each host	platform	REQUIRED HOST CONTRACT

ONE-WRITER RULE

A consumer may retain bytes and translate meaning locally. It may not rename itself as the producer, republish a modified V1 body under the original Messageld or extend the producer payload in place.

NAMING AND OWNERSHIP GRAMMAR

Names expose the owner and the version

NAMING PROFILE C-5

Surface	Grammar	Example
public HTTP route	/api/v{major}/{owner-language resource}	/api/v1/alarm-observations
internal HTTP route	/internal/v{major}/{derived resource}	/internal/v1/root-cause-case-projections/{caseld}
event type	nexus1.{producer}.fact-kebab.v{major}	nexus1.root-cause.root-cause-case-opened.v1
routing key	{producer}.fact-kebab.v{major}	root-cause.root-cause-case-opened.v1
producer stream	{producer}/{aggregate-or-scope}/{id}	root-cause/case/{caseld}

NO IMPLICIT LATEST

A route, media type, event type or routing key without an explicit supported major version is not an alias for the newest contract.

HTTP V1 TRANSPORT PROFILE

Version, media type and cache policy are explicit

HTTP BASELINE C-6

Coordinate	Normative rule
protocol	HTTPS in deployed environments; HTTP semantics remain versioned independently
request JSON	application/json; charset=utf-8 unless a stricter vendor type is declared
response JSON	UTF-8; exact V1 DTO; stable Problem Details on failure
major version	URI segment /v1; no silent negotiation to V2
unknown members	rejected by the strict V1 reader
dates	RFC 3339 UTC with explicit Z; no locale-dependent values
caching	commands no-store; private operator projections private, no-store

READER PROFILE

Because V1 contracts use strict unknown-member rejection, adding an otherwise optional JSON member to the same V1 object is breaking for this contract set. Evolve by a new accepted schema/version, not by assumption.

AUTHENTICATION AND AUTHORIZATION

Identity is checked before resource disclosure

HTTP SECURITY OUTCOMES C-7

Condition	HTTP	Invariant
missing or invalid authentication	401	no owner handler or row lookup
authenticated but missing capability	403	no existence disclosure
resource outside admitted scope	403	same safe shape as capability denial
valid principal and scope	continue	authorization context reaches Application
unsupported media type	415	no deserialization side effect
oversized body	413	bounded rejection before allocation-heavy work

PRINCIPAL SEPARATION

External user identity, workload identity, migration authority and broker producer identity are separate credentials. A trace ID or correlation ID never authorizes a request.

PROBLEM DETAILS PROFILE

Failures are stable, safe and machine-readable

NORMATIVE V1 PROBLEM SHAPE

```
public sealed record NexusProblemV1(
    string Type, string Title, int Status,
    string Code, bool Retryable,
    string? Detail, string? Instance,
    string TraceId);

// Type: urn:nexus1:problem:{owner}:{code}
// Detail is bounded and safe; internal exception text is excluded.
```

PROBLEM VOCABULARY C-8

Code family	Typical HTTP	Meaning
request-invalid / semantic-invalid	400 / 422	syntax or typed business validation failed
not-authenticated / not-authorized	401 / 403	identity/capability/scope failure
identity-conflict	409	same stable key names different meaning
precondition-failed	412	explicit version/ETag basis no longer matches
rate-limited	429	bounded retry may be advised
owner-not-ready / dependency-unavailable	503	no false success or stale authority

SECRET BOUNDARY

Problem responses never include connection strings, token claims, stack traces, SQL text, broker credentials, raw payloads or hidden resource values.

HTTP IDENTITY LAYERS

Attempt, operation and business meaning have different keys

IDENTITY SEPARATION C-9

Identity	Authority	Conflict rule
Idempotency-Key header	client transport operation	same key + different request digest -> 409
OperationId body field	application operation	stable across retries and commit ambiguity
SourceSystem + SourceEventId	upstream semantic occurrence	unique inside AlarmManagement
PayloadSha256	submitted evidence bytes	same source identity + changed digest -> 409
MessageId	event producer	same ID + changed event body -> quarantine/conflict

NO SUBSTITUTION

An Idempotency-Key is not a MessageId. An OperationId is not a trace ID. A producer fact identity is not a broker delivery tag.

ALARM HTTP CONTRACT CARD

Observation acceptance ends at owner-local durability

ROUTE C-10

Coordinate	Exact decision
method + route	POST /api/v1/alarm-observations
owner	AlarmManagement
capability	alarm.observation.submit
request media	application/json; charset=utf-8
idempotency	Idempotency-Key + stable OperationId + semantic source identity
success boundary	local AlarmManagementDb acceptance; 202 when newly accepted
not implied	broker delivery, RootCause case, verdict or Reporting convergence

FINAL ROUTE

The earlier illustrative /api/v1/alarm-flood-detections route is superseded for the final slice. Flood detection is an owner decision produced by alarm-observation acceptance, not the public command name.

ALARM REQUEST DTO

The V1 request carries replayable source evidence

SUBMITALARMOBSERVATIONV1

```
public sealed record SubmitAlarmObservationV1(
    Guid OperationId,
    string SourceSystem,
    string SourceEventId,
    string AlarmKey,
    string DetectorScope,
    string Severity,
    DateTimeOffset SourceOccurredAtUtc,
    string PayloadSha256);
```

FIELD RULES C-11

Field	Rule
OperationId	nonempty and stable across retries
SourceSystem	allowlisted ASCII; maximum 80 characters
SourceEventId	nonempty; maximum 120; unique with SourceSystem
AlarmKey / DetectorScope	normalized, bounded and nonsecret
Severity	exact V1 enum; no unknown-to-default coercion
SourceOccurredAtUtc	UTC and inside the admitted horizon
PayloadSha256	exactly 64 lowercase hexadecimal characters

ALARM RESPONSE AND OUTCOMES

The receipt states what AlarmManagement committed

ALARMOBSERVATIONRECEIPTV1

```
public sealed record AlarmObservationReceiptV1(
    Guid OperationId,
    Guid ObservationId,
    string Outcome,
    Guid? FloodDetectionId,
    string DecisionIdentity,
    DateTimeOffset AcceptedAtUtc);
```

OUTCOME MATRIX C-12

Outcome	HTTP	Response meaning
Accepted	202	new local acceptance committed
Replay	200	matching prior outcome returned
IdentityConflict	409	stable identity names different meaning
Rejected	400 / 422	syntax or semantic validation failed
NotReady	503	owner cannot safely accept
Unknown	503	commit ambiguity; retry/lookup with same identity

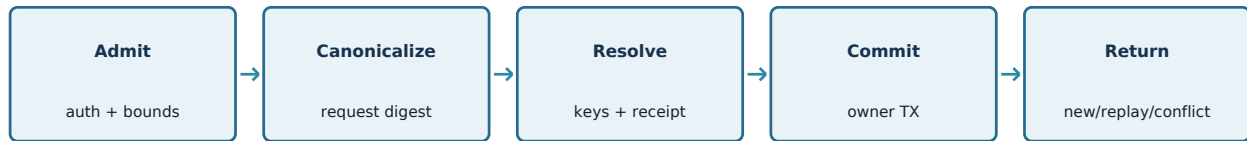
DECISIONIDENTITY

The value binds canonical decision inputs and policy identity. It is not an authorization token, row version or substitute for the operation receipt.

ALARM IDEMPOTENCY PROTOCOL

Retries return history; conflicts never overwrite it

ACCEPTANCE PATH C-13



REQUIRED ORACLES C-13

Observation	Required result
same Idempotency-Key, same OperationId and same request digest	same receipt; no extra effect
same Idempotency-Key, changed digest	409 identity-conflict; zero mutation
same source identity, new OperationId, same payload	semantic duplicate resolution
same source identity, changed PayloadSha256	409 evidence conflict
commit outcome unknown	fresh read by stable identity; never mint replacement history

ROOTCAUSE HTTP ROUTE MAP

Case opening is event-driven; owner mutations remain explicit

CANONICAL TARGET ROUTES C-14

Method + route	Capability	Meaning
GET /api/v1/root-cause-cases/{caseld}	root-cause.case.read	owner-local case truth
POST /api/v1/root-cause-cases/{caseld}/evidence	root-cause.evidence.attach	attach admitted evidence
POST /api/v1/root-cause-cases/{caseld}/verdicts	root-cause.verdict.issue	issue or abstain under policy
no public create route	n/a	AlarmFloodDetected.v1 opens the final-slice case

EVIDENCE STATUS

These routes resolve the exact canonical target left generic in Chapter 60. Physical controller presence and runtime conformance remain PENDING; the catalogue must fail if a different route is discovered.

ROOTCAUSE EVIDENCE COMMAND

Evidence enters under an explicit revision precondition

ATTACHROOTCAUSEEVIDENCEV1

```

public sealed record AttachRootCauseEvidenceV1(
    Guid OperationId,
    string EvidenceKey,
    string EvidenceType,
    string SourceReference,
    string ContentSha256,
    DateTimeOffset ObservedAtUtc,
    int ExpectedEvidenceRevision);

public sealed record EvidenceMutationReceiptV1(
    Guid OperationId, string CaseId, string Outcome,
    int EvidenceRevision, int EvidenceCount,
    string EvidenceMembershipSha256);
  
```

EVIDENCE OUTCOMES C-15

Condition	HTTP / outcome
new valid evidence on expected revision	200 Attached
same semantic EvidenceKey	200 Duplicate; same receipt meaning
revision no longer matches	412 precondition-failed
closed/frozen case	409 case-state-conflict
invalid type, digest or source reference	422 semantic-invalid

ROOTCAUSE VERDICT COMMAND

The issue request binds policy and exact evidence membership

ISSUEROOTCAUSEVERDICTV1

```
public sealed record IssueRootCauseVerdictV1(  
    Guid OperationId,  
    string VerdictCode,  
    decimal Confidence,  
    string PolicyIdentity,  
    int ExpectedEvidenceRevision,  
    string EvidenceMembershipSha256);  
  
public sealed record RootCauseVerdictReceiptV1(  
    Guid OperationId, string CaseId, string Outcome,  
    string? VerdictId, string? VerdictIdentity,  
    int EvidenceRevision, DateTimeOffset ObservedAtUtc);
```

VERDICT OUTCOMES C-16

Outcome	HTTP	Meaning
Issued	200	verdict and Pending public fact committed
Existing	200	matching stable operation resolved
Abstained	200	policy refused to issue; no verdict fact
EvidenceChanged	412	revision or membership digest mismatch
IdentityConflict	409	same operation names different request
Rejected	422	typed policy/request validation failed

ROOTCAUSE OWNER QUERY

Owner truth and projected observation are not interchangeable

ROOTCAUSECASEV1

```
public sealed record RootCauseCaseV1(  
    string CaseId,  
    string Status,  
    string SourceDecisionId,  
    string SiteId,  
    string LineId,  
    int EvidenceRevision,  
    int EvidenceCount,  
    string EvidenceMembershipSha256,  
    RootCauseVerdictSummaryV1? Verdict,  
    DateTimeOffset OpenedAtUtc,  
    long RepresentationVersion);
```

QUERY SEMANTICS C-17

Condition	Result
authorized existing case	200 exact owner-local representation
authorized absent case	404 owner-local not-found
invalid identifier	400 request-invalid
wrong scope or capability	403 with no existence disclosure
owner unavailable	503; caller must not substitute a projection as authority

REPORTING INTERNAL QUERY

The BFF reads one derived owner through a bounded contract

CANONICAL INTERNAL ROUTE C-18

Coordinate	Decision
method + route	GET /internal/v1/root-cause-case-projections/{caseId}
owner	Reporting
caller	Query BFF workload identity only
capability	reporting.root-cause-case-projection.read
scope	site/tenant predicate applied before row disclosure
timeout	bounded by the BFF request budget
data source	active Reporting projection generation only

NO DATABASE SHORTCUT

Query BFF has no ReportingDb credential. Reporting's database schema is not an internal API contract.

PUBLIC BFF QUERY

The final route returns an authorized observation

CANONICAL PUBLIC ROUTE C-19

Coordinate	Exact decision
method + route	GET /api/v1/root-cause-cases/{caseId}
owner	Query BFF edge
capability	reporting.root-cause-case.read
Accept	application/vnd.nexus1.operator-case.v1+json
success media	application/vnd.nexus1.operator-case.v1+json
cache	Cache-Control: private, no-store
source	Reporting contract only; no synchronous owner/database fan-in

SUPERSEDED ROUTE

The Chapter 35 reference route `/api/operator/v1/root-cause-cases/{caseId}` is superseded by the Chapter 60 final route above. Clients and tests must not support both as implicit aliases.

BFF RESPONSE DTO

Value, watermark and freshness travel together

OPERATORCASEV1

```
public sealed record OperatorCaseV1(
    string ObservationState,
    RootCauseCaseSummaryV1? Data,
    ProjectionFreshnessV1 Freshness,
    DateTimeOffset ObservedAtUtc);

public sealed record StreamWatermarkV1(
    string StreamId,
    long LastContiguousPosition);
```

STABLE FIELDS C-20

Field	Meaning
ObservationState	observed not-observed-yet
Data	Reporting-owned case summary; null is legal
Freshness	evidence qualifying the observation
ObservedAtUtc	one BFF clock sample used for response ages
Watermark	source stream and last contiguous position; nullable when unobserved

PROJECTION FRESHNESS DTO

Current is earned after every negative gate passes

PROJECTIONFRESHNESSV1

```
public sealed record ProjectionFreshnessV1(
    string ProjectionName,
    string ContractSetIdentity,
    string ProjectionGenerationId,
    string UpdateStatus,
    string ReasonCode,
    long? ProjectionAgeMs,
    long? RepresentedEventAgeMs,
    long StatusObservationAgeMs,
    int GapCount,
    StreamWatermarkV1? Watermark,
    DateTimeOffset? LastAppliedAtUtc,
    DateTimeOffset StatusObservedAtUtc);
```

UPDATESTATUS VOCABULARY C-21

Value	Meaning
current-as-observed	fresh sample; no known failure, gap or backlog
catching-up	known work remains
delayed	age/backlog exceeded warning policy
gap-detected	contiguous source history is incomplete
failed	terminal projector or contract conflict known
unknown	evidence too old or insufficient to classify

PROJECTION ABSENCE AND FAULT SEMANTICS

Not observed yet is not a business 404

BFF OUTCOME MATRIX C-22

Condition	HTTP / body
row absent; path healthy	200 + not-observed-yet/current-as-observed
row absent; backlog or gap	200 + not-observed-yet/catching-up or gap-detected
row present; delayed status	200 + value and explicit delayed evidence
Reporting unavailable	503 Problem Details; no fabricated stale body
invalid UUID	400 versioned Problem Details
unauthorized scope	403 without projection disclosure

TRUTH WORDING

The UI may say Current as observed with the timestamp and watermark. It may not say globally current, live source truth or complete merely because HTTP returned 200.

HTTP NEGATIVE CONTRACT CORPUS

A route is defined by its refusals as well as its success

REQUIRED MUTATIONS C-23

Mutation	Expected result	Side effects
unknown JSON member	400 request-invalid	0
wrong content type	415 unsupported-media-type	0
oversized body	413 request-too-large	0
unknown enum value	422 semantic-invalid	0
changed request under stable idempotency key	409 identity-conflict	0
stale evidence revision	412 precondition-failed	0
missing capability / wrong scope	403 not-authorized	0
owner not ready	503 owner-not-ready	0

ZERO MEANS ZERO

Negative corpus tests assert no Domain invocation, no database command, no outgoing fact and no resource-disclosing telemetry.

LIFECYCLE ENDPOINTS

Startup, liveness and readiness answer different questions

HOST ROUTES C-24

Route	Success	Failure	Never checks
GET /health/startup	startup completed	503 initializing/failed	ongoing downstream health
GET /health/live	process can make progress	503 local fatal state	database, broker or peer
GET /health/ready	role can accept assigned work	503 safely withdrawn	unrelated downstream services

SAFE HEALTH BODY

```
public sealed record HealthResponseV1(  
    string Status,  
    string Role,  
    string BuildIdentity,  
    DateTimeOffset ObservedAtUtc,  
    IReadOnlyList<string> SafeReasonCodes);
```

EXPOSURE RULE

Health responses reveal no exception, endpoint, credential, connection string, migration SQL or tenant/resource identifier.

PUBLIC EVENT ENVELOPE V1

Business identity is inside the canonical body

PUBLICFACTENVELOPEV1

```
public sealed record PublicFactEnvelopeV1<TPayload>(  
    int EnvelopeVersion,  
    string MessageId,  
    string EventType,  
    int SchemaVersion,  
    DateTimeOffset OccurredAtUtc,  
    string Producer,  
    string CorrelationId,  
    string CausationId,  
    string ContentType,  
    string FingerprintAlgorithm,  
    TPayload Payload,  
    string ContentFingerprint);
```

ENVELOPE CONSTANTS C-25

Coordinate	V1 value
EnvelopeVersion	1
ContentType	application/json
FingerprintAlgorithm	sha-256+jcs
encoding	UTF-8 canonical JSON
unknown top-level member	reject

ENVELOPE FIELD SEMANTICS

Every field answers one bounded question

ENVELOPE LEDGER C-26

Field	Authority	Meaning
MessageId	producer	immutable public-fact identity
EventType	producer contract	fully qualified fact and major version
SchemaVersion	producer contract	payload reader coordinate
OccurredAtUtc	producer	when the represented fact occurred
Producer	producer contract	semantic owner, not a host instance
CorrelationId	journey	groups related work; never uniqueness/authorization
CausationId	producer	direct business cause
ContentFingerprint	codec profile	canonical meaning without this field
Payload	producer	versioned public evidence

OCCURRENCE IS NOT OBSERVATION

Consumers record ReceivedAtUtc, RecordedAtUtc and AppliedAtUtc separately. They do not overwrite OccurredAtUtc or use arrival time to reorder producer history.

BUSINESS IDENTITY VERSUS DELIVERY METADATA

A retry changes the attempt, not the historical fact

SEPARATION C-27

Business envelope	Delivery metadata outside body
MessageId	delivery tag / redelivered flag
EventType + SchemaVersion	exchange / queue / binding observation
CorrelationId + CausationId	traceparent / tracestate / span ID
OccurredAtUtc	published, received and settlement timestamps
ContentFingerprint + exact body digest	retry count / lease / channel number
ProducerStream	prefetch, consumer tag and client headers

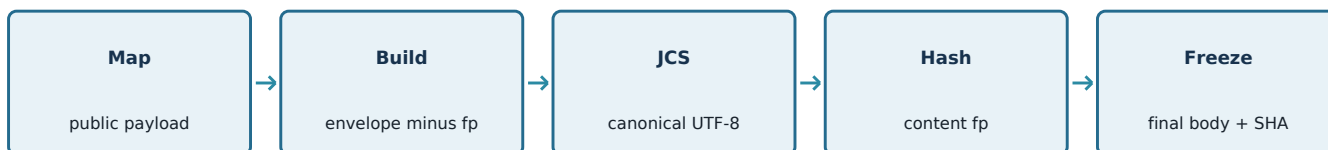
RETRY INVARIANT

Every dispatch attempt for one stored fact preserves the same MessageId, event coordinates, ContentFingerprint and exact serialized bytes. Only bounded attempt evidence changes.

CANONICAL SERIALIZATION AND FINGERPRINTS

Meaning digest and final-byte digest answer different questions

ENCODING PATH C-28



DIGEST RULES C-28

Artifact	Calculation	Use
fingerprint input	RFC 8785-compatible canonical envelope without ContentFingerprint	semantic integrity
ContentFingerprint	sha256:{lowercase hex of fingerprint input}	self-verifying contract
final body	canonical envelope including ContentFingerprint	stored and transported bytes
EnvelopeSha256	SHA-256 of final body	outbox, capture and replay identity

PRETTY JSON

Readable examples are documentation only. Tests compare retained one-line bytes, exact length and both digests.

PRODUCER FREEZE AND REPLAY

The owner commits bytes before transport begins

PRODUCER OBLIGATIONS C-29

Moment	Required invariant
mapping	only reviewed public fields; deterministic order
encoding	descriptor, document and stored coordinates self-verify
owner transaction	domain result and exact Pending body commit together
dispatch	load immutable stored bytes; do not rebuild from current rows
unknown confirm	retry same MessageId and bytes
published result	record observed mandatory route + positive confirm separately

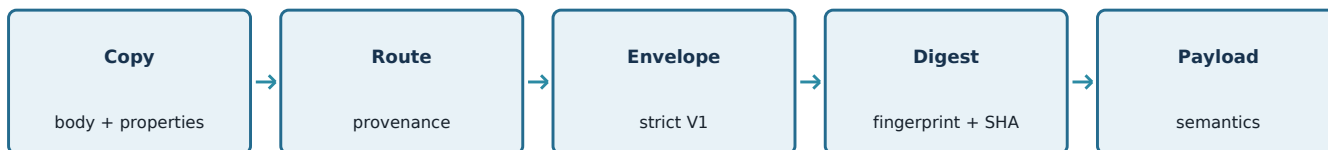
FORBIDDEN

A relay never invokes a public mapper, reads the current aggregate to reconstruct history, changes correlation from Activity.Current or allocates a replacement MessageId.

CONSUMER ADMISSION ORDER

Bytes are trusted in a fail-closed sequence

ADMISSION PATH C-30



ADMISSION ORDER C-30

Step	Failure example	Before Domain?
physical bounds	body-size-invalid	yes
exchange/route/type/media	unexpected-route	yes
producer path authority	producer-path-not-authorized	yes
strict envelope	envelope-unknown-member	yes
identity equality	message-id-mismatch	yes
fingerprint and body	fingerprint-mismatch	yes
typed payload semantics	invalid-window/order/digest	yes

ALARMFLOODDETECTED.V1 CONTRACT CARD

AlarmManagement publishes sufficient flood evidence

COORDINATES C-31

Coordinate	Exact value
producer	alarm-management
event type	nexus1.alarm-management.alarm-flood-detected.v1
schema / envelope	1 / 1
exchange	nexus.events
routing key	alarm-management.alarm-flood-detected.v1
content / fingerprint	application/json; UTF-8 / sha-256+jcs
core consumer	RootCause only
status	ACTIVATED SPECIFICATION; runtime PENDING

PUBLIC MEANING

One AlarmManagement-owned flood decision occurred under a named policy and carries enough immutable evidence for RootCause to open a case without reading AlarmManagementDb.

ALARMFLOODDETECTED.V1 PAYLOAD

The public fact exports evidence, not the aggregate

EXACT V1 PAYLOAD

```
public sealed record AlarmFloodDetectedV1(  
    string DecisionId,  
    string SiteId,  
    string LineId,  
    DateTimeOffset WindowStartUtc,  
    DateTimeOffset WindowEndUtc,  
    IReadOnlyList<AlarmReferenceV1> Alarms,  
    string PolicyId,  
    DateTimeOffset DetectedAtUtc,  
    ProducerStreamPositionV1 ProducerStream);  
  
public sealed record AlarmReferenceV1(string AlarmId, int Position);  
public sealed record ProducerStreamPositionV1(string StreamId, long Position);
```

SEMANTIC RULES C-32

Field family	Invariant
window	start < end <= detected time; UTC
alarms	nonempty; positions exactly 0..N-1; public IDs only
DecisionId	producer semantic key; not a RootCause Caseld
PolicyId	stable public policy identity; not implementation state
ProducerStream	owner-local sequence; never global order

ALARMFLOODDETECTED.V1 GOLDEN CORPUS

The first activated fact has one frozen byte identity

GOLDEN VECTOR C-33

Coordinate	Exact result
MessageId	019bc4d4-5dc1-7b8d-9d48-542d25315356
correlation / causation	request-am-20482 / decision-am-9184
OccurredAtUtc	2026-07-19T07:43:11.482Z
stream	alarm-management/site-ath-17/line-02 @ 84
ContentFingerprint	sha256:ef70dfaafe0e08599eb82877a77e60fe3d94b2ae0aa1456d2a175af9e2712b97
canonical length	1003 UTF-8 bytes
final body SHA-256	7cf98319c20a5915d5710bbcb5878fe51cdae89a5b68081c135fd0e962c276a

GOLDEN PAYLOAD

Decision flood-am-6031; site-ath-17; line-02; five ordered alarm references; policy alarm-flood-policy/3. Any one-byte drift fails the corpus gate.

ROOTCAUSECASEOPENED.V1 CONTRACT CARD

RootCause publishes creation, not conclusion

COORDINATES C-34

Coordinate	Exact value
producer	root-cause
event type	nexus1.root-cause.root-cause-case-opened.v1
schema / envelope	1 / 1
exchange	nexus.events
routing key	root-cause.root-cause-case-opened.v1
content / fingerprint	application/json; UTF-8 / sha-256+jcs
core consumer	Reporting
status	ACTIVATED SPECIFICATION; runtime PENDING

PUBLIC MEANING

RootCause created one owner-local case from the named AlarmFloodDetected message. The fact contains no hypothesis, verdict, mutable case graph or private evidence payload.

ROOTCAUSECASEOPENED.V1 PAYLOAD

The first case fact preserves the causal edge

EXACT V1 PAYLOAD

```
public sealed record RootCauseCaseOpenedV1(  
    string CaseId,  
    string SourceAlarmFloodMessageId,  
    string SourceDecisionId,  
    string SiteId,  
    string LineId,  
    DateTimeOffset OpenedAtUtc,  
    int InitialEvidenceCount,  
    string CasePolicyId,  
    ProducerStreamPositionV1 ProducerStream);
```

SEMANTIC RULES C-35

Field	Invariant
Caseld	allocated once by RootCause; stable public reference
SourceAlarmFloodMessageId	equals envelope CausationId
InitialEvidenceCount	positive; golden opening contains five
CasePolicyId	root-cause/open-case/1 in golden corpus
ProducerStream	root-cause/case/{Caseld} at position 1

ROOTCAUSECASEOPENED.V1 GOLDEN CORPUS

Case opening is the second frozen body

GOLDEN VECTOR C-36

Coordinate	Exact result
MessageId	019bc4d4-6ea0-7d14-a57a-3e8643c55f13
Caseld	019bc4d4-6dfe-7bcb-9de8-3af71979285a
correlation / causation	request-am-20482 / 019bc4d4-5dc1-7b8d-9d48-542d25315356
OccurredAtUtc	2026-07-19T07:43:12.017Z
stream	root-cause/case/019bc4d4-6dfe-7bcb-9de8-3af71979285a @ 1
ContentFingerprint	sha256:4a04853244adcdbc8f35834949ea0532fdd43d3cc24fe225e71104c8a581ce27
canonical length	876 UTF-8 bytes
final body SHA-256	454f51cf4b0250b9c855346a6648a7d73530f19687ee3b52cda2edf9d0339c5b

ROOTCAUSEVERDICTISSUED.V1 CONTRACT CARD

The verdict fact carries its evidence basis

COORDINATES C-37

Coordinate	Exact value
producer	root-cause
event type	nexus1.root-cause.root-cause-verdict-issued.v1
schema / envelope	1 / 1
exchange	nexus.events
routing key	root-cause.root-cause-verdict-issued.v1
content / fingerprint	application/json; UTF-8 / sha-256+jcs
core consumer	Reporting; Audit/Compliance remain candidate consumers
status	ACTIVATED SPECIFICATION; runtime PENDING

FAN-OUT BOUNDARY

One producer body may reach several independently owned queues. Subscriber-specific mutations or envelopes are forbidden.

ROOTCAUSEVERDICTISSUED.V1 PAYLOAD

Policy, confidence and evidence membership are frozen

EXACT V1 PAYLOAD

```
public sealed record RootCauseVerdictIssuedV1(  
    string VerdictId, string RootCauseCaseId,  
    string SourceAlarmFloodMessageId,  
    string SiteId, string LineId,  
    string VerdictCode, decimal Confidence,  
    string PolicyIdentity, string VerdictIdentity,  
    int EvidenceRevision, int EvidenceCount,  
    string EvidenceMembershipSha256,  
    DateTimeOffset IssuedAtUtc,  
    ProducerStreamPositionV1 ProducerStream);
```

SEMANTIC RULES C-38

Coordinate	Invariant
Confidence	canonical decimal 0..1; no binary-float reinterpretation
VerdictIdentity	digest of decision meaning and policy basis
EvidenceMembershipSha256	digest of ordered evidence membership
EvidenceRevision / Count	must agree with frozen verdict basis
ProducerStream	same RootCause case stream, position 2

ROOTCAUSEVERDICTISSUED.V1 GOLDEN CORPUS

The verdict is the third activated frozen body

GOLDEN VECTOR C-39

Coordinate	Exact result
MessageId	019bc4d4-82a4-77f2-ae06-09b172aa59db
VerdictId	019bc4d4-81c0-7ea1-82d5-4f9b8719d0a1
correlation / causation	request-am-20482 / decision-rc-7184
OccurredAtUtc	2026-07-19T07:47:38.442Z
evidence membership SHA	af3f3ad8da7edfaa50d14ebf422d424e2dd4e49767a2f4a056ce7682a801b5b7
verdict identity	a2451a72990d5933e175396c86fd96854ff1db0e6571e0ae85afc37fb1734c17
ContentFingerprint	sha256:07d530ef636be3d2af8dcc3b20cd91b212a076a4e70c1a9c55730930fbfe1e2e
canonical length / SHA-256	1157 / 192e5e73c5a35d438e2ee9edb0a9e7044293c29bf321ba1abe34f191d8b44c00

GOLDEN MEANING

COMMON_POWER_SUPPLY_INSTABILITY at confidence 0.94 under root-cause/verdict/3, with evidence revision 1 and count 5.

PRODUCER STREAM AND ORDERING

Contiguous owner order is not delivery order

STREAM RULES C-40

Rule	Required behaviour
scope	one producer-owned stream; no cross-service global sequence
position	positive, immutable and never reused for changed meaning
consumer checkpoint	advance only through the contiguous admitted prefix
future position	durably buffer exact bytes; record the gap
same position + same meaning	semantic duplicate; no second effect
same position + changed meaning	terminal position conflict

GOLDEN CASE STREAM C-40

Position	Fact
1	RootCauseCaseOpened.v1
2	RootCauseVerdictIssued.v1

CONSUMER ALLOWLISTS AND ROUTES

A broad wildcard must not activate a new contract

CORE ALLOWLIST C-41

Consumer	Exact accepted event	Action
RootCause	nexus1.alarm-management.alarm-flood-detected.v1	translate and open/resolve case
Reporting	nexus1.root-cause.root-cause-case-opened.v1	apply stream position 1
Reporting	nexus1.root-cause.root-cause-verdict-issued.v1	apply stream position 2

ROUTING ALLOWLIST C-41

Producer	Publish family	Denied example
alarm-management	alarm-management.#	root-cause.root-cause-case-opened.v1
root-cause	root-cause.#	alarm-management.alarm-flood-detected.v1
reporting	reporting.# when required	root-cause.#

ADMISSION CHANGE

Adding a reader or binding is a reviewed contract-set change. Deploying a consumer that happens to deserialize the payload does not make it approved.

CONSUMER SETTLEMENT AND DURABLE EFFECTS

Acknowledgement follows a known local outcome

SETTLEMENT MATRIX C-42

Local result	Broker action	Contract claim
committed new effect	ack	inbox + owner effect durable
matching duplicate	ack	prior durable result proved
durably buffered gap	ack	exact bytes now owned locally
transient / commit unknown	retry or remain unsettled	no success inferred
invalid contract	reject to terminal evidence	zero Domain effect
identity/position conflict	reject + incident evidence	history unchanged

AT-LEAST-ONCE BOUNDARY

Transport may redeliver. The contract requires effectively-once business consequences only where durable transport and semantic identities are both proved.

DUPLICATE AND CONFLICT GRAMMAR

Identity equality is accepted only after digest equality

RESOLUTION C-43

Observation	Classification	Disposition
same MessageId + same body SHA	transport duplicate	resolve stored receipt
same MessageId + changed body SHA	identity conflict	quarantine; no effect
new MessageId + same semantic key + same meaning	semantic duplicate	one effect
new MessageId + same semantic key + changed meaning	semantic conflict	terminal evidence
same stream position + same semantic digest	position duplicate	no view change
same stream position + changed digest	position conflict	fail projection generation

NEVER BY AGE

A message is not a duplicate because it is old, redelivered or correlated with prior work. Durable matching identities and digests are required.

COMPATIBILITY CLASSIFICATION

Strict V1 readers make in-place field growth breaking

CHANGE MATRIX C-44

Change	V1 classification	Required action
add JSON member	BREAKING under strict unknown-member rejection	new schema/version + reader rollout
remove or rename member	BREAKING	new major/schema
change type, nullability or numeric grammar	BREAKING	new contract
add enum value	BREAKING unless reader explicitly admits it	new reader set/version
change field semantics or authority	BREAKING	new contract + migration decision
change producer, event type or route	BREAKING coordinate	side-by-side publication
change whitespace outside canonical corpus	NOT A VALID CHANGE	canonical bytes remain exact

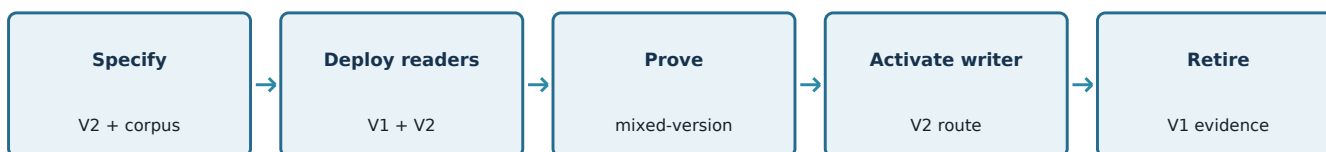
READER-FIRST RULE

Compatibility is evaluated against retained producer and consumer corpora, not against a serializer's claim that two object graphs look equivalent.

VERSION ROLLOUT PROTOCOL

Readers advance before writers and versions coexist

BREAKING ROLLOUT C-45



ROLLOUT GATES C-45

Gate	Required evidence
contract set	new schema ID, type/route, owner and change rationale
consumer readiness	every allowlisted consumer accepts retained V2 positive/negative corpus
side-by-side operation	V1 and V2 metrics, no accidental wildcard admission
writer activation	producer publishes only after readers and topology are ready
retirement	zero admitted V1 backlog, retention/replay decision and rollback boundary

INACTIVE AND CANDIDATE FACTS

Designed language does not enter the final contract set

INACTIVE CATALOGUE C-46

Fact / family	Placement	Final status
RecommendationGenerated.v1	RL Advisory candidate	NOT ACTIVATED; no core consumers
ComplianceReviewOpened.v1	Compliance extension	DESIGNED EXTENSION; not a final core fact
RootCauseVerdictCorrected.v1	later correction design	NOT IN FINAL THREE-FACT CORE
ComplianceReviewCorrectionApplied.v1	later correction design	NOT IN FINAL THREE-FACT CORE
Process Manager workflow contracts	coordination extension	NOT IN SEVEN-ROLE CORE

ACTIVATION GATE

A candidate needs an admitted owner, independent lifecycle rationale, credentials, topology, consumer allowlist, compatibility corpus, fault evidence and updated Appendix B/C control totals.

HISTORICAL CONTRACT RECONCILIATION

Earlier mechanism examples remain useful without remaining normative

RESOLUTION LEDGER C-47

Earlier shape	Final decision
reduced AlarmFloodDetectedV1 records	superseded by Chapter 32 exact payload and 1003-byte corpus
/api/v1/alarm-flood-detections	superseded by POST /api/v1/alarm-observations
/api/operator/v1/root-cause-cases/{caseld}	superseded by GET /api/v1/root-cause-cases/{caseld}
generic RootCause command APIs	resolved to evidence and verdict target routes in C-14
Audit/Compliance as normal verdict subscribers	candidate consumers; Reporting is the only final core consumer
four/six-fact workflow catalogues	extension designs; final core activates exactly three facts

INTERPRETATION RULE

Superseded examples still teach their local mechanism. They cannot override the final route, payload, consumer or activation decisions in Chapters 60 and Appendices B-C.

CONTRACT CORPUS LAYOUT

Positive and negative artifacts are versioned together

REQUIRED REPOSITORY SHAPE

```
contracts/  
  http/alarm-observation/v1/  
    request.accepted.json  
    response.accepted.json  
    rejected/*.json  
  events/alarm-flood-detected/v1/  
    canonical.json  
    manifest.json  
    rejected/*.json  
  events/root-cause-case-opened/v1/...  
  events/root-cause-verdict-issued/v1/...  
  bff/operator-case/v1/...  
  catalogue/contracts.v1.json
```

CORPUS MANIFEST C-48

Field	Purpose
contractId / schemald	stable coordinate
owner / consumers	authority and allowlist
bodyLength / bodySha256	exact bytes
contentFingerprint	canonical meaning
expectedDisposition	positive or stable rejection
sourceCommit / toolchain	reproducibility

GOLDEN EVENT REGISTER

Three bodies form the canonical activated corpus

ACTIVATED CORPUS C-49

Fact	Bytes	Final SHA-256
AlarmFloodDetected.v1	1003	7cf98319c20a5915d5710bbccb5878fe51cdae89a5b68081c135fd0e962c276a
RootCauseCaseOpened.v1	876	454f51cf4b0250b9c855346a6648a7d73530f19687ee3b52cda2edf9d0339c5b
RootCauseVerdictIssued.v1	1157	192e5e73c5a35d438e2ee9edb0a9e7044293c29bf321ba1abe34f191d8b44c00

JOURNEY IDENTITY C-49

Coordinate	Golden value
CorrelationId	request-am-20482 across all three facts
direct causation	decision-am-9184 -> flood MessageId -> decision-rc-7184
RootCause case stream	CaseOpened @ 1; VerdictIssued @ 2

CONTROL TOTAL

Exactly three activated facts means exactly three positive canonical event bodies in the final core register. Extra discovered active routes fail the catalogue gate.

EVENT MUTATION CORPUS

One mutation must cause one stable refusal

NEGATIVE VECTORS C-50

Mutation	Expected code	Effect
unknown envelope member	envelope-unknown-member	0 Domain calls
MessageId property/body mismatch	message-id-mismatch	0
event type or schema mismatch	unsupported-contract	0
payload changed; old fingerprint	fingerprint-mismatch	0
fingerprint updated; stale body digest	body-digest-mismatch	0
wrong producer route authority	producer-path-not-authorized	0
duplicate/gapped alarm positions	invalid-public-order	0
invalid verdict evidence digest	verdict-evidence-summary-invalid	0
stream position conflict	stream-position-conflict	0 visible change

CONTRACT TEST LAYERS

Different tests prove different parts of the promise

PROOF MATRIX C-51

Layer	Proves	Cannot prove alone
DTO/schema unit	required members, enum and strict-reader rules	transport coordinates
mapper test	public field lineage, determinism and leakage absence	stored bytes
codec corpus	canonical bytes, length and digests	broker routing
producer component	owner transaction stages exact body	consumer effect
broker capture	published properties/route/body preserved	business idempotency
consumer component	admission, inbox and owner-local effect	deployed cross-service path
end-to-end harness	same corpus crosses final topology	production readiness

ZERO DISCOVERY

A test run that discovers zero contract cases fails. Every receipt reports expected, discovered, passed, failed and skipped counts.

MACHINE-READABLE CATALOGUE SCHEMA

The document and the build consume the same coordinates

CONTRACTCATALOGUEENTRYV1

```
public sealed record ContractCatalogueEntryV1(  
    string ContractId,  
    string Kind,  
    string Owner,  
    string Version,  
    string Coordinate,  
    string SchemaId,  
    string Status,  
    IReadOnlyList<string> Consumers,  
    string CorpusPath,  
    string CorpusSha256,  
    string CompatibilityProfile,  
    string EvidenceStatus);
```

CLOSED VOCABULARIES C-52

Field	Allowed values
Kind	HTTP_COMMAND HTTP_QUERY EVENT HEALTH
Status	ACTIVATED_SPECIFICATION CANONICAL_TARGET DESIGNED_INACTIVE
CompatibilityProfile	strict-v1 strict-envelope-v1+jcs
EvidenceStatus	PENDING DEMONSTRATED FALSIFIED

CATALOGUE ENTRY EXAMPLES

Coordinates are data, not comments

VALUE-FREE EXAMPLES

```
{  
  "contractId": "event.alarm-flood-detected.v1",  
  "kind": "EVENT",  
  "owner": "AlarmManagement",  
  "coordinate": "nexus1.alarm-management.alarm-flood-detected.v1",  
  "status": "ACTIVATED_SPECIFICATION",  
  "consumers": ["RootCause"],  
  "compatibilityProfile": "strict-envelope-v1+jcs",  
  "evidenceStatus": "PENDING"  
}  
{  
  "contractId": "event.recommendation-generated.v1",  
  "status": "DESIGNED_INACTIVE",  
  "consumers": [],  
  "evidenceStatus": "PENDING"  
}
```

NO ACTIVATION BY FILE

A schema file for an inactive contract may exist for review. Its presence does not authorize publishing, binding, consuming or counting it as a final-core fact.

CATALOGUE VALIDATOR

The build fails closed on ownership, count and drift

CORE VALIDATION SHAPE

```
Require(entries.Count(x => x.Kind == "EVENT" &&
    x.Status == "ACTIVATED_SPECIFICATION") == 3,
    "ACTIVATED_EVENT_COUNT_DRIFT");

RequireUnique(entries, x => x.ContractId);
RequireEvery(entries, x => ExactlyOneOwner(x));
RequireEvery(activatedEvents, x => x.Consumers.Count > 0);
RequireExact(activatedEvents.Select(x => x.ContractId),
    "event.alarm-flood-detected.v1",
    "event.root-cause-case-opened.v1",
    "event.root-cause-verdict-issued.v1");
RejectDiscoveredRouteNotInCatalogue(routeInventory);
RejectCatalogueEntryWithoutCorpusOrStatus();
```

FAIL CODES C-53

Code	Meaning
CONTRACT_OWNER_AMBIGUOUS	zero or multiple semantic owners
ACTIVE_ROUTE_UNCATALOGUED	runtime/repository route has no approved entry
ACTIVATED_EVENT_COUNT_DRIFT	final three-fact invariant changed
CORPUS_DIGEST_MISMATCH	retained bytes changed
INACTIVE_BINDING_DISCOVERED	candidate contract entered topology

EVIDENCE RECORD

A pass is bound to identities and retained artifacts

CONTRACTEVIDENCERECORDV1

```
public sealed record ContractEvidenceRecordV1(
    string EvidenceId,
    string BuildIdentity,
    string RunId,
    string CatalogueSha256,
    string ContractCorpusSha256,
    string ProducerArtifactSha256,
    IReadOnlyList<string> ConsumerArtifactSha256,
    int Expected, int Discovered,
    int Passed, int Failed, int Skipped,
    string Status,
    DateTimeOffset ObservedAtUtc,
    IReadOnlyList<string> ArtifactReferences);
```

ACCEPTANCE CONDITIONS C-54

Condition	Required value
Expected == Discovered	true and nonzero
Failed / Skipped	0 / 0 for the admitted gate
identities	same build, catalogue and corpus across producer/consumers
artifacts	exist, hash correctly and are retained
status	DEMONSTRATED only after all validations pass

DRIFT AND RELEASE GATES

Contract drift blocks promotion before traffic

RELEASE COMPARISON C-55

Compared artifact	Blocking drift
OpenAPI / endpoint inventory	method, route, media, auth or status differs
public API manifest	DTO member/type/nullability changed
schema/corpus	bytes, digest, required member or rejection changed
broker topology	exchange, route, queue or producer permission differs
consumer allowlist	reader added/removed without compatibility decision
catalogue control totals	owner, role, database or activated-fact count drifts

PROMOTION RULE

An approved change updates the contract, corpus, reader rollout plan, catalogue digest and evidence expectations together. Regenerating a baseline solely to make a diff disappear is forbidden.

SECURITY AND PRIVACY REVIEW

Public does not mean unrestricted or non-sensitive

REVIEW QUESTIONS C-56

Surface	Required answer
identifier	is it a reviewed public reference rather than a private database key?
payload	is every field required by an approved consumer use case?
scope	can the caller observe only its authorized site/tenant?
error	can denial or validation leak resource existence or internal values?
telemetry	are body, token, key and evidence values excluded?
retention	who retains canonical bodies, digests and replay evidence?
candidate	can an inactive fact be accidentally routed or deserialized?

DATA MINIMIZATION

The catalogue contains contract identities and fixture values needed for reproducibility. Runtime logs and metrics retain bounded IDs/digests, not arbitrary payloads.

CONTRACT ANTI-PATTERNS

Convenient equivalence is where drift hides

REJECTED PATTERNS C-57

Anti-pattern	Why rejected
serialize the aggregate	leaks private model and rewrites historical meaning
accept unknown fields for future proofing	silently changes strict V1 compatibility policy
use correlation as idempotency	groups a journey but does not identify one effect
rebuild outbox body on retry	same MessageId can name changed bytes
wildcard consumer binding	new event becomes accidental input
HTTP 200 as global success	owner-local response cannot prove downstream convergence
404 for absent projection	conflates not observed with nonexistent
candidate schema means active service	design presence is not runtime admission

IMPLEMENTATION CHECKLIST

A contract is ready only when every layer agrees

PER-CONTRACT CHECKLIST C-58

Area	Required deliverable
ownership	Appendix B owner, decision authority and explicit exclusions
coordinates	exact route/type, major version, media and caller/consumer
types	strict DTO/schema with closed vocabulary and bounds
identity	idempotency, MessageId, semantic key and conflict behaviour
bytes	canonical positive corpus, length, fingerprint and body SHA
negative corpus	stable rejection code and zero side effects
evolution	compatibility classification, rollout and retirement
security	principal, capability, scope, safe error and minimization
evidence	nonzero tests, artifacts, identities and truthful status

APPENDIX C CHECKPOINT

The final core now has one contract authority

COMPLETION LEDGER C-59

Claim	Result
HTTP routes and DTOs	CANONICAL TARGETS FIXED
event envelope and payloads	THREE ACTIVATED V1 SPECIFICATIONS FIXED
golden bytes and digests	INTERNALLY FROZEN CORPUS
producer/consumer ownership	EXACT AND ALLOWLISTED
compatibility and rollout	STRICT V1 POLICY FIXED
candidate contracts	SEPARATED; NOT ACTIVATED
repository presence	PENDING
runtime and cross-language evidence	PENDING
production readiness	NOT CLAIMED

HONEST STATUS

CATALOGUE COMPLETE AS SPECIFICATION | FINAL CORE CONTRACT COUNT FIXED | PHYSICAL AND RUNTIME CONFORMANCE PENDING

HANDOFF TO APPENDIX D

The next appendix must bind configuration to these contracts

Appendix D - Configuration Reference will catalogue the exact configuration surfaces that allow these contracts to run without moving business meaning into environment variables. It must cover provider precedence, closed options schemas, validation, secret references, broker coordinates, route activation, policy identities, readiness dependencies, reload rules and value-free evidence manifests.

CARRY-FORWARD OBLIGATIONS C-60

From Appendix C	Appendix D must preserve
three activated event types	exact broker route/activation options; no candidate defaults
HTTP route ownership	host binding and authorization policy identity
strict reader profiles	explicit serializer/canonicalizer selection
consumer allowlists	closed configuration with startup validation
corpus and catalogue digests	immutable configuration/evidence identity references
PENDING evidence	no environment may be labelled verified without receipts

APPENDIX C LAST WORD

A contract is a promise that can fail a build

The catalogue becomes valuable when it refuses convenient ambiguity: one owner, one coordinate, one version, one exact corpus and one explicit status. That discipline lets implementations change while public history remains trustworthy.

FINAL STATUS

API AND EVENT CONTRACT CATALOGUE COMPLETE | THREE ACTIVATED FACTS | CANONICAL HTTP TARGETS FIXED |
REPOSITORY AND RUNTIME EVIDENCE PENDING

APPENDIX D

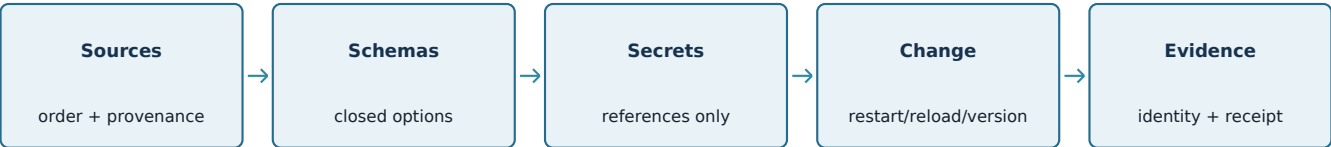
Configuration Reference

The final configuration authority for the NEXUS-1 Phase-0 core: provider order, strict schemas, validation, secret references, immutable policy identity, contract activation, readiness dependencies, safe reload and value-free evidence.

APPENDIX DECISION

Configuration may select technical composition and bounded operational mechanics. It may not move ownership, weaken invariants, rewrite public meaning or activate a candidate contract. Every admitted snapshot has a source order, schema, validation result, change contract and value-free identity.

REFERENCE READING PATH



CONTROL TOTALS D-1

Coordinate	Final value
core service / edge boundaries	3 / 1
long-running roles / migration jobs	7 / 3
core databases / activated public facts	3 / 3
candidate activation by default	0 - forbidden
evidence posture	SPECIFIED; repository and runtime receipts PENDING

PURPOSE AND BOUNDARY

Configuration is executable input with limited authority

A configuration value can choose an endpoint, resource budget, diagnostic exporter, secret reference or accepted policy artifact. That power must be bounded. Host owns provider composition and binding; Edge and Infrastructure consume narrow technical options; Application accepts immutable policy identities; Domain receives typed values only.

LAYER AUTHORITY D-2

Layer	May receive	Must never receive
Host	providers, environment identity, composition	business truth
Edge	HTTP/auth/presentation options	database credentials for foreign owners
Infrastructure	owned DB, broker and adapter options	Domain invariant switches
Application	accepted policy snapshot/identity	IConfiguration, IOptions, environment name
Domain	typed decision input + ConfigurationIdentity	framework types, endpoints, secrets, reload

BOUNDARY TEST

If a class asks IConfiguration for a key, it owns provider spelling, precedence, conversion, fallback and secret handling. Only Host and narrowly reviewed adapter bootstrap code may own that responsibility.

CONFIGURATION CLASSIFICATION

Every key has an owner, sensitivity and change contract

CLASSIFICATION AXES D-3

Axis	Closed values
owner	Host Edge Infrastructure Application policy
sensitivity	PUBLIC INTERNAL SECRET_REFERENCE DECISION_POLICY
mutability	RESTART_BOUND VALIDATED_RELOAD VERSIONED_IMMUTABLE
source	base JSON environment JSON development secret NEXUS1_env command line
failure	startup reject readiness withdraw retain last-known-good explicit replay failure
evidence	schema version + source category + identity/status; never unrestricted values

ADMISSION RULE

No key enters appsettings, deployment YAML or a secret store before these axes and its validation rule are recorded in the machine catalogue.

CONFIGURATION INVENTORY AT A GLANCE

Sections follow runtime responsibility, not convenience

SECTION CATALOGUE D-4

Section	Owner	Change
RuntimeIdentity	Host	restart-bound
Contracts	Host/Edge	restart-bound
Serialization	Infrastructure	restart-bound
Broker + Messaging	Infrastructure	restart/reload split
AlarmManagement:*	Alarm owner adapters	restart/reload/policy split
RootCause:*	RootCause adapters/Application	restart/reload/version split
Reporting:*	Reporting adapters	restart/reload split
QueryBff:*	Edge	restart-bound
Authentication + Authorization	Edge/Host	restart-bound policy identity
Health + Telemetry	Host/Infrastructure	restart/reload split
Migrations	migration jobs only	one-shot immutable plan

APPROVED PROVIDER ORDER

Later sources override earlier sources only by policy

PROVIDER PRECEDENCE D-5

Order	Provider	Permitted purpose
0	appsettings.json	non-secret, environment-neutral defaults
1	appsettings.{Environment}.json	environment mechanics and bounded tuning
2	User Secrets - Development only	non-production sensitive development material
3	NEXUS1_environment variables	deployment injection using __ hierarchy
4	command-line arguments	explicit, recorded operational override

EXPLICIT COMPOSITION

```
builder.Configuration.Sources.Clear();
builder.Configuration
    .AddJsonFile("appsettings.json", optional: false, reloadOnChange: false)
    .AddJsonFile($"appsettings.{env}.json", optional: true, reloadOnChange: false);
if (builder.Environment.IsDevelopment())
    builder.Configuration.AddUserSecrets<Program>(optional: true, reloadOnChange: false);
builder.Configuration
    .AddEnvironmentVariables(prefix: "NEXUS1_")
    .AddCommandLine(args);
```

PROVIDER PROVENANCE

The effective source order is inspected, not assumed

VALUE-FREE SOURCE RECORD

```
public sealed record ConfigurationSourceRecord(
    int Precedence,
    string ProviderKind,
    bool CanReload,
    string ApprovedPurpose);

// Records provider category and order only.
// It never enumerates configuration values.
```

PROVIDER GATES D-6

Finding	Failure code
unapproved provider type	CFG_SOURCE_PROVIDER_UNKNOWN
provider order differs	CFG_SOURCE_PRECEDENCE_DRIFT
command line not admitted for environment	CFG_SOURCE_OVERRIDE_FORBIDDEN
unprefixed environment-variable provider	CFG_SOURCE_ENV_PREFIX_MISSING
reload enabled on restart-bound file	CFG_SOURCE_RELOAD_CONTRACT_MISMATCH

NO ENVIRONMENT DUMP

The manifest records provider kinds/order and approved section names. It never calls `IConfiguration.AsEnumerable()` for evidence.

KEY GRAMMAR AND ENVIRONMENT MAPPING

One canonical key has one cross-platform spelling

KEY GRAMMAR D-7

Surface	Form	Example
JSON	Section:Group:Key	Messaging:OutboxDispatcher:BatchSize
environment	NEXUS1_Section_Group_Key	NEXUS1_Messaging__OutboxDispatcher__BatchSize
command line	--Section:Group:Key=value	--Messaging:OutboxDispatcher:BatchSize=32
secret reference	scheme://owner/purpose	secret://alarm-api/alarm-db-dsn
identity digest	sha256:{64 lowercase hex}	configuration/catalogue/corpus identity

NAMING RULE

Environment variables use double underscore for hierarchy. Environment identity selects mechanics only; it never appears inside a decision-policy digest.

CLOSED SCHEMA BINDING

Unknown, missing and malformed keys fail startup

STRICT OPTIONS REGISTRATION

```
services.AddOptions<TOptions>()
    .Bind(configuration.GetRequiredSection(TOptions.Section),
        binder => binder.ErrorOnUnknownConfiguration = true)
    .ValidateDataAnnotations()
    .ValidateOnStart();

// A custom IValidateOptions<TOptions> adds cross-field,
// ownership and contract-coordinate rules.
```

BINDING OUTCOMES D-8

Input	Outcome
required section absent	startup rejected
unknown member	startup rejected with safe key/rule
conversion failure	startup rejected; no fallback to default
valid shape; invalid relation	custom validator rejects
valid options; dependency unreachable	binding passes; startup/readiness policy decides

VALIDATION HIERARCHY

Binding success is not semantic acceptance

VALIDATION LEVELS D-9

Level	Example	Owner
presence/conversion	missing section, invalid enum, unknown key	Host binder
range/format	zero timeout, malformed URI/digest	options validator
cross-field	confirm timeout >= lease; retry budget >= request budget	adapter validator
ownership	RootCause DSN points to Alarm DB	deployment/architecture gate
contract	route/type/corpus identity differs from Appendix C	contract validator
decision policy	unrecognized version or digest mismatch	Application mapper
capability	credential cannot connect/use required object	startup/readiness probe

VALIDATEONSTART BOUNDARY

It proves that registered options can be created and validated. It does not prove database reachability, schema presence, broker topology, permissions, network path or downstream health.

STABLE VALIDATION DIAGNOSTICS

A startup failure names the rule without leaking the value

CONFIGURATION ERROR CATALOGUE D-10

Code	Meaning
CFG_SECTION_REQUIRED	required closed section absent
CFG_KEY_UNKNOWN	key not admitted by schema
CFG_VALUE_MALFORMED	conversion or format failure
CFG_RANGE_INVALID	bounded numeric/time value outside profile
CFG_CROSS_FIELD_INVALID	internally incompatible option combination
CFG_OWNER_MISMATCH	resource belongs to another boundary
CFG_CONTRACT_DRIFT	route/type/schema/corpus differs from Appendix C
CFG_SECRET_REFERENCE_INVALID	missing/unsupported secret reference
CFG_POLICY_IDENTITY_MISMATCH	versioned artifact digest differs

SAFE DIAGNOSTIC

Evidence may contain section, key, classification and rule code. It contains no value, substring, length, hash of secret material, DSN or exception dump.

SECRET CLASSIFICATION

Secrets are capabilities, not configuration evidence

SECRET CONTROLS D-11

Moment	Required behaviour	Forbidden
source	placeholder/schema only	credential in repository/history
development	User Secrets with isolated test value	production trust claim
deployment	approved provider/workload identity	image, ConfigMap or transcript value
application	resolve late inside adapter scope	Domain/Application propagation
telemetry	classification + present/missing only	value, hash, length, claims
rotation	explicit reconnect/restart contract	assume options notification rotates client

NO SECRET FINGERPRINT

Hashing a password, key or structured credential is not redaction. The value-free manifest never stores a secret-derived digest.

SECRET REFERENCES

The repository names requirements, never credentials

REFERENCE AND RESOLVER SEAM

```
public readonly record struct SecretReference(string Value)
{
    public override string ToString() => "[secret-reference]";
}

public interface ISecretResolver
{
    ValueTask<SecretLease> ResolveAsync(
        SecretReference reference,
        CancellationToken cancellationToken);
}
```

REFERENCE RULES D-12

Rule	Required result
scheme	approved development/provider scheme only
owner	reference belongs to exactly one workload role
purpose	named database, broker, OIDC or key-ring capability
ToString/logging	constant redacted representation
resolution	late, bounded, cancellable and value-free telemetry

CORE SECRET REQUIREMENT CATALOGUE

Runtime and migration authority never share a credential

REQUIRED REFERENCES D-13

Role	Required capability references
alarm-api	AlarmManagementDb runtime DML
alarm-worker	AlarmManagementDb runtime DML; broker publish
rootcause-api	RootCauseDb runtime DML
rootcause-worker	RootCauseDb runtime DML; broker consume/publish
projection-api	ReportingDb read-only
projection-worker	ReportingDb write; broker consume
query-bff	OIDC/client or protected key-ring capability; no database
alarm/rootcause/reporting migrator	one owned database DDL credential each

AUTHORITY RULE

Long-running runtime graphs contain no migration credential. Query BFF contains no database credential. No role receives a foreign owner's DB or queue capability.

SECRET CANARY AND ROTATION EVIDENCE

Synthetic values prove containment without using real credentials

CANARY SURFACES D-14

Surface	Required assertion
logs and exception responses	canary absent
metrics labels/attributes	canary absent
trace attributes/events	canary absent
health responses	canary absent
configuration/evidence manifests	canary absent; presence only
HTTP and event contracts	canary absent

ROTATION OUTCOMES D-14

Adapter	Change contract
database connection	restart or explicitly tested pool replacement
broker credential	new authenticated channel; bounded old/new overlap
OIDC/client secret	restart/rebuild authenticated client unless tested otherwise
data-protection key	persistent overlap and recovery policy

ENVIRONMENT BOUNDARY

Environment changes mechanics, never business meaning

ALLOWED DIFFERENCES D-15

Difference	Allowed?	Reason
database/broker endpoint	YES	technical location
secret resolver implementation	YES	deployment capability
telemetry exporter/detail	YES	observation mechanics
bounded timeout/concurrency	YES	validated operational capacity
explicit test adapter	TEST PROFILE	composition substitute, not Domain branch
disable invariant in Development	NO	truth cannot depend on host label
production-only verdict threshold	NO	hidden decision input
skip outbox/inbox in tests	NO	different correctness boundary

ENVIRONMENT IDENTITY

Only Development, CI/Test, Phase0Compose and Phase0Kubernetes profiles admitted by their own manifests may start. An unknown identity fails bounded startup.

ENVIRONMENT EQUIVALENCE

Different adapters must preserve the same accepted policy

EQUIVALENCE GATE

```
var identity = PolicyFixtures.AcceptedIdentity;
var input = DecisionFixture.StableInput(identity);

var development = EnvironmentFixture.For("Development").Decide(input);
var compose = EnvironmentFixture.For("Phase0Compose").Decide(input);
var kubernetes = EnvironmentFixture.For("Phase0Kubernetes").Decide(input);

compose.Should().BeEquivalentTo(development);
kubernetes.Should().BeEquivalentTo(development);
```

DIFF ALLOWLIST D-16

May differ	Must agree
secret resolver, endpoints, diagnostics/exporters	Domain/Application graph
timeouts/concurrency inside admitted ranges	contract types/routes/schema/corpus
resource requests/limits	policy identity and deterministic outcome
platform object kind	logical role, owner, health semantics and credentials

CONFIGURATION IDENTITY

One admitted snapshot has a reproducible, value-safe identity

IDENTITY RECORDS

```
public sealed record RuntimeConfigurationIdentity(
    string SchemaVersion,
    string SafeValuesSha256,
    string ProviderManifestSha256,
    string ContractCatalogueSha256,
    string ContractCorpusSha256,
    IReadOnlyList<PolicyIdentity> Policies);

public readonly record struct PolicyIdentity(
    string PolicyId, int Version, string Sha256);
```

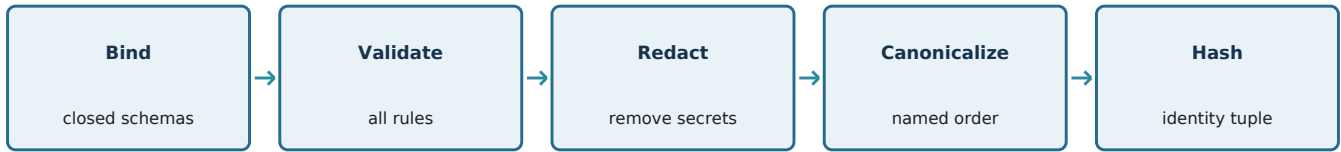
IDENTITY INPUTS D-17

Included	Excluded
canonical non-secret values and closed section versions	secret values or secret-derived hashes
provider kinds/order	absolute file paths
contract catalogue/corpus digests	runtime delivery attempts
accepted policy artifact identities	environment label as causal input
role activation and safe resource references	host instance IDs and timestamps

CANONICAL IDENTITY ALGORITHM

Schema, safe values and provenance are hashed separately

IDENTITY PATH D-18



DIGEST SEPARATION D-18

Digest	Protects
SafeValuesSha256	canonical non-secret effective configuration
ProviderManifestSha256	source kinds, order and reload attributes
ContractCatalogueSha256	approved HTTP/event coordinates and activation
ContractCorpusSha256	exact V1 positive/negative contract artifacts
PolicyIdentity.Sha256	one immutable decision-policy document

NO SINGLE OPAQUE HASH

Separate digests let evidence state whether values, provenance, contracts or policy changed without exposing confidential material.

DECISION-POLICY CONFIGURATION

Business meaning changes only through a new immutable artifact

ROOTCAUSE DECISION POLICY

```
public sealed record RootCauseDecisionPolicyDocument(  
    string PolicyId,  
    int Version,  
    int MinimumEvidenceCount,  
    decimal IssueThreshold,  
    string[] EvidenceOrder,  
    string ExpectedSha256);  
  
// Acceptance normalizes the public fields, computes SHA-256,  
// compares ExpectedSha256, then creates ConfigurationIdentity.
```

POLICY RULES D-19

Rule	Failure
canonical named non-secret fields only	CFG_POLICY_NONCANONICAL
PolicyId + Version immutable	CFG_POLICY_VERSION_REUSED
computed digest equals expected	CFG_POLICY_IDENTITY_MISMATCH
identity persisted with decision	CFG_POLICY_RECEIPT_MISSING
unknown historical identity	explicit replay failure; never latest fallback

HISTORICAL REPLAY

Current configuration cannot rewrite past decisions

REPLAY INPUTS D-20

Required coordinate	Rule
ordered evidence identity	exact historical membership/order
engine version	pinned implementation identity
ConfigurationIdentity	exact retained policy artifact
contract identity	public mapping/corpus version used then
environment	excluded from decision basis

FAIL-CLOSED LOOKUP

```
public RootCauseDecisionPolicy Require(ConfigurationIdentity id) =>
    retained.TryGetValue(id, out var policy)
        ? policy
        : throw new ConfigurationIdentityUnavailable(id);

// There is no RequireLatest() fallback on replay.
```

CONSERVATION RULE

Same ordered evidence, engine version and ConfigurationIdentity must produce the same issue-or-abstain result. Missing history is an evidence failure, not permission to use today's settings.

CHANGE CONTRACTS

Restart, reload and new policy version are distinct protocols

MUTABILITY CLASSES D-21

Class	Examples	Acceptance
RESTART_BOUND	endpoints, providers, serializer, credentials, contract activation	new admitted process snapshot
VALIDATED_RELOAD	bounded batch/poll tuning, safe log level	atomic candidate validation; last-known-good
VERSIONED_IMMUTABLE	decision policy, topology/contract catalogue, rollout plan	new identity and explicit rollout

FORBIDDEN TRANSITIONS D-21

Attempt	Reason
reload database endpoint in-place	active client/pool semantics unproved
monitor CurrentValue for Domain decision	same business case can see mixed meaning
edit immutable version under same ID	historical replay becomes ambiguous
partial multi-key reload	mixed snapshot violates cross-field validation

ATOMIC VALIDATED RELOAD

A candidate snapshot is published only as one accepted object

ACCEPTED SNAPSHOT SHAPE

```
public sealed class AcceptedSnapshot<T> where T : class
{
    private T _current;

    public T Current => Volatile.Read(ref _current);

    private void OnChange(T candidate)
    {
        var result = validator.Validate(null, candidate);
        if (result.Succeeded)
            Interlocked.Exchange(ref _current, candidate);
        else
            rejected.Record(typeof(T).Name, result.Failures);
    }
}
```

RELOAD OUTCOMES D-22

Candidate	Result
complete and valid	atomic swap + accepted identity/telemetry
invalid range/cross-field	reject; retain current
partial document	reject; retain current
notification duplicated	same accepted digest; no second semantic effect
provider notifications stale/unreliable	restart-bound deployment policy

RUNTIME IDENTITY SECTION

Every role starts under one admitted runtime coordinate

RUNTIMEIDENTITYOPTIONS

```
public sealed record RuntimeIdentityOptions
{
    public const string Section = "RuntimeIdentity";
    public required string Environment { get; init; }
    public required string LogicalRole { get; init; }
    public required string BuildIdentity { get; init; }
    public required string ConfigurationSha256 { get; init; }
    public required string DeploymentCandidate { get; init; }
}
```

IDENTITY RULES D-23

Field	Validation
Environment	closed admitted profile
LogicalRole	exact seven-role inventory; no module/candidate role
BuildIdentity	matches immutable image/artifact identity
ConfigurationSha256	matches generated safe configuration
DeploymentCandidate	matches signed rollout/evidence plan

CONTRACT CATALOGUE BINDING

Configuration references the Appendix C authority by digest

CONTRACTOPTIONS

```
public sealed record ContractOptions
{
    public const string Section = "Contracts";
    public required string CatalogueSha256 { get; init; }
    public required string CorpusSha256 { get; init; }
    public required int EnvelopeVersion { get; init; }
    public required string CompatibilityProfile { get; init; }
}
```

CONTRACT CONSTANTS D-24

Coordinate	Required value
EnvelopeVersion	1
CompatibilityProfile	strict-envelope-v1+jcs
CatalogueSha256	exact Appendix C machine catalogue digest
CorpusSha256	exact three-fact positive/negative corpus digest

NO INLINE CONTRACT DEFINITION

Event types, HTTP routes, DTO fields and compatibility rules are not duplicated as free-form settings. Configuration references the versioned catalogue and validates discovered routes against it.

SERIALIZATION SECTION

Serializer and canonicalizer are restart-bound contract mechanics

SERIALIZATIONOPTIONS

```
public sealed record SerializationOptions
{
    public const string Section = "Serialization";
    public int EnvelopeVersion { get; init; } = 1;
    public string ContentType { get; init; } = "application/json";
    public string Encoding { get; init; } = "utf-8";
    public string Canonicalization { get; init; } = "rfc8785-jcs";
    public string FingerprintAlgorithm { get; init; } = "sha-256+jcs";
    public bool RejectUnknownMembers { get; init; } = true;
}
```

FIXED RULES D-25

Attempted override	Result
RejectUnknownMembers=false	CFG_CONTRACT_STRICTNESS_WEAKENED
non-UTF-8 encoding	CFG_CONTRACT_ENCODING_UNSUPPORTED
pretty JSON as canonical	CFG_CANONICALIZER_UNSUPPORTED
different fingerprint profile	CFG_FINGERPRINT_PROFILE_DRIFT

BROKER CONNECTION SECTION

Connection mechanics are separate from topology authority

BROKERCONNECTIONOPTIONS

```
public sealed record BrokerConnectionOptions
{
    public const string Section = "Broker";
    public required string Host { get; init; }
    [Range(1, 65535)] public int Port { get; init; } = 5672;
    public required string VirtualHost { get; init; }
    public required SecretReference CredentialReference { get; init; }
    public bool TlsRequired { get; init; } = true;
    [Range(1, 60)] public int ConnectTimeoutSeconds { get; init; } = 10;
}
```

BROKER RULES D-26

Rule	Failure
TLS profile required outside isolated local lab	CFG_BROKER_TLS_REQUIRED
credential reference owner matches role	CFG_BROKER_CREDENTIAL_OWNER
runtime has no configure permission	readiness/security evidence
endpoint change	restart-bound; no live mixed channels

BROKER TOPOLOGY ACTIVATION

Exactly three facts enter the final configured contract set

ACTIVATED ROUTES D-27

Producer	Route	Core consumer queue
alarm-management	alarm-management.alarm-flood-detected.v1	rootcause.alarm-events.v1
root-cause	root-cause.root-cause-case-opened.v1	reporting.integration-events.v1
root-cause	root-cause.root-cause-verdict-issued.v1	reporting.integration-events.v1

FIXED EXCHANGE D-27

Coordinate	Value
exchange	nexus.events
type	topic; durable
publish	mandatory + publisher confirms
consumer acknowledgement	manual after known durable outcome

CANDIDATE DEFAULT

RecommendationGenerated.v1, ComplianceReviewOpened.v1, correction facts and Process Manager bindings are absent. Discovering any as active produces CFG_INACTIVE_CONTRACT_ENABLED.

TOPOLOGY DECLARATION VERSUS RUNTIME

A provisioner configures; applications use only admitted routes

PRINCIPAL CONFIGURATION D-28

Principal	Configure	Write	Read
topology provisioner	approved exchanges/queues/bindings	declaration-required only	inspection only
alarm-worker	none	nexus.events / alarm route	none
rootcause-worker	none	nexus.events / root-cause routes	rootcause.alarm-events.v1
projection-worker	none	none	reporting.integration-events.v1
APIs / query-bff	none	none	none

READINESS

Runtime checks passive topology and its own permitted capability. It does not gain configure permission merely to make startup convenient.

ALARM HTTP SECTION

Input bounds protect the exact Appendix C command

ALARMHTTPOPTIONS

```
public sealed record AlarmHttpOptions
{
    public const string Section = "AlarmManagement:Http";
    public int MaxBodyBytes { get; init; } = 262_144;
    public string SubmitRoute { get; init; } = "/api/v1/alarm-observations";
    public string RequiredCapability { get; init; } = "alarm.observation.submit";
    public int MaxSourceSystemLength { get; init; } = 80;
    public int MaxSourceEventIdLength { get; init; } = 120;
}
```

NON-OVERRIDABLE RULES D-29

Coordinate	Required
route	exact Appendix C route
unknown JSON members	rejected by serializer profile
capability	cannot be weakened/disabled by environment
body bound	1..262144 bytes
Problem Details	safe V1 profile; no developer dump

ALARM DATABASE SECTION

The runtime receives DML for AlarmManagementDb only

ALARMDATABASEOPTIONS

```
public sealed record AlarmDatabaseOptions
{
    public const string Section = "AlarmManagement:Database";
    public required string Provider { get; init; }
    public required SecretReference RuntimeConnection { get; init; }
    [Range(1, 300)] public int CommandTimeoutSeconds { get; init; } = 30;
    [Range(0, 10)] public int MaxRetryCount { get; init; } = 3;
    public bool ApplyMigrationsOnStartup { get; init; } = false;
}
```

OWNERSHIP GATES D-30

Rule	Required result
provider in admitted list	provider-specific validator selected
connection reference owner	alarm-api/alarm-worker role only
foreign database token/name	CFG_OWNER_MISMATCH
ApplyMigrationsOnStartup	must remain false
retry budget	below outer operation timeout

ALARM FLOOD POLICY

Detection meaning is a versioned policy, not an environment switch

ALARMFLOODPOLICYDOCUMENT

```
public sealed record AlarmFloodPolicyDocument(  
    string PolicyId,  
    int Version,  
    TimeSpan Window,  
    int MinimumAlarmCount,  
    string EvidenceOrdering,  
    string ExpectedSha256);
```

POLICY OBLIGATIONS D-31

Obligation	Reason
canonical ordered inputs	ALM-DET-001 deterministic replay
PolicyId/version/digest stored	historical decision identity
same identity in all environments	environment equivalence
no secret or endpoint fields	business identity remains value-safe
new semantics -> new version/digest	no silent history rewrite

ALARM OUTBOX DISPATCHER

One batch has bounded size, lease, confirmation and shutdown

OUTBOXDISPATCHEROPTIONS

```
public sealed record OutboxDispatcherOptions  
{  
    public const string Section = "Messaging:OutboxDispatcher";  
    public int BatchSize { get; init; } = 64; // 1..500  
    public int RecoveryBatchSize { get; init; } = 128;  
    public TimeSpan LeaseDuration { get; init; } = TimeSpan.FromSeconds(45);  
    public TimeSpan ConfirmTimeout { get; init; } = TimeSpan.FromSeconds(10);  
    public TimeSpan IdleDelay { get; init; } = TimeSpan.FromMilliseconds(250);  
    public TimeSpan ShutdownDrainBudget { get; init; } = TimeSpan.FromSeconds(20);  
}
```

CROSS-FIELD RULES D-32

Rule	Failure
ConfirmTimeout < LeaseDuration	CFG_OUTBOX_CONFIRM_EXCEEDS_LEASE
ShutdownDrainBudget < LeaseDuration	CFG_OUTBOX_DRAIN_EXCEEDS_LEASE
BatchSize 1..500	CFG_OUTBOX_BATCH_INVALID
reload changes full accepted snapshot	partial reload rejected

ALARM PUBLICATION CONTRACT

The relay is configured to publish stored history only

PUBLICATION SECTION D-33

Key	Required value / rule
Enabled	true for alarm-worker; false/absent elsewhere
Exchange	nexus.events
RoutingKey	alarm-management.alarm-flood-detected.v1
EventType	nexus1.alarm-management.alarm-flood-detected.v1
SchemaVersion	1
GoldenBodyLength	1003
GoldenBodySha256	7cf98319c20a5915d5710bbcbb5878fe51cdae89a5b68081c135fd0e962c276a

SELF-VERIFICATION

Configuration coordinates must equal the Appendix C catalogue. The dispatcher still publishes the exact body stored in the outbox; it never serializes from these keys.

ROOTCAUSE DATABASE SECTION

RootCause runtime and migrator authority remain separate

ROOTCAUSEDATABASEOPTIONS

```
public sealed record RootCauseDatabaseOptions
{
    public const string Section = "RootCause:Database";
    public required string Provider { get; init; }
    public required SecretReference RuntimeConnection { get; init; }
    [Range(1, 300)] public int CommandTimeoutSeconds { get; init; } = 30;
    [Range(0, 10)] public int MaxRetryCount { get; init; } = 3;
    public bool ApplyMigrationsOnStartup { get; init; } = false;
}
```

DATABASE PROHIBITIONS D-34

Finding	Failure
AlarmManagementDb or ReportingDb coordinate	CFG_OWNER_MISMATCH
migration secret in API/worker	CFG_MIGRATION_AUTHORITY_IN_RUNTIME
unknown provider	CFG_DATABASE_PROVIDER_UNSUPPORTED
startup migration enabled	CFG_RUNTIME_MIGRATION_FORBIDDEN

ROOTCAUSE CONSUMER SECTION

One strict input route opens the owner-local case

ROOTCAUSECONSUMEROPTIONS

```
public sealed record RootCauseConsumerOptions
{
    public const string Section = "RootCause:AlarmFloodConsumer";
    public bool Enabled { get; init; } = true;
    public string Queue { get; init; } = "rootcause.alarm-events.v1";
    public string Exchange { get; init; } = "nexus.events";
    public string RoutingKey { get; init; } =
        "alarm-management.alarm-flood-detected.v1";
    [Range(1, 512)] public ushort Prefetch { get; init; } = 32;
    public int MaxBodyBytes { get; init; } = 262_144;
}
```

CONSUMER GATES D-35

Rule	Required
auto acknowledgement	false
allowed event type	exact AlarmFloodDetected V1
unknown route/schema	reject before Domain
prefetch	bounded by proved concurrency/capacity
worker disabled	not an accepted Phase0Core profile

ROOTCAUSE PUBLIC FACTS

Two exact routes represent the case stream

PRODUCED CONTRACTS D-36

Position	Fact	Route
1	RootCauseCaseOpened.v1	root-cause.root-cause-case-opened.v1
2	RootCauseVerdictIssued.v1	root-cause.root-cause-verdict-issued.v1

GOLDEN COORDINATES D-36

Fact	Length	Body SHA-256
CaseOpened	876	454f51cf4b0250b9c855346a6648a7d73530f19687ee3b52cda2edf9d0339c5b
VerdictIssued	1157	192e5e73c5a35d438e2ee9edb0a9e7044293c29bf321ba1abe34f191d8b44c00

ACTIVATION INVARIANT

Both contracts are required in RootCause's producer set; no Compliance, correction or workflow-manager fact enters the final profile.

ROOTCAUSE DECISION POLICY

Issue-or-abstain uses an accepted immutable snapshot

POLICY FIELDS D-37

Field	Rule
PolicyId	stable RootCause-owned public identity
Version	monotonic semantic version coordinate
MinimumEvidenceCount	positive; invariant-compatible
IssueThreshold	canonical decimal within admitted range
EvidenceOrder	closed, normalized and deterministic
ExpectedSha256	matches canonical policy document

NOT RELOADABLE

The policy store may receive a new artifact, but an in-flight or historical decision selects one ConfigurationIdentity. IOptionsMonitor.CurrentValue is not a decision input.

REPORTING DATABASE SECTION

Read and write roles use Reporting-owned capabilities only

ROLE GRANTS D-38

Role	Reference	Authority
projection-api	Reporting read connection	active generation + freshness read
projection-worker	Reporting write connection	inbox/ledger/view/checkpoint/status
reporting-migrator	Reporting migration connection	DDL through migration job
query-bff	none	HTTP only; no ReportingDb

FIXED RULES D-38

Key	Requirement
ApplyMigrationsOnStartup	false for API and worker
ActiveGeneration	stored pointer; not environment default
foreign DB coordinate	rejected
provider retry	bounded and below settlement budget

REPORTING PROJECTION CONSUMER

Two exact RootCause facts advance one contiguous stream

REPORTINGPROJECTIONOPTIONS

```
public sealed record ReportingProjectionOptions
{
    public const string Section = "Reporting:RootCauseCaseProjection";
    public bool Enabled { get; init; } = true;
    public string Queue { get; init; } = "reporting.integration-events.v1";
    public string ContractSetIdentity { get; init; } =
        "reporting/root-cause-case-summary/1";
    public int MaxBodyBytes { get; init; } = 262_144;
    public int MaxGapDrainBatch { get; init; } = 128;
}
```

EXACT READER SET D-39

Fact	Stream position
RootCauseCaseOpened.v1	1
RootCauseVerdictIssued.v1	2

NO WILDCARD ACTIVATION

The broker may route through a shared queue, but the configured reader allowlist is exact. Unknown RootCause facts are not implicitly admitted.

PROJECTION FRESHNESS POLICY

Current-as-observed is earned from bounded evidence

PROJECTIONFRESHNESSOPTIONS

```
public sealed record ProjectionFreshnessOptions
{
    public const string Section = "Reporting:Freshness";
    public TimeSpan MaxStatusAge { get; init; } = TimeSpan.FromSeconds(15);
    public TimeSpan MaxHeartbeatAge { get; init; } = TimeSpan.FromSeconds(15);
    public TimeSpan WarningAge { get; init; } = TimeSpan.FromSeconds(30);
    public TimeSpan SamplingInterval { get; init; } = TimeSpan.FromSeconds(5);
}
```

FRESHNESS RULES D-40

Rule	Reason
SamplingInterval < MaxStatusAge	sample can remain valid between polls
stale sample -> unknown	never default current
gap count > 0 -> gap-detected	semantic completeness wins
backlog/age -> catching-up or delayed	operational evidence visible

QUERY BFF UPSTREAM SECTION

The edge calls Reporting through one bounded internal contract

QUERYBFFUPSTREAMOPTIONS

```
public sealed record QueryBffUpstreamOptions
{
    public const string Section = "QueryBff:Reporting";
    public required Uri BaseAddress { get; init; }
    public string RouteTemplate { get; init; } =
        "/internal/v1/root-cause-case-projections/{caseId}";
    public TimeSpan Timeout { get; init; } = TimeSpan.FromSeconds(2);
    public required string WorkloadAudience { get; init; }
}
```

EDGE RULES D-41

Rule	Failure
route equals Appendix C internal target	CFG_CONTRACT_DRIFT
base address uses approved service discovery	CFG_UPSTREAM_UNAPPROVED
timeout below public request budget	CFG_TIMEOUT_BUDGET_INVALID
no RootCause/DB fallback	architecture gate

QUERY BFF PUBLIC SECTION

The public route and media profile cannot drift by environment

PUBLIC QUERY CONSTANTS D-42

Coordinate	Required value
route	/api/v1/root-cause-cases/{caseId}
media	application/vnd.nexus1.operator-case.v1+json
capability	reporting.root-cause-case.read
cache	private, no-store
absence	200 not-observed-yet; never inferred owner 404
upstream unavailable	503 safe Problem Details

SUPERSEDED ROUTE

Configuration must reject /api/operator/v1/root-cause-cases/{caseId}. Supporting both as aliases would reopen the Appendix C reconciliation.

AUTHENTICATION SECTION

Issuer, audience and token mechanics are restart-bound

AUTHENTICATIONOPTIONS

```
public sealed record AuthenticationOptions
{
    public const string Section = "Authentication";
    public required Uri Authority { get; init; }
    public required string Audience { get; init; }
    public required string MetadataIdentity { get; init; }
    public bool RequireHttpsMetadata { get; init; } = true;
    public TimeSpan ClockSkew { get; init; } = TimeSpan.FromSeconds(30);
}
```

SECURITY RULES D-43

Rule	Required
RequireHttpsMetadata	true outside isolated local test
issuer/audience	closed per role and environment manifest
clock skew	bounded; never business occurrence correction
token detail	never returned or logged

AUTHORIZATION POLICY IDENTITY

Capabilities and scopes are versioned policy, not ad-hoc strings

FINAL CAPABILITY MAP D-44

Surface	Capability
Alarm observation POST	alarm.observation.submit
RootCause owner GET	root-cause.case.read
RootCause evidence POST	root-cause.evidence.attach
RootCause verdict POST	root-cause.verdict.issue
Reporting internal GET	reporting.root-cause-case-projection.read
Query BFF public GET	reporting.root-cause-case.read

POLICY CONFIGURATION D-44

Coordinate	Rule
PolicyId/version/digest	immutable accepted artifact
scope source	trusted claims mapped through closed vocabulary
environment override	cannot remove capability/scope enforcement
unknown capability	startup/configuration failure

HEALTH SECTION

Probe configuration preserves action semantics

FIXED ENDPOINTS D-45

Route	Action	Configurable bounds
/health/startup	wait/stop on bounded initialization	initialization timeout
/health/live	restart only for terminal in-process fault	stale local heartbeat bound
/health/ready	admit/withdraw new work	owned dependency/readiness budget

FORBIDDEN HEALTH CONFIG D-45

Attempt	Reason
liveness checks database/broker	dependency outage becomes restart storm
one generic /health endpoint	platform actions collapse
readiness requires downstream service	transitive availability coupling
secret/internal topology in response	exposure boundary violation

TELEMETRY SECTION

Export mechanics may vary; semantic vocabulary may not

TELEMETRY KEYS D-46

Key	Class	Change
ServiceName / LogicalRole	identity	restart-bound
OTLP endpoint	technical	restart-bound
sampling profile	operational	validated reload or rollout
log level	operational	validated reload
metric/attribute allowlist	security policy	versioned immutable
RunId / BuildIdentity	evidence context	process immutable

CARDINALITY BOUNDARY

MessageId, Caseld, tenant, payload and arbitrary exception values are not metric labels. Configuration can reduce sampling or detail; it cannot turn telemetry into a business ledger.

MIGRATION JOB CONFIGURATION

Three one-shot jobs own schema authority

MIGRATION ROLES D-47

Job	Database	Required identity
alarm-migrator	AlarmManagementDb	migration set + DDL secret reference
rootcause-migrator	RootCauseDb	migration set + DDL secret reference
reporting-migrator	ReportingDb	migration set + DDL secret reference

JOB-ONLY OPTIONS D-47

Key	Rule
ExpectedMigrationSetSha256	matches image/plan
TargetDatabaseOwner	exact one owner
LockTimeout / CommandTimeout	bounded and provider validated
AllowDestructive	false unless separately approved irreversible plan
ExitAfterCompletion	true; no hosted long-running migrator

ENVIRONMENT PROFILES

Profiles differ by mechanics and evidence expectations

PROFILE MATRIX D-48

Profile	Adapters	Secret source	Evidence
Development	local approved dependencies	User Secrets	component evidence only
CI/Test	disposable real provider/broker	synthetic injected references	automated gate receipts
Phase0Compose	pinned Compose services	mounted/injected references	reproducible runtime dossier
Phase0Kubernetes	minimal rendered objects	Secret refs/workload identity	platform mapping evidence

SAME CORE

Every profile preserves three owners, one edge, seven roles, three migrators, three databases and exactly three activated facts.

COMPOSE PROJECTION

Environment variables project one admitted snapshot

COMPOSE MAPPING D-49

Configuration	Compose projection
safe non-secret values	environment or mounted immutable file
secret references	Compose secret/file references; no inline value in evidence
ConfigurationSha256	label/environment identity checked by host
role	one service entry per logical runtime role
health	startup/live/ready semantics retained
migration	explicit one-shot profile/job before runtime admission

EQUIVALENCE

Compose service name, container ID and environment location may differ from Kubernetes objects. Logical role, image, configuration identity, contracts, credentials and probe meaning must agree.

KUBERNETES PROJECTION

Immutable ConfigMap identity causes a deliberate rollout

VALUE-SAFE CONFIGMAP SHAPE

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: nexus1-runtime-@@CONFIG_SHA12@@
  annotations:
    nexus1.io/configuration-sha256: "@@CONFIG_SHA256@@"
immutable: true
data:
  RuntimeIdentity_Environment: "Phase0Kubernetes"
  Broker__Host: "rabbitmq.dependencies.svc.cluster.local"
  Broker__Port: "5672"
  Telemetry_OtlpEndpoint: "http://otel.dependencies.svc.cluster.local:4317"

```

PROJECTION RULES D-50

Rule	Required
configuration object	digest-suffixed and immutable
Pod template	references exact object and digest
secret	Secret/provider reference only
mounted update	not used as atomic multi-key reload
change	new identity -> new rollout receipt

PLATFORM EQUIVALENCE

Compose and Kubernetes must express the same architecture

EQUIVALENCE LEDGER D-51

Coordinate	Compose	Kubernetes	Must agree
role	service	Deployment/Job	logical role
image	digest	digest	immutable artifact
config	env/secret refs	ConfigMap/Secret refs	configuration identity
probe	healthcheck	startup/live/ready	meaning + bounds
dependency	condition/order	policy/readiness	admission semantics
resource	limits profile	requests/limits	declared budget identity

NO PLATFORM TRUTH

A ConfigMap, Secret, Deployment or Compose service proves mechanism presence only. It does not prove configuration acceptance until the role emits a matching receipt.

VALUE-FREE CONFIGURATION MANIFEST

Evidence exposes shape, provenance and status without values

MANIFEST ENTRY

```
public sealed record SafeConfigurationEntry(  
    string Section,  
    string Key,  
    string Owner,  
    string Classification,  
    string Presence,  
    string SourceCategory,  
    string ChangeContract,  
    string ValidationStatus,  
    string? NonSecretIdentity);  
  
// Value is deliberately absent.
```

MANIFEST INCLUDES D-52

Included	Excluded
approved key names and schema versions	unapproved discovered keys
provider kinds/order	provider file paths or values
secret present/missing + classification	secret hashes/lengths/substrings
non-secret policy/contract identities	payload/credential material
restart/reload/version contract	raw options object dump

MACHINE-READABLE CATALOGUE

The build and the appendix consume one closed schema

CONFIGURATIONCATALOGUEENTRYV1

```
public sealed record ConfigurationCatalogueEntryV1(  
    string Key,  
    string SectionSchema,  
    string Owner,  
    string Classification,  
    string ChangeContract,  
    IReadOnlyList<string> AllowedSources,  
    string ValidatorId,  
    string FailureMode,  
    bool Required,  
    string EvidenceStatus);
```

CLOSED VOCABULARIES D-53

Field	Values
Classification	PUBLIC INTERNAL SECRET_REFERENCE DECISION_POLICY
ChangeContract	RESTART_BOUND VALIDATED_RELOAD VERSIONED_IMMUTABLE
EvidenceStatus	PENDING DEMONSTRATED FALSIFIED
AllowedSources	approved provider categories only

CONFIGURATION VALIDATOR

Unknown keys, inactive contracts and authority drift fail closed

GLOBAL GATE SHAPE

```
RequireExact(runtimeRoles, FinalCore.SevenRoles);
RequireExact(databaseOwners, FinalCore.ThreeDatabases);
RequireExact(activatedFacts, AppendixC.ThreeFacts);
RejectUnknownKeys(boundTree, catalogue);
RejectSecretValuesInSafeSources(boundTree, catalogue);
RejectCandidateActivation(boundTree);
RequireProviderOrder(actualSources, approvedSources);
RequireContractDigests(options.Contracts, AppendixC.Identity);
RequireNoMigrationCredentialInRuntime(composition);
RequireNoConfigurationApiInInnerLayers(assemblies);
```

GLOBAL FAILURES D-54

Code	Meaning
CFG_KEY_UNCATALOGUED	discovered key has no owner/schema
CFG_INACTIVE_CONTRACT_ENABLED	candidate contract activated
CFG_ROLE_COUNT_DRIFT	final seven-role inventory changed
CFG_SECRET_IN_SAFE_SOURCE	confidential value in non-secret channel
CFG_MIGRATION_AUTHORITY_IN_RUNTIME	DDL capability leaked

NEGATIVE STARTUP MATRIX

Each unsafe mutation stops before traffic

REQUIRED CASES D-55

Mutation	Expected failure	Side effects
required section missing	CFG_SECTION_REQUIRED	host not ready
unknown key	CFG_KEY_UNKNOWN	host not ready
timeout zero	CFG_RANGE_INVALID	host not ready
confirm >= lease	CFG_CROSS_FIELD_INVALID	worker not started
foreign database reference	CFG_OWNER_MISMATCH	no DB call
wrong event route/digest	CFG_CONTRACT_DRIFT	no publish/consume
candidate event Enabled=true	CFG_INACTIVE_CONTRACT_ENABLED	no binding
secret value in manifest	CFG_SECRET_IN_SAFE_SOURCE	artifact rejected
policy digest mismatch	CFG_POLICY_IDENTITY_MISMATCH	no Domain decision

RELOAD FAULT MATRIX

Invalid change preserves the last accepted snapshot

FAULT CASES D-56

Change	Required outcome
batch size 64 -> 0	reject; remain 64
two-key document observed partially	reject until complete identity
duplicate notification same digest	one accepted observation
provider stops notifying	status ages; restart/rollout policy
secret rotation; client still old	readiness withdraw or bounded restart
database endpoint changed via reload	reject as restart-bound
decision policy edited under same ID	reject immutable identity

TELEMETRY

Reload events expose section, old/new non-secret identity, outcome and safe reason. They never expose changed values.

ARCHITECTURE CONFIGURATION GATES

Inner layers remain independent of providers and environments

FORBIDDEN API SCAN D-57

Assembly	Rejected types
*.Domain	IConfiguration; all IOptions forms; host environment; secret provider
*.Application	IConfiguration; IOptions/IOptionsMonitor; host environment; raw secret
*.Contracts	configuration framework types and environment names
Query BFF	DbContext, owner database connection or migration API
runtime hosts	migration credential and migrator service

POSITIVE PLACEMENT D-57

Concern	Allowed placement
provider order/binding	Host
technical options	own Edge/Infrastructure adapter
policy mapping/identity	Application boundary
typed accepted decision values	Domain input

CONFIGURATION EVIDENCE RECORD

A configuration claim is bound to build and deployment identity

CONFIGURATIONEVIDENCEV1

```
public sealed record ConfigurationEvidenceV1(  
    string EvidenceId,  
    string BuildIdentity,  
    string DeploymentCandidate,  
    string ConfigurationSha256,  
    string ProviderManifestSha256,  
    string CatalogueSha256,  
    string ContractCatalogueSha256,  
    string ContractCorpusSha256,  
    IReadOnlyList<string> PolicyIdentities,  
    int Expected, int Discovered, int Passed, int Failed, int Skipped,  
    string Status,  
    IReadOnlyList<string> ArtifactReferences);
```

ACCEPTANCE D-58

Condition	Required
Expected == Discovered	true and nonzero
Failed / Skipped	0 / 0
manifest	value-free and digest-valid
identities	same across host, platform renderer and evidence plan
Status	DEMONSTRATED only after attached artifacts pass

CONFIGURATION EVIDENCE SUITES

Different tests prove different boundaries

PROOF CATALOGUE D-59

Evidence ID	Primary proof
CFG-OWN-001	configuration APIs stay out of inner layers
CFG-SRC-001	provider kinds/order and override semantics
CFG-BIND-001	closed sections bind; unknown/malformed fail
CFG-VAL-001	range/cross-field/ownership rules
CFG-ENV-001	environment mechanics preserve Domain result
CFG-DEC-001	policy identity and historical replay
CFG-SEC-001	secret canary containment
CFG-REL-001	atomic reload acceptance/rejection
CFG-MAN-001	value-free manifest identity
CFG-REVIEW-001	review bound to exact candidate

CURRENT RESULT

All suites are specified. No repository run and retained evidence bundle exist in this manuscript environment; status remains PENDING.

CONFIGURATION DRIFT AND ROLLOUT

A changed identity requires an explicit candidate transition

DRIFT COMPARISON D-60

Changed coordinate	Required action
safe operational value	validate, classify reload/restart, update digest
provider kind/order	review source policy; new ProviderManifestSha256
secret reference	rotation/restart receipt; value remains absent
policy artifact	new version/identity + replay/decision equivalence
contract catalogue/corpus	Appendix C compatibility rollout first
role/database/active-fact count	architecture decision and Appendix B-D updates

NO IN-PLACE EDIT

Compose and Kubernetes use revision-by-identity. A new admitted digest changes the runtime candidate deliberately; mounted or edited values never create a silent partial transition.

CONFIGURATION ANTI-PATTERNS

Convenient defaults can silently redesign the system

REJECTED PATTERNS D-61

Anti-pattern	Why rejected
inject IConfiguration everywhere	spreads provider authority into business code
environment-specific invariant	host label changes truth
connection string in appsettings	credential leaks into source/artifacts
hash a secret for evidence	digest remains a comparison/leak oracle
reload all options	restart-bound clients see mixed state
activate route with Enabled default	candidate contract enters final traffic
migrate on API startup	every replica gains DDL authority
config dump during incident	response bundle leaks values
change policy under same version	history cannot replay honestly

IMPLEMENTATION CHECKLIST

Every section must be closed, validated and evidenced

CHECKLIST D-62

Area	Required deliverable
inventory	one owner/class/source/change rule per key
providers	explicit order and precedence tests
binding	strict options schema + unknown-key rejection
validation	range, format, cross-field, owner and contract gates
secrets	references/resolver/canary/rotation without values
policy	canonical artifact, digest, persistence and replay
reload	atomic accepted snapshot and last-known-good behavior
platform	Compose/Kubernetes projection with same identity
evidence	nonzero suites, value-free manifest and attached receipts

APPENDIX D CHECKPOINT

Configuration can now be audited as an architectural boundary

COMPLETION LEDGER D-63

Claim	Result
provider order and key grammar	FIXED
closed section schemas and validation	SPECIFIED
secret-reference and rotation rules	SPECIFIED; values excluded
decision-policy identity and replay	SPECIFIED
three contract routes and active-fact bindings	FIXED FROM APPENDIX C
restart/reload/version protocols	FIXED
Compose/Kubernetes equivalence	SPECIFIED
repository implementation	PENDING
runtime evidence / production readiness	PENDING / NOT CLAIMED

HONEST STATUS

CONFIGURATION REFERENCE COMPLETE AS SPECIFICATION | FINAL CORE COUNTS PRESERVED | VALUE-FREE EVIDENCE DEFINED | REPOSITORY AND RUNTIME CONFORMANCE PENDING

HANDOFF TO APPENDIX E

The next appendix must catalogue failure and recovery evidence

Appendix E - Failure Scenario and Recovery Catalogue will turn the failure policies already distributed across the book into one runnable disturbance inventory. Each scenario must name the initiating fault, injection boundary, conserved owner truth, expected degradation, recovery action, evidence oracle, stopping condition and honest status.

CARRY-FORWARD OBLIGATIONS D-64

From Appendix D	Appendix E must preserve
one admitted configuration identity	fault begins from known snapshot
restart/reload/version classification	recovery action cannot contradict change contract
secret-free evidence	failure bundles remain value-free
three activated facts and exact routes	fault targets final core only
probe semantics and time budgets	degradation/recovery oracle
PENDING status	no campaign passes without retained execution receipt

APPENDIX D LAST WORD

Configuration may change mechanics, never ownership

A trustworthy configuration system makes change explicit: where a value came from, who owns it, what validates it, whether it restarts, reloads or creates a new identity, and what evidence can prove the transition. Anything less lets environment become hidden architecture.

FINAL STATUS

CONFIGURATION REFERENCE COMPLETE | CONTRACT ACTIVATION FIXED | SECRET VALUES EXCLUDED | CHANGE PROTOCOLS EXPLICIT | EXECUTABLE EVIDENCE PENDING

APPENDIX E

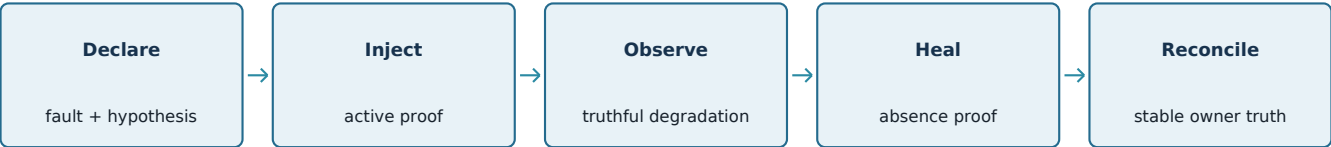
Failure Scenario and Recovery Catalogue

The runnable disturbance inventory for the NEXUS-1 Phase-0 core: six mandatory campaigns, bounded supporting faults, conserved owner truth, explicit degradation, idempotent healing, multi-source oracles and fail-closed evidence verdicts.

APPENDIX DECISION

A restart, reconnect or green endpoint is an action - not proof of recovery. A scenario becomes evidence only when its fault is proven active, degradation is observed truthfully, healing is proven complete, owner truth is conserved and progress stabilizes within a declared local-evidence budget.

CATALOGUE READING PATH



CONTROL TOTALS E-1

Coordinate	Final value
mandatory campaigns	6
supporting catalogue cards	20
core owners / edge / roles / migrators	3 / 1 / 7 / 3
core databases / activated public facts	3 / 3
evidence posture	SPECIFIED; repository and runtime receipts PENDING

PURPOSE AND AUTHORITY

The catalogue freezes recovery claims, not tool commands

Chapter 54 defines the fail-closed harness and demonstrates the shape of six experiments. Appendix E is the single lookup authority for what each disturbance means, which owner truth must survive, what degradation is acceptable, which observation ends recovery and which evidence status may be claimed.

AUTHORITY SPLIT E-2

Artifact	Owns	Does not own
Chapter 53	timeout, retry, circuit and capacity semantics	recovery verdict
Chapter 54	runner, actuator, snapshots and campaign method	final catalogue inventory
Appendix E	scenario identity, hypothesis and oracle	platform-specific command syntax
Appendix G	exact commands and reproduction sequence	scenario meaning

NO INVENTED EXECUTION

Every card is a complete specification. None is marked passed until a retained result binds the exact build, package lock, configuration, resilience profile, catalogue digest and executed receipts.

CLAIM MODEL

Model, property, check and evidence remain distinct

PROOF VOCABULARY E-3

Term	Meaning	Failure mode
model	declared fault boundary and state sources	wrong abstraction
property	truth/progress that should survive	false hypothesis
check	executable observation or comparison	weak oracle
evidence	retained fresh result of the check	missing/stale receipt
claim	bounded conclusion admitted by evidence	overstatement

CONDITIONAL CONCLUSION

A passing campaign supports only its named build, configuration identity, workload, observation boundary and fault mechanism. It is not absolute implementation correctness and never becomes a production RTO/RPO claim.

CATALOGUE ENTRY SCHEMA

Eight fields make every disturbance reviewable

FAILURESCENARIOCATALOGUEENTRYV1

```
public sealed record FailureScenarioCatalogueEntryV1(
    string ScenarioId,
    string Title,
    FaultSpecification InitiatingFault,
    InjectionBoundary InjectionBoundary,
    ConservedTruth[] ConservedOwnerTruth,
    ExpectedDegradation ExpectedDuringFault,
    HealingProtocol Healing,
    EvidenceOracle[] Oracles,
    StoppingCondition Stop,
    ScenarioStatus Status);
```

CLOSED STATUS VOCABULARY E-4

Status	Meaning
SPECIFIED_RUNTIME_PENDING	card and checks exist; no admitted run
PASSED	all mandatory receipts and postconditions pass
FAILED_EXPECTATION	valid evidence falsifies the hypothesis
INVALID_EVIDENCE	run cannot support a conclusion
ABORTED_HEALED	campaign aborted; cleanup proven complete
ABORTED_UNHEALED	cleanup unproved; runtime quarantined

FIXED RUNTIME BOUNDARY

Faults target only admitted owners and roles

CORE TARGET CENSUS E-5

Boundary	Admitted targets
AlarmManagement	alarm-api, alarm-worker, AlarmManagementDb
RootCause	rootcause-api, rootcause-worker, RootCauseDb
Reporting	projection-api, projection-worker, ReportingDb
Edge	query-bff; no database ownership
shared dependency	nexus.events broker and telemetry pipeline
schema authority	three separate service-owned migrator jobs

CANDIDATE EXCLUSION

Audit, Compliance, RL Advisory, Process Manager and correction-fact routes remain outside the final Phase-0 core. A campaign that discovers them enabled is invalid, not expanded automatically.

CONTRACT-BOUND DISTURBANCE

Only three activated public facts enter the campaigns

ACTIVATED BINDINGS E-6

Fact	Route	Consumer queue
AlarmFloodDetected.v1	alarm-management.alarm-flood-detected.v1	rootcause.alarm-events.v1
RootCauseCaseOpened.v1	root-cause.root-cause-case-opened.v1	reporting.integration-events.v1
RootCauseVerdictIssued.v1	root-cause.root-cause-verdict-issued.v1	reporting.integration-events.v1

TRANSPORT AUTHORITY E-7

Coordinate	Fixed value
exchange	nexus.events topic durable
consumer allowlists	exact event type + schema version; no wildcard activation
duplicate identity	same MessageId plus canonical payload digest
settlement	ack only after durable Inbox/effect outcome

EXPERIMENT IDENTITY

One identity joins plan, runtime and evidence

EXPERIMENTIDENTITYV1

```
public sealed record ExperimentIdentityV1(  
    Guid ExperimentId,  
    string ScenarioId,  
    string BuildSha256,  
    string PackageLockSha256,  
    string ComposeOrManifestSha256,  
    string RuntimeConfigurationSha256,  
    string ResilienceProfileSha256,  
    string ContractCatalogueSha256,  
    string FailureCatalogueSha256,  
    int Seed,  
    DateTimeOffset AuthorizedUtc);
```

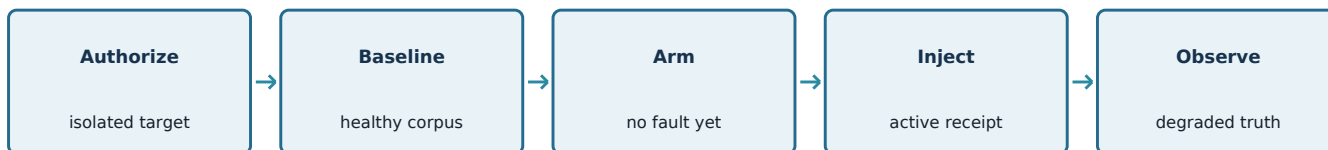
IDENTITY RULE

A run is not replayed against 'latest'. Every digest is resolved before authorization. A changed catalogue, configuration or contract corpus creates a new experiment identity.

SEVEN-PHASE LIFECYCLE

Failure and recovery cannot skip a phase

LIFECYCLE E-8



LIFECYCLE CONTINUATION E-9

Phase	Required outcome
6. heal	fault removed through an idempotent lease
7. reconcile	owner truth and progress stable within budget

NO SKIPPED PHASE

An unhealthy baseline aborts before injection. Missing active-fault proof invalidates causal attribution. Missing absence proof makes recovery unproved. A single green sample cannot close reconciliation.

SAFETY ENVELOPE

Destructive campaigns run only in isolated evidence runtimes

AUTHORIZATION GATES E-10

Gate	Required value
environment class	LocalEvidence or EphemeralCI only
Compose project	nexus1-evidence-<run> exact prefix
target labels	book, environment and experiment labels match
targets	closed service/container allowlist
concurrency	one destructive campaign per runtime
watchdog	hard lifetime plus independent emergency healing
unhealed result	quarantine; never reuse runtime

PROHIBITED TARGET

Shared development, staging, production-like and production environments are not eligible. A scenario plan never becomes an unrestricted shell command.

FAULT LEASE

Injection and healing are one idempotent resource

FAULTLEASE

```
public abstract class FaultLease : IAsyncDisposable
{
    public abstract FaultReceipt Receipt { get; }
    public abstract Task ProveActiveAsync(Cancellation token);
    public abstract Task HealAsync(Cancellation token);
    public abstract Task ProveAbsentAsync(Cancellation token);

    public async ValueTask DisposeAsync()
    {
        using var emergency = new CancellationTokenSource(
            TimeSpan.FromSeconds(20));
        await HealAsync(emergency.Token);
        await ProveAbsentAsync(emergency.Token);
    }
}
```

CLEANUP TRUTH

A failed assertion does not cancel healing. If healing or absence proof fails, the result is ABORTED_UNHEALED and the entire runtime is quarantined.

TIMING VOCABULARY

Detection, degradation, healing and convergence are separate

TIMING MODEL E-11

Interval	Start	End
activation	inject command	active-fault proof
detection	active proof	first correct degraded observation
fault duration	active proof	absence proof
readiness recovery	absence proof	readiness stable
state recovery	absence proof	durable invariants stable
campaign	authorized baseline	sealed evidence result

NOT AN SLO

Reference budgets bound a local evidence job. They are not production availability objectives, RTOs or RPOs and cannot be published as such.

REFERENCE CAMPAIGN BUDGETS

Every wait is bounded and interpreted locally

EVIDENCE-HARNESS DEFAULTS E-12

Scenario	Observe	Heal	Recover	Stable
E-CORE-001 consumer kill	15 s	20 s	60 s	3 x 500 ms
E-CORE-002 broker stop	20 s	30 s	90 s	3 x 500 ms
E-CORE-003 database pause	15 s	20 s	60 s	3 x 500 ms
E-CORE-004 duplicate	15 s	n/a	30 s	3 x 500 ms
E-CORE-005 projection delay	20 s	10 s	90 s	3 x 500 ms
E-CORE-006 service restart	20 s	30 s	90 s	3 x 500 ms

OVERRIDE PROTOCOL

A repository may pin different local-evidence values only through a versioned profile whose digest enters ExperimentIdentityV1. Silent timeout changes invalidate comparison.

STABLE CONVERGENCE

A single successful poll never ends recovery

STABLE OBSERVATION GATE

```

public async Task<T> AwaitStableAsync<T>(
    Func<CancellationToken, Task<T>> probe,
    Func<T, bool> predicate,
    int consecutive,
    TimeSpan interval,
    TimeSpan timeout,
    CancellationToken token)
{
    using var budget = CancellationTokenSource.CreateLinkedTokenSource(token);
    budget.CancelAfter(timeout);
    var stable = 0;
    while (true)
    {
        var value = await probe(budget.Token);
        stable = predicate(value) ? stable + 1 : 0;
        if (stable >= consecutive) return value;
        await Task.Delay(interval, budget.Token);
    }
}

```

DISTRIBUTED STATE ORACLE

Recovery is judged across owners, not one table

DISTRIBUTEDSTATESNAPSHOTV1

```

public sealed record DistributedStateSnapshotV1(
    DateTimeOffset ObservedUtc,
    long AcceptedAlarmFacts,
    long OutboxPending,
    long OutboxDispatched,
    long BrokerReady,
    long BrokerUnacked,
    long InboxProcessing,
    long InboxCompleted,
    long InboxDuplicates,
    long RootCauseEffects,
    long ReportingCheckpoint,
    long SourceHighWater,
    FreshnessReceipt[] Freshness);

```

NO GLOBAL SNAPSHOT CLAIM

Observations are not atomic across databases or broker state. Each source retains its observation time and freshness; the verifier reasons about a bounded stable window.

CONSERVATION FOR THE CONTROLLED CORPUS

The test corpus has explicit terminal equations

TERMINAL INVARIANTS E-13

Invariant	Qualification
accepted operation receipts = owner facts	AlarmManagement owner truth
unique confirmed public facts = admitted outbox messages	stable MessageId corpus
one RootCause effect per admitted AlarmFlood MessageId	controlled one-to-one fixture only
Inbox completed + duplicate terminal dispositions	no processing lease remains
Reporting checkpoint = source high-water	bounded reporting corpus
outbox pending + broker ready/unacked = 0	post-recovery terminal drain

SCOPE GUARD

These equations describe the deliberately bounded evidence corpus. They are not universal domain cardinality claims for arbitrary future NEXUS-1 flows.

TRACE AND TELEMETRY QUALIFICATION

A crash may end a span; missing evidence stays explicit

TRACE EVIDENCE CLASSES E-14

Class	Meaning	Effect
COMPLETE	expected spans/events retained	trace obligation may pass
PARTIAL_AT_CRASH	prefix plus kill chronology retained	qualified with durable milestone
SAMPLED_OUT	policy excluded trace	state/metrics still required
PIPELINE_LOSS	collector/export not fresh	trace obligation invalid
NOT_FOUND	no matching trace	cause unresolved

TELEMETRY IS NOT TRUTH OWNER

Logs and traces explain chronology. Owner databases, broker observations and durable receipts decide conservation and effect uniqueness.

EVIDENCE DOSSIER

Every run seals a value-free artifact set

REQUIRED ARTIFACTS E-15

Artifact	Contains
plan.json	scenario, digests, seed, budgets, required oracles
baseline.json	fresh control snapshot and workload receipts
fault-receipt.json	target, mechanism, active proof - no secret values
during-fault.json	actual degraded state and freshness
heal-receipt.json	healing action and absence proof
recovery.json	stable snapshots and durations
trace-query.json	query, result class and pipeline freshness
result.json + hashes.json	verdict and byte digests

VALUE-FREE RULE

Secret values, secret-derived hashes, connection strings, tokens, raw authorization headers and unrestricted payloads are forbidden. Evidence records references, identities, classifications and bounded facts only.

MANDATORY CAMPAIGN INDEX

Six disturbances form the minimum recovery suite

CORE CATALOGUE E-16

ID	Disturbance	Primary owner	Principal proof
E-CORE-001	kill consumer	RootCause	redelivery; one effect
E-CORE-002	stop broker	AlarmManagement	outbox conservation and drain
E-CORE-003	pause database	selected DB owner	bounded wait; honest outcome
E-CORE-004	duplicate message	RootCause	Inbox idempotency
E-CORE-005	delay projection	Reporting	visible staleness; catch-up
E-CORE-006	restart service	selected service	durable resume; new epoch

MINIMUM SUITE

A runtime checkpoint cannot claim recovery coverage if any of these six cards has zero admitted executions. Supporting scenarios add discrimination; they never replace the core six.

E-CORE-001 - KILL ROOTCAUSE CONSUMER AFTER COMMIT, BEFORE ACK**Kill RootCause consumer after commit, before ack****CATALOGUE CARD E-CORE-001**

Field	Declared value
initiating fault	SIGKILL rootcause-worker at evidence barrier consumer.after-commit.before-ack
injection boundary	rootcause-worker process epoch and rootcause.alarm-events.v1 delivery
conserved owner truth	RootCauseDb effect and completed Inbox row remain authoritative and unique
expected degradation	worker readiness absent; delivery becomes eligible for redelivery; no second effect
healing action	start replacement worker; wait for readiness and redelivery settlement
evidence oracle	container chronology + barrier receipt + delivery attempts + one effect + completed Inbox
stopping condition	new epoch ready; same MessageId settled; effect count 1; broker unacked 0 for 3 samples

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-CORE-001 CRASH-WINDOW MATRIX**Different kill windows require different oracles****CONSUMER WINDOWS E-17**

Window	First attempt state	Recovery proof
before Inbox claim	no durable admission	redelivery processes normally
after claim, before effect	processing lease exists	lease expiry/repair then retry
after effect, before ack	effect + completed Inbox durable	duplicate-completed then ack
after ack	broker owns no delivery	no redelivery expected

NO EXACTLY-ONCE CLAIM

The campaign proves one owner effect for the controlled corpus under at-least-once delivery and Inbox deduplication. It does not prove exactly-once network delivery.

E-CORE-002 - STOP BROKER WHILE ALARMMANAGEMENT ACCEPTS WORK**Stop broker while AlarmManagement accepts work****CATALOGUE CARD E-CORE-002**

Field	Declared value
initiating fault	stop broker container after a healthy control corpus
injection boundary	nexus.events connection path used by alarm-worker publisher
conserved owner truth	AlarmManagementDb accepted fact and outbox row commit atomically
expected degradation	publication unavailable; pending count and oldest age rise within workload bound
healing action	start broker; allow normal reconnect/circuit behavior; drain through confirms
evidence oracle	owner receipts + outbox states + broker lifecycle + confirms + downstream Inbox/effects
stopping condition	all accepted facts confirmed and consumed; outbox/ready/unacked/processing zero for 3 samples

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-CORE-002 RECOVERY CHRONOLOGY

Broker process recovery is not backlog recovery

MILESTONES E-18

Order	Milestone	Evidence
1	broker process started	container state
2	broker ready	management/health probe
3	publisher connected	client observation
4	circuit permits attempt	transition corpus
5	first confirm	publisher receipt
6	outbox zero	AlarmManagement snapshot
7	consumers caught up	Inbox/effect/source positions

ACCEPTANCE POLICY

This card uses the fixed outbox contract: owner database acceptance may continue while dispatch is delayed. A different rejection policy would be a different versioned scenario.

E-CORE-003 - PAUSE A SERVICE-OWNED DATABASE

Pause a service-owned database

CATALOGUE CARD E-CORE-003

Field	Declared value
initiating fault	pause one allowlisted SQL Server database container while a named operation is in flight
injection boundary	owned adapter call, connection pool, command timeout and service readiness
conserved owner truth	no cross-owner mutation; commit state is classified from durable receipts, never guessed
expected degradation	calls hang until bounds; readiness withdraws; liveness may remain healthy; excess load sheds
healing action	unpause database; prove fresh query, readiness, classification and pool/capacity return
evidence oracle	container pause state + timeout ledger + capacity census + receipt/effect reconciliation
stopping condition	no paused target; DB query and service ready; unknown outcomes reconciled; pool stable 3 samples

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-CORE-003 COMMIT-OUTCOME MATRIX

Lost acknowledgement never decides the database outcome

COMMIT WINDOWS E-19

Injection point	Required classification	Next action
before transaction begins	UNAVAILABLE; no owner rows	safe normal retry if policy allows
before commit sent	NOT_COMMITTED	bounded replay if operation is idempotent
after commit sent, before ack	UNKNOWN_COMMIT	receipt/effect reconciliation
after confirmed commit	COMPLETED	never replay as new operation

TRUTH AUTHORITY

Caller timeout and connection loss bound waiting; the operation receipt and owner effect determine whether the mutation exists.

E-CORE-004 - PUBLISH ONE ALARMFLOOD ENVELOPE TWICE

Publish one AlarmFlood envelope twice

CATALOGUE CARD E-CORE-004

Field	Declared value
initiating fault	publish a canonical byte-equivalent envelope twice with identical Messageld
injection boundary	nexus.events route alarm-management.alarm-flood-detected.v1
conserved owner truth	one RootCause Inbox identity and one domain effect exist for that Messageld
expected degradation	two deliveries and two processing attempts are visible; only one logical effect
healing action	no external healing; consumer reaches duplicate-completed terminal disposition
evidence oracle	broker deliveries + Inbox row/state + duplicate disposition + domain effect + settlements
stopping condition	two deliveries settled; one Inbox row; one effect; zero unacked for 3 samples

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-CORE-004 IDENTITY CONFLICT

Same Messageld with a different digest is not a duplicate

DUPLICATE INTERPRETATIONS E-20

Finding	Verdict
same ID + same canonical digest	benign redelivery path
same ID + different canonical digest	identity conflict; reject and retain security/defect evidence
two Inbox rows	uniqueness failure
two owner effects	idempotency failure
duplicate delivery unacked	settlement failure
zero owner effects	processing/recovery failure

METRIC SEPARATION

Delivery attempts, Inbox dispositions and domain effects remain separate measurements. Combining them would hide whether deduplication worked.

E-CORE-005 - DELAY THE REPORTING PROJECTION

Delay the Reporting projection

CATALOGUE CARD E-CORE-005

Field	Declared value
initiating fault	enable an Evidence-only latency toxic or barrier on projection apply
injection boundary	projection-worker path from reporting.integration-events.v1 to ReportingDb
conserved owner truth	RootCause owner facts continue; Reporting read state remains derived and provenance-bound
expected degradation	checkpoint/age lag rises; API returns old value with current-as-observed staleness metadata
healing action	remove toxic/release barrier; resume worker from durable checkpoint
evidence oracle	source high-water + monotonic checkpoint series + missing-position census + API staleness fields
stopping condition	checkpoint equals source high-water; no missing positions; lag returns to baseline for 3 samples

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-CORE-005 STALE-READ CONTRACT

Delayed truth is visible; it is never rewritten as freshness

PROJECTIONREADV1

```
public sealed record ProjectionReadV1<T>(  
    T Value,  
    long AppliedPosition,  
    long SourceHighWater,  
    DateTimeOffset ObservedUtc,  
    DateTimeOffset LastAppliedUtc)  
{  
    public long PositionLag => Math.Max(0, SourceHighWater - AppliedPosition);  
    public bool IsStale => PositionLag > 0;  
}
```

CATCH-UP BOUNDARY

Removing latency begins healing. Recovery ends only after monotonic checkpoint equality, zero missing source positions and a stable window.

E-CORE-006 - RESTART A ROLE WITH DURABLE WORK IN FLIGHT

Restart a role with durable work in flight

CATALOGUE CARD E-CORE-006

Field	Declared value
initiating fault	restart --no-deps one allowlisted API or worker using the admitted configuration snapshot
injection boundary	selected role process epoch, readiness and owned durable checkpoint/queue state
conserved owner truth	domain/database state is unchanged; pending outbox/inbox/projection work remains durable
expected degradation	readiness false during initialization; process-local cache/circuit state resets under contract
healing action	start same configuration; wait for readiness; resume from durable work; issue control operation
evidence oracle	epoch change + readiness transition + owner counts + backlog drain + fresh operation receipt
stopping condition	new epoch ready; no abandoned processing; checkpoint caught up; control succeeds for 3 samples

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-CORE-006 STATE-CLASS MATRIX

Restart resets process memory, not owner truth

EXPECTED STATE E-21

State class	Expected after restart
domain/database	unchanged and authoritative
outbox/inbox	pending work resumes from durable rows
projection cursor	resumes from ReportingDb checkpoint
in-memory cache	cold/refilled only under its contract
circuit state	new process-local state unless explicitly externalized
readiness	false during startup; true after initialization
instance identity	new service epoch

CONFIGURATION CAVEAT

Compose restart reuses the existing service configuration. A configuration change requires its Appendix D restart/reload/version protocol and is not smuggled into this scenario.

E-SUP-007 - KILL BEFORE INBOX CLAIM**Kill before Inbox claim****CATALOGUE CARD E-SUP-007**

Field	Declared value
initiating fault	SIGKILL rootcause-worker before durable Inbox admission
injection boundary	rootcause-worker receive boundary
conserved owner truth	broker still owns the delivery; no RootCause effect exists
expected degradation	delivery returns ready; worker unavailable
healing action	start worker
evidence oracle	broker chronology + zero pre-kill Inbox/effect + later completed effect
stopping condition	delivery settled once logically; one effect; stable drain

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-SUP-008 - EXPIRE A PROCESSING LEASE**Expire a processing lease****CATALOGUE CARD E-SUP-008**

Field	Declared value
initiating fault	kill after Inbox claim but before owner effect
injection boundary	RootCause Inbox processing lease
conserved owner truth	claim is durable; effect is absent
expected degradation	message cannot progress until bounded lease/repair policy
healing action	run owner-approved lease repair then redeliver
evidence oracle	lease timestamps + repair receipt + one terminal effect
stopping condition	Inbox completed; no processing lease; one effect

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-SUP-009 - REJECT POISON CONTRACT**Reject poison contract****CATALOGUE CARD E-SUP-009**

Field	Declared value
initiating fault	deliver unsupported event type/schema to closed consumer allowlist
injection boundary	rootcause.alarm-events.v1 parser boundary
conserved owner truth	no Domain call and no owner mutation
expected degradation	delivery is quarantined/dead-lettered visibly; consumer continues
healing action	none beyond terminal quarantine handling
evidence oracle	contract failure code + DLQ/quarantine receipt + zero owner effects
stopping condition	poison delivery terminal and visible; healthy control message succeeds

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-SUP-010 - REJECT MESSAGEID DIGEST CONFLICT

Reject Messageld digest conflict

CATALOGUE CARD E-SUP-010

Field	Declared value
initiating fault	deliver same Messageld with different canonical payload digest
injection boundary	RootCause Inbox identity gate
conserved owner truth	first accepted identity/effect remains unchanged
expected degradation	conflicting delivery rejected as defect/security evidence
healing action	none; operator review outside automatic replay
evidence oracle	stored digest + conflict receipt + zero second effect
stopping condition	conflict terminal; first effect unchanged; no nackd delivery

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-011 - LOSE PUBLISHER CONFIRMATION

Lose publisher confirmation

CATALOGUE CARD E-SUP-011

Field	Declared value
initiating fault	interrupt broker path after publish send but before confirm observation
injection boundary	alarm-worker publisher confirm boundary
conserved owner truth	outbox row remains authoritative and not falsely dispatched
expected degradation	publication outcome is pending/unknown; row retained for safe redelivery
healing action	restore path; publish same Messageld; await confirm
evidence oracle	outbox state + confirm chronology + downstream Inbox deduplication
stopping condition	row dispatched only after confirm; one downstream effect

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-012 - EXPIRE AN OUTBOX LEASE

Expire an outbox lease

CATALOGUE CARD E-SUP-012

Field	Declared value
initiating fault	stop alarm-worker after leasing a pending outbox batch
injection boundary	AlarmManagement outbox dispatcher lease
conserved owner truth	outbox messages remain durable and undispached without confirm
expected degradation	leased rows wait until bounded expiry; no second owner fact
healing action	start worker; recover expired lease; republish stable IDs
evidence oracle	lease owner/expiry + confirms + dispatch states
stopping condition	no live expired lease; all rows confirmed/dispatched

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-013 - ALARM DATABASE UNAVAILABLE BEFORE TRANSACTION**Alarm database unavailable before transaction****CATALOGUE CARD E-SUP-013**

Field	Declared value
initiating fault	stop/pause AlarmManagementDb before command transaction begins
injection boundary	alarm-api owned database adapter
conserved owner truth	no accepted fact and no outbox row are invented
expected degradation	request is unavailable; readiness withdraws
healing action	restore DB; prove readiness; retry only same idempotent operation contract
evidence oracle	operation receipt + owner row/outbox absence/presence reconciliation
stopping condition	one terminal receipt; accepted pair atomic or both absent

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-SUP-014 - ROOTCAUSE COMMIT ACKNOWLEDGEMENT LOST**RootCause commit acknowledgement lost****CATALOGUE CARD E-SUP-014**

Field	Declared value
initiating fault	cut DB response after commit is sent
injection boundary	rootcause-api command transaction
conserved owner truth	RootCause owner truth decides completion
expected degradation	client receives UNKNOWN_COMMIT rather than false success/failure
healing action	restore path and reconcile by OperationId
evidence oracle	operation receipt + effect row + response classification
stopping condition	Completed or NotCommitted; never inconsistent; no blind replay

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-SUP-015 - DETECT A REPORTING POSITION GAP**Detect a Reporting position gap****CATALOGUE CARD E-SUP-015**

Field	Declared value
initiating fault	withhold one allowlisted fact position in Evidence adapter
injection boundary	projection-worker ordered apply/checkpoint boundary
conserved owner truth	Reporting never advances checkpoint beyond an unapplied required position
expected degradation	projection stalls visibly; read remains stale
healing action	release missing position; apply in order
evidence oracle	source positions + checkpoint series + missing-position ledger
stopping condition	no gap; checkpoint equals high-water; stable lag zero

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-SUP-016 - PAUSE REPORTING DATABASE**Pause Reporting database****CATALOGUE CARD E-SUP-016**

Field	Declared value
initiating fault	pause ReportingDb while projection-worker applies
injection boundary	Reporting-owned write adapter
conserved owner truth	RootCause source facts remain untouched
expected degradation	projection lag rises; projection API exposes stale state
healing action	unpause DB; restore worker readiness and resume checkpoint
evidence oracle	DB state + Inbox/checkpoint + lag + API provenance
stopping condition	checkpoint caught up; no processing lease; stable readiness

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-017 - QUERY BFF UPSTREAM TIMEOUT**Query BFF upstream timeout****CATALOGUE CARD E-SUP-017**

Field	Declared value
initiating fault	delay projection-api beyond BFF total timeout
injection boundary	query-bff to projection-api read boundary
conserved owner truth	BFF owns no copied truth and does not manufacture freshness
expected degradation	bounded unavailable/degraded response with provenance
healing action	remove delay; allow circuit and endpoint to recover naturally
evidence oracle	attempt ledger + response contract + zero BFF database writes
stopping condition	fresh request succeeds; circuit stable; no hidden cache lie

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-018 - RESTART QUERY BFF**Restart Query BFF****CATALOGUE CARD E-SUP-018**

Field	Declared value
initiating fault	restart query-bff during concurrent reads
injection boundary	edge process and authentication/session technical state
conserved owner truth	owner services and databases remain unchanged
expected degradation	edge temporarily unavailable; no owner truth loss
healing action	start BFF with same admitted identity; wait readiness
evidence oracle	epoch/readiness + downstream owner snapshots + control read
stopping condition	new epoch ready; control read returns current-as-observed state

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-019 - LOSE TELEMETRY EXPORT**Lose telemetry export****CATALOGUE CARD E-SUP-019**

Field	Declared value
initiating fault	disable collector/export route during one campaign
injection boundary	OpenTelemetry pipeline
conserved owner truth	owner truth and fault receipts remain intact
expected degradation	trace evidence class becomes PIPELINE_LOSS
healing action	restore exporter and prove freshness
evidence oracle	collector health + export counters + durable state
stopping condition	campaign verdict INVALID_EVIDENCE when trace is mandatory; no false pass

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-020 - EXPOSE FALSE READINESS**Expose false readiness****CATALOGUE CARD E-SUP-020**

Field	Declared value
initiating fault	feed a fresh green readiness sample with a deliberately stale owner snapshot
injection boundary	recovery verifier input
conserved owner truth	stale state cannot be promoted to recovered truth
expected degradation	process appears ready while state evidence is stale
healing action	none; this is a deliberately broken evidence control
evidence oracle	freshness validator must reject the dossier
stopping condition	verdict INVALID_EVIDENCE

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-021 - SATURATE A DEPENDENCY BULKHEAD**Saturate a dependency bulkhead****CATALOGUE CARD E-SUP-021**

Field	Declared value
initiating fault	hold all permits at one admitted remote boundary
injection boundary	named concurrency limiter partition
conserved owner truth	accepted durable work is not duplicated or dropped
expected degradation	excess calls shed with bounded overload outcome
healing action	release held calls; allow queue to drain
evidence oracle	permit/queue census + attempt counts + receipts
stopping condition	active/queued return baseline; no local retry amplification

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-022 - DETECT NESTED RETRY AMPLIFICATION**Detect nested retry amplification****CATALOGUE CARD E-SUP-022**

Field	Declared value
initiating fault	compose two retry owners in Evidence-only descriptor
injection boundary	one logical remote boundary
conserved owner truth	one logical operation cannot exceed its versioned attempt ceiling
expected degradation	attempt count exceeds policy if gate fails
healing action	remove invalid composition; no runtime healing claim
evidence oracle	descriptor scan + attempt ledger + amplification formula
stopping condition	structural gate detects duplicate retry owner; campaign fails as designed

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-023 - REJECT INVALID CONFIGURATION RELOAD**Reject invalid configuration reload****CATALOGUE CARD E-SUP-023**

Field	Declared value
initiating fault	publish a partial/invalid reload candidate to an explicitly reloadable group
injection boundary	Appendix D validated reload boundary
conserved owner truth	last-known-good accepted snapshot remains active
expected degradation	candidate rejected; service stays on prior identity
healing action	correct complete candidate or retain prior snapshot
evidence oracle	candidate validation + accepted identity chronology
stopping condition	prior identity still active; no partial field mix; readiness honest

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-024 - START SERVICE BEFORE DEPENDENCY READINESS**Start service before dependency readiness****CATALOGUE CARD E-SUP-024**

Field	Declared value
initiating fault	start a role while its required DB/broker is running but not ready
injection boundary	Compose readiness-aware startup path
conserved owner truth	no owner mutation before required initialization
expected degradation	startup/readiness remains false; bounded initializer records cause
healing action	make dependency ready; allow bounded initialization
evidence oracle	startup/live/ready chronology + zero premature effects
stopping condition	role ready only after dependency and initialization; control operation succeeds

HONEST STATUS
SPECIFIED | RUNTIME PENDING

E-SUP-025 - FLAP READINESS DURING EDUCATIONAL ROLLOUT

Flap readiness during educational rollout

CATALOGUE CARD E-SUP-025

Field	Declared value
initiating fault	toggle dependency readiness during minimal Kubernetes rollout translation
injection boundary	Phase0Kubernetes probe and deployment mapping
conserved owner truth	service ownership and contracts remain identical to Compose
expected degradation	pod not admitted ready; traffic shifts only under probe contract
healing action	restore dependency and observe controlled readiness
evidence oracle	rendered manifest digest + probe events + owner snapshots
stopping condition	new replica stable ready; old/new owner truth conserved; educational evidence only

HONEST STATUS

SPECIFIED | RUNTIME PENDING

E-SUP-026 - FAIL A SERVICE-OWNED MIGRATION JOB

Fail a service-owned migration job

CATALOGUE CARD E-SUP-026

Field	Declared value
initiating fault	inject provider failure into one migrator before schema completion
injection boundary	one of three separate migration principals/jobs
conserved owner truth	runtime principal gains no DDL and foreign databases remain untouched
expected degradation	migration job fails visibly; dependent runtime stays unready
healing action	repair cause; rerun idempotent/forward-safe migration protocol
evidence oracle	migration history + principal audit + schema digest + readiness
stopping condition	expected schema digest present; job terminal; runtime starts; no foreign DDL

HONEST STATUS

SPECIFIED | RUNTIME PENDING

SUPPORTING CATALOGUE INDEX

Twenty cards discriminate failures around the core six

SUPPORTING FAMILIES E-22

Family	Scenario IDs	Boundary
consumer/inbox	E-SUP-007..010	claim, lease, poison, identity conflict
outbox/publisher	E-SUP-011..012	unknown confirm and lease recovery
owner databases	E-SUP-013..014	pre-transaction and unknown commit
reporting/read path	E-SUP-015..018	gap, DB pause, BFF timeout/restart
evidence/resilience	E-SUP-019..022	pipeline loss, stale proof, capacity, amplification
configuration/deployment	E-SUP-023..026	reload, startup, rollout, migrator

COVERAGE MEANING

Supporting cards test diagnostic discrimination and boundary preservation. They do not enlarge the admitted architecture or replace the six required recovery campaigns.

READINESS AND RECOVERY DEPTH

A scenario passes at its deepest required layer

RECOVERY LAYERS E-23

Layer	Postcondition
control plane	no fault, toxic or barrier remains
process	target has expected same/new epoch
liveness	process-local progress check healthy
readiness	dependencies and initialization ready
messaging	pending/ready/unacked/processing within terminal bounds
durable state	conservation and uniqueness pass
projection	checkpoint and lag converge
telemetry	pipeline fresh or result explicitly qualified

DEEPEST LAYER

A service may restore readiness quickly and still fail because backlog does not drain, a duplicate effect exists, or projection truth never catches up.

MACHINE-READABLE CATALOGUE

The repository must serialize the same closed inventory

FAILURECATALOGUEV1

```
public sealed record FailureCatalogueV1(
    string SchemaVersion,
    string ContractCatalogueSha256,
    string ConfigurationCatalogueSha256,
    FailureScenarioCatalogueEntryV1[] Entries,
    int MandatoryCoreCount,
    string CatalogueSha256);

public static class FailureCatalogueRules
{
    public const int MandatoryCoreCount = 6;
    public const int SupportingCount = 20;
    public const int ActivatedPublicFactCount = 3;
}
```

DIGEST AUTHORITY

Canonical serialization and the exact reader profile are pinned by the repository. A hand-edited PDF table cannot silently override the machine catalogue.

EVIDENCE MANIFEST

One record binds all mandatory results to one checkpoint

RECOVERYEVIDENCEMANIFESTV1

```
public sealed record RecoveryEvidenceManifestV1(
    string BuildSha256,
    string PackageLockSha256,
    string RuntimeManifestSha256,
    string ConfigurationSha256,
    string ResilienceProfileSha256,
    string FailureCatalogueSha256,
    DateTimeOffset StartedUtc,
    DateTimeOffset EndedUtc,
    ExperimentResultV1[] Results,
    EvidenceArtifactV1[] SharedArtifacts,
    EvidenceVerdict Verdict);
```

MANIFEST RULES E-24

Rule	Failure code
exactly 6 unique mandatory IDs discovered	REC_CORE_DISCOVERY_DRIFT
catalogue and runtime digests match	REC_IDENTITY_MISMATCH
all required artifacts present and fresh	REC_ARTIFACT_INVALID
every injected fault has active and absent proof	REC_FAULT_PROOF_MISSING
no unhealed runtime reused	REC_QUARANTINE_BREACH

FAIL-CLOSED VERIFIER

Missing evidence can never become a pass

RECOVERYEVIDENCEVERIFIER

```
public static EvidenceVerdict Verify(
    FailureCatalogueV1 catalogue,
    RecoveryEvidenceManifestV1 manifest)
{
    Require(catalogue.MandatoryCoreCount == 6, "REC_CORE_COUNT");
    Require(catalogue.Entries.Select(x => x.ScenarioId).Distinct().Count()
        == catalogue.Entries.Length, "REC_DUPLICATE_ID");
    Require(manifest.FailureCatalogueSha256 == catalogue.CatalogueSha256,
        "REC_IDENTITY_MISMATCH");

    var core = manifest.Results.Where(x => x.ScenarioId.StartsWith("E-CORE-"));
    Require(core.Select(x => x.ScenarioId).Distinct().Count() == 6,
        "REC_ZERO_OR_PARTIAL_DISCOVERY");
    Require(core.All(x => x.HasActiveProof && x.HasAbsentProof),
        "REC_FAULT_PROOF_MISSING");
    Require(core.All(x => x.ArtifactsFresh && x.StablePostconditions),
        "REC_ARTIFACT_OR_STATE_INVALID");
    return core.All(x => x.Verdict == ExperimentVerdict.Passed)
        ? EvidenceVerdict.Demonstrated
        : EvidenceVerdict.FalsifiedOrInvalid;
}
```

NEGATIVE EVIDENCE CONTROLS

The verifier must reject attractive false recoveries

BROKEN DOSSIERS E-25

Injected defect	Required verdict
green readiness plus stale owner snapshot	INVALID_EVIDENCE
fault command succeeded but active probe absent	INVALID_EVIDENCE
healing command returned but absence proof absent	ABORTED_UNHEALED
one successful poll without stable window	FAILED_EXPECTATION
trace required while pipeline is stale/lost	INVALID_EVIDENCE
duplicate effect with otherwise empty queues	FAILED_EXPECTATION
first failed run overwritten by green rerun	INVALID_EVIDENCE
candidate contract route appears active	INVALID_EVIDENCE

MUTATION OBLIGATION

A verifier that accepts any broken dossier above is not evidence infrastructure; its own result is rejected.

FIRST-FAILURE RETENTION

Rerun until green is forbidden

ATTEMPT CHRONOLOGY E-26

Attempt	Retention rule
first run	immutable result and full artifacts always retained
rerun	new ExperimentId; links to predecessor and stated reason
flaky diagnosis	compares identities, timing and observed differences
checkpoint claim	reports every attempt; cannot select only latest green

ATTEMPT LINEAGE

```
public sealed record CampaignAttemptLineage(  
    Guid ExperimentId,  
    Guid? PreviousExperimentId,  
    string RerunReasonCode,  
    string ImmutableResultSha256);
```

HONEST FLAKINESS

A later pass may show nondeterminism or repaired infrastructure. It never erases what the earlier evidence falsified or failed to observe.

CI EXECUTION DISCIPLINE

Destructive campaigns are serial inside one isolated runtime

CI GATES E-27

Rule	Reason
ephemeral project per job	no shared target or inherited backlog
baseline control before faults	separate pre-existing failure
serial destructive collection	faults do not overlap
hard job and fault timeouts	bound hazard lifetime
always-run healing	assertion/cancellation cannot skip cleanup
artifact upload on every result	first failure remains diagnosable
quarantine on unhealed	unsafe runtime never reused
zero-discovery fails	disabled tests cannot produce green

EVIDENCE ORDER

Control campaign -> six core campaigns -> supporting discrimination -> manifest verification. Parallel test speed is not worth ambiguous causation.

DOSSIER DIRECTORY CONTRACT

Artifacts remain locatable without becoming secret-bearing

EXPECTED EVIDENCE TREE

```
evidence/recovery/<build-sha12>/<experiment-id>/
  plan.json
  baseline.json
  fault-receipt.json
  during-fault.json
  heal-receipt.json
  recovery.json
  state-snapshots.ndjson
  trace-query.json
  container-events.ndjson
  result.json
  hashes.json
```

RETENTION METADATA E-28

Field	Permitted content
target	catalogue target ID, never connection endpoint with credentials
configuration	accepted digest and safe manifest reference
identity	build/contract/catalogue digests
state	bounded counts, IDs and classifications
telemetry	query result and freshness; no raw secret-bearing headers

ARCHITECTURE GATES

Failure code cannot weaken ownership while testing failure

STATIC AND DESCRIPTOR GATES E-29

Gate	Required result
REC-OWN-001	no cross-service DbContext or connection use
REC-MSG-002	exact three active facts and exact allowlists
REC-BAR-003	Evidence barrier absent from production composition
REC-ACT-004	actuator targets closed allowlist; no arbitrary command
REC-SEC-005	dossiers contain no secret or secret-derived value
REC-CFG-006	experiment begins from one admitted configuration identity
REC-RTR-007	restart/reload/version action matches Appendix D contract
REC-MIG-008	DDL occurs only under one of three migrator principals

EVIDENCE-ONLY SEAMS

Barriers, chaos hooks and container actuators must be absent from production binaries or production composition. Structural tests prove that exclusion.

CROSS-APPENDIX IDENTITY

Recovery evidence reuses the frozen catalogues

DIGEST REFERENCES E-30

Source	Required reference
Appendix B	service/role/database ownership catalogue digest
Appendix C	API/event contract catalogue and golden corpus digest
Appendix D	configuration catalogue and accepted snapshot digest
Appendix E	failure catalogue digest
Chapter 53	resilience profile digest
Chapter 54	runner/evidence schema version

DRIFT RESULT

A mismatched digest does not mean the scenario failed. It means the run is incomparable and its evidence verdict is `INVALID_EVIDENCE`.

PLATFORM EQUIVALENCE

Compose and Kubernetes project the same recovery semantics

MECHANISM MAPPING E-31

Semantic action	Phase0Compose	Phase0Kubernetes translation
abrupt role loss	kill container	delete/terminate one pod in isolated namespace
dependency unavailable	stop dependency	isolated network/dependency fixture only
dependency hung	pause/toxic	network fault fixture, not ordinary manifest
restart	restart same service config	replace pod under same admitted template
readiness	health-aware dependency	readiness probe and endpoint
evidence	Compose events + snapshots	events + pod identity + same owner snapshots

EDUCATIONAL BOUNDARY

Kubernetes cards validate translation of already-proved needs. They do not claim a production operations platform, multi-region recovery or cluster-level resilience maturity.

OPERATOR-READABLE RUNBOOK CARD

Humans see the same state machine as the verifier

RUNBOOK FIELDS E-32

Field	Question answered
scenario and hypothesis	what are we trying to falsify?
authorized target	what may be disturbed?
baseline gate	was the runtime healthy first?
active-fault proof	did the intended fault affect the target?
expected/actual degradation	did the system fail truthfully?
healing and absence proof	is the disturbance gone?
stable owner truth	did durable correctness and progress return?
verdict and residual	what may now be claimed?

NO MANUAL OVERRIDE TO PASS

An operator may abort and heal. They cannot turn missing machine evidence into PASSED through narrative approval.

ANTI-PATTERN REVIEW

Ten attractive recovery claims remain invalid

REJECTED CLAIMS E-33

Claim	Why invalid
container running means recovered	process state is shallower than durable truth
sleep 10 seconds is a convergence test	time passed without a predicate
queue zero proves no loss	accepted/effect conservation absent
HTTP 200 proves no duplicate effect	first attempt may have committed
connection restored means broker recovered	backlog may remain
readiness green means projection current	checkpoint may lag
one trace proves the whole path	state/coverage may be missing
timeout means rollback	remote outcome may be unknown
rerun green replaces failure	first evidence was erased
tool documentation proves runtime behavior	mechanism is not executed evidence

EVIDENCE STATUS BOARD

Specification completeness and runtime proof stay separate

APPENDIX E STATUS E-34

Obligation	Asset	Status
catalogue schema	typed entry and closed status	SPECIFIED
six core scenarios	fault/degradation/heal/oracle/stop cards	SPECIFIED
twenty supporting scenarios	diagnostic discrimination cards	SPECIFIED
safety and cleanup	allowlist, lease, watchdog, quarantine	SPECIFIED
state and trace evidence	snapshots, classes, dossier	SPECIFIED
fail-closed verdict	machine manifest and verifier	SPECIFIED
repository implementation	discoverable tests and schemas	PENDING
executed Compose/Kubernetes receipts	retained campaigns	PENDING

HONEST BOUNDARY

The catalogue is complete as a specification. No real fault run, repository build or retained campaign bundle was executed in this manuscript environment; runtime recovery remains unproved.

APPENDIX E CHECKPOINT

Recovery can now be reviewed as a closed evidence claim

COMPLETION LEDGER E-35

Question	Required answer
Was the baseline fresh?	control corpus passed before injection
Was the named fault active?	independent active-fault receipt
Was degradation truthful?	expected and actual state retained
Was healing complete?	idempotent action plus absence proof
Did owner truth survive?	conservation and uniqueness oracles
Did progress stabilize?	bounded consecutive stable observations
Is the dossier admissible?	fresh, value-free, digest-bound artifacts
Is the claim honest?	status reflects result; no missing receipt promoted

FINAL APPENDIX E RULE

Recovery is proved only when truth and progress stabilize after a proven fault. The catalogue forbids process health, elapsed time or narrative confidence from substituting for that argument.

HANDOFF TO APPENDIX F

The next appendix freezes security and trust boundaries

Appendix F - Security and Trust Boundary Notes will consolidate the trust model already distributed across Chapters 43-48: operator identity at the edge, capability enforcement at owner services, service identity, claims minimization, secret references, broker and database principals, telemetry exposure, evidence-only administrative paths and the explicit Phase-0 residual risk ledger.

CARRY-FORWARD OBLIGATIONS E-36

From Appendix E	Appendix F must preserve
closed actuator allowlist	administrative fault path cannot become general remote control
value-free dossiers	no tokens, credentials or secret-derived fingerprints
owner-state probes	least privilege and read-only evidence principals
three owners and exact contracts	security boundaries cannot invent new services/routes
identity-conflict and poison evidence	security-relevant terminal handling
honest PENDING status	no security maturity claim without retained receipts

APPENDIX E LAST WORD

A recovery claim must survive its own evidence loss

The strongest recovery catalogue is not the one with the most faults. It is the one that refuses to pass when the target, chronology, owner truth, healing proof or retained evidence is incomplete.

FINAL STATUS

FAILURE AND RECOVERY CATALOGUE COMPLETE | SIX CORE CAMPAIGNS FIXED | SUPPORTING ORACLES CLOSED | VALUE-FREE DOSSIERS DEFINED | EXECUTABLE RUNTIME EVIDENCE PENDING

APPENDIX F

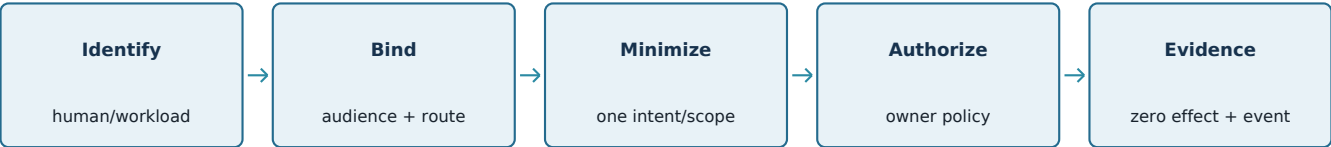
Security and Trust Boundary Notes

The final Phase-0 trust authority for NEXUS-1: human and workload identity, local capability enforcement, claims minimization, broker and database principals, security material, telemetry exposure, administrative separation and bounded evidence.

APPENDIX DECISION

Trust is accepted only for one principal type, audience, capability, resource, time and policy identity at one named boundary. Authentication never becomes universal authority; network location, certificate validity, a broker connection, an upstream allow decision or a green test cannot bypass owner-local enforcement.

TRUST DECISION PATH



CONTROL TOTALS F-1

Coordinate	Final value
trust zones / named crossings	9 / 12
security requirements / named threats / ID tombstones	21 / 24 / 4
active core owners / edge / roles / migrators	3 / 1 / 7 / 3
activated public facts	3
repair-admin crossings	2 - DESIGNED_INACTIVE
evidence posture	SPECIFIED; repository/runtime/security campaign PENDING

PURPOSE AND AUTHORITY

The appendix freezes trust decisions, not a product checklist

Chapters 43-48 define threat modelling, edge identity, workload authentication, claims minimization, material handling and adversarial evidence. Appendix F is the lookup authority that reconciles those decisions with the final three-owner runtime, exact Appendix C contracts, Appendix D configuration identities and Appendix E evidence-only failure paths.

AUTHORITY SPLIT F-2

Source	Owns	Appendix F contributes
Chapters 43-48	models, mechanisms, tests and limitations	one closed cross-reference
Appendix B	active owners, edge and roles	principal census
Appendix C	HTTP/event promises	security metadata
Appendix D	identity/material configuration	restart/rotation binding
Appendix E	failure actuator and evidence paths	administrative containment

NO MATURITY INFLATION

These notes define an educational Phase-0 security contract. They do not claim penetration testing, independent audit, production PKI, HSM custody, SOC/SIEM operation, zero-trust maturity or safety qualification.

FINAL-CORE RECONCILIATION

The threat model is broader than the active runtime

ACTIVATION LEDGER F-3

Element	Final status	Security treatment
AlarmManagement	ADMITTED	owner-local API/DB/broker policies
RootCause	ADMITTED	owner-local API/DB/broker policies
Reporting	ADMITTED	projection API/worker/DB policies
Query BFF	ADMITTED EDGE	human session + workload identity; no DB
Audit / Compliance / RL Advisory	CANDIDATE OR MODULE	no active credential/route/queue
Process Manager / Repair Control	DESIGNED_INACTIVE	threat entries retained; no runtime authority

INVENTORY GATE

A certificate, broker user, database principal, route or queue for an inactive candidate is a security and architecture drift failure - not automatic activation.

TWO-PLANE MODEL

Normal runtime and administrative authority remain separate

PLANE COMPARISON F-4

Plane	Flows	Authority rule
runtime	operator -> BFF -> owner APIs; producers -> broker -> consumers; roles -> owned DB	business action ends at owner policy
delivery	artifact/config/secret authority -> hosts; migrator -> one DB	deployment identity never becomes runtime business authority
evidence	diagnostic reader/fault harness -> bounded probes/actuators	isolated, allowlisted, value-free and run-scoped
repair	admin device -> repair control -> owner port	model retained but DESIGNED_INACTIVE in final core

NO UNIVERSAL CREDENTIAL

No user token, workload certificate, broker credential, telemetry identity, deployment identity, migrator principal or evidence credential is universal authority.

TRUST ZONES

A zone groups one control posture, not one subnet

ZONE CATALOGUE F-5

Zone	Primary contents	Default stance
Z0 User device	browser/session/UI	untrusted input and storage
Z1 Edge	Query BFF/public edge	authenticate and mediate
Z2 Service runtime	owner APIs/workers	local authorization
Z3 Messaging	exchange/queues	route-scoped access
Z4 Owner data	three service databases	owner-only DML
Z5 Observability	collector/backend/evidence	sensitive diagnostic
Z6 Admin control	repair/evidence control	highest scrutiny; repair inactive
Z7 Delivery/config	artifacts/secrets/policies	supply authority
Z8 External authority	OIDC/time/package/trust sources	explicit dependency contract

TRUST BOUNDARY SCHEMA

Every crossing names accepted trust and denied authority

TRUSTBOUNDARYCATALOGUEENTRYV1

```
public sealed record TrustBoundaryCatalogueEntryV1(  
    string BoundaryId,  
    string SourceZone,  
    string DestinationZone,  
    string Flow,  
    IReadOnlySet<PrincipalType> AcceptedPrincipals,  
    string Audience,  
    IReadOnlySet<string> Capabilities,  
    DataClass MaximumDataClass,  
    string FailureBehavior,  
    IReadOnlyList<string> RequiredEvidence,  
    BoundaryActivation Activation,  
    string EvidenceStatus);
```

ACTIVATION VOCABULARY F-6

Value	Meaning
ACTIVE_SPECIFICATION	belongs to the final runtime target; execution still pending
DESIGNED_INACTIVE	threat requirement retained; no runtime path or credential
REMOVED	tombstoned identity/route; negative probe required

BOUNDARY INDEX - RUNTIME AND DATA

TB-01 through TB-06 preserve identity, messaging and ownership

CROSSINGS F-7

ID	Source -> destination	Flow	Activation
TB-01	operator device -> Query BFF	interactive request/session	ACTIVE_SPECIFICATION
TB-02	Query BFF -> owner API	bounded human context + workload call	ACTIVE_SPECIFICATION
TB-03	producer worker -> RabbitMQ	public fact publication	ACTIVE_SPECIFICATION
TB-04	RabbitMQ -> consumer worker	at-least-once delivery	ACTIVE_SPECIFICATION
TB-05	runtime role -> owned database	owner-local DML/read	ACTIVE_SPECIFICATION
TB-06	migrator -> owned database	bounded DDL	ACTIVE_SPECIFICATION

BOUNDARY INDEX - OPERATIONS AND SUPPLY

TB-07 through TB-12 keep side paths visible

CROSSINGS F-8

ID	Source -> destination	Flow	Activation
TB-07	service/collector -> telemetry	logs/metrics/traces	ACTIVE_SPECIFICATION
TB-08	diagnostic caller -> health/evidence	bounded operational read	ACTIVE_SPECIFICATION
TB-09	admin workstation -> repair API	request/approve/execute	DESIGNED_INACTIVE
TB-10	repair control -> owner repair port	typed privileged action	DESIGNED_INACTIVE
TB-11	delivery pipeline -> runtime host	artifact/config deployment	ACTIVE_SPECIFICATION
TB-12	secret/config authority -> runtime	credential/policy material	ACTIVE_SPECIFICATION

NO INVISIBLE MANAGEMENT PLANE

A path used only during failure evidence, migration, deployment or rotation still belongs in the trust catalogue.

TB-01 - OPERATOR DEVICE TO QUERY BFF

Operator device to Query BFF

TRUST CARD TB-01

Coordinate	Decision
source -> destination	Z0 User device -> Z1 Edge
flow	interactive session/request
accepted principal	authenticated current human session; anonymous only at protocol routes
local authorization	session, anti-forgery, capability and server-owned scope
protected authority/data	operator context and bounded response
failure behavior	challenge/deny/no-store; zero owner call
required evidence	OIDC/session/cookie/XSRF and endpoint negative cases

BOUNDARY STATUS

SPECIFIED | RUNTIME PENDING

TB-02 - QUERY BFF TO OWNER SERVICE

Query BFF to owner service

TRUST CARD TB-02

Coordinate	Decision
source -> destination	Z1 Edge -> Z2 Service runtime
flow	mTLS workload call plus minimized delegated context
accepted principal	known Query BFF WorkloadId; bounded pairwise human context where required
local authorization	destination workload capability plus owner resource policy
protected authority/data	owner command/query authority
failure behavior	deny/unavailable; zero Domain/DB/outbox effect
required evidence	fresh TLS + audience/capability/scope matrix + effect probes

BOUNDARY STATUS

SPECIFIED | RUNTIME PENDING

TB-03 - PRODUCER WORKER TO RABBITMQ

Producer worker to RabbitMQ

TRUST CARD TB-03

Coordinate	Decision
source -> destination	Z2 Service runtime -> Z3 Messaging
flow	publish one declared public fact route
accepted principal	role-specific EXTERNAL broker user
local authorization	vhost/resource/topic ACL; stable MessageId
protected authority/data	producer-owned public fact
failure behavior	broker denial/pending outbox; no foreign publish
required evidence	real broker permission matrix + confirms + outbox state

BOUNDARY STATUS
SPECIFIED | RUNTIME PENDING

TB-04 - RABBITMQ TO CONSUMER WORKER

RabbitMQ to consumer worker

TRUST CARD TB-04

Coordinate	Decision
source -> destination	Z3 Messaging -> Z2 Service runtime
flow	at-least-once delivery
accepted principal	expected broker origin evidence and closed contract allowlist
local authorization	envelope/producer/route/schema/digest/Inbox admission
protected authority/data	consumer owner effect and settlement
failure behavior	reject/quarantine/redeliver; no invalid Domain effect
required evidence	broker origin + corpus + Inbox/effect/ack probes

BOUNDARY STATUS
SPECIFIED | RUNTIME PENDING

TB-05 - RUNTIME ROLE TO OWNED DATABASE

Runtime role to owned database

TRUST CARD TB-05

Coordinate	Decision
source -> destination	Z2 Service runtime -> Z4 Owner data
flow	owner-local DML/read
accepted principal	one role-specific runtime principal
local authorization	database grant plus repository/Application contract
protected authority/data	service-owned source or derived state
failure behavior	connection/grant denial; readiness honest
required evidence	grant export + forbidden probes + transaction effect snapshot

BOUNDARY STATUS
SPECIFIED | RUNTIME PENDING

TB-06 - MIGRATOR TO OWNED DATABASE**Migrator to owned database****TRUST CARD TB-06**

Coordinate	Decision
source -> destination	Z7 Delivery/config -> Z4 Owner data
flow	bounded schema DDL
accepted principal	one service-owned migration principal
local authorization	artifact/migration/target/environment approval
protected authority/data	schema and migration history
failure behavior	job fails; runtime remains unready; no foreign DDL
required evidence	principal, history, schema digest and deployment receipt

BOUNDARY STATUS
SPECIFIED | RUNTIME PENDING

TB-07 - RUNTIME TO TELEMETRY PIPELINE**Runtime to telemetry pipeline****TRUST CARD TB-07**

Coordinate	Decision
source -> destination	Z2 Service runtime -> Z5 Observability
flow	allowlisted logs/metrics/traces
accepted principal	known workload telemetry exporter
local authorization	attribute/event allowlist, limits and redaction
protected authority/data	diagnostic evidence only
failure behavior	bounded drop/backpressure; no business authority
required evidence	canary scan + freshness + cardinality and spoof cases

BOUNDARY STATUS
SPECIFIED | RUNTIME PENDING

TB-08 - DIAGNOSTIC CALLER TO HEALTH/EVIDENCE**Diagnostic caller to health/evidence****TRUST CARD TB-08**

Coordinate	Decision
source -> destination	Z6 Admin control -> Z2/Z5 runtime evidence
flow	bounded operational read
accepted principal	public safe probe or run-scoped evidence reader
local authorization	endpoint profile, purpose and artifact/run scope
protected authority/data	safe status/count/digest references
failure behavior	deny/404/unavailable; no topology/material leak
required evidence	endpoint inventory + disclosure and forbidden-write probes

BOUNDARY STATUS
SPECIFIED | RUNTIME PENDING

TB-09 - ADMIN DEVICE TO REPAIR CONTROL**Admin device to repair control****TRUST CARD TB-09**

Coordinate	Decision
source -> destination	Z0/Z6 admin device -> Z6 Admin control
flow	request/approve/execute repair plan
accepted principal	separate repair human roles
local authorization	freshness, separation of duties, exact immutable plan
protected authority/data	repair intent and approval
failure behavior	deny and record; zero owner effect
required evidence	model and adversarial obligations only

BOUNDARY STATUS

DESIGNED_INACTIVE | NO FINAL-CORE ENDPOINT/CREDENTIAL

TB-10 - REPAIR CONTROL TO OWNER REPAIR PORT**Repair control to owner repair port****TRUST CARD TB-10**

Coordinate	Decision
source -> destination	Z6 Admin control -> Z2 Service runtime
flow	typed privileged owner action
accepted principal	repair-control workload plus approved plan context
local authorization	owner-local repair capability, digest, precondition and RepairKey
protected authority/data	owner-local repair receipt
failure behavior	deny/stale/conflict; no generic SQL/broker path
required evidence	model and owner-port negative obligations only

BOUNDARY STATUS

DESIGNED_INACTIVE | NO FINAL-CORE ENDPOINT/CREDENTIAL

TB-11 - DELIVERY PIPELINE TO RUNTIME HOST**Delivery pipeline to runtime host****TRUST CARD TB-11**

Coordinate	Decision
source -> destination	Z7 Delivery/config -> Z2 Service runtime
flow	artifact/config deployment
accepted principal	approved delivery workload/actor
local authorization	artifact/config/environment identity and rollout gate
protected authority/data	which code and public policy run
failure behavior	deployment blocked/rolled back with evidence
required evidence	artifact, config, actor, target, rollout and reversal receipts

BOUNDARY STATUS

SPECIFIED | RUNTIME PENDING

TB-12 - SECRET/CONFIG AUTHORITY TO RUNTIME**Secret/config authority to runtime****TRUST CARD TB-12**

Coordinate	Decision
source -> destination	Z7 Delivery/config -> Z2 Service runtime
flow	credential, certificate and policy material
accepted principal	scoped provider/deployment identity
local authorization	strict reference, source, permission, compatibility and rotation policy
protected authority/data	workload capability and security policy
failure behavior	startup/readiness fails closed; last-known-good only while valid
required evidence	value-free material manifest + fault/rotation/leak cases

BOUNDARY STATUS
SPECIFIED | RUNTIME PENDING

SECURITY REQUIREMENTS I**Identity, integrity and accountability stay mechanism-neutral****SR-043-001..006 F-9**

ID	Required outcome
SR-043-001	accepted principal type, authority, audience and freshness per boundary
SR-043-002	decision binds every integrity-sensitive input
SR-043-003	durable accountability for consequential action
SR-043-004	outward data allowlisted by classification
SR-043-005	cost budgets protect normal-path capacity
SR-043-006	local least privilege and forbidden-capability absence

SECURITY REQUIREMENTS II**Delegation and failure never weaken local authority****SR-043-007..013 F-10**

ID	Required outcome
SR-043-007	fresh protected context for privileged human action
SR-043-008	server-owned resource/site/tenant scope
SR-043-009	owner service reauthorizes delegated human action
SR-043-010	human identity is never sufficient workload trust
SR-043-011	identity/policy uncertainty fails closed and visible
SR-043-012	publisher/consumer routes match declared ownership
SR-043-013	repair plan binding, dual control and owner receipt

SECURITY REQUIREMENTS III

Data, telemetry, delivery and material remain bounded

SR-043-014..021 F-11

ID	Required outcome
SR-043-014	runtime credentials cannot cross owner databases
SR-043-015	migration DDL is separate, bounded and auditable
SR-043-016	strict route/envelope/inbox admission before Domain work
SR-043-017	sensitive data excluded/minimized in telemetry
SR-043-018	telemetry never becomes business/audit authority
SR-043-019	runtime binds reviewed artifact/config identity
SR-043-020	secret/certificate lifecycle prevents logging/reuse
SR-043-021	security context has freshness/revocation semantics

HUMAN PRINCIPAL SEPARATION

A person receives one bounded action posture

HUMAN CATALOGUE F-12

Principal	Allowed	Must not inherit	Status
operator reader	bounded query	command/repair	ACTIVE
business actor	declared owner command	DB/broker admin	ACTIVE
repair requester	prepare exact plan	approval/execution	DESIGNED_INACTIVE
repair approver	approve immutable digest	edit/execute	DESIGNED_INACTIVE
repair executor	start approved job	expand scope	DESIGNED_INACTIVE
auditor	read protected dossier	runtime mutation	EXTERNAL/PROCEDURAL
platform admin	platform operation	owner business decision	EXTERNAL/PROCEDURAL

SEPARATION RULE

Repair.Execute never implies Repair.Approve. Platform access never becomes business authority. A UI role label alone authorizes nothing.

ACTIVE WORKLOAD PRINCIPALS

Seven runtime roles receive seven narrow identities

RUNTIME PRINCIPAL MATRIX F-13

Role	Network/API	Data/broker authority
query-bff	OIDC edge; approved owner APIs	no DB; no broker
alarm-api	Alarm owner HTTP	AlarmManagementDb DML only
alarm-worker	no public HTTP	Alarm DB DML + alarm fact publish
rootcause-api	RootCause owner HTTP	RootCauseDb DML only
rootcause-worker	no public HTTP	RootCause DB + flood consume + case/verdict publish
projection-api	Reporting internal HTTP	ReportingDb read only
projection-worker	no public HTTP	ReportingDb write + reporting queue consume

NO CO-LOCATION INHERITANCE

Sharing a host, namespace, image base or network does not merge workload identities or grants.

NON-RUNTIME PRINCIPALS

DDL, topology, evidence and secret authority stay separate

ADMINISTRATIVE PRINCIPALS F-14

Principal class	Allowed	Denied
three migrators	one owner DB DDL each	runtime requests and foreign DBs
broker provisioner	declared topology/permissions	business fact publication
delivery identity	approved artifact/config deployment	owner DML
secret authority	issue/rotate scoped material	business reads
telemetry collector	receive/export safe telemetry	owner writes
evidence reader	named run-scoped artifacts/probes	repair or broad DB access
failure actuator	Appendix E allowlisted isolated targets	shared/prod/general shell

BFF AUTHENTICATION CONTRACT

OIDC authenticates the human into a server-side session

EDGE MECHANISM F-15

Coordinate	Fixed decision
client	Query BFF is confidential OIDC client
flow	Authorization Code with PKCE
validation	issuer, client audience, state, nonce, lifetime and callback
token handling	no access/id/refresh token in browser or ordinary session ticket
session	server-side, revocable, absolute expiry
browser credential	opaque protected cookie reference only
claim mapping	MapInboundClaims=false; exact provider profiles

AUTHENTICATION LIMIT

A valid BFF session is evidence of a current human context. It is not owner permission and never authenticates the BFF workload to a service.

COOKIE AND REQUEST BINDING

Session and anti-forgery protect different things

COOKIE CONTRACT F-16

Cookie	Attributes	Purpose
__Host-NEXUS1-Operator	Path=/; Secure; HttpOnly; SameSite=Lax; no Domain	opaque server session reference
__Host-NEXUS1-XSRF	Path=/; Secure; readable; SameSite=Strict; no Domain	request-token echo only

MUTATION GATES F-17

Gate	Failure outcome
current session	401; no owner call
anti-forgery token	400/403; no owner call
capability and scope	403 safe denial; no existence disclosure
owner reauthorization	owner denies; zero Domain/DB/outbox effect

XSS RESIDUAL

HttpOnly blocks ordinary cookie extraction; injected same-origin script may still act as the operator. CSP, encoding, dependency hygiene and explicit intent remain separate controls.

SESSION FRESHNESS

Cache success is never indefinite trust

OPERATORSESSIONSTATEV2

```
public sealed record OperatorSessionStateV2(
    string SessionReference,
    string PairwiseSubjectSeedReference,
    DateTimeOffset AuthenticatedAtUtc,
    DateTimeOffset AbsoluteExpiresUtc,
    long IdentityRevision,
    EdgeCapabilitySnapshot EdgeCapabilities,
    string EdgePolicyRevision,
    bool Revoked);
```

FRESHNESS TRIGGERS F-18

Change	Required behavior
logout/revocation	revoke server record before clearing cookie
capability revision	reload current mapping; old session cannot widen
scope removal	old target becomes denied/not-found
policy unavailable/stale	fail closed and visible
key-ring generation change	bounded overlap and restart evidence

FINAL HTTP CAPABILITY MAP

The exact Appendix C routes carry exact owner policies

PROTECTED SURFACES F-19

Surface	Capability	Owner decision
POST /api/v1/alarm-observations	alarm.observation.submit	AlarmManagement local command policy
GET /api/v1/root-cause-cases/{caselid}	root-cause.case.read	RootCause resource/scope policy
POST /api/v1/root-cause-cases/{caselid}/evidence	root-cause.evidence.attach	RootCause resource/revision policy
POST /api/v1/root-cause-cases/{caselid}/verdicts	root-cause.verdict.issue	RootCause policy/evidence precondition
GET /internal/v1/root-cause-case-projections/{caselid}	reporting.root-cause-case-projection.read	Reporting allows Query BFF only
Query BFF GET /api/v1/root-cause-cases/{caselid}	reporting.root-cause-case.read	edge session/capability/scope

SUPERSEDED SURFACE

/api/operator/v1/root-cause-cases/{caselid} and the earlier /ui-api/v1 owner examples are not aliases for final runtime contracts.

DUAL-PRINCIPAL OWNER GATE

Workload and delegated human facts answer different questions

OWNER DECISION INPUT F-20

Coordinate	Question
workload principal	which authenticated service opened this call?
destination audience	was the context built for this owner?
workload capability	may that caller invoke this endpoint class?
human subject reference	on whose behalf is the call made?
operator intent	which one action was requested?
resource scope	does current owner state admit this target?
policy identities/time	were all inputs current for one decision?

OWNER AUTHORITY

The BFF allow decision is upstream evidence. The owning service reloads its resource and performs the final capability, audience, scope and freshness decision.

WORKLOAD IDENTITY MECHANISM

Direct mTLS authenticates one named Phase-0 role

CERTIFICATE PROFILE F-21

Coordinate	Requirement
trust	custom Phase-0 workload root; environment bound
identity	exactly one recognized URI SAN
grammar	urn:nexus1:workload:v1:phase0:<role>
purpose	client-auth EKU; key possession; validity window
destination	server DNS SAN and expected audience
fallback	no CN, substring, network address or forwarded-cert header
authorization	destination-local capability after authentication

MECHANISM LIMIT

A valid certificate proves a recognized workload at this boundary. It is not global internal trust, business authority or instance attestation.

CANONICAL WORKLOAD IDENTIFIERS

Identity strings are exact, versioned and tombstoned

WORKLOADID

```
public readonly record struct WorkloadId(string Value)
{
    private const string Prefix = "urn:nexus1:workload:v1:phase0:";

    public static WorkloadId ParseExact(string value)
    {
        if (!value.StartsWith(Prefix, StringComparison.Ordinal))
            throw new WorkloadIdentityException("unsupported-workload-id");
        if (value != value.Trim().ToLowerInvariant())
            throw new WorkloadIdentityException("non-canonical-workload-id");
        return new(value);
    }
}
```

ACTIVE SUFFIXES F-22

Class	Suffixes
edge/API	nexus1-bff, alarm-api, rootcause-api, projection-api
workers	alarm-worker, rootcause-worker, projection-worker
migrations	alarm-migrator, rootcause-migrator, reporting-migrator

DESTINATION CAPABILITY POLICY

The receiver decides; the caller cannot mint permission**WORKLOADCAPABILITYREQUIREMENT**

```
public sealed record WorkloadCapabilityRequirement(
    ServiceAudience Audience,
    WorkloadCapability Capability) : IAuthorizationRequirement;

public sealed record WorkloadPolicyDecisionInput(
    WorkloadId Caller,
    ServiceAudience Destination,
    WorkloadCapability Capability,
    string EndpointIdentity,
    string PolicyRevision,
    DateTimeOffset ObservedUtc);
```

NO WILDCARD POSTURE F-23

Caller	Destination	Allowed class
query-bff	AlarmManagement/RootCause/Reporting APIs	only catalogued operator proxy/query capabilities
workers	broker	own publish/consume routes only
migrators	one owner DB	schema DDL only
unknown/disabled	any	DENY

RABBITMQ AUTHENTICATION

TLS EXTERNAL identifies transport authority, not event meaning**BROKER POSTURE F-24**

Coordinate	Fixed decision
listener	TLS 5671; plaintext 5672 disabled in admitted profile
peer	verify_peer + fail_if_no_peer_cert
mechanism	EXTERNAL only; no PLAIN fallback/shared password
namespace	Phase-0 vhost, environment bound
user mapping	one broker user -> one canonical WorkloadId
permissions	configure/write/read regex per role/resource
User ID	must agree with authenticated broker user

SEMANTIC SPLIT

Broker acceptance proves connection identity and permitted topology operation. Consumers still validate exchange, route, schema, producer and message identity before Domain work.

BROKER PERMISSION MATRIX

Each active worker has only its declared message posture**ACTIVE GRANTS F-25**

Principal	Publish	Consume	Configure
alarm-worker	alarm-management.alarm-flood-detected.v1	none	none
rootcause-worker	root-cause case-opened + verdict-issued routes	rootcause.alarm-events.v1	none
projection-worker	none	reporting.integration-events.v1	none
APIs/query-bff	none	none	none
broker provisioner	none as business producer	none	declared topology only

CANDIDATE EXCLUSION

No active grant exists for Audit, Compliance, RecommendationGenerated, correction facts or Process Manager queues.

CONSUMER SEMANTIC ADMISSION

A broker-delivered message is still untrusted input

ADMISSION ORDER F-26

Step	Gate	Failure
1	authenticated broker user and User ID agree	reject/terminal evidence
2	expected exchange and exact route	reject foreign route
3	envelope producer/event/schema allowlist	poison/quarantine
4	canonical body digest and MessageId identity	identity conflict
5	Inbox admission/idempotency	duplicate-completed or processing
6	Domain/Application translation	typed rejection; no defaults
7	durable Inbox/effect commit before ack	redelivery allowed

REPAIR LIMIT

Poison replay, if ever activated, must re-enter through the normal broker/consumer admission path. No admin path may invoke a Domain handler directly.

ACTIVATED PUBLIC-FACT AUTHORITY

Three facts define the complete Phase-0 broker trust surface

PUBLIC-FACT PRINCIPAL MATRIX F-26A

Fact / route	Only publisher	Only admitted consumer
AlarmFloodDetected.v1 alarm-management.alarm-flood-detected.v1	alarm-worker	rootcause-worker via rootcause.alarm-events.v1
RootCauseCaseOpened.v1 root-cause.root-cause-case-opened.v1	rootcause-worker	projection-worker via reporting.integration-events.v1
RootCauseVerdictIssued.v1 root-cause.root-cause-verdict-issued.v1	rootcause-worker	projection-worker via reporting.integration-events.v1

SHARED TRANSPORT, SEPARATE AUTHORITY

Coordinate	Final rule
exchange	nexus.events carries only the three declared public-fact routes
publisher grant	write only the role's declared routing keys; no configure/read
consumer grant	read/ack only the declared queue; no foreign publish/configure
semantic admission	producer, event type/version, route, MessageId and digest must agree
identity conflict	same MessageId with different bytes is poison/conflict, never duplicate success

CLOSED ACTIVATION

Any fourth event type, new route, queue, publisher or consumer is contract and security drift until Appendices B-D and the evidence catalogue admit it together.

DATABASE PRINCIPALS

A connection string is an ownership decision

RUNTIME GRANTS F-27

Role	Database authority	Forbidden
alarm-api/worker	AlarmManagementDb required DML	RootCauseDb/ReportingDb/DDL
rootcause-api/worker	RootCauseDb required DML	AlarmManagementDb/ReportingDb/DDL
projection-api	ReportingDb read	write/DDL/owner DBs
projection-worker	ReportingDb projection write	owner DBs/DDL
query-bff	none	every DbContext/connection

NEGATIVE EVIDENCE

The permission inventory and forbidden connection probes must prove absence of foreign authority without exporting credential values or secret-derived fingerprints.

MIGRATION AND SCHEMA AUTHORITY

Three one-shot jobs own DDL windows

MIGRATOR CONTRACT F-28

Job	Allowed	Required evidence
alarm-migrator	AlarmManagementDb DDL only	artifact, migration list, target, principal, history
rootcause-migrator	RootCauseDb DDL only	artifact, migration list, target, principal, history
reporting-migrator	ReportingDb DDL only	artifact, migration list, target, principal, history

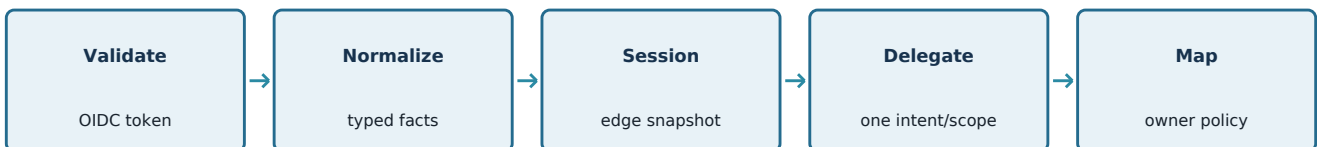
DDL PROHIBITIONS F-29

Forbidden	Gate
ApplyMigrationsOnStartup=true in runtime	configuration/architecture failure
shared migrator credential	principal inventory failure
wrong database/environment	binding and digest failure
green startup as migration proof	history/schema evidence still required

CLAIMS MINIMIZATION PIPELINE

Each boundary emits a smaller, more local type

IDENTITY REDUCTION F-30



WHAT STOPS WHERE F-31

Transition	Discarded	Retained
token -> subject	signature bytes and unused claims	issuer/subject/auth evidence
subject -> session	provider syntax and raw entitlements	edge revision/capability/scope
session -> owner	global sets and all destinations	one audience/intent/scope
context -> decision	external-policy authority	owner facts and decision refs

NO RAW BAG

No owner receives the browser token, cookie, ClaimsPrincipal, general claim bag, full group list or upstream precomputed allow decision.

EXTERNAL CLAIM CONTRACT

A claim is an asserted fact, never an instruction

PROVIDER PROFILE F-32

Coordinate	Rule
issuer	one exact enabled profile
subject/auth-time/assurance	exact names, types and cardinality
entitlements	bounded count/length; unknown values grant nothing
unknown claim name	ignore + value-free drift observation
required missing/wrong type	mapping failure; no default grant
duplicate singleton	reject; never first-wins
forwarding-forbidden claims	never persisted or delegated

ROLE BYPASS BLOCKED

Framework `IsInRole` and a generic role claim are not owner authorization paths. Capability values arise only from versioned local maps.

DELEGATED HUMAN CONTEXT

One request carries one audience, intent and scope

DELEGATEDOPERATORCONTEXTV1

```
public sealed record DelegatedOperatorContextV1(  
    AudienceSubjectReference Subject,  
    OwnerService Audience,  
    OperatorIntent Intent,  
    ScopeCoordinate Scope,  
    DateTimeOffset AuthenticatedAtUtc,  
    DateTimeOffset IssuedUtc,  
    DateTimeOffset ExpiresUtc,  
    string IdentityRevision,  
    string EdgePolicyRevision,  
    string DecisionReference);
```

FIELD BOUNDARIES F-33

Field	Cannot mean
Audience	set of reusable destinations
Intent	granted capability without owner decision
Scope	membership list or client-widened region
Subject	globally correlatable raw external subject
revisions/reference	policy contents or bearer proof

AUDIENCE-PSEUDONYMOUS REFERENCE

Different owners do not receive one global subject key

DERIVATION CONTRACT F-34

Coordinate	Requirement
input	validated external issuer+subject and one owner audience
key	protected audience-specific key revision
domain separation	versioned NEXUS-1 context + audience
output	bounded non-reversible reference for that owner
rotation	single-write, bounded dual-read compatibility
evidence	revision and behavior only; no key/value/digest of subject

STANDARDS BOUNDARY

This is a NEXUS-1 correlation-minimizing reference. The book does not claim OpenID Provider pairwise-sub conformance.

OWNER POLICY SNAPSHOT

Authorization mapping is local, immutable and current

POLICY ADMISSION F-35

Input	Required behavior
PolicyId/version/digest	exact accepted artifact
destination profile	known audience/intent/scope kind
resource	loaded from owner store after coarse admission
scope	client input may narrow, never widen
unknown entitlement/capability	deny; no generic role
partial/corrupt reload	candidate rejected; current stays
stale/unavailable required policy	fail closed and visible

NO GLOBAL AUTHORIZATION DB

Each owner maps the minimized context to its own language and current resource. A central subject-to-permission service is not introduced.

SECRET-REFERENCE BOUNDARY

Secrets grant capabilities; configuration stores references

MATERIAL CLASSES F-36

Material	Reference owner	Never evidence
database credential	role-specific DB adapter	connection string/value/hash
broker credential/certificate	publisher/consumer adapter	key/password/certificate password
OIDC client secret	Query BFF identity adapter	secret or derived fingerprint
workload private key	TLS host/client adapter	PEM/key bytes
pairwise key ring	BFF reference derivation	key/value/subject digest
policy artifact	owner-local policy reader	secret values or unapproved body export

RUNTIME ABSENCE

No public endpoint returns secrets, uploads keys, edits trust roots or writes policy files. Material is prepared by the deployment/provider boundary.

SECURITY MATERIAL SNAPSHOT

Certificates, trust, keys and policy activate atomically

SECURITYMATERIALSNAPSHOTIDENTITY

```
public sealed record SecurityMaterialSnapshotIdentity(  
    string EnvironmentProfile,  
    string CertificatePolicyIdentity,  
    string TrustBundleIdentity,  
    string WorkloadMapIdentity,  
    string OidcProfileIdentity,  
    string PairwiseKeyRingRevision,  
    string AuthorizationPolicyIdentity,  
    DateTimeOffset ActivatedUtc);  
  
public enum SnapshotState  
{  
    Candidate, Validated, Active, Retiring, Rejected  
}
```

ATOMICITY RULE

An isolated candidate resolves every reference and validates cross-material compatibility before one accepted snapshot swaps. Partial activation is forbidden.

MATERIAL VALIDATION

Readable material is not automatically acceptable material

VALIDATION GATES F-37

Kind	Required checks
secret	source allowed, exact permissions, bounded size, purpose
certificate/key	PEM parse, key match, chain, SAN, EKU, time, policy
trust bundle	approved roots, bounded chain/count, environment
workload map	one canonical identity/binding, no duplicate/tombstone reuse
policy JSON	strict schema, closed values, canonical digest, compatibility
OIDC profile	authority/audience/redirect/source identity
key ring	revision, owner audience coverage, rotation state

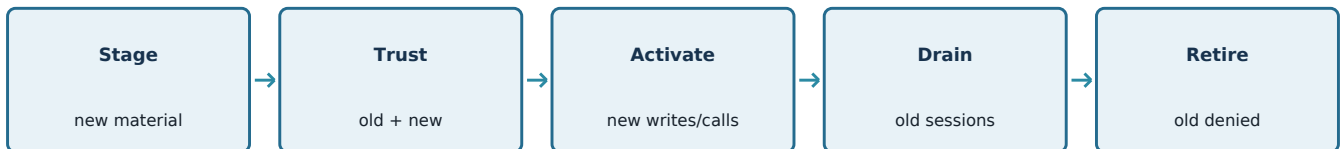
FAILURE BEHAVIOR

Missing, malformed, expired, revoked, over-permitted or incompatible required material blocks readiness and may block startup. It never activates an empty default.

ROTATION PROTOCOL

Fresh handshakes prove old/new/retired behavior

ROTATION F-38



ROTATION EVIDENCE F-39

Phase	Required oracle
overlap	fresh old and new handshakes succeed only if policy permits
activation	new identity observed at destination
drain	old connections gone within explicit local bound
retirement	fresh old credential/cert/key use fails
recovery	crash phase reconstructed from durable attempt evidence

NO TIME-ONLY SUCCESS

Waiting for an overlap interval does not prove rotation. Connection reuse must not hide certificate rejection or retired credential acceptance.

MATERIAL HEALTH AND DIAGNOSTICS

Health reports state, never material

PROBE SURFACES F-40

Probe	May expose	Must not expose
/health/startup	initialization disposition	material path/value
/health/live	process-local progress	secret-source call/details
/health/ready	accepted snapshot usable	issuer/cert/path/grants
rotation evidence	safe phase + attempt reference	key/cert inventory
configuration evidence	identity/status/reference	secret/body/policy values

SAFE FAILURE

External responses use bounded remediation classes. Internal evidence stores references and reason codes, never secret-bearing exceptions or configuration objects.

TELEMETRY DATA BOUNDARY

Observability explains decisions without becoming authority

ALLOWLIST F-41

Surface	Allowed	Forbidden
trace	safe lineage IDs, route templates, state classes	tokens, cookies, bodies, secrets
structured log	event/reason/policy refs, bounded actor ref	raw subject, email, arbitrary exception Data
metric	small route/result/principal classes	resource IDs, subjects, tenant/site IDs
Problem Details	safe type/title/status/correlation	topology, claims, credential detail
evidence export	named artifact + SHA-256 + status	unapproved raw logs/dumps

TRUTH HIERARCHY

Telemetry may be absent, sampled or spoofed. Owner rows, Inbox/outbox state, broker evidence and security audit receipts remain the authoritative effect oracles.

DIAGNOSTICS AND EVIDENCE CALLERS

Read-only evidence is still privileged authority

EVIDENCE PRINCIPAL F-42

Capability	Allowed scope	Denied
health read	safe public/internal dispositions	dependency secrets/topology
owner snapshot	run-scoped bounded counts/digests	business payload/foreign owner
broker observation	named queue/route state	publish/configure/admin
telemetry query	RunId/CasId bounded search	unrestricted export
artifact read	one evidence dossier	other runs/material roots
failure actuator	Appendix E isolated target action	business/repair mutation

CLOSED ACTUATOR

Scenario input resolves through an allowlisted target and action catalogue. It never becomes an arbitrary process, SQL, broker-management or shell command.

ENDPOINT SECURITY INVENTORY

Every route declares authentication and policy metadata

INVENTORY CLASSES F-43

Surface	Required metadata
BFF login/callback	anonymous protocol entry; exact OIDC handler
BFF logout	session + anti-forgery + revoke-before-clear
BFF session summary	session; safe value-free response
public/owner API	human/workload mode + capability + audience + scope
internal Reporting API	known Query BFF workload + internal capability
health	explicit safe anonymous/internal profile
evidence	run-scoped evidence principal; no material export

DISCOVERY GATE

An unclassified endpoint, unexpected AllowAnonymous marker, protected route without a policy or superseded route blocks startup/build evidence.

ADVERSARIAL IDENTITY MATRIX

One coordinate changes at a time

NEGATIVE CASES F-44

Mutation	Expected boundary result	Effect oracle
missing/expired session	401	zero owner calls
valid identity, missing capability	403	zero Domain/DB/outbox
wrong audience	401/403 at destination	zero owner effect
cross-owner scope	safe 403/404	zero disclosure/write
user token without workload mTLS	challenge/deny	zero owner call
known mTLS caller, wrong capability	403	zero Domain effect
unknown/ambiguous URI SAN	authentication reject	zero application call
stale/unknown policy revision	deny/unavailable	zero write

TRANSPORT, STORAGE AND MATERIAL MATRIX

Positive connectivity cannot prove least privilege

ADVERSARIAL CASES F-45

Mutation	Expected result	Zero-effect proof
foreign broker route	broker ACL denial; consumer also rejects	no foreign fact/effect
schema-invalid/poison body	terminal quarantine	no Domain invocation
same MessageId/different digest	identity conflict	first effect unchanged
foreign DB credential probe	connection/grant denial	no foreign DML
runtime DDL attempt	grant denial	schema unchanged
malformed/over-permitted material	startup/readiness reject	no empty/default activation
wrong SAN/EKU/time/trust	fresh TLS rejection	zero destination action
retired secret/cert	fresh use rejected	new material remains active

ZERO-EFFECT ORACLE

A denial proves both decision and absence of consequence

PROTECTED-EFFECT PROBES F-46

Surface	Required zero effect
HTTP owner command	no handler call, transaction, owner row or outbox row
owner query denial	no resource body or existence disclosure
broker publish denial	no accepted publish/confirm/consumer effect
consumer admission denial	no Inbox effect/Domain call; terminal evidence only
database grant denial	no DML/DDI/history change
material candidate rejection	active snapshot identity unchanged
evidence/repair path denial	no owner mutation or expanded artifact read

BOUNDED SECURITY EVENT

One safe denial/security event is permitted. Its existence cannot substitute for the protected-effect probes.

SECURITY EVENT CONTRACT

Accountability remains reconstructable and minimally sensitive

SECURITYDECISIONOBSERVEDV1

```
public sealed record SecurityDecisionObservedV1(  
    Guid RunId,  
    string BoundaryId,  
    string PrincipalClass,  
    string SafePrincipalReference,  
    string ActionClass,  
    string ResourceClass,  
    string Outcome,  
    string ReasonCode,  
    string PolicyIdentity,  
    string CorrelationReference,  
    string ZeroEffectReference,  
    DateTimeOffset ObservedUtc);
```

EXCLUDED FIELDS

Bearer tokens, cookies, raw claims, names, email addresses, secret references, resource bodies and unrestricted exception text are structurally absent.

LEAK AND CANARY EVIDENCE

Protected values must be absent from every outward artifact

SCAN CORPUS F-47

Target	Required result
source/config/fixtures	no committed secret/private key/reusable test material
container layers/image	no material in any layer; runtime paths absent
stdout/structured logs	zero canary/value/token patterns
traces/metrics	zero canary and forbidden attribute names/cardinality
Problem Details/health	zero topology/material/identity detail
evidence dossier	value-free typed artifacts only
dumps/support bundles	not collected by default; protected workflow if needed

NO SECRET FINGERPRINT

Hashing a token, password or low-entropy secret is not redaction. The dossier contains no secret-derived value.

SECURITY EVIDENCE SCHEMA

Claims and adversarial cases remain machine-checkable

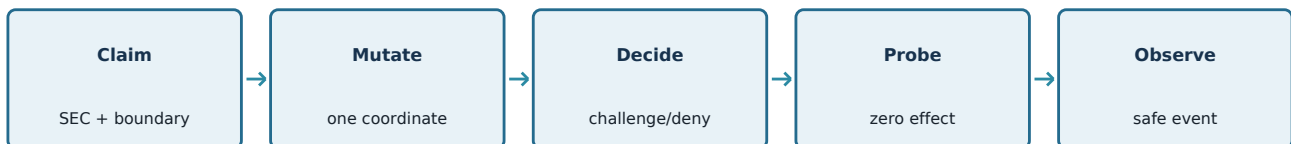
SECURITYCLAIM AND ADVERSARIALCASE

```
public sealed record SecurityClaim(  
    string ClaimId,  
    string DecisionRef,  
    string BoundaryId,  
    string ProtectedEffect,  
    IReadOnlySet<TestLayer> RequiredLayers,  
    string Phase0Limit);  
  
public sealed record AdversarialCase(  
    string CaseId,  
    string ClaimId,  
    string Mutation,  
    ExpectedDecision Decision,  
    string EffectOracle,  
    string EventClass);  
  
public enum TestLayer { PolicyUnit, Host, Component, Deployment, Fault }  
public enum ExpectedDecision { Challenge, Deny, Reject, Unavailable }
```

SAME-RUN EVIDENCE DOSSIER

A security claim needs six joins

EVIDENCE PATH F-48



DOSSIER MEMBERS F-49

Artifact	Purpose
campaign.json	RunId, build, time, toolchain, environment
catalogue.json	claims, cases, boundaries and digest
endpoint-policy.json	route/method/auth/policy inventory
principal-grants.json	accepted/forbidden workload, broker and DB rights
results/	typed case status, effect and event references
leak-scan.json	zero forbidden pattern result
manifest.json	path/length/SHA-256/media type; same run

DOSSIER FAIL-CLOSED VALIDATOR

Zero discovery, stale artifacts and mixed runs fail

SECURITYDOSSIERVERIFIER

```
public static SecurityVerdict Verify(SecurityDossier d)
{
    Require(d.RequiredClaimCount > 0, "SEC_ZERO_CLAIMS");
    Require(d.DiscoveredCaseIds.SetEquals(d.RequiredCaseIds),
        "SEC_DISCOVERY_DRIFT");
    Require(d.Results.All(x => x.RunId == d.RunId), "SEC_MIXED_RUN");
    Require(d.Results.All(x => x.Status == SecurityCaseStatus.Pass),
        "SEC_REQUIRED_CASE_NOT_PASS");
    Require(d.Results.All(x => x.ZeroEffectRef is not null),
        "SEC_EFFECT_PROOF_MISSING");
    Require(d.Artifacts.All(x => x.IsFresh && x.HashVerified),
        "SEC_ARTIFACT_INVALID");
    Require(d.LeakScan.ForbiddenMatches == 0, "SEC_LEAK_FOUND");
    return SecurityVerdict.BoundedPhase0Evidence;
}
```

STATUS VOCABULARY

PASS, FAIL, FLAKY, NOT_RUN and INCONCLUSIVE are distinct. Only PASS satisfies a required case; unknown status is invalid.

EVIDENCE EXPIRY

Relevant change invalidates affected security claims

INVALIDATION MAP F-50

Change	Minimum rerun
service build/endpoint	affected route, policy and zero-effect cases
authorization policy	mapping/resource matrix
certificate/trust profile	fresh handshake and rotation
broker user/permission/topology	origin, ACL and consumer admission
database grant/migration	forbidden probes and schema authority
secret/material/config identity	validation, startup, leak and rotation
evidence catalogue/verifier	full impacted claim set
unknown/unclassified change	full security campaign

NO TIMELESS PROOF

A previous PASS does not survive a changed build, policy, trust bundle, permission set, environment, catalogue or expired material without classified impact evidence.

THREAT LEDGER COVERAGE

Twenty-four named threats remain connected to boundaries and tests

THREAT GROUPS F-51

Group	Representative threats	Primary boundaries
operator/edge	stolen session, scope tamper, hidden action, confused deputy	TB-01/02
workload/messaging	user-as-service, foreign publish, overbroad broker, poison replay	TB-02/03/04
owner/schema	cross-owner DML, runtime DDL, wrong migration target	TB-05/06
telemetry/diagnostics	leak, spoof, cardinality, telemetry-as-audit	TB-07/08
admin/repair	stolen admin, self approval, owner-port misuse	TB-09/10 inactive
supply/material	artifact substitution, secret reuse, emergency path	TB-11/12

THREAT STATUS

Chapter 43 defines 24 named THR-043 entries. THR-043-007..010 have no defining ledger row and are therefore TOMBSTONED_IDENTIFIER coordinates, not invented threats. Mitigation cannot be claimed until every named entry has required cases and same-run effect evidence.

NAMED THREAT TRACEABILITY I

Operator, workload, messaging and owner-data entries stay exact

THR-043 NAMED LEDGER F-51A

ID	Threat	Boundary	Requirement
001	stolen/forged user context	TB-01	SR-001,007
002	client scope tamper	TB-01	SR-002,008
003	hidden-action bypass	TB-01	SR-006
004	BFF confused deputy	TB-02	SR-006,009
005	user token used as service	TB-02	SR-001,010
006	identity outage fails open	TB-01/02	SR-011
011	foreign event publication	TB-03	SR-006,012
012	broker over-privilege	TB-03/04	SR-006
013	unsafe poison replay	TB-04/09/10	SR-013
014	cross-owner database write	TB-05	SR-006,014
015	runtime receives DDL authority	TB-06	SR-015
022	message/body identity conflict	TB-04	SR-002,016

IDENTITY RULE

The discontinuous IDs preserve Chapter 43 identity. Renumbering them would silently break evidence, test and review references.

NAMED THREAT TRACEABILITY II

Telemetry, administration, supply and material entries stay exact

THR-043 NAMED LEDGER F-51B

ID	Threat	Boundary	Requirement
016	sensitive telemetry leak	TB-07	SR-004,017
017	telemetry tamper/spoof	TB-07	SR-002,018
018	diagnostic disclosure	TB-08	SR-004
023	cardinality exhaustion	TB-07	SR-005
024	collector lateral movement	TB-07	SR-006
025	telemetry mistaken for audit	TB-07	SR-003,018
019	stolen admin context	TB-09	SR-007,013
020	owner repair-port misuse	TB-10	SR-006,013
021	artifact/config substitution	TB-11/12	SR-002,019
026	self/duplicate approval	TB-09	SR-013
027	secret disclosure/reuse	TB-12	SR-020
028	emergency path unrecorded	TB-09/10	SR-003,013

RESERVED COORDINATES

IDs	Final disposition
THR-043-007..010	TOMBSTONED_IDENTIFIER - no defining Chapter 43 row; cannot be cited as a threat or silently reused

PLATFORM PROJECTION

Compose and Kubernetes preserve the same trust semantics

MECHANISM MAPPING F-52

Concern	Phase0Compose	Phase0Kubernetes translation
workload identity	mounted role certificate + direct mTLS	Secret ref/workload identity mapping; same URI SAN policy
network service	named Compose network/HTTPS	Service/DNS; network is not identity
DB/broker grants	per-role injected references	per-workload refs; same permission manifest
config/material	mounted references/profile digest	ConfigMap/Secret refs; same accepted identity
evidence	container/broker/DB/HTTP laboratory	pod/manifest events plus same owner probes

EDUCATIONAL LIMIT

Kubernetes manifests translate proved needs. They do not prove cluster segmentation, admission policy, network-policy maturity, secret-store custody or production operations.

RESIDUAL RISK REGISTER

A complete Phase-0 dossier may retain unresolved risks

REQUIRED RESIDUALS F-53

Residual	Why it remains	Future evidence decision
privileged broker substitution	no end-to-end event signature	authenticity design/campaign
manifest rewrite	local SHA-256 is not independent signature	retention/signing authority
synthetic/test issuer	does not prove production identity provider	real provider scoped evidence
single-host isolation	Compose boundaries share one host	platform isolation campaign
host compromise	process-readable material may be exposed	custody/hardening decision
organizational operations	no SOC/SIEM/incident team evidence	people/process/tool operation

RESIDUALS ARE OUTPUTS

A passing bounded campaign may coexist with explicit residual risk. Hiding the residual would make the security claim weaker, not stronger.

OUTSIDE PHASE-0

The dossier is not certification or an audit opinion

NON-CLAIMS F-54

Not claimed	What would be needed
penetration/red-team coverage	independent scoped engagement
independent audit/certification	criteria, auditor and opinion
production PKI/HSM custody	operational issuance/key evidence
zero-trust maturity	organization-wide identity/policy/operations
SIEM/SOC operation	people, process, tools and retained incidents
multi-region resilience	regional architecture and campaigns
safety/plant security	outside educational software demonstrator
vulnerability-free dependencies	ongoing supply-chain assurance

ARCHITECTURE AND DRIFT GATES

Forbidden authority is as important as granted authority

FAIL-FAST GATES F-55

Code	Rejected observation
SEC_BOUNDARY_DRIFT	unknown/missing/duplicate TB crossing
SEC_INACTIVE_PRINCIPAL	credential/route/queue for inactive candidate
SEC_OWNER_DB_CROSSING	foreign DB or BFF DbContext
SEC_RUNTIME_DDL	migration capability in long-running role
SEC_BROKER_OVERGRANT	foreign publish/consume/configure right
SEC_USER_AS_WORKLOAD	owner accepts human token without workload identity
SEC_RAW_CLAIM_FORWARD	token/cookie/claim bag crosses BFF
SEC_SECRET_EVIDENCE	secret or secret-derived value serialized
SEC_EVIDENCE_ACTUATOR_BROAD	general SQL/broker/shell admin path
SEC_ZERO_DISCOVERY	required adversarial suite not discovered

EVIDENCE STATUS BOARD

Specification completeness and security proof stay separate

APPENDIX F STATUS F-56

Obligation	Asset	Status
zones/crossings	9 zones + 12 trust cards	SPECIFIED
requirements/threats	21 SR + 24 named THR + 4 ID tombstones	SPECIFIED
human/workload identity	OIDC session + direct mTLS/EXTERNAL	SPECIFIED
capability/claims	final route map + minimized context	SPECIFIED
DB/broker/material	least privilege + rotation rules	SPECIFIED
telemetry/evidence	value-free allowlists and dossier	SPECIFIED
repository implementation	inventories, tests, manifests	PENDING
same-run security campaign	non-zero retained dossier	PENDING

HONEST BOUNDARY

No repository test run, real broker-permission export, database-grant probe, leak scan, certificate campaign or same-run dossier was executed in this manuscript environment. Security runtime evidence remains PENDING.

APPENDIX F CHECKPOINT

Trust can now be reviewed without hidden global authority

COMPLETION LEDGER F-57

Question	Required answer
Which principal crossed?	one known human/workload/admin class
For which audience/action?	exact route, capability and destination
What human context moved?	one minimized intent/scope/reference
Who decided?	receiving owner policy and current resource
Which capability was absent?	forbidden-grant inventory and probe
What effect was prevented?	zero Domain/DB/outbox/broker oracle
What evidence is safe?	same-run value-free event/artifact refs
What remains unproved?	explicit residual and PENDING runtime status

FINAL APPENDIX F RULE

Trust is local, typed, minimized, expiring and evidenced. No upstream success, credential class or operational convenience may widen it.

HANDOFF TO APPENDIX G

The next appendix must make reproduction exact

Appendix G - Commands, Scripts, and Reproduction Guide will turn the book's specified build, database, broker, Compose, evidence and verification steps into one ordered command authority. It must distinguish safe read-only inspection, ordinary build/test execution and destructive isolated campaigns; pin working directories, inputs, expected outputs, exit codes, artifact identities and cleanup.

CARRY-FORWARD OBLIGATIONS F-58

From Appendix F	Appendix G must preserve
closed principals and targets	commands cannot accept arbitrary service/DB/broker authority
secret references only	no credential appears on command line/output
three owners/seven roles/three facts	reproduction cannot activate candidates
migration separation	DDL commands run only under one-shot migrator identity
evidence isolation	destructive commands require isolated evidence environment
same-run dossiers	commands seal manifests and retain failures

APPENDIX F LAST WORD

A trusted path ends at a local decision

A secure distributed design does not ask whether a caller is trusted in general. It asks which identity was accepted, for which destination and action, under which current policy, what effect was prevented or admitted, and what evidence can still disagree.

FINAL STATUS

SECURITY AND TRUST NOTES COMPLETE | 12 CROSSINGS FIXED | 21 REQUIREMENTS + 24 NAMED THREATS PRESERVED | 4 ID TOMBSTONES EXPLICIT | FINAL CORE PRINCIPALS RECONCILED | RUNTIME SECURITY EVIDENCE PENDING

APPENDIX G

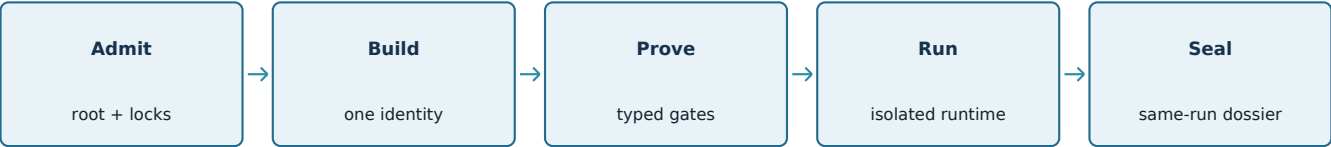
Commands, Scripts, and Reproduction Guide

The ordered Phase-0 command authority for a clean checkout: repository admission, locked restore, build, tests, Compose projection, service-owned migrations, deterministic seed, runtime verification, final scenario, isolated failure campaigns, evidence sealing, cleanup and honest claim language.

APPENDIX DECISION

A command is part of the specification until the named repository path exists and a retained receipt binds its inputs, identity, exit code, outputs and cleanup. This appendix fixes syntax and order; it does not manufacture execution evidence.

REPRODUCTION CHAIN



CONTROL TOTALS G-1

Coordinate	Final value
public wrappers	2 equivalent surfaces: nexus.sh / nexus.ps1
safety classes	5 closed classes
runtime roles / migrators / facts	7 / 3 / 3
database owners	3
failure core	6 mandatory scenarios
command/runtime evidence	SPECIFIED / PENDING

PURPOSE AND AUTHORITY**One guide closes many chapter-local command examples**

Chapters 55-60 introduced launcher, CI, migration, Kubernetes and final-demonstrator commands while the final Appendices fixed repository, runtime, contract, configuration, failure and trust inventories. Appendix G reconciles them into one public grammar rooted at artifacts/from-services-to-runtime/.

AUTHORITY ORDER G-2

Authority	Owns	Appendix G use
Appendix A	final paths and output tree	working directory and canonical entry points
Appendices B-C	roles, owners, HTTP/events	closed command targets
Appendix D	configuration identities	profile and material admission
Appendix E	failure/recovery catalogue	destructive campaign plan
Appendix F	principals and trust boundaries	credential and evidence containment
Chapters 55-60	mechanisms and proposed assets	underlying implementation obligations

CONFLICT RULE

When a chapter-local path conflicts with the later Appendix A map, Appendix A wins. Appendix G records the superseded spelling instead of leaving two apparent authorities.

SPECIFICATION VERSUS EXECUTION**Exact syntax is not a transcript****CLAIM VOCABULARY G-3**

State	Permitted sentence	Evidence requirement
SPECIFIED	the command must...	normative grammar and postcondition
IMPLEMENTED	the repository contains...	path + source identity
EXECUTED	the process ran...	same-run process receipt
DEMONSTRATED	the recorded run showed...	admitted evidence for one profile
PENDING	execution has not been retained	Missing collection remains nonempty

CURRENT STATUS

No clean-checkout run, provider campaign, Compose runtime, broker/database permission probe, security campaign or final reproduction dossier was executed while producing this appendix. Every runtime result remains PENDING.

CANONICAL PATH RECONCILIATION**The final repository map replaces provisional spellings****PATH AUTHORITY G-4**

Canonical Appendix A path	Superseded chapter example	Disposition
Nexus1.Runtime.sln	NEXUS1.sln	TOMBSTONED_PATH
deploy/compose/compose.yaml	deploy/compose.yaml	TOMBSTONED_PATH
deploy/compose/compose.override.development.yaml	deploy/compose.local.yaml	TOMBSTONED_PATH
deploy/compose/compose.evidence.yaml	deploy/compose.evidence.yaml	TOMBSTONED_PATH
src/Services/<Owner>/Nexus1.<Owner>.*	src/<Owner>/*	TOMBSTONED_TEMPLATE
eng/scripts/nexus.sh nexus.ps1	direct chapter-local scripts	PUBLIC SURFACE REPLACED

NO COMPATIBILITY ALIAS

The superseded paths are documentation tombstones, not required aliases. A repository gate must fail if commands or workflows still treat them as canonical.

REPOSITORY ROOT

Every command begins from one resolved companion root

CANONICAL CHECKOUT COORDINATE

```
upstream: https://github.com/gregory82gr/Nexus-1-phase-0
checkout: Nexus-1-phase-0/
book root: artifacts/from-services-to-runtime/
sentinels:
  Nexus1.Runtime.sln
  global.json
  Directory.Build.props
  Directory.Packages.props
  deploy/topology/runtime-inventory.yaml
  eng/scripts/nexus.sh
  eng/scripts/nexus.ps1
```

ROOT ADMISSION G-5

Check	Failure
real path is below resolved checkout	ROOT_OUTSIDE_CHECKOUT
all sentinels exist with admitted case	ROOT_SENTINEL_MISSING
no path traversal or symlink escape	ROOT_ESCAPE
source revision is exactly 40 lowercase hex	SOURCE_REVISION_INVALID
dirty state classified by profile	SOURCE_STATE_NOT_ADMITTED

PUBLIC ENTRY POINTS

Bash and PowerShell expose the same typed grammar

LINUX/MACOS AND CI

```
cd artifacts/from-services-to-runtime
./eng/scripts/nexus.sh doctor --profile LocalEvidence
./eng/scripts/nexus.sh reproduce \
  --profile LocalEvidence \
  --scenario final-phase0-v1 \
  --seed phase0-v1
```

WINDOWS POWERSHELL

```
Set-Location artifacts/from-services-to-runtime
./eng/scripts/nexus.ps1 doctor --profile LocalEvidence
./eng/scripts/nexus.ps1 reproduce `
  --profile LocalEvidence `
  --scenario final-phase0-v1 `
  --seed phase0-v1
```

WRAPPER RULE

The wrappers only locate the pinned .NET orchestration projects and pass argument tokens. They do not contain a second policy, invoke eval, concatenate a shell command or reinterpret exit codes.

CLOSED COMMAND GRAMMAR

Verbs and targets are enumerated, not free-form

PUBLIC VERBS G-6

Verb	Mutation	Declared postcondition
doctor	none	host/root/toolchain findings retained
plan	artifact directory only	immutable admitted plan or explicit rejection
reproduce	build/provider/runtime resources	sealed dossier or retained failure
status	none	actual inventory compared with one run
verify	derived summary only	manifest admitted or rejected
down	stop/remove containers	named volumes retained
reset	isolated synthetic resources	resolved target removed under guard

NO ESCAPE HATCH

There is no public exec, sql, rabbitmqadmin, shell, arbitrary project, arbitrary database, arbitrary route or arbitrary path verb.

SAFETY CLASSES

Every command declares how much authority it needs

COMMANDSAFETYCLASS G-7

Class	Examples	Allowed effects
INSPECT	doctor, status, inventory	read host/repository/runtime metadata
BUILD_LOCAL	restore, build, test, package-local	write run artifacts and isolated provider leases
RUNTIME_LOCAL	compose render/up/down, scenario	named local synthetic project
DESTRUCTIVE_EVIDENCE	fault campaign, guarded reset	isolated nexus1-evidence-<run> only
DELIVERY_PRIVILEGED	candidate push, migration, rollout	trusted CI/deployment identity only

FAIL CLOSED

A verb cannot infer or upgrade its safety class from credentials found on the host. Profile, target and required principal are part of the admitted plan.

PROFILE ADMISSION

A profile fixes environment, dirtiness and destructive authority

PROFILES G-8

Profile	Source state	Destructive faults	Candidate authority
LocalDeveloper	dirty allowed + recorded	forbidden	none
LocalEvidence	clean required	isolated project only	none
EphemeralCI	clean exact revision	isolated lease only	no registry promotion
CandidateCI	clean exact revision	policy-selected	digest push/attestation
KubernetesTranslation	admitted candidate	separate disposable lab	render/preflight by default

EVIDENCE RULE

Only LocalEvidence and EphemeralCI may run the Appendix E destructive local campaign. CandidateCI requires the delivery policy; Kubernetes translation cannot borrow local fault credentials.

CLOSED ARCHITECTURE INPUTS

Reproduction cannot activate a candidate service

FINAL RUNTIME CENSUS G-9

Category	Admitted set
owners	AlarmManagement, RootCause, Reporting
edge	Query BFF
roles	alarm-api, alarm-worker, rootcause-api, rootcause-worker, projection-api, projection-worker, query-bff
migrators	alarm-migrator, rootcause-migrator, reporting-migrator
facts	AlarmFloodDetected.v1, RootCauseCaseOpened.v1, RootCauseVerdictIssued.v1
inactive	Audit, Compliance, Process Manager, Repair Control, RL Advisory

INVENTORY FAILURE

Any extra role, database, migrator, event type, publisher, consumer, route or inactive-candidate credential rejects the plan before mutation.

RUN IDENTITY

One immutable identity joins every stage

REPRODUCTIONIDENTITYV1

```
public sealed record ReproductionIdentityV1(  
    string RunId,  
    string SourceRevision,  
    string SourceTreeSha256,  
    string ToolchainLockSha256,  
    string PackageLocksSha256,  
    string RuntimeInventorySha256,  
    string ComposeModelSha256,  
    string ImageLockSha256,  
    string ConfigurationSha256,  
    string MigrationSetSha256,  
    string SeedManifestSha256,  
    string ContractCorpusSha256,  
    string CommandCatalogueSha256);
```

JOIN RULE

A receipt with a different identity digest is foreign evidence even if its filename, branch, semantic version or human-readable scenario name matches.

RUN IDENTIFIER AND OUTPUT ROOT

The tool creates a safe RunId before consequential work

RUNID CONTRACT G-10

Coordinate	Rule
grammar	r-YYYYMMDDThhmmssZ-<12 lowercase hex>
creator	orchestrator only; user cannot inject separators
output	artifacts/evidence/<RunId>/
collision	fail; never merge or overwrite
permissions	owner read/write; evidence readers least privilege
finalization	manifest sealed once; raw artifacts immutable

NO BROAD PATH

The command resolves the output beneath artifacts/evidence, creates one new directory and refuses symlink escape, existing content, wildcard targets, workspace roots and home directories.

EVIDENCE DIRECTORY

Every plane has one named home

CANONICAL EVIDENCE TREE

```
artifacts/evidence/<RunId>/
  plan.json
  manifest.json
  identity/{source,build,packages,images,config,migrations}.json
  commands/{index,receipts,stdout,stderr}/
  build/{restore,compile,inventory}/
  tests/{unit,architecture,component,contract,security,e2e,failure,deployment}/
  runtime/{topology,health,lifecycle}/
  database/{principals,transactions,convergence}/
  broker/{topology,deliveries,settlements,quarantine}/
  http/{requests,responses,authorization}/
  telemetry/{logs,metrics,traces}/
  failures/{leases,disturbances,recovery}/
  summary/{claims,missing,verdict}.json
```

ARTIFACT RULE

Raw observations are append-only. Derived summaries name and hash their inputs; no summary may replace missing raw evidence.

IMMUTABLE RUN PLAN

The plan is admitted before restore, containers or providers

REPRODUCTIONPLANV1

```
public sealed record ReproductionPlanV1(
    string PlanId,
    ReproductionProfile Profile,
    string ScenarioId,
    string SeedId,
    IReadOnlyList<StageId> OrderedStages,
    IReadOnlySet<RoleId> Roles,
    IReadOnlySet<OwnerId> Owners,
    IReadOnlySet<FactType> Facts,
    TimeSpan OverallBudget,
    RetentionPolicy Retention,
    ReproductionIdentityV1 Identity,
    string Signature);
```

PLAN ADMISSION G-11

Input	Gate
scenario/seed/profile	closed catalogue lookup
roles/owners/facts	exact Appendix B-C set
stages	dependency-closed topological order
budgets	positive, bounded and stage-specific
identity	all digests recomputed before execution
signature	required only where profile policy declares it

COMMAND RECEIPT

Every process records intent and bounded result

COMMANDRECEIPTV1

```
public sealed record CommandReceiptV1(  
    string RunId,  
    string StageId,  
    int Sequence,  
    CommandSafetyClass SafetyClass,  
    string ProgramIdentity,  
    IReadOnlyList<string> RedactedArguments,  
    DateTimeOffset StartedUtc,  
    DateTimeOffset CompletedUtc,  
    int ChildExitCode,  
    string StdoutSha256,  
    string StderrSha256,  
    CleanupDisposition Cleanup,  
    StageVerdict Verdict,  
    string IdentityDigest);
```

MEANING

A child exit code is an observation. The stage passes only when output inventory, identity, required postconditions and cleanup receipts are also admitted.

PROCESS EXECUTION

Arguments stay tokens and output stays bounded

RUNNER CONSTRAINTS G-12

Concern	Required behavior
launch	UseShellExecute=false; ArgumentList tokenization
shell	no eval, command substitution from user input or concatenated command
stdout/stderr	concurrent bounded capture; truncation is explicit failure/diagnostic
timeout	linked cancellation with stage budget
redaction	allowlist fields and forbidden-pattern scan before persistence
working directory	resolved repository root or exact isolated lease
environment	closed allowlist; values classified before inheritance

NO SUCCESS LAUNDERING

An exception, timeout, cancellation, truncated output, missing receipt or cleanup uncertainty cannot be converted into a successful aggregate exit code.

EXIT-CODE CONTRACT

Stable families make automation and triage deterministic

PUBLIC EXIT CODES G-13

Code	Family	Meaning
0	ADMITTED	declared verb postcondition satisfied
2	USAGE	unknown verb/option or invalid closed value
10-19	PREFLIGHT	root, toolchain, source or capacity rejected
20-29	IDENTITY	lock, digest, plan or artifact mismatch
30-39	BUILD	restore/build/inventory failure
40-49	TEST	discovery/execution/oracle failure
50-59	RUNTIME	Compose/migration/readiness failure
60-69	SCENARIO	meaning/convergence/recovery failure
70-79	EVIDENCE	capture/redaction/join/seal failure
80-89	CLEANUP	bounded cleanup uncertain or failed
130	CANCELLED	external cancellation retained

SECRET AND MATERIAL BOUNDARY

Commands carry references, never credential values

MATERIAL RULES G-14

Surface	Allowed	Forbidden
CLI	reference ID or mounted path coordinate	secret/token/DSN/password/private key
environment	provider bootstrap reference	unclassified inherited variables
process list	non-sensitive IDs	bearer material
logs/receipts	revision, provider, permission class	value or secret-derived fingerprint
Compose/Kubernetes	secret reference name	stringData/value
evidence	material identity and validation state	contents or recoverable transform

LEAK GATE

Every retained stdout, stderr, manifest and configuration projection passes the Appendix F canary and forbidden-pattern scan. A match fails the run and restricts the contaminated artifact.

STAGE 00 - DOCTOR

Preflight fails before consequential mutation

CANONICAL COMMAND

```
./eng/scripts/nexus.sh doctor --profile LocalEvidence  
./eng/scripts/nexus.ps1 doctor --profile LocalEvidence
```

DOCTOR INVENTORY G-15

Check	Retained identity/failure
root sentinels and source	paths/digests; ROOT_* or SOURCE_*
SDK and package tools	global.json match; TOOLCHAIN_*
Docker Engine/Compose features	parsed versions/features; ENGINE_*
host architecture/capacity	value-free class; PLATFORM_*
ports	allowlisted ownership only; PORT_CONFLICT
material references	metadata and permissions; MATERIAL_*
clock	UTC/monotonic capability; CLOCK_UNSUITABLE

STAGE 01 - SOURCE IDENTITY

The source tree is classified before it is built

READ-ONLY OBSERVATIONS

```
git rev-parse --verify HEAD  
git status --porcelain=v1 --untracked-files=all  
git submodule status --recursive  
./eng/scripts/nexus.sh inventory --plane source --output <RunId>
```

SOURCE RULES G-16

Observation	Decision
clean exact 40-hex revision	eligible for evidence profiles
dirty tracked/untracked tree	LocalDeveloper label only
uninitialized/foreign submodule	reject
case-colliding path	reject
source changes after plan	cancel and retain SOURCE_CHANGED

STAGE 02 - TOOLCHAIN AND LOCKS

Versions are read from reviewed files, not copied into prose

LOCK INPUTS G-17

Input	Authority	Gate
.NET SDK	global.json	exact admitted feature band
packages	Directory.Packages.props + lock files	locked restore only
NuGet feeds	nuget.config	allowlisted sources
base images	eng/container/base-images.lock.json	digest references
Compose	runtime contract feature floor	feature admission
scanners	tool/image lock + policy	identified result

NO AMBIENT UPGRADE

Restore, build, image creation or scanning cannot update a lock while judging an unrelated source revision.

STAGE 03 - LOCKED RESTORE

One restore establishes the dependency graph

UNDERLYING RESTORE

```
dotnet restore Nexus1.Runtime.sln \  
  --locked-mode \  
  --configfile nuget.config \  
  -p:RestoreIgnoreFailedSources=false \  
  -p:RestoreNoCache=true
```

RESTORE POSTCONDITIONS G-18

Required	Failure
all expected projects restored	PROJECT_SET_CHANGED
no lock file changed	PACKAGE_LOCK_DRIFT
all feeds admitted and reachable	PACKAGE_SOURCE_UNTRUSTED/UNAVAILABLE
package graph receipt retained	PACKAGE_EVIDENCE_MISSING
later stages use --no-restore	PIPELINE_GRAPH_AMBIGUOUS

STAGE 04 - COMPILE

The solution builds once under the run identity

UNDERLYING BUILD

```
dotnet build Nexus1.Runtime.sln \  
  --configuration Release \  
  --no-restore \  
  -p:SourceRevisionId=<40-hex> \  
  -p:ContinuousIntegrationBuild=true \  
  -bl:artifacts/evidence/<RunId>/build/compile/nexus1.binlog
```

COMPILE RECEIPT G-19

Field	Admission
sourceRevisionId	equals plan revision
warnings	zero except admitted unexpired exceptions
assembly inventory	matches solution catalogue
publish tree digests	retained per executable role
binlog	diagnostic digest retained; not public evidence body

STAGE 05 - ARCHITECTURE INVENTORY

Build outputs must still match the closed architecture

DISCOVERED INVENTORY G-20

Plane	Expected
projects	Domain/Application/Infrastructure + seven roles + three migrators
dependencies	layer and owner boundaries
HTTP	six final Appendix C protected surfaces
events	three facts, exact publisher/consumer routes
stores	three owner databases and principal classes
deployment	seven long-running roles + three one-shot jobs
inactive	zero candidate endpoint/queue/credential

DRIFT GATE

A successfully compiled extra service is a failure, not progress. Inventory admission precedes provider, image and runtime work.

STAGE 06 - TEST CATALOGUE

Every family proves a different claim surface

REQUIRED FAMILIES G-21

Family	Primary oracle	Infrastructure
unit	Domain invariants/policies	none
application	ports, order and zero effects	fakes
architecture	forbidden references/inventory	assemblies
component	transactions/concurrency/grants	real provider lease
contract	HTTP/event golden and rejection corpus	serializer/host/broker
security	deny, redaction and least privilege	identity/provider profiles
runtime/e2e	ownership, delay, recovery and convergence	digest-pinned Compose

ZERO-DISCOVERY GATE

Exit code zero cannot hide an empty suite

TESTRUN ADMISSION

```
Require(run.ExitCode == 0, "TEST_PROCESS_FAILED");
Require(run.Discovered >= expected.MinimumDiscovered,
  "TEST_DISCOVERY_BELOW_MINIMUM");
Require(run.Executed == run.Discovered - run.ExplicitlySkipped,
  "TEST_EXECUTION_INCOMPLETE");
Require(run.Failed == 0, "TEST_FAILURE");
Require(run.ResultFiles.Count > 0, "TEST_EVIDENCE_MISSING");
Require(run.IdentityDigest == plan.IdentityDigest,
  "TEST_IDENTITY_MISMATCH");
```

FALSE-GREEN CASES G-22

Case	Failure
wrong project glob	TEST_PROJECT_SET_CHANGED
filter selects zero	TEST_FILTER_EMPTY
provider adapter omitted	TEST_DISCOVERY_BELOW_MINIMUM
stale TRX copied	TEST_IDENTITY_MISMATCH
shard missing	TEST_SHARD_INCOMPLETE

FAST TEST COMMAND

Cheap deterministic gates run first

CANONICAL WRAPPER

```
./eng/scripts/nexus.sh test \  
--run-id <RunId> \  
--families unit,application,architecture
```

REPRESENTATIVE UNDERLYING INVOCATION

```
dotnet test <catalogue-resolved-project> \  
--configuration Release --no-restore --no-build \  
--logger "trx;LogFileName=<family>.trx" \  
--results-directory artifacts/evidence/<RunId>/tests/<family>
```

CATALOGUE AUTHORITY

The wrapper resolves project paths from the admitted test catalogue. A user cannot replace the catalogue with a glob or silently omit a required family.

PROVIDER-BACKED COMPONENT STAGE

Real semantics run inside one isolated lease

PROVIDER LEASE G-23

Coordinate	Requirement
provider image	digest from admitted lock
database/schema	unique RunId-owned coordinate
migration	same owner artifact planned for runtime
seed	deterministic fixture digest
principal	test class with no foreign owner authority
cleanup	bounded drop/remove with receipt
evidence	provider/version/permission class; no DSN

NO SUBSTITUTION

In-memory and SQLite adapters cannot satisfy SQL Server transaction, collation, locking, execution-strategy, migration or grant claims.

CONTRACT STAGE

Exact bytes and rejection cases travel together

CONTRACT INPUTS G-24

Surface	Required corpus
HTTP	six final routes, methods, status/media types and negative bodies
events	three fact types, exact schema versions and canonical bytes
envelope	producer, route, MessageId, correlation/causation and digest
compatibility	declared reader/writer matrix
rejection	unknown field/type/version, invalid order and identity conflict
receipt	corpus digest, discovered cases and every verdict

NO REGENERATED GOLDEN

The test stage cannot rewrite golden files before comparing them. Corpus update is a separate reviewed source change.

SECURITY STAGE

Positive connectivity never replaces negative authority probes

SECURITY COMMAND OUTPUTS G-25

Asset	Required observation
endpoint-policy inventory	each route has exact principal/capability
principal grants	accepted and forbidden workload rights
broker ACL	three facts/routes; foreign publish/consume denied
database grants	owner DML, migrator DDL, all crossings denied
material manifest	references/identities only
leak scan	zero forbidden patterns/canaries
adversarial report	nonzero cases, same run/build and effect oracles

APPENDIX F CARRY-FORWARD

THR-043-007..010 remain tombstoned identifiers. A command or test cannot invent cases for them merely to make the count reach 28.

IMAGES AND CANDIDATE AUTHORITY

Local images are diagnostic; candidate images require trusted CI

IMAGE MODES G-26

Mode	Output	Authority
image-local	OCI archive + local receipt	never candidate or promotion evidence
CandidateCI	registry digest + SBOM + provenance	trusted job with admitted source
rebuild compare	two clean-builder digests	bounded repeatability result
promotion	existing admitted digest	no rebuild in deployment

IMAGE PROHIBITIONS G-27

Forbidden	Reason
image:latest	mutable subject
tag as rollout authority	tag can move
secret through ARG	material exposure
deployment-time docker build	artifact changes after proof

COMPOSE PROJECTION

The merged model is rendered before Docker mutates state

CANONICAL FILES

```

deploy/topology/runtime-inventory.yaml
deploy/compose/compose.yaml
deploy/compose/compose.override.development.yaml
deploy/compose/compose.evidence.yaml
deploy/compose/images.lock.json

```

READ-ONLY RENDER

```

./eng/scripts/nexus.sh compose-render \
  --run-id <RunId> \
  --profile LocalEvidence \
  --output artifacts/evidence/<RunId>/runtime/topology

```

PROJECTION GATE

The rendered service, image, port, network, volume, secret and profile inventories must equal runtime-inventory.yaml and the admitted plan before compose up.

COMPOSE PROJECT ISOLATION

Project names bind ownership and cleanup

PROJECT GRAMMAR G-28

Profile	Exact project pattern	Volumes
LocalDeveloper	nexus1-local	synthetic labels required
LocalEvidence	nexus1-evidence-<run-suffix>	new, isolated, synthetic
EphemeralCI	nexus1-ci-<run-suffix>	new, isolated, disposable

ISOLATION GATES G-29

Gate	Failure
project absent before destructive run	PROJECT_NOT_ISOLATED
every resource has RunId/project label	RESOURCE_OWNERSHIP_UNKNOWN
no external/shared network	NETWORK_SCOPE_BROAD
no pre-existing volume	VOLUME_PREEXISTING
fault actuator available only in evidence profile	ACTUATOR_SCOPE_BROAD

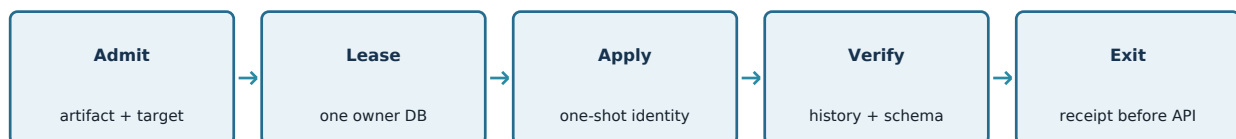
SERVICE-OWNED MIGRATION ORDER

Three one-shot identities run before long-lived roles

MIGRATION JOBS G-30

Owner	Job	Database	Authority
AlarmManagement	alarm-migrator	AlarmManagementDb	owned DDL/history
RootCause	rootcause-migrator	RootCauseDb	owned DDL/history
Reporting	reporting-migrator	ReportingDb	owned DDL/history

MIGRATION SEQUENCE G-31



NO UNIVERSAL MIGRATOR

The public command accepts only the three owner enum values and resolves the exact executable, credential reference and target internally. Runtime API/worker material never contains DDL authority.

MIGRATION COMMAND

DDL is explicit and never hidden in startup

CANONICAL WRAPPER

```
./eng/scripts/nexus.sh migrate \  
--run-id <RunId> \  
--owners AlarmManagement,RootCause,Reporting \  
--mode apply-admitted
```

MIGRATION RECEIPT G-32

Field	Required value
owner/database	exact admitted pair
principal class	one owner migrator
artifact digest	matches build identity
before/after history	ordered known migration set
schema digest	accepted owner schema
lease	acquired/released on same session
exit	job completed before long-running roles

DETERMINISTIC SEED

Seed identity is data input, not hidden setup

PHASE0-V1 ADMISSION G-33

Coordinate	Rule
catalogue	seed ID exists once
bytes	canonical digest matches plan
owner segments	each owner accepts only its records
order/randomness	explicit deterministic seed and ordering
reapply	idempotent or explicit conflict; never duplicate silently
environment	synthetic profiles only
receipt	counts and digests; no sensitive values

NO PRODUCTION SEED

The reproduction command cannot select or connect to an unclassified external dataset. phase0-v1 is synthetic educational input.

RUNTIME START

Compose wait is necessary; the runtime verifier finishes the claim

CANONICAL WRAPPER

```
./eng/scripts/nexus.sh up \  
--run-id <RunId> \  
--profile LocalEvidence \  
--seed phase0-v1 \  
--start-budget 180s
```

START POSTCONDITIONS G-34

Layer	Required observation
Compose	expected containers running/healthy
migrations	three receipts; no pending migration
identity	image/config/topology/seed digests equal plan
broker	topology + publisher/consumer controls
projection	control checkpoint reaches high-water
telemetry	fresh control trace when profile enabled
stability	three consecutive fresh passing snapshots

READINESS BUDGETS

No arbitrary sleep can satisfy a predicate

BUDGET MODEL G-35

Phase	Maximum	Failure evidence
Compose start	plan value, default 180s	last container state/log tail
migration	per owner plan budget	lease/history/schema snapshot
application readiness	dependency-specific	last typed readiness result
projection convergence	scenario-specific	lag/high-water snapshots
cleanup	separate bounded budget	remaining resource inventory

TIMEOUT SEMANTICS

A timeout means the required predicate was not observed within the admitted budget. It never means the component is unhealthy forever, and it never becomes a pass by sleeping longer outside the plan.

STATUS, DOWN AND RESET

Inspection, stopping and destruction are different verbs

LIFECYCLE COMMANDS

```
./eng/scripts/nexus.sh status --run-id <RunId>  
./eng/scripts/nexus.sh down --run-id <RunId> --budget 60s  
./eng/scripts/nexus.sh reset --run-id <RunId> \  
--confirm-synthetic-project nexus1-evidence-<run-suffix>
```

LIFECYCLE SEMANTICS G-36

Verb	Data	Guard
status	unchanged	read-only inventory
down	named volumes retained	exact project labels
reset	synthetic volumes removed	project + RunId + synthetic labels + explicit confirmation

NO BROAD DELETION

Reset resolves and prints the exact resource set before deletion. Empty variables, wildcards, label-only global queries, workspace roots and home directories are forbidden targets.

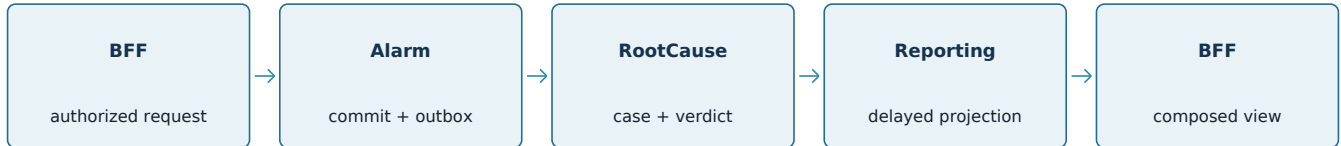
FINAL SCENARIO

One operator action traverses the admitted owner chain

SCENARIO COMMAND

```
./eng/scripts/nexus.sh demonstrate \  
--run-id <RunId> \  
--scenario final-phase0-v1 \  
--golden tests/Nexus.GoldenCorpus/final-v1.json
```

OBSERVED PATH G-37



MEANING BOUNDARY

The scenario observes each owner through its public or evidence-only contract. It never writes an owner database, calls a Domain handler directly or treats telemetry as business truth.

SCENARIO CHECKPOINTS

Success is a join of distinct durable observations

FINAL-PHASE0-V1 G-38

Checkpoint	Required observation
operator command	stable BFF request accepted once
Alarm commit	owner state + pending AlarmFloodDetected.v1
dispatch	confirmed route + sent outbox
RootCause consume	Inbox + case effect + settlement
RootCause facts	CaseOpened and VerdictIssued pending/sent
Reporting	Inbox + projection high-water
BFF query	authorized composed response with freshness
convergence	expected terminal digests and no hidden rewrite

HTTP EVIDENCE

Requests retain meaning without retaining credentials

HTTP RECEIPT G-39

Retain	Exclude
method + route template + endpoint ID	raw URL identifiers when unnecessary
capability + owner decision reference	token, cookie, claim bag
status + media type + contract digest	full sensitive body by default
correlation/causation references	globally correlatable external subject
zero-effect snapshot for denied calls	secret/configuration dump

AUTHORIZATION EVIDENCE

A denied request passes its case only when the expected status and the absence of Domain, database, outbox and broker effects are all observed.

BROKER EVIDENCE

Three routes and two queues form the closed transport surface

BROKER PROBES G-40

Probe	Required result
alarm-worker publish AlarmFloodDetected.v1	allowed exact route
rootcause-worker publish CaseOpened/VerdictIssued	allowed exact routes
rootcause-worker consume rootcause.alarm-events.v1	allowed read/ack
projection-worker consume reporting.integration-events.v1	allowed read/ack
foreign publish/consume/configure	denied
same Messageld/different digest	identity conflict, never duplicate pass

TRANSPORT LIMIT

TLS, EXTERNAL, publisher confirm and acknowledgement each prove a transport fact. None proves producer Domain authority or consumer business completion.

DATABASE EVIDENCE

Owner truth and derived truth are compared through typed probes

DATABASE PLANES G-41

Store	Allowed probe	Forbidden
AlarmManagementDb	owner state/outbox/transaction snapshot	foreign write or DDL under runtime
RootCauseDb	case/evidence/verdict + inbox/outbox	Alarm/Reporting mutation
ReportingDb	projection/inbox/high-water	source verdict manufacture

CONVERGENCE RECEIPT G-42

Coordinate	Rule
owner digest	canonical semantic projection, not raw page/file hash
projection digest	derived state with source high-water
lag	observed and bounded; never rewritten as freshness
principal	probe identity class retained without DSN

TELEMETRY EVIDENCE

Observability explains the path but never decides it

TELEMETRY CHECKS G-43

Signal	Required	Cannot prove
logs	allowlisted events + correlation	owner state
traces	HTTP/broker spans and causal links	unsampled absence
metrics	rates, errors, lag and cardinality	individual outcome
collector	freshness/backpressure/drop status	business success
leak scan	zero canary/forbidden attributes	all future configurations

EVIDENCE HIERARCHY

Owner databases and public contracts decide business outcome. Telemetry corroborates sequence, timing and failure; it is neither audit authority nor repair authority.

DESTRUCTIVE EVIDENCE CAMPAIGN

Faults require an isolated runtime and a signed lease

CAMPAIGN COMMAND

```
./eng/scripts/nexus.sh campaign \  
--run-id <RunId> \  
--catalogue core-recovery-v1 \  
--confirm-isolated-runtime nexus1-evidence-<run-suffix>
```

CAMPAIGN GATES G-44

Gate	Required
profile	LocalEvidence or EphemeralCI
project	new exact RunId-owned project
catalogue	closed Appendix E IDs and targets
actuators	allowlisted typed operations only
baseline	stable same-run snapshot before disturbance
lease	one active disturbance coordinate
cleanup	bounded and independently verified

CORE RECOVERY SUITE

Six disturbances are mandatory before the campaign can seal

CORE-RECOVERY-V1 G-45

ID	Disturbance	Deepest oracle
E-CORE-001	consumer crash after commit, before ack	one durable effect + duplicate completion
E-CORE-002	broker unavailable during outbox dispatch	pending intent survives and later dispatches
E-CORE-003	lost publisher confirmation	stable MessageId retry; no new fact
E-CORE-004	same MessageId with changed bytes	identity conflict + zero business effect
E-CORE-005	projection consumer delayed/restarted	visible lag then convergence
E-CORE-006	service restart with durable state	memory loss; owner truth preserved

NO RERUN LAUNDERING

Every attempt is retained. A later pass does not erase an earlier failure, mixed identity or cleanup uncertainty.

FAULT ACTUATOR BOUNDARY

A catalogue entry resolves one typed operation

ALLOWED ACTUATOR CLASSES G-46

Class	Example	Forbidden widening
process	pause/kill one labelled worker	arbitrary host PID
network	block one project-local dependency edge	host/global firewall
broker	pause project broker or bounded route path	management-wide mutation
resource	apply one declared CPU/memory pressure	unbounded host exhaustion
clock/schedule	test-controlled delay source	host clock mutation

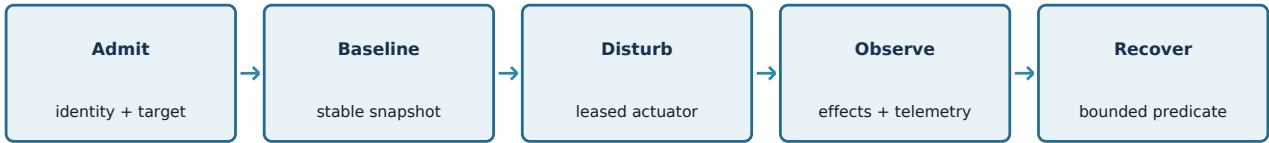
NO GENERAL ADMIN

The evidence profile exposes no SQL console, arbitrary broker command, shell, host process selector or unrestricted network tool.

FAILURE LIFECYCLE

Every scenario follows seven retained phases

SCENARIO LIFECYCLE G-47



REMAINING PHASES

Phase	Required result
Reconcile	owner/broker/projection state agrees with oracle
Cleanup	actuator released; isolated resources known
Seal	manifest retains every attempt and residual

ABORT SEMANTICS

Cancellation or harness failure enters ABORTED_UNHEALED until cleanup and recovery state are explicitly observed. It never skips directly to NotRun or Passed.

EVIDENCE CAPTURE

Capture is a named stage, not scattered shell redirection

COLLECTOR INPUTS G-48

Plane	Capture
identity	source/build/package/image/config/migration digests
commands	ordered redacted receipts and child outputs
tests	discovery, results, corpus and provider identities
runtime	topology, health, lifecycle and readiness snapshots
data/broker	principal classes, effects, deliveries and settlements
telemetry	allowlisted logs/metrics/traces + freshness
failures	leases, disturbance, recovery, reconciliation, cleanup

FAILURE RETENTION

Capture runs on pass, failure, timeout and cancellation. Missing capture is itself a retained evidence failure, not permission to discard the run.

EVIDENCE MANIFEST

One value-free manifest joins immutable artifacts

REPRODUCTIONEVIDENCEMANIFESTV1

```
public sealed record ReproductionEvidenceManifestV1(  
    string SchemaVersion,  
    string RunId,  
    string IdentityDigest,  
    string PlanSha256,  
    IReadOnlyList<StageReceiptReference> Stages,  
    IReadOnlyList<ArtifactReference> Artifacts,  
    IReadOnlyList<ClaimVerdict> Claims,  
    IReadOnlyList<string> Missing,  
    IReadOnlyList<ResidualRisk> Residuals,  
    ReproductionVerdict Verdict,  
    DateTimeOffset SealedUtc);
```

NO HAND-AUTHORED PASS

Verdict is derived from required stages, artifacts and claims. The manifest writer cannot accept Passed as user input.

EVIDENCE VERIFIER

Missing, stale or mixed evidence rejects the dossier

FAIL-CLOSED ADMISSION

```
Require(m.RunId == expected.RunId, "RUN_ID_MISMATCH");
Require(m.IdentityDigest == expected.IdentityDigest,
  "EVIDENCE_IDENTITY_MISMATCH");
Require(m.Stages.Select(x => x.StageId).ToHashSet()
  .SetEquals(expected.RequiredStages), "STAGE_SET_INCOMPLETE");
Require(m.Artifacts.All(HashMatches), "ARTIFACT_DIGEST_MISMATCH");
Require(m.Claims.Count >= expected.MinimumClaims, "ZERO_CLAIMS");
Require(m.Missing.Count == 0, "REQUIRED_EVIDENCE_MISSING");
Require(m.Claims.All(x => x.Verdict == ClaimVerdict.Proved),
  "CLAIM_NOT_PROVED");
```

HONEST OUTPUT

A failed verifier preserves the dossier and its Missing/failure collection. It never deletes inconvenient artifacts or rewrites PENDING as PASS.

SEAL COMMAND

Verification and sealing are separate from execution

CANONICAL VERIFICATION

```
./eng/scripts/nexus.sh verify \
  --run-id <RunId> \
  --policy docs/evidence/reproduction-policy.v1.json

./eng/scripts/nexus.sh seal \
  --run-id <RunId> \
  --require-verdict ADMITTED
```

SEAL GATES G-49

Gate	Failure
all required planes present	REQUIRED_EVIDENCE_MISSING
all digests recompute	ARTIFACT_DIGEST_MISMATCH
same RunId/identity	CROSS_RUN_JOIN
zero secret/canary matches	EVIDENCE_LEAK
cleanup disposition known	CLEANUP_UNCERTAIN
manifest absent before seal	MANIFEST_ALREADY_SEALED

REDACTION AND LEAK RESPONSE

Redaction reduces exposure; it cannot certify absence alone

LEAK WORKFLOW G-50

Step	Action
1 detect	stop sealing and restrict affected artifact
2 classify	identify material class without repeating value
3 revoke/rotate	follow provider-specific material lifecycle
4 purge	remove exposed copy under retention policy; retain incident reference
5 rerun	new RunId/identity after remediation
6 record	failed run remains visible with safe incident metadata

NO HASH ESCAPE

A raw secret hash or stable fingerprint may still enable correlation or attack. Secret-derived values are not safe evidence merely because they are hashed.

RESUME AND REUSE

Only identity-bound non-destructive stages may be reused

REUSE MATRIX G-51

Stage	Reuse rule
doctor	rerun; host state may change
source/locks	reuse only exact recomputed digest
restore/build	cache allowed; receipt identity must match
tests	rerun when subject/policy/provider/corpus changed
Compose/migration/scenario	never copy receipt; observe current runtime
destructive failure	never resume after disturbance; new attempt/lease
sealed manifest	immutable; no append or verdict rewrite

RESUME COMMAND

resume --run-id is admitted only for a failed, unsealed run whose identity still matches and whose cleanup state is known. It creates new attempt receipts; it does not overwrite earlier ones.

CLEANUP TRUTH

Cleanup has its own result and cannot hide uncertainty

CLEANUPDISPOSITION G-52

Value	Meaning	Aggregate effect
NOT_REQUIRED	read-only stage	neutral
COMPLETED	resolved resources reached expected state	eligible
PARTIAL	some owned resources remain	run fails
UNKNOWN	inventory/probe unavailable	run fails; manual isolation
FORBIDDEN	cleanup target not safely owned	stop without mutation

DATA PRESERVATION

Ordinary down retains named volumes. Only guarded reset removes synthetic volumes. A failed run is not permission to destroy evidence or unknown data.

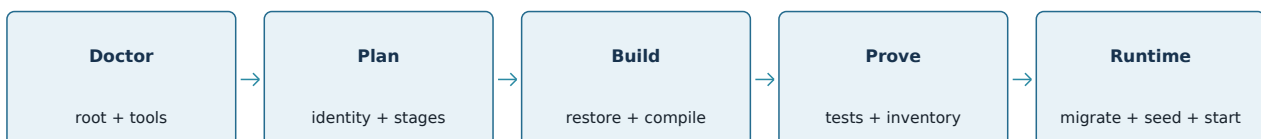
ONE-COMMAND REPRODUCTION

The public happy path is one bounded invocation

LOCAL EVIDENCE RUN

```
./eng/scripts/nexus.sh reproduce \  
--profile LocalEvidence \  
--scenario final-phase0-v1 \  
--seed phase0-v1 \  
--failure-catalogue core-recovery-v1 \  
--retention retain-always
```

INTERNAL ORDER G-53



SECOND HALF

The orchestrator then runs the final scenario, isolated recovery campaign, capture, verification, bounded down and seal. Any failed stage stops forward mutation but still enters capture and cleanup.

STAGE-BY-STAGE REPRODUCTION

Experts may expose phases without changing policy

ORDERED COMMANDS

```
nexus doctor      --profile LocalEvidence
nexus plan        --profile LocalEvidence --scenario final-phase0-v1 --seed phase0-v1
nexus restore     --run-id <RunId>
nexus build       --run-id <RunId>
nexus inventory   --run-id <RunId>
nexus test        --run-id <RunId> --families required
nexus compose-render --run-id <RunId>
nexus migrate     --run-id <RunId> --owners required
nexus up          --run-id <RunId>
nexus demonstrate --run-id <RunId> --scenario final-phase0-v1
nexus campaign    --run-id <RunId> --catalogue core-recovery-v1
nexus capture     --run-id <RunId>
nexus down        --run-id <RunId>
nexus verify      --run-id <RunId>
nexus seal        --run-id <RunId>
```

SAME ENGINE

Here nexus denotes either public wrapper. Each phase reads the immutable plan and refuses omitted dependencies, reordered stages or a different identity.

CLEAN-ROOM COMPARISON

Two independent runs test reproducibility without demanding equal noise

COMPARISON G-54

Must equal	May differ
source/toolchain/package/contract digests	RunId and wall-clock timestamps
runtime inventory/Compose/image/config identities	container IDs and IP addresses
migration/seed/topology identities	trace/span IDs
owner terminal semantic digests	elapsed milliseconds within budget
projection terminal digest/high-water relation	scheduler/interleaving details

QUALIFIED RESULT

Equality of two admitted runs is evidence for those inputs and profiles. It is not a timeless guarantee that every future host, dependency or rebuild will be identical.

CI MAPPING

Local and CI use one command grammar with different authority

PIPELINE MAPPING G-55

Gate	Local	CandidateCI
doctor/identity	developer/evidence profile	clean trusted runner
restore/build/test	same policies; local labels	candidate receipts
image	local OCI only	push + SBOM/provenance
runtime	nexus1-local/evidence	isolated ephemeral project
promotion	forbidden	admitted digest only
attestation	draft	platform-published

NO CI MAGIC

A hosted runner, green workflow badge or trusted branch does not replace command receipts, nonzero discovery, provider identities, runtime observations or dossier verification.

KUBERNETES TRANSLATION

The proved runtime is rendered; Kubernetes is not a new architecture

CANONICAL PROJECTION COMMANDS

```
./eng/scripts/nexus.sh kubernetes-render \  
  --run-id <RunId> \  
  --candidate artifacts/candidate/manifest.json \  
  --profile deploy/kubernetes/profile/kubernetes-capabilities.json  
  
./eng/scripts/nexus.sh kubernetes-verify --run-id <RunId>
```

DEFAULT SCOPE G-56

Action	Default
render	allowed from admitted inputs
schema/policy verify	required
server dry-run/diff	separate target-cluster authority
apply/rollout	not part of ordinary local reproduction
rollback	Chapter 58 decision required; never blind rollout undo

WINDOWS EQUIVALENCE

PowerShell changes syntax, not semantics

CROSS-PLATFORM INVARIANTS G-57

Coordinate	Must agree
verb/options	same closed grammar and defaults
path identity	repository-relative slash-normalized manifest paths
argument handling	token arrays; no Invoke-Expression
exit codes	same public family
text encoding	UTF-8 without command-dependent mutation
line endings	source policy; canonical evidence normalization
digests	defined over canonical bytes/tree rules

WRAPPER CONFORMANCE

A shared golden command corpus feeds both wrappers. Any difference in resolved plan, arguments, safety class, output paths or exit code is a deployment-test failure.

TROUBLESHOOTING ORDER

Triage starts at the earliest failed contract

FAILURE ROUTING G-58

Failure family	Inspect first	Do not do
PREFLIGHT	doctor findings/root/tool versions	run containers anyway
IDENTITY	plan and recomputed digest	copy older receipt
BUILD	restore/compile receipt + binlog	enable implicit restore
TEST	discovery/shard/provider/corpus	remove filter/family silently
RUNTIME	rendered model/migration/readiness	sleep and retry blindly
SCENARIO	deepest owner/broker/projection oracle	treat green health as success
EVIDENCE	Missing/digest/redaction/join	hand-edit manifest
CLEANUP	actual owned resource inventory	broad delete

ANTI-PATTERN CATALOGUE

Convenient commands can destroy the evidence model

FORBIDDEN SHORTCUTS G-59

Shortcut	Why rejected
docker compose up from memory	unknown files/profiles/locks
dotnet test without inventory	zero discovery can pass
Database.Migrate in API startup	runtime receives DDL
rabbitmqadmin with broad credential	management plane escapes role policy
curl with token on CLI	material exposed in history/process list
kubectrl apply before admitted render/dry-run	target mutation precedes policy
rerun until green	failed attempts disappear
copy yesterday's artifacts	cross-run identity
edit manifest verdict	summary manufactures proof

MANUSCRIPT ANTI-HALLUCINATION GATE

Expected output is a schema, never a fabricated transcript

OUTPUT LANGUAGE G-60

May publish	Must not publish
field names, types, allowed states	invented IDs, counts, timings or hashes
example labelled ILLUSTRATIVE	example styled as captured output
PENDING Missing collection	green manifest without retained artifacts
command postcondition	claim that the repository already implements it
failure-code semantics	claim that an unexecuted failure occurred

PUBLICATION RULE

The final edition may replace PENDING only with a dossier produced by these commands and admitted by the verifier. Prose cannot promote itself.

EXPECTED ARTIFACT SCHEMAS

A reader knows what to find without being shown fake values

MINIMUM FILES G-61

Path	Required fields
plan.json	RunId/profile/stages/identity/signature state
identity/*.json	input class, source reference, digest, schema
commands/index.json	ordered receipt refs and verdicts
tests/*/receipt.json	discovered/executed/failed/identity/artifacts
runtime/topology/*.json	expected/actual/drift
failures/*/receipt.json	baseline/disturbance/recovery/cleanup
summary/missing.json	all unsatisfied required coordinates
manifest.json	sealed joins, claims, residuals and verdict

REPRODUCTION ACCEPTANCE

A complete run satisfies more than process success

ACCEPTANCE CHECKLIST G-62

Question	Required answer
Was the root exact?	canonical path and sentinels admitted
Were inputs immutable?	one ReproductionIdentity across all stages
Did required tests run?	nonzero complete catalogue
Was the runtime exact?	roles/images/config/migrations/seed/topology equal plan
Did business meaning converge?	owner/broker/projection oracles agree
Did failures recover?	six core scenarios with cleanup
Is evidence safe?	same-run, value-free, digest-valid
What remains?	explicit Missing/residuals; no inflated claim

IMPLEMENTATION ASSETS

The repository must provide one coherent command product

REQUIRED SOURCE ASSETS G-63

Asset	Responsibility	Status
eng/scripts/nexus.sh	POSIX public wrapper	PENDING
eng/scripts/nexus.ps1	PowerShell public wrapper	PENDING
eng/Nexus.Build	restore/build/test/package orchestration	PENDING
eng/Nexus.Inventory	closed architecture discovery	PENDING
eng/Nexus.Contracts	corpus and compatibility gates	PENDING
eng/Nexus.Deployment	Compose/Kubernetes render/compare	PENDING
eng/Nexus.Evidence	capture/join/redact/verify/seal	PENDING
docs/evidence/*.json	schemas, catalogue and policies	PENDING

NO IMPLEMENTATION CLAIM

These exact paths are the target contract inherited from Appendix A. Their presence, compilation and execution have not been established in this manuscript environment.

APPENDIX G STATUS BOARD

Specification completeness and execution remain separate

STATUS G-64

Obligation	Asset	State
root/path reconciliation	canonical + tombstoned path ledger	SPECIFIED
command grammar	verbs, profiles, safety and exit codes	SPECIFIED
build/test sequence	locked ordered stage contracts	SPECIFIED
runtime/migration	Compose, owners, seed and readiness	SPECIFIED
scenario/failure	final-phase0-v1 + core-recovery-v1	SPECIFIED
evidence/cleanup	manifest, verifier, redaction and leases	SPECIFIED
repository implementation	scripts/projects/policies	PENDING
clean-room execution	two complete retained dossiers	PENDING

HONEST BOUNDARY

The guide is complete as a command specification. It is not evidence that a clean checkout currently restores, builds, runs, recovers or reproduces.

APPENDIX G CHECKPOINT

Reproduction is now one reviewable engineering contract

COMPLETION LEDGER G-65

Coordinate	Final rule
where	canonical companion root and output tree
what	closed roles, stores, facts, scenarios and profiles
who	least-privilege runtime, migrator, evidence and delivery identities
how	ordered typed verbs with stable postconditions
failure	retained attempt, bounded recovery and cleanup
evidence	same-run immutable value-free dossier
claim	SPECIFIED/PENDING until admitted execution

FINAL APPENDIX G RULE

A reproducible command is not merely one that can be rerun. It is one whose identity, authority, inputs, effects, failure state, cleanup and evidence can all be independently challenged.

HANDOFF TO APPENDIX H

The final matrix must connect every technical obligation

Appendix H - Technical Material Traceability Matrix will join every chapter claim and back-matter authority to its proposed repository asset, executable check, retained artifact, claim state and residual limitation. It must expose missing implementation, duplicate authority, superseded paths and unsupported evidence rather than summarize them away.

CARRY-FORWARD OBLIGATIONS G-66

From Appendix G	Appendix H must preserve
canonical path ledger	one asset identity and explicit tombstones
closed runtime census	no candidate activation through traceability rows
command/stage IDs	exact check and output coordinates
SPECIFIED vs PENDING	no unexecuted asset upgraded to demonstrated
same-run evidence	artifact references bind RunId and identity
Missing/residuals	unproved obligations stay visible

APPENDIX G LAST WORD

A command earns authority through its postcondition

NEXUS-1 does not trust a command because it is familiar, short or green. It trusts only the bounded observation that the command was designed to produce, under the exact identity and authority recorded for that run.

FINAL STATUS

COMMAND AND REPRODUCTION GUIDE COMPLETE | CANONICAL PATHS RECONCILED | ORDERED COMMAND AUTHORITY FIXED | REPOSITORY IMPLEMENTATION + CLEAN-ROOM DOSSIERS PENDING

APPENDIX H

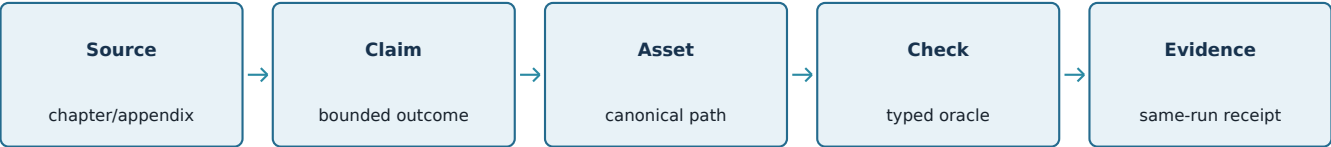
Technical Material Traceability Matrix

The final Phase-0 traceability authority connecting all sixty chapters and Appendices A-G to their material class, canonical asset, executable check, retained evidence, claim state, final architectural disposition and residual limitation.

APPENDIX DECISION

Traceability does not turn prose into implementation or a proposed check into evidence. Every row must preserve source identity, canonical path, check semantics, same-run artifact identity and an honest state; missing links remain visible.

TRACEABILITY CHAIN



CONTROL TOTALS H-1

Coordinate	Final value
Parts / chapters / epilogue	10 / 60 / 1
prior appendices	7
active owners / edge / roles / migrators	3 / 1 / 7 / 3
public facts / HTTP surfaces	3 / 6
chapter anchors in this matrix	60 / 60
implementation/runtime evidence	PENDING / PENDING

PURPOSE AND AUTHORITY

The matrix exposes gaps instead of averaging them away

The project description defines BASE, EXTEND, NEW and VERIFY obligations. The completed manuscript adds precise code, models, catalogues and commands, while the appendices freeze the final architecture. Appendix H joins these sources without claiming that proposed repository assets or runtime campaigns were executed.

AUTHORITY ORDER H-2

Source	Owns	Matrix responsibility
project description	chapter purpose and material flags	preserve scope
Chapters 1-60	mechanisms, listings and limitations	primary claim anchor
Appendices A-G	final catalogues and command authority	canonical asset/check identity
Epilogue	claim-state honesty	prevent evidence inflation
future repository dossier	implemented/executed evidence	may advance state after verification

NO WEIGHTED COMPLETENESS

Fifty-nine complete chapter anchors cannot hide one missing chapter. A green aggregate cannot hide one PENDING required asset or one unsupported claim.

TRACEABILITY ROW SCHEMA

One row is a typed, versioned relationship

TECHNICALMATERIALTRACEV1

```
public sealed record TechnicalMaterialTraceV1(  
    string TraceId,  
    SourceCoordinate Source,  
    IReadOnlySet<MaterialClass> Material,  
    string ClaimId,  
    string BoundedClaim,  
    CanonicalAssetReference Asset,  
    CheckReference Check,  
    EvidenceReference? Evidence,  
    ClaimState State,  
    ArchitecturalDisposition Disposition,  
    IReadOnlyList<string> Residuals,  
    string TraceIdentityDigest);
```

MINIMUM GRANULARITY

The chapter tables provide one primary anchor per chapter. A repository ledger may expand a chapter into many claim rows, but it may not merge unrelated claims into one pass/fail result.

TECHNICAL-MATERIAL LEGEND

Material flags describe work, not completion

PROJECT-DESCRIPTION CLASSES H-3

Code	Meaning	Traceability obligation
B	BASE - reuse established NEXUS-1 material	identify inherited source and changes
E	EXTEND - adapt an existing asset	bind old/new identity and compatibility
N	NEW - create code/schema/contract/diagram	canonical path + build/check
V	VERIFY - executable lab/current-source check	same-run evidence or PENDING

IMPORTANT DISTINCTION H-4

Material	Not equivalent to
N	asset exists or compiles
V	experiment executed
B/E	inheritance remains correct after distribution

CLAIM AND EVIDENCE STATES

State advances only through retained evidence

CLAIMSTATE H-5

State	Required support	Current use
SPECIFIED	bounded claim + asset/check contract	all chapter anchors
IMPLEMENTED	repository path under exact source identity	PENDING
EXECUTED	same-run process/test receipt	PENDING
DEMONSTRATED	admitted evidence proves one profile	PENDING
FALSIFIED	retained counterexample	none claimed here
SUPERSEDED	replacement plus tombstone	selected paths/decisions
INACTIVE	designed but no final-core authority	selected candidates

CURRENT BOUNDARY

Manuscript presence supports SPECIFIED. It does not support IMPLEMENTED, EXECUTED or DEMONSTRATED for the proposed companion repository.

INTEGRITY INVARIANTS

A trace is complete only when every link is challengeable

TRACE INVARIANTS H-6

Invariant	Failure
one stable TraceId and ClaimId	TRACE_ID_DUPLICATE/CHANGED
one canonical asset or explicit no-asset rationale	ASSET_UNBOUND
one executable check or explicit documentary check	CHECK_UNBOUND
evidence binds RunId and identity	CROSS_RUN_EVIDENCE
state is derivable from linked support	STATE_INFLATED
superseded identity has replacement/tombstone	TOMBSTONE_MISSING
inactive design has no runtime credential/route	INACTIVE_AUTHORITY_PRESENT
residuals remain attached	LIMITATION_DROPPED

PART I TRACEABILITY

Crossing Into Distribution - Chapters 1-6

CHAPTER ANCHORS H-7

Ch / material	Primary obligation	Canonical asset + check	State
1 B	inherit catalogues; name unproved decisions	docs/catalogs/handoff + catalogue diff	SPECIFIED/PENDING
2 E,N	preserve invariants, language, auth and determinism	docs/traceability/preservation + invariant gate	SPECIFIED/PENDING
3 E,N	select AlarmManagement -> RootCause first slice	candidate ledger + extraction decision gate	SPECIFIED/PENDING
4 N,V	canonical solution/project/build/test/deploy topology	Appendix A root + clean-checkout gate	SPECIFIED/PENDING
5 E,N	share primitives/contracts, never Domain/EF/repositories	dependency inventory + architecture tests	SPECIFIED/PENDING
6 N	service definition-of-done scorecard	docs/catalogs/service-scorecard + claim completeness	SPECIFIED/PENDING

FINAL DISPOSITION

The first slice admits three owners and one BFF in the final core; subscriber candidates are not activated by the original selection narrative.

FROM SERVICES TO RUNTIME

PART II TRACEABILITY

The Service Foundation - Chapters 7-12

CHAPTER ANCHORS H-8

Ch / material	Primary obligation	Canonical asset + check	State
7 N,V	repeatable Domain/Application/Infrastructure/host template	owner project template + generated-host smoke	SPECIFIED/PENDING
8 E,N,V	composition root and strict dependency direction	DI inventory + captive/duplicate/architecture gates	SPECIFIED/PENDING
9 E,N	owner-local layered implementation	service projects + end-to-end application slice	SPECIFIED/PENDING
10 E,N,V	typed config, precedence, validation and secret refs	Appendix D profiles + config equivalence	SPECIFIED/PENDING
11 N,V	startup/liveness/readiness meanings and budgets	health catalogue + dependency fault probes	SPECIFIED/PENDING
12 E,N,V	distinct test families and evidence claims	test catalogue + zero-discovery verifier	SPECIFIED/PENDING

PART III TRACEABILITY

Data Ownership in Executable Form - Chapters 13-18

CHAPTER ANCHORS H-9

Ch / material	Primary obligation	Canonical asset + check	State
13 E,N	map 17 schemas to three owner databases	ownership catalogue + cross-store dependency graph	SPECIFIED/PENDING
14 E,N,V	AlarmManagement model, mapping, outbox and grants	AlarmManagement.Infrastructure + provider component tests	SPECIFIED/PENDING
15 E,N,V	RootCause aggregate persistence and invariants	RootCause.Infrastructure + materialization/concurrency tests	SPECIFIED/PENDING
16 N,V	three independent migration histories and windows	three migrators + history/schema receipts	SPECIFIED/PENDING
17 N,V	typed remote identity without FK/join/write	reference records + stale/unknown identity tests	SPECIFIED/PENDING
18 E,N,V	local transaction/concurrency/outbox atomicity	provider transaction and competing-update lab	SPECIFIED/PENDING

PART IV TRACEABILITY

Public Contracts and the Synchronous Edge - Chapters 19-24

CHAPTER ANCHORS H-10

Ch / material	Primary obligation	Canonical asset + check	State
19 B,E	separate Domain, integration and HTTP vocabularies	src/Contracts + contract ownership inventory	SPECIFIED/PENDING
20 N,V	canonical versioned message envelope	envelope schema/serializer + golden/rejection corpus	SPECIFIED/PENDING
21 E,N,V	public commands/queries, validation and failure contracts	Appendix C HTTP catalogue + host contract tests	SPECIFIED/PENDING
22 N,V	thin Query BFF; browser never orchestrates workflow	src/Edge/Query.Bff + production-path host tests	SPECIFIED/PENDING
23 E,N,V	versioning, Problem Details and idempotency keys	HTTP corpus + replay/conflict tests	SPECIFIED/PENDING
24 N,V	producer/consumer compatibility	contracts/v1/corpus + additive/breaking matrix	SPECIFIED/PENDING

PART V TRACEABILITY

The Message Backbone - Chapters 25-30

CHAPTER ANCHORS H-11

Ch / material	Primary obligation	Canonical asset + check	State
25 N,V	closed RabbitMQ topology, ACL and DL routes	topology manifest + real broker permission probes	SPECIFIED/PENDING
26 E,N,V	outbox in producer transaction	owner outbox mapping + broker-down commit test	SPECIFIED/PENDING
27 N,V	dispatch leases, confirms and restart recovery	dispatcher + crash-point/provider tests	SPECIFIED/PENDING
28 E,N,V	Inbox admission and once-only local effect	Inbox mapping + duplicate/redelivery tests	SPECIFIED/PENDING
29 E,N,V	bounded retry, DLQ, poison visibility and replay	failure policy + quarantine/replay evidence	SPECIFIED/PENDING
30 E,N,V	order/duplication/evolution without broker myths	sequence/conflict/compatibility corpus	SPECIFIED/PENDING

FROM SERVICES TO RUNTIME

PART VI TRACEABILITY

The First End-to-End Flow - Chapters 31-36

CHAPTER ANCHORS H-12

Ch / material	Primary obligation	Canonical asset + check	State
31 E,N,V	admit AlarmManagement through API, DB and outbox	alarm-api/worker + provider/broker/telemetry checks	SPECIFIED/PENDING
32 N,V	publish AlarmFloodDetected.v1 from committed alarm state	contract + tx/dispatch/trace/recovery corpus	SPECIFIED/PENDING
33 E,N,V	consume once, open RootCauseCase and emit CaseOpened	rootcause-worker + inbox/owner-effect tests	SPECIFIED/PENDING
34 N,V	publish RootCauseVerdictIssued.v1 after valid verdict	verdict contract + fan-out compatibility checks	SPECIFIED/PENDING
35 N,V	project delayed truth for Reporting and BFF	projection worker/API + high-water/convergence tests	SPECIFIED/PENDING
36 E,N,V	prove complete slice; keep RL advisory optional	final-phase0-v1 + required/optional branch oracles	SPECIFIED/PENDING

FINAL DISPOSITION

Audit and Compliance are subscriber candidates, not final-core services. Chapter 34 defines a publishable fact and compatibility surface; it does not create their hosts, queues, credentials or databases. RL Advisory remains DESIGN_INACTIVE.

PART VII TRACEABILITY

Long-Running Workflows - Chapters 37-42

CHAPTER ANCHORS H-13

Ch / material	Primary obligation	Canonical asset + check	State
37 E,N,V	specify durable process-manager state and decisions	workflow model + transition/replay tests	SPECIFIED / DESIGN_INACTIVE
38 B,E,N	record choreography/orchestration ownership decision	ADR + authority and coupling review	SPECIFIED/PENDING
39 E,N,V	model deadlines, escalation and human review	clock abstraction + timeout/race corpus	SPECIFIED / DESIGN_INACTIVE
40 E,N,V	compensate with new history, never fake rollback	compensation fact/model + duplicate/failure tests	SPECIFIED/PENDING
41 E,N,V	show delayed/pending/failed states honestly	BFF read model + stale/high-water UI tests	SPECIFIED/PENDING
42 N,V	reconcile, diagnose and repair under bounded authority	repair plan/receipt + detection/authorization checks	SPECIFIED / DESIGN_INACTIVE

NO HIDDEN ACTIVATION

The Process Manager and Repair Control remain design authorities only. The final Phase-0 runtime contains no workflow database, workflow host, repair-control endpoint, runtime credential or broker authority for either candidate.

PART VIII TRACEABILITY

Security as Architecture - Chapters 43-48

CHAPTER ANCHORS H-14

Ch / material	Primary obligation	Canonical asset + check	State
43 E,N	model assets, nine zones, twelve boundaries and threats	Appendix F inventories + referential-integrity gate	SPECIFIED/PENDING
44 E,N,V	OIDC at BFF; local authorization and reauthentication	auth policy + anonymous/expired/privilege tests	SPECIFIED/PENDING
45 N,V	direct workload identity with mTLS	certificate/SAN/trust policy + negative handshakes	SPECIFIED/PENDING
46 E,N,V	minimize delegated context; map local policy	claim schema + overclaim/tenant/role corpus	SPECIFIED/PENDING
47 E,N,V	reference secrets/certificates and rotate safely	material snapshot + rotation/redaction probes	SPECIFIED/PENDING
48 N,V	bind adversarial tests to retained audit evidence	security catalogue + same-run evidence verifier	SPECIFIED/PENDING

THREAT IDENTITY

The final inventory contains 24 named active threats. THR-043-007..010 remain tombstoned identifiers; they are not renumbered, reused or counted as active findings.

PART IX TRACEABILITY

Observability and Resilience - Chapters 49-54

CHAPTER ANCHORS H-15

Ch / material	Primary obligation	Canonical asset + check	State
49 E,N,V	carry correlation and causation across boundaries	envelope/activity mapping + broken-link tests	SPECIFIED/PENDING
50 E,N,V	emit structured, allowlisted, domain-meaningful logs	logging policy + schema/redaction/cardinality tests	SPECIFIED/PENDING
51 N,V	trace HTTP, broker, DB and background work with OTel	instrumentation map + parent/link/export checks	SPECIFIED/PENDING
52 N,V	define metrics, objectives and dashboards honestly	metric catalogue + query/cardinality/freshness gates	SPECIFIED/PENDING
53 E,N,V	bound timeouts/retries/circuits/bulkheads by budget	resilience profiles + timing/concurrency fault lab	SPECIFIED/PENDING
54 N,V	inject failure and retain recovery evidence	core-recovery-v1 + six scenario dossiers	SPECIFIED/PENDING

TELEMETRY BOUNDARY

Logs, traces and metrics help explain observations; they do not decide business truth. Production SLO attainment, performance capacity and recovery success remain unclaimed until measured under an admitted run identity.

PART X TRACEABILITY

Delivery, Migration, and the Final Demonstrator - Chapters 55-60

CHAPTER ANCHORS H-16

Ch / material	Primary obligation	Canonical asset + check	State
55 E,N,V	reproducible Compose runtime and readiness	canonical compose files + rendered/runtime equality	SPECIFIED/PENDING
56 B,E,N,V	reversible strangler extraction and coexistence	cutover ledger + one-writer/reversal checks	SPECIFIED/PENDING
57 N,V	locked CI gates and immutable container images	pipeline policy + clean-checkout/image identity	SPECIFIED/PENDING
58 N,V	expand/contract DB change, rollout and rollback	release plan + compatibility/reversal evidence	SPECIFIED/PENDING
59 N,V	translate proven needs to Kubernetes	rendered manifests + policy/dry-run comparison	SPECIFIED/PENDING
60 E,N,V	execute the final distributed demonstrator	final-phase0-v1 + final evidence manifest	SPECIFIED/PENDING

COMPLETION BOUNDARY

The manuscript specifies the deliverable, command path and evidence admission rules. It does not claim that a companion checkout currently builds, that containers ran, or that the final demonstrator succeeded.

EPILOGUE AND FINAL REFERENCE OBLIGATIONS

The ending preserves epistemic honesty

BACK-MATTER ANCHORS H-17

Coordinate	Obligation	Check	State
Epilogue	separate designed, implemented, executed and demonstrated	claim language/state audit	PRESENT
Glossary	define stable book vocabulary and cross-references	term/use/link audit	PRESENT
Notation	normalize timing, probability, sets and identifiers	symbol/domain/dimension audit	PRESENT
Sources	identify standards and primary references without fake precision	citation/provenance audit	PRESENT
Working index	index concepts, contracts, roles, threats and commands	coverage/collision/link audit	PRESENT
Final consistency audit	re-run all editorial and architecture joins	closed audit ledger	EXECUTED
Reproduction readiness	publish final immutable file identity and residuals	digest/render/link/metadata gate	DECLARED

SEQUENCE

Appendix H closes technical-material traceability. The glossary, notation, sources, working index, final consistency audit and final readiness checkpoint follow as final reference back matter; they are not a Chapter 61.

APPENDICES A-G TRACEABILITY

Each prior appendix owns one final authority

APPENDIX ANCHORS H-18

Appendix	Final authority	Canonical check	State
A	solution/project/deployment/evidence path map	root/path/project inventory equality	SPECIFIED/PENDING
B	owners, roles, data, dependencies and candidates	closed census + referential integrity	SPECIFIED/PENDING
C	six HTTP surfaces and three public facts	contract inventory/corpus compatibility	SPECIFIED/PENDING
D	configuration keys, profiles, validation and secrecy	value-free manifest + equivalence tests	SPECIFIED/PENDING
E	six core recovery scenarios and support cases	fault/recovery/cleanup dossiers	SPECIFIED/PENDING
F	zones, boundaries, requirements, threats and controls	security inventory/adversarial evidence	SPECIFIED/PENDING
G	public command grammar, stages and reproduction	postcondition/receipt/manifest verifier	SPECIFIED/PENDING

APPENDIX H AUTHORITY

H owns joins and state derivation. It may point to an authority in A-G, but it may not silently redefine that authority or use a chapter-local example as a competing canonical value.

CANONICAL PATH REGISTRY

Every repository trace starts at one root

CANONICAL ROOTS H-19

Coordinate	Canonical identity	Validation
companion root	artifacts/from-services-to-runtime/	sentinel/root resolver
solution	Nexus1.Runtime.sln	closed project inventory
source	src/Services + src/Edge + src/Contracts + src/BuildingBlocks	allowlisted tree diff
tests	tests/Architecture Unit Component Contract Integration Deployment	test-family discovery
deployment	deploy/compose + deploy/kubernetes	rendered model comparison
engineering	eng/scripts + eng/Nexus.*	wrapper/command conformance
evidence	artifacts/evidence/<RunId>/	same-run manifest join

PATH RULE

A trace stores paths relative to the canonical companion root. Absolute machine paths, shell-dependent aliases and copied artifact locations are observations, not source identities.

SUPERSEDED-PATH REGISTRY

Old examples remain searchable but lose authority

TOMBSTONES H-20

Superseded	Replacement	Required result
NEXUS1.sln	Nexus1.Runtime.sln	SUPERSEDED; no active script reference
deploy/compose.yaml	deploy/compose/compose.yaml	SUPERSEDED; render only replacement
deploy/compose.local.yaml	deploy/compose/compose.override.development.yaml	SUPERSEDED
deploy/compose.evidence.yaml	deploy/compose/compose.evidence.yaml	SUPERSEDED
src/<Owner>.*	src/Services/<Owner>/Nexus1.<Owner>.*	SUPERSEDED; inventory rejects old root
chapter-local helper scripts	eng/scripts/nexus.sh nexus.ps1	SUPERSEDED; public verbs only

TOMBSTONE SEMANTICS

A tombstone preserves history and diagnostic value. It is not an alias that both paths may continue to use; canonical-reference tests must fail on a live superseded path.

FINAL-CORE RUNTIME CENSUS

Traceability cannot manufacture a service

OWNER AND ROLE CLOSURE H-21

Owner	Long-running roles	Migrator / store	Disposition
AlarmManagement	alarm-api; alarm-worker	alarm-migrator / AlarmManagementDb	ACTIVE FINAL CORE
RootCause	rootcause-api; rootcause-worker	rootcause-migrator / RootCauseDb	ACTIVE FINAL CORE
Reporting	projection-api; projection-worker	reporting-migrator / ReportingDb	ACTIVE FINAL CORE
Edge	query-bff	none / none	ACTIVE FINAL CORE

CONTROL TOTALS H-22

Inventory	Expected
owners / edge	3 / 1
long-running roles	7
migrators / owner stores	3 / 3

CANDIDATE DISPOSITION REGISTRY

Design depth does not imply runtime authority

INACTIVE CANDIDATES H-23

Candidate	What the book specifies	Forbidden final-core residue	Disposition
Audit	possible VerdictIssued subscriber	host/queue/store/credential	CANDIDATE_INACTIVE
Compliance	possible VerdictIssued subscriber	host/queue/store/credential	CANDIDATE_INACTIVE
Process Manager	durable workflow semantics	host/workflow DB/queue/credential	DESIGN_INACTIVE
Repair Control	repair plan/approval/receipt model	admin route/host/credential/actuator	DESIGN_INACTIVE
RL Advisory	optional non-authoritative advice branch	decision authority/required route/queue	DESIGN_INACTIVE

NEGATIVE EVIDENCE

Closed inventories, absent credentials, absent topology and forbidden-path tests are required. Merely omitting a candidate from a diagram is not proof that it cannot acquire authority.

HTTP SURFACE TRACE

Six protected surfaces, one owning edge

FINAL HTTP CATALOGUE H-24

Surface	Owner	Primary trace	Required checks
alarm command	AlarmManagement API	Ch21,23,31 / App C	auth, validation, idempotency, Problem Details
alarm query	AlarmManagement API	Ch21,31 / App C	auth, not-found, versioning
case query	RootCause API	Ch21,33 / App C	auth, typed identity, not-found
verdict command	RootCause API	Ch21,34 / App C	auth, concurrency, idempotency
projection query	Reporting API	Ch35,41 / App C	auth, age/high-water/degraded state
operator composite	Query BFF	Ch22,41 / App C	OIDC session, aggregation budget, partial truth

HTTP BOUNDARY

Routes, methods and schemas are owned by Appendix C. This matrix binds them to chapters and checks; it does not add a seventh surface or activate an administrative repair API.

FROM SERVICES TO RUNTIME

PUBLIC-FACT TRACE

Three active facts carry the business flow

FINAL EVENT CATALOGUE H-25

Public fact	Producer / local cause	Active consumer	Required checks
AlarmFloodDetected.v1	AlarmManagement / committed flood	RootCause worker	envelope, tx-outbox, confirm, inbox, replay
RootCauseCaseOpened.v1	RootCause / committed case	Reporting projection worker	envelope, tx-outbox, compatibility, convergence
RootCauseVerdictIssued.v1	RootCause / committed verdict	Reporting projection worker	invariant, tx-outbox, replay, projection age

FAN-OUT BOUNDARY

A public fact may be compatible with future subscribers without making them active. The active fact count remains three; candidate Audit and Compliance subscriptions remain outside final-core topology.

OWNER-STORE TRACE

Local ACID and public convergence have separate proof

DATABASE OBLIGATIONS H-26

Store	Writer	Atomic boundary	Checks
AlarmManagementDb	alarm-api/worker owner code	alarm aggregate + outbox	migration, constraints, commit/rollback, concurrency
RootCauseDb	rootcause-api/worker owner code	case/evidence/verdict + outbox + inbox	materialize, dedupe, invariant, competing update
ReportingDb	projection-worker owner code	inbox + projection + high-water	replay, monotonicity, convergence, rebuild

FORBIDDEN CROSS-OWNER OPERATIONS H-27

Operation	Expected
cross-database FK/join/write	ABSENT; architecture/provider tests reject
API performs DDL	ABSENT; negative grant test rejects
shared EF model/repository	ABSENT; dependency inventory rejects

BROKER AND DELIVERY TRACE

One fact crosses several independently fallible seams

DELIVERY CHAIN H-28

Stage	Canonical mechanism	Oracle / evidence
commit	owner state + outbox in one local transaction	provider transaction receipt
claim	bounded lease with attempt/next-attempt	lease race and expiry receipt
publish	publisher confirm under message identity	confirm/nack/channel-loss receipt
route	declared exchange/queue/binding/ACL/DLX	expected/actual topology diff
admit	Inbox unique message identity	duplicate/redelivery receipt
effect	one local owner transaction	owner state plus outbox/inbox oracle
recover	bounded retry then quarantine/replay	failure/recovery/cleanup dossier

NO EXACTLY-ONCE TRANSPORT CLAIM

The proof target is at-least-once delivery plus idempotent admission and one committed local effect, not a broker promise of global exactly-once execution.

CONFIGURATION TRACE

Values stay private while policy remains reviewable

CONFIGURATION JOINS H-29

Concern	Authority	Check / retained output
key/type/default/required	Appendix D	schema and missing/unknown-key validation
provider precedence	Chapter 10 + Appendix D	value-free provider manifest
cross-field policy	owner options validators	startup failure matrix
profile equivalence	Development/Evidence/CI/translation profiles	canonical policy digest comparison
secret reference	file/env/provider reference metadata	no-value scan and authorization probe
reload	immutable validated snapshot	invalid-change retention and atomic swap
runtime identity	rendered config/material snapshot digests	same-run identity join

NON-CLAIM

A value-free manifest proves configuration shape and source identity, not that a secret value is correct or that production policy has been approved.

FAILURE AND RECOVERY TRACE

Six core campaigns are mandatory

CORE SCENARIOS H-30

ID	Disturbance	Deepest oracle
E-CORE-001	broker unavailable during producer commit/dispatch	owner state retained; outbox later confirmed
E-CORE-002	dispatcher crash around publish confirm	safe duplicate; no lost business fact
E-CORE-003	consumer crash/redelivery	one Inbox admission and one local effect
E-CORE-004	projection unavailable/lagging	owner truth intact; age visible; convergence later
E-CORE-005	database interruption/concurrency	bounded failure; no partial owner commit
E-CORE-006	identity/certificate/config disturbance	fail closed; bounded recovery; material not leaked

EVIDENCE CONTRACT

Each campaign retains baseline, disturbance, bounded observation, recovery, cleanup and residuals under one RunId. Health becoming green is not a substitute for the deepest business oracle.

SECURITY TRACE

Threats, requirements and negative tests share identity

SECURITY CONTROL TOTALS H-31

Inventory	Final count	Join gate
trust zones	9	every component/data/control surface has one zone
trust boundaries	12	every crossing names protocol, identity and protection
security requirements	21	each maps threat/boundary/control/check
named active threats	24	each maps owner, mitigation, residual and test
tombstoned threat IDs	4 (THR-043-007..010)	never reused or counted active

IDENTITY BOUNDARIES H-32

Boundary	Required identity
browser -> BFF	operator OIDC session; reauth for sensitive action
workload -> workload	direct mTLS workload identity; no forwarded user token as service identity
runtime -> store/broker	least-privilege runtime principal; migrator separate

COMMAND AND EVIDENCE TRACE

A command earns authority through a typed postcondition

PUBLIC COMMAND JOINS H-33

Stage	Canonical entry	Trace output
doctor/plan	eng/scripts/nexus.sh nexus.ps1	root/tool/input/plan identity
restore/build	Nexus1.Runtime.sin --locked-mode	command receipt + binlog/artifact inventory
test	closed family/shard/provider/corpus catalogue	nonzero discovery/execution receipts
migrate	three explicit migrators	schema/history/grant receipts
up/ready	canonical Compose render and predicates	runtime/topology/readiness snapshot
scenario/failure	final-phase0-v1 / core-recovery-v1	business and recovery dossiers
verify/seal	evidence verifier	Missing/residuals + immutable manifest

NO FABRICATED OUTPUT

Appendix G defines schemas and postconditions. Appendix H links those contracts to claims. Neither appendix publishes invented Runlds, hashes, durations, counts or green transcripts.

CLAIM-TO-EVIDENCE LAYERS

Different claims require different witnesses

EVIDENCE LAYER MATRIX H-34

Claim family	Minimum check	Minimum retained support	Cannot prove
Domain invariant	pure deterministic unit test	case vector + outcome receipt	provider mapping
Application order/effects	handler/port interaction test	call/effect ledger	transaction atomicity
Persistence	real provider component test	migration/query/transaction receipt	broker delivery
Contract	golden/rejection/compat corpus	corpus identity + verdicts	running topology
Host boundary	production-path host test	route/auth/problem receipt	deployment recovery
Broker delivery	real broker redelivery/ACL test	topology + delivery receipt	business convergence alone
Deployment/recovery	fault lab with deepest oracle	baseline/fault/recovery/cleanup dossier	production SLO

EVIDENCE SPECIFICITY

A stronger-looking but mismatched artifact does not substitute for the witness required by the claim. A trace screenshot cannot prove a database rollback; a unit test cannot prove broker permissions.

EVIDENCE-PLANE JOINS

Every artifact belongs to one attempt and one identity

REQUIRED JOIN KEYS H-35

Object	Required identity	Verifier rule
Trace row	TraceId + TraceIdentityDigest	stable under editorial ordering
Claim	ClaimId + source revision	unique bounded meaning
Asset	canonical path + tree/file digest	exists under exact root or PENDING
Check	CheckId + command/tool/policy identity	discoverable and nonempty or PENDING
Run	RunId + ReproductionIdentity	one attempt; no cross-run substitution
Evidence	artifact path + digest + schema	inside run root; digest/schema valid
Verdict	derived state + Missing + residuals	not hand-edited; fail closed

JOIN FAILURE

If a report cannot be joined back to its exact source, command, run, input identity and artifact digest, it is diagnostic material only and cannot advance the claim state.

TRACEABILITY GRAPH

The matrix is a graph with typed edges

TRACEGRAPHV1

```
public sealed record TraceGraphV1(
    IReadOnlyDictionary<string, SourceNode> Sources,
    IReadOnlyDictionary<string, ClaimNode> Claims,
    IReadOnlyDictionary<string, AssetNode> Assets,
    IReadOnlyDictionary<string, CheckNode> Checks,
    IReadOnlyDictionary<string, EvidenceNode> Evidence,
    IReadOnlyList<TraceEdge> Edges);

public enum TraceEdgeKind
{
    Defines, Extends, Implements, Verifies,
    Produces, Supports, Supersedes, Limits
}
```

ALLOWED MINIMUM PATH H-36

Source material	Required graph path
B	source --Defines--> claim; inherited source identity retained
E	source --Extends--> asset --Verifies--> check
N	source --Defines--> asset --Verifies--> check
V	check --Produces--> evidence --Supports--> claim, or state PENDING

STABLE TRACE IDENTITY

Editorial movement must not change a claim

CANONICALTRACEIDENTITY

```
static string TraceIdentity(TechnicalMaterialTraceV1 row)
{
    var canonical = string.Join("\n", new[]
    {
        "trace-v1", Normalize(row.TraceId),
        Normalize(row.Source.StableCoordinate),
        string.Join(",", row.Material.Order()),
        Normalize(row.ClaimId), Normalize(row.BoundedClaim),
        Normalize(row.Asset.CanonicalIdentity),
        Normalize(row.Check.CheckId),
        Normalize(row.Disposition.ToString()),
        string.Join("|", row.Residuals.Order())
    });
    return Sha256Utf8(canonical);
}
```

EXCLUDED FROM IDENTITY

Page number, table position, line wrapping, generated-at time and current evidence verdict do not define the claim. They may change without manufacturing a new obligation.

INCLUDED IN IDENTITY

Source coordinate, material class, bounded meaning, canonical asset/check, architectural disposition and residual limits define what the row actually claims.

STRUCTURAL VALIDATOR

Duplicates and orphan edges fail before execution

VALIDATESTRUCTURE

```
IEnumerable<Finding> ValidateStructure(TraceGraphV1 graph)
{
    foreach (var id in DuplicateIds(graph))
        yield return Error("TRACE_ID_DUPLICATE", id);

    foreach (var edge in graph.Edges)
        if (!graph.Contains(edge.From) || !graph.Contains(edge.To))
            yield return Error("TRACE_EDGE_ORPHAN", edge.Id);

    foreach (var claim in graph.Claims.Values)
        if (!graph.Edges.Any(e => e.To == claim.Id &&
            e.Kind is TraceEdgeKind.Defines or TraceEdgeKind.Supports))
            yield return Error("CLAIM_UNANCHORED", claim.Id);

    foreach (var source in RequiredSources())
        if (!graph.Sources.ContainsKey(source))
            yield return Error("SOURCE_COORDINATE_MISSING", source);
}
```

CLOSED DISCOVERY

RequiredSources is generated from the approved contents: Chapters 1-60, the Epilogue and Appendices A-G. A missing coordinate fails even when every discovered row is green.

MATERIAL-CLASS VALIDATOR

BASE, EXTEND, NEW and VERIFY impose different edges

VALIDATEMATERIAL

```
IEnumerable<Finding> ValidateMaterial(TechnicalMaterialTraceV1 row)
{
    if (row.Material.Contains(MaterialClass.Base) &&
        row.Source.InheritedSource is null)
        yield return Error("BASE_SOURCE_MISSING", row.TraceId);

    if (row.Material.Contains(MaterialClass.Extend) &&
        row.Asset.Predecessor is null)
        yield return Error("EXTENSION_PREDECESSOR_MISSING", row.TraceId);

    if (row.Material.Contains(MaterialClass.New) &&
        row.Asset.CanonicalIdentity is null)
        yield return Error("NEW_ASSET_UNBOUND", row.TraceId);

    if (row.Material.Contains(MaterialClass.Verify) &&
        row.Check is null)
        yield return Error("VERIFY_CHECK_UNBOUND", row.TraceId);
}
```

PENDING IS VALID

A VERIFY row may remain PENDING when the check or evidence has not been executed. It may not omit the intended oracle or be marked EXECUTED because prose describes an expected outcome.

EVIDENCE ADMISSION VALIDATOR

A state is derived, never selected

DERIVECLAIMSTATE

```
ClaimState DeriveClaimState(TraceRow row, RunDossier? run)
{
    if (row.Disposition is ArchitecturalDisposition.Inactive)
        return ClaimState.Inactive;
    if (row.Asset.IsSuperseded)
        return ClaimState.Superseded;
    if (run is null || !run.IdentityMatches(row))
        return ClaimState.Specified;
    if (!run.AssetInventory.ContainsExact(row.Asset))
        return ClaimState.Specified;
    if (!run.Checks.TryGetComplete(row.Check, out var receipt))
        return ClaimState.Implemented;
    if (!receipt.Passed)
        return ClaimState.Falsified;
    return run.DemonstrationAdmits(row)
        ? ClaimState.Demonstrated : ClaimState.Executed;
}
```

FAIL CLOSED

Missing, stale, cross-run, zero-discovery, schema-invalid or digest-invalid support cannot promote a claim. A failed relevant check yields FALSIFIED, not a silently retried green state.

INACTIVE-AUTHORITY VALIDATOR

Candidate designs must remain powerless

VALIDATEINACTIVE

```
IEnumerable<Finding> ValidateInactive(
    Candidate candidate,
    RuntimeInventory runtime,
    SecurityInventory security,
    BrokerInventory broker)
{
    if (runtime.Roles.Any(r => r.Owner == candidate.Id))
        yield return Error("INACTIVE_ROLE_PRESENT", candidate.Id);
    if (runtime.Stores.Any(s => s.Owner == candidate.Id))
        yield return Error("INACTIVE_STORE_PRESENT", candidate.Id);
    if (security.Principals.Any(p => p.Owner == candidate.Id))
        yield return Error("INACTIVE_CREDENTIAL_PRESENT", candidate.Id);
    if (broker.Consumers.Any(c => c.Owner == candidate.Id))
        yield return Error("INACTIVE_SUBSCRIPTION_PRESENT", candidate.Id);
    if (runtime.HttpSurfaces.Any(h => h.Owner == candidate.Id))
        yield return Error("INACTIVE_ROUTE_PRESENT", candidate.Id);
}
```

REQUIRED SET

The validator runs for Audit, Compliance, Process Manager, Repair Control and RL Advisory. Documentation may describe them; final-core executable inventories must not grant them authority.

CANONICAL-REFERENCE VALIDATOR

One replacement closes every superseded path

VALIDATECANONICALREFERENCES

```
foreach (var tombstone in tombstones)
{
    Require(graph.Contains(tombstone.Replacement));
    foreach (var reference in ScanActiveReferences(tombstone.OldIdentity))
        findings.Add(Error("TOMBSTONED_PATH_ACTIVE", reference));
}

RequireOnly("solution", "Nexus1.Runtime.sln");
RequireOnly("compose-base", "deploy/compose/compose.yaml");
RequireOnly("compose-dev",
    "deploy/compose/compose.override.development.yaml");
RequireOnly("public-posix", "eng/scripts/nexus.sh");
RequireOnly("public-powershell", "eng/scripts/nexus.ps1");
```

SCOPE OF SCAN

Active references include solution files, project references, build scripts, workflows, Compose/Kubernetes inputs, test data, documentation commands and manifest schemas. Quoted tombstone registries are excluded by type, not by text guessing.

COVERAGE VALIDATOR

Completeness is set equality

VALIDATECOVERAGE

```
var expectedChapters = Enumerable.Range(1, 60)
    .Select(n => $"CH-{n:000}").ToHashSet();
var expectedAppendices = new[] { "APP-A", "APP-B", "APP-C",
    "APP-D", "APP-E", "APP-F", "APP-G" }.ToHashSet();

RequireSetEqual(expectedChapters,
    graph.Sources.Keys.Where(k => k.StartsWith("CH-")),
    "CHAPTER_TRACE_COVERAGE");
RequireSetEqual(expectedAppendices,
    graph.Sources.Keys.Where(k => k.StartsWith("APP-")),
    "APPENDIX_TRACE_COVERAGE");
Require(graph.Sources.ContainsKey("EPILOGUE"));
```

NO DENOMINATOR DRIFT

Discovery cannot decide the expected count. The approved contents own the denominator: 60 chapters, seven prior appendices and one epilogue coordinate.

CROSS-CATALOGUE RECONCILIATION

Final counts must agree everywhere

EQUALITY GATES H-37

Join	Expected equality	Failure
roles	Appendix B = Compose = command plan = evidence runtime	ROLE_CENSUS_DRIFT
migrators/stores	A/B/D/G = rendered deployment = migration receipts	STORE_AUTHORITY_DRIFT
HTTP	C = host inventory = BFF policy = security tests	HTTP_SURFACE_DRIFT
facts	C = contracts = broker topology = producers/consumers	PUBLIC_FACT_DRIFT
failures	E = G campaign plan = evidence dossiers	FAILURE_CAMPAIGN_DRIFT
security	F = deployment identity = negative test catalogue	SECURITY_JOIN_DRIFT
paths	A = G command inputs = H asset references	CANONICAL_PATH_DRIFT

REQUIRED ACTION

A mismatch is resolved at the owning authority, followed by regeneration of dependent views. It is never resolved by accepting multiple 'current' values.

STATE-TRANSITION POLICY

Progress is monotonic only when support remains valid

ALLOWED TRANSITIONS H-38

From	To	Condition
SPECIFIED	IMPLEMENTED	exact canonical asset exists under admitted source identity
IMPLEMENTED	EXECUTED	complete nonzero check receipt in same-run dossier
EXECUTED	DEMONSTRATED	declared demonstration oracle admits claim
any active	FALSIFIED	retained relevant counterexample
SPECIFIED/active	SUPERSEDED	replacement and tombstone admitted
SPECIFIED	INACTIVE	candidate explicitly denied final-core authority
any	SPECIFIED	support invalidated, identity changed or evidence lost

NO FALSE MONOTONICITY

A changed contract, source tree, toolchain, topology or security policy can invalidate earlier support. Historical dossiers stay immutable; the current claim state may regress until reverified.

TRACEABILITY ACCEPTANCE GATES

The matrix passes only as a closed system

ACCEPTANCE H-39

Gate	Required result
source coverage	60/60 chapters, Epilogue, 7/7 prior appendices
material coverage	every B/E/N/V flag preserved and structurally satisfied
asset/check binding	one canonical reference or explicit documentary rationale
state derivation	no state exceeds admitted support
architecture closure	3 owners, 1 edge, 7 roles, 3 migrators/stores
contract closure	3 active facts and 6 protected HTTP surfaces
candidate closure	five inactive candidates have zero final-core authority
path closure	canonical identities only; six path families tombstoned
evidence closure	same-run digest/schema joins or explicit PENDING/Missing

COVERAGE RESULT

The manuscript trace is complete; execution is not

COVERAGE LEDGER H-40

Coordinate	Expected	Present	Current result
Parts	10	10	COVERED
Chapters	60	60	COVERED
Epilogue	1	1	COVERED
Appendices A-G	7	7	COVERED
Material flags	all approved	all approved	COVERED
canonical asset/check contracts	all anchors	all anchors	SPECIFIED
repository assets	exact paths	not inspected here	PENDING
runtime evidence	same-run dossiers	not executed here	PENDING

MEANING OF COMPLETE

COMPLETE in Appendix H means every planned technical-material source has a bounded trace and an honest disposition. It does not mean every proposed source file exists or every VERIFY campaign has executed.

RESIDUALS AND NON-CLAIMS

The final matrix keeps uncertainty attached

RESIDUAL LEDGER H-41

Residual	Current handling
companion repository not delivered in this PDF	all implementation states remain PENDING
provider/broker/container behavior not executed here	VERIFY rows remain SPECIFIED/PENDING
performance/capacity/production SLO unmeasured	no production-readiness or capacity claim
Kubernetes translation not deployed	educational target; validation PENDING
candidate subscribers/workflow/repair/advisory	explicitly inactive with negative-authority checks
final reference back matter not yet authored	named ordered sequence after Appendix H
future implementation may falsify assumptions	retain failures; update state and residuals

FORBIDDEN CONCLUSION

The phrase 'all traces present' must never be rewritten as 'the distributed system works.' Traceability makes the remaining work precise; it does not perform that work.

APPENDIX H STATUS BOARD

Final technical-material authority is closed

STATUS H-42

Obligation	Result
project-description material flags	reconciled for Chapters 1-60
chapter and appendix coverage	60/60 chapters; A-G 7/7; Epilogue anchored
canonical architecture joins	owners/edge/roles/stores/contracts equal final catalogues
inactive candidate control	five candidates named; zero final-core authority required
canonical and superseded paths	replacement/tombstone rules fixed
claim/evidence state model	typed derivation and regression rules fixed
validators and acceptance gates	specified; implementation PENDING
same-run implementation/runtime evidence	PENDING

HONEST CHECKPOINT

Appendix H is complete as the Phase-0 technical-material traceability specification. Repository implementation, clean-room execution and runtime demonstration remain separate future evidence work.

APPENDIX H CHECKPOINT

Every obligation now has a place to be proved or refused

COMPLETION LEDGER H-43

Question	Final answer
where did the obligation originate?	stable chapter/appendix/epilogue coordinate
what work class applies?	BASE, EXTEND, NEW and/or VERIFY
what exactly is claimed?	bounded ClaimId and residuals
where would it live?	one canonical asset identity
how would it be challenged?	typed check and deepest oracle
what supports it?	same-run evidence reference and digest
what is its current truth state?	derived SPECIFIED/PENDING unless admitted support exists
what architecture does it authorize?	explicit active/inactive/superseded disposition

FINAL APPENDIX H RULE

A trace is trustworthy only when it remains useful on a failed run: it must identify the exact obligation, the missing or falsifying support, the owner of correction and the limit of the surviving claim.

HANDOFF TO FINAL REFERENCE BACK MATTER

The technical argument is closed; navigation and verification remain

The next phase does not add Chapter 61 or expand the runtime. It makes the completed book easier to interpret, cite, navigate and reproduce while applying a final consistency audit across every established authority.

ORDERED FINAL SEQUENCE H-44

Next coordinate	Purpose
Glossary of Microservice and Distributed-Systems Terms	freeze vocabulary and cross-references
Mathematical and Timing Notation Used in the Book	normalize symbols, units, clocks and probability
Sources and Reference Notes	record primary standards and provenance
Working Index	navigate concepts, code, roles, facts, threats and commands
Final Consistency Audit	re-run architecture, trace, editorial and visual joins
Final Reproduction Readiness / Deferred Evidence	record absence + future evidence contract

APPENDIX H LAST WORD

Traceability is the refusal to let a green summary hide a missing proof

FINAL STATUS

TECHNICAL MATERIAL TRACEABILITY COMPLETE | 60/60 CHAPTER ANCHORS | APPENDICES A-G JOINED | FINAL CORE RECONCILED | IMPLEMENTATION + EXECUTION EVIDENCE PENDING

FINAL REFERENCE BACK MATTER

Glossary of Microservice and Distributed-Systems Terms

The controlled vocabulary for the completed NEXUS-1 runtime volume. Each definition states how the term is used in this book and points to the principal chapter or appendix authority.

GLOSSARY AUTHORITY

A glossary entry freezes language; it does not create a new service, contract, runtime claim or source of evidence. When an entry conflicts with an earlier illustrative snippet, the final catalogue and the cited authority own the current meaning.

GLOSSARY CONTROL TOTALS GL-1

Coordinate	Value
defined terms	176
alphabetical groups	24
chapter range	1-60
appendix authorities	A-H
final-core constants	3 owners / 1 edge / 7 roles / 3 stores / 3 facts / 6 HTTP surfaces
implementation and runtime evidence	still PENDING where not admitted

HOW TO READ THE GLOSSARY

Definition, NEXUS-1 meaning, then authority

ENTRY CONTRACT GL-2

Field	Meaning
Term	preferred spelling or code identity used by this volume
Definition	bounded meaning in the NEXUS-1 distributed runtime
Primary anchor	chapter/appendix where the term receives its main engineering treatment
Internal link	the term cell links to the primary anchor in the final PDF

CROSS-REFERENCE RULE

A primary anchor is a navigational starting point, not the only occurrence. The Working Index will later enumerate significant secondary appearances, code identities, threats, commands and aliases.

CONTROLLED DISTINCTIONS

Similar words do not share one proof

COLLISION REGISTER GL-3

Do not collapse	Required distinction
Domain event / integration event / public fact	internal occurrence / versioned external message / committed owner truth carried publicly
retry / redelivery / replay	caller-policy attempt / broker delivery attempt / authorized recovery action
startup / liveness / readiness	finite initialization / process viability / ability to receive intended work
correlation / causation	shared scenario grouping / directed production relationship
specified / implemented / executed / demonstrated	defined / asset present / check ran / declared oracle admitted
owner truth / projection / BFF composite	authoritative state / delayed read model / bounded edge composition
candidate / inactive / superseded	possible future / denied current authority / replaced identity

CANONICAL NEXUS-1 NAMES

Code identities keep exact spelling

IDENTITY REGISTER GL-4

Kind	Canonical identities
owners	AlarmManagement; RootCause; Reporting
edge	Query BFF / query-bff
stores	AlarmManagementDb; RootCauseDb; ReportingDb
facts	AlarmFloodDetected.v1; RootCauseCaseOpened.v1; RootCauseVerdictIssued.v1
solution	Nexus1.Runtime.sln
companion root	artifacts/from-services-to-runtime/

CASE AND VERSION RULE

Code identifiers, role IDs, paths and .v1 suffixes are not editorial variants. The glossary preserves their exact current identity; superseded spellings remain tombstones only.

FROM SERVICES TO RUNTIME

A - ARCHITECTURE, AGGREGATES AND AUTHORITY

GLOSSARY ENTRIES A-1

Term	Definition in this book	Primary anchor
Abstention	A deliberate refusal to issue a verdict when evidence, identity, policy or confidence requirements are not satisfied. Abstention is a valid safe outcome, not an exception path.	Ch 2, 9; Epilogue
Active final core	The closed Phase-0 runtime admitted by the final catalogues: three owners, one Query BFF, seven long-running roles and three migrators/stores.	Ch 60; App B, H
Aggregate	A consistency boundary whose state changes are accepted only through its invariant-preserving behavior. It is not a synonym for a database table or service.	Ch 9, 15, 18
Aggregate root	The entity that controls access to an aggregate and is the only entry point for invariant-preserving transitions inside that boundary.	Ch 9, 15
AlarmManagement	The final-core owner of alarm admission, flood detection, alarm state, AlarmManagementDb and publication of AlarmFloodDetected.v1.	Ch 14, 31-32; App B

A - ARCHITECTURE, AGGREGATES AND AUTHORITY - CONTINUED

GLOSSARY ENTRIES A-2

Term	Definition in this book	Primary anchor
AlarmManagementDb	The service-owned database written by AlarmManagement runtime code and migrated only by alarm-migrator. Its local transaction may include alarm state and outbox rows.	Ch 14, 16, 18; App B
Architectural disposition	The explicit status that says whether a described component is active, inactive, a candidate or superseded. Design detail alone cannot promote disposition.	Ch 60; App B, H
Architecture test	An executable dependency, ownership, project or registration rule that fails when code topology contradicts the declared architecture.	Ch 5, 8, 12
At-least-once delivery	A delivery model in which a message may be observed more than once. Correctness therefore depends on idempotent admission and owner-local transactional effects.	Ch 25, 28-30
Atomicity	The all-or-nothing property of one local transaction. In this book it never means a global transaction across services, stores and broker.	Ch 18, 26

A - ARCHITECTURE, AGGREGATES AND AUTHORITY - CONTINUED

GLOSSARY ENTRIES A-3

Term	Definition in this book	Primary anchor
Audit evidence	Retained, identity-bound material from which a reviewer can derive a claim state. Logs alone are evidence only when their schema, run and integrity are admitted.	Ch 48, 54; App G-H
Authority	The bounded permission to decide, write, migrate, publish, subscribe or operate. Every authority must have one named owner and a testable negative boundary.	Ch 13, 43-48; App F

B - BOUNDARIES, BROKERS AND BOUNDED BEHAVIOR

GLOSSARY ENTRIES B-1

Term	Definition in this book	Primary anchor
Backoff	A policy that increases delay between eligible retry attempts within a bounded budget. It does not make an unsafe operation idempotent.	Ch 29, 53
BASE	A project-description material class requiring reuse of established NEXUS-1 material with its inherited identity and changed assumptions made explicit.	App H
BFF (Backend for Frontend)	The Query BFF is the browser-facing edge that authenticates the operator and composes bounded reads. It does not own domain workflow or data.	Ch 22, 41, 44; App B-C
Bounded context	A semantic boundary within which a model and ubiquitous language have one coherent meaning. It is not automatically one deployable service.	Ch 3, 13
Bounded shutdown	A host-stop contract that stops admission, drains or relinquishes work safely, honors cancellation and reaches a known state within a declared time budget.	Ch 8, 11, 27

B - BOUNDARIES, BROKERS AND BOUNDED BEHAVIOR - CONTINUED

GLOSSARY ENTRIES B-2

Term	Definition in this book	Primary anchor
Broker topology	The declared exchanges, queues, bindings, dead-letter routes, consumers and permissions required for the three active public facts.	Ch 25; App B-C
Bulkhead	A concurrency or resource partition that prevents one dependency or work class from exhausting capacity needed by unrelated work.	Ch 53
Business oracle	The deepest domain observation that decides whether a scenario succeeded. Health, telemetry and process exit are supporting observations, not substitutes.	Ch 36, 54, 60; App E

FROM SERVICES TO RUNTIME

C - CLAIMS, CONTRACTS, CAUSATION AND CHANGE

GLOSSARY ENTRIES C-1

Term	Definition in this book	Primary anchor
Candidate inactive	A described capability that receives no final-core host, route, queue, store, principal or decision authority. Audit and Compliance use this disposition.	Ch 34; App B, H
Canonical asset	The one current repository, contract, deployment or evidence identity authorized by the final catalogues. Older examples require explicit tombstones.	App A, G-H
Canonical path	A path relative to artifacts/from-services-to-runtime/ that owns the current implementation identity. Machine-specific absolute paths are observations only.	App A, G-H
Captive dependency	A shorter-lived dependency held by a longer-lived object, commonly a singleton retaining scoped state. Descriptor and scope tests must reject it.	Ch 8
Causation	The directed relationship that one operation or message was produced because of another. It is distinct from shared correlation.	Ch 49

C - CLAIMS, CONTRACTS, CAUSATION AND CHANGE - CONTINUED

GLOSSARY ENTRIES C-2

Term	Definition in this book	Primary anchor
Circuit breaker	A stateful resilience policy that temporarily rejects calls after a qualifying failure pattern so a dependency and caller can recover within defined budgets.	Ch 53
Claim	A bounded statement whose truth state is derived from a source, canonical asset, check, evidence and residual limitations.	Epilogue; App H
Claim state	The evidence-derived status SPECIFIED, IMPLEMENTED, EXECUTED, DEMONSTRATED, FALSIFIED, SUPERSEDED or INACTIVE. It is never chosen for presentation convenience.	Epilogue; App H
Clean-room run	A reproduction attempt from an admitted fresh checkout and isolated runtime lease, without relying on undeclared local build, container, data or evidence residue.	Ch 57, 60; App G
Command	A request to attempt a state change at an owning boundary. Acceptance, rejection and idempotent replay are explicit outcomes.	Ch 21, 23

C - CLAIMS, CONTRACTS, CAUSATION AND CHANGE - CONTINUED

GLOSSARY ENTRIES C-3

Term	Definition in this book	Primary anchor
Compensation	A new, explicit business action or fact that mitigates a committed outcome. It does not erase history or pretend a distributed rollback occurred.	Ch 40
Competing update	Two or more attempts to change the same concurrency-protected owner state. The losing attempt must receive a semantic conflict rather than overwrite silently.	Ch 18
Concurrency token	A provider-persisted value used to detect that owner state changed after it was read and before an attempted commit.	Ch 14-15, 18
Configuration identity	A digest-bound description of validated configuration shape, provider order, profile and referenced material metadata, excluding secret values.	Ch 10; App D, G
Contract	A versioned public schema and behavioral obligation at an HTTP or message boundary. Internal domain types are not public contracts.	Ch 19-24; App C

C - CLAIMS, CONTRACTS, CAUSATION AND CHANGE - CONTINUED

GLOSSARY ENTRIES C-4

Term	Definition in this book	Primary anchor
Contract corpus	A retained set of accepted, rejected, old and new payload examples used by producers and consumers to test compatibility and failure semantics.	Ch 20, 24, 30
Correlation	A shared identifier grouping work that belongs to one observed scenario. Correlation aids navigation but does not establish causal order.	Ch 49
CQRS	Separation of command and query models when their responsibilities justify it. It does not require separate services, stores or frameworks.	Ch 21-22, 35
Cross-run join	An invalid evidence relationship that combines assets or receipts from different RunIds or reproduction identities to manufacture one apparently complete dossier.	App G-H
Cutover	The controlled change that transfers one specific read or write authority from the legacy path to the extracted owner under measured preconditions and reversal rules.	Ch 13, 56, 58

FROM SERVICES TO RUNTIME

D - DELIVERY, DATA AND DOMAIN MEANING

GLOSSARY ENTRIES D-1

Term	Definition in this book	Primary anchor
Dead-letter exchange	A broker exchange that receives messages rejected or expired under declared topology. It supports quarantine routing; it is not itself an investigation workflow.	Ch 25, 29
Dead-letter queue	A bounded queue for messages that exhausted delivery policy or were explicitly quarantined. Replay requires authorization and a new receipt.	Ch 25, 29; App E
Deduplication	Recognition that an already-admitted message or idempotency key must not create another local effect. It is scoped to a named identity and owner.	Ch 23, 28
Delivery attempt	One broker or dispatcher attempt with its own timing and outcome under a stable message identity. Attempts are not new business facts.	Ch 27, 29
DEMONSTRATED	A claim state reached only when admitted evidence from the declared demonstration profile satisfies the bounded business oracle. It is narrower than production readiness.	Epilogue; App G-H

D - DELIVERY, DATA AND DOMAIN MEANING - CONTINUED

GLOSSARY ENTRIES D-2

Term	Definition in this book	Primary anchor
Dependency direction	The rule that source dependencies point toward domain and application policy, while infrastructure and edge implement outward concerns.	Ch 8-9
Determinism	The property that the same admitted inputs and policy identity produce the same decision. It does not imply that distributed timing is deterministic.	Ch 2, 9, 20
Distributed transaction	A transaction spanning multiple owners or resources. The final design does not use one; it combines local transactions with messages, idempotency and reconciliation.	Ch 18, 26, 40
Domain event	An owner-internal record that something meaningful occurred inside an aggregate or local domain model. It is not automatically published outside the owner.	Ch 19
Domain invariant	A rule that must be true before and after every accepted aggregate transition, regardless of API, persistence or message transport.	Ch 2, 9, 15

D - DELIVERY, DATA AND DOMAIN MEANING - CONTINUED

GLOSSARY ENTRIES D-3

Term	Definition in this book	Primary anchor
Dossier	The immutable collection of plan, identity, command, check, runtime, failure, cleanup, Missing and residual artifacts for one RunId.	Ch 54, 60; App G

E - EVIDENCE, EVENTS AND EXECUTION STATES

GLOSSARY ENTRIES E-1

Term	Definition in this book	Primary anchor
Edge	The externally facing boundary that adapts operator/browser interaction to owner contracts. In the final core this is the Query BFF, not a domain owner.	Ch 22, 44; App B
Effect	A committed observable change inside one owner, such as state mutation, Inbox admission or Outbox insertion. Intent and delivery are not effects by themselves.	Ch 9, 18, 28
Envelope	The stable transport metadata surrounding a versioned public fact, including message, correlation, causation, producer, schema and occurrence identities.	Ch 20; App C
Epistemic honesty	The discipline of saying exactly what is designed, implemented, executed, demonstrated, falsified or still unknown, with residual limits attached.	Epilogue; App H
Eventual consistency	A model in which independent owners may temporarily disagree but converge through declared messages, retries, repair and observable freshness.	Ch 30, 35, 41-42

FROM SERVICES TO RUNTIME

E - EVIDENCE, EVENTS AND EXECUTION STATES - CONTINUED

GLOSSARY ENTRIES E-2

Term	Definition in this book	Primary anchor
Evidence	A retained, integrity-checked observation tied to an exact check, source identity and RunId. Expected output and prose are not evidence.	Ch 48, 54; App G-H
Evidence manifest	The machine-verifiable summary joining required artifacts, identities, claim verdicts, Missing coordinates and residuals for one run.	Ch 60; App G-H
Exactly-once local effect	The target achieved when duplicate delivery is admitted once through Inbox identity and one owner transaction commits one effect. It is not exactly-once transport.	Ch 28, 30
EXECUTED	A claim state indicating that the exact bound check ran completely under admitted identity and produced a retained result, whether passing or falsifying.	Epilogue; App H
EXTEND	A material class requiring an inherited asset to be adapted while predecessor identity, semantic change and compatibility remain traceable.	App H

F - FAILURE, FINAL-CORE TRUTH AND FRESHNESS

GLOSSARY ENTRIES F-1

Term	Definition in this book	Primary anchor
Fail closed	A failure mode that denies admission or authority when identity, configuration, contract or evidence cannot be validated, instead of silently weakening policy.	Ch 10, 43-48; App F
Failure campaign	A controlled sequence of baseline, disturbance, bounded observation, recovery, cleanup and residual capture under one scenario and RunId.	Ch 54; App E, G
Failure domain	The set of components likely to fail together because they share process, host, network, store, broker, credential or operational authority.	Ch 3, 53-55
FALSIFIED	A claim state assigned when retained relevant evidence contradicts the bounded claim. The failed attempt remains part of history.	Epilogue; App H
Final core	The authoritative Phase-0 service, role, data and contract census admitted for the final demonstrator. Candidate designs remain outside it.	Ch 60; App B-C, H

F - FAILURE, FINAL-CORE TRUTH AND FRESHNESS - CONTINUED

GLOSSARY ENTRIES F-2

Term	Definition in this book	Primary anchor
Freshness	The age and progress information that tells an operator how far a read model may lag owner truth. It must be visible with the projection result.	Ch 35, 41, 52

G - GOLDEN PATHS AND GROUND TRUTH

GLOSSARY ENTRIES G-1

Term	Definition in this book	Primary anchor
Golden path	The repeatable service and delivery pattern chosen for common NEXUS-1 work, including layered projects, host composition, tests, configuration and evidence hooks.	Ch 7-12
Graceful degradation	A bounded response that preserves truthful partial service when a dependency is unavailable. It must expose missing or stale information rather than fabricate completeness.	Ch 22, 41, 53
Ground truth	The authoritative observation for a particular claim. For business state it is owner truth; for deployment identity it is the admitted rendered/runtime inventory.	Ch 35-36, 55, 60

H - HEALTH, HOSTS AND HUMAN REVIEW

GLOSSARY ENTRIES H-1

Term	Definition in this book	Primary anchor
Health check	A bounded observation used for startup, liveness or readiness. It must state which meaning it serves and must not perform business repair.	Ch 11
High-water mark	A monotonic projection coordinate showing the latest admitted source position or event processed by a read model.	Ch 35, 41-42
Host	The process composition boundary that registers services, endpoints, workers, health checks, configuration and shutdown behavior.	Ch 7-8, 11
Human review	An explicit workflow state in which automated progress pauses for authorized operator judgment under deadline, evidence and audit rules.	Ch 39

FROM SERVICES TO RUNTIME

I - IDENTITY, IDEMPOTENCY AND INVARIANTS

GLOSSARY ENTRIES I-1

Term	Definition in this book	Primary anchor
Idempotency	The property that repeated admission of the same declared operation identity produces the same accepted result without another business effect.	Ch 23, 28-30
Idempotency key	A client-supplied operation identity scoped to one route, owner and request meaning. Reuse with conflicting content must be rejected.	Ch 23; App C
Identity digest	A cryptographic digest over canonicalized identity fields used to detect that source, configuration, contract, plan or trace meaning changed.	Ch 10, 20; App G-H
IMPLEMENTED	A claim state showing that the exact canonical asset exists under the admitted source identity. It does not show that its check executed.	Epilogue; App H
INACTIVE	A disposition and claim state for a described design that receives no final-core runtime authority. Absence is verified across roles, routes, stores, credentials and topology.	App B, H

I - IDENTITY, IDEMPOTENCY AND INVARIANTS - CONTINUED

GLOSSARY ENTRIES I-2

Term	Definition in this book	Primary anchor
Inbox	An owner-local persistence record that atomically admits a message identity with its resulting effect, preventing duplicate redelivery from repeating that effect.	Ch 28
Integration event	A versioned owner-public message designed for external consumers. It is derived from committed owner meaning and is distinct from a domain event.	Ch 19-20; App C
Invariant	A condition that all accepted transitions inside a declared consistency boundary must preserve. Tests challenge both success and refusal paths.	Ch 2, 9, 15
Isolated lease	A bounded allocation of provider, broker, network and runtime resources owned by one test or reproduction attempt and cleaned by exact inventory.	Ch 12, 54-55; App G

J - JOIN SEMANTICS AND HTTP FAILURE LANGUAGE

GLOSSARY ENTRIES J-1

Term	Definition in this book	Primary anchor
Join key	A stable typed identity used to relate source, claim, asset, check, run and evidence without ambiguous name or timestamp matching.	App G-H
JSON Problem Details	The standardized HTTP error representation used to expose machine-readable type, status, title and bounded extensions without leaking internals.	Ch 23; App C

K - KUBERNETES AS A TRANSLATION TARGET

GLOSSARY ENTRIES K-1

Term	Definition in this book	Primary anchor
Kubernetes translation	The educational mapping of already-proven Compose/runtime needs into manifests, probes, identities and policies. It is not claimed as a deployed platform.	Ch 59

L - LEASES, LIVENESS AND LOCAL AUTHORITY

GLOSSARY ENTRIES L-1

Term	Definition in this book	Primary anchor
Lease	A time-bounded claim on work or resources that can expire and be safely recovered. Lease expiry must not create a second non-idempotent effect.	Ch 27, 42
Least privilege	The rule that each workload, migrator, operator and evidence identity receives only the actions and resources required for its bounded role.	Ch 13, 43-48; App F
Linearized PDF	A PDF organized for fast web preview while preserving searchable vector text, internal links, page identity and visual rendering.	Final readiness checkpoint
Liveness	A process-local answer to whether the host can continue executing. It must not fail merely because a remote dependency is unavailable.	Ch 11
Local transaction	One atomic commit inside one service-owned database. Messages and remote owner changes occur outside this boundary.	Ch 18, 26, 28

FROM SERVICES TO RUNTIME

L - LEASES, LIVENESS AND LOCAL AUTHORITY - CONTINUED

GLOSSARY ENTRIES L-2

Term	Definition in this book	Primary anchor
Long-running role	A continuously hosted API, worker, projection or edge process. The final core has exactly seven; migrators are finite jobs, not long-running roles.	Ch 8, 55, 60; App B

M - MESSAGES, MIGRATIONS AND MATERIAL CLASSES

GLOSSARY ENTRIES M-1

Term	Definition in this book	Primary anchor
Material class	One of BASE, EXTEND, NEW or VERIFY, describing the kind of technical work a chapter requires rather than its completion state.	App H
Message identity	The stable identifier of one public fact instance across dispatcher attempts, broker redelivery, Inbox admission, telemetry and evidence.	Ch 20, 27-30
Message ordering	The limited sequence guarantees available within a declared key, queue and consumer design. The system never assumes one global order.	Ch 30
Migration principal	A finite privileged identity allowed to apply one owner's schema migrations. Runtime API and worker principals do not receive DDL authority.	Ch 13, 16, 47; App F
Migrator	A finite owner-specific job that validates and applies one independent migration history before runtime readiness. The final core has three migrators.	Ch 16, 55; App B

M - MESSAGES, MIGRATIONS AND MATERIAL CLASSES - CONTINUED

GLOSSARY ENTRIES M-2

Term	Definition in this book	Primary anchor
Missing	The explicit collection of required source, asset, check or evidence coordinates not satisfied by a run. Missing cannot be averaged away by green results elsewhere.	App G-H
Monotonic projection	A read model whose admitted source progress never moves backward for the same projection identity, even across duplicate or out-of-order delivery.	Ch 35, 42
mTLS	Mutual TLS that authenticates both endpoints and protects transport. Workload identity and authorization still require explicit mapping and negative tests.	Ch 45, 47; App F

N - NEGATIVE AUTHORITY AND NON-CLAIMS

GLOSSARY ENTRIES N-1

Term	Definition in this book	Primary anchor
Negative authority	Executable evidence that an identity or inactive component cannot perform an action, access a resource or acquire a route that the architecture denies.	Ch 13, 43-48; App F-H
NEW	A material class requiring a new canonical code, schema, contract, diagram or policy asset together with its owning path and check.	App H
NEXUS-1 Phase-0	The educational distributed demonstrator used by this volume. It is not connected to a real facility and is not safety-class operational software.	Front matter; Ch 60
Non-claim	A statement that explicitly excludes an interpretation the evidence cannot support, such as production readiness, capacity, global exactly-once or deployed Kubernetes.	Epilogue; App H

O - OBSERVABILITY, OPERATORS AND OWNERSHIP

GLOSSARY ENTRIES O-1

Term	Definition in this book	Primary anchor
Observability	The capability to explain runtime behavior through correlated logs, traces, metrics and state evidence. It supports diagnosis but does not decide business truth.	Ch 49-52
One-writer rule	At any migration stage, exactly one authority may commit the owned business state for a capability. Shadow comparison and dual reads do not grant a second writer.	Ch 13, 56, 58
OpenTelemetry	The instrumentation and semantic context used for distributed traces, metrics and log correlation across HTTP, broker, database and background work.	Ch 51
Operator	An authenticated human using the Query BFF and owner contracts under local authorization. Operator identity is not forwarded as workload identity.	Ch 22, 39, 44-46
Outbox	An owner-local table written atomically with business state and later dispatched to the broker with confirm, lease and recovery semantics.	Ch 26-27

FROM SERVICES TO RUNTIME

O - OBSERVABILITY, OPERATORS AND OWNERSHIP - CONTINUED**GLOSSARY ENTRIES O-2**

Term	Definition in this book	Primary anchor
Owner	The single bounded authority responsible for a domain model, writes, database, migrations and public contract production. The final core has three owners.	Ch 13; App B
Owner truth	The authoritative committed state inside a service-owned database. Projections and BFF composites are delayed views and must expose freshness.	Ch 35, 41-42

P - PENDING WORK, PROJECTIONS AND PUBLIC FACTS**GLOSSARY ENTRIES P-1**

Term	Definition in this book	Primary anchor
PENDING	A visible status showing that required implementation, execution or evidence has not yet been established. It is an honest result, not unfinished formatting.	Epilogue; App G-H
Poison message	A message that repeatedly fails deterministic admission or handling under the declared contract and policy, requiring bounded quarantine and diagnosis.	Ch 29
Policy digest	A digest over canonicalized non-secret policy fields used to compare environments and bind a check to the exact decision rules it exercised.	Ch 10; App D, G
Problem Details	See JSON Problem Details. The book uses a stable type and bounded extensions to keep HTTP failures compatible and non-secret.	Ch 23; App C
Process manager	A durable coordinator model for workflow decisions, deadlines and compensations. It remains DESIGN_INACTIVE in the final Phase-0 runtime.	Ch 37-40; App B, H

P - PENDING WORK, PROJECTIONS AND PUBLIC FACTS - CONTINUED**GLOSSARY ENTRIES P-2**

Term	Definition in this book	Primary anchor
Projection	A read-optimized view built from admitted public facts. It may lag owner truth and therefore carries high-water and freshness information.	Ch 35, 41-42
Projection rebuild	A controlled recreation of read-model state from an admitted source corpus under an isolated identity, without changing owner truth.	Ch 35, 42
Provider-backed test	A component test against the actual database provider semantics used by the design, including constraints, transactions, migration behavior and concurrency.	Ch 12, 14-18
Public fact	A versioned integration event representing a committed owner truth. The final core activates exactly three public facts.	Ch 19-20; App C
Publisher confirm	The broker acknowledgement that a published message reached the broker under the configured publish path. It does not prove consumer effect.	Ch 27

Q - QUERIES, QUARANTINE AND THE QUERY BFF**GLOSSARY ENTRIES Q-1**

Term	Definition in this book	Primary anchor
Quarantine	The controlled isolation of a poison or exhausted message with reason, identity and authorization requirements for investigation and replay.	Ch 29; App E
Query	A request for information that does not command a business state change. Its response may include freshness, partial-result or degraded-state metadata.	Ch 21-22, 35, 41
Query BFF	The final-core edge role query-bff that authenticates browser sessions and composes bounded operator reads without owning workflow or persistence.	Ch 22, 41, 44; App B-C

R - READINESS, RECOVERY, RECONCILIATION AND RETRIES**GLOSSARY ENTRIES R-1**

Term	Definition in this book	Primary anchor
Readiness	A bounded answer to whether a role should receive its intended work now, based on startup completion and only its required dependencies.	Ch 11, 55
Reauthentication	A fresh operator authentication requirement before a sensitive action even when an existing session is otherwise valid.	Ch 44; App F
Reconciliation	Detection of divergence between independently committed owner, broker and projection observations, followed by a bounded plan to explain or repair it.	Ch 42
Redelivery	A broker delivery of the same message identity after rejection, timeout, channel loss or consumer failure. It must not create a second local effect.	Ch 28-30
Repair	A controlled, authorized action derived from a reconciliation finding. Repair adds explicit history and evidence; it never silently rewrites owner truth.	Ch 42; App E

FROM SERVICES TO RUNTIME

R - READINESS, RECOVERY, RECONCILIATION AND RETRIES - CONTINUED

GLOSSARY ENTRIES R-2

Term	Definition in this book	Primary anchor
Repair Control	The proposed approval and actuation boundary for repairs. It remains DESIGN_INACTIVE with no final-core route, host, principal or actuator.	Ch 42; App B, H
Replay	An authorized new attempt to deliver a quarantined or retained message under a recorded plan. Replay is distinct from automatic redelivery.	Ch 29-30; App E
Reproduction identity	The digest-bound set of source, dependency, tool, policy, contract, deployment and scenario inputs shared by all stages of one reproducible run.	Ch 55, 57, 60; App G
Residual	A known limitation or uncertainty that survives a check and restricts the interpretation of its claim. Residuals remain attached to evidence and verdicts.	Epilogue; App G-H
Retry	A new invocation attempt initiated by caller or policy after a qualifying failure. It is safe only when operation semantics, budget and idempotency permit it.	Ch 29, 53

R - READINESS, RECOVERY, RECONCILIATION AND RETRIES - CONTINUED

GLOSSARY ENTRIES R-3

Term	Definition in this book	Primary anchor
Retry budget	The total bounded attempts and elapsed time allowed for retries inside a larger request, shutdown or recovery budget.	Ch 29, 39, 53
RootCause	The final-core owner of RootCauseCase, evidence, hypothesis, verdict, RootCauseDb and publication of case-opened and verdict-issued public facts.	Ch 15, 33-34; App B
RootCauseCase	The aggregate that controls case identity, evidence, hypotheses, verdict and closure invariants inside the RootCause owner.	Ch 9, 15, 33-34
RootCauseDb	The RootCause-owned database migrated by rootcause-migrator and written by RootCause runtime roles, including Inbox, case state and Outbox boundaries.	Ch 15-16, 18, 28
RunId	The unique identifier for one reproduction or failure attempt. Every admissible receipt and artifact resides under and binds back to this identity.	App G-H

R - READINESS, RECOVERY, RECONCILIATION AND RETRIES - CONTINUED

GLOSSARY ENTRIES R-4

Term	Definition in this book	Primary anchor
Runtime role	A named deployable process with bounded identity, configuration, health, dependency and authority. Role count is closed by the final catalogue.	Ch 55, 60; App B

S - SERVICES, STATE AND SCHEMA EVOLUTION

GLOSSARY ENTRIES S-1

Term	Definition in this book	Primary anchor
Same-run evidence	Evidence whose source, check, runtime and artifact identities all bind to the same RunId and reproduction identity, with no borrowed receipt from another attempt.	Ch 54, 60; App G-H
Schema evolution	A compatible sequence of additive, transitional and removal changes coordinated across producers, consumers, stores and deployment windows.	Ch 24, 30, 58
Secret reference	A configuration value that identifies how authorized runtime code obtains secret material without placing the material itself in source, manifests, logs or evidence.	Ch 10, 47; App D, F
Semantic conflict	A stable application or HTTP outcome showing that an attempted transition conflicts with current owner state, identity or idempotency history.	Ch 18, 21, 23
Semantic retry	A retry that creates a fresh unit of work, rereads current owner state and reevaluates policy after a recognized transient provider failure.	Ch 14, 18, 53

FROM SERVICES TO RUNTIME

S - SERVICES, STATE AND SCHEMA EVOLUTION - CONTINUED

GLOSSARY ENTRIES S-2

Term	Definition in this book	Primary anchor
Service	A deployable boundary with owned behavior, data, contracts, runtime identity and operations. The book does not equate every class, schema or bounded context with a service.	Ch 3, 6-9, 13
Service catalogue	The closed authority listing owners, roles, stores, dependencies, contracts and inactive candidates for the final runtime.	Ch 60; App B
Service-owned database	A database whose schema, writes, migration history and runtime grants belong to exactly one service owner. Other owners use contracts, not cross-database access.	Ch 13-18
SLO (Service Level Objective)	A measurable target over a defined service indicator and window. This manuscript defines candidate objectives but does not claim production attainment.	Ch 52
SPECIFIED	A claim state showing that bounded meaning, intended asset, check and residuals are defined. It does not assert repository presence or execution.	Epilogue; App H

S - SERVICES, STATE AND SCHEMA EVOLUTION - CONTINUED

GLOSSARY ENTRIES S-3

Term	Definition in this book	Primary anchor
Startup	The finite initialization phase that validates configuration and establishes required local conditions before a role may become ready.	Ch 11
Strangler migration	Incremental extraction that routes one bounded capability at a time to a new owner while coexistence, one-writer rules, comparison and reversal remain explicit.	Ch 56
Structured logging	Logging through named, typed, allowlisted fields that preserve domain meaning while controlling secrets, personal data and unbounded cardinality.	Ch 50
SUPERSEDED	A state for an identity replaced by one canonical successor and retained as a tombstone. Superseded paths must not remain active aliases.	App A, G-H

T - TIME, TRACES, TRANSACTIONS AND TRUST

GLOSSARY ENTRIES T-1

Term	Definition in this book	Primary anchor
Timeout	A local upper bound on waiting for one operation. It must fit inside caller, retry, workflow, shutdown and recovery budgets and does not cancel remote effects automatically.	Ch 39, 53
Tombstone	A retained record that an old path, identifier or decision is no longer authoritative, including its replacement and non-reuse rule.	App A, F-H
Trace	A distributed observation composed of spans and links. It explains execution shape but does not by itself prove a business invariant or causal truth.	Ch 49, 51
Trace context	The propagated identifiers and flags used to relate spans across HTTP, broker and background work without placing business decisions inside telemetry metadata.	Ch 49, 51
Traceability	The typed relationship from source to bounded claim, canonical asset, check, same-run evidence, derived state, disposition and residual.	App H

T - TIME, TRACES, TRANSACTIONS AND TRUST - CONTINUED

GLOSSARY ENTRIES T-2

Term	Definition in this book	Primary anchor
Transaction boundary	The exact state and records committed atomically by one owner operation. Remote calls and broker delivery remain outside it.	Ch 18, 26, 28
Transactional outbox	The Outbox pattern in which business state and a pending public fact record commit in one owner-local transaction before asynchronous dispatch.	Ch 26
Trust boundary	A crossing where identity, authority, data classification, protocol or operational control changes and therefore requires explicit protection and tests.	Ch 43; App F
Trust zone	A set of components operating under the same relevant trust assumptions. Zone membership does not remove the need to validate boundary crossings.	Ch 43; App F
Typed identity	A domain or contract-specific identifier type that prevents accidental interchange of semantically different IDs even when their wire representation is similar.	Ch 9, 17, 20

FROM SERVICES TO RUNTIME

U - UBIQUITOUS LANGUAGE AND UNITS OF WORK

GLOSSARY ENTRIES U-1

Term	Definition in this book	Primary anchor
Ubiquitous language	The stable vocabulary shared by domain experts, engineers, contracts and tests inside a bounded context. This glossary freezes the cross-book terms used by the runtime volume.	Ch 2, 9, 13
Unit of work	The owner-local scope that tracks one transaction's reads, changes and commit. A semantic retry must create a new unit of work.	Ch 14, 18
Unknown outcome	A state in which the caller cannot yet establish whether a remote or broker operation completed. Recovery uses identity and reconciliation rather than blind repetition.	Ch 27, 30, 53

V - VALIDATION, VERSIONS AND VERDICTS

GLOSSARY ENTRIES V-1

Term	Definition in this book	Primary anchor
Value-free manifest	A manifest that records configuration keys, sources, types, policy and material-reference metadata without recording secret or environment-specific values.	Ch 10; App D, G
Verdict	The RootCause aggregate's evidence-constrained conclusion. A final verdict requires the declared evidence and invariant conditions and may be replaced only by explicit history.	Ch 9, 15, 34
VERIFY	A material class requiring an executable check against current source, provider, broker, host or deployment semantics, with same-run evidence or explicit PENDING.	App H
Versioned contract	A public HTTP or message schema with explicit compatibility policy, stable identity and corpus tests across supported producer and consumer versions.	Ch 20, 23-24, 30

W - WORKERS, WORKLOADS AND ONE-WRITER MIGRATION

GLOSSARY ENTRIES W-1

Term	Definition in this book	Primary anchor
Worker	A long-running host role that consumes messages, dispatches Outbox records or builds projections using fresh scopes, bounded batches and cancellation-aware shutdown.	Ch 8, 27-28, 35
Working index	The forthcoming navigational map from concepts, contracts, roles, threats, commands and aliases to their principal pages. It follows this glossary and the notation/sources sections.	Final back matter
Workload identity	The direct identity of a running service role, normally mapped from mTLS certificate or platform identity. It is distinct from operator claims.	Ch 45-46; App F

Z - ZERO-DISCOVERY AND ZERO-TRUST GATES

GLOSSARY ENTRIES Z-1

Term	Definition in this book	Primary anchor
Zero-discovery gate	A test-runner guard that fails when required tests, cases, contracts or scenarios were not discovered, preventing an empty run from appearing green.	Ch 12, 24, 57; App G
Zero-trust stance	The assumption that network location alone grants no trust; every boundary requires explicit identity, authorization, protection and negative verification.	Ch 43-48; App F

TERM/USE/LINK AUDIT

Every entry is bounded and navigable

GLOSSARY ACCEPTANCE GL-5

Gate	Required result
uniqueness	no duplicate preferred term under case-insensitive comparison
definition	nonempty NEXUS-1-specific meaning; no circular entry-only definition
primary anchor	one valid chapter, appendix, front-matter or final-back-matter target
link	every term cell has an internal destination inside the same PDF
architecture	definitions preserve final-core counts and inactive-candidate boundaries
claim language	no PENDING work promoted to implemented, executed or demonstrated
identity	canonical spellings and tombstones remain distinct

GLOSSARY CHECKPOINT

The book now has one stable language for its final audit

COMPLETION LEDGER GL-6

Question	Answer
What does the term mean here?	one bounded definition
Where is it engineered?	one primary internal link
Which similar term differs?	collision register and explicit non-equivalence
Does it activate architecture?	no; final catalogues remain authoritative
Does it prove execution?	no; evidence state remains independently derived
What comes next?	Mathematical and Timing Notation Used in the Book

FINAL GLOSSARY RULE

A stable term reduces ambiguity only when its boundary is preserved in code, contracts, tests, evidence and operator language. Vocabulary without enforcement is documentation drift.

GLOSSARY LAST WORD

Name the boundary before asking the system to defend it

FINAL STATUS

GLOSSARY COMPLETE | 176 CONTROLLED TERMS | 24 ALPHABETICAL GROUPS | INTERNAL PRIMARY-ANCHOR LINKS
| FINAL-CORE AND CLAIM-STATE LANGUAGE PRESERVED

FINAL REFERENCE BACK MATTER

Mathematical and Timing Notation Used in the Book

The canonical symbol, unit, clock, ordering, metric and evidence notation for the completed NEXUS-1 runtime volume.

NOTATION AUTHORITY

A formula defines a model or check contract. It becomes an observation only when its inputs, units, clock domain, identity and retained run evidence are admitted. This section does not promote PENDING implementation or runtime claims.

NOTATION CONTROL TOTALS NT-1

Coordinate	Value
stable notation entries	140
notation families	12
clock domains	UTC wall clock / process-local monotonic clock
ratio domain	[0,1]; percent = 100 * ratio
attempt indexing	zero-based; upper bound excluded
final-core constants	unchanged: 3 owners / 1 edge / 7 roles / 3 stores / 3 facts / 6 HTTP

READING AND PRECEDENCE RULES

Meaning comes from scope, domain and unit

EXPRESSION CONTRACT NT-2

Rule	Required interpretation
parentheses	evaluate inner expression first
multiplication/division	before addition/subtraction
comparison	after arithmetic and unit conversion
! then && then	Boolean precedence used by code-oriented expressions
left-to-right	only among operators of equal precedence unless policy says otherwise
declared domain	an identifier, instant, duration, count and ratio are never interchangeable

ASCII SOURCE RULE

The source notation uses `:=`, `=>`, `<=>`, `<=`, `>=`, `!=`, `&&` and `||` so PDF extraction, code review and cross-platform reproduction retain the same characters.

CLOCK SELECTION

Use wall time for facts and monotonic time for elapsed budgets

CLOCK MATRIX NT-3

Question	Use	Do not use
When did a public fact occur?	UTC <code>t_occurred</code>	process stopwatch ticks
How long did local work take?	monotonic <code>m1-m0</code>	UTC subtraction after clock adjustment
Has an absolute credential expired?	UTC instant with <code>epsilon_clock</code>	process uptime
Is a local lease still valid?	one monotonic clock domain	cross-host monotonic values
What is projection freshness?	source UTC age plus skew assumptions	queue depth alone
Did recovery meet its bound?	monotonic elapsed plus business oracle	health timestamp only

CLOCK NON-EQUIVALENCE

UTC establishes externally meaningful instants but may jump; monotonic time measures local elapsed duration but cannot be compared across processes without an explicit synchronization model.

TIMING DISTINCTIONS

Timeout, deadline and budget answer different questions

TIMING COLLISION REGISTER NT-4

Term	Answers	Canonical form
duration	how much time	$d \geq 0$
timeout	maximum wait for one attempt	T_i
deadline	latest absolute completion instant	$t_0 + B$
budget	total elapsed allowance for an operation family	B_{total}
backoff	delay before another eligible attempt	D_i
TTL	lifetime from issue/creation	duration, not instant
freshness age	how old the represented source truth is	$t_{now} - t_{source}$

DIMENSIONAL CONSISTENCY

Operators cannot repair incompatible units

DIMENSION GATES NT-5

Expression	Result
instant + duration	instant
instant - instant	duration when clocks are comparable
duration + duration	duration after unit conversion
count / duration	rate
rate * duration	expected count under stated assumptions
good / eligible	dimensionless ratio
duration + count	INVALID_DIMENSION
UTC instant - monotonic tick	INVALID_CLOCK_DOMAIN

FAIL CLOSED

A check with a missing unit, clock domain, aggregation window or identity is incomplete. The verifier must reject the result rather than assume a convenient interpretation.

ILLUSTRATIVE DERIVATIONS

Equations expose the budget that prose can hide

ILLUSTRATIVE RETRY BUDGET - NOT CAPTURED RUNTIME OUTPUT

```

B_total = 12 s
n_max = 3 attempts
T = [2 s, 2 s, 2 s]
D = [0 s, 0.5 s, 1 s]
overhead_margin = 1 s

required = sum(T) + sum(D) + overhead_margin
          = 6 s + 1.5 s + 1 s
          = 8.5 s

required <= B_total      // 8.5 s <= 12 s
remaining_margin = 3.5 s

```

INTERPRETATION

The arithmetic shows only that the declared allocations fit. It does not prove the dependency will recover, that three attempts are safe, or that the production latency objective is met.

ILLUSTRATIVE CONVERGENCE AND EVIDENCE

A predicate names the oracle without pretending it ran

ILLUSTRATIVE PREDICATES

```

projection_current(k) :=
  h_P >= h_required
  && age <= F_max
  && Diff(0_k, P_k) == empty

claim_executed(c, r) :=
  Implemented(c)
  && Run(check(c)) == r
  && RequiredTests(c) == ExecutedTests(c, r)
  && digest(evidence(c, r)) == manifest.digest

```

EVIDENCE BOUNDARY

A well-formed predicate is a specification. It advances state only when an admitted implementation evaluates it using retained same-run inputs.

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NT-01 - OPERATORS AND LOGICAL STATEMENTS

Editorial equations use ASCII operators so source, code and PDF extraction preserve the same identity.

NOTATION ENTRIES NT-01.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-001 $x := y$	Define x to mean y in the current scope; this is not an equality observation.	definition	Ch 2; Glossary
N-002 $x = y$	Mathematical equality of values after canonical interpretation.	predicate	Ch 2, 20
N-003 $x == y$	Executable equality in C#-oriented listings or pseudocode.	Boolean	Ch 9, 20
N-004 $x != y$	Executable or tabular inequality; the two values must not compare equal.	Boolean	Ch 9, 20
N-005 $P \Rightarrow Q$	Implication: whenever predicate P holds, Q must hold; the reverse is not implied.	logic	Ch 2, 9
N-006 $P \Leftrightarrow Q$	Equivalence: P and Q imply each other under the declared model.	logic	Ch 2, 9

NT-01 - OPERATORS AND LOGICAL STATEMENTS - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-01.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-007 $P \&\& Q$	Logical conjunction; both predicates are required.	Boolean	Ch 9, 12
N-008 $P \ \ Q$	Logical disjunction; at least one predicate is required unless stated exclusive.	Boolean	Ch 9, 12
N-009 $!P$	Logical negation of predicate P.	Boolean	Ch 9, 12
N-010 forall x in S: P(x)	Universal requirement over every admitted member of set S.	logic/set	Ch 12; App H
N-011 exists x in S: P(x)	Existence requirement for at least one admitted member of S.	logic/set	Ch 12; App H
N-012 clamp(x, lo, hi)	Return min(max(x, lo), hi); used only when bounded clipping is part of policy.	same as x	Ch 10, 53

NT-02 - SETS, SEQUENCES AND MAPPINGS

Closed inventories, ordered attempts and typed lookups use different collection notation.

NOTATION ENTRIES NT-02.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-013 S	A finite set unless the surrounding text explicitly defines an infinite domain.	set	App B-H
N-014 S	Cardinality: the number of distinct members in set S.	count ≥ 0	Ch 60; App B-H
N-015 empty	The empty set with no admitted members.	set	App H
N-016 A subset B	Every member of A is also a member of B; equality is allowed.	set relation	Ch 24, 43
N-017 A union B	Members present in A, B or both.	set	Ch 12, 43
N-018 A intersect B	Members present in both A and B.	set	Ch 12, 43

NT-02 - SETS, SEQUENCES AND MAPPINGS - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-02.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-019 A minus B	Members present in A but not B.	set	Ch 12; App H
N-020 x[0..n-1]	A zero-based ordered sequence with n elements; order is meaningful.	sequence	Ch 27-30
N-021 len(x)	Number of elements in a sequence, including duplicates.	count ≥ 0	Ch 12, 30
N-022 $K \rightarrow V$	A mapping from a key domain K to value domain V; uniqueness applies to keys.	map	Ch 10, 20
N-023 (a, b, c)	An ordered tuple whose position and type define meaning.	product type	Ch 9, 20
N-024 Expected == Actual	Closed-inventory equality; no expected item missing and no unexpected item present.	set predicate	Ch 12, 55; App H

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NT-03 - IDENTITY AND CANONICALIZATION

An identifier names an object; a digest binds exact canonical content. Neither substitutes for the other.

NOTATION ENTRIES NT-03.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-025 OwnerId	Typed identity of one service owner; final values are AlarmManagement, RootCause and Reporting.	identifier	Ch 13; App B
N-026 RunId	Identity of one reproduction or failure attempt and its evidence root.	identifier	Ch 54, 60; App G
N-027 MessageId	Stable identity of one public fact instance across publish attempts and redeliveries.	identifier	Ch 20, 27-30
N-028 CorrelationId	Identifier grouping observations from one scenario or request family; it does not assert causation.	identifier	Ch 49
N-029 CausationId	Identifier of the direct operation or message that caused the current public fact or activity.	identifier	Ch 20, 49
N-030 IdempotencyKey	Client operation identity scoped to a route, owner and canonical request meaning.	identifier	Ch 23; App C

NT-03 - IDENTITY AND CANONICALIZATION - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-03.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-031 TraceId / SpanId	Telemetry identifiers for one trace and one span; they are not business identifiers.	identifier	Ch 49, 51
N-032 ClaimId / TraceRowId	Stable identities for one bounded claim and one technical-material trace row.	identifier	Epilogue; App H
N-033 Canonical(x)	Deterministic byte or text representation of x under a declared schema and normalization policy.	bytes	Ch 10, 20; App G-H
N-034 H(x)	SHA-256 over Canonical(x), written as lowercase hexadecimal unless a schema states otherwise.	256-bit digest	Ch 10, 57; App G
N-035 TreeDigest(root)	Digest over a sorted, path-aware canonical tree inventory, not a hash of archive timestamps.	256-bit digest	Ch 57; App G
N-036 IdentityMatch(a,b)	True only when the identity schema, canonicalization version and digest all agree.	Boolean	App G-H

NT-04 - CLOCKS, INSTANTS AND DURATIONS

Wall-clock time orders external facts; monotonic time measures elapsed budgets inside a running process.

NOTATION ENTRIES NT-04.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-037 t	A UTC wall-clock instant used for externally meaningful timestamps.	UTC instant	Ch 20, 39, 49
N-038 m	A monotonic clock reading meaningful only inside one clock/process domain.	monotonic ticks	Ch 11, 39, 53
N-039 d	A non-negative duration; never a calendar timestamp.	time ≥ 0	Ch 11, 39, 53
N-040 $t_0 < t_1$	Wall-clock ordering when both instants use the same admitted time basis.	predicate	Ch 20, 39
N-041 elapsed := $m_1 - m_0$	Elapsed duration measured with one monotonic clock; wall-clock subtraction is not used for local budgets.	time ≥ 0	Ch 11, 39, 53
N-042 deadline := $t_0 + B$	Absolute UTC deadline formed from a start instant and duration budget B.	UTC instant	Ch 39, 53

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NT-04 - CLOCKS, INSTANTS AND DURATIONS - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-04.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-043 remaining := max(0,B-elapsed)	Budget left after monotonic elapsed time; child work may not exceed remaining.	time >= 0	Ch 39, 53
N-044 [a,b)	Half-open interval including a and excluding b; adjacent windows do not double-count their boundary.	interval	Ch 39, 52
N-045 epsilon_clock	Declared upper bound on accepted clock uncertainty or skew for cross-host comparisons.	time >= 0	Ch 45, 47, 49
N-046 t_occurred	Owner-assigned instant at which the represented domain fact occurred.	UTC instant	Ch 20; App C
N-047 t_published	Instant a producer dispatch attempt received broker confirmation; not occurrence time.	UTC instant	Ch 27, 49
N-048 t_received	Instant a consumer observed a delivery attempt; duplicates may have different values.	UTC instant	Ch 28-30, 49

NT-04 - CLOCKS, INSTANTS AND DURATIONS - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-04.3

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-049 t_processed	Instant the consumer committed the owner-local effect for the admitted message.	UTC instant	Ch 28, 35, 49
N-050 UTC ISO-8601	External timestamps render as ISO-8601 UTC with Z; display-local time is presentation only.	timestamp text	Ch 20, 49; App C

NT-05 - TIMEOUTS, BUDGETS, RETRIES AND LEASES

A timeout bounds one wait; a deadline bounds an absolute end; a budget bounds the complete operation family.

NOTATION ENTRIES NT-05.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-051 B_total	Total elapsed-time budget owned by the caller, workflow or shutdown operation.	time > 0	Ch 39, 53
N-052 T_i	Timeout allocated to attempt i; it must not exceed the remaining total budget.	time > 0	Ch 29, 53
N-053 D_i	Backoff delay before retry attempt i.	time >= 0	Ch 29, 53
N-054 J_i	Bounded jitter sampled or derived for attempt i under the declared policy.	time	Ch 29, 53
N-055 D_i=min(Dmax,D0*2^i)+J_i	Exponential-backoff family; exact jitter distribution and cap belong to the policy identity.	time	Ch 29, 53
N-056 i in [0,n_max)	Zero-based attempt index; n_max counts all allowed attempts, not retries after the first.	integer	Ch 29, 53

NT-05 - TIMEOUTS, BUDGETS, RETRIES AND LEASES - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-05.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-057 sum(T_i+D_i) <= B_total	Necessary retry-budget condition; processing overhead and cleanup margins must also fit.	time predicate	Ch 29, 39, 53
N-058 T_child <= remaining	A nested operation may not claim more time than its caller can still spend.	time predicate	Ch 39, 53
N-059 cancel_at <= deadline	Cancellation is requested no later than the operation deadline; remote effects still require reconciliation.	instant predicate	Ch 8, 39, 53
N-060 B_shutdown	Maximum host-stop budget covering admission stop, drain/relinquish and cleanup.	time > 0	Ch 8, 11
N-061 lease_valid := m_now < m_expiry	Local lease validity evaluated with one monotonic clock domain.	Boolean	Ch 27, 42
N-062 f_new > f_old	Fencing-token order required before a newer lease holder may supersede an older writer.	integer order	Ch 27, 42

FROM SERVICES TO RUNTIME

NT-06 - MESSAGING, CAUSATION AND LOCAL EFFECTS

Transport attempts may duplicate; business identity and owner-local admission define effect cardinality.

NOTATION ENTRIES NT-06.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-063 $a \rightarrow b$	a causally precedes b because b was produced from a; this is a partial, not global, order.	relation	Ch 20, 30, 49
N-064 $a \sim b$	a and b share correlation context; neither causal direction is implied.	relation	Ch 49
N-065 $\text{seq}(\text{owner}, \text{key})$	Monotonic owner-assigned sequence for one declared ordering key; there is no system-wide sequence.	integer ≥ 0	Ch 30
N-066 $\text{attempt}(\text{MessageId}, i)$	Delivery or dispatch attempt i for one stable public-fact identity.	attempt identity	Ch 27-30
N-067 $\text{duplicate}(a, b)$	True when two attempts carry the same MessageId under the same contract identity.	Boolean	Ch 28-30
N-068 $\text{Apply}(k, p, S)$	Apply operation key k and payload meaning p to owner state S under idempotency rules.	state transition	Ch 23, 28

NT-06 - MESSAGING, CAUSATION AND LOCAL EFFECTS - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-06.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-069 $\text{Apply}(k, p, \text{Apply}(k, p, S)) = \text{Apply}(k, p, S)$	Idempotent replay equation for the same key and canonical payload meaning.	state equality	Ch 23, 28
N-070 $H(p1) \neq H(p2)$ with same k	Conflicting reuse of one idempotency key; reject instead of returning the previous result.	conflict predicate	Ch 23
N-071 $ \text{effects}(\text{MessageId}) \leq 1$	At most one committed owner-local effect for all duplicate deliveries of one message identity.	cardinality	Ch 28, 30
N-072 owner state + Outbox	Atomic producer commit set for one public fact; broker delivery is outside the transaction.	local transaction	Ch 18, 26
N-073 Inbox + owner effect	Atomic consumer commit set for message admission and its local effect.	local transaction	Ch 28
N-074 confirm != consumer effect	Publisher confirmation proves broker acceptance only, not consumer admission or business convergence.	non-equivalence	Ch 27, 36

NT-07 - PROJECTION LAG AND EVENTUAL CONVERGENCE

Owner truth and read models are named separately so delayed truth never appears synchronous.

NOTATION ENTRIES NT-07.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-075 $O_k(t)$	Committed owner truth for key k at instant t.	owner state	Ch 35, 41-42
N-076 $P_k(t)$	Projection state for key k at instant t; it may lag $O_k(t)$.	read-model state	Ch 35, 41-42
N-077 h_O / h_P	Owner and projection high-water positions under the same declared source coordinate.	position ≥ 0	Ch 35, 41-42
N-078 $\text{lag_pos} := \max(0, h_O - h_P)$	Non-negative positional projection lag when the positions are comparable.	count	Ch 35, 41, 52
N-079 $\text{age} := \max(0, t_{\text{now}} - t_{\text{source}})$	Age of the newest source fact represented by a read model; clock uncertainty remains explicit.	time ≥ 0	Ch 35, 41, 52
N-080 $\text{fresh} := \text{age} \leq F_{\text{max}}$	Freshness predicate under threshold F_{max} for one query profile.	Boolean	Ch 35, 41, 52

NT-07 - PROJECTION LAG AND EVENTUAL CONVERGENCE - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-07.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-081 $h_P(\text{next}) \geq h_P(\text{now})$	Monotonic high-water requirement for one projection identity.	position predicate	Ch 35, 42
N-082 eventually $P_k = \text{Project}(O_k)$	Conditional convergence goal under delivery, repair and stable-policy assumptions; no fixed time is implied.	temporal predicate	Ch 35, 41-42
N-083 $\text{Diff}(O, P)$	Typed reconciliation difference between owner truth and the projection expected from it.	difference set	Ch 42
N-084 $ \text{writers}(\text{capability}, t) = 1$	One-writer migration invariant at every admitted cutover instant.	cardinality	Ch 13, 56, 58
N-085 partial != complete	A BFF response missing one dependency remains explicitly partial; it is never relabeled complete.	non-equivalence	Ch 22, 41

FROM SERVICES TO RUNTIME

NT-08 - METRICS, OBJECTIVES AND PROBABILITY

Metrics define calculations and windows; they do not assert production measurements that were not retained.

NOTATION ENTRIES NT-08.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-086 C(t)	Monotonic counter value at instant t; reset or process identity must be represented separately.	count	Ch 52
N-087 rate := $\Delta C / \Delta t$	Observed event rate over a declared interval; never infer from one instantaneous sample.	count/time	Ch 52
N-088 G(t)	Gauge sampled at t, such as current queue depth or active leases.	declared unit	Ch 52
N-089 H_bucket(x)	Cumulative histogram bucket count for observations $\leq x$ under one instrument identity.	count	Ch 52
N-090 mean(x) := $\sum(x_i)/n$	Arithmetic mean for $n > 0$; it does not describe tail latency.	same as x	Ch 52
N-091 q_p(x)	Empirical p-quantile under declared aggregation and window; p is in [0,1].	same as x	Ch 52

NT-08 - METRICS, OBJECTIVES AND PROBABILITY - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-08.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-092 availability := good/eligible	Ratio of good eligible observations to all eligible observations in one window.	ratio [0,1]	Ch 52
N-093 error_ratio := bad/eligible	Ratio of qualifying bad observations to eligible observations in one window.	ratio [0,1]	Ch 52
N-094 error_budget := 1-SLO	Allowed bad fraction implied by one ratio SLO; it is not a retry allowance.	ratio [0,1]	Ch 52
N-095 burn := observed_bad/error_budget	Error-budget burn ratio over a named window; interpretation requires nonzero budget.	dimensionless	Ch 52
N-096 L = $\lambda * W$	Little's Law for a stable observed system: average items L equals arrival rate λ times average time W.	count	Ch 52-53
N-097 utilization := busy/capacity	Fraction of declared capacity occupied; capacity identity and time window are required.	ratio [0,1]	Ch 52-53

NT-08 - METRICS, OBJECTIVES AND PROBABILITY - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-08.3

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-098 P(A)	Probability assigned to event A under a declared model or empirical estimator.	ratio [0,1]	Ch 36, 52
N-099 P(A B)	Conditional probability of A given B; it does not establish that B causes A.	ratio [0,1]	Ch 36, 49
N-100 frequency(A) := count(A)/n	Observed relative frequency in n admitted trials; distinguish it from model probability.	ratio [0,1]	Ch 36, 52
N-101 score in [0,1]	Normalized advisory score only when its calibration and semantics are declared; it is not authorization.	dimensionless	Ch 36

NT-09 - SECURITY AND VALIDITY WINDOWS

Identity validity uses wall-clock instants plus explicit skew and rotation assumptions.

NOTATION ENTRIES NT-09.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-102 $t_{\text{nbf}} \leq t < t_{\text{exp}}$	Token or credential is inside its half-open validity interval before skew handling.	instant predicate	Ch 44-47; App F
N-103 $t_{\text{nbf-eps}} \leq t < t_{\text{exp+eps}}$	Validation interval with explicit accepted clock skew eps; eps belongs to policy identity.	instant predicate	Ch 44-47; App F
N-104 cert.NotBefore / NotAfter	Certificate wall-clock validity bounds; successful time validation still requires chain, SAN and policy checks.	UTC instant	Ch 45, 47
N-105 overlap := old_valid intersect new_valid	Rotation interval in which both old and new material may be accepted under bounded policy.	time interval	Ch 47
N-106 TTL	Time-to-live duration from issue or creation under a named object and policy; not an absolute timestamp.	time > 0	Ch 44, 47
N-107 operator != workload	Operator session identity and direct service workload identity are separate principals.	non-equivalence	Ch 44-46; App F

FROM SERVICES TO RUNTIME

NT-09 - SECURITY AND VALIDITY WINDOWS - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-09.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-108 authorize(subject,action,resource)	Local policy decision over one authenticated subject, bounded action and resource identity.	Boolean decision	Ch 44-46
N-109 claims_min	Minimum delegated claims required for the local decision; extra claims are not harmless by default.	claim set	Ch 46

NT-10 - EVIDENCE PREDICATES AND CLAIM STATES

Evidence notation derives state; it never treats manuscript prose as runtime execution.

NOTATION ENTRIES NT-10.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-110 Specified(c)	Claim c has bounded meaning, source, intended asset/check and residuals.	Boolean	Epilogue; App H
N-111 Implemented(c)	The exact canonical asset for c exists under the admitted source identity.	Boolean	Epilogue; App H
N-112 Executed(c,r)	The bound check for c ran completely in run r with retained result and identity.	Boolean	Epilogue; App G-H
N-113 Demonstrated(c,r)	The declared demonstration oracle admitted c using same-run evidence from r.	Boolean	Epilogue; App G-H
N-114 Falsified(c,e)	Relevant retained evidence e contradicts the bounded claim c.	Boolean	Epilogue; App H
N-115 Supports(e,c)	Evidence e is relevant, identity-bound and sufficient for the part of c it is cited to support.	relation	App G-H

NT-10 - EVIDENCE PREDICATES AND CLAIM STATES - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-10.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-116 Run(e)=Run(check)=r	Same-run join required before e may support the result of check in run r.	identity predicate	App G-H
N-117 Missing(r)	Required coordinates not satisfied by run r; an empty set is necessary but not sufficient for success.	set	App G-H
N-118 Residuals(c,r)	Known limits that constrain interpretation of c after run r.	set	Epilogue; App G-H
N-119 Required == Discovered == Executed	Closed test-family completeness before pass/fail verdicts are aggregated.	set equality	Ch 12, 57; App G
N-120 digest(e)==manifest.digest	Artifact integrity predicate after canonical byte/schema validation.	Boolean	Ch 48, 60; App G
N-121 state(c,t1) may regress	Changed identity, invalidated support or lost evidence may return a claim to SPECIFIED.	state transition	App H

NT-11 - FAILURE-CAMPAIGN TIMELINE

A recovery experiment is an ordered set of bounded phases, not a screenshot taken after green health.

NOTATION ENTRIES NT-11.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-122 I_base=[t0,t1)	Baseline observation interval before disturbance.	UTC interval	Ch 54; App E
N-123 t_fault	Admitted instant at which the controlled disturbance begins.	UTC instant	Ch 54; App E
N-124 I_fault=[t_fault,t_release)	Interval during which the declared disturbance remains active.	UTC interval	Ch 54; App E
N-125 t_detect	First admitted observation satisfying the scenario detection predicate.	UTC instant	Ch 54; App E
N-126 t_recovered	First instant the deepest business recovery oracle is satisfied, not merely health.	UTC instant	Ch 54; App E
N-127 recovery_elapsed	Monotonic duration from disturbance release or declared start coordinate to recovered oracle.	time >= 0	Ch 54; App E

FROM SERVICES TO RUNTIME

NT-11 - FAILURE-CAMPAIGN TIMELINE - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-11.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-128 B_recovery	Scenario-specific maximum observation budget; exceeding it is a failed or incomplete campaign.	time > 0	Ch 54; App E
N-129 t_cleanup	Instant cleanup completes and exact owned-resource inventory reaches its postcondition.	UTC instant	Ch 54-55; App E-G
N-130 t0<=t_fault<=t_detect<=t_rec overed<=t_cleanup	Expected phase order when all coordinates exist; absent phases remain Missing, not fabricated.	instant predicate	Ch 54; App E-G

NT-12 - UNITS AND RENDERING CONVENTIONS

A number without a declared unit, scale, aggregation window or identity is not a complete engineering observation.

NOTATION ENTRIES NT-12.1

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-131 ms / s / min / h	Milliseconds, seconds, minutes and hours. Convert before addition or comparison.	time	Ch 11, 39, 52-54
N-132 B / KiB / MiB / GiB	Bytes and binary multiples 2 ¹⁰ , 2 ²⁰ and 2 ³⁰ ; use KB/MB/GB only when decimal units are intended.	data size	Ch 52, 55, 57
N-133 count	Non-negative integer cardinality; a count is not a rate until divided by a time window.	integer >= 0	Ch 52
N-134 x per s	Rate of x per second over a declared window.	x/time	Ch 52
N-135 ratio	Dimensionless value from zero through one, inclusive: [0,1].	dimensionless	Ch 52
N-136 percent	100 times a ratio, written with %; do not compare 0.99 directly with 99% without conversion.	percentage	Ch 52

NT-12 - UNITS AND RENDERING CONVENTIONS - CONTINUED

The same domain, unit and primary-anchor rules continue.

NOTATION ENTRIES NT-12.2

ID / notation	Meaning in this book	Domain / unit	Primary anchor
N-137 v1	First public schema major identity in this volume; the suffix belongs to the contract name.	version identity	Ch 20, 24, 30; App C
N-138 p. n / pp. a-b	One page n or inclusive page range a through b in final-reference navigation.	page reference	Contents; Working Index
N-139 SPECIFIED / PENDING	Specification is complete while implementation or execution support remains outstanding.	status pair	Epilogue; App G-H
N-140 ILLUSTRATIVE	Example values or transcripts not captured from an admitted run; never styled as evidence.	editorial label	App G

NOTATION AUDIT

Every expression must be typeable and challengeable

ACCEPTANCE GATES NT-6

Gate	Required result
stable ID	N-001 through N-140, unique and contiguous
symbol	one preferred ASCII expression
domain/unit	explicit for every entry
clock	UTC instant, monotonic reading or duration distinguished
anchor	one valid internal destination for every notation row
dimension	invalid additions/comparisons fail closed
probability	never presented as causation or authority
evidence	formula does not promote an unexecuted claim

NOTATION CHECKPOINT

The final audit can now compare meaning, unit and time

COMPLETION LEDGER NT-7

Question	Canonical answer
Is this a fact time or elapsed duration?	UTC instant or monotonic difference, never implicit
What does the number measure?	declared domain and unit
Which boundary owns the budget?	caller, workflow, host or scenario
Does ordering imply cause?	only the explicit $a \rightarrow b$ relation does
Is the probability authoritative?	no; model/estimator and decision policy remain separate
Did the formula execute?	only same-run retained evidence can answer
What comes next?	Sources and Reference Notes

FINAL NOTATION RULE

A distributed-system equation is incomplete until its identities, units, clock domains, intervals, assumptions and failure interpretation are explicit.

NOTATION LAST WORD

A deadline without a clock and a metric without a window are only numbers

FINAL STATUS

MATHEMATICAL AND TIMING NOTATION COMPLETE | 140 STABLE ENTRIES | 12 FAMILIES | DIMENSION + CLOCK GATES FIXED | IMPLEMENTATION AND RUNTIME EVIDENCE STATES UNCHANGED

FINAL REFERENCE BACK MATTER

Sources and Reference Notes

Primary standards, official documentation, foundational papers and NEXUS-1 author sources behind the completed runtime volume.

CITATION BOUNDARY

A source establishes lineage, vocabulary or platform semantics. It does not prove that NEXUS-1 implemented, deployed, executed or satisfied the cited obligation. Claim state advances only through the evidence model already defined in the book.

SOURCE CONTROL TOTALS SR-1

Coordinate	Value
stable source entries	74
source families	11
external URI verification	2026-08-11
preferred authority	author source -> normative standard -> official documentation -> foundational paper
version rule	pin contract-affecting specifications; never substitute latest silently
final-core constants	unchanged: 3 owners / 1 edge / 7 roles / 3 stores / 3 facts / 6 HTTP

HOW TO READ THE SOURCE REGISTER

Authority, applicability and evidence are separate coordinates

REFERENCE INTERPRETATION SR-2

Question	Required answer
What does the source control?	Only the named protocol, platform behavior, model or project scope.
Which version applies?	The printed version or the version retained in the run manifest.
Does citation prove conformance?	No. Conformance needs a mapped requirement, implementation and executed verifier.
Does official documentation override project scope?	No. Author-approved project description and contents control NEXUS-1 scope.
Can a foundational paper be treated as a recipe?	No. Its assumptions and abstraction must be mapped before use.
What if a URI changes?	Recover by stable SRC ID, exact title, issuer and version; do not guess a replacement.

LINK BEHAVIOR

Each external SRC identifier and printed URI is clickable in the PDF. Internal author artifacts are deliberately not assigned invented public locations.

CLAIM-STATE PROTECTION

A bibliography cannot manufacture runtime evidence

EVIDENCE LADDER SR-3

Available material	Maximum honest conclusion
standard or specification	required or defined semantics under the source's scope
official framework documentation	documented platform behavior for a named version
project repository path	artifact exists at an identified revision
compile/test source	an executable verifier is specified or implemented
same-run retained output	the verifier executed for one admitted run
production observation	only when environment, interval, identity and capture are retained

STATE INVARIANT

SPECIFIED/PENDING remains distinct from IMPLEMENTED, EXECUTED and DEMONSTRATED. None of the references below changes the existing state ledger.

SR-01 - AUTHOR SOURCES AND PROJECT PROVENANCE

These artifacts define NEXUS-1 scope and editorial lineage. They outrank generic guidance when describing this project, but they do not prove execution.

SOURCE ENTRIES SR-01.1

ID / authority	Primary source / URI	Reference note	Scope
SRC-001 AUTHOR SOURCE	NEXUS-1 Project Description - author-supplied attachment AUTHOR-SUPPLIED / CANONICAL INTERNAL ARTIFACT	Controls the target system, business purpose, modernization boundary and exclusions.	Whole volume
SRC-002 AUTHOR SOURCE	NEXUS-1 approved Contents and chapter sequence - author-supplied attachment AUTHOR-SUPPLIED / CANONICAL INTERNAL ARTIFACT	Controls part/chapter order and the obligation assigned to each chapter.	Contents; Ch 1-60
SRC-003 CANONICAL ARTIFACT	From Services to Runtime - current canonical PDF and trace appendices AUTHOR-SUPPLIED / CANONICAL INTERNAL ARTIFACT	Records the editorial checkpoint, claim states, inventories, notation and retained cross-references.	Whole volume; Apps A-H
SRC-004 PROJECT REPOSITORY	NEXUS-1 Phase 0 repository - Grigorios Agathangelidis https://github.com/gregory82gr/Nexus-1-phase-0	Primary codebase lineage for inventory and migration analysis; repository presence is not runtime evidence.	Ch 1-7, 54-60

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SR-02 - .NET PLATFORM, HOSTING AND CONFIGURATION

Official Microsoft sources define framework behavior and supported APIs. The book still owns every NEXUS-1 composition and policy choice.

SOURCE ENTRIES SR-02.1

ID / authority	Primary source / URI	Reference note	Scope
SRC-005 OFFICIAL POLICY	.NET and .NET Core Support Policy - Microsoft https://dotnet.microsoft.com/en-us/platform/support/policy/dotnet-core	Support lifecycle anchor for the declared .NET baseline; deployment manifests must pin the actual runtime.	Ch 5, 8, 54
SRC-006 LANGUAGE SPEC	C# Language Specification - Microsoft https://learn.microsoft.com/en-us/dotnet/csharp/language-reference/language-specification/	Normative language semantics behind compile-oriented listings; package and framework APIs are separate authorities.	Ch 6-12; code listings
SRC-007 OFFICIAL DOCS	.NET Generic Host - Microsoft https://learn.microsoft.com/en-us/dotnet/core/extensions/generic-host	Host lifetime, configuration, logging and dependency-injection foundation for executable composition roots.	Ch 8, 11, 26-30
SRC-008 OFFICIAL DOCS	Dependency injection guidelines - Microsoft https://learn.microsoft.com/en-us/dotnet/core/extensions/dependency-injection-guidelines	Service lifetime, disposal and container guidance used by composition and captive-dependency checks.	Ch 8-9, 12

SR-02 - .NET PLATFORM, HOSTING AND CONFIGURATION - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-02.2

ID / authority	Primary source / URI	Reference note	Scope
SRC-009 OFFICIAL DOCS	Options pattern in .NET - Microsoft https://learn.microsoft.com/en-us/dotnet/core/extensions/options	Binding, validation and reload semantics underlying strict configuration boundaries.	Ch 10, 44-47
SRC-010 OFFICIAL DOCS	Configuration providers in .NET - Microsoft https://learn.microsoft.com/en-us/dotnet/core/extensions/configuration-providers	Provider ordering and precedence foundation; NEXUS-1 adds value-free manifests and equivalence gates.	Ch 10, 54
SRC-011 OFFICIAL DOCS	Worker Services in .NET - Microsoft https://learn.microsoft.com/en-us/dotnet/core/extensions/workers	Framework basis for long-running hosted processes and scoped work units.	Ch 8, 26-30, 53
SRC-012 OFFICIAL DOCS	Background tasks with hosted services in ASP.NET Core - Microsoft (.NET 8) https://learn.microsoft.com/en-us/aspnet/core/fundamentals/host/hosted-services?view=aspnetcore-8.0	Hosted-service and scoped-background-task mechanics; bounded shutdown remains a NEXUS-1 obligation.	Ch 8, 11, 27-30

SR-02 - .NET PLATFORM, HOSTING AND CONFIGURATION - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-02.3

ID / authority	Primary source / URI	Reference note	Scope
SRC-013 OFFICIAL DOCS	Health checks in ASP.NET Core - Microsoft (.NET 8) https://learn.microsoft.com/en-us/aspnet/core/host-and-deploy/health-checks?view=aspnetcore-8.0	API basis for health checks; the book separately defines startup, liveness and readiness controller meaning.	Ch 11, 39, 53

SR-03 - PERSISTENCE, TRANSACTIONS AND MIGRATIONS

Provider-backed behavior is distinguished from in-memory approximations. These sources define platform mechanisms, not NEXUS-1 ownership proof.

SOURCE ENTRIES SR-03.1

ID / authority	Primary source / URI	Reference note	Scope
SRC-014 OFFICIAL DOCS	DbContext lifetime, configuration and initialization - EF Core https://learn.microsoft.com/en-us/ef/core/dbcontext-configuration/	Single-unit-of-work lifetime and configuration foundation for owner-local persistence.	Ch 9, 14-18
SRC-015 OFFICIAL DOCS	Using transactions - EF Core https://learn.microsoft.com/en-us/ef/core/saving/transactions	Transaction and savepoint behavior supporting atomic owner-state plus pending-fact commits.	Ch 14-20, 26, 28
SRC-016 OFFICIAL DOCS	Handling concurrency conflicts - EF Core https://learn.microsoft.com/en-us/ef/core/saving/concurrency	Optimistic concurrency token and conflict handling semantics behind semantic retries.	Ch 14-18, 42
SRC-017 OFFICIAL DOCS	Applying migrations - EF Core https://learn.microsoft.com/en-us/ef/core/managing-schemas/migrations/applying	Migration application choices and production cautions; NEXUS-1 separately restricts migration authority.	Ch 13-18, 56-58

FROM SERVICES TO RUNTIME

SR-03 - PERSISTENCE, TRANSACTIONS AND MIGRATIONS - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-03.2

ID / authority	Primary source / URI	Reference note	Scope
SRC-018 OFFICIAL DOCS	Choosing a testing strategy - EF Core https://learn.microsoft.com/en-us/ef/core/testing/choosing-a-testing-strategy	Rationale for testing important behavior against the production database system or equivalent provider.	Ch 12, 14-18, 55
SRC-019 OFFICIAL DOCS	Efficient querying - EF Core https://learn.microsoft.com/en-us/ef/core/performance/efficient-querying	Query-shape and result-size guidance used when assessing projections and bounded reads.	Ch 19, 22, 35, 41
SRC-020 OFFICIAL DOCS	Snapshot isolation in SQL Server - Microsoft https://learn.microsoft.com/en-us/dotnet/framework/data/adonet/sql/snapshot-isolation-in-sql-server	Provider isolation and row-versioning reference; actual database settings remain environment evidence.	Ch 14-18, 42, 56

SR-04 - HTTP, JSON AND PUBLIC CONTRACT SEMANTICS

Pinned standards define wire meaning. NEXUS-1 contract identity also requires its own schema, compatibility policy and retained corpus.

SOURCE ENTRIES SR-04.1

ID / authority	Primary source / URI	Reference note	Scope
SRC-021 IETF STANDARD	RFC 9110 - HTTP Semantics (2022) https://www.rfc-editor.org/rfc/rfc9110.html	Method, status, representation, conditional request and idempotent-method semantics.	Ch 7, 9, 21-24
SRC-022 IETF STANDARD	RFC 9457 - Problem Details for HTTP APIs (2023) https://www.rfc-editor.org/rfc/rfc9457.html	Current Problem Details media types and extension model; supersedes RFC 7807.	Ch 9, 21-24, 46
SRC-023 IETF STANDARD	RFC 8259 - The JavaScript Object Notation Data Interchange Format (2017) https://www.rfc-editor.org/rfc/rfc8259.html	JSON grammar and interoperability baseline for HTTP contracts and manifests.	Ch 9-10, 20, 24
SRC-024 IETF STANDARD	RFC 8785 - JSON Canonicalization Scheme (2020) https://www.rfc-editor.org/rfc/rfc8785.html	Deterministic JSON canonicalization reference for digestable contract and policy identities.	Ch 10, 20, 57; Apps G-H

SR-04 - HTTP, JSON AND PUBLIC CONTRACT SEMANTICS - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-04.2

ID / authority	Primary source / URI	Reference note	Scope
SRC-025 IETF STANDARD	RFC 3339 - Date and Time on the Internet: Timestamps (2002) https://www.rfc-editor.org/rfc/rfc3339.html	Internet timestamp profile used for externally meaningful UTC facts.	Ch 20, 39, 49; Notation
SRC-026 IETF STANDARD	RFC 9562 - Universally Unique Identifiers (UUIDs) (2024) https://www.rfc-editor.org/rfc/rfc9562.html	Current UUID formats and generation considerations; typed business identity remains application-owned.	Ch 20, 23, 27-30
SRC-027 PINNED SPEC	OpenAPI Specification 3.1.1 - OpenAPI Initiative (2024) https://spec.openapis.org/oas/v3.1.1.html	Pinned interface-description grammar used by contract catalogues; newer OAS versions do not silently replace it.	Ch 7, 21-24; App C
SRC-028 PINNED SPEC	JSON Schema Draft 2020-12 Core https://json-schema.org/draft/2020-12/json-schema-core	Schema vocabulary, identifiers, references and instance evaluation model.	Ch 20, 24; Apps C, H

SR-04 - HTTP, JSON AND PUBLIC CONTRACT SEMANTICS - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-04.3

ID / authority	Primary source / URI	Reference note	Scope
SRC-029 PINNED SPEC	JSON Schema Draft 2020-12 Validation https://json-schema.org/draft/2020-12/json-schema-validation	Validation keywords and assertion semantics for executable contract checks.	Ch 20, 24; Apps C, H

SR-05 - MESSAGING AND DISTRIBUTED CORRECTNESS

Broker documentation defines transport behavior. Local atomicity, idempotency, convergence and compensation remain explicit application obligations.

SOURCE ENTRIES SR-05.1

ID / authority	Primary source / URI	Reference note	Scope
SRC-030 OFFICIAL DOCS	RabbitMQ Reliability Guide https://www.rabbitmq.com/docs/reliability	Connection failure, retransmission, acknowledgement and replicated-queue reliability boundaries.	Ch 20, 26-30, 36
SRC-031 OFFICIAL DOCS	RabbitMQ Consumer Acknowledgements and Publisher Confirms https://www.rabbitmq.com/docs/confirms	Separates broker acceptance from consumer processing and business convergence.	Ch 20, 27-30, 36
SRC-032 OFFICIAL DOCS	RabbitMQ Quorum Queues https://www.rabbitmq.com/docs/quorum-queues	Replicated queue and publisher-confirm behavior relevant to declared broker topology.	Ch 27, 36, 54
SRC-033 OFFICIAL DOCS	RabbitMQ Dead Letter Exchanges https://www.rabbitmq.com/docs/dlx	Dead-letter routing, safety limitations and quorum at-least-once distinctions.	Ch 29-30, 36, 53

FROM SERVICES TO RUNTIME

SR-05 - MESSAGING AND DISTRIBUTED CORRECTNESS - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-05.2

ID / authority	Primary source / URI	Reference note	Scope
SRC-034 OFFICIAL DOCS	RabbitMQ Time-To-Live and Expiration https://www.rabbitmq.com/docs/ttl	Queue/message TTL behavior and races; TTL is not a business deadline.	Ch 29, 39, 53
SRC-035 PROTOCOL SPEC	AMQP 0-9-1 Protocol Specification https://www.rabbitmq.com/amqp-0-9-1-protocol	Wire protocol and AMQ model for the RabbitMQ path; AMQP 1.0 is not treated as the same protocol revision.	Ch 26-30
SRC-036 FOUNDATIONAL PAPER	Sagas - Garcia-Molina and Salem, ACM SIGMOD (1987) https://dl.acm.org/doi/10.1145/38713.38742	Foundational long-lived transaction and compensation model; NEXUS-1 workflows still require explicit semantics.	Ch 32-38, 41-42
SRC-037 FOUNDATIONAL PAPER	Time, Clocks, and the Ordering of Events in a Distributed System - Lamport (1978) https://lamport.azurewebsites.net/pubs/time-clocks.pdf	Causal partial order and logical-clock foundation; it does not create a global physical clock.	Ch 20, 30, 39, 49

SR-05 - MESSAGING AND DISTRIBUTED CORRECTNESS - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-05.3

ID / authority	Primary source / URI	Reference note	Scope
SRC-038 FOUNDATIONAL PAPER	Dynamo: Amazon's Highly Available Key-value Store - DeCandia et al. (2007) https://www.allthingsdistributed.com/files/amazon-dynamo-sosp2007.pdf	Foundational availability, partitioning and eventual-consistency context; not a template for NEXUS-1 storage ownership.	Ch 35, 41-42, 52

SR-06 - IDENTITY, AUTHENTICATION AND TRUST

Security standards define protocol and verification obligations. A valid token, certificate or channel never grants implicit business authority.

SOURCE ENTRIES SR-06.1

ID / authority	Primary source / URI	Reference note	Scope
SRC-039 IETF STANDARD	RFC 8446 - The Transport Layer Security Protocol Version 1.3 (2018) https://www.rfc-editor.org/rfc/rfc8446.html	TLS 1.3 protocol reference for protected channels and handshake semantics.	Ch 44-47
SRC-040 IETF STANDARD	RFC 7519 - JSON Web Token (JWT) (2015) https://www.rfc-editor.org/rfc/rfc7519.html	JWT claim-set representation and registered claims; validation policy remains deployment-specific.	Ch 44-46
SRC-041 IETF BCP	RFC 8725 - JSON Web Token Best Current Practices (2020) https://www.rfc-editor.org/rfc/rfc8725.html	Algorithm, key and validation threat guidance for secure JWT use.	Ch 44-46
SRC-042 IETF STANDARD	RFC 6749 - The OAuth 2.0 Authorization Framework (2012) https://www.rfc-editor.org/rfc/rfc6749.html	Base OAuth 2.0 roles and flows; consult the current security BCP for hardened use.	Ch 44-46

SR-06 - IDENTITY, AUTHENTICATION AND TRUST - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-06.2

ID / authority	Primary source / URI	Reference note	Scope
SRC-043 IETF BCP	RFC 9700 - Best Current Practice for OAuth 2.0 Security (2025) https://www.rfc-editor.org/rfc/rfc9700.html	Current OAuth 2.0 security guidance and deprecations; source version is explicit.	Ch 44-46
SRC-044 IETF STANDARD	RFC 8705 - OAuth 2.0 Mutual-TLS Client Authentication and Certificate-Bound Access Tokens (2020) https://www.rfc-editor.org/rfc/rfc8705.html	mTLS client authentication and certificate-bound token semantics.	Ch 45-47
SRC-045 FINAL SPEC	OpenID Connect Core 1.0 incorporating errata set 2 (2023) https://openid.net/specs/openid-connect-core-1_0.html	Identity layer and ID Token semantics; authentication and authorization remain distinct.	Ch 44-46
SRC-046 NIST STANDARD	NIST SP 800-207 - Zero Trust Architecture (2020) https://csrc.nist.gov/pubs/sp/800/207/final	Resource-centered trust and explicit authentication/authorization context; not a product architecture.	Ch 44-48

SR-06 - IDENTITY, AUTHENTICATION AND TRUST - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-06.3

ID / authority	Primary source / URI	Reference note	Scope
SRC-047 OWASP STANDARD	OWASP Application Security Verification Standard 5.0.0 https://owasp.org/www-project-application-security-verification-standard/	Versioned verification requirements used as security-review input, not a claim that NEXUS-1 passed ASVS.	Ch 44-48, 55

FROM SERVICES TO RUNTIME

SR-07 - OBSERVABILITY AND OPERATIONAL SIGNALS

Signal formats make correlation possible. They do not prove causation, business completion or production objectives without admitted evidence.

SOURCE ENTRIES SR-07.1

ID / authority	Primary source / URI	Reference note	Scope
SRC-048 W3C STANDARD	W3C Trace Context Recommendation (2021) https://www.w3.org/TR/trace-context/	traceparent/tracestate propagation and interoperability for distributed tracing.	Ch 49-51
SRC-049 W3C CANDIDATE	W3C Baggage Candidate Recommendation Snapshot (2024) https://www.w3.org/TR/baggage/	Application-defined context propagation; candidate status and privacy risk remain explicit.	Ch 49-51
SRC-050 PROJECT SPEC	OpenTelemetry Specification https://opentelemetry.io/docs/specs/otel/	API, SDK and data-model baseline; the deployment must pin the emitted specification identity.	Ch 49-52, 54
SRC-051 EVOLVING SPEC	OpenTelemetry Semantic Conventions https://opentelemetry.io/docs/specs/semconv/	Names and values for observed operations; stability labels must be checked before a contract depends on them.	Ch 49-52

SR-07 - OBSERVABILITY AND OPERATIONAL SIGNALS - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-07.2

ID / authority	Primary source / URI	Reference note	Scope
SRC-052 STABLE SPEC	OpenTelemetry Metrics Data Model https://opentelemetry.io/docs/specs/otel/metrics/data-model/	Aggregation temporality and metric data representation supporting signal identity checks.	Ch 52
SRC-053 OFFICIAL DOCS	Prometheus Data Model https://prometheus.io/docs/concepts/data_model/	Time-series, sample and label identity foundation for metric inventory.	Ch 52
SRC-054 OFFICIAL DOCS	Prometheus Histograms and Summaries https://prometheus.io/docs/practices/histograms/	Bucket, sum, count and quantile trade-offs; aggregation/window meaning must remain explicit.	Ch 52-53; Notation

SR-08 - CONTAINERS, COMPOSITION AND ORCHESTRATION

Deployment specifications describe artifact and controller behavior. They do not establish that the described environment was deployed or observed.

SOURCE ENTRIES SR-08.1

ID / authority	Primary source / URI	Reference note	Scope
SRC-055 OCI SPEC	OCI Image Format Specification https://github.com/opencontainers/image-spec/blob/main/spec.md	Content-addressed image manifest, configuration and layer model for artifact identity.	Ch 54, 57, 60
SRC-056 OCI SPEC	OCI Runtime Specification https://github.com/opencontainers/runtime-spec/blob/main/spec.md	Runtime bundle and lifecycle reference for container execution boundaries.	Ch 54, 57
SRC-057 COMPOSE SPEC	Compose Specification https://github.com/compose-spec/compose-spec/blob/main/spec.md	Service, network, volume, dependency and configuration model for local composition.	Ch 54, 60
SRC-058 OFFICIAL DOCS	Kubernetes Liveness, Readiness and Startup Probes https://kubernetes.io/docs/concepts/workloads/pods/probes/	Controller interpretation of probes; NEXUS-1 health semantics must still be correctly mapped.	Ch 11, 39, 54

SR-08 - CONTAINERS, COMPOSITION AND ORCHESTRATION - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-08.2

ID / authority	Primary source / URI	Reference note	Scope
SRC-059 OFFICIAL DOCS	Kubernetes ConfigMaps https://kubernetes.io/docs/concepts/configuration/configmap/	Non-confidential configuration distribution model and update behavior.	Ch 10, 54
SRC-060 OFFICIAL DOCS	Kubernetes Secrets https://kubernetes.io/docs/concepts/configuration/secret/	Secret object mechanics and cautions; encryption, access and rotation controls remain required.	Ch 10, 44-47, 54
SRC-061 OFFICIAL DOCS	Kubernetes Deployments https://kubernetes.io/docs/concepts/workloads/controllers/deployment/	Declarative rollout and availability-controller behavior for long-running services.	Ch 53-54, 58
SRC-062 OFFICIAL DOCS	Kubernetes Jobs https://kubernetes.io/docs/concepts/workloads/controllers/job/	Run-to-completion controller semantics relevant to migration and verification jobs.	Ch 54, 56-60

FROM SERVICES TO RUNTIME

SR-09 - INTEGRITY, CANONICALIZATION AND SUPPLY-CHAIN PROVENANCE

Digests and attestations bind bytes and statements. Trust still depends on canonicalization, signer identity, verification policy and retained inputs.

SOURCE ENTRIES SR-09.1

ID / authority	Primary source / URI	Reference note	Scope
SRC-063 NIST STANDARD	FIPS PUB 180-4 - Secure Hash Standard https://csrc.nist.gov/pubs/fips/180-4/upd1/final	SHA-256 algorithm authority for canonical digests; hashing alone does not establish authenticity.	Ch 10, 57, 60; App G
SRC-064 SLSA SPEC	SLSA Specification 1.2 https://slsa.dev/spec/v1.2/	Versioned supply-chain security levels and provenance model used as comparative guidance.	Ch 54, 57, 60; App G
SRC-065 SLSA SPEC	SLSA Build Provenance 1.2 https://slsa.dev/spec/v1.2/build-provenance	Provenance predicate semantics for who built an artifact, from which inputs and by what process.	Ch 57, 60; App G
SRC-066 IN-TOTO SPEC	in-toto Attestation Framework - Statement v1 https://in-toto.io/Statement/v1	Subject digest plus typed predicate envelope for machine-verifiable claims.	Ch 55, 57, 60; Apps G-H

SR-09 - INTEGRITY, CANONICALIZATION AND SUPPLY-CHAIN PROVENANCE - CONTINUED

The same authority, version and evidence boundaries continue.

SOURCE ENTRIES SR-09.2

ID / authority	Primary source / URI	Reference note	Scope
SRC-067 SPDX SPEC	SPDX Specification 3.0.1 https://spdx.github.io/spdx-spec/v3.0.1/	Pinned SBOM and system-package metadata model; producing an SPDX document remains separate work.	Ch 54, 57, 60

SR-10 - TESTING, FAULT INJECTION AND CONTROLLED EXPERIMENTS

Test tools enable observations only when discovery, execution, environment identity, oracle and retained output are all admitted.

SOURCE ENTRIES SR-10.1

ID / authority	Primary source / URI	Reference note	Scope
SRC-068 OFFICIAL DOCS	Integration tests in ASP.NET Core - Microsoft (.NET 8) https://learn.microsoft.com/en-us/aspnet/core/test/integration-tests?view=aspnetcore-8.0	Production-path host and HTTP pipeline test foundation using WebApplicationFactory.	Ch 12, 21-24, 55
SRC-069 PROJECT DOCS	Testcontainers for .NET https://dotnet.testcontainers.org/	Throwaway real-provider test environments; image, version and topology must be pinned in evidence.	Ch 12, 14-18, 55
SRC-070 PROJECT DOCS	Toxiproxy - Shopify https://github.com/Shopify/toxiproxy	Programmable network fault injection for bounded test scenarios; tool presence is not a completed experiment.	Ch 39, 53, 55
SRC-071 FOUNDATIONAL GUIDANCE	Principles of Chaos Engineering https://principlesofchaos.org/	Controlled experimentation vocabulary; this book requires safer predeclared blast radius and reversal gates.	Ch 53, 55, 60

SR-11 - TIME, UNITS AND QUANTITATIVE REASONING

Units, clocks and queueing equations are model inputs. They cannot promote unmeasured or incompatible values into runtime facts.

SOURCE ENTRIES SR-11.1

ID / authority	Primary source / URI	Reference note	Scope
SRC-072 BIPM STANDARD	The International System of Units (SI), 9th edition, version 4.01 (2026) https://doi.org/10.59161/AUEZ1291	Authoritative SI unit and symbol reference for durations, rates and derived quantities.	Ch 39, 52-53; Notation
SRC-073 FOUNDATIONAL PAPER	A Proof for the Queueing Formula $L = \lambda W$ - John D. C. Little (1961) https://doi.org/10.1287/opre.9.3.383	Conditions and proof lineage for Little's Law; applicability requires a stable admitted system and window.	Ch 52-53; Notation
SRC-074 OFFICIAL API	System.Diagnostics.Stopwatch - Microsoft .NET API https://learn.microsoft.com/en-us/dotnet/api/system.diagnostics.stopwatch	Process-local high-resolution elapsed-time API reference; values are not comparable across processes.	Ch 11, 39, 53; Notation

SOURCE AUDIT

Every external dependency must remain recoverable and bounded

ACCEPTANCE GATES SR-4

Gate	Required result
stable identity	SRC-001 through SRC-074, unique and contiguous
authority	source class visible for every entry
locator	exact URI or explicit author-supplied/internal status
version	pinned when wire, security, telemetry or provenance behavior may change
applicability	reference note states what the source contributes
scope	chapter family or appendix anchor present
links	one valid external URI annotation for every public source
claim state	no citation promotes implementation or runtime evidence

SOURCES CHECKPOINT

The final audit can now trace assertions to their authority

COMPLETION LEDGER SR-5

Question	Canonical answer
What controls NEXUS-1 scope?	author-supplied project description and approved contents
What controls protocol meaning?	the pinned normative specification
What controls framework behavior?	official documentation for the admitted version
What controls implementation status?	repository identity plus executable checks
What controls runtime truth?	same-run retained observations under declared assumptions
What remains unchanged?	3/1/7/3/3/6 inventories and all PENDING evidence states
What comes next?	Working Index

FINAL SOURCE RULE

Use a source to locate authority, then use the book's own obligation, oracle and evidence record to decide whether the claim is satisfied.

SOURCES LAST WORD

Lineage explains a claim; retained evidence decides it

FINAL STATUS

SOURCES AND REFERENCE NOTES COMPLETE | 74 STABLE ENTRIES | 11 FAMILIES | PRIMARY AUTHORITIES + PINNED VERSIONS | IMPLEMENTATION AND RUNTIME EVIDENCE STATES UNCHANGED

FINAL REFERENCE BACK MATTER

Working Index

A practical concept-to-page map for the current 2,017-page body. Entries cover engineering concepts, chapter obligations, exact runtime identities, protected surfaces, commands, failure scenarios, sources, aliases and tombstones.

INDEX AUTHORITY

This index reports the current canonical pagination. It does not create implementation or execution evidence, admit a candidate service, reactivate a superseded identity or replace the Glossary, Notation, Sources and final catalogues.

WORKING-INDEX CONTROL TOTALS IX-1

Coordinate	Value
indexed entries	381
alphabetical groups	24
chapter obligations	60
runtime constants	3 owners / 1 edge / 7 roles / 3 stores / 3 facts / 6 protected HTTP surfaces
recovery core	6 mandatory Appendix E scenarios
claim boundary	SPECIFIED/PENDING remains distinct from executed or demonstrated

HOW TO USE THE WORKING INDEX

Start with the exact identity, then follow the authority

ENTRY CONTRACT IX-2

Field	Meaning
ID and term	stable working-index identity and preferred spelling
Principal pages	curated engineering locations in the current PDF; every printed number is an internal link
Locator	chapter, appendix, source, role or boundary reason for the reference
see	alias or tombstone that redirects to one preferred index term
see also	related authority named without collapsing distinct concepts

PAGINATION RULE

The references use the final page labels visible in this checkpoint. Later consistency and reproduction audits append after the index, so they do not renumber the indexed body.

COVERAGE AND EXCLUSIONS

Navigation is broad; authority stays narrow

COVERAGE REGISTER IX-3

Indexed	Treatment
concepts	176 controlled Glossary terms with principal chapter/appendix pages
chapters	all 60 chapter obligations by exact title and opening page
runtime	owners, role IDs, stores, migrators, public facts and protected surfaces
operations	closed reproduction verbs, wrapper identities and mandatory failure scenarios
sources	all 74 stable Sources entries at their actual reference-note pages
aliases	preferred-term redirects for legacy names, shorthand and superseded paths/routes

NO PROMOTION BY INDEXING

Audit, Compliance, RL Advisory, Process Manager and Repair Control remain inactive candidates or protected modules. Their index presence records disposition; it grants zero runtime authority.

A - WORKING INDEX

INDEX ENTRIES A-1

ID / preferred term	Principal pages	Locator / cross-reference
IX-001 A Proof for the Queuing Formula $L = \lambda W$ - John D. C. Little (1961)	2016	Source SRC-073; FOUNDATIONAL PAPER; Ch 52-53; Notation
IX-002 A Verdict Fans Out to Audit and Compliance	814	Chapter 34; chapter obligation
IX-003 Abstention	49 / 33 / 175 / 1774	Ch 2, 9; Epilogue; controlled definition p. 1986
IX-004 Active final core	1751 / 1806 / 1968	Ch 60; App B, H; controlled definition p. 1986
IX-005 Aggregate	175 / 318 / 387	Ch 9, 15, 18; controlled definition p. 1986

A - WORKING INDEX - CONTINUED

INDEX ENTRIES A-2

ID / preferred term	Principal pages	Locator / cross-reference
IX-006 Aggregate root	175 / 318	Ch 9, 15; controlled definition p. 1986
IX-007 Alarm service	see	see AlarmManagement; canonical owner name
IX-008 alarm-api	712 / 1810 / 1945 / 1611 / 1727 / 1761	Appendix B/G runtime role; see also AlarmManagement
IX-009 alarm-migrator	341 / 1945 / 1954 / 1729 / 1790 / 1810	finite migration job for AlarmManagementDb
IX-010 alarm-worker	576 / 1810 / 1945 / 1611 / 1727 / 1761	Appendix B/G runtime role; outbox dispatcher

FROM SERVICES TO RUNTIME

A - WORKING INDEX - CONTINUED

INDEX ENTRIES A-3

ID / preferred term	Principal pages	Locator / cross-reference
IX-011 AlarmFloodDetected.v1	744 / 1809 / 1834 / 36 / 49 / 65	active public fact owned by AlarmManagement
IX-012 AlarmManagement	295 / 712 / 744 / 1806	Ch 14, 31-32; App B; controlled definition p. 1986
IX-013 AlarmManagement Enters the Runtime	712	Chapter 31; chapter obligation
IX-014 AlarmManagementDb	295 / 341 / 387 / 1806	Ch 14, 16, 18; App B; controlled definition p. 1986
IX-015 AMQP 0-9-1 Protocol Specification	2014	Source SRC-035; PROTOCOL SPEC; Ch 26-30

A - WORKING INDEX - CONTINUED

INDEX ENTRIES A-4

ID / preferred term	Principal pages	Locator / cross-reference
IX-016 /api/operator/v1/root-cause-cases/{caseld}	see	see GET /api/v1/root-cause-cases/{caseld} - edge; superseded route tombstone
IX-017 Applying migrations - EF Core	2012	Source SRC-017; OFFICIAL DOCS; Ch 13-18, 56-58
IX-018 Architectural disposition	1751 / 1806 / 1968	Ch 60; App B, H; controlled definition p. 1986
IX-019 Architecture test	87 / 152 / 243	Ch 5, 8, 12; controlled definition p. 1986
IX-020 artifacts/from-services-to-runtime/	1786 / 1942 / 1944 / 1788 / 1943 / 1974	canonical companion root

A - WORKING INDEX - CONTINUED

INDEX ENTRIES A-5

ID / preferred term	Principal pages	Locator / cross-reference
IX-021 At-least-once delivery	555 / 619 / 645 / 680	Ch 25, 28-30; controlled definition p. 1986
IX-022 Atomicity	387 / 576	Ch 18, 26; controlled definition p. 1986
IX-023 Audit	814 / 1808 / 1945	inactive module/service candidate; zero final-core authority
IX-024 Audit evidence	1375 / 1578 / 1942 / 1968	Ch 48, 54; App G-H; controlled definition p. 1986
IX-025 Authority	266 / 1132 / 1182 / 1231 / 1280 / 1334 / 1375 / 1914	Ch 13, 43-48; App F; controlled definition p. 1986

B - WORKING INDEX

INDEX ENTRIES B-1

ID / preferred term	Principal pages	Locator / cross-reference
IX-026 Backend for Frontend	see	see BFF (Backend for Frontend); preferred term
IX-027 Background tasks with hosted services in ASP.NET Core - Microsoft (.NET 8)	2012	Source SRC-012; OFFICIAL DOCS; Ch 8, 11, 27-30
IX-028 Backoff	645 / 1549	Ch 29, 53; controlled definition p. 1986
IX-029 BASE	1968	App H; controlled definition p. 1986
IX-030 BFF	see	see Query BFF; final edge identity

B - WORKING INDEX - CONTINUED

INDEX ENTRIES B-2

ID / preferred term	Principal pages	Locator / cross-reference
IX-031 BFF (Backend for Frontend)	486 / 1047 / 1182 / 1806 / 1832	Ch 22, 41, 44; App B-C; controlled definition p. 1986
IX-032 Bounded context	49 / 266	Ch 3, 13; controlled definition p. 1986
IX-033 Bounded shutdown	152 / 221 / 594	Ch 8, 11, 27; controlled definition p. 1986
IX-034 Broker topology	555 / 1806 / 1832	Ch 25; App B-C; controlled definition p. 1986
IX-035 Building AlarmManagementDb	295	Chapter 14; chapter obligation

FROM SERVICES TO RUNTIME

B - WORKING INDEX - CONTINUED

INDEX ENTRIES B-3

ID / preferred term	Principal pages	Locator / cross-reference
IX-036 Building RootCauseDb	318	Chapter 15; chapter obligation
IX-037 Bulkhead	1549	Ch 53; controlled definition p. 1986
IX-038 Business oracle	894 / 1578 / 1751 / 1888	Ch 36, 54, 60; App E; controlled definition p. 1986

C - WORKING INDEX

INDEX ENTRIES C-1

ID / preferred term	Principal pages	Locator / cross-reference
IX-039 C# Language Specification - Microsoft	2012	Source SRC-006; LANGUAGE SPEC; Ch 6-12; code listings
IX-040 Candidate inactive	850 / 814 / 1806 / 1968	Ch 34; App B, H; controlled definition p. 1986
IX-041 Canonical asset	1786 / 1942 / 1968	App A, G-H; controlled definition p. 1986
IX-042 Canonical path	1786 / 1942 / 1968	App A, G-H; controlled definition p. 1986
IX-043 Captive dependency	152	Ch 8; controlled definition p. 1986

C - WORKING INDEX - CONTINUED

INDEX ENTRIES C-2

ID / preferred term	Principal pages	Locator / cross-reference
IX-044 CaseOpened.v1	see	see RootCauseCaseOpened.v1; canonical active fact
IX-045 Causation	1417	Ch 49; controlled definition p. 1986
IX-046 Choosing a testing strategy - EF Core	2013	Source SRC-018; OFFICIAL DOCS; Ch 12, 14-18, 55
IX-047 Choosing the First Distributed Slice	49	Chapter 3; chapter obligation
IX-048 Choreography, Orchestration, and the Decision Record	956	Chapter 38; chapter obligation

C - WORKING INDEX - CONTINUED

INDEX ENTRIES C-3

ID / preferred term	Principal pages	Locator / cross-reference
IX-049 CI Quality Gates and Container Images	1669	Chapter 57; chapter obligation
IX-050 Circuit breaker	1549	Ch 53; controlled definition p. 1986
IX-051 Claim	1774 / 1968	Epilogue; App H; controlled definition p. 1986
IX-052 Claim state	1774 / 1968	Epilogue; App H; controlled definition p. 1986
IX-053 Claims Minimization and Policy Mapping	1280	Chapter 46; chapter obligation

C - WORKING INDEX - CONTINUED

INDEX ENTRIES C-4

ID / preferred term	Principal pages	Locator / cross-reference
IX-054 Clean-room run	1669 / 1751 / 1942	Ch 57, 60; App G; controlled definition p. 1986
IX-055 Command	464 / 509	Ch 21, 23; controlled definition p. 1986
IX-056 Commands and Queries at a Service Boundary	464	Chapter 21; chapter obligation
IX-057 Compensation	1016	Ch 40; controlled definition p. 1986
IX-058 Compensation Without Pretending to Undo History	1016	Chapter 40; chapter obligation

FROM SERVICES TO RUNTIME

C - WORKING INDEX - CONTINUED

INDEX ENTRIES C-5

ID / preferred term	Principal pages	Locator / cross-reference
IX-059 Competing update	387	Ch 18; controlled definition p. 1986
IX-060 Complete Slice and the Optional RL Advisory Branch	894	Chapter 36; chapter obligation
IX-061 Compliance	814 / 1808 / 1945	inactive module/service candidate; zero final-core authority
IX-062 Compose Specification	2015	Source SRC-057; COMPOSE SPEC; Ch 54, 60
IX-063 Concurrency token	295 / 318 / 387	Ch 14-15, 18; controlled definition p. 1986

C - WORKING INDEX - CONTINUED

INDEX ENTRIES C-6

ID / preferred term	Principal pages	Locator / cross-reference
IX-064 Configuration and Environment Boundaries	199	Chapter 10; chapter obligation
IX-065 Configuration identity	199 / 1860 / 1942	Ch 10; App D, G; controlled definition p. 1986
IX-066 Configuration providers in .NET - Microsoft	2012	Source SRC-010; OFFICIAL DOCS; Ch 10, 54
IX-067 Contract	416 / 442 / 464 / 486 / 509 / 532 / 1832	Ch 19-24; App C; controlled definition p. 1986
IX-068 Contract Compatibility Tests	532	Chapter 24; chapter obligation

C - WORKING INDEX - CONTINUED

INDEX ENTRIES C-7

ID / preferred term	Principal pages	Locator / cross-reference
IX-069 Contract corpus	442 / 532 / 680	Ch 20, 24, 30; controlled definition p. 1986
IX-070 Correlation	1417	Ch 49; controlled definition p. 1986
IX-071 Correlation and Causation Across Every Boundary	1417	Chapter 49; chapter obligation
IX-072 CQRS	464 / 486 / 850	Ch 21-22, 35; controlled definition p. 1986
IX-073 Cross-run join	1942 / 1968	App G-H; controlled definition p. 1986

C - WORKING INDEX - CONTINUED

INDEX ENTRIES C-8

ID / preferred term	Principal pages	Locator / cross-reference
IX-074 Cross-Service Identity Without Cross-Database Foreign Keys	363	Chapter 17; chapter obligation
IX-075 Cutover	266 / 1637 / 1698	Ch 13, 56, 58; controlled definition p. 1986

D - WORKING INDEX

INDEX ENTRIES D-1

ID / preferred term	Principal pages	Locator / cross-reference
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IX-097 Dynamo: Amazon's Highly Available Key-value Store - DeCandia et al. (2007)	2014	Source SRC-038; FOUNDATIONAL PAPER; Ch 35, 41-42, 52

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IX-295 RFC 8446 - The Transport Layer Security Protocol Version 1.3 (2018)	2014	Source SRC-039; IETF STANDARD; Ch 44-47
IX-296 RFC 8705 - OAuth 2.0 Mutual-TLS Client Authentication and Certificate-Bound Access Tokens (2020)	2014	Source SRC-044; IETF STANDARD; Ch 45-47
IX-297 RFC 8725 - JSON Web Token Best Current Practices (2020)	2014	Source SRC-041; IETF BCP; Ch 44-46
IX-298 RFC 8785 - JSON Canonicalization Scheme (2020)	2013	Source SRC-024; IETF STANDARD; Ch 10, 20, 57; Apps G-H
IX-299 RFC 9110 - HTTP Semantics (2022)	2013	Source SRC-021; IETF STANDARD; Ch 7, 9, 21-24

R - WORKING INDEX - CONTINUED

INDEX ENTRIES R-8

ID / preferred term	Principal pages	Locator / cross-reference
IX-300 RFC 9457 - Problem Details for HTTP APIs (2023)	2013	Source SRC-022; IETF STANDARD; Ch 9, 21-24, 46
IX-301 RFC 9562 - Universally Unique Identifiers (UUIDs) (2024)	2013	Source SRC-026; IETF STANDARD; Ch 20, 23, 27-30
IX-302 RFC 9700 - Best Current Practice for OAuth 2.0 Security (2025)	2014	Source SRC-043; IETF BCP; Ch 44-46
IX-303 RL Advisory	894 / 1808 / 1945	inactive optional module; RecommendationGenerated.v1 not activated
IX-304 Rollout, Database Change, and Rollback	1698	Chapter 58; chapter obligation

R - WORKING INDEX - CONTINUED

INDEX ENTRIES R-9

ID / preferred term	Principal pages	Locator / cross-reference
IX-305 Root Cause service	see	see RootCause; canonical owner name
IX-306 RootCause	318 / 778 / 814 / 1806	Ch 15, 33-34; App B; controlled definition p. 1986
IX-307 RootCause Consumes and Opens a Case	778	Chapter 33; chapter obligation
IX-308 rootcause-api	778 / 1814 / 1945 / 1589 / 1611 / 1727	Appendix B/G runtime role; see also RootCause
IX-309 rootcause-migrator	341 / 1945 / 1954 / 1790 / 1813 / 1880	finite migration job for RootCauseDb

FROM SERVICES TO RUNTIME

R - WORKING INDEX - CONTINUED

INDEX ENTRIES R-10

ID / preferred term	Principal pages	Locator / cross-reference
IX-310 rootcause-worker	619 / 1814 / 1945 / 1611 / 1727 / 1761	Appendix B/G runtime role; consumer and outbox relay
IX-311 RootCauseCase	175 / 318 / 778 / 814	Ch 9, 15, 33-34; controlled definition p. 1986
IX-312 RootCauseCaseOpened.v1	778 / 1809 / 1834 / 777 / 859 / 895	active public fact owned by RootCause
IX-313 RootCauseDb	318 / 341 / 387 / 619	Ch 15-16, 18, 28; controlled definition p. 1986
IX-314 RootCauseVerdictIssued.v1	814 / 1809 / 1834 / 36 / 49 / 70	active public fact owned by RootCause

R - WORKING INDEX - CONTINUED

INDEX ENTRIES R-11

ID / preferred term	Principal pages	Locator / cross-reference
IX-315 RunId	1942 / 1968	App G-H; controlled definition p. 1986
IX-316 Runtime role	1608 / 1751 / 1806	Ch 55, 60; App B; controlled definition p. 1986

S - WORKING INDEX

INDEX ENTRIES S-1

ID / preferred term	Principal pages	Locator / cross-reference
IX-317 Sagas - Garcia-Molina and Salem, ACM SIGMOD (1987)	2014	Source SRC-036; FOUNDATIONAL PAPER; Ch 32-38, 41-42
IX-318 Same-run evidence	1578 / 1751 / 1942 / 1968	Ch 54, 60; App G-H; controlled definition p. 1986
IX-319 Schema evolution	532 / 680 / 1698	Ch 24, 30, 58; controlled definition p. 1986
IX-320 Secret reference	199 / 1334 / 1860 / 1914	Ch 10, 47; App D, F; controlled definition p. 1986
IX-321 Secrets, Certificates, and Configuration Handling	1334	Chapter 47; chapter obligation

S - WORKING INDEX - CONTINUED

INDEX ENTRIES S-2

ID / preferred term	Principal pages	Locator / cross-reference
IX-322 Security Tests and Audit Evidence	1375	Chapter 48; chapter obligation
IX-323 Semantic conflict	387 / 464 / 509	Ch 18, 21, 23; controlled definition p. 1986
IX-324 Semantic retry	295 / 387 / 1549	Ch 14, 18, 53; controlled definition p. 1986
IX-325 Service	49 / 108 / 128 / 152 / 175 / 266	Ch 3, 6-9, 13; controlled definition p. 1986
IX-326 Service catalogue	1751 / 1806	Ch 60; App B; controlled definition p. 1986

S - WORKING INDEX - CONTINUED

INDEX ENTRIES S-3

ID / preferred term	Principal pages	Locator / cross-reference
IX-327 Service-owned database	266 / 295 / 318 / 341 / 363 / 387	Ch 13-18; controlled definition p. 1986
IX-328 Service-to-Service Identity	1231	Chapter 45; chapter obligation
IX-329 Shared Nothing, Shared Little	87	Chapter 5; chapter obligation
IX-330 SLO (Service Level Objective)	1518	Ch 52; controlled definition p. 1986
IX-331 SLSA Build Provenance 1.2	2016	Source SRC-065; SLSA SPEC; Ch 57, 60; App G

S - WORKING INDEX - CONTINUED

INDEX ENTRIES S-4

ID / preferred term	Principal pages	Locator / cross-reference
IX-332 SLSA Specification 1.2	2016	Source SRC-064; SLSA SPEC; Ch 54, 57, 60; App G
IX-333 Snapshot isolation in SQL Server - Microsoft	2013	Source SRC-020; OFFICIAL DOCS; Ch 14-18, 42, 56
IX-334 SPDX Specification 3.0.1	2016	Source SRC-067; SPDX SPEC; Ch 54, 57, 60
IX-335 SPECIFIED	1774 / 1968	Epilogue; App H; controlled definition p. 1986
IX-336 Startup	221	Ch 11; controlled definition p. 1986

S - WORKING INDEX - CONTINUED

INDEX ENTRIES S-5

ID / preferred term	Principal pages	Locator / cross-reference
IX-337 Strangler migration	1637	Ch 56; controlled definition p. 1986
IX-338 Structured logging	1457	Ch 50; controlled definition p. 1986
IX-339 Structured Logs That Preserve Domain Meaning	1457	Chapter 50; chapter obligation
IX-340 SUPERSEDED	1786 / 1942 / 1968	App A, G-H; controlled definition p. 1986
IX-341 System.Diagnostics.Stopwatch - Microsoft .NET API	2016 / 2017	Source SRC-074; OFFICIAL API; Ch 11, 39, 53; Notation

T - WORKING INDEX

INDEX ENTRIES T-1

ID / preferred term	Principal pages	Locator / cross-reference
IX-342 Testcontainers for .NET	2016	Source SRC-069; PROJECT DOCS; Ch 12, 14-18, 55
IX-343 Testing Pyramid for a Distributed Service	243	Chapter 12; chapter obligation
IX-344 The International System of Units (SI), 9th edition, version 4.01 (2026)	2016	Source SRC-072; BIPM STANDARD; Ch 39, 52-53; Notation
IX-345 Threat and Trust Boundaries Before Mechanisms	1132	Chapter 43; chapter obligation
IX-346 Time, Clocks, and the Ordering of Events in a Distributed System - Lamport (1978)	2014	Source SRC-037; FOUNDATIONAL PAPER; Ch 20, 30, 39, 49

T - WORKING INDEX - CONTINUED

INDEX ENTRIES T-2

ID / preferred term	Principal pages	Locator / cross-reference
IX-347 Timeout	985 / 1549	Ch 39, 53; controlled definition p. 1986
IX-348 Timeout, Retry, Circuit Breaker, and Bulkhead	1549	Chapter 53; chapter obligation
IX-349 Timeouts, Escalations, and Human Review	985	Chapter 39; chapter obligation
IX-350 Tombstone	1786 / 1914 / 1942 / 1968	App A, F-H; controlled definition p. 1986
IX-351 Toxiproxy - Shopify	2016	Source SRC-070; PROJECT DOCS; Ch 39, 53, 55

T - WORKING INDEX - CONTINUED

INDEX ENTRIES T-3

ID / preferred term	Principal pages	Locator / cross-reference
IX-352 Trace	1417 / 1489	Ch 49, 51; controlled definition p. 1986
IX-353 Trace context	1417 / 1489	Ch 49, 51; controlled definition p. 1986
IX-354 Traceability	1968	App H; controlled definition p. 1986
IX-355 Transaction boundary	387 / 576 / 619	Ch 18, 26, 28; controlled definition p. 1986
IX-356 Transactional outbox	576	Ch 26; controlled definition p. 1986

FROM SERVICES TO RUNTIME

T - WORKING INDEX - CONTINUED

INDEX ENTRIES T-4

ID / preferred term	Principal pages	Locator / cross-reference
IX-357 Transactional Outbox in Each Producer	576	Chapter 26; chapter obligation
IX-358 Translating Proven Needs to Kubernetes	1725	Chapter 59; chapter obligation
IX-359 Trust boundary	1132 / 1914	Ch 43; App F; controlled definition p. 1986
IX-360 Trust zone	1132 / 1914	Ch 43; App F; controlled definition p. 1986
IX-361 Typed identity	175 / 363 / 442	Ch 9, 17, 20; controlled definition p. 1986

U - WORKING INDEX

INDEX ENTRIES U-1

ID / preferred term	Principal pages	Locator / cross-reference
IX-362 Ubiquitous language	49 / 33 / 175 / 266	Ch 2, 9, 13; controlled definition p. 1986
IX-363 Unit of work	295 / 387	Ch 14, 18; controlled definition p. 1986
IX-364 Unknown outcome	594 / 680 / 1549	Ch 27, 30, 53; controlled definition p. 1986
IX-365 Using transactions - EF Core	2012	Source SRC-015; OFFICIAL DOCS; Ch 14-20, 26, 28

V - WORKING INDEX

INDEX ENTRIES V-1

ID / preferred term	Principal pages	Locator / cross-reference
IX-366 Value-free manifest	199 / 1860 / 1942	Ch 10; App D, G; controlled definition p. 1986
IX-367 Verdict	175 / 318 / 814	Ch 9, 15, 34; controlled definition p. 1986
IX-368 VerdictIssued.v1	see	see RootCauseVerdictIssued.v1; canonical active fact
IX-369 VERIFY	1968	App H; controlled definition p. 1986
IX-370 Versioned contract	442 / 509 / 532 / 680	Ch 20, 23-24, 30; controlled definition p. 1986

V - WORKING INDEX - CONTINUED

INDEX ENTRIES V-2

ID / preferred term	Principal pages	Locator / cross-reference
IX-371 Versioned Message Envelope	442	Chapter 20; chapter obligation
IX-372 Versioning, Problem Details, and Idempotency Keys	509	Chapter 23; chapter obligation

W - WORKING INDEX

INDEX ENTRIES W-1

ID / preferred term	Principal pages	Locator / cross-reference
IX-373 W3C Baggage Candidate Recommendation Snapshot (2024)	2015	Source SRC-049; W3C CANDIDATE; Ch 49-51
IX-374 W3C Trace Context Recommendation (2021)	2015	Source SRC-048; W3C STANDARD; Ch 49-51
IX-375 What Must Remain True After Extraction	33	Chapter 2; chapter obligation
IX-376 Worker	152 / 594 / 619 / 850	Ch 8, 27-28, 35; controlled definition p. 1986
IX-377 Worker Services in .NET - Microsoft	2012	Source SRC-011; OFFICIAL DOCS; Ch 8, 26-30, 53

W - WORKING INDEX - CONTINUED

INDEX ENTRIES W-2

ID / preferred term	Principal pages	Locator / cross-reference
IX-378 Working index	2018	Final back matter; controlled definition p. 1986
IX-379 Workload identity	1231 / 1280 / 1914	Ch 45-46; App F; controlled definition p. 1986

Z - WORKING INDEX

INDEX ENTRIES Z-1

ID / preferred term	Principal pages	Locator / cross-reference
IX-380 Zero-discovery gate	243 / 532 / 1669 / 1942	Ch 12, 24, 57; App G; controlled definition p. 1986
IX-381 Zero-trust stance	1132 / 1182 / 1231 / 1280 / 1334 / 1375 / 1914	Ch 43-48; App F; controlled definition p. 1986

INDEX INTEGRITY AUDIT

Every number, alias and census claim is checked

WORKING-INDEX ACCEPTANCE IX-4

Gate	Required result
identity	contiguous IX identifiers; no case-insensitive duplicate preferred term
destination	every direct entry has at least one in-range page and every printed page is linked
cross-reference	every see target exists once and links to the target index entry
coverage	60 chapter obligations, 74 source records, seven roles, three migrators, three facts and six protected HTTP surfaces
architecture	three owners, one edge and inactive candidates remain unchanged
claim state	no PENDING asset is described as implemented, executed or demonstrated
pagination	all indexed pages precede the Working Index and use final visible labels

WORKING INDEX CHECKPOINT

The current volume can now be navigated by concept and exact identity

COMPLETION LEDGER IX-5

Question	Answer
Where is the concept engineered?	curated principal pages
Where is the exact role, route, fact or command fixed?	catalogue or reproduction-guide page
Where is an external authority documented?	stable Sources entry
What happens to old names?	see redirect and explicit tombstone language
Does the index prove runtime behavior?	no; evidence state remains independently derived
What closes the manuscript?	

Final Consistency Audit plus the Reproduction Readiness and Deferred Evidence Checkpoint

FINAL INDEX RULE

An index is trustworthy only when every locator is both useful and falsifiable. A plausible term without an exact page is omitted; a page without a bounded reason is not promoted to a principal reference.

WORKING INDEX LAST WORD

Navigate by authority, not by vocabulary density

FINAL STATUS

WORKING INDEX COMPLETE | 381 ENTRIES | 24 GROUPS | 60 CHAPTER OBLIGATIONS | 74 SOURCES | INTERNAL PAGE AND SEE LINKS | FINAL-CORE AND CLAIM-STATE BOUNDARIES PRESERVED

FINAL REFERENCE BACK MATTER

Final Consistency Audit

A claim-aware reconciliation of the project description, the ten-Part contents, the final catalogues and the 2,036-page manuscript inherited at this checkpoint.

AUDIT DECISION

The book is structurally and semantically coherent after two bounded editorial corrections: the missing Part X bookmark is restored and two obsolete ProjectionDb labels are corrected to ReportingDb. Earlier candidate designs remain visible, but later catalogues explicitly deny them final-core authority.

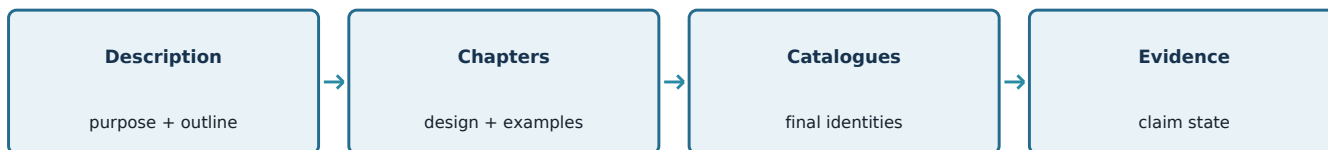
AUDIT CONTROL TOTALS CA-1

Coordinate	Audited value
Parts / chapters / appendices	10 / 60 / 8
consistency checks	72
final-core census	3 owners / 1 edge / 7 roles / 3 migrators / 3 stores / 3 facts / 6 protected HTTP surfaces
inactive candidates	Audit / Compliance / Process Manager / Repair Control / RL Advisory
editorial corrections	2 bounded corrections; no chapter removed or renumbered
runtime posture	SPECIFIED / PENDING; no clean-run evidence invented

AUDIT METHOD AND AUTHORITY ORDER

Later evidence narrows earlier design without erasing it

CONSISTENCY DECISION PATH



AUTHORITY ORDER CA-2

Priority	Authority	Use in this audit
1	standing boundary + non-goals	deny safety, production and hidden-control claims
2	Appendices A-G	fix final paths, roles, stores, contracts, trust, failures and commands
3	Appendix H	reconcile material class, status, tombstones and residual obligations
4	Epilogue	limit what the architecture and evidence may claim
5	Working Index	navigate preferred terms, aliases and principal pages

INTERPRETATION RULE

An earlier chapter may contain a complete candidate design. That design becomes current runtime authority only if the final service, contract, database, security and reproduction catalogues admit it under the same identity.

PROJECT-DESCRIPTION RECONCILIATION

The manuscript preserves purpose while narrowing admission

DESCRIPTION-TO-FINAL CROSSWALK CA-3

Description coordinate	Final manuscript decision	Audit result
five strongest service candidates	AlarmManagement, RootCause and Reporting admitted; Audit/Compliance retained as inactive candidates	RECONCILED
Process Manager implementation	complete extension design; no final role, store, fact or credential	RECONCILED
optional RL branch	module-only advisory snapshot; no activated fact	PASS
Appendix D database catalogue	ownership moved to Chapters 13-18 + Appendix B/G; D owns configuration	RECONCILED
600-750 page target	superseded as a size constraint; completeness and atomic layout govern	ACCEPTED EVOLUTION
compile/runtime evidence	requirements retained as PENDING rather than converted into narrative success	PASS

WHY THIS IS COHERENT

The project description selected candidates before execution. The final catalogues apply the description's own rule: when admission evidence is absent, narrow the runtime instead of pretending the candidate exists.

CLOSED FINAL-CORE CENSUS

One authoritative set controls every later claim

RUNTIME IDENTITIES CA-4

Kind	Exact admitted identities
owners	AlarmManagement; RootCause; Reporting
edge	Query BFF / query-bff
roles	alarm-api; alarm-worker; rootcause-api; rootcause-worker; projection-api; projection-worker; query-bff
migrators	alarm-migrator; rootcause-migrator; reporting-migrator
stores	AlarmManagementDb; RootCauseDb; ReportingDb
facts	AlarmFloodDetected.v1; RootCauseCaseOpened.v1; RootCauseVerdictIssued.v1

CLOSED-WORLD RULE

Any additional role, database, migrator, publisher, consumer, route or inactive-candidate credential rejects the final plan before mutation.

PROTECTED CONTRACT CENSUS

Six business HTTP surfaces, three lifecycle surfaces

BUSINESS SURFACES CA-5

Owner	Method + route	Authority
AlarmManagement	POST /api/v1/alarm-observations	authorized observation admission
RootCause	GET /api/v1/root-cause-cases/{caseld}	owner-local case truth
RootCause	POST /api/v1/root-cause-cases/{caseld}/evidence	attach admitted evidence
RootCause	POST /api/v1/root-cause-cases/{caseld}/verdicts	issue or abstain under policy
Reporting	GET /internal/v1/root-cause-case-projections/{caseld}	bounded projection query for Query BFF
Query BFF	GET /api/v1/root-cause-cases/{caseld}	authorized delayed observation

LIFECYCLE SURFACES CA-6

Route	Meaning	Must not become
GET /health/startup	finite initialization completed	ongoing dependency monitor
GET /health/live	process can continue executing	remote-dependency test
GET /health/ready	role can accept intended work	unrelated-service census

EVIDENCE POSTURE

Editorial verification and runtime demonstration are separate

CLAIM-STATE LEDGER CA-7

Coordinate	Current state	Allowed sentence
architecture + catalogues	SPECIFIED	the final design requires...
PDF structure/navigation	EXECUTED	the current artifact passed the named manuscript checks...
repository presence	PENDING	the canonical paths and assets are required...
clean restore/build/test	PENDING	the command chain must prove...
Compose + provider runtime	PENDING	the admitted run must show...
failure/security campaigns	PENDING	the scenario remains unexecuted until retained evidence exists...
production readiness	NOT CLAIMED	the Phase-0 demonstrator does not establish it

AUDIT HONESTY

This section executes editorial, structural and semantic consistency checks on the manuscript. It does not execute the repository, SQL Server, RabbitMQ, Compose, Kubernetes or failure campaigns.

FROM SERVICES TO RUNTIME

FA-01 - IDENTITY, PURPOSE AND NON-GOALS

CONSISTENCY CHECKS FA-01

ID / gate	Pages	Verdict	Finding
CA-001 Title and implementation-blueprint identity	3 / PASS / From Services to Runtime remains the		NEXUS-1 companion implementation blueprint.
CA-002 Exact handoff from architecture to runtime	16 / 1774	PASS	The manuscript begins at the From Flow to Services handoff and ends with an evidence-limited judgement.
CA-003 Educational Phase-0 boundary	3 / 1774	PASS	Front matter and conclusion deny safety-class, certified, operational and real-facility claims.
CA-004 Implementation difficulty is not hidden	9 / 49	PASS	Ownership, delayed truth, partial failure and operational cost remain explicit.
CA-005 Production readiness is a non-claim	3 / 1608 / 1774	PASS	Compose and Kubernetes material is educational and runtime evidence remains pending.
CA-006 Deterministic decisions keep human authority	33 / 894	PASS	RL/AI remains advisory or explanatory and receives no silent decision authority.

FA-02 - OUTLINE, PARTS AND BACK MATTER

CONSISTENCY CHECKS FA-02

ID / gate	Pages	Verdict	Finding
CA-007 Ten-Part master structure	8 / 15 / 1608	PASS	Parts I-X are present in content order.
CA-008 Sixty-chapter outline	8 / 16 / 1751	PASS	Chapters 1-60 are present once and follow the approved titles.
CA-009 Six chapters per Part	8	PASS	Every Part owns one contiguous six-chapter range.
CA-010 Epilogue and Appendices A-H	1774 / 1786 / 1968	PASS	The epilogue and all eight final catalogues are present.
CA-011 Reference-back-matter order	1986 / 1999 / 2010 / 2018	PASS	Glossary, Notation, Sources and Working Index follow Appendix H in the approved order.
CA-012 Part X bookmark hierarchy	1608	CORRECTED	A missing Part X bookmark was inserted before Chapters 55-60; content and pagination were unchanged.

FA-03 - FINAL ARCHITECTURE CENSUS

CONSISTENCY CHECKS FA-03

ID / gate	Pages	Verdict	Finding
CA-013 Service-owner count	1808 / 1945	PASS	Exactly three admitted owners: AlarmManagement, RootCause and Reporting.
CA-014 Edge count and authority	1808 / 1840	PASS	Exactly one Query BFF edge; it owns no domain truth or database.
CA-015 Long-running runtime-role count	1795 / 1945	PASS	Exactly seven roles: two Alarm, two RootCause, two Reporting and one Query BFF.
CA-016 Finite migrator count	1795 / 1945 / 1954	PASS	Exactly three owner-specific migration jobs; they are not long-running services.
CA-017 Core store count	1825 / 1954	PASS	Exactly three core service-owned databases.
CA-018 Activated public-fact count	1832 / 1945	PASS	Exactly three producer-owned public facts are active specifications.

FA-04 - ARCHITECTURAL EVOLUTION AND CANDIDATE DISPOSITION

CONSISTENCY CHECKS FA-04

ID / gate	Pages	Verdict	Finding
CA-019 Audit planned service versus final disposition	814 / 1758 / 1820	RECONCILED	Chapter 34 preserves the candidate design; final catalogues admit no Audit role, store, queue or credential.
CA-020 Compliance planned service versus final disposition	814 / 1758 / 1821	RECONCILED	Compliance semantics remain designed, but the final runtime grants zero authority.
CA-021 Process Manager design versus final disposition	924 / 1758 / 1823	RECONCILED	WorkflowDb and the fourth fact remain an extension design, not a final-core runtime.
CA-022 RL Advisory optional branch	894 / 917 / 1822	PASS	RecommendationGenerated.v1 remains inactive and advisory-only.
CA-023 Repair Control protected module	1082 / 1824	PASS	Repair semantics are retained without an admitted independent role or store.
CA-024 Appendix D scope evolution	266 / 1806 / 1860	RECONCILED	Database ownership is fixed by Chapters 13-18 and Appendix B/G; Appendix D now owns configuration identity.

FROM SERVICES TO RUNTIME

FA-05 - DATA OWNERSHIP AND TRANSACTION BOUNDARIES

CONSISTENCY CHECKS FA-05

ID / gate	Pages	Verdict	Finding
CA-025 AlarmManagementDb ownership	295 / 1825 / 1954	PASS	Alarm runtime owns DML; alarm-migrator alone owns its DDL/history.
CA-026 RootCauseDb ownership	318 / 1825 / 1954	PASS	RootCause runtime owns case/evidence/verdict state and its inbox/outbox.
CA-027 ReportingDb ownership	850 / 1825 / 1954	PASS	Reporting owns projections, inbox, ledger and high-water state.
CA-028 Runtime principals cannot migrate	341 / 1914 / 1954	PASS	Runtime roles receive bounded DML or read authority and no DDL.
CA-029 No cross-database foreign keys or writes	363 / 1825	PASS	Cross-owner references are immutable identifiers; foreign mutation remains forbidden.
CA-030 Local outbox/inbox atomicity	576 / 619 / 1825	PASS	Each local effect and its publication/admission record commit in one owner database.

FA-06 - HTTP CONTRACTS AND LIFECYCLE SURFACES

CONSISTENCY CHECKS FA-06

ID / gate	Pages	Verdict	Finding
CA-031 Alarm observation command surface	1834 / 1836	PASS	POST /api/v1/alarm-observations is the final AlarmManagement route.
CA-032 RootCause owner HTTP surfaces	1838 / 1839	PASS	Owner query, evidence attachment and verdict issue are explicit protected contracts.
CA-033 Reporting internal query	1840	PASS	The Query BFF reads Reporting through one bounded internal contract, never ReportingDb.
CA-034 Public Query BFF route	1840	PASS	GET /api/v1/root-cause-cases/{caseId} returns an authorized delayed observation.
CA-035 Startup/live/ready lifecycle surfaces	221 / 1842	PASS	Three host routes keep initialization, process viability and work admission distinct.
CA-036 Legacy route tombstones	1836 / 1840 / 1853	PASS	Earlier alarm and operator routes remain documented only as superseded spellings.

FA-07 - PUBLIC FACTS AND MESSAGING SEMANTICS

CONSISTENCY CHECKS FA-07

ID / gate	Pages	Verdict	Finding
CA-037 Three exact active fact identities	1832 / 1843 / 1846	PASS	AlarmFloodDetected.v1, RootCauseCaseOpened.v1 and RootCauseVerdictIssued.v1 are canonical.
CA-038 Candidate facts remain inactive	917 / 1821 / 1852	PASS	RecommendationGenerated.v1 and ComplianceReviewOpened.v1 receive no final routing authority.
CA-039 Canonical V1 envelope	442 / 1842	PASS	Message, schema, time, correlation, causation and producer identities are explicit.
CA-040 At-least-once transport boundary	555 / 619	PASS	The manuscript never upgrades broker delivery to global exactly-once execution.
CA-041 Publisher-confirm and unknown-outcome handling	594 / 645	PASS	Confirmation uncertainty retains stable identity and safe retry state.
CA-042 Bounded retry, quarantine and replay	645 / 1888	PASS	Poison messages become visible evidence; replay is authorized and receipted.

FA-08 - SECURITY, IDENTITY AND LEAST PRIVILEGE

CONSISTENCY CHECKS FA-08

ID / gate	Pages	Verdict	Finding
CA-043 Edge authentication	1182 / 1840	PASS	The BFF authenticates browser/operator identity before composition.
CA-044 Owner-local authorization	1182 / 1280 / 1838	PASS	Owning services enforce capability and scope; edge success is not delegated authority.
CA-045 Service workload identity	1231 / 1914	PASS	A user token alone is never treated as service trust.
CA-046 Claims minimization	1280 / 1914	PASS	Only bounded actor context crosses owner boundaries.
CA-047 Secret and certificate references	1334 / 1860 / 1914	PASS	Configuration records references and metadata, never secret values or derived fingerprints.
CA-048 Security execution evidence	1375 / 1942	DECLARED PENDING	Negative campaigns and broker/database permission probes are specified but not claimed as executed.

FROM SERVICES TO RUNTIME

FA-09 - OBSERVABILITY, FAILURE AND RECOVERY

CONSISTENCY CHECKS FA-09

ID / gate	Pages	Verdict	Finding
CA-049 Correlation versus causation	1417	PASS	Shared scenario identity and directed causal edges remain distinct.
CA-050 Structured logs preserve bounded meaning	1457	PASS	Stable event names avoid unbounded payloads and secret-bearing fields.
CA-051 Trace context is diagnostic	1489 / 1842	PASS	Trace/span identity never replaces durable business lineage.
CA-052 Metrics and objectives are educational	1518 / 1892	PASS	Evidence budgets are not published as production SLO, RTO or RPO claims.
CA-053 Six mandatory failure scenarios	1578 / 1892 / 1894	PASS	Consumer kill, broker stop, DB pause, duplicate, projection delay and service restart are fixed.
CA-054 Recovery means truth plus stable progress	1607 / 1894	PASS	A restart, reconnect or green probe alone is explicitly insufficient.

FA-10 - DELIVERY, MIGRATION AND REPRODUCTION

CONSISTENCY CHECKS FA-10

ID / gate	Pages	Verdict	Finding
CA-055 Canonical repository and solution paths	1786 / 1943	PASS	artifacts/from-services-to-runtime/ and Nexus1.Runtime.sln own final identity.
CA-056 Two equivalent public wrappers	1944 / 1962	PASS	nexus.sh and nexus.ps1 expose one closed command grammar.
CA-057 Five safety classes	1947	PASS	Mutation and destructive evidence require an admitted profile and isolated target.
CA-058 Docker Compose runtime	1608 / 1942	DECLARED PENDING	The topology and commands are specified; no retained clean runtime run is fabricated.
CA-059 Kubernetes remains a translation	1725 / 1963	PASS	Manifests translate already-proved needs and do not claim a deployed platform.
CA-060 Clean-room reproduction	1669 / 1942 / 1962	DECLARED PENDING	No clean checkout/build/test/runtime dossier is claimed before the final checkpoint.

FA-11 - CLAIM STATES, EVIDENCE AND SOURCE AUTHORITY

CONSISTENCY CHECKS FA-11

ID / gate	Pages	Verdict	Finding
CA-061 Claim-state vocabulary	1774 / 1942 / 1968	PASS	SPECIFIED, IMPLEMENTED, EXECUTED, DEMONSTRATED, FALSIFIED, SUPERSEDED and INACTIVE remain distinct.
CA-062 No promotion from PENDING	1751 / 1774 / 1942	PASS	Architecture detail and expected output never manufacture repository or runtime evidence.
CA-063 Sources versus conformance	2010	PASS	Authoritative documentation supports semantics but does not prove this runtime executed.
CA-064 Diagram validity rule	3 / 1968	PASS	A diagram is accepted only when repository, contracts, ownership, tests and runtime agree.
CA-065 Same-run evidence	1578 / 1942 / 1968	PASS	Cross-run joins and stale receipts cannot manufacture a complete dossier.
CA-066 Current implementation/runtime posture	1751 / 1774 / 1968	DECLARED PENDING	The book closes architecture and evidence specifications while preserving missing execution obligations.

FA-12 - NAVIGATION, LAYOUT AND FINAL HANDOFF

CONSISTENCY CHECKS FA-12

ID / gate	Pages	Verdict	Finding
CA-067 Inherited body page census	2036	PASS	The audited pre-section body contains 2,036 pages with one graphical cover and no true blank page.
CA-068 Controlled reference back matter	1986 / 1999 / 2010 / 2018	PASS	176 glossary terms, 140 notation entries, 74 sources and 381 working-index entries are present.
CA-069 Contents, bookmarks and internal links	8 / 2018	PASS	All destinations are in range; Part X hierarchy is corrected; reference sections remain navigable.
CA-070 Cover, headers, footers and page labels	1 / 8	PASS	The cover remains full-bleed and every text page retains its expected frame and visible page label.
CA-071 Preview and text behavior	1 / 2018	PASS	The delivered PDF remains searchable, unencrypted and linearized for preview.
CA-072 Handoff to honest reproduction readiness	1942 / 2037	PASS	Runtime claims remain pending; the final checkpoint must record absence without promotion.

BOUNDED CORRECTION REGISTER

Two real defects were corrected; design history was preserved

CORRECTION LEDGER CA-8

Correction	Before	After	Impact
CA-COR-001	bookmark tree jumped from Part IX directly through Chapters 55-60	Part X bookmark inserted at p. 1608	navigation only; zero page renumbering
CA-COR-002	two Ch 59 workload-store cells said ProjectionDb	both cells now say ReportingDb	aligns Kubernetes translation with final store census

PRESERVATION RULE

Audit/Compliance, WorkflowDb, candidate facts, earlier routes and provisional paths were not deleted. Their later tombstones and inactive dispositions are architecturally meaningful history, not accidental clutter.

RESIDUAL REGISTER

What remains unresolved is operational, not editorial

DECLARED RESIDUALS CA-9

Residual	Required next evidence	Current claim
canonical repository presence	exact paths + admitted source revision	PENDING
locked restore/build/test	same-run process receipts and complete test discovery	PENDING
three provider migrations	owner-specific DDL/history receipts	PENDING
seven-role runtime	rendered inventory + readiness + durable progress	PENDING
three public facts	broker topology, corpus digest and owner effects	PENDING
six failure scenarios	baseline/fault/heal/recovery/cleanup evidence	PENDING
security boundaries	negative probes with protected-effect oracles	PENDING
clean-room comparison	two admitted runs with equal semantic identities	PENDING

NO EDITORIAL WORKAROUND

The readiness checkpoint records these residuals as deferred. It does not mark them complete merely because the manuscript is internally consistent.

FINAL AUDIT VERDICT

Coherent, navigable and claim-honest - with runtime evidence still pending

FINAL VERDICT CA-10

Question	Answer
Does the book follow its purpose and ten-Part contents?	yes; all 60 chapter obligations and final back matter are present
Does one final architecture control the manuscript?	yes; 3/1/7/3/3/3/6 plus five inactive candidates
Are earlier candidate designs misleadingly active?	no; final catalogues and this audit deny their runtime authority
Were actual defects corrected?	yes; Part X bookmark and ReportingDb workload labels
Does this audit prove the runtime?	no; implementation and execution residuals remain PENDING
What closes the book-production sequence?	Final Reproduction Readiness and Deferred Evidence Checkpoint

VERDICT

FINAL CONSISTENCY AUDIT: PASS WITH DECLARED RUNTIME RESIDUALS. The appended readiness checkpoint must preserve those residuals as NOT IMPLEMENTED, NOT EXECUTED and NOT DEMONSTRATED.

FINAL CONSISTENCY AUDIT LAST WORD

Consistency closes the manuscript; implementation must earn any stronger claim

FINAL STATUS

FINAL CONSISTENCY AUDIT COMPLETE | 72 CHECKS | 12 FAMILIES | 2 CORRECTIONS | 10 PARTS / 60 CHAPTERS / 8 APPENDICES | FINAL-CORE CENSUS CLOSED | RUNTIME EVIDENCE STILL PENDING

FINAL REFERENCE BACK MATTER

Final Reproduction Readiness and Deferred Evidence Checkpoint

An explicit stop condition for a book whose architecture is specified but whose reference application, repository and runtime have not been implemented.

CHECKPOINT DECISION

No reproduction run is reported. No repository, build, test, migration, container, broker, failure-injection or security receipt exists for this manuscript edition. The honest result is readiness specified, implementation absent and execution deferred.

FINAL CLAIM BOUNDARY RR-1

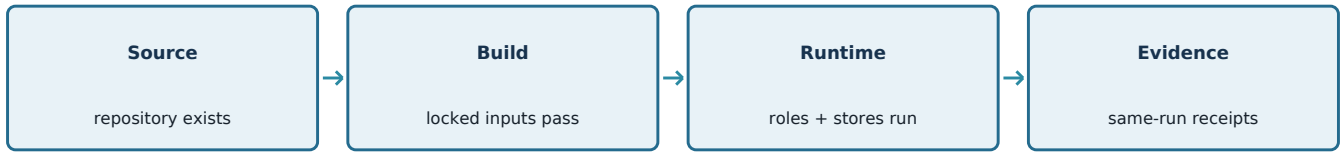
Coordinate	Final state
manuscript consistency	PASS - editorial and semantic checks executed
declared reference-slice architecture	SPECIFIED - bounded design, not running software
reference application and repository	NOT IMPLEMENTED
clean restore/build/test	NOT EXECUTED
end-to-end runtime behaviour	NOT DEMONSTRATED
production readiness	NOT CLAIMED

FROM SERVICES TO RUNTIME

WHY THE CHECKPOINT STOPS

An absent implementation cannot produce a reproduction receipt

EVIDENCE ADMISSION PATH



GATE DISPOSITION RR-2

Gate	Required evidence	This edition
R0 - source identity	immutable revision + exact artifact path	ABSENT
R1 - dependency restore	lock-respecting restore receipt	NOT EXECUTED
R2 - compilation	solution build receipt + binary digests	NOT EXECUTED
R3 - tests	discovery inventory + passing result corpus	NOT EXECUTED
R4 - migrations	three owner-specific provider receipts	NOT EXECUTED
R5 - runtime	seven-role inventory + readiness + durable progress	NOT EXECUTED
R6 - behaviour	correlated command/fact/projection evidence	NOT EXECUTED
R7 - recovery/security	fault, heal, negative-probe and cleanup receipts	NOT EXECUTED

NO SUBSTITUTE

Detailed code listings, expected console output, topology diagrams and command examples are not execution evidence. They specify what a future implementation must satisfy.

DECLARED LEGACY SCALE

654 tables and 17 schemas describe an estate, not a service map

SOURCE-ESTATE INTERPRETATION RR-3

Statement	What it establishes	What it does not establish
654 tables	declared scale of the legacy relational estate	654 independently classified ownership decisions
17 schemas	declared organizational/database structure	17 bounded contexts or 17 services
existing foreign keys	current coupling and integrity relationships	future service boundaries
current transactions	legacy atomicity assumptions	safe distributed transaction cuts
current application modules	candidate capability groupings	admitted independently deployable services

INVENTORY QUALIFICATION

The 654/17 figures are project inputs at this checkpoint. This manuscript does not claim that every table, key, procedure, job, view, write path or data-retention obligation has been machine-inventoried and dispositioned.

FROM SERVICES TO RUNTIME

WHY THE FINAL SERVICE COUNT IS UNKNOWN

Service boundaries follow business authority, not table arithmetic

BOUNDARY EVIDENCE RR-4

Decision coordinate	Question that must be answered
business capability	Which outcomes and language change together?
invariants	Which rules require one consistency boundary?
write authority	Who is the single admitted writer for each fact?
transaction scope	Which commits must remain atomic and which may become delayed?
identity lifecycle	Who creates, retires and resolves each identifier?
change cadence	Which capability must deploy independently for a real reason?
failure isolation	Which failure should stop, delay or spare which work?
security authority	Which principal may perform which protected effect?
operational ownership	Which team can build, observe, recover and evolve the boundary?

NO INVENTED NUMBER

The manuscript therefore does not estimate a final enterprise service count. Three owners are admitted for the reference slice only; the wider estate may justify more, fewer or differently shaped boundaries after evidence-led discovery.

THE BOUNDED REFERENCE SLICE

Three owners demonstrate a migration pattern, not the whole migration

SPECIFIED SLICE RR-5

Kind	Admitted design identity	Status
owners	AlarmManagement; RootCause; Reporting	SPECIFIED
edge	Query BFF	SPECIFIED
long-running roles	alarm-api; alarm-worker; rootcause-api; rootcause-worker; projection-api; projection-worker; query-bff	SPECIFIED
migrators	alarm-migrator; rootcause-migrator; reporting-migrator	SPECIFIED
stores	AlarmManagementDb; RootCauseDb; ReportingDb	SPECIFIED
active public facts	AlarmFloodDetected.v1; RootCauseCaseOpened.v1; RootCauseVerdictIssued.v1	SPECIFIED

EXPLICIT EXCLUSIONS RR-6

Coordinate	Final disposition
remaining legacy tables/schemas	NOT CLASSIFIED INTO FINAL SERVICE OWNERSHIP
Audit / Compliance	INACTIVE CANDIDATES
Process Manager / WorkflowDb	INACTIVE EXTENSION DESIGN
Repair Control	PROTECTED MODULE; NO ADMITTED ROLE OR STORE
RL Advisory	OPTIONAL ADVISORY BRANCH; NO ACTIVATED FACT

FROM SERVICES TO RUNTIME

FINAL CLAIM-STATE LEDGER

Each statement is limited by the strongest retained evidence

CLAIM VOCABULARY RR-7

State	Meaning in this edition	Permitted wording
SPECIFIED	architecture, contract or procedure is described	the blueprint requires...
ILLUSTRATED	listing demonstrates intended shape	the example shows...
IMPLEMENTED	artifact exists at an exact revision/path	not available for the reference application
EXECUTED	named command ran with retained receipt	only manuscript/PDF QA qualifies
DEMONSTRATED	runtime behaviour survived a named oracle	not available for the reference runtime
FALSIFIED	evidence contradicted a claim	no runtime campaign was run
SUPERSEDED	later identity replaces earlier spelling/design	use the final catalogue identity
INACTIVE	candidate receives no final runtime authority	design retained; deployment denied
NOT CLAIMED	the edition deliberately refuses the assertion	production/safety readiness is not established

PROMOTION RULE

A future edition may advance a coordinate only by naming the artifact, immutable build identity, command, environment, oracle and retained receipt that earned the stronger state.

HOW TO READ CODE AND COMMANDS

Listings are blueprint material unless execution is explicitly receipted

LISTING CLASSES RR-8

Material class	Reader interpretation	Evidence ceiling
reference code	compile-oriented design for a future repository	ILLUSTRATED / SPECIFIED
SQL or migration example	provider-oriented design, not applied DDL	SPECIFIED
Compose/Kubernetes manifest	required topology shape, not a deployed environment	SPECIFIED
shell/PowerShell command	future reproduction interface	SPECIFIED
expected output	acceptance shape or oracle example	ILLUSTRATED
test code	required executable claim, not a retained result	SPECIFIED
diagram	architecture hypothesis constrained by final catalogues	SPECIFIED

READER WARNING

The words reference, example, expected and proposed never mean compiled, deployed, executed, measured or demonstrated.

FROM SERVICES TO RUNTIME

NULL EVIDENCE MANIFEST

Absence is recorded instead of being hidden

EVIDENCE MANIFEST RR-9

Evidence coordinate	Expected artifact	Recorded value
repository identity	immutable revision + repository URL	ABSENT
artifact root	exact source path	ABSENT
solution identity	Nexus1.Runtime.sln at admitted revision	ABSENT
dependency locks	package/tool/container lock digests	ABSENT
restore/build	same-run process receipts	ABSENT
test inventory/results	discovered tests + result corpus digest	ABSENT
database migrations	three provider DDL/history receipts	ABSENT
container images	seven role digests + three migrator digests	ABSENT
runtime inventory	roles/stores/broker/readiness receipt	ABSENT
business flow	correlated command/fact/projection dossier	ABSENT
failure/security campaigns	fault/negative-probe/cleanup receipts	ABSENT
reproduction comparison	two admitted clean-run semantic digests	ABSENT

MANIFEST INTERPRETATION

ABSENT does not mean a command failed. It means no admissible artifact was available to run or inspect. Converting ABSENT to PASS would be fabrication.

DEFERRED EVIDENCE REGISTER

The checkpoint records why each runtime claim remains open

DEFERRED OBLIGATIONS RR-10

Obligation	Blocking fact	Current outcome
repository inspection	reference repository not implemented	DEFERRED
locked build	no admitted source revision or dependency locks	DEFERRED
provider migration	no owner database projects or migration bundles	DEFERRED
message topology	no executable broker declarations or deployed principals	DEFERRED
end-to-end flow	no running producer, consumer, stores or BFF	DEFERRED
failure recovery	no baseline runtime to fault and heal	DEFERRED
security campaign	no protected effects or workload identities to probe	DEFERRED
clean-room equality	no first admitted run exists	DEFERRED

CHECKPOINT SEMANTICS

This is a completed editorial checkpoint with a deferred runtime outcome. The section itself is complete; the implementation work it describes is not.

FROM SERVICES TO RUNTIME

FUTURE ESTATE-DISCOVERY TRANCHE

Before more services, every legacy object needs a disposition

REQUIRED INVENTORY RR-11

Work product	Minimum controlled content
object catalogue	654 table identities plus views, procedures, jobs, triggers and functions
write-path map	application, integration, job and operator writers per object
transaction graph	commands and batches that cross object boundaries atomically
foreign-key graph	direction, optionality, cascade behaviour and orphan risk
data classification	sensitivity, retention, residency and audit obligations
volume/churn profile	size, write rate, hot paths, archival and recovery needs
capability map	business language, outcomes, policies and responsible owner
disposition ledger	retain, split, replicate, project, migrate, retire or investigate

COMPLETENESS GATE

No claim of full decomposition is admissible until every inventoried object has exactly one controlled disposition, an accountable decision owner and an unresolved-state reason when evidence is missing.

TABLE-TO-OWNER DISPOSITION LEDGER

A machine-readable ledger must replace intuition

REQUIRED COLUMNS RR-12

Column	Purpose
legacy_object_id	stable schema-qualified identity
object_kind	table/view/procedure/job/trigger/function/other
current_writers	all observed write authorities
transaction_cluster	atomic legacy work that may constrain a cut
candidate_capability	business capability hypothesis
proposed_owner	single future truth owner or UNRESOLVED
disposition	retain/split/replicate/project/migrate/retire/investigate
integration_contract	identifier, API or public fact replacing direct access
migration_state	inventory/classified/proved/cutover/retired
evidence_refs	queries, traces, interviews, decisions and receipts
risk/rollback	data-loss, downtime, reversal and repair constraints

CLOSED-WORLD LEDGER RULE

Unknown is a valid temporary value; an omitted row is not. The ledger must preserve unresolved objects visibly instead of forcing premature service ownership.

FROM SERVICES TO RUNTIME

FUTURE SERVICE-ADMISSION GATE

A candidate becomes a service only after it earns operational independence

ADMISSION QUESTIONS RR-13

Gate	Required answer before admission
capability	coherent language, outcomes and policies
truth ownership	single writer and explicit read/integration contracts
transaction boundary	local invariants survive the proposed cut
data migration	reconciliation, cutover, reversal and retirement are defined
failure model	timeouts, duplicates, delay, poison and recovery have bounded behaviour
security	workload identity, least privilege and negative probes are specified
observability	logs, metrics, traces and durable business lineage expose behaviour
operability	deployment, migration, backup, restore, support and ownership are affordable
evidence	component, contract, architecture and runtime tests can falsify the claim

ADMISSION VERDICT

A useful module is not automatically a service. Independent deployment is accepted only when independent ownership, failure, security, data and recovery responsibilities are real and supportable.

FUTURE REFERENCE-IMPLEMENTATION TRANCHE

If implementation begins, prove one narrow slice before broadening it

ORDERED WORK RR-14

Stage	Deliverable	Do not broaden until
1	immutable repository skeleton and locked toolchain	clean restore/build is receipted
2	AlarmManagement owner database + observation command	provider component tests pass
3	local outbox + AlarmFloodDetected.v1 publisher	unknown outcome and retry are proved
4	RootCause inbox + case/evidence/verdict owner	duplicates and concurrency are proved
5	Reporting projection + delayed-truth query	replay and high-water behaviour are proved
6	Query BFF authentication/authorization/composition	owner-local authorization remains enforced
7	seven-role Compose runtime + three migrators	inventory and readiness match catalogues
8	failure/security campaigns	truth, progress, cleanup and protected effects recover

SLICE BOUNDARY

Even a successful reference slice would demonstrate only its declared contracts and failure envelope. It would not prove a complete migration of the 654-table, 17-schema estate.

FUTURE EVIDENCE-BUNDLE CONTRACT

Reproduction begins only when the dossier can identify one exact run

MINIMUM BUNDLE RR-15

Bundle coordinate	Required value
source	repository URL, immutable revision, dirty-state receipt and artifact root
toolchain	.NET SDK, package locks, database, broker, container engine and OS identities
commands	exact public command sequence with arguments and safety profile
build	restore/build/test process receipts and discovered-test census
images	role/migrator image digests and provenance
databases	migration IDs, schema fingerprints, principals and ownership receipts
broker	exchange/queue/binding/principal inventory and policy fingerprints
runtime	role inventory, readiness transitions, correlation/run identity and timestamps
behaviour	command, fact, inbox/outbox, owner state and projection conservation evidence
fault/security	baseline, injection, observation, heal, recovery, negative probes and cleanup
comparison	two admitted clean runs with equal semantic identities

SAME-RUN RULE

Receipts from different revisions, runs or environments may not be assembled into a synthetic success. Every cross-coordinate join carries the same immutable run identity.

WHAT THE READER MAY TRUST

The manuscript is useful without pretending that the runtime exists

READER TRUST MAP RR-16

The reader may rely on	The reader must not infer
the final declared architecture is internally reconciled	that the architecture has been implemented
service/data/contract boundaries are specified precisely	that 654 tables have been fully assigned
code and command listings define intended responsibilities	that listings compile or commands were run
tests and evidence oracles define falsifiable claims	that those tests discovered or passed
the reference slice is bounded to three owners	that three is the final enterprise service count
the PDF passed editorial, navigation and rendering QA	that SQL Server, RabbitMQ or containers were exercised
all runtime residuals remain visible	that production, safety or certification readiness follows

VALUE WITHOUT EXAGGERATION

The book contributes a disciplined implementation blueprint, decision system and evidence contract. Its value does not depend on relabelling a future implementation as present fact.

EDITORIAL IDENTITY CORRECTION

The edition now names itself according to what it can prove

IDENTITY CHANGES RR-17

Coordinate	Previous wording	Current wording
cover/header identity	NEXUS-1 Companion Implementation Volume	NEXUS-1 Companion Implementation Blueprint
cover subtitle	Implementing the NEXUS-1 Microservice Architecture in .NET	An Architecture and Implementation Blueprint for NEXUS-1 in .NET
cover architecture nodes	Audit / Compliance shown as peer runtime nodes	Query BFF / Evidence; inactive candidates removed from the cover runtime picture
final step	Final Reproduction Checkpoint	Final Reproduction Readiness and Deferred Evidence Checkpoint
runtime outcome	future checkpoint implied possible closure	NOT IMPLEMENTED / NOT EXECUTED / NOT DEMONSTRATED

MEANING OF BLUEPRINT

Blueprint is not a weaker engineering term. It is a more accurate one: the manuscript defines boundaries, responsibilities, examples, acceptance oracles and evidence obligations without claiming an artifact that does not exist.

FINAL HONEST VERDICT

The book closes; the implementation obligation remains visible

FINAL VERDICT RR-18

Decision	Verdict	Basis
manuscript consistency	PASS	72-check Final Consistency Audit plus full PDF QA
architecture specification	COMPLETE FOR THE DECLARED SLICE	three-owner reference design and final catalogues agree
full legacy decomposition	NOT PERFORMED	654-table/17-schema disposition ledger does not exist
reference application	NOT IMPLEMENTED	no admitted repository revision or artifact root exists
runtime reproduction	NOT EXECUTED	no build, migration, runtime or campaign receipt exists
end-to-end behaviour	NOT DEMONSTRATED	no running system was available to challenge
production/safety readiness	NOT CLAIMED	Phase-0 educational boundary remains controlling

FINAL CHECKPOINT RESULT

FINAL REPRODUCTION READINESS: SPECIFIED. IMPLEMENTATION: ABSENT. EXECUTION: DEFERRED. EVIDENCE: NOT FABRICATED. The next legitimate state change belongs to a future implementation project or evidence-bearing edition, not to further editorial assertion.

LAST WORD

Nothing claims to exist that does not

EDITION CLOSURE

THIS MANUSCRIPT IS COMPLETE AS AN ARCHITECTURE AND IMPLEMENTATION BLUEPRINT. IT IS NOT A COMPLETED SOFTWARE IMPLEMENTATION, A REPRODUCED RUNTIME, OR PRODUCTION-READY GUIDANCE.

DEFINITIVE PUBLICATION RECORD

Final Publication Artifact and Build Identity

The exact identity, paths, toolchain and evidentiary boundary of the definitive book edition.

PUBLICATION DECISION

This edition is final as a book artifact. Its editorial sources, PDF, cover image and external artifact manifest are retained as one publication set. The reference application remains NOT IMPLEMENTED; runtime reproduction remains NOT EXECUTED; production readiness remains NOT CLAIMED.

FINAL EDITION STATE FP-1

Coordinate	Definitive value
book title	From Services to Runtime - Definitive Final Edition
companion identity	NEXUS-1 Companion Implementation Blueprint
author	Grigorios Agathangelidis
copyright	Copyright (c) 2026 Grigorios Kyriakos Agathangelidis. All rights reserved.
publication build ID	NEXUS1-FSTR-FINAL-20260811-C158929B
source checkpoint SHA-256	c158929b277c1e0af5984fb7c533036684150208cbbf12b296ba0b607dc7b154
application state	NOT IMPLEMENTED
runtime evidence state	NOT EXECUTED / NOT DEMONSTRATED

EXACT ARTIFACT PATHS

One publication set, four controlled deliverables

ARTIFACT CONTRACT FP-2

Artifact	Canonical path	Role
final PDF	output/pdf/NEXUS1_Runtime_Definitive_Final_Edition.pdf	reader edition; searchable, bookmarked and page-aware
editable source	output/source/NEXUS1_Runtime_Final_Editable_Source.zip	code-based editorial source, inputs and build instructions
cover PNG	output/cover/NEXUS1_Runtime_Final_Cover_2100x3000.png	full-bleed 2100 x 3000 publication cover
artifact manifest	output/manifest/NEXUS1_Runtime_Final_Artifact_Manifest.json	final sizes, SHA-256 values, build identity and toolchain
application artifact	ABSENT	no reference application or runtime receipt is claimed

PUBLICATION TOOLCHAIN FP-3

Component	Pinned identity for this build
Python	3.12.13
PyMuPDF	1.26.6
ReportLab	4.4.9
pypdf	6.10.0
pdfplumber	0.11.8
Node.js	v24.14.0
qpdf-wasm	0.1.0
Poppler pdftoppm	26.05.0

EDITABLE-SOURCE MEANING

The editable source is the authentic programmatic editorial source used to assemble the book. A synthetic DOCX is not supplied because it would not be the source from which this edition was built.

BUILD AND EVIDENCE BOUNDARY**Publication reproduction is not application reproduction****CONTROLLED BOOK COMMANDS FP-4**

Stage	Exact command
assemble	python3 assemble_definitive_final_edition.py
linearize	node linearize_qpdf.mjs <prelinear-pdf> <final-pdf>
quality assurance	python3 qa_definitive_final_edition.py
artifact manifest	python3 build_artifact_manifest.py

EVIDENCE SEPARATION FP-5

Claim	Admissible result in this edition
book source can be edited	YES - retained code-based editorial source
book PDF can be rebuilt	YES - publication pipeline and versions are recorded
book PDF passed visual/mechanical QA	YES - final QA report is generated against the delivered PDF
reference application can be restored or built	NO CLAIM - application artifact is absent
databases, broker or containers were exercised	NO - runtime reproduction was not executed
end-to-end behavior or production readiness is proved	NO - not demonstrated and not claimed

WHY THE FINAL PDF HASH IS EXTERNAL

A PDF cannot contain its own final SHA-256 without changing the bytes being hashed. The immutable final hashes therefore belong to the adjacent artifact manifest; this page records the build identity and the SHA-256 of the admitted source checkpoint.

DEFINITIVE PUBLICATION VERDICT

BOOK ARTIFACT: FINAL. EDITABLE SOURCE: RETAINED. COVER: FULL-BLEED. NAVIGATION: PAGE-AWARE. APPLICATION: NOT IMPLEMENTED. RUNTIME REPRODUCTION: NOT EXECUTED. PRODUCTION READINESS: NOT CLAIMED.

DEFINITIVE REFERENCE BACK MATTER

Final Page-Aware Index

The definitive concept-to-page map for the complete final edition, including final audits, publication identity and delivered artifacts.

INDEX AUTHORITY

This index preserves the complete Working Index corpus and adds the final evidence, publication and artifact coordinates. It navigates claims; it does not promote SPECIFIED material to IMPLEMENTED, EXECUTED or DEMONSTRATED.

FINAL-INDEX CONTROL TOTALS FX-1

Coordinate	Value
indexed entries	409
alphabetical groups	25
chapter obligations	60
source records	74
final additions	28
architecture constant	3 owners / 1 edge / 7 roles / 3 stores / 3 facts / 6 protected HTTP surfaces
evidence ceiling	application NOT IMPLEMENTED; runtime NOT EXECUTED; production readiness NOT CLAIMED

HOW TO USE THE FINAL INDEX

Follow the printed page, then verify the claim state

ENTRY CONTRACT FX-2

Field	Meaning
FX identity and term	contiguous final-index identity and preferred spelling
principal pages	curated final-edition locations; every printed number is an internal link
locator	the chapter, appendix, source, artifact or boundary reason for the reference
see	an alias or tombstone linked to exactly one preferred index term
final additions	publication, dedication, audit and explicit absence/status coordinates

PAGE-AWARE RULE

The index is appended last. Its locators therefore use the exact visible page labels of the delivered definitive PDF and are not renumbered by later manuscript content.

COVERAGE AND EVIDENTIARY BOUNDARY

Complete navigation without invented implementation evidence

COVERAGE REGISTER FX-3

Indexed coordinate	Treatment
body and back matter	all 60 chapters, Epilogue, Appendices A-H, Glossary, Notation, Sources and prior Working Index corpus
final audits	Final Consistency Audit and Final Reproduction Readiness checkpoint
publication set	PDF, editable source, full-bleed cover, manifest and build identity
legacy scale	654 tables and 17 schemas remain declared input scale, not a completed service map
application state	ABSENT / NOT IMPLEMENTED / NOT EXECUTED / NOT DEMONSTRATED / NOT CLAIMED remain explicit

NO PROMOTION BY PUBLICATION

Producing a finished book proves only the book artifact. It does not prove that the reference application exists or that any database, broker, container or business flow ran.

- FINAL PAGE-AWARE INDEX

FINAL INDEX ENTRIES #-1

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-001 17 schemas	2046 / 2053 / 2055	declared legacy-estate scale; not a service count
FX-002 654 tables	2046 / 2053 / 2055	declared legacy-estate scale; not a completed ownership map

A - FINAL PAGE-AWARE INDEX

FINAL INDEX ENTRIES A-1

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-003 A Proof for the Queuing Formula $L = \lambda W$ - John D. C. Little (1961)	2016	Source SRC-073; FOUNDATIONAL PAPER; Ch 52-53; Notation
FX-004 A Verdict Fans Out to Audit and Compliance	814	Chapter 34; chapter obligation
FX-005 Abstention	49 / 33 / 175 / 1774	Ch 2, 9; Epilogue; controlled definition p. 1986
FX-006 Active final core	1751 / 1806 / 1968	Ch 60; App B, H; controlled definition p. 1986
FX-007 Aggregate	175 / 318 / 387	Ch 9, 15, 18; controlled definition p. 1986
FX-008 Aggregate root	175 / 318	Ch 9, 15; controlled definition p. 1986
FX-009 Alarm service	see	see AlarmManagement; canonical owner name

FROM SERVICES TO RUNTIME

A - FINAL PAGE-AWARE INDEX - CONTINUED

FINAL INDEX ENTRIES A-2

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-010 alarm-api	712 / 1810 / 1945 / 1611 / 1727 / 1761	Appendix B/G runtime role; see also AlarmManagement
FX-011 alarm-migrator	341 / 1945 / 1954 / 1729 / 1790 / 1810	finite migration job for AlarmManagementDb
FX-012 alarm-worker	576 / 1810 / 1945 / 1611 / 1727 / 1761	Appendix B/G runtime role; outbox dispatcher
FX-013 AlarmFloodDetected.v1	744 / 1809 / 1834 / 36 / 49 / 65	active public fact owned by AlarmManagement
FX-014 AlarmManagement	295 / 712 / 744 / 1806	Ch 14, 31-32; App B; controlled definition p. 1986
FX-015 AlarmManagement Enters the Runtime	712	Chapter 31; chapter obligation
FX-016 AlarmManagementDb	295 / 341 / 387 / 1806	Ch 14, 16, 18; App B; controlled definition p. 1986

A - FINAL PAGE-AWARE INDEX - CONTINUED

FINAL INDEX ENTRIES A-3

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-017 AMQP 0-9-1 Protocol Specification	2014	Source SRC-035; PROTOCOL SPEC; Ch 26-30
FX-018 /api/operator/v1/root-cause-cases/{caseld}	see	see GET /api/v1/root-cause-cases/{caseld} - edge; superseded route tombstone
FX-019 Application artifact - ABSENT	2049 / 2055	no admitted reference-application artifact exists
FX-020 Application implementation - NOT IMPLEMENTED	2045 / 2049 / 2053 / 2056	final evidence ceiling
FX-021 Applying migrations - EF Core	2012	Source SRC-017; OFFICIAL DOCS; Ch 13-18, 56-58
FX-022 Architectural disposition	1751 / 1806 / 1968	Ch 60; App B, H; controlled definition p. 1986
FX-023 Architecture test	87 / 152 / 243	Ch 5, 8, 12; controlled definition p. 1986

A - FINAL PAGE-AWARE INDEX - CONTINUED

FINAL INDEX ENTRIES A-4

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-024 Artifact manifest	2055 / 2056	external final size and SHA-256 record
FX-025 artifacts/from-services-to-runtime/	1786 / 1942 / 1944 / 1788 / 1943 / 1974	canonical companion root
FX-026 At-least-once delivery	555 / 619 / 645 / 680	Ch 25, 28-30; controlled definition p. 1986
FX-027 Atomicity	387 / 576	Ch 18, 26; controlled definition p. 1986
FX-028 Audit	814 / 1808 / 1945	inactive module/service candidate; zero final-core authority
FX-029 Audit evidence	1375 / 1578 / 1942 / 1968	Ch 48, 54; App G-H; controlled definition p. 1986
FX-030 Authority	266 / 1132 / 1182 / 1231 / 1280 / 1334 / 1375 / 1914	Ch 13, 43-48; App F; controlled definition p. 1986

B - FINAL PAGE-AWARE INDEX

FINAL INDEX ENTRIES B-1

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-031 Backend for Frontend	see	see BFF (Backend for Frontend); preferred term
FX-032 Background tasks with hosted services in ASP.NET Core - Microsoft (.NET 8)	2012	Source SRC-012; OFFICIAL DOCS; Ch 8, 11, 27-30
FX-033 Backoff	645 / 1549	Ch 29, 53; controlled definition p. 1986
FX-034 BASE	1968	App H; controlled definition p. 1986
FX-035 BFF	see	see Query BFF; final edge identity
FX-036 BFF (Backend for Frontend)	486 / 1047 / 1182 / 1806 / 1832	Ch 22, 41, 44; App B-C; controlled definition p. 1986
FX-037 Bounded context	49 / 266	Ch 3, 13; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-038 Bounded shutdown	152 / 221 / 594	Ch 8, 11, 27; controlled definition p. 1986
FX-039 Broker topology	555 / 1806 / 1832	Ch 25; App B-C; controlled definition p. 1986
FX-040 Build identity	2054 / 2055 / 2056	publication identity NEXUS1-FSTR-FINAL-20260811-C158929B
FX-041 Building AlarmManagementDb	295	Chapter 14; chapter obligation
FX-042 Building RootCauseDb	318	Chapter 15; chapter obligation
FX-043 Bulkhead	1549	Ch 53; controlled definition p. 1986
FX-044 Business oracle	894 / 1578 / 1751 / 1888	Ch 36, 54, 60; App E; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-045 C# Language Specification - Microsoft	2012	Source SRC-006; LANGUAGE SPEC; Ch 6-12; code listings
FX-046 Candidate inactive	850 / 814 / 1806 / 1968	Ch 34; App B, H; controlled definition p. 1986
FX-047 Canonical asset	1786 / 1942 / 1968	App A, G-H; controlled definition p. 1986
FX-048 Canonical path	1786 / 1942 / 1968	App A, G-H; controlled definition p. 1986
FX-049 Captive dependency	152	Ch 8; controlled definition p. 1986
FX-050 CaseOpened.v1	see	see RootCauseCaseOpened.v1; canonical active fact
FX-051 Causation	1417	Ch 49; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-052 Choosing a testing strategy - EF Core	2013	Source SRC-018; OFFICIAL DOCS; Ch 12, 14-18, 55
FX-053 Choosing the First Distributed Slice	49	Chapter 3; chapter obligation
FX-054 Choreography, Orchestration, and the Decision Record	956	Chapter 38; chapter obligation
FX-055 CI Quality Gates and Container Images	1669	Chapter 57; chapter obligation
FX-056 Circuit breaker	1549	Ch 53; controlled definition p. 1986
FX-057 Claim	1774 / 1968	Epilogue; App H; controlled definition p. 1986
FX-058 Claim state	1774 / 1968	Epilogue; App H; controlled definition p. 1986

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FINAL INDEX ENTRIES C-3

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-059 Claims Minimization and Policy Mapping	1280	Chapter 46; chapter obligation
FX-060 Clean-room run	1669 / 1751 / 1942	Ch 57, 60; App G; controlled definition p. 1986
FX-061 Command	464 / 509	Ch 21, 23; controlled definition p. 1986
FX-062 Commands and Queries at a Service Boundary	464	Chapter 21; chapter obligation
FX-063 Compensation	1016	Ch 40; controlled definition p. 1986
FX-064 Compensation Without Pretending to Undo History	1016	Chapter 40; chapter obligation
FX-065 Competing update	387	Ch 18; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-066 Complete Slice and the Optional RL Advisory Branch	894	Chapter 36; chapter obligation
FX-067 Compliance	814 / 1808 / 1945	inactive module/service candidate; zero final-core authority
FX-068 Compose Specification	2015	Source SRC-057; COMPOSE SPEC; Ch 54, 60
FX-069 Concurrency token	295 / 318 / 387	Ch 14-15, 18; controlled definition p. 1986
FX-070 Configuration and Environment Boundaries	199	Chapter 10; chapter obligation
FX-071 Configuration identity	199 / 1860 / 1942	Ch 10; App D, G; controlled definition p. 1986
FX-072 Configuration providers in .NET - Microsoft	2012	Source SRC-010; OFFICIAL DOCS; Ch 10, 54

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-073 Contract	416 / 442 / 464 / 486 / 509 / 532 / 1832	Ch 19-24; App C; controlled definition p. 1986
FX-074 Contract Compatibility Tests	532	Chapter 24; chapter obligation
FX-075 Contract corpus	442 / 532 / 680	Ch 20, 24, 30; controlled definition p. 1986
FX-076 Correlation	1417	Ch 49; controlled definition p. 1986
FX-077 Correlation and Causation Across Every Boundary	1417	Chapter 49; chapter obligation
FX-078 Cover - full-bleed	1 / 2055	definitive cover and 2100 x 3000 derivative
FX-079 CQRS	464 / 486 / 850	Ch 21-22, 35; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-080 Cross-run join	1942 / 1968	App G-H; controlled definition p. 1986
FX-081 Cross-Service Identity Without Cross-Database Foreign Keys	363	Chapter 17; chapter obligation
FX-082 Cutover	266 / 1637 / 1698	Ch 13, 56, 58; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-083 DbContext lifetime, configuration and initialization - EF Core	2012	Source SRC-014; OFFICIAL DOCS; Ch 9, 14-18
FX-084 Dead-letter exchange	555 / 645	Ch 25, 29; controlled definition p. 1986
FX-085 Dead-letter queue	555 / 645 / 1888	Ch 25, 29; App E; controlled definition p. 1986
FX-086 Dedication	2	dedication to Vladimírka
FX-087 Deduplication	509 / 619	Ch 23, 28; controlled definition p. 1986
FX-088 Definition of Done for a NEXUS-1 Service	108	Chapter 6; chapter obligation
FX-089 Definitive Final Edition	3 / 2054 / 2056	final publication state of the book

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-090 Delivery attempt	594 / 645	Ch 27, 29; controlled definition p. 1986
FX-091 DEMONSTRATED	1774 / 1942 / 1968	Epilogue; App G-H; controlled definition p. 1986
FX-092 Dependency direction	152 / 175	Ch 8-9; controlled definition p. 1986
FX-093 Dependency injection guidelines - Microsoft	2012	Source SRC-008; OFFICIAL DOCS; Ch 8-9, 12
FX-094 deploy/compose.yaml	see	see deploy/compose/compose.yaml; superseded path tombstone
FX-095 deploy/compose/compose.yaml	1608 / 1794 / 1943 / 1974 / 1954 / 1981	canonical Compose model path
FX-096 Determinism	49 / 33 / 175 / 442	Ch 2, 9, 20; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-097 Dispatch, Publisher Confirms, and Recovery	594	Chapter 27; chapter obligation
FX-098 Distributed Tracing with OpenTelemetry	1489	Chapter 51; chapter obligation
FX-099 Distributed transaction	387 / 576 / 1016	Ch 18, 26, 40; controlled definition p. 1986
FX-100 DLQ	see	see Dead-letter queue; expanded term
FX-101 Domain event	416	Ch 19; controlled definition p. 1986
FX-102 Domain Events, Integration Events, and API Contracts	416	Chapter 19; chapter obligation
FX-103 Domain invariant	49 / 33 / 175 / 318	Ch 2, 9, 15; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-104 Domain, Application, Infrastructure, and Edge Inside a Service	175	Chapter 9; chapter obligation
FX-105 Dossier	1578 / 1751 / 1942	Ch 54, 60; App G; controlled definition p. 1986
FX-106 Dynamo: Amazon's Highly Available Key-value Store - DeCandia et al. (2007)	2014	Source SRC-038; FOUNDATIONAL PAPER; Ch 35, 41-42, 52

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-107 E-CORE-001 consumer kill	1892 / 1894	Appendix E mandatory recovery scenario; RootCause
FX-108 E-CORE-002 broker stop	1892 / 1894	Appendix E mandatory recovery scenario; AlarmManagement
FX-109 E-CORE-003 database pause	1892 / 1894	Appendix E mandatory recovery scenario; selected database owner
FX-110 E-CORE-004 duplicate message	1892 / 1894	Appendix E mandatory recovery scenario; RootCause
FX-111 E-CORE-005 projection delay	1892 / 1894	Appendix E mandatory recovery scenario; Reporting
FX-112 E-CORE-006 service restart	1892 / 1894 / 1959	Appendix E mandatory recovery scenario; selected service
FX-113 Edge	486 / 1182 / 1806	Ch 22, 44; App B; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-114 Edge Authentication and Operator Authorization	1182	Chapter 44; chapter obligation
FX-115 Editable source bundle	2055 / 2056	authentic code-based editorial source
FX-116 Effect	175 / 387 / 619	Ch 9, 18, 28; controlled definition p. 1986
FX-117 Efficient querying - EF Core	2013	Source SRC-019; OFFICIAL DOCS; Ch 19, 22, 35, 41
FX-118 Envelope	442 / 1832	Ch 20; App C; controlled definition p. 1986
FX-119 Epistemic honesty	1774 / 1968	Epilogue; App H; controlled definition p. 1986
FX-120 Eventual consistency	680 / 850 / 1047 / 1082	Ch 30, 35, 41-42; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-121 Eventual Consistency in the Operator Experience	1047	Chapter 41; chapter obligation
FX-122 Evidence	1375 / 1578 / 1942 / 1968	Ch 48, 54; App G-H; controlled definition p. 1986
FX-123 Evidence manifest	1751 / 1942 / 1968	Ch 60; App G-H; controlled definition p. 1986
FX-124 Exactly-once local effect	619 / 680	Ch 28, 30; controlled definition p. 1986
FX-125 EXECUTED	1774 / 1968	Epilogue; App H; controlled definition p. 1986
FX-126 EXTEND	1968	App H; controlled definition p. 1986
FX-127 Extracting Without a Big-Bang Rewrite	1637	Chapter 56; chapter obligation

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-128 Fail closed	199 / 1132 / 1182 / 1231 / 1280 / 1334 / 1375 / 1914	Ch 10, 43-48; App F; controlled definition p. 1986
FX-129 Failure campaign	1578 / 1888 / 1942	Ch 54; App E, G; controlled definition p. 1986
FX-130 Failure domain	49 / 1549 / 1578 / 1608	Ch 3, 53-55; controlled definition p. 1986
FX-131 Failure Injection and Recovery Evidence	1578	Chapter 54; chapter obligation
FX-132 FALSIFIED	1774 / 1968	Epilogue; App H; controlled definition p. 1986
FX-133 Final Consistency Audit	2037 / 2044 / 2053	72-check manuscript consistency audit
FX-134 Final core	1751 / 1806 / 1832 / 1968	Ch 60; App B-C, H; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-135 Final NEXUS-1 Distributed Demonstrator	1751	Chapter 60; chapter obligation
FX-136 Final page-aware index	2057	definitive concept-to-page navigation layer
FX-137 Final publication artifact and build identity	2054 / 2055 / 2056	definitive artifact contract
FX-138 Final Reproduction Readiness and Deferred Evidence Checkpoint	2045 / 2053	honest runtime stop condition
FX-139 FIPS PUB 180-4 - Secure Hash Standard	2016	Source SRC-063; NIST STANDARD; Ch 10, 57, 60; App G
FX-140 FloodDetected	see	see AlarmFloodDetected.v1; canonical active fact
FX-141 Freshness	850 / 1047 / 1518	Ch 35, 41, 52; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-142 From Services to Runtime - current canonical PDF and trace appendices	2011	Source SRC-003; CANONICAL ARTIFACT; Whole volume; Apps A-H
FX-143 From Seventeen Schemas to Service-Owned Databases	266	Chapter 13; chapter obligation
FX-144 Full legacy decomposition - NOT PERFORMED	2046 / 2053	654-table and 17-schema disposition remains future work

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-145 GET /api/v1/root-cause-cases/{caselId} - edge	1840 / 1834 / 1924 / 1838 / 1853	Appendix C protected public Query BFF surface
FX-146 GET /api/v1/root-cause-cases/{caselId} - owner	1838 / 1839 / 1834 / 1924 / 1840 / 1853	Appendix C protected RootCause owner query
FX-147 GET /health/live	221 / 1842 / 1843	required process-local liveness surface
FX-148 GET /health/ready	221 / 1842 / 1843	required role-readiness lifecycle surface
FX-149 GET /health/startup	221 / 1842 / 1843	required finite-startup lifecycle surface
FX-150 GET /internal/v1/root-cause-case-projections/{caselId}	1840 / 1924	Appendix C protected Reporting query for Query BFF
FX-151 Golden path	128 / 152 / 175 / 199 / 221 / 243	Ch 7-12; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-152 Golden-Path Service Template	128	Chapter 7; chapter obligation
FX-153 Graceful degradation	486 / 1047 / 1549	Ch 22, 41, 53; controlled definition p. 1986
FX-154 Ground truth	850 / 894 / 1608 / 1751	Ch 35-36, 55, 60; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-155 Handling concurrency conflicts - EF Core	2012	Source SRC-016; OFFICIAL DOCS; Ch 14-18, 42
FX-156 Handoff from From Flow to Services	16	Chapter 1; chapter obligation
FX-157 Health check	221	Ch 11; controlled definition p. 1986
FX-158 Health checks in ASP.NET Core - Microsoft (.NET 8)	2012	Source SRC-013; OFFICIAL DOCS; Ch 11, 39, 53
FX-159 High-water mark	850 / 1047 / 1082	Ch 35, 41-42; controlled definition p. 1986
FX-160 Host	128 / 152 / 221	Ch 7-8, 11; controlled definition p. 1986
FX-161 Host Composition and Dependency Direction	152	Chapter 8; chapter obligation

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-162 Human review	985	Ch 39; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-163 Idempotency	509 / 619 / 645 / 680	Ch 23, 28-30; controlled definition p. 1986
FX-164 Idempotency key	509 / 1832	Ch 23; App C; controlled definition p. 1986
FX-165 Identity digest	199 / 442 / 1942 / 1968	Ch 10, 20; App G-H; controlled definition p. 1986
FX-166 Implementation Blueprint	1 / 3 / 2045 / 2054	accurate identity for the book's evidentiary ceiling
FX-167 IMPLEMENTED	1774 / 1968	Epilogue; App H; controlled definition p. 1986
FX-168 in-toto Attestation Framework - Statement v1	2016	Source SRC-066; IN-TOTO SPEC; Ch 55, 57, 60; Apps G-H
FX-169 INACTIVE	1806 / 1968	App B, H; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-170 Inbox	619	Ch 28; controlled definition p. 1986
FX-171 Inbox and Idempotent Consumers	619	Chapter 28; chapter obligation
FX-172 Independent Migrations and Database Lifecycle	341	Chapter 16; chapter obligation
FX-173 Integration event	416 / 442 / 1832	Ch 19-20; App C; controlled definition p. 1986
FX-174 Integration tests in ASP.NET Core - Microsoft (.NET 8)	2016	Source SRC-068; OFFICIAL DOCS; Ch 12, 21-24, 55
FX-175 Invariant	49 / 33 / 175 / 318	Ch 2, 9, 15; controlled definition p. 1986
FX-176 Isolated lease	243 / 1578 / 1608 / 1942	Ch 12, 54-55; App G; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-177 Join key	1942 / 1968	App G-H; controlled definition p. 1986
FX-178 JSON Problem Details	509 / 1832	Ch 23; App C; controlled definition p. 1986
FX-179 JSON Schema Draft 2020-12 Core	2013	Source SRC-028; PINNED SPEC; Ch 20, 24; Apps C, H
FX-180 JSON Schema Draft 2020-12 Validation	2013	Source SRC-029; PINNED SPEC; Ch 20, 24; Apps C, H

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-181 Kubernetes ConfigMaps	2015	Source SRC-059; OFFICIAL DOCS; Ch 10, 54
FX-182 Kubernetes Deployments	2015	Source SRC-061; OFFICIAL DOCS; Ch 53-54, 58
FX-183 Kubernetes Jobs	2015	Source SRC-062; OFFICIAL DOCS; Ch 54, 56-60
FX-184 Kubernetes Liveness, Readiness and Startup Probes	2015	Source SRC-058; OFFICIAL DOCS; Ch 11, 39, 54
FX-185 Kubernetes Secrets	2015	Source SRC-060; OFFICIAL DOCS; Ch 10, 44-47, 54
FX-186 Kubernetes translation	1725	Ch 59; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-187 Lease	594 / 1082	Ch 27, 42; controlled definition p. 1986
FX-188 Least privilege	266 / 1132 / 1182 / 1231 / 1280 / 1334 / 1375 / 1914	Ch 13, 43-48; App F; controlled definition p. 1986
FX-189 Legacy service count - unknown	2047 / 2053	service boundaries require evidence, not table arithmetic
FX-190 Linearized PDF	1986 / 1942	Final reproduction checkpoint; controlled definition p. 1986
FX-191 Liveness	221	Ch 11; controlled definition p. 1986
FX-192 Liveness, Readiness, and Startup Dependencies	221	Chapter 11; chapter obligation
FX-193 Local transaction	387 / 576 / 619	Ch 18, 26, 28; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-194 Local Transactions, Concurrency, and Commit Boundaries	387	Chapter 18; chapter obligation
FX-195 Long-running role	152 / 1608 / 1751 / 1806	Ch 8, 55, 60; App B; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-196 Material class	1968	App H; controlled definition p. 1986
FX-197 Message identity	442 / 594 / 619 / 645 / 680	Ch 20, 27-30; controlled definition p. 1986
FX-198 Message ordering	680	Ch 30; controlled definition p. 1986
FX-199 Metrics, Service Objectives, and Dashboards	1518	Chapter 52; chapter obligation
FX-200 Migration principal	266 / 341 / 1334 / 1914	Ch 13, 16, 47; App F; controlled definition p. 1986
FX-201 Migrator	341 / 1608 / 1806	Ch 16, 55; App B; controlled definition p. 1986
FX-202 Missing	1942 / 1968	App G-H; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-203 Monotonic projection	850 / 1082	Ch 35, 42; controlled definition p. 1986
FX-204 mTLS	1231 / 1334 / 1914	Ch 45, 47; App F; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-205 Negative authority	266 / 1132 / 1182 / 1231 / 1280 / 1334 / 1375 / 1914	Ch 13, 43-48; App F-H; controlled definition p. 1986
FX-206 .NET and .NET Core Support Policy - Microsoft	2012	Source SRC-005; OFFICIAL POLICY; Ch 5, 8, 54
FX-207 .NET Generic Host - Microsoft	2012	Source SRC-007; OFFICIAL DOCS; Ch 8, 11, 26-30
FX-208 NEW	1968	App H; controlled definition p. 1986
FX-209 nexus build	1962 / 1963	Appendix G closed reproduction command
FX-210 nexus campaign	1962 / 1963	Appendix G closed reproduction command
FX-211 nexus capture	1962 / 1963	Appendix G closed reproduction command

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-212 nexus compose-render	1962 / 1963	Appendix G closed reproduction command
FX-213 nexus demonstrate	1962 / 1963	Appendix G closed reproduction command
FX-214 nexus doctor	1962 / 1963	Appendix G closed reproduction command
FX-215 nexus down	1962 / 1963	Appendix G closed reproduction command
FX-216 nexus inventory	1962 / 1963	Appendix G closed reproduction command
FX-217 nexus migrate	1962 / 1963	Appendix G closed reproduction command
FX-218 nexus plan	1962 / 1963	Appendix G closed reproduction command

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-219 nexus reproduce	1962	Appendix G closed reproduction command
FX-220 nexus restore	1962 / 1963	Appendix G closed reproduction command
FX-221 nexus seal	1962 / 1963	Appendix G closed reproduction command
FX-222 nexus test	1962 / 1963	Appendix G closed reproduction command
FX-223 nexus up	1962 / 1963	Appendix G closed reproduction command
FX-224 nexus verify	1962 / 1963	Appendix G closed reproduction command
FX-225 NEXUS-1 approved Contents and chapter sequence - author-supplied attachment	2011	Source SRC-002; AUTHOR SOURCE; Contents; Ch 1-60

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-226 NEXUS-1 Distributed Process Manager	924	Chapter 37; chapter obligation
FX-227 NEXUS-1 Phase 0 repository - Grigorios Agathangelidis	2011	Source SRC-004; PROJECT REPOSITORY; Ch 1-7, 54-60
FX-228 NEXUS-1 Phase-0	3 / 1751	Front matter; Ch 60; controlled definition p. 1986
FX-229 NEXUS-1 Project Description - author-supplied attachment	207 / 209 / 218 / 1885 / 2011 / 2017	Source SRC-001; AUTHOR SOURCE; Whole volume
FX-230 nexus.ps1	1942 / 1944 / 1962 / 1798 / 1974 / 1943	PowerShell public reproduction wrapper
FX-231 nexus.sh	1942 / 1944 / 1962 / 1309 / 1798 / 1974	POSIX public reproduction wrapper
FX-232 nexus1.alarm-management.alarm-flood-detected.v1	1843 / 444 / 747 / 782 / 1756 / 1846	Appendix C canonical event type

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-233 nexus1.root-cause.root-cause-case-opened.v1	1845 / 788 / 854 / 1834 / 789 / 790	Appendix C canonical event type
FX-234 nexus1.root-cause.root-cause-verdict-issued.v1	1846 / 822 / 854 / 913 / 1849 / 823	Appendix C canonical event type
FX-235 Nexus1.Runtime.sln	1787 / 1943 / 68 / 134 / 251 / 1788	canonical companion solution identity
FX-236 NEXUS1.sln	see	see Nexus1.Runtime.sln; superseded path tombstone
FX-237 NIST SP 800-207 - Zero Trust Architecture (2020)	2014	Source SRC-046; NIST STANDARD; Ch 44-48
FX-238 Non-claim	1774 / 1968	Epilogue; App H; controlled definition p. 1986
FX-239 NOT CLAIMED	2048 / 2053 / 2056	deliberate refusal of production-readiness assertion

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-240 NOT DEMONSTRATED	2045 / 2048 / 2053 / 2056	no runtime behavior survived an execution oracle
FX-241 NOT EXECUTED	2045 / 2046 / 2053 / 2056	no application restore, build, migration or runtime campaign ran
FX-242 NOT IMPLEMENTED	2045 / 2049 / 2053 / 2056	reference application and repository are absent
FX-243 Null evidence manifest	2049	absence recorded instead of converted to PASS

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-244 Observability	1417 / 1457 / 1489 / 1518	Ch 49-52; controlled definition p. 1986
FX-245 OCI Image Format Specification	2015	Source SRC-055; OCI SPEC; Ch 54, 57, 60
FX-246 OCI Runtime Specification	2015	Source SRC-056; OCI SPEC; Ch 54, 57
FX-247 One-writer rule	266 / 1637 / 1698	Ch 13, 56, 58; controlled definition p. 1986
FX-248 OpenAPI Specification 3.1.1 - OpenAPI Initiative (2024)	2013	Source SRC-027; PINNED SPEC; Ch 7, 21-24; App C
FX-249 OpenID Connect Core 1.0 incorporating errata set 2 (2023)	2014	Source SRC-045; FINAL SPEC; Ch 44-46
FX-250 OpenTelemetry	1489	Ch 51; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-251 OpenTelemetry Metrics Data Model	2015	Source SRC-052; STABLE SPEC; Ch 52
FX-252 OpenTelemetry Semantic Conventions	2015	Source SRC-051; EVOLVING SPEC; Ch 49-52
FX-253 OpenTelemetry Specification	2015	Source SRC-050; PROJECT SPEC; Ch 49-52, 54
FX-254 Operator	486 / 985 / 1182 / 1231 / 1280	Ch 22, 39, 44-46; controlled definition p. 1986
FX-255 Operator console	see	see Query BFF; final edge route
FX-256 Operator Console BFF	486	Chapter 22; chapter obligation
FX-257 Options pattern in .NET - Microsoft	2012	Source SRC-009; OFFICIAL DOCS; Ch 10, 44-47

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-258 Orchestrator	see	see Process manager; workflow term; not a final-core role
FX-259 Ordering, Duplication, and Contract Evolution	680	Chapter 30; chapter obligation
FX-260 Outbox	576 / 594	Ch 26-27; controlled definition p. 1986
FX-261 OWASP Application Security Verification Standard 5.0.0	2014	Source SRC-047; OWASP STANDARD; Ch 44-48, 55
FX-262 Owner	266 / 1806	Ch 13; App B; controlled definition p. 1986
FX-263 Owner truth	850 / 1047 / 1082	Ch 35, 41-42; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-264 PENDING	1774 / 1942 / 1968	Epilogue; App G-H; controlled definition p. 1986
FX-265 Poison message	645	Ch 29; controlled definition p. 1986
FX-266 Policy digest	199 / 1860 / 1942	Ch 10; App D, G; controlled definition p. 1986
FX-267 POST /api/v1/alarm-observations	509 / 1834 / 1837 / 714 / 1760 / 1810	Appendix C protected owner command surface
FX-268 POST /api/v1/root-cause-cases/{caseld}/evidence	1838 / 1924	Appendix C protected RootCause command surface
FX-269 POST /api/v1/root-cause-cases/{caseld}/verdicts	1838 / 1839 / 1924	Appendix C protected RootCause command surface
FX-270 Principles of Chaos Engineering	2016	Source SRC-071; FOUNDATIONAL GUIDANCE; Ch 53, 55, 60

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FINAL INDEX ENTRIES P-2

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-271 Problem Details	509 / 1832	Ch 23; App C; controlled definition p. 1986
FX-272 Process manager	924 / 956 / 985 / 1016 / 1806 / 1968	Ch 37-40; App B, H; controlled definition p. 1986
FX-273 Production readiness - NOT CLAIMED	2045 / 2053 / 2056	Phase-0 educational boundary remains controlling
FX-274 Projection	850 / 1047 / 1082	Ch 35, 41-42; controlled definition p. 1986
FX-275 Projection rebuild	850 / 1082	Ch 35, 42; controlled definition p. 1986
FX-276 projection-api	850 / 1818 / 1945 / 1611 / 1727 / 1761	Appendix B/G runtime role; Reporting read surface
FX-277 projection-worker	850 / 1818 / 1945 / 1611 / 1727 / 1761	Appendix B/G runtime role; Reporting projection writer

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FINAL INDEX ENTRIES P-3

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-278 Prometheus Data Model	2015	Source SRC-053; OFFICIAL DOCS; Ch 52
FX-279 Prometheus Histograms and Summaries	2015	Source SRC-054; OFFICIAL DOCS; Ch 52-53; Notation
FX-280 Provider-backed test	243 / 295 / 318 / 341 / 363 / 387	Ch 12, 14-18; controlled definition p. 1986
FX-281 Public fact	416 / 442 / 1832	Ch 19-20; App C; controlled definition p. 1986
FX-282 Publication reproduction	2056	book build and QA only; distinct from application reproduction
FX-283 Publisher confirm	594	Ch 27; controlled definition p. 1986
FX-284 Publishing AlarmFloodDetected.v1	744	Chapter 32; chapter obligation

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-285 Quarantine	645 / 1888	Ch 29; App E; controlled definition p. 1986
FX-286 Query	464 / 486 / 850 / 1047	Ch 21-22, 35, 41; controlled definition p. 1986
FX-287 Query BFF	486 / 1047 / 1182 / 1806 / 1832	Ch 22, 41, 44; App B-C; controlled definition p. 1986
FX-288 query-bff	486 / 1822 / 1945 / 1611 / 1688 / 1727	Appendix B/G edge role; see also Query BFF

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-289 RabbitMQ Consumer Acknowledgements and Publisher Confirms	2013	Source SRC-031; OFFICIAL DOCS; Ch 20, 27-30, 36
FX-290 RabbitMQ Dead Letter Exchanges	2013	Source SRC-033; OFFICIAL DOCS; Ch 29-30, 36, 53
FX-291 RabbitMQ Quorum Queues	2013	Source SRC-032; OFFICIAL DOCS; Ch 27, 36, 54
FX-292 RabbitMQ Reliability Guide	2013	Source SRC-030; OFFICIAL DOCS; Ch 20, 26-30, 36
FX-293 RabbitMQ Time-To-Live and Expiration	2014	Source SRC-034; OFFICIAL DOCS; Ch 29, 39, 53
FX-294 RabbitMQ Topology for NEXUS-1	555	Chapter 25; chapter obligation
FX-295 Readiness	221 / 1608	Ch 11, 55; controlled definition p. 1986

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FINAL INDEX ENTRIES R-2

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-296 Reauthentication	1182 / 1914	Ch 44; App F; controlled definition p. 1986
FX-297 Reconciliation	1082	Ch 42; controlled definition p. 1986
FX-298 Reconciliation and Repair Jobs	1082	Chapter 42; chapter obligation
FX-299 Redelivery	619 / 645 / 680	Ch 28-30; controlled definition p. 1986
FX-300 Reference slice	2047 / 2053	three owners, one edge, seven roles, three stores and three facts
FX-301 Repair	1082 / 1888	Ch 42; App E; controlled definition p. 1986
FX-302 Repair Control	1082 / 1806 / 1968	Ch 42; App B, H; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-303 Repair service	see	see Repair Control; inactive protected control
FX-304 Replay	645 / 680 / 1888	Ch 29-30; App E; controlled definition p. 1986
FX-305 Reporting	850 / 1808 / 1818	final-core derived owner of ledger, checkpoint and freshness-bearing projection truth
FX-306 Reporting Builds Delayed Truth	850	Chapter 35; chapter obligation
FX-307 Reporting service	see	see Reporting; canonical derived owner name
FX-308 reporting-migrator	341 / 1945 / 1954 / 1791 / 1807 / 1877	finite migration job for ReportingDb
FX-309 Repository and Solution Topology	65	Chapter 4; chapter obligation

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FINAL INDEX ENTRIES R-4

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-310 Reproducible Docker Compose Runtime	1608	Chapter 55; chapter obligation
FX-311 Reproduction identity	1608 / 1669 / 1751 / 1942	Ch 55, 57, 60; App G; controlled definition p. 1986
FX-312 Residual	1774 / 1942 / 1968	Epilogue; App G-H; controlled definition p. 1986
FX-313 Retry	645 / 1549	Ch 29, 53; controlled definition p. 1986
FX-314 Retry budget	645 / 985 / 1549	Ch 29, 39, 53; controlled definition p. 1986
FX-315 Retry, Backoff, Dead Letters, and Poison Messages	645	Chapter 29; chapter obligation
FX-316 RFC 3339 - Date and Time on the Internet: Timestamps (2002)	2013	Source SRC-025; IETF STANDARD; Ch 20, 39, 49; Notation

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FINAL INDEX ENTRIES R-5

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-317 RFC 6749 - The OAuth 2.0 Authorization Framework (2012)	2014	Source SRC-042; IETF STANDARD; Ch 44-46
FX-318 RFC 7519 - JSON Web Token (JWT) (2015)	2014	Source SRC-040; IETF STANDARD; Ch 44-46
FX-319 RFC 8259 - The JavaScript Object Notation Data Interchange Format (2017)	2013	Source SRC-023; IETF STANDARD; Ch 9-10, 20, 24
FX-320 RFC 8446 - The Transport Layer Security Protocol Version 1.3 (2018)	2014	Source SRC-039; IETF STANDARD; Ch 44-47
FX-321 RFC 8705 - OAuth 2.0 Mutual-TLS Client Authentication and Certificate-Bound Access Tokens (2020)	2014	Source SRC-044; IETF STANDARD; Ch 45-47
FX-322 RFC 8725 - JSON Web Token Best Current Practices (2020)	2014	Source SRC-041; IETF BCP; Ch 44-46
FX-323 RFC 8785 - JSON Canonicalization Scheme (2020)	2013	Source SRC-024; IETF STANDARD; Ch 10, 20, 57; Apps G-H

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FINAL INDEX ENTRIES R-6

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-324 RFC 9110 - HTTP Semantics (2022)	2013	Source SRC-021; IETF STANDARD; Ch 7, 9, 21-24
FX-325 RFC 9457 - Problem Details for HTTP APIs (2023)	2013	Source SRC-022; IETF STANDARD; Ch 9, 21-24, 46
FX-326 RFC 9562 - Universally Unique IDentifiers (UUIDs) (2024)	2013	Source SRC-026; IETF STANDARD; Ch 20, 23, 27-30
FX-327 RFC 9700 - Best Current Practice for OAuth 2.0 Security (2025)	2014	Source SRC-043; IETF BCP; Ch 44-46
FX-328 RL Advisory	894 / 1808 / 1945	inactive optional module; RecommendationGenerated.v1 not activated
FX-329 Rollout, Database Change, and Rollback	1698	Chapter 58; chapter obligation
FX-330 Root Cause service	see	see RootCause; canonical owner name

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FINAL INDEX ENTRIES R-7

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-331 RootCause	318 / 778 / 814 / 1806	Ch 15, 33-34; App B; controlled definition p. 1986
FX-332 RootCause Consumes and Opens a Case	778	Chapter 33; chapter obligation
FX-333 rootcause-api	778 / 1814 / 1945 / 1589 / 1611 / 1727	Appendix B/G runtime role; see also RootCause
FX-334 rootcause-migrator	341 / 1945 / 1954 / 1790 / 1813 / 1880	finite migration job for RootCauseDb
FX-335 rootcause-worker	619 / 1814 / 1945 / 1611 / 1727 / 1761	Appendix B/G runtime role; consumer and outbox relay
FX-336 RootCauseCase	175 / 318 / 778 / 814	Ch 9, 15, 33-34; controlled definition p. 1986
FX-337 RootCauseCaseOpened.v1	778 / 1809 / 1834 / 777 / 859 / 895	active public fact owned by RootCause

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FINAL INDEX ENTRIES R-8

FX ID / preferred term	Principal pages	Locator / cross-reference
FX-338 RootCauseDb	318 / 341 / 387 / 619	Ch 15-16, 18, 28; controlled definition p. 1986
FX-339 RootCauseVerdictIssued.v1	814 / 1809 / 1834 / 36 / 49 / 70	active public fact owned by RootCause
FX-340 RunId	1942 / 1968	App G-H; controlled definition p. 1986
FX-341 Runtime reproduction - NOT EXECUTED	2045 / 2053 / 2056	no build, database, broker or container receipt
FX-342 Runtime role	1608 / 1751 / 1806	Ch 55, 60; App B; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-343 Sagas - Garcia-Molina and Salem, ACM SIGMOD (1987)	2014	Source SRC-036; FOUNDATIONAL PAPER; Ch 32-38, 41-42
FX-344 Same-run evidence	1578 / 1751 / 1942 / 1968	Ch 54, 60; App G-H; controlled definition p. 1986
FX-345 Schema evolution	532 / 680 / 1698	Ch 24, 30, 58; controlled definition p. 1986
FX-346 Secret reference	199 / 1334 / 1860 / 1914	Ch 10, 47; App D, F; controlled definition p. 1986
FX-347 Secrets, Certificates, and Configuration Handling	1334	Chapter 47; chapter obligation
FX-348 Security Tests and Audit Evidence	1375	Chapter 48; chapter obligation
FX-349 Semantic conflict	387 / 464 / 509	Ch 18, 21, 23; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-350 Semantic retry	295 / 387 / 1549	Ch 14, 18, 53; controlled definition p. 1986
FX-351 Service	49 / 108 / 128 / 152 / 175 / 266	Ch 3, 6-9, 13; controlled definition p. 1986
FX-352 Service catalogue	1751 / 1806	Ch 60; App B; controlled definition p. 1986
FX-353 Service count	2047 / 2053	not inferred from 654 tables or 17 schemas
FX-354 Service-owned database	266 / 295 / 318 / 341 / 363 / 387	Ch 13-18; controlled definition p. 1986
FX-355 Service-to-Service Identity	1231	Chapter 45; chapter obligation
FX-356 Shared Nothing, Shared Little	87	Chapter 5; chapter obligation

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-357 SLO (Service Level Objective)	1518	Ch 52; controlled definition p. 1986
FX-358 SLSA Build Provenance 1.2	2016	Source SRC-065; SLSA SPEC; Ch 57, 60; App G
FX-359 SLSA Specification 1.2	2016	Source SRC-064; SLSA SPEC; Ch 54, 57, 60; App G
FX-360 Snapshot isolation in SQL Server - Microsoft	2013	Source SRC-020; OFFICIAL DOCS; Ch 14-18, 42, 56
FX-361 Source checkpoint SHA-256	2054 / 2056	hash of the admitted 2,053-page source checkpoint
FX-362 SPDX Specification 3.0.1	2016	Source SRC-067; SPDX SPEC; Ch 54, 57, 60
FX-363 SPECIFIED	1774 / 1968	Epilogue; App H; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-364 Startup	221	Ch 11; controlled definition p. 1986
FX-365 Strangler migration	1637	Ch 56; controlled definition p. 1986
FX-366 Structured logging	1457	Ch 50; controlled definition p. 1986
FX-367 Structured Logs That Preserve Domain Meaning	1457	Chapter 50; chapter obligation
FX-368 SUPERSEDED	1786 / 1942 / 1968	App A, G-H; controlled definition p. 1986
FX-369 System.Diagnostics.Stopwatch - Microsoft .NET API	2016 / 2017	Source SRC-074; OFFICIAL API; Ch 11, 39, 53; Notation

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-370 Testcontainers for .NET	2016	Source SRC-069; PROJECT DOCS; Ch 12, 14-18, 55
FX-371 Testing Pyramid for a Distributed Service	243	Chapter 12; chapter obligation
FX-372 The International System of Units (SI), 9th edition, version 4.01 (2026)	2016	Source SRC-072; BIPM STANDARD; Ch 39, 52-53; Notation
FX-373 Threat and Trust Boundaries Before Mechanisms	1132	Chapter 43; chapter obligation
FX-374 Time, Clocks, and the Ordering of Events in a Distributed System - Lamport (1978)	2014	Source SRC-037; FOUNDATIONAL PAPER; Ch 20, 30, 39, 49
FX-375 Timeout	985 / 1549	Ch 39, 53; controlled definition p. 1986
FX-376 Timeout, Retry, Circuit Breaker, and Bulkhead	1549	Chapter 53; chapter obligation

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-377 Timeouts, Escalations, and Human Review	985	Chapter 39; chapter obligation
FX-378 Tombstone	1786 / 1914 / 1942 / 1968	App A, F-H; controlled definition p. 1986
FX-379 Toxiproxy - Shopify	2016	Source SRC-070; PROJECT DOCS; Ch 39, 53, 55
FX-380 Trace	1417 / 1489	Ch 49, 51; controlled definition p. 1986
FX-381 Trace context	1417 / 1489	Ch 49, 51; controlled definition p. 1986
FX-382 Traceability	1968	App H; controlled definition p. 1986
FX-383 Transaction boundary	387 / 576 / 619	Ch 18, 26, 28; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-384 Transactional outbox	576	Ch 26; controlled definition p. 1986
FX-385 Transactional Outbox in Each Producer	576	Chapter 26; chapter obligation
FX-386 Translating Proven Needs to Kubernetes	1725	Chapter 59; chapter obligation
FX-387 Trust boundary	1132 / 1914	Ch 43; App F; controlled definition p. 1986
FX-388 Trust zone	1132 / 1914	Ch 43; App F; controlled definition p. 1986
FX-389 Typed identity	175 / 363 / 442	Ch 9, 17, 20; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-390 Ubiquitous language	49 / 33 / 175 / 266	Ch 2, 9, 13; controlled definition p. 1986
FX-391 Unit of work	295 / 387	Ch 14, 18; controlled definition p. 1986
FX-392 Unknown outcome	594 / 680 / 1549	Ch 27, 30, 53; controlled definition p. 1986
FX-393 Using transactions - EF Core	2012	Source SRC-015; OFFICIAL DOCS; Ch 14-20, 26, 28

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-394 Value-free manifest	199 / 1860 / 1942	Ch 10; App D, G; controlled definition p. 1986
FX-395 Verdict	175 / 318 / 814	Ch 9, 15, 34; controlled definition p. 1986
FX-396 VerdictIssued.v1	see	see RootCauseVerdictIssued.v1; canonical active fact
FX-397 VERIFY	1968	App H; controlled definition p. 1986
FX-398 Versioned contract	442 / 509 / 532 / 680	Ch 20, 23-24, 30; controlled definition p. 1986
FX-399 Versioned Message Envelope	442	Chapter 20; chapter obligation
FX-400 Versioning, Problem Details, and Idempotency Keys	509	Chapter 23; chapter obligation

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-401 W3C Baggage Candidate Recommendation Snapshot (2024)	2015	Source SRC-049; W3C CANDIDATE; Ch 49-51
FX-402 W3C Trace Context Recommendation (2021)	2015	Source SRC-048; W3C STANDARD; Ch 49-51
FX-403 What Must Remain True After Extraction	33	Chapter 2; chapter obligation
FX-404 Worker	152 / 594 / 619 / 850	Ch 8, 27-28, 35; controlled definition p. 1986
FX-405 Worker Services in .NET - Microsoft	2012	Source SRC-011; OFFICIAL DOCS; Ch 8, 26-30, 53
FX-406 Working index	2018	Final back matter; controlled definition p. 1986
FX-407 Workload identity	1231 / 1280 / 1914	Ch 45-46; App F; controlled definition p. 1986

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FX ID / preferred term	Principal pages	Locator / cross-reference
FX-408 Zero-discovery gate	243 / 532 / 1669 / 1942	Ch 12, 24, 57; App G; controlled definition p. 1986
FX-409 Zero-trust stance	1132 / 1182 / 1231 / 1280 / 1334 / 1375 / 1914	Ch 43-48; App F; controlled definition p. 1986

FINAL INDEX INTEGRITY AUDIT

Every final locator and redirect is mechanically checked

FINAL-INDEX ACCEPTANCE FX-4

Gate	Required result
identity	contiguous FX identifiers and no case-insensitive duplicate preferred term
destination	every direct entry has an in-range page and every printed page is internally linked
cross-reference	every see target exists exactly once and links to its preferred final-index entry
coverage	381 inherited Working Index entries plus all declared final additions
pagination	all destination pages use the final visible page labels
architecture	three owners, one edge and all inactive candidates retain their final dispositions
claim state	publication completion does not promote application or runtime evidence

DEFINITIVE INDEX CLOSURE

The final edition closes with exact, falsifiable navigation

COMPLETION LEDGER FX-5

Question	Final answer
Where is the concept engineered?	curated principal pages
Where is the exact role, route, fact or command fixed?	catalogue or controlled-reference page
Where is an external authority documented?	stable Sources entry
Where is the publication identity fixed?	Final Publication Artifact and Build Identity
Where are runtime absences recorded?	Final Reproduction Readiness checkpoint and final status entries
Does the final index prove runtime behavior?	no; evidence state remains independently derived

FINAL INDEX RULE

The definitive index ends the manuscript because any later content would change the pagination it certifies. Final artifact hashes remain in the external manifest, where they can identify the delivered bytes without circularity.

FINAL INDEX LAST WORD

Navigate by authority; trust only retained evidence

FINAL STATUS

FINAL PAGE-AWARE INDEX COMPLETE | 409 ENTRIES | 25 GROUPS | INTERNAL PAGE AND SEE LINKS | FINAL-CORE AND CLAIM-STATE BOUNDARIES PRESERVED